Hecke Module Extensions in $H^*(\mathrm{SL}_2(\mathbb{Z}); \mathrm{Sym}^n \mathbb{F}_\ell^2)$

Computational classification of all extension classes, in all computable degrees, over $\mathbb{F}_\ell$ and over $\mathbb{F}_{\ell^r}$ — a concise mathematical specification

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Abstract

This document is the programme specification that accompanies the binding mathematical contract `gl2_contract.tex`, cited as [C]. In [C]'s notation and normalisations it states what it means to compute and classify *all* extension classes of Hecke modules occurring in the cohomology groups the project computes, and how to do it with deterministic certificates at every step. The question which T_q are needed is reduced to a certified local statement, one maximal ideal at a time: the operators computed must generate the local Hecke algebra $\mathbb{T}_\mathfrak{m}$. The $\mathbb{T}_\mathfrak{m}$-modules $H^\bullet_\mathfrak{m}$ are recorded by exhibiting them, with the invariants that make two records comparable (Loewy and socle series, active tangent directions, Jordan types, positions of the canonical submodules): the classification data (D1)–(D7). Taking the six cohomology groups and the sequences between them from [C], it treats the Hecke algebra acting, with its separation and generation bounds; primary decomposition and the local algebras; the classification data and the predictions; the algorithms stage by stage, each with its certificate; scalar extension to $\mathbb{F}_{\ell^r}$; the output record schema, over $\mathbb{F}_\ell$ and $\mathbb{Z}/\ell^k$; first test cases; and an interface for L-values, with its Eisenstein instance, measures and truncated ℓ-adic L-functions. Every claim carries a STATUS tag; the tags are indexed in Appendix A.

Contents

0 Conventions and standing assumptions **2**

1 The cohomologies computable now, and the sequences relating them **3**

2 The Hecke algebra acting **8**

3 Primary decomposition and the local algebras $\mathbb{T}_\mathfrak{m}$ **12**

4 What "all extension classes" means **14**

5 The algorithms, stage by stage, with certificates **36**

6 Scalar extension to $\mathbb{F}_{\ell^r}$ **53**

7 Output schema **56**

8 First test cases and expected outcomes **105**

9 The L-value interface, and its Eisenstein instance **108**

A Index of claims with STATUS **141**

B Audit response **150**

C OQ2 verdict response and the owner's decisions **151**

D Revision history **154**

0 Conventions and standing assumptions

0.1 Notation dictionary

The two source notes use p for the coefficient prime and k for the symmetric-power degree. This document follows the contract [C] instead, because the implementation does:

object	this document / [C]	[M], [S]
coefficient prime (characteristic of the field)	ℓ (`ell`)	p
Hecke prime (index of T_q)	q, or p (`p_hecke`)	q
symmetric-power degree	n (`n_sym`)	k
modular weight	$k := n+2$	$k+2$
coefficient module	$M_n := \mathrm{Sym}^n \mathbb{F}_\ell^2$	$M = M_k(\mathbb{F}_p)$

Every formula quoted from [M] is rewritten with $k \mapsto n$, $p \mapsto \ell$.

0.2 The standing conventions of [C]

[C] is `gl2_contract.tex`, the unified binding mathematical contract (this revision cites it at **v1.7.3**, 2026-09-22 — the slot-6 close, status only, on top of v1.7.2; before that slot 5's X revision v1.7.0, its referee pass v1.7.1 and the slot close v1.7.2, which is status only, on top of slot 4's X revision v1.6.0, which resolves the eight items [C] v1.5.2 deferred, its referee pass v1.6.1 and slot 4's close v1.6.2, and the slot-3 chain v1.5.0 / v1.5.1 / v1.5.2; *none of the eight adds, renames or removes a label*, so every label cited here exists there and the v1.3.3 labels are all preserved. What v1.6.0 changes that this document depends on: [C, `obl:coneexact`](i) is a statement on the stored equivariant-cochain basis with the flat identities recorded, (iv)'s explicit im = ker check is degree-generic, (v) and [C, `eq:conedims`]'s first identity no longer need $H^2_{\mathrm{eq}} = 0$, and [C, `sec:recordschema`] gains item 6, the extension record and its six rules — which §7 of this document implements. What v1.7.0 changes: [C, `obl:coneexact`](iii)'s $\ell \in \{2,3\}$ carve-out on the restriction half is *dropped*, the identity asserted at every ℓ with the proof written in, and [C, `sec:recordschema`]'s items are restated for the record schema `hecke_ext/v2`, which §7 now names), which supersedes `hecke_action.tex` (the document [C] named through v1.2.2) and `monodromy_v5.tex` (cited here as [v5]); both are frozen. Every label this document cites as [C, ·] is defined there.

- $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, $\Delta = \{g \in M_2(\mathbb{Z}) : \det g > 0\}$; $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, ST, TS as in [C, `sec:model`]. $\Gamma_\infty = \langle T, -I \rangle$.
- $\rho_n(A) = \mathrm{Sym}^n(A)$ in the *column* convention $A \cdot x = ax+cy$, $A \cdot y = bx+dy$, basis $u_j = x^{n-j}y^j$, no determinant twist, extended multiplicatively to the monoid Δ [C, `def:rhon`], [C, `prop:symconv`]. This coincides with the action of [M §0].
- $\ell \geq 5$ throughout (hypothesis (H3) of [C, `sec:conventions`]), so that $H^\bullet(K, R; L) \cong H^\bullet(\Gamma; M_n)$ [C, `rmk:orbifold`]. n is *even*: for odd n the central relation forces $H^\bullet = 0$ [C, `rmk:negI`], and the pipeline returns empty matrices.
- $T_p = T_{\Gamma\,\mathrm{diag}(1,p)\Gamma}$ is the operator of [C, `def:hecke`], assembled by [C, `alg:assembly`], normalised by $T_p|_{H^0(\mathrm{Sym}^0)} = p+1$, the Eisenstein eigenvalue $1 + p^{n+1}$ on H^1, and the recursion $T_{p^{r+1}} = T_p T_{p^r} - p^{n+1} T_{p^{r-1}}$ for the *full* operators [C, `thm:relations`] ($T_{p^2} = T_{\Delta^{\mathrm{prim}}_{p^2}} + p^n I$). The bad-prime operator is $U_\ell := T_\ell$ [C, `cor:ulrecursion`]. Prime indices suffice: $T_{p^r} \in \mathbb{F}_\ell[T_p]$ and $T_{MN} = T_M T_N$.
- The *convention warning* of [M §4] (operator vs. its transpose) is settled by [C]: the oracles $a_2(\Delta) = -24$ at $n = 10$ and $1 + p^{n+1}$ on the Eisenstein class are part of the acceptance gate

(`HeckeOracleValidation`, `HeckeEndToEndValidation`). Consequently the cuspidal eigenvalues on H^1 are $a_q(f)$ and the boundary characters of §1.3 are as stated there, not swapped.

0.3 Sym versus divided powers

For $n \geq \ell$ the Γ-module M_n is not self-dual; its dual is the divided-power module $D_n = (\mathrm{Sym}^n \mathbb{F}_\ell^2)^\vee$ [M §0]. The pipeline computes M_n today; D_n enters through the duality *predictions* of [C, `cl:pd`] until it is *computed* — as a first-class contragredient local system $\rho_n^\vee(g) = \rho_n(\mathrm{adj}\, g)^T$ with the pairing map at cochain and cohomology level, [C, `def:contragredient`], [C, `prop:pairing`] — in phase E7 of [X] v1.5.1 (slots 6–7; owner decision OQ4, decided). The adjointness $\langle A_c(T_p)x, y\rangle = \langle x, A(D_n)(T_p)y\rangle$ of [C, `cl:pd`](2) is then the acceptance gate for X_3 at $\ell \geq 5$ (owner decision OP3), and the socle↔Loewy reversal of Prediction 4.6 is asserted by `HeckeDualityValidation` (v1.4.0; through v1.3.3 this sentence read "D_n enters solely through the predictions" and cited Open Question 4 as open).

0.4 What "certified" means

Repository policy (`FpPoly.h`, `HeckeModuleDecomposition.h`): randomisation may *find* an object (a factor, a witness, a cyclic vector) but never *certifies* it; every reported statement is re-derived by a deterministic, unconditional check that a third party can repeat from the emitted data alone. Every stage below names its certificate.

1 The cohomologies computable now, and the sequences relating them

1.1 The six groups, and what [C] proves about them

Recalled, not claimed here. For $M = M_n$ the **six groups** are

$$X_1 = H^0(\Gamma; M), \qquad X_2 = H^0(\partial; M), \qquad X_3 = H^1_c(\Gamma; M),$$
$$X_4 = H^1(\Gamma; M), \qquad X_5 = H^1(\partial; M), \qquad X_6 = H^2_c(\Gamma; M),$$

with $H_{\mathrm{tot}} := \bigoplus_i X_i$, $d := \dim H_{\mathrm{tot}}$, $H^\bullet(\partial; M) := H^\bullet(\Gamma_\infty; M)$ and $H^\bullet_c$ the cohomology of the (shifted) cone of the restriction $r^\bullet$. This is [C, `eq:sixgroups`] of [C, `sec:six`] verbatim; the two defining pointers are [C, `def:bdrygroup`] for the boundary groups and [C, `def:pair`] with [C, `lem:conerelative`](3) for the compactly supported ones. The six-term exact sequence relating them is [C, `eq:les`], its dimension bookkeeping [C, `cl:les`](2), and $H^1_! := \ker r^1 = \mathrm{im}(X_3 \to X_4)$ is defined there. [C] writes the restriction maps r^0, r^1 with the same letter and the T-invariant $N = y^\ell - x^{\ell-1}y$ as Nm, reserving $N = p^\nu$ for the Hecke level (its notation collision C18).

Nine statements moved into [C], and why this document no longer states them. The amalgam presentation, the dimension of H^1 with its Dickson correction, the torsion origin of the Dickson classes, the boundary groups, the six-term sequence, the boundary characters and Poincaré duality are *facts about the objects* [C] constructs, not part of the classification programme this document specifies. The scope cut that moved them is recorded in [C] as erratum E18 ([C, `err:e18`]). Until v1.3 this document kept its own copies, so nine labels were defined in both places and a citation of the form `[C, cl:pd]` did not resolve to a unique statement; v1.3 removes the copies. Nothing was lost: every one of the nine is preserved in [C] under the same label, five of them now carrying a proof this document did not give — the amalgam presentation, the Bockstein, the boundary groups, the six-term sequence, and clauses (1) and (3) of the boundary characters.

statement, as this document stated it through v1.2.2	now cited as
§1.1 "Six groups" — the six X_i, H_{tot}, d	[C, `sec:six`], [C, `eq:sixgroups`]
Claim "Amalgam presentation"	[C, `cl:amalgam`]
Claim "Dimension of H^1"	[C, `cl:dimH1`]
Claim "Torsion origin of the Dickson classes"	[C, `cl:bockstein`]
Claim "Boundary groups"	[C, `cl:boundary`]
Claim "The sequence (3.1)"	[C, `cl:les`]
the displayed sequence of that claim	[C, `eq:les`]
Claim "Boundary eigenvalues"	[C, `cl:chi`]
Claim "Poincaré duality"	[C, `cl:pd`]

Four of the nine are corrected in [C]; the v1.2.2 forms are superseded. A reader holding an older PDF of this document should read these four in [C] and not here.

1. **The Dickson-locus sentence of the boundary-eigenvalue claim is false** ([C] erratum E21, [C, `rmk:chicoincide`]). Through v1.2.2 this document concluded "$\chi_1 \equiv \chi_2$ at every Dickson locus". On the Dickson locus $n = a(\ell+1) + b(\ell^2-\ell)$ one has $n \equiv 2a \pmod{\ell-1}$, so the criterion is $(\ell-1) \mid (n-2)$, *not* always: it holds at the first Dickson class $n = \ell+1$ — this document's own example $(n,\ell) = (6,5)$ — and fails already at $(12,5)$, where $\chi_1(2) \equiv 3$ and $\chi_2(2) \equiv 2$ in $\mathbb{F}_5$. What survives is the use it was put to: the first Dickson class never separates the left- from the right-coset indexing, so the separation comes from the Eisenstein eigenvalue $1+q^{n+1}$ ([C, `prop:bdrychar`](3)), which is how (A1) settles the convention.
2. **"$H^i(\partial; M) = 0$ for $i \geq 2$" is false at $\ell = 2$** ([C, `rmk:bdryrepairs`](i), [C, `lem:ell2unbounded`]). The hypothesis that carries it is $2 \in R^\times$, and [C, `cl:boundary`] now states it. This is the same $\ell = 2$ unboundedness this document records in (A2); under its standing hypothesis $\ell \geq 5$ nothing here is affected.
3. **The hypotheses of the six-term sequence changed in both directions** ([C, `rmk:sixscope`]). $H^0_c = 0$ is *unconditional* — [C, `lem:conerelative`](4) proves it with no hypothesis on ℓ and none on n — so [C] does not list it, where this document did; and "n even" is needed for the six groups to be non-zero, not for exactness. But $H^2(\Gamma; M) = 0$ is *kept*, and is not removable: [C, `lem:h2torsion`](2) exhibits $(\ell, n) = (2, 0)$, a pair with n even, at which $H^q(\Gamma; \mathbb{F}_2) \neq 0$ for every $q \geq 0$.
4. **Poincaré duality holds in the adjointness form only** ([C, `rmk:pdscope`](1)). This document's "T_q self-adjoint once D_n carries the adjugate Δ-structure $g \mapsto \det(g) g^{-1}$" is true of the adjointness identity $\langle \rho_n(g)v, \varphi\rangle = \langle v, \rho_n^\vee(\det(g)g^{-1})\varphi\rangle$, which is exact, integral and carries no determinant factor, and *false* of the diagonal one $\langle \rho_n(g)v, \rho_n^\vee(g)\varphi\rangle = (\det g)^n \langle v, \varphi\rangle$, which carries the twist q^n for $g = \mathrm{diag}(1, q)$. [C, `cl:pd`](2) is stated in the former, and it is that form Prediction 4.6 and identity 5 of §5.1 rest on. [C] also carries the claim as OPEN (owner Open Question 4): its coefficient-level input [C, `prop:pairing`] is proved and its cohomological content is not.

1.2 What the code computes today

The following is a statement about the tree at commit `d54c589` (branch `matrix-engine`), checked against `Hecke/README.md` and the headers, and agreeing with the independent audit `docs/implementation/hecke_extensions_capability_inventory.md`.

- X_1, X_4 and $H^2(\Gamma; M)$ $(= 0$, asserted as a 0×0 block) on the 7-vertex model, with bases, projectors and the Hecke matrices $A^{(q)}(T_p)$ for $q \in \{0, 1, 2\}$ and every prime p (including $p = \ell$), plus $T_{p^r}, T_n, T(p,p)$ (`HeckeOperatorFamily`). This is the whole *level (b)* of the plan.
- X_2, X_5 as equivariant cohomology of the circle with monodromy $\rho_n(T)$ (`LocalSystemPresets::CreateS1Generic`), and the restriction maps r^0, r^1 as matrices between the stored bases (`EquivariantCochainComplexMorphismOnK`, exercised by `TestSL2Z_To_S1Cusp_RestrictionMorphism`). **No Hecke action is implemented on** X_2, X_5: `HeckeOperatorOnK` is bound to the 7-vertex model (`RequireModel(SL2Z7VertexModel())`).
- X_3, X_6 **are not computable yet**: the mapping cone exists only for the non-equivariant `CochainComplex` (`SimplicialCohomology.h`, `ConeComplex()`), not for `EquivariantCochain`

ComplexMorphismOnK. The algebraic cone for equivariant complexes is [C, def:pair] and [C, lem:conerelative], and its Hecke action [C, def:coneaction], [C, thm:conewelldef] ([C, sec:cone]); building it is a mechanical addition. ([v5] §8 carried the construction before the scope cut and is now frozen.)

- The involution $\iota = \mathrm{diag}(-1, 1)$ [M §0] has **no production implementation**; it is built, and [C] [C, obl:iota] is discharged, by the two validations HeckeIotaTauSymmetryValidation and HeckeIotaSignValidation.
- Level (c): the primary decomposition of H^q under one operator or a commuting family, with certified irreducible factors, generalized-Jordan partitions, residue degrees (§3), and the JSON report. No local algebra, tangent space, Loewy or socle series, commutant, or $\mathbb{F}_{\ell^r}$ arithmetic exists: Stages 3–5 of §5 are entirely new code (inventory §1); the substrate they need (Matrix products / RREF / kernel, SolveColumnsExact, FpPolyT) is present.

Addendum, v1.3.3 (2026-09-16, the close of slot 1 of the extension stage; branch hecke-extensions, 77 registered validations). The statement above is kept as the record of d54c589; what slot 1 changed, with no golden value moved:

- **A Hecke action on X_2, X_5 now exists**: Hecke/HeckeBoundaryOperator computes Θ^{∂}_g of [C] [C, def:bdryop] on the circle model from the certified Γ-level τ_∂ of Hecke/HeckeBoundaryDatum ([C] [C, def:taubdry]), with [C] [C, obl:taubdry] discharged in full by HeckeBoundaryTauValidation and HeckeBoundaryOperatorValidation; the record fields boundary_model = circle and boundary_is_group_cohomology = ($\ell \neq 2$) say which object is computed. The operator is not yet integrated into the report of §7 (E4b, slot 2), and the groups H0_bd, H1_bd stay unavailable in the record until then.
- X_3, X_6 **are still not computable**: no η, no equivariant cone.
- **The locality certificate of Certificate 5.2 exists as a read-only function** (hecke::CertifyLocality, called by nothing in compute() until E0b): on the whole golden grid (236 decompositions, 348 blocks) every block certifies local with the shipped witness, so E0b's fix moves no decomposition hash (§3.2's defect is real — 48 constructed non-local families that the shipped predicate accepts are rejected — but does not occur on the grid); and the $(160, 691)$ probe verdict is at Claim 2.13.
- **The Sturm bound $B'(k, \ell)$ is computed** (HeckeExt/HeckeSturmBound, the first unit of the extension layer HeckeExt/; Claim 2.9), and **the identity certificate** of the plan's §1.3 in its slot-1 subset (HeckeExt/HeckeIdentityCertificate: the $\dim H^0$-robust Eichler–Shimura count, commutativity, $d_2 = 0$, the q^2 recursion against the independent double-coset route) runs on ten pipeline points (HeckeIdentityValidation).
- **A reach limit of U_ℓ, recorded**: at $n = 16$ the exact-integer coset construction of HeckeCosetTables completes for U_ℓ at $\ell \in \{691, 1009, 2003, 3617\}$ (1–34 s, 0.1–0.7 GB) and overflows its int64 tier ("exact multiply overflow") at $\ell \geq 5003$, independently of n; the plan's $(16, 43867)$ RSS point was a placeholder-model prediction that cannot be run. The two measured points $(128, 691)$: 2717 MB and $(256, 131)$: 1677 MB fit $\mathrm{rss} = (\ell + 1)(c_0 + c_1(n + 1)^2)$ with $c_1 \approx 1.8$–$1.9 \cdot 10^{-4}$ and a c_0 the two fits disagree on by 4.4×; the constants are re-fitted before E1b freezes them.

Addendum, v1.4.0 (2026-09-16, slot 2 of the extension stage, as reported by its packages WP-1, WP-2, WP-4, WP-5 and WP-7). What slot 2 built, and the one golden re-baseline it made:

- **E0b — the locality certificate runs inside the refinement loop** (WP-1; HeckeModuleDecomposition: = RefineToFixedPoint + RunLocalityStage + FinalizeChangeOfBasis): one candidate sequence — the single-operator witnesses with $\deg \pi_j = f$ first, the all-ones combination, then the fixed-seed combinations — capped by the budget; a candidate with ≥ 2 distinct irreducible factors *splits* the block at the parent's position in CanonicalLess order; the first clause-(1) candidate fixes residue_witness exactly as before (the shipped semantics, byte for byte), so on the golden grid, where candidate 0 certifies, no block list moves. The mathematical settings layer Hecke/HeckeSettings ([X] §1.5, owner decision R1) exists with the key locality (budget=<N>, default 64, or strict); strict parses and validates and *throws*

`not_implemented_in_this_build` until slot 3 (WP-6), never degrading to budget mode (slot-2 owner confirmation 3). `hecke_report/v2` (WP-2): the per-block keys `local`, `witness_poly`, `coefficient_field` = {`w_hat_exponent`, `P`} with `P` keyed by prime, `level_c.locality` = {`mode`, `budget`, `source`, `complete`, `splits`}, the corrected multiplicity note, and the four per-complex degree fields of §7 — the stage's *one* re-baseline, the 384 report texts and nothing else, explained key path by key path by `golden_explain.py`.

- **E1a — the local algebra and certificates (a), (b)** (WP-4, WP-5; `HeckeExt/HeckeLocalAlgebra`, `HeckeExt/HeckeGenerationCertificates`, namespace `hecke_ext`): Algorithm 5.5 as implemented (its "as implemented" paragraph), Certificate 5.6 with the canonical generator, Certificate 5.7 with the socle bound, the structured system on a cyclic-most element and the resumable `CommutantState` (R3 instance 3), the cross-block evaluation test `SameMaximalIdeal`, and the pipeline-free map `generation_verdict` of the default configuration; `HeckeExtAlgebraValidation` on constructed truths (0.10 s). The real-block arms are E1b's.
- **E4b — $H^1_!$, Harder and the diagnostics** (WP-7; `Hecke/HeckeBoundaryDiagnostics`, `HeckeExt/HeckeBoundary`, `HeckeBoundaryParabolicValidation`, record `hecke_boundary_parabolic/v1` with the per-complex degree fields): $H^1_! = \ker r^1$ computed on 315 (cell, p, ν) pairs; Harder's criterion asserted on the Kummer list in the form "$r^1 = 0$ on the joint eigenspace" (Claim 1.1); `cochain_commutes` and `eta_zero` both false on 315/315 ((A2)); and the two refutations of this revision (§8, (D6)). X_2, X_5 are still not in the record of §7 (E2).
- **Still not computable**: X_3, X_6 (the η-table is WP-8, the first part of E5), the production ι (E3, slot 3), D_n (E7). The U_ℓ reach repair (E1b-0) and the RSS constants are WP-3's and are not recorded here.

Addendum, v1.4.2 (2026-09-17, slot 3 of the extension stage, as reported by its packages WP-1, WP-2, WP-3, WP-5 and WP-6). What slot 3 built; it made **no** golden re-baseline and no golden emission:

- **E1b-0 — the U_ℓ reach** (WP-1): the overflow at $\ell \geq 5003$ was in [C] [C, `alg:tau`] step 1's fundamental-domain reduction, not in the coset construction; step 1 now escalates per operation to an `__int128` tier, bit-identically wherever the `int64` tier completes, and the reach is $\ell \leq 23170$, past which the τ-table is refused by its own certification ([C] [C, `lem:bounds`], [C, `thm:backend`]). The RSS model is frozen: $\text{rss} = (\ell{+}1)(c_0{+}c_1(n{+}1)^2)$ MB with $c_0 = 0.123152$, $c_1 = 2.24985{\cdot}10^{-4}$ (`u_ell_rss/v2-2026-09-17`, 16 measured points, residuals $+17.9\,\%/-15.9\,\%$); at $M = 2048$ it admits U_ℓ for $n \leq 804$ at $\ell \leq 13$, $n \leq 110$ at $\ell = 691$, $n \leq 18$ at $\ell = 10007$, and the ten Kummer verdicts are unchanged (seven admitted, three refused). The setting `u_ell` is registered.
- **E1b — generation certificate (c) on real blocks** (WP-2, `HeckeExt/HeckeSturmSweep`): Certificate 5.8 sweeps every prime $q \leq B'(k, \ell)$ by a membership test in A_S; at $(10, 691)$ the algebra is $\mathbb{F}_{691}[\varepsilon]/\varepsilon^2$, (a) fails, (b) is `excluded_by_socle_bound` with the naive $\dim C = 5$, and (c) is `sturm(`$B' = 2$`), final = true`; at $(n, \ell) = (22, 5), (26, 7)$ the $\{2, 3\}$ sweep is rejected and the B' sweep accepted; both unavailable modes run on a forced cap with the real $A^{(1)}(T_q)$; Stage 1 consults the frozen U_ℓ admission. The settings are registered as `sturm` and `sturm_unavailable`, which is how §7 names them since v1.5.0; the default configuration never produces `"heuristic"`.
- **E0b-S — strict locality mode** (WP-3): Certificate 5.2's strict mode as specified, the covering example verified, the golden grid identical under it. The implementation reports a strict-mode witness as a matrix and checks that it commutes with the family; it asserts $\dim_{\mathbb{F}_\ell} \bar{A} = \text{lcm}_p \deg \pi_p$ on a local block. **Both are now statements of this document** (v1.5.0): the hypothesis a *reported* witness must satisfy is stated in Certificate 5.2, and the sharpness identity is Claim 5.3. With them, the mode-dependence of the sub-block order after a split (Certificate 5.2, closing paragraph) and the two `phase_reached` keys of §7.
- **E5-2, E5-3 — the cone and its Hecke action** (WP-5, WP-6; `EquivariantMappingCone`, `Hecke/HeckeConeAction`): $H^\bullet_c$ with the long exact sequence and its maps in stored bases, and the operator on it induced by Φ_g with the derived $h = \tilde{\eta}$ ([C] [C, `def:coneaction`]); every clause of [C] [C, `obl:coneexact`] but the cross-routes (vii) is evaluated and [C, `obl:eta`] is discharged. At $(10, 691)$: $\dim H^q_c = [0, 3, 0]$, $\text{rank}\, r^1 = 1$, $\dim H^1_! = 2$; at the ten Kummer cells

$\dim H^1_c = 1 + 2\dim S_{n+2}$ and $H^2_c = 0$. So X_3, X_6 are *computable* on the model complexes; they are not yet in the record of §7 (E2).

- **Still not computable**: the production ι (E3, moved to slot 4), D_n (E7); the record `hecke_ext/v1` (E2, from slot 4). *Read as of the slot-3 close* (v1.6.1, the referee pass): this list is a snapshot of that date and it is the last addendum in this section, so a reader arriving here should not take it for the present state — slot 4 built the production ι (E3) and wrote the record, and slot 5 moved it to `hecke_ext/v2` with the boundary, cone and per-sign objects; *only* D_n (E7) is still not computable. The version block above carries what slots 4 and 5 built; the slot-4 and slot-5 addenda are the close's. *(v1.7.2, the slot-6 close:)* D_n is computable too, with its Hecke action and the pairings (slot-6 WP-1, WP-3), so nothing on this list remains.

DECISION. "All cohomologies the project can compute at this stage" means, with Hecke action: X_1, X_4 (and the vanishing H^2). The classification programme of §5 is stated for all six X_i; its Stages 0–1 list the three additions that make the other four available (boundary Hecke action, equivariant cone, ι), all of which are formula-level. Two of them need no new mathematics; the cone needs a *definition*, now fixed: the Hecke action on X_3, X_6 is the one induced by Φ_g from the certified triple $(\tau, \tau_\partial, \eta)$ over $\mathbb{Z}$ with the cochain homotopy *derived* ((A2)), which is what makes the operator on X_3 well defined at all — with a freely chosen cochain homotopy it is not ((A2), third bullet). Until the three additions land, the record of §7 carries the four missing groups as `null` with an explicit `"unavailable"` marker, and the extension classes "in the cone" are reported as *predicted* from r^1 ([C, `cl:les`]), not computed.

1.3 Boundary characters

The characters $\chi_a(q) := q^a + q^{n+1-a}$ by which the Hecke algebra acts on X_2 and X_5, their eigenvectors $x^{n-b\ell}\mathrm{Nm}^b$ on $X_2 = M^T$, the Eisenstein value $\chi_0(q) = 1+q^{n+1}$, the right-coset variant χ_{b+1} that settles the convention hedge of [M §4], and the coincidence criterion $\chi_a \equiv \chi_{a'}$ iff $a \equiv a'$ or $a \equiv n+1-a'$ (mod $\ell-1$) are [C, `cl:chi`], with [C, `prop:bdrychar`] for the eigenvector computation. The Dickson-locus corollary this document drew from the criterion through v1.2.2 is *false* and is corrected in [C, `rmk:chicoincide`]; see item 1 of §1.1.

Claim 1.1 (Harder's denominator, Ribet's converse)**.** *[standard, reference] For $n < \ell - 1$: $r^1 = 0$ on the Eisenstein class of $H^1(\Gamma; M_n)$ iff $\ell \mid \mathrm{num}(B_k/k)$, $k = n+2$ (Harder [20]; Ribet [29]). Then the reduced Eisenstein class lies in $H^1_!$ and $H^1_{\mathfrak{m}_{\mathrm{Eis}}}$ is a non-split $\mathbb{T}$-module — the prototype of the extensions classified here [M §5]. All cases with $n < \ell - 1$ and $k \le 26$: (k, ℓ) one of $(12, 691)$, $(16, 3617)$, $(18, 43867)$, $(20, 283)$, $(20, 617)$, $(22, 131)$, $(22, 593)$, $(24, 103)$, $(24, 2294797)$, $(26, 657931)$ [37] ($B_{24} = -236364091/2730$, $236364091 = 103 \cdot 2294797$).* The mod-$\ell$ reading (v1.4.0). *"The Eisenstein class" is the integral class — the reduction of the Eisenstein eigenline of $M_k(\mathbb{Z}_\ell)$ into $H^1(\Gamma; M_n(\mathbb{F}_\ell))$ — and the [standard, reference] tag covers that reading. Mod ℓ it is* not *a Hecke datum at a Kummer prime: the joint eigenspace E of the T_q for the eigenvalue $1 + q^{n+1}$ on H^1 is two-dimensional there (the socle of the non-split block, (D6)), so the statement a mod-ℓ computation can check, and does check, is that r^1 vanishes on* all *of E; the state* `inconclusive` *is reserved for $0 < \mathrm{rank}(r^1 E) < \dim E$ (not observed).* ***Verified 2026-09-16*** *(`HeckeBoundaryParabolicValidation`, slot-2 WP-7): $\mathrm{rank}(r^1 E) = 0$ on all ten cells of the list, and the converse $\mathrm{rank}(r^1 E) = 1$ on the 27 non-Kummer cells with $n < \ell - 1$ of its grid — 37/37 consistent, 0 inconclusive. What this does* not *say is $H^1_! = H^1$: see §8.*

The criterion's hypothesis and the divisibility predicate's domain are different things *(v1.5.0; slot-4 WP-7 finding 4). This claim is stated for $n < \ell - 1$, i.e. $k = n + 2 \le \ell$; the* question *"does ℓ divide $\mathrm{num}(B_k/k)$" is meaningful for every even $k \ge 2$ and every ℓ, and a predicate answering it must therefore be total where the criterion is not. A naive evaluation of $B_k \bmod \ell$ by the $\mathbb{F}_\ell$-recursion $\sum_j \binom{k+1}{j} B_j = 0$ is valid only for $k < \ell - 1$, and two provable branches cover the rest (v1.5.1, correcting v1.5.0, which named only the two instances $k = \ell - 1$ and $k = \ell$ and so left every even k with $(\ell - 1) \mid k$*

and $k > \ell-1$ — at $\ell \in \{2,3\}$ that is every *even k — with no branch at all).* ***Whenever*** *$(\ell-1) \mid k$, von Staudt–Clausen puts ℓ into the* denominator *of B_k (indeed $\ell\, B_k \equiv -1 \pmod{\ell}$), so $\ell \nmid \mathrm{num}(B_k)$ and a fortiori $\ell \nmid \mathrm{num}(B_k/k)$, which divides $\mathrm{num}(B_k)$: the answer is* `false` *outright and no recursion is run. That branch subsumes $k = \ell - 1$, the instance at the top of this claim's own range; and since $\ell - 1 \in \{1,2\}$ divides every even k, it is the branch that decides* every *even k at $\ell \in \{2,3\}$ — in particular this claim's own edge cell $(n,\ell) = (0,2)$, where $k = 2 = \ell$ and $\mathrm{num}(B_2/2) = \mathrm{num}(1/12) = 1$, so the answer is* `false`*. (The remaining degenerate case belongs to* odd *$k \geq 3$, which is outside this claim: there $B_k = 0$, the numerator vanishes and every prime divides it, a degenerate* `true`*.)* ***Otherwise*** *$k \not\equiv 0 \pmod{\ell - 1}$, and the Kummer congruence $B_k/k \equiv B_{k'}/k' \pmod{\ell}$ for $k \equiv k' \pmod{\ell - 1}$ reduces the question to $k' = k \bmod (\ell - 1) < \ell - 1$, where the recursion is valid.* ***Conformance****: a shipped predicate must handle those branches explicitly and say which one it took, and an implementation that pins a table of $\mathrm{num}(B_k/k)$ for small k (*`HeckeExt/HeckeBoundary`*, $k \leq 26$) is answering the same question over a finite range, not a different one. Implemented as* `HeckeExt/HeckePredictions'` `bernoulli_mod` */* `kummer_divides` *(slot-4 WP-7, with* `HeckePredictionsValidation`*); the rewiring of the pinned table to it is E6's, because it changes what* `harder_applicable` *records for $k > 26$.*

2 The Hecke algebra acting

2.1 Definitions

Let $\mathbb{T}_\infty \subseteq \mathrm{End}_{\mathbb{F}_\ell}(H_{\mathrm{tot}})$ be the $\mathbb{F}_\ell$-algebra generated by $\{T_q\}_{q \text{ prime}}$ (all q, $q = \ell$ included), and for a finite set S of primes $\mathbb{T}_S \subseteq \mathbb{T}_\infty$ the subalgebra generated by $\{T_q\}_{q \in S}$. Both are commutative artinian ([C, `obl:algebra`]; finite-dimensional), so

$$\mathbb{T}_\infty = \prod_{\mathfrak{m} \in \mathrm{MaxSpec}\, \mathbb{T}_\infty} \mathbb{T}_\mathfrak{m}, \qquad H_{\mathrm{tot}} = \bigoplus_{\mathfrak{m}} H_\mathfrak{m}, \qquad H_\mathfrak{m} := \{v : \mathfrak{m}^N v = 0,\ N \gg 0\},$$

with residue fields $\mathbb{T}_\infty/\mathfrak{m} \cong \mathbb{F}_{\ell^r}$, $r = r_\mathfrak{m}$. The *character* of $\mathfrak{m}$ is the Galois orbit of $(a_q(\mathfrak{m}))_q \in \mathbb{F}_{\ell^r}^{\mathbb{P}}$, $a_q(\mathfrak{m}) := T_q \bmod \mathfrak{m}$. Equivariance of [C, `eq:les`] makes each X_i a $\mathbb{T}_\infty$-module and $X_{i,\mathfrak{m}} := X_i \cap H_\mathfrak{m}$.

Claim 2.1 (Locality of extensions)**.** *[proved here]* $\mathrm{Ext}^1_{\mathbb{T}_\infty}(A,B) = 0$ *for $\mathbb{T}_\infty$-modules A, B supported at different maximal ideals. Hence every non-split extension of Hecke modules occurring in any X_i, or between terms of [C,* `eq:les`*], lives inside a single $H_\mathfrak{m}$, and $H_\mathfrak{m}$ is a module over the local ring $\mathbb{T}_\mathfrak{m}$.*

Proof. $\mathbb{T}_\infty = \prod \mathbb{T}_\mathfrak{m}$ is a product of rings; the idempotent $e_\mathfrak{m}$ acts as 1 on A and 0 on B, so any extension $0 \to B \to E \to A \to 0$ splits via $e_\mathfrak{m} E$. □

Claim 2.2 (Generation criterion)**.** *[proved here]* $\mathbb{T}_S = \mathbb{T}_\infty$ *iff (a) the natural map* $\mathrm{MaxSpec}\, \mathbb{T}_\infty \to \mathrm{MaxSpec}\, \mathbb{T}_S$ *is a bijection ("S separates"), and (b) for every $\mathfrak{m}$ the images of $T_q - a_q(\mathfrak{m})$, $q \in S$, span $\mathfrak{m}\mathbb{T}_\mathfrak{m}/\mathfrak{m}^2\mathbb{T}_\mathfrak{m}$ over $\mathbb{T}_\infty/\mathfrak{m}$ (here $a_q(\mathfrak{m})$ is read inside $\mathbb{T}_\mathfrak{m}$ through its coefficient field, Remark 3.2). When (b) holds at $\mathfrak{m}$, the $\mathbb{T}_S$-module structure of $H_\mathfrak{m}$* is *its $\mathbb{T}_\infty$-module structure, and $\dim_{\mathbb{F}_{\ell^r}} \mathfrak{m}/\mathfrak{m}^2$ is the minimal number of operators that can generate $\mathbb{T}_\mathfrak{m}$ over its residue field.*

Proof. $\mathbb{T}_S \subseteq \mathbb{T}_\infty$ is a finite (integral) extension of artinian rings. If (a) holds, $\mathbb{T}_S = \prod \mathbb{T}_{S,\mathfrak{m}'}$ with $\mathfrak{m}' = \mathfrak{m} \cap \mathbb{T}_S$ and $\mathbb{T}_{S,\mathfrak{m}'} \to \mathbb{T}_\mathfrak{m}$ local. Nakayama for the finite $\mathbb{T}_{S,\mathfrak{m}'}$-module $\mathbb{T}_\mathfrak{m}$: surjectivity iff $\mathbb{T}_{S,\mathfrak{m}'} \to \mathbb{T}_\mathfrak{m}/\mathfrak{m}'\mathbb{T}_\mathfrak{m}$ is onto; since $\mathbb{T}_\mathfrak{m}/\mathfrak{m}'\mathbb{T}_\mathfrak{m}$ is local with residue field $\mathbb{T}_\mathfrak{m}/\mathfrak{m}$ and (b) says $\mathfrak{m}'\mathbb{T}_\mathfrak{m} + \mathfrak{m}^2 = \mathfrak{m}$, Nakayama again gives $\mathfrak{m}'\mathbb{T}_\mathfrak{m} = \mathfrak{m}$ and then $\mathbb{T}_\mathfrak{m}/\mathfrak{m}'\mathbb{T}_\mathfrak{m} = \mathbb{T}_\infty/\mathfrak{m}$, which is the image of $\mathbb{T}_{S,\mathfrak{m}'}$ iff the residue field extension is generated by the a_q, $q \in S$ — part of (a) (the residue field of $\mathfrak{m}'$ equals that of $\mathfrak{m}$ exactly when MaxSpec maps bijectively with the same residue degrees; see Remark 2.3). Conversely $\mathbb{T}_S = \mathbb{T}_\infty$ trivially gives (a),(b). □

Remark 2.3. Condition (a) contains a residue-degree clause: S must separate $\mathfrak{m}$ from its own Galois conjugates' images, i.e. $\mathbb{F}_{\ell^r} = \mathbb{F}_\ell(a_q(\mathfrak{m}) : q \in S)$. This is exactly what the certified `residue_degree` witness of Stage S11 (§3) establishes.

2.2 Separation

Claim 2.4 (Finiteness of the anemic spectrum)**.** *[standard, reference] For fixed $\ell \geq 5$ the set $\Sigma(\ell)$ of anemic eigensystems $\{a_q\}_{q\neq\ell}$ occurring in $H^1(\Gamma; M_n)$ for* some *n is finite: every level-1 mod-ℓ eigensystem is a θ-twist of one in weight $\leq \ell+1$ (Serre [31], Jochnowitz [23, Thm. 4.1, p. 276], finiteness Ash–Stevens [1, Thm. 2.2, p. 204]; both locators verified on page images, slot-17 R17-4, v1.17.0-pre.5); equivalently $\Sigma(\ell)$ is the set of semisimple $\bar\rho : G_{\mathbb{Q},\{\ell\}} \to \mathrm{GL}_2(\bar{\mathbb{F}}_\ell)$ that are modular of level 1. Together with the Eisenstein-type characters χ_a, $a \in \mathbb{Z}/(\ell-1)$ ([C, `cl:chi`]), this is the whole list of characters of $\mathbb{T}_\infty$ on H_{tot} for every n.*

Claim 2.5 (Separating set)**.** *[proved here] (given Claim 2.4) There is a finite set $S_{\mathrm{sep}}(\ell)$ of primes $q \neq \ell$, depending only on ℓ, such that $\{T_q\}_{q\in S_{\mathrm{sep}}(\ell)}$ separates all pairs in $\Sigma(\ell) \cup \{\chi_a\}$* as Galois orbits*. It is computable once per ℓ from the weights $n \leq \ell-1$ and their θ-twists (§5).*

Proof. Finitely many distinct systems; two distinct systems differ at some $q \neq \ell$; take the union. □

Claim 2.6 (T_ℓ is indispensable)**.** *[standard, reference] No ℓ-independent set of primes $q \neq \ell$ separates the* full *spectrum: for a normalised eigenform f of weight w with $a_\ell(f) \neq 0$, the system of $\theta^{\ell-1}f$ (weight $w+\ell^2-1$, $a_\ell = 0$) coexists in that weight with $fE_{\ell-1}^{\ell+1}$ ($a_\ell = a_\ell(f)$) and agrees with it at every $q \neq \ell$ [M §6.1]. If $a_\ell(f) = 0$ the two full systems coincide: multiplicity ≥ 2, and in weight ℓw the span of f, $f|V$ carries $U_\ell = \left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)$, a non-split self-extension. Hence T_ℓ (i.e. U_ℓ, computed by the same pipeline [C, `cor:ulrecursion`]) is always in S.*

2.3 Generation bounds

Claim 2.7 (Sturm)**.** *[standard, reference] (a, first sentence) / [conjectural] ($B = B(k)$ in the second) / [proved here] (b) Let $k = n+2$ and $B(k) := \lfloor k/12 \rfloor + 1$.*

- *(a) The $\mathbb{Z}$-algebra $\mathbb{T}_\mathbb{Z}(S_k)$ generated by all T_m on $S_k(\mathbb{Z})$ is spanned as a $\mathbb{Z}$-module by T_m, $m \leq \lfloor k/12 \rfloor$ (Sturm [36]; Stein [35, §9.4]; verified on the Miller basis for $12 \leq k \leq 36$). Hence, by the pairing $(T, g) \mapsto a_1(Tg)$, which identifies $\mathbb{T}_\mathbb{Q}(M_k)$ with $M_k(\mathbb{Q})^\vee$ (a form with $a_m = 0$ for all $m \geq 1$ is constant, hence 0), $\mathbb{T}_\mathbb{Z}(M_k) \otimes \mathbb{Q}$ is spanned by T_m, $m \leq B$, for every B such that no nonzero $g \in M_k$ has $a_1(g) = \cdots = a_B(g) = 0$; such a B exists. That $B = B(k)$ suffices is equivalent to $a_d(f_0) \neq 0$, resp. (for $k \equiv 2 \bmod 12$, where $B(k) = d+1$) to $(a_d(f_0), a_{d+1}(f_0)) \neq (0,0)$, for the Miller element $f_0 = 1 + O(q^d)$, $d = \dim M_k$; this is [conjectural], verified (exact rational arithmetic on E_4, E_6, Δ) for even $4 \leq k \leq 70$ (v1.24.0-pre.2: "equivalent" corrected for $k \equiv 2 \bmod 12$ and the range extended, HeckeZ referee A7-m6, N8; v1.24.0-pre.1 erratum, owner ruling Q9 of 2026-11-04: the earlier argument via Δg assumed $a_0(g) = 0$; for $g = 1 + 196560q^2 + \cdots \in M_{12}$, $\Delta g = q - 24q^2 + \ldots$ vanishes to order 1 only).*
- *(b) The $\mathbb{Z}$-module statement for M_k — $\mathbb{T}_\mathbb{Z}(M_k)$ spanned by T_m, $m \leq B(k)$ — is* false*, and false at primes ≥ 5. On the Miller basis of $M_k(\mathbb{Z})$ the index $[\mathbb{T}_\mathbb{Z}(M_k) : \langle T_m : m \leq B(k)\rangle_\mathbb{Z}]$ is divisible by 5^2 at $k = 24$, by $5 \cdot 7$ at $k = 28$, by $5 \cdot 7 \cdot 131$ at $k = 34$ (lower bounds 400, 2240, 36680 from $\langle T_m : m \leq 3B+3\rangle$; recomputed for this revision), and the reductions of $T_1, \ldots, T_{B(k)}$ span only 2 of the 3 dimensions of the image of $\mathbb{T}_\mathbb{Z}(M_k) \otimes \mathbb{F}_\ell$ in $\mathrm{End}M_k(\mathbb{F}_\ell)$ at $(k,\ell) = (24,5), (28,7), (34,131)$. The obstruction is the constant term: the pairing of (a) sees a_m, $m \geq 1$, only, and Sturm's theorem mod ℓ for M_k needs a_0. At $(12, 691)$ the statement happens to hold (the index 3 is prime to 691), which is why the test case of §8 does not expose it.*

Definition 2.8 (The computed bound $B'(k,\ell)$). For $B \geq 2$ let $W_B := \mathbb{Z}[T_q : q \leq B \text{ prime}] \subseteq \mathbb{T}_\mathbb{Z}(M_k) \subseteq \mathrm{End}_\mathbb{Z}M_k(\mathbb{Z})$ (the $\mathbb{Z}$-subalgebra; a finitely generated $\mathbb{Z}$-module) and $\mathbb{T}^{\mathrm{sat}} := \mathbb{T}_\mathbb{Q}(M_k) \cap \mathrm{End}_\mathbb{Z}M_k(\mathbb{Z})$, the saturation (computable as the saturation of $\langle T_m : m \leq B\rangle_\mathbb{Z}$ for any B at which this span has rank $\dim M_k$; by Claim 2.7(a) such B exist, and $B = B(k)$ works in the verified range; v1.24.0-pre.1 erratum). $B'(k,\ell)$ is the least $B \leq B_{\max}$ such that the $\mathbb{F}_\ell$-algebra generated in $\mathrm{End}M_k(\mathbb{F}_\ell)$ by the reductions $T_q \bmod \ell$, $q \leq B$ prime, computed on the Miller basis, has dimension $\dim M_k$. The cap $B_{\max}$ is carried by the mathematical setting `sturm` (as `cap=<B>`; v1.5.0, the registered spelling —

through v1.4.2 this document wrote `sturm.cap`, which no registry has) (default $12\,B(k)$; [X] §1.5, slot-1 owner confirmation 6), and the outcome — `found(`B'`)` or `none_at_cap` — is recorded either way together with the resumable state of [X] §1.3.1 R3 (`budget_history`; `resume(cap)` refuses a lower cap and is a no-op on a complete state). At $n = 0$ ($k = 2$, $M_2 = 0$) no bound is defined: the record carries the outcome `not_applicable` at record level — a record-layer state that E2 implements — while the implementation's `SturmOutcome` stays three-valued {`found`, `none_at_cap`, `in_progress`} (v1.4.0; slot-1 WP-3 §7 item 7, slot-2 plan §3.16).

Claim 2.9 (B' certifies generation over $\mathbb{Z}_\ell$)**.** *[proved here] If $B' = B'(k,\ell)$ exists then $W_{B'} + \ell\,\mathbb{T}_{\mathbb{Z}}(M_k) = \mathbb{T}_{\mathbb{Z}}(M_k)$: every T_q is a polynomial with $\mathbb{Z}$-coefficients in the $T_{q'}$, $q' \le B'$ prime, plus ℓ times an element of $\mathbb{T}_{\mathbb{Z}}(M_k)$. It exists iff $\ell \nmid [\mathbb{T}^{\mathrm{sat}} : \mathbb{T}_{\mathbb{Z}}(M_k)]$.*

Proof. $\mathbb{T}^{\mathrm{sat}}$ is saturated, so $\mathbb{T}^{\mathrm{sat}} \otimes \mathbb{F}_\ell \hookrightarrow \mathrm{End}M_k(\mathbb{F}_\ell)$ and the dimension condition says $W_{B'} + \ell\mathbb{T}^{\mathrm{sat}} = \mathbb{T}^{\mathrm{sat}}$; by Nakayama $W_{B'} \otimes \mathbb{Z}_\ell = \mathbb{T}^{\mathrm{sat}} \otimes \mathbb{Z}_\ell \supseteq \mathbb{T}_{\mathbb{Z}} \otimes \mathbb{Z}_\ell \supseteq W_{B'} \otimes \mathbb{Z}_\ell$, so $\mathbb{T}_{\mathbb{Z}}/W_{B'}$ is finite of order prime to ℓ and $W_{B'} + \ell\mathbb{T}_{\mathbb{Z}} = \mathbb{T}_{\mathbb{Z}}$. Conversely the dimension condition forces $\ell \nmid [\mathbb{T}^{\mathrm{sat}} : \mathbb{T}_{\mathbb{Z}}]$, and if $\ell \nmid [\mathbb{T}^{\mathrm{sat}} : \mathbb{T}_{\mathbb{Z}}]$ then, since $W_B \uparrow \mathbb{T}_{\mathbb{Z}}$, it holds for B large. Cost: $\dim M_k \times B'$ integer q-expansion coefficients and one algebra closure mod ℓ — independent of the cohomology pipeline. Computed values (this revision): $B' = 2$ at $(12, 691)$ and at $(34, 131)$ ($\mathbb{F}_{131}[T_2]$ is already everything, although $\langle T_1, T_2, T_3\rangle_{\mathbb{Z}}$ is not); $B' = 5$ at $(24, 5), (30, 5), (32, 5), (36, 5)$; $B' = 7$ at $(28, 7)$. At $(24, 5)$ and $(28, 7)$ the $\mathbb{F}_\ell$-algebra generated by T_2, T_3 has dimension $2 < 3$: the primes $\le B(k)$ do *not* generate. **Implemented** (v1.3.3; slot 1, `HeckeExt/HeckeSturmBound`, pipeline-free — q-expansions of E_4, E_6 and $\Delta = q\prod(1-q^n)^{24}$ reduced mod ℓ as they are formed, so $\ell \in \{2, 3\}$ work too; the cap $12\,B(k)$ as the setting `sturm`; a certificate with `verify()`; the resumable state of the implementation plan's R3): `HeckeSturmBoundValidation` pins every value above and, in addition, $B' = 2$ on all ten Kummer cells of Claim 1.1's list and at $(162, 691)$, $(200, 691)$ and $(258, 691)$ (the last recorded, 1.4 ms, the slowest case of the list), $B' = 5$ at $(60, 5)$, $B' = 3$ at $(36, 7)$, and the proper subalgebra $\dim\langle T_2, T_3\rangle = 2 < 3$ at $(24, 5)$ and $(28, 7)$. □

Claim 2.10 (What Sturm certifies mod ℓ)**.** *[proved here] (from [C, `cl:bockstein`] and Claim 2.9, given a recorded $B'(k,\ell)$) The $\mathbb{Z}$-algebras generated by the T_m on $M_k(\mathbb{Z})$ and on $H^1(\Gamma; M_n(\mathbb{Z}))/\mathrm{tors}$ are canonically isomorphic (both torsion-free with the same relations after $\otimes\mathbb{Q}$, Eichler–Shimura). Let q be a prime and write $T_q = F_q(T_{q'} : q' \le B') + \ell t'$ with $F_q \in \mathbb{Z}[X_{q'}]$, $t' \in \mathbb{T}_{\mathbb{Z}}(M_k)$ (Claim 2.9). Then $t := T_q - F_q(T_{q'})$ acts on $H^1(\Gamma; M_n(\mathbb{F}_\ell))$ with image inside the image of $\mathrm{tors}\,H^1(\Gamma; M_n(\mathbb{Z}))$, which is $\cong H^0(\Gamma; M_n(\mathbb{F}_\ell))$ (the Dickson part). Hence $t = 0$ on every $H^1_{\mathfrak{m}}$ whose character is not that of a Dickson class, and on those $\mathfrak{m}$ the operators $\{T_q\}_{q \le B' \text{ prime}}$ generate $\mathbb{T}_{\mathfrak{m}}|_{H^1}$. On Dickson-type $\mathfrak{m}$, on X_2, X_3, X_5, X_6, and when $B'(k,\ell)$ does not exist, Sturm gives no certificate. In the record this is the tri-state `final` of §7 (v1.4.0): `sturm(B')` on a non-Dickson $\mathfrak{m}$ of X_4 makes the ideal `final = true`; off that range the sweep is a heuristic and the ideal is `final = false` with reason `sturm_heuristic` in the default configuration, or `final = "heuristic"` under `final_mode = sturm_heuristic`; a missing B' is reason `sturm_unavailable`, or `"heuristic"` under `sturm_unavailable = heuristic` ([X] §1.3.1 R3.2). With $B(k)$ in place of B' the conclusion fails inside these hypotheses: at $(n,\ell) = (22, 5)$ and $(26, 7)$ there are no Dickson classes, and T_2, T_3 ($B(k) = 3$) generate a proper subalgebra of $\mathbb{T}_{\mathbb{Z}}(M_k) \otimes \mathbb{F}_\ell$ (Claim 2.9).*

Proof. t preserves $H^1(\Gamma; M_n(\mathbb{Z}))$ and acts as $\ell t'$ on the torsion-free quotient, so $t\,H^1(\mathbb{Z}) \subseteq \ell H^1(\mathbb{Z}) + \mathrm{tors}$; reduce mod ℓ ($H^1(\mathbb{F}_\ell) = H^1(\mathbb{Z}) \otimes \mathbb{F}_\ell$, [C, `cl:bockstein`]). $(\mathrm{tors} \otimes \mathbb{F}_\ell)_{\mathfrak{m}} = 0$ unless $\mathfrak{m}$ is Dickson-type. □

(v1.15.1-pre.4; slot-15 WP-19) *What the dimension condition of Definition 2.8 certifies.* By Claim 2.11, $\mathbb{T}_{\mathbb{Z}}(M_k) \otimes \mathbb{F}_\ell$ acts on $M_k(\mathbb{F}_\ell)$ with image of dimension $\dim M_k$ at every $\ell \ge 5$. So an $\mathbb{F}_\ell$-algebra generated by $T_q \bmod \ell$ that reaches dimension $\dim M_k$ *is* that image. $\mathbb{T}_{\mathbb{Z}}(M_k) \otimes \mathbb{Z}_\ell$ is saturated, and $B'(k,\ell)$ exists at every $\ell \ge 5$ (Claim 2.9's criterion $\ell \nmid [\mathbb{T}^{\mathrm{sat}} : \mathbb{T}_{\mathbb{Z}}(M_k)]$ always holds); `none_at_cap` can still be recorded when the cap is too small. The statement of this claim is unchanged.

Claim 2.11 (The forms' Hecke algebra mod ℓ is faithful of full dimension on $M_k(\mathbb{F}_\ell)$)**.** *[proved here] (v1.15.1-pre.4, slot-15 WP-19, proof in `docs/implementation/slot15/WP-19_forms_proofs.md` §1;*

refereed in the WP-4 window) Let $\ell \geq 5$, $k \geq 4$ even, $M_k(\mathbb{F}_\ell) := M_k(\mathbb{Z}) \otimes \mathbb{F}_\ell$, $S_k(\mathbb{F}_\ell) := S_k(\mathbb{Z}) \otimes \mathbb{F}_\ell$, and let $\mathbb{T}(X)$ be the $\mathbb{F}_\ell$-algebra generated by all T_m in $\mathrm{End}(X)$.

(a) $\dim \mathbb{T}(M_k(\mathbb{F}_\ell)) = \dim M_k$ *and* $M_k(\mathbb{F}_\ell) \cong \mathrm{Hom}(\mathbb{T}(M_k(\mathbb{F}_\ell)), \mathbb{F}_\ell)$ *as $\mathbb{T}$-modules; hence* $\dim \mathbb{T}_\mathfrak{m} = \dim M_k(\mathbb{F}_\ell)_\mathfrak{m}$ *for every maximal ideal $\mathfrak{m}$, and $\mathbb{T}_\mathbb{Z}(M_k) \otimes \mathbb{F}_\ell \to \mathrm{End} M_k(\mathbb{F}_\ell)$ is injective. The same holds for S_k at every prime ℓ.*

(b) If $(\ell-1) \mid k$, then $M_k(\mathbb{F}_\ell) = \mathbb{F}_\ell \cdot 1 \oplus S_k(\mathbb{F}_\ell)$ as $\mathbb{T}$-modules, and the system $m \mapsto \sigma_{k-1}(m) \bmod \ell$ of the constant form $1 = E_{\ell-1}^{k/(\ell-1)}$ occurs in no cusp form mod ℓ; so $\mathbb{T}(M_k(\mathbb{F}_\ell)) = \mathbb{F}_\ell \times \mathbb{T}(S_k(\mathbb{F}_\ell))$.

Proof. The pairing $(t, f) \mapsto a_1(tf)$ is $\mathbb{T}$-balanced, and $a_1(T_m f) = a_m(f)$. Its kernel on the form side consists of the constants in X, and its kernel on the algebra side is 0 when X has no non-zero constant. A non-zero constant lies in $M_k(\mathbb{F}_\ell)$ iff $(\ell-1) \mid k$: if $f \equiv c \neq 0$, then $fE_4 \equiv cE_4$, and Serre's weight congruence [31] (LNM 350, n° 1.3, Théorème 1, p. 198, verified (slot-16 WP-8): for $p \geq 3$ and f, f' with rational coefficients of weights k, k', $f \neq 0$ and $v_p(f - f') \geq v_p(f) + m$ force $k \equiv k' \pmod{(p-1)p^{m-1}}$) with $m = 1$ gives $(\ell-1) \mid k$. Conversely $E_{\ell-1} \equiv 1$ (von Staudt–Clausen). So the pairing is perfect for S_k, and for M_k when $(\ell-1) \nmid k$, and (a) follows by localising the pairing.

For (b), 1 is a T_m-eigenvector with eigenvalue $\sigma_{k-1}(m)$ (read on a_0), and $f - a_0(f) \cdot 1 \in S_k(\mathbb{F}_\ell)$ (Miller basis). Suppose a cusp form mod ℓ carries that system. Then it has $a_m = \sigma_{k-1}(m)$ for $m \geq 1$. Lift it to $\tilde{F} \in S_k(\mathbb{Z})$ and put $c := -B_k/2k$, which has $v_\ell(c) = -1 - v_\ell(k)$ by von Staudt–Clausen. Then $h := (G_k - \tilde{F})/c \in M_k(\mathbb{Q})$ satisfies $h \equiv 1 \pmod{\ell^{2+v_\ell(k)}}$, and $hE_4 \equiv E_4$ gives, by the same theorem, $(\ell-1)\ell^{1+v_\ell(k)} \mid k$, which is absurd. So the idempotent of the constant's system lies in $\mathbb{T}(M_k(\mathbb{F}_\ell))$ and projects onto $\mathbb{F}_\ell \cdot 1$, and (a) follows in this case as well.

$\ell \geq 5$ enters through Serre's theorem (its modulus for $p \geq 5$) and through $E_{\ell-1}$ being a modular form of level one. □

Claim 2.12 (F9: the Eichler–Shimura count on a block, with its torsion term)**.** *[proved here] (v1.15.1-pre.4, slot-15 WP-19, proof in `WP-19_forms_proofs.md` §2; refereed in the WP-4 window) Let $\ell \geq 5$ and $n > 0$ even, $k = n + 2$. Let P' be a finite set of primes, where T_ℓ is the level-one operator (the pipeline's U_ℓ). Let $\lambda \in \mathbb{F}_\ell^{P'}$, and let X_λ be the joint generalised λ-eigenspace. Then*

$$\dim H^1(\Gamma; M_n)_\lambda = \dim S_k(\mathbb{F}_\ell)_\lambda + \dim M_k(\mathbb{F}_\ell)_\lambda + \dim H^0(\Gamma; M_n)_\lambda.$$

By [C, `prop:heckeH0`](1),(2), $\dim H^0(\Gamma; M_n)_\lambda$ is 0 if $\ell \in P'$ and $\lambda_\ell \neq 0$. Otherwise it is the number of (a, b) with $a(\ell+1) + b(\ell^2 - \ell) = n$ and $(1+p)p^a \equiv \lambda_p$ for all $p \in P' \setminus \{\ell\}$. If $\{T_p\}_{p \in P}$, $P \supseteq P'$, generates $\mathbb{T}(S_k(\mathbb{F}_\ell))$, resp. $\mathbb{T}(M_k(\mathbb{F}_\ell))$, the first two terms equal the dimensions of the algebras $\mathbb{F}_\ell[T_p|_{X_\lambda} : p \in P]$ (Claim 2.11). `HeckeFormsHeckeAlgebraValidation`'s F9 identity $D = \dim \mathbb{T}(S_k)_\mathfrak{m} + \dim \mathbb{T}(M_k)_\mathfrak{m}$ is therefore exact on every block with $\dim H^0(\Gamma; M_n)_\lambda = 0$ (the torsion guard), and is off by exactly that term otherwise.

Proof. $H^1(\Gamma; M_n) = \mathbb{L}/\ell$ is an extension of $\mathbb{L}'/\ell$ by $\mathbb{L}_{\mathrm{tors}}/\ell$ [C, `prop:cohforms`](1). Localising at λ is exact. $\dim(\mathbb{L}_{\mathrm{tors}}/\ell)_\lambda = \dim \mathbb{L}[\ell]_\lambda = \dim H^0(\Gamma; M_n)_\lambda$, by counting cyclic summands and by the Hecke-equivariant Bockstein [C, `cl:bockstein`](2),(3). For a lattice, $\dim(L/\ell)_\lambda$ counts the $\bar{\mathbb{Q}}_\ell$-systems reducing to λ. By [C, `lem:heckerankone`]'s proof, $\mathbb{L}' \otimes \bar{\mathbb{Q}}_\ell \cong M_k \oplus \overline{S_k}$, semisimply, so the free term is $2s + e = s + (s + e)$, where s counts the cuspidal and e the Eisenstein systems over λ; and $S_k(\mathbb{Z}_\ell)$, $M_k(\mathbb{Z}_\ell)$ give s and $s + e$. On H^0, T_p is diagonal on Dickson monomials and U_ℓ is nilpotent [C, `prop:heckeH0`]. □

Claim 2.13 (Bellaïche–Khare)**.** *[standard, reference] (recalled; to be checked against the paper before use as a default) For $\ell \geq 5$ and an irreducible unobstructed modular $\bar{\rho}$ of level 1, the completed cuspidal anemic Hecke algebra $\mathbb{T}_{\bar{\rho}}$ is isomorphic to $\mathbb{F}_\ell[[x, y]]$, with x, y realisable as $T_q - a_q$, $T_{q'} - a_{q'}$ for suitable primes q, q' (Bellaïche–Khare [3]; Deo [13] for U_ℓ and higher level; Bellaïche [4]). Every fixed-weight $\mathbb{T}_{n,\mathfrak{m}}$ is a quotient, so two well-chosen operators away from ℓ plus T_ℓ generate $\mathbb{T}_{n,\mathfrak{m}}$ for all n simultaneously; which pair works is a tangent-space condition checkable in any one weight. For $n \gg \ell$ no single T_q suffices mod ℓ (finitely many values in $\mathbb{F}_{\ell^r}$ against $\sim n/6$ eigensystems); for small n one operator can (T_2 alone generates at $(10, 691)$, Claim 8.1). The campaign row $(n, \ell) = (160, 691)$ is a* candidate *example (§8): T_5 alone has a* 4:1 *block where T_2 and T_3 have*

2:1, 2:1, 2:1, 4:2*; whether that* 4:1 *is two* $\mathbb{F}_\ell$*-rational systems merged at* a_5 *or is* T_2*'s* 4:2 *block with an* $\mathbb{F}_\ell$*-rational* a_5 *was not determined by the single-operator rows of* `results_table.tsv`. ***Verified computationally on 2026-09-16*** *(v1.3.3; the slot-1 probe* `HeckeLocalityProbe`*, one* `ComputeMultiOp` *run on* $\{T_2, T_3, T_5\}$ *with the locality certificate on every block): the simultaneous decomposition has seven blocks* 2:1, 1:1, 2:1, 2:1, 4:2, 6:3, 10:5 *(dimension:residue degree, in that order), every one local,* `certify() = 0`*; the* 4:2 *block is a genuine residue-degree-*2 *system.* ***Which alternative holds*** *(v1.7.0; measured by* `HeckeExtSeparationValidation`*, its* `SEPARATION_SINGLE` *line, slot-5 WP-8): the* second. *Run with* $\{T_2\}$, $\{T_3\}$, $\{T_5\}$ *and* $\{T_2, T_3\}$ *alone, every family gives seven blocks with the same dimension multiset* $\{1, 2, 2, 2, 4, 6, 10\}$*;* T_2 *alone,* T_3 *alone and* $\{T_2, T_3\}$ *each give the family's list exactly, and* T_5 *alone gives* $4{:}1, 2{:}1, 2{:}1, 2{:}1, 1{:}1, 6{:}3, 10{:}5$ *— its* 4:1 *is the family's* 4:2 *block, with* a_5 *rational on it, seen with a coarser invariant. It is not two* $\mathbb{F}_{691}$*-rational systems merged at* a_5 *and split by the refinement, as v1.3.3–v1.6.2 said: that would give eight blocks, and every family gives seven. (v1.7.1: the validation compares the* $(\dim : f)$ *lists and the block counts, not subspaces; that the partitions* coincide *follows from the counts, since the family's decomposition refines each single operator's and every run has seven blocks.) So the row is evidence for the refinement logic, but it is* not *an instance of this claim's "no single* T_q *suffices":* ***the block partition at*** $(160, 691)$***, with its residue degrees, is already*** T_2***'s (and*** T_3***'s);*** T_5 ***alone gives the same partition and underreports one residue degree.*** *(Through v1.6.2 this sentence read "as this claim predicts for* $n \gg \ell$*, no single* T_q *separates everything while* $\{T_2, T_3\}$ *does", which the measurement refutes; v1.7.0's "no single* T_q *certifies the residue degrees that* $\{T_2, T_3\}$ *does" is refuted by the same line,* T_2 *alone giving the family's residue degrees; the general statement for* $n \gg \ell$ *above is a counting argument and is not affected.) The row does not by itself verify the isomorphism* $\mathbb{T}_{\bar\rho} \cong \mathbb{F}_\ell[[x, y]]$*, which stays [standard, reference] (recalled).*

Claim 2.14 (T_ℓ nilpotent on θ-images)**.** *[(1): proved in [C]; (2): standard, reference] (v1.12.0-pre.1)*

1. *U_ℓ kills θ-images over $\mathbb{F}_\ell$: on H^1 the image of $L\cdot : H^1(\Gamma; M_{n-\ell-1}) \to H^1(\Gamma; M_n)$ (multiplication by the Dickson invariant, $\mathrm{SL}_2(\mathbb{F}_\ell)$-equivariant, Hecke-equivariant up to a* det*-twist [M §2]) is killed by $U_\ell = T_\ell$, i.e. $A(U_\ell)_n \circ \theta = 0$, for every $n \geq \ell + 1$ and every prime ℓ* — already on cochains. Pointer *(v1.12.0-pre.1): this sentence is [C,* `prop:thetatwist`*] (in [C,* `sec:dicksontheta`*], [C] v1.11.0-pre.1, slot-11 WP-2), proved there in the code's conventions ([C,* `def:rhon`*], [C,* `H2`*]): every coset representative of $\Gamma\,\mathrm{diag}(1, \ell)\,\Gamma$ has determinant ℓ, so the Hecke operator U_ℓ on $H^\bullet(\Gamma; M_{n-\ell-1} \otimes \det)$ is $\ell \cdot A(U_\ell)_{n-\ell-1} = 0$ over $\mathbb{F}_\ell$, and θ intertwines the twisted operator with $A(U_\ell)_n$. This document keeps no proof of its own; where the two differ, [C] wins. The computational check is (6e) of §5.1 (slot-11 WP-6); that θ-images are non-ordinary is [C,* `cor:thetanonordinary`*].* Not *claimed: that U_ℓ vanishes on all of H^0 — it does not in general ([C,* `prop:heckeH0`*](4); (6f) of §5.1).* Through v1.11.1 *the sentence read "U_ℓ kills θ-images; … is killed by T_ℓ after the twist" and carried [standard, reference].*
2. *For $n \geq \ell + 1$, T_ℓ has large kernel and typically nilpotent Jordan blocks on $H_{\mathfrak{m}}$; its Jordan type is a basic extension invariant [M §6.3]. [standard, reference] (unchanged).*

3 Primary decomposition and the local algebras $\mathbb{T}_{\mathfrak{m}}$

3.1 What exists: Stages S10–S11

`HeckeModuleDecomposition` (header comment is the binding definition) computes, for a degree q and a commuting family $\{A^{(q)}(T_{p_j})\}_j$ over $\mathbb{F}_\ell$:

(i) the common refinement of the single-operator primary decompositions: blocks $V = \{v : \pi_j(T_{p_j})^N v = 0\ \forall j\}$ with $\pi_j \in \mathbb{F}_\ell[X]$ irreducible;

(ii) per block and operator, the minimal polynomial $\pi_j^{e_j}$, the characteristic polynomial, and the generalized-Jordan partition of $\dim V / \deg \pi_j$ from $\mathrm{rank}\,\pi_j(T_{p_j})^r$;

(iii) the residue degree $f := \mathrm{lcm}_j \deg \pi_j$, certified as $[\mathbb{T}_S/\mathfrak{m} : \mathbb{F}_\ell]$ by a *witness* $w = \sum_j c_j T_{p_j}$ whose minimal polynomial on the block has squarefree part a single irreducible of degree exactly f (deterministic-seed search; reported as `residue_degree_lower_bound` if no witness is found);

(iv) an adapted basis U, and `certify()`: all of the above re-derived deterministically (Berkowitz charpoly, Frobenius/gcd irreducibility proof, product identity, block-diagonality of every $U^{-1}A_jU$, nilpotency exponents, partitions from ranks).

Factorisation is Cantor–Zassenhaus with a fixed seed, always followed by `CertifyFactorization`.

3.2 A gap: per-operator refinement is not $\operatorname{MaxSpec}\mathbb{T}_S$

Claim 3.1. *[proved here] The common refinement (i) can leave a block on which $\mathbb{T}_S$ is* not *local. It is local iff the residue-degree witness (iii) exists with* $\deg\pi_w = f$ and $\dim_{\mathbb{F}_\ell}(\mathbb{T}_S|_V/\operatorname{rad}) = f$.

Proof. Example: ℓ arbitrary, $\alpha \in \mathbb{F}_{\ell^2} \setminus \mathbb{F}_\ell$, $V = \mathbb{F}_{\ell^2} \oplus \mathbb{F}_{\ell^2}$, $t_1 = (\alpha, \alpha^\ell)$, $t_2 = (\alpha, \alpha)$ acting diagonally. Each t_j has minimal polynomial the irreducible $\pi = \min(\alpha)$, so (i) does not split V, yet $\mathbb{F}_\ell[t_1, t_2] \ni t_1 - t_2 = (0, \alpha^\ell - \alpha)$ is a non-nilpotent zero-divisor: two maximal ideals, characters (α, α) and (α^ℓ, α) — not Galois conjugate. In general $\mathbb{T}_S|_V/\operatorname{rad}$ is a product of fields each containing a root of every π_j; it is a single field iff its $\mathbb{F}_\ell$-dimension is f; equivalently, iff some (then a generic) element w has irreducible squarefree minimal polynomial π_w of degree f *and* every t_j is congruent modulo nilpotents to a polynomial in w — the two clauses of Certificate 5.2; the first alone is not sufficient (in the example $t_2 = (\alpha, \alpha)$ has irreducible squarefree minimal polynomial of degree $f = 2$). Until slot 2 (E0b) the code accepted a single operator with $\deg\pi_j = f$ as the witness (`HeckeModuleDecomposition.cpp:687--692`), so in the example it reported V as one block with `residue_degree_certified=true`: the witness search certified a *primitive element of* $\mathbb{T}_S/\operatorname{rad}$ *assuming locality*, not locality itself (independently found in `hecke_extensions_capability_inventory.md` §2.3.4). Since E0b (slot-2 WP-1, 2026-09-16) the refinement loop enforces both clauses of Certificate 5.2 and the example splits; the 48 constructed non-local families of `HeckeLocalityValidation`, which a frozen copy of the old predicate accepts, are all split and certified, and no block of the golden grid moved (every one was local, slot 1). □

> **DECISION.** A block is a maximal-ideal component of $\mathbb{T}_S$ only when certified local (Certificate 5.2); Stage 2 of §5 refines by the primary decomposition of the witnesses $w = \sum c_j T_{p_j}$ themselves until that certificate passes, within a bounded number of attempts (the budget mode, the default); the strict mode of Certificate 5.2 continues to termination.

3.3 The local algebra and its invariants

Fix a certified block $V = H_{\mathfrak{m}}$ (over $\mathbb{F}_\ell$; $\dim_{\mathbb{F}_\ell} V = d_{\mathfrak{m}}$) and generators $t_j := T_{p_j}|_V$, $j \in S$. Let $A_S := \mathbb{F}_\ell[t_j : j \in S] \subseteq \operatorname{End}_{\mathbb{F}_\ell}(V)$ (computed as an $\mathbb{F}_\ell$-subspace of $\operatorname{End}(V)$, §5 Stage 3). Its invariants:

- $\dim_{\mathbb{F}_\ell} A_S$; the radical $\operatorname{rad} A_S = \{t \in A_S : t \text{ nilpotent}\} = \mathfrak{m}_S$ (the unique maximal ideal once local), $A_S/\mathfrak{m}_S \cong \mathbb{F}_{\ell^r}$;
- the *embedding dimension* $\dim_{\mathbb{F}_{\ell^r}} \mathfrak{m}_S/\mathfrak{m}_S^2$ (Claim 2.2(b));
- the nilpotency index ν ($\mathfrak{m}_S^\nu = 0 \neq \mathfrak{m}_S^{\nu-1}$), i.e. the Loewy length of A_S; the socle $\operatorname{soc} A_S = \operatorname{Ann}_{A_S}(\mathfrak{m}_S)$ and whether A_S is Gorenstein ($\dim_{\mathbb{F}_{\ell^r}} \operatorname{soc} A_S = 1$);
- the Hilbert function $h(i) = \dim_{\mathbb{F}_{\ell^r}} \mathfrak{m}_S^i/\mathfrak{m}_S^{i+1}$.

All are $\mathbb{F}_\ell$-linear algebra on A_S (§6: no field extension is needed; the $\mathbb{F}_{\ell^r}$-dimensions are the $\mathbb{F}_\ell$-dimensions divided by r).

Remark 3.2 (The coefficient field). A local artinian $\mathbb{F}_\ell$-algebra A with residue field $\mathbb{F} = \mathbb{F}_{\ell^r}$ contains a unique subfield K mapping isomorphically onto $\mathbb{F}$ (Hensel; $\mathbb{F}/\mathbb{F}_\ell$ is separable): $K = \{x \in A : x^{\ell^r} = x\}$, the Teichmüller representatives. Every $x \in A$ is uniquely $x = s + n$ with $s \in K$, n nilpotent, and $s = x^{\ell^{rN}}$ once ℓ^{rN} exceeds the nilpotency index. Claims 2.2(b), 4.2 and 6.1 read "$T_q - a_q(\mathfrak{m})$" and "$\mathbb{F}$-vector space" through K; Stage 3 constructs it explicitly.

Claim 3.3 (Residue field acts through A_S)**.** *[proved here] Let $w \in A_S$ be a residue witness (π_w irreducible of degree r, $\min_V(w) = \pi_w^e$) and put $\hat{w} := w^{\ell^N}$ with N a multiple of r and $\ell^N \geq D$. Then $\pi_w(\hat{w}) = \pi_w(w)^{\ell^N} = 0$, $\hat{w} \equiv w \pmod{\mathfrak{m}_S}$, and $\mathbb{F}_\ell[\hat{w}] = K \cong \mathbb{F}_\ell[X]/(\pi_w)$ is the coefficient field of*

A_S. For every j there is a unique $P_j \in \mathbb{F}_\ell[X]_{<r}$ with $n_j := t_j - P_j(\hat{w})$ nilpotent, namely the one with $P_j(\hat{w})$ the K-component of t_j (for $j = w$: $n_w = w - \hat{w}$); the tuple $(P_j(\bar{X}))_j \in \mathbb{F}_{\ell^r}^S$ is the character $a(\mathfrak{m})$ in the coordinates $\mathbb{F}_{\ell^r} = \mathbb{F}_\ell[X]/(\pi_w)$. In general $P_j(w) \notin K$ and w itself must not be used: for $A_S = \mathbb{F}[\varepsilon]/(\varepsilon^2)$, $t = \alpha + \varepsilon$, $w = t$, one has $t - P(w) = 0$ with the unreduced *$P = X$ although the nilpotent part of t is $\varepsilon \neq 0$; with $\hat{w} = \alpha$ one gets $n_t = \varepsilon$. (The character is always the reduced representative $P_j = X \bmod \pi_w \in \mathbb{F}_\ell[X]_{<r}$: here $r = 1$ and $P_t = \alpha$, so $P_t(w) = P_t(\hat{w}) = \alpha$ and the two recipes agree — the substitution of w for $\hat{w}$ is wrong only at $r \geq 2$, where $P_j(w) \notin K$ in general. Both facts are pinned by* `HeckeLocalityValidation`*, slot 1: the trap arm at $r \geq 2$, and the $r = 1$ twin arm on which the analytic and the matrix path agree verdict field for verdict field.)*

Proof. $A_S/\mathfrak{m}_S$ is a field of degree r containing $\mathbb{F}_\ell[\bar{w}]$ of degree r, so $\bar{w}$ generates it; $\hat{w}$ has the same residue ($r \mid N$) and satisfies $\pi_w(\hat{w}) = 0$ because Frobenius is a ring endomorphism of A_S fixing $\mathbb{F}_\ell$, so $\mathbb{F}_\ell[\hat{w}]$ is a field of degree r, i.e. K. Uniqueness of P_j: $K \to \mathbb{F}$ is an isomorphism and $\bar{t}_j = \overline{P_j(\hat{w})}$. □

4 What "all extension classes" means

4.1 Simple modules and Ext^1 over $\mathbb{T}_\mathfrak{m}$

Over the local artinian ring $\mathbb{T}_\mathfrak{m}$ with residue field $\mathbb{F} := \mathbb{F}_{\ell^r}$ there is exactly one simple module, $\mathbb{F}$ itself ($\dim_{\mathbb{F}_\ell} \mathbb{F} = r$). Every finite $\mathbb{T}_\mathfrak{m}$-module X has a composition series with all factors $\mathbb{F}$; its length is $\dim_{\mathbb{F}_\ell} X/r$. Over $\mathbb{F}_\ell$ (before scalar extension) $H_\mathfrak{m}$ is $\mathbb{T}_\mathfrak{m}$-primary with all Jordan–Hölder factors isomorphic to $\mathbb{F}$.

Claim 4.1 (Ext^1 between simples)**.** *[proved here]* $\mathrm{Ext}^1_{\mathbb{T}_\mathfrak{m}}(\mathbb{F}, \mathbb{F}) \cong \mathrm{Hom}_\mathbb{F}(\mathfrak{m}/\mathfrak{m}^2, \mathbb{F})$*, of $\mathbb{F}$-dimension the embedding dimension. An extension $0 \to \mathbb{F} \to E \to \mathbb{F} \to 0$ is classified by the $\mathbb{F}$-linear map $\mathfrak{m}/\mathfrak{m}^2 \to \mathrm{Hom}_\mathbb{F}(\mathbb{F}, \mathbb{F}) = \mathbb{F}$, $t \mapsto (\bar{b} \mapsto t\,b)$.*

Proof. Apply $\mathrm{Hom}_{\mathbb{T}_\mathfrak{m}}(-, \mathbb{F})$ to $0 \to \mathfrak{m} \to \mathbb{T}_\mathfrak{m} \to \mathbb{F} \to 0$: $\mathrm{Hom}(\mathbb{T}_\mathfrak{m}, \mathbb{F}) \to \mathrm{Hom}(\mathfrak{m}, \mathbb{F}) \to \mathrm{Ext}^1(\mathbb{F}, \mathbb{F}) \to 0$, and $\mathrm{Hom}_{\mathbb{T}_\mathfrak{m}}(\mathfrak{m}, \mathbb{F}) = \mathrm{Hom}_\mathbb{F}(\mathfrak{m}/\mathfrak{m}^2, \mathbb{F})$; the first map is zero because $\mathfrak{m} \cdot \mathbb{F} = 0$. The description of the class is the standard one for a two-step module: $E/\mathfrak{m}E$ and $\mathfrak{m}E$ are $\mathbb{F}$-spaces and $\mathfrak{m}$ acts through $\mathfrak{m}/\mathfrak{m}^2$. □

Claim 4.2 (Yoneda versus module-theoretic data)**.** *[proved here] For a two-step $\mathbb{T}_\mathfrak{m}$-module E ($\mathfrak{m}^2 E = 0$) with $A := \mathfrak{m}E \subseteq E$, $B := E/\mathfrak{m}E$ (both $\mathbb{F}$-spaces through the coefficient field, Remark 3.2), the Yoneda class of $0 \to A \to E \to B \to 0$ in $\mathrm{Ext}^1_{\mathbb{T}_\mathfrak{m}}(B, A) = \mathrm{Hom}_\mathbb{F}(\mathfrak{m}/\mathfrak{m}^2, \mathrm{Hom}_\mathbb{F}(B, A))$ is the* structure map *$\theta_E : \mathfrak{m}/\mathfrak{m}^2 \to \mathrm{Hom}_\mathbb{F}(B, A)$, $t \mapsto (b \mapsto tb)$; it is a complete invariant of the isomorphism class of E up to $\mathrm{GL}(A) \times \mathrm{GL}(B)$, and E is determined by $(\dim B, \dim A, \theta_E)$. The image of θ_E is the space of* active directions*; $\operatorname{rank} \theta_E = 0$ iff E is semisimple.*

Proof. E is the $\mathbb{T}_\mathfrak{m}$-module $B \oplus A$ with $t \cdot (b, a) = (0, \theta_E(t)b)$; isomorphisms of such modules that are the identity on the filtration are exactly pairs (g_B, g_A) conjugating θ_E; the Yoneda class is by definition the obstruction to splitting the same data. □

4.2 The classification data (binding)

For a longer Loewy length the isomorphism type of a $\mathbb{T}_\mathfrak{m}$-module is *not* captured by a finite list of numbers: for embedding dimension ≥ 2 the module category of $\mathbb{F}[x, y]/(x, y)^\nu$ is of wild representation type for $\nu \geq 3$ (Drozd [15]; Ringel [30]; $\nu = 2$ is the tame Kronecker case) [standard, reference]. "Classifying all extension classes" is therefore taken to mean the following *certified record* per $(n, \ell, \mathfrak{m})$, which determines the modules up to isomorphism by exhibiting them, together with the invariants that make two records comparable:

(D1) $\mathbb{T}_\mathfrak{m}$ (in the form of the matrices of the certified generators t_j on $H_\mathfrak{m}$, an $\mathbb{F}_\ell$-basis of A_S, and the residue witness), its invariants of §3.3, and the certificate that $A_S = \mathbb{T}_\mathfrak{m}$ (Stage 4, §5).

(D2) For each of the six X_i: the Loewy (radical) series $X_{i,\mathfrak{m}} \supseteq \mathfrak{m}X_{i,\mathfrak{m}} \supseteq \mathfrak{m}^2 X_{i,\mathfrak{m}} \supseteq \cdots$ and the socle series $0 \subseteq \operatorname{soc} X \subseteq \operatorname{soc}^2 X \subseteq \cdots$ ($\operatorname{soc}^{j+1} X/\operatorname{soc}^j X = \operatorname{soc}(X/\operatorname{soc}^j X)$), with layer dimensions in units of r; X is semisimple iff $\mathfrak{m}X = 0$.

(D3) For each generator t_j and each X_i: the Jordan type of $t_j - a_{p_j}(\mathfrak{m})$ over $\mathbb{F}$, i.e. the partition read from the nullities of $\pi_j(t_j)^s$ on $X_{i,\mathfrak{m}}$ divided by f (Claim 6.1: the S10/S11 generalized-Jordan partition with every multiplicity divided by $f/\deg \pi_j$), and, when ℓ-adic lifts are relevant, the same for T_ℓ specifically (Claim 2.14).

(D4) For every consecutive two-step piece $\mathfrak{m}^i X/\mathfrak{m}^{i+2}X$ the structure map θ from $\mathfrak{m}/\mathfrak{m}^2$ to $\mathrm{Hom}_{\mathbb{F}}(\mathfrak{m}^i X/\mathfrak{m}^{i+1}X,\ \mathfrak{m}^{i+1}X/\mathfrak{m}^{i+2}X)$ ($\mathbb{F}$-linear, $\mathbb{F}$ acting through the coefficient field), its rank and its matrices; the subset of S whose $n_j = t_j - P_j(\hat{w})$ (Claim 3.3) span the active directions *over* $\mathbb{F}$ is the minimal set needed to *see* the extensions at $\mathfrak{m}$.

(D5) Cyclicity and freeness: is $X_{i,\mathfrak{m}}$ cyclic over $\mathbb{T}_{\mathfrak{m}}$ (a quotient of $\mathbb{T}_{\mathfrak{m}}$), is it free of rank one ("multiplicity one"), and the minimal number of generators $\dim_{\mathbb{F}} X/\mathfrak{m}X$.

(D6) The position of the canonical submodules relative to the Loewy layers: $H^1_!$, the *Eisenstein line* at $\mathfrak{m}$ — defined as $\mathrm{soc}(X^+_{4,\mathfrak{m}})$ when that socle is one-dimensional over $\mathbb{F}$ and undefined otherwise (the reductions of the integral Eisenstein and cuspidal eigenlines are not mod-ℓ Hecke data: they reduce into the same joint eigenspace $E = \mathrm{soc}\, X_{4,\mathfrak{m}}$ of the T_q, §8. *Sharpened, v1.4.0*: through v1.3.3 this read "at $(10, 691)$ both reduce to the same line"; measured on the unsplit H^1, E is *two-dimensional* at $(10, 691)$ and at every one of the ten Kummer cells — the socle of the non-split block, $\varepsilon\mathbb{T}_{\mathfrak{m}} \oplus \mathbb{F}$ in the notation of Claim 8.1, one line per ι-sign expected — and a line on every off-Dickson non-Kummer cell with $k \le 26$ (43 cells) (`HeckeBoundaryParabolicValidation`, slot-2 WP-7, 2026-09-16); so without the sign split no "Eisenstein line" exists at a Kummer cell, and $\mathrm{soc}(X^+_{4,\mathfrak{m}})$ is a line there only after E3 ([to be verified computationally] per sign). Only a mod-ℓ^2 run separates the two reductions: the $\mathbb{Z}/\ell^2$-based definition of E8 is this datum's *refinement*, never a replacement, and the mod-ℓ socle semantics stays the published one until an E8 run exists — slot-1 owner confirmation 9(a); v1.9.0: an E8 run exists since slot 8, but only in the validations and the golden harness, never in a record, so the published semantics is unchanged; what the run measured at $(10, 691)$ is recorded in §7.1 (d)) — the Dickson classes (X_1 and $r^0 X_1 \subseteq X_2$), $L \cdot H^1(M_{n-\ell-1})$ when $n \ge \ell + 1$, and the images/kernels of the maps of [C, `eq:les`]; for each: a Loewy layer, a sum of layers, or skew.

(D7) The list of *predicted* extensions (§4.3) with a confirmed/refuted flag, and the list of unpredicted non-split pieces.

"X_i carries a non-trivial Hecke module extension at $\mathfrak{m}$" $:\Leftrightarrow \mathfrak{m}X_{i,\mathfrak{m}} \neq 0$.

4.3 What to expect

Prediction 4.3 (Eisenstein–cuspidal). [standard, reference] (Ribet [29], Mazur [25]; Kummer congruences) *Reworded at v1.12.0.* Let $k = n + 2 \not\equiv 0 \pmod{\ell - 1}$ and let χ_0 be the system of E_k's own class: the anemic eigenvalues $1 + p^{k-1}$ ($p \neq \ell$) *together with* the U_ℓ-eigenvalue 1. Then $\chi_0 \equiv$ a cuspidal system at $\mathfrak{m}$ iff $\ell \mid \mathrm{num}(B_k/k)$, i.e. (Kummer) iff $\ell \mid \mathrm{num}(B_w/w)$ for the $w \equiv k \pmod{\ell - 1}$ with $2 < w \le \ell - 1$; the "usual exceptional residues" are $w \equiv 0, 2 \pmod{\ell - 1}$, where no such w exists (at $w \equiv 2$ the answer is "never", $B_2/2$ being an ℓ-unit at $\ell \ge 5$). Then r^1 vanishes on the relevant class and $X_{4,\mathfrak{m}}$ is non-split. For $n < \ell - 1$ this is Claim 1.1; the Galois side is the class of Ribet's lemma in $H^1(\mathbb{Q}_{\{\ell\}}, \omega^{-(n+1)})$. *What the prediction does not say.* Through v1.12.0-pre.2 it read "$\chi_a \equiv$ a cuspidal system at $\mathfrak{m}$ iff $\ell \mid \mathrm{num}(B_w/w)$ for some $w \equiv n + 2 - 2a \pmod{\ell - 1}$, $2 < w \le \ell - 1$ (up to the usual exceptional residues)", for every twist a and on the anemic system. Read literally that is false: at $(10, 7)$, $a = 1$, $\chi_1(p) = p + p^{10} \equiv p + p^4 \equiv \tau(p) \bmod 7$ (Ramanujan's $\tau(n) \equiv n\sigma_3(n) \bmod 7$), a cuspidal system, while $w \equiv 4$ and $B_4/4 = -1/120$ is a 7-unit; for $a \ge 1$ the θ-twist of an Eisenstein system is a mod-ℓ cusp form at every ℓ ([C, `rmk:thetaclassical`]), so no Bernoulli condition governs it. At $a = 0$ the anemic system alone does not suffice either: the congruence census (§7.2, edition 1) has 58 non-reduced `mixed` rows at the regular primes $7, 11, 13$ carrying E_k's anemic system with U_ℓ-eigenvalue 0 (the system of $\theta^{\ell-1}E_k$ modulo ℓ, whose a_ℓ is 0), and 592 `mixed` rows at those primes in all. (Found by slot-11 WP-19, confirmed by its referee WP-20.) The [standard, reference] tag is for the statement as reworded. *Which "cuspidal" (v1.12.1, the slot-11 WP-15 referee's m-6).* "A cuspidal system" above means the system of a characteristic-zero cusp eigenform (as the $(198, 691)$ item of §8 and the examples document read it); under that reading both directions are the classical ones (Ribet, Mazur). If it is read as "the system of a mod-ℓ cusp form", the direction ($\Rightarrow$) still holds, but the

"iff" is then the Serre / Swinnerton-Dyer constant-term criterion (E_k mod ℓ with vanishing constant term is a mod-ℓ cusp form of weight k, and lifts by Deligne–Serre with $a_\ell \equiv 1$), not the Ribet–Mazur statement. The truth of the claim does not depend on the reading.

Prediction 4.4 (Cuspidal–cuspidal). [conjectural] (expected on the evidence of [M §7]; to be confirmed case by case) Congruent forms in one weight arise from $E_{\ell-1}$-multiplication, from $\theta^{\ell-1}$ (Claim 2.6), and accidentally. Multiplicity ≥ 2 forces the split/non-split question; T_ℓ is usually the active direction, but anemic nilpotents occur (large $\mathbb{T}_\mathfrak{m}$).

Prediction 4.5 (Boundary and cone). [conjectural] When $(\ell-1) \mid n$ or more generally $\chi_a \equiv \chi_{a'}$, the cone can glue $H^0(\partial)$- and $H^1(\partial)$-type classes; when a Dickson class has the character of a cusp form or of another boundary class, X_3 and X_4 can glue them. These are the extension classes that are "in the cone" rather than in H^1; exactness of [C, `eq:les`] determines $X_{3,\mathfrak{m}}$ only up to extension, the cone determines it. *Which* cone module is meant is therefore part of the prediction, and it is fixed by (A2): the $\mathbb{T}_\mathfrak{m}$-module structure on $X_{3,\mathfrak{m}}$ is the one induced by $\Phi_g(a,b) = (\Theta_g a, \Theta_g^\partial b + ha)$ with h derived from the certified pair homotopy η. It is *not* determined by exactness plus a freely chosen cochain homotopy: the two would differ by an arbitrary element of $\mathrm{Hom}(\ker r^1, \mathrm{coker}\, r^0)$, which is non-zero in exactly the situations this prediction is about — at $(10,691)$ it has dimension 3 while $\dim X_3 = 4$ ((A2), third bullet). The gluing is therefore a statement about a well-defined module, and the computation reports it as such.

Prediction 4.6 (Duality). [standard, reference] ([C, `cl:pd`]) The Loewy layers of $H^{2-i}(\Gamma; D_n)_\mathfrak{m}$ are the socle layers of $X_{i,\mathfrak{m}} = H^i_c(\Gamma; M_n)_\mathfrak{m}$ in reverse order, and vice versa ($\mathrm{rad}^i(V^\vee) = (\mathrm{soc}^i V)^\perp$; both series are in (D2), so nothing new is computed — but "reversed Loewy series" would fail on any non-rigid module); a module cyclic for M_n need not be cyclic for D_n. Computed in E7 (slots 6–7 of [X] v1.5.1, scheduled; asserted at $(10,691)$ and on the Kummer cells by `HeckeDualityValidation`, v1.4.0).

Prediction 4.7 (Galois interpretation). [conjectural] (Bellaïche–Chenevier [2]) Non-split $\mathbb{T}_\mathfrak{m}$-extensions between a cuspidal and an Eisenstein-type factor correspond to non-zero classes in $H^1(\mathbb{Q}_{\{\ell\}}, \bar\rho_1 \otimes \bar\rho_2^\vee)$; the computed active-direction ranks are to be compared with those Galois cohomology dimensions.

Prediction 4.8 (Concentration for $n \gg \ell$). [standard, reference] (from Claim 2.4) $\dim H^1 \sim n/6$ while $\#\Sigma(\ell)$ is bounded: for fixed ℓ the module is dominated by a bounded number of $\mathfrak{m}$ with $\dim H_\mathfrak{m} \to \infty$; the invariants of interest become the embedding dimension and Loewy length of $\mathbb{T}_\mathfrak{m}$ as functions of n (compare the Krull-dimension-2 picture of Claim 2.13).

Prediction 4.9 (The lattice is nearly split on congruence-free classes (R); formerly C-F9). [conjectural] (restated a fifth time at v1.25.0-pre.1, **owner ruling of 2026-11-06**: on irreducible classes the hypothesis "not dihedral" is replaced by "of one weight"; the v1.22.0-pre.1 form is **refuted on cohomology** at a non-ordinary class at $\ell = 59$, see below (`docs/implementation/slot25/R59_verification.md`); it is the statement below with "when $\bar\rho_\mathfrak{m}$ is irreducible, of one weight" read as "not dihedral" and without the definition of *of one weight*, and is not repeated. Restated a fourth time at v1.22.0-pre.1 (dihedral classes excluded; owner ruling of 2026-10-09 on census finding F33-2, `docs/implementation/slot22/owner_ruling_F33-2_1009.md`, wording of the slot-22 Fable referee `docs/implementation/slot22/F33-2_referee.md` §3.3): the v1.17.0-pre.4 form is **refuted on cohomology** at the dihedral classes at $\ell = 23$ and 31, see below; it is the statement below with "congruence-free and not dihedral" read as "congruence-free" and without the definition of *dihedral*, and is not repeated. Restated a third time at v1.17.0-pre.4, owner ruling of 2026-10-02 on finding B-1 of the slot-17 Fable referee F1, `docs/implementation/slot17/WP-3_referee_F1.md`: the v1.17.0-pre.2 form, asserted on *Kummer-free* classes, is **contradicted on the forms side** at irreducible congruence classes, the regular prime $\ell = 139$ among them, see below. Restated a second time at v1.17.0-pre.2, owner ruling O-R17-2a of 2026-10-02, before the referee: the v1.16.0 form, which excluded only the Kummer–Frobenius locus, is **refuted as stated** on cohomology at $(608,547)$ and contradicted on the forms side at three further primes by slot-17 R17-2, see below. The label is kept. Below the new form stand, as published and marked contradicted or refuted, the v1.17.0-pre.2 form, the v1.16.0 form (restated v1.15.1-pre.7: slot-15 WP-12, proposal (E-1) of `docs/implementation/slot15/WP-12_U_E_proposals.md`; the

Fable referee `docs/implementation/slot15/WP-4_referee.md` §8; owner ruling S15-O2 of 2026-10-01) with its evidence, and the v1.14.0 statement with its evidence.) Let $\ell \geq 5$, $n \geq 1$ even, $k = n+2$, $\Lambda := H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}_\ell)$, and $\mathfrak{m}$ a maximal ideal of $\mathbb{T}$ on Λ with residue field $\mathbb{F}_\ell$, not of Dickson type; write $a_q \in \mathbb{F}_\ell$ for the eigenvalue of T_q at $\mathfrak{m}$, and $\bar\rho_\mathfrak{m}$ for its semisimple residual Galois representation. When $\bar\rho_\mathfrak{m} \cong \omega^i \oplus \omega^j$ is reducible, let $d \in [1, \ell-2]$ with $d \equiv j - i \pmod{\ell-1}$ (d is odd) and call $w_1 = d+1$, $w_2 = \ell - d$ (so $w_1 + w_2 = \ell + 1$, both even and, $\mathfrak{m}$ not being of Dickson type, both in $[4, \ell-3]$) the *seeds* of $\mathfrak{m}$; the pair $\{w_1, w_2\}$ does not depend on the order of i, j. Call $\mathfrak{m}$ a *Kummer class* if $\bar\rho_\mathfrak{m}$ is reducible and ℓ divides B_{w_1} or B_{w_2}, and *Kummer-free* otherwise — that is, if $\bar\rho_\mathfrak{m}$ is irreducible, or reducible with both seeds ℓ-regular. Call $\mathfrak{m}$ *congruence-free* if for every even w with $2 \leq w \leq \ell+1$ and every $0 \leq j \leq \ell-2$ the generalised eigenspace of the system $(p^{-j}a_p)_{p \neq \ell}$ in $M_w(\mathbb{F}_\ell)$ has dimension at most 1. On a reducible non-Dickson class this is Kummer-free (at a seed w the eigenspace has dimension $1 + g_w$, below); on an irreducible class it says that no two level-1 eigenforms of a weight $\leq \ell+1$ in the class are congruent. Call $\mathfrak{m}$ *dihedral* if $\bar\rho_\mathfrak{m}$ is irreducible and $\bar\rho_\mathfrak{m} \cong \bar\rho_\mathfrak{m} \otimes \omega^{(\ell-1)/2}$ (equivalently, by Brauer–Nesbitt, $a_p = 0$ for every prime $p \neq \ell$ with $\left(\frac{p}{\ell}\right) = -1$; equivalently $\bar\rho_\mathfrak{m}$ is induced from $\mathbb{Q}(\sqrt{\ell^*})$, $\ell^* = (-1)^{(\ell-1)/2}\ell$, i.e. $\bar\rho_\mathfrak{m}|_{G_{\mathbb{Q}(\zeta_\ell)}}$ is reducible, the case the Taylor–Wiles hypothesis excludes). At level 1 the dihedral classes are, for $\ell \equiv 3 \pmod 4$ with $h(-\ell) > 1$, the twists $\omega^j \otimes \bar\rho_f$ of the classes of the cusp forms f of weight $(\ell+1)/2$ with $\bar\rho_f|_{I_\ell} \cong 1 \oplus \omega^{(\ell-1)/2}$ — f is its own companion, the test above sees its line at the twists j and $j + (\ell-1)/2$ — and there are none at $\ell \equiv 1 \pmod 4$ (parity: $\mathbb{Q}(\sqrt{\ell})$ is real and, ℓ prime, has $h^+ = h$, so an induced class unramified outside ℓ is even). A reducible class is never called dihedral here, although $\omega^i \oplus \omega^{i+(\ell-1)/2}$ is its own quadratic twist: on those classes the statement is kept (evidence below). (v1.22.0-pre.1.) Call an irreducible $\mathfrak{m}$ *of one weight* if there is exactly one pair (w, j), w even, $2 \leq w \leq \ell+1$, $0 \leq j \leq \ell-2$, for which the generalised eigenspace of the system $(p^{-j}a_p)_{p\neq\ell}$ in $M_w(\mathbb{F}_\ell)$ is non-zero — the pairs and the eigenspaces of the test that defines *congruence-free* above; so an irreducible class that is congruence-free and of one weight has, over all pairs, exactly one non-zero eigenspace, and it is a line. An irreducible class of one weight is *not dihedral*: since $p^{(\ell-1)/2} \equiv \left(\frac{p}{\ell}\right) \pmod \ell$, the system at $(w, j + \frac{\ell-1}{2})$ (twist read modulo $\ell-1$) is $\left(\left(\frac{p}{\ell}\right)p^{-j}a_p\right)_{p\neq\ell}$, which on a dihedral class (where $a_p = 0$ whenever $\left(\frac{p}{\ell}\right) = -1$) equals the system at (w, j); so a dihedral class has either no such pair or at least two. The new hypothesis therefore implies the old one, and the statement below asserts less than the v1.22.0-pre.1 form. Not of one weight are, besides the dihedral classes (pairs (w, j), $(w, j + \frac{\ell-1}{2})$, $w = \frac{\ell+1}{2}$, above), every non-ordinary class (pairs (k_0, j) and $(\ell + 3 - k_0, j + k_0 - 2)$, by the θ-cycle; [standard, reference], literature, checked at the eight $\mathbb{F}_\ell$-rational non-ordinary classes at the census primes $\ell \leq 199$ in R59 §6) and every ordinary class with a companion form (pairs (k_0, j) and $(\ell + 1 - k_0, j + k_0 - 1)$, Gross's companion forms; [standard, reference], literature, no instance other than the dihedral ones computed). The expected meaning (the weight part of Serre's conjecture; not verified here): an irreducible $\mathfrak{m}$ is of one weight exactly when $\bar\rho_\mathfrak{m}|_{I_\ell}$ is a non-split extension of two characters, ordinary without companion. (v1.25.0-pre.1, owner ruling of 2026-11-06.) If $\mathfrak{m}$ is congruence-free and, when $\bar\rho_\mathfrak{m}$ is irreducible, of one weight, then

$$(T_q - a_q)(T_{q'} - a_{q'})\,\Lambda_\mathfrak{m} \subseteq \ell\,\Lambda_\mathfrak{m} \qquad \text{for all primes } q, q' \quad (T_\ell = U_\ell);$$

equivalently (as for the v1.14.0 statement below), the local algebra A of $\mathbb{T}$ on $V = H^1(\Gamma; M_n)_\mathfrak{m}$ has $\mathfrak{m}_A^2 = 0$. Nothing is asserted on the other classes. (Regularity of ℓ no longer implies the hypothesis: the v1.17.0-pre.2 sentence "at a regular prime every $\mathfrak{m}$ is Kummer-free, so in particular the statement holds at every regular ℓ" is withdrawn, contradicted at $\ell = 139$ below.)

Where the excluded locus sits (v1.17.0-pre.2; the bookkeeping of slot-17 R17-1, `docs/implementation/slot17/` 5, not binding). Fix an irregular pair (ℓ, w_K), $\ell \mid B_{w_K}$, and put $w_O = \ell + 1 - w_K$. Its Kummer classes are $\omega^a(1 \oplus \omega^{w_O-1}) = \omega^b(1 \oplus \omega^{w_K-1})$; R17-1 splits the cuspidal ones into the *standard* family $a \in [1, w_K - 2]$ and the *swapped* family $b \in [1, w_O - 2]$, besides the two Eisenstein ends $a = 0$ (the class of E_{w_O}) and $b = 0$ (that of E_{w_K}). A Kummer–Frobenius ideal of the v1.16.0 form (seed $w = w_O$, twist $0 \leq t \leq J(w) = w_K - 1$) is a Kummer class from its onset on, of the standard family ($t = a$) or at one of the two Eisenstein ends ($t = 0$, or $t = J(w)$, the class $b = 0$); the swapped family is *never* Kummer–Frobenius, since that would need $\ell \mid B_{w_O}$ as well, a double divisibility (R17-1 Lemma 3.1, proved there; none for $\ell < 2 \cdot 10^4$, by scan; since v1.18.0-pre.1 Claim 4.14 of this document). By

R17-1's Theorem 4.1 (proved there from literature inputs, which are VERIFY there; not a statement of [C] — since this document now argues from it, it is to become a statement of [C] or of this document in slot 18; v1.17.0-pre.3, the referee F1's m-E5; **since v1.18.0-pre.1 it is Claim 4.16** of this document, with its inputs located and tagged there), $\dim V = m_1(1+2g_{w_1}) + m_2(1+2g_{w_2})$ on a reducible non-Dickson class, with m_1, m_2 the Jordan–Hölder multiplicities of the two Serre weights in $\mathrm{Sym}^n \mathbb{F}_\ell^2$ and g_w the dimension of the generalised eigenspace of E_w's system ($a_p = 1 + p^{w-1}$) in $S_w(\mathbb{F}_\ell)$, which is 0 unless $\ell \mid B_w$. So on a reducible congruence-free (equivalently, Kummer-free) class the coefficients $1 + 2g$ never exceed 1 ($\dim V = m_1 + m_2$): on reducible classes the restated form asserts $\mathfrak{m}_A^2 = 0$ exactly where the Kummer coefficient does not enter. Its irreducible analogue — two congruent level-1 eigenforms of one weight $\leq \ell + 1$ — is what the hypothesis now also excludes (v1.17.0-pre.4).

Contradicted on the forms side: the v1.17.0-pre.2 form (v1.17.0-pre.4; the slot-17 referee F1, finding B-1, reproduced with tracked tooling by slot-17 R17-2, `docs/implementation/slot17/R17-2_report.md` §7: `r172/r172_forms.py --seed`, `r172/r172_seedcheck.py`; computed by script, forms side only, nothing here is cohomology). The v1.17.0-pre.2 form asserted (R) on *Kummer-free* classes (residual representation irreducible, or reducible with both seeds ℓ-regular), and added that it therefore holds at every regular ℓ. At $(\ell, k_0) = (67, 32)$, $(131, 28)$, $(139, 28)$ the space S_{k_0} has dimension 2 and is one Galois orbit over a quadratic Hecke field in which ℓ ramifies (the discriminant of the T_2-charpoly is $2^6 \cdot 3^2 \cdot 67 \cdot 273067$, resp. $2^6 \cdot 3^6 \cdot 131 \cdot 139$); the two eigenforms are congruent modulo the prime above ℓ, confirmed over $\mathbb{Z}$ ($(T_p - \lambda_p)^2 = 0 \neq T_p - \lambda_p$ on $S_{k_0}(\mathbb{F}_\ell)$ for every $p \leq 23$), and no $p^a + p^b$ fits the system, so the class is irreducible, hence Kummer-free; $67 \mid B_{58}$ and $131 \mid B_{22}$, while 139 is **regular**. At the θ^j-twists $k = \ell k_0 + j(\ell+1)$, $j = 1, 2$, i.e. the six cells $n = 2210$, 2278 ($\ell = 67$), 3798, 3930 ($\ell = 131$), 4030, 4170 ($\ell = 139$), the generalised eigenspace $W \subseteq S_k(\mathbb{F}_\ell)$ has $\dim W = 4$, Hilbert function $[1, 1, 1, 1]$, $\dim A = \dim W$ (generated; the same with the anemic set $\{2, 3, 5, 7\}$ and with $p \leq 43$), T_2 of Jordan type [4] and T_ℓ nilpotent: $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell \cong \mathbb{F}_\ell[x]/x^4$. Six further pairs, $(73, 40)$, $(71, 54)$, $(89, 68)$, $(181, 38)$, $(227, 46)$, $(349, 38)$, each first confirmed over $\mathbb{Z}$, show the same $\mathbb{F}_\ell[x]/x^4$ at both onset cells (12 cells). By [C, `prop:cohforms`](5) these blocks are torsion-free and A is the quotient of $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell$ by the saturation defect ([C, `rmk:sattor`]), so the pre.2 form fails there unless $\dim I \geq 2$; A itself was not computed (the cheapest cohomology candidate, $(2210, 67)$ — prediction $\dim V = 8$, $A \cong \mathbb{F}_{67}[x]/x^4$, saturated — is cluster-only; predict RSS first). Since 139 is regular, the sentence "(R) holds at every regular ℓ" (slot 15, `WP-4_referee.md` §8, kept in the v1.16.0 and pre.2 forms) is contradicted on the forms side, and so is the v1.16.0 form at this second, independent locus (an irreducible class is never Kummer–Frobenius). None of these classes is congruence-free (at $w = k_0$, $j = 0$ the eigenspace has dimension 2), so none is a counterexample to the restated form.

Refuted on cohomology: the v1.17.0-pre.4 form, at dihedral classes (v1.22.0-pre.1; census 3.3 finding F33-2, `docs/implementation/slot22/C33_followups_report.md`, the Fable referee `F33-2_referee.md`; computed). At $\ell = 23$ the class of Δ and at $\ell = 31$ that of the weight-16 form are dihedral by $\mathbb{Q}(\sqrt{-\ell})$: $\bar\rho \cong \bar\rho \otimes \omega^{(\ell-1)/2}$, $\bar\rho|_{I_\ell} \cong 1 \oplus \omega^{(\ell-1)/2}$, the form its own companion. Every twist is congruence-free in the sense above (the only non-zero eigenspaces are the form's own line at $(w, j) = ((\ell+1)/2, 0)$ and $((\ell+1)/2, (\ell-1)/2)$, each of dimension 1), and the v1.17.0-pre.4 form asserted $\mathfrak{m}_A^2 = 0$ there. Census 3.3 (`rows33.tsv`, C1a local run, cells certified, 24 cells recomputed byte for byte) reads $A \cong \mathbb{F}_\ell[x]/x^3$, Hilbert function $[1, 1, 1]$, on 161 rows: 140 at $\ell = 23$ ($n \geq 298$, $\dim V = 8$ or 12, module Loewy layers $[4, 2, 2]$ and $[6, 4, 2]$) and 21 at $\ell = 31$ ($n \geq 526$, $\dim V = 8$), at every twist s, with $(T_2 - a_2)^2 \neq 0$ on V on each (recomputed from the records' matrices); the operators of the primes inert in $\mathbb{Q}(\sqrt{-\ell})$ and U_ℓ lie in $\mathfrak{m}_A^2$. The forms side agrees: at $(298, 23)$, $(540, 23)$ and $(526, 31)$, $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell \cong \mathbb{F}_\ell[x]/x^4$ on the four-dimensional generalised eigenspace (by script, modulo ℓ^6: every eigenform of the class is ramified at ℓ, Newton slope $\frac{1}{2}$ at the split primes and 1 at the inert ones, $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}$ monogenic), and A is its quotient by a saturation defect of length 1 ([C, `prop:cohforms`](5)); at $(540, 23)$ the defect has length 2 and $A \cong \mathbb{F}_\ell[x]/x^2$, so the seven rows $n = 540 + 24t$ (among the 768 consistent dihedral rows, not among the 161) hold the statement only through the defect. The onset is $n = \ell(\ell+3)/2 - 1$ (298, 526); for $\ell = 59, 71, 79, 83$ it lies beyond $n \leq 700$. No non-dihedral class in census 3.3 reaches Loewy length 3 (the referee's §4: of the 100 025 non-Dickson, non-dihedral rows of residue degree 1, Loewy length 1 on 85 123, 2 on 14 818,

4 on the 84 Kummer–Frobenius rows at $(\ell, w_K) = (37, 32)$, which are not congruence-free, and 3 on none). The hypothesis now excludes dihedral classes (929 rows of residue degree 1 in census 3.3: the 161 above, and 768 consistent with $\mathfrak{m}_A^2 = 0$, below their onsets); the reducible self-twist classes ($\omega^i \oplus \omega^{i+(\ell-1)/2}$, both seeds $(\ell+1)/2$) stay inside: 1770 rows at $\ell = 7, 11, 19, 23, 31, 43, \ldots$, $\dim V$ up to 24, all with $\mathfrak{m}_A^2 = 0$. Whether the true obstruction is the self-twist or, more generally, two cuspidal Serre weights cannot be decided on census 3.3 (no ordinary class with a cuspidal companion other than the self-twist ones occurs at level 1 and $\ell \le 103$; the referee's §§2.2, 2.6): a non-dihedral class of that shape with $\mathfrak{m}_A^2 \neq 0$ would refute the present form. (v1.25.0-pre.1: one does, at $\ell = 59$, a non-ordinary class — the slot-22 scan took "ordinary" as $a_\ell \neq 0$ and did not look at non-ordinary classes, every one of which has that shape; next paragraph.)

Refuted on cohomology: the v1.22.0-pre.1 form, at a non-ordinary class (v1.25.0-pre.1, owner ruling of 2026-11-06; census 4.0 rows found by the gallery referee, `docs/implementation/slot25/PG140_referee.md` verified independently from scratch, the data source `docs/implementation/slot25/R59_verification.md`, scripts and outputs in its `r59/`; computed). At $\ell = 59$ the class of $f_{16} = \Delta E_4$, system $(a_2, a_3, a_5, a_7, a_{59}) \equiv (39, 15, 13, 14, 0)$, is irreducible (no $p^i + p^j$ fits the system at the 11 primes $p \le 31$), so neither of Dickson type nor a Kummer class; *not dihedral* ($59 \equiv 3 \bmod 8$, so $\left(\frac{2}{59}\right) = -1$, while $a_2 = 39 \neq 0$; also $a_{11} = 41 \neq 0$); and *congruence-free*: over all even $w \le 60$ and all 58 twists the eigenspace in $M_w(\mathbb{F}_{59})$ is non-zero at exactly the two pairs $(16, 0)$ and $(46, 14)$, a line at both (exact at this ℓ: T_2, T_3, T_5 span the Hecke algebra of $M_w(\mathbb{F}_{59})$ for $w \le 60$). It is *non-ordinary*, $a_{59} \equiv 0$, and so has two cuspidal Serre weights, 16 and $46 = \ell + 3 - 16$: it is not of one weight. Census 4.0 (main section) has four Loewy-3 rows of the class, residue degree 1, `final`, `sturm(B′)`: $n = 884$ and 944 (θ^0, resp. θ^1; $\dim V = 6$, $A \cong \mathbb{F}_{59}[x]/x^3$, module Loewy layers $[2, 2, 2]$) and $n = 942$, 1000 (θ^0; $\dim V = 8$, Hilbert function $(1, 2, 1)$, layers $[4, 2, 2]$). Recomputed with the shared binary from the Hecke matrices on H^1: $(T_2 - a_2)^2 \neq 0$ on V at all four cells (ranks of $T_2 - a_2$, its square and cube $4, 2, 0$; block dimensions 6, 8, 6, 8), and at $n = 884$ by a second, independent linear-algebra route on the dumped matrices. On the forms side $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell$ has the census Hilbert function at all four cells, so the saturation defect is 0 there ([C, `prop:cohforms`](5)); the blocks are torsion-free ($\dim V = 2 \dim W$). So the v1.22.0-pre.1 form, which asserted $\mathfrak{m}_A^2 = 0$ on this congruence-free, non-dihedral class, fails; nothing proved in this document and no data sentence becomes false (the census-3.3 sentences above are scoped to $n \le 700$; the onset here is $n = \ell(k_0 - 1) - 1 = 884$). The forms side repeats the pattern at a non-ordinary class with large image, $\ell = 79$, weight 38 (companion weight 44): $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell$ of Hilbert function $(1, 1, 1)$ at $n = 2922 = 79 \cdot 37 - 1$ and $(1, 2, 1)$ at $n = 3000$ (by script; cohomology not computed), so the obstruction is non-ordinarity, not the S_4-type image of the class at 59. *Consistency of the restated form* (R59 §7.1): of the 672 rows of census 4.0 (main and b1; section b2 has none) with Loewy length ≥ 3, all of residue degree 1, the 418 dihedral rows at $\ell = 23$, 31 and the 4 rows at 59 are not of one weight, the 235 Kummer rows at $\ell = 37$, 67, 547 and the 15 rows at $\ell = 139$ ($\theta^j(f_{138})$, a congruence in weight 138) are not congruence-free; and the other irreducible, non-dihedral rows of residue degree 1 with $\dim V \ge 6$ (where (R) has content) are 5805 rows at $11 \le \ell \le 31$, 23 ordinary classes (of Δ, f_{16}, f_{18}, f_{20}, f_{22}, f_{24}, f_{26}) each at exactly one pair (w, j), hence of one weight, with Loewy length 1 on 2008 and 2 on 3797. So no row of census 4.0 contradicts the restated form, and within the census it newly excludes, among the rows with $\dim V \ge 6$, only the four at $\ell = 59$. The narrower alternative "congruence-free, not dihedral and ordinary" (R59 §7.3) would keep the ordinary classes with a companion other than themselves inside, the shape warned about above, of which no instance has data; the owner chose "of one weight".

Refuted as stated: the v1.16.0 form (v1.17.0-pre.2; slot-17 R17-2, `docs/implementation/slot17/R17-2_repor` every item at a Kummer class that is *not* Kummer–Frobenius, so a counterexample to the v1.16.0 form and none to the restated one):

1. *On cohomology* (computed; R17-2 §6.1). At $\ell = 547$, $w_K = 486$ has $g_{486} = 2$ (two cusp forms of weight 486 congruent to E_{486} modulo 547, while $v_{547}(B_{486}) = 1$; R17-1 §4, by script), so every cuspidal Kummer class at $(547, 486)$ already has rank $1 + g = 3$ in its first window. The class $a = 1$ at $n = 608$ ($k = 610$, $a_p \equiv p + p^{62}$) lies far below its Kummer–Frobenius onset $k \ge 547 \cdot 62 + 548 = 34462$. `HeckeSaturationDefectValidation` with `HECKE_SAT_DEFECT_CELLS="608:547"` (the slot-17 shared binary; operators $T_2, T_3, T_5, T_7, U_{547}$; $e_{\max} = 0$; 23 min, 3.7 GiB): on the block

of the system $(a_2, a_3, a_5, a_7, a_{547}) = (17, 185, 281, 442, 0)$, $\dim V = 6$, $\operatorname{rank} \mathbb{T}_{\mathfrak{m}} = 3$, the census algebra A and $\mathbb{T}_{\mathfrak{m}}/547$ both have Hilbert function $(1, 1, 1)$, so $A \cong \mathbb{F}_{547}[x]/x^3$; the saturation defect is $I_{\mathfrak{m}} = 0$ ($\mathbb{T}_{\mathfrak{m}}$ saturated, [C, `prop:satdefect`]), and the `stable` verdict is certified at $K = 2$ ([C, `cor:stablecert`]). Hence $\mathfrak{m}_A^2 \neq 0$ on V. The forms side reads the same algebra ($\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/547 \cong \mathbb{F}_{547}[x]/x^3$, $\dim W = 3$, by script; the run's own forms cross-route agrees), as [C, `prop:cohforms`](5) requires on a saturated torsion-free block.

2. *On the forms side only* (computed by script, R17-2 §§2–3: the algebra generated by the T_p, $p \leq 23$, on the generalised eigenspace W of $S_k(\mathbb{F}_\ell)$, with $\dim A = \dim W$ as its generation certificate, unchanged with the anemic set $\{2, 3, 5, 7\}$ and with $p \leq 43$): $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell \cong \mathbb{F}_\ell[x]/x^4$ at the swapped classes $b = 1, \ldots, 4$ at $\ell = 37$ ($k = 1222$, 1260, 1298, 1336; $n = 1220, \ldots, 1334$), $b = 1, 2$ at $\ell = 103$ ($k = 2576$, 2680) and $b = 1, 2$ at $\ell = 691$ ($k = 8984$, 9676; the Kummer seed 12 of Δ); and $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/547 \cong \mathbb{F}_{547}[x]/x^3$ at $(1156, 547)$ (class $a = 2$). On these torsion-free blocks [C, `prop:cohforms`](5) makes A the quotient of $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell$ by the saturation defect, so the census-algebra form fails there unless $\dim I \geq 2$ (≥ 1 at 547), [C, `rmk:sattor`] (v1.17.0-pre.3 wording, the referee F1's m-E4). A itself was not computed at these cells ($(1220, 37)$ needs an exclusive memory window, $(2574, 103)$ and $(8982, 691)$ are beyond local reach); every saturation defect measured so far at the Kummer classes has length ≤ 1 (elsewhere defects are longer: length 2 at $(540, 23)$ above; δ up to 190 in census 4.0; v1.24.0-rc.2), and every Loewy-4 Kummer block measured is saturated, so the expectation (not a claim) is that cohomology agrees.

What held at the Kummer classes tested (now outside the assertion; recorded as data). The Eisenstein ends: $\mathfrak{m}^2 = 0$ on the cusp side of the $b = 0$ classes $(1182, 37)$ (E_{32}), $(484, 547)$ (E_{486}) and $(8290, 691)$ (Δ) (forms side; the Eisenstein part needs cohomology). The pre-jump cells of R17-1 §5.2 on cohomology (R17-2 §6.2, console route with U_ℓ): $(738, 67)$, $(944, 59)$, $(1184, 37)$ have module Loewy layers $[3, 1]$ and $A \cong \mathbb{F}_\ell[x]/x^2$ ($(670, 67)$, the census control, reproduced); $(738, 67)$ is saturated and `stable` certified.

Observed on the Kummer locus (not asserted here; a candidate for a later restatement of Prediction 4.10, slot 18 — taken up at v1.18.0-pre.1 as Prediction 4.10 (B), (C) and Prediction 4.18, both [conjectural]). Every cuspidal Kummer-class block computed below the second growth of the multiplicities ($(m_1, m_2) \leq (2, 2)$) is curvilinear of the largest length its rank allows, A (or, where only the forms side was computed, $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell$) $\cong \mathbb{F}_\ell[x]/x^{\dim V/2}$: the 24 Kummer–Frobenius onset cells at 37 and the slot-17 onset cells at 67, 59, 101 (Prediction 4.10); the swapped cells and the $g = 2$ first windows of item 2 and of $(608, 547)$; and, at rank 2, the Kummer rows of edition 3 with $\dim V = 4$ (modules $[2, 2]$ and the pre-jump $[3, 1]$). At the second growth $(3, 3)$ it fails on the forms side ($(1, 2, 2, 1)$ at $k = 1412$, 1450, $\ell = 37$; Prediction 4.10).

Evidence for the restated form. Slot 17 tested only Kummer classes and irreducible congruence classes, none of them congruence-free, and adds none. The evidence is the v1.16.0 form's table below, read on its congruence-free rows: at $\ell \in \{7, 11, 13\}$ every row is congruence-free (M_w, $w \leq \ell + 1$, has dimension ≤ 2, and its one cusp form Δ, at $w = 12$, is irreducible modulo 11 and 13); at the other primes the rows mix congruence-free classes, Kummer classes and irreducible congruence classes, every row with Loewy length ≥ 3 is a Kummer class (the 18 Kummer–Frobenius records and six weight-shift cells at 37), and the irreducible congruence classes with $\ell \leq 107$ ($\ell = 67$, 71, 73, 89) have their onsets at $n \geq 2210$, beyond every row, so passing from Kummer-free to congruence-free loses no evidence (v1.17.0-pre.4; the referee F1's B-1, item 4). No row in the table violates the restated form. The 161 dihedral rows of census 3.3 (above) violate the v1.17.0-pre.4 form, which did not exclude dihedral classes; they lie outside the present form (v1.22.0-pre.1, the slot-22 Fable referee `docs/implementation/slot22/F33-2_referee.md`, owner ruling of 2026-10-09). (v1.17.0-pre.3, the referee F1's m-E3.) On a torsion-free block with $\dim V \leq 4$, $\mathfrak{m}_A^2 = 0$ is automatic: $\dim A \leq \operatorname{rank} \mathbb{T}_{\mathfrak{m}} = \lceil \dim V/2 \rceil \leq 2$ ([C, `lem:heckerankone`], [C, `prop:cohforms`](5)); so the $120 + 229$ below-onset Kummer rows were never evidence, and the content of the prediction lies at $\dim V \geq 5$ (the rows at $\ell \leq 13$ with $\dim V$ up to 26, and the three-copy rows at 13). Whether it holds on every congruence-free class is open; the C++ readings keep *reporting*, never gating.

The v1.16.0 form (restated v1.15.1-pre.7; refuted as stated at v1.17.0-pre.2 by $(608, 547)$, above; kept as published, with its evidence and its refutation of the v1.14.0 statement).

The definition of *Kummer–Frobenius* made here is kept: Prediction 4.10 uses it. Let $\ell \geq 5$, $n \geq 1$ even, $k = n + 2$, $\Lambda := H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}_\ell)$, and $\mathfrak{m}$ a maximal ideal of $\mathbb{T}$ on Λ with residue field $\mathbb{F}_\ell$, not of Dickson type; write $a_q \in \mathbb{F}_\ell$ for the eigenvalue of T_q at $\mathfrak{m}$. Call $\mathfrak{m}$ *Kummer–Frobenius* if

- its residual representation is $\omega^t(1 \oplus \omega^{w-1})$ for an even w with $4 \leq w \leq \ell - 3$;
- ℓ divides the companion Bernoulli number $B_{\ell+1-w}$;
- $k \geq \ell w + t(\ell + 1)$, with $0 \leq t \leq J(w)$, where $J(w) \equiv -w \pmod{\ell}$ (the twist $\theta^t E_w$ has not dropped filtration).

If $\mathfrak{m}$ is not Kummer–Frobenius, then

$$(T_q - a_q)(T_{q'} - a_{q'})\,\Lambda_{\mathfrak{m}} \subseteq \ell\,\Lambda_{\mathfrak{m}} \qquad \text{for all primes } q, q' \quad (T_\ell = U_\ell).$$

Equivalently, the local algebra A of $\mathbb{T}$ on $V = H^1(\Gamma; M_n)_{\mathfrak{m}}$ has $\mathfrak{m}_A^2 = 0$. In particular this holds at every regular ℓ (the referee's form (R), `WP-4_referee.md` §8).

Refuted as stated (v1.15.1-pre.7). The v1.14.0 statement (no Kummer–Frobenius exclusion) fails at $\ell = 37$, by three independent routes: (1) *certified $\mathbb{F}_\ell$ records*: the ideals $\mathrm{eis}(a, a+5)$ (residual $\omega^a(1 \oplus \omega^5)$, $37 \,\|\, B_{32}$) of the slot-14 C1a records (first-cluster job `34567994`, staged data set `np_dataset_c15`, not published) at $n = 654, 656, \ldots, 676$ ($a = 1, \ldots, 12$) and $n = 690, \ldots, 700$ ($a = 1, \ldots, 6$) — 18 records, each `verify()`$= 0$, `final = true` by `sturm(`B'`)`, `sturm_proof_range = true` — have `loewy_A` $= 4$, embedding dimension 1, Hilbert function $[1, 1, 1, 1]$, i.e. $A \cong \mathbb{F}_{37}[x]/x^4$, and module Loewy layers $[3, 2, 2, 1]$ ($V \cong A \oplus A/x^3 \oplus A/x$, $\dim V = 8$); since $\mathfrak{m}_A$ is generated by the $T_q - a_q$, some product $(T_q - a_q)(T_{q'} - a_{q'})$ is non-zero on V, whatever the full generator set (referee §8, point 2); (2) *the forms side* (slot-15 WP-12 §1.2): $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell \cong \mathbb{F}_{37}[x]/x^4$ on $S_k(\mathbb{F}_{37})$, $k = 656, \ldots, 702$, consistent with [C, `prop:cohforms`](5) on these torsion-free blocks (none of the 18 is of Dickson type; the Dickson locus in the window is $n = 684$); (3) *the lattice mod* 37^2 (slot-15 WP-13, `HeckeCensusLatticeValidation` at $(654, 37)$, `docs/implementation/slot15/WP-13_report.md`): ideal 8 — the system $a_{2,3,5,7} \equiv (29, 29, 16, 33)$, $U_{37} \equiv 0$, $\dim V = 8$, the cluster record's ideal 8 — prints `CENSUS_LATTICE 654 37 8 ... status=failed cf9_lattice=false cf9_field=false first_fail=2,2 m_nonzero=true lattice_rank=8 loewy_V=4`: $(T_2 - 29)^2 \Lambda_{\mathfrak{m}} \not\subseteq 37\Lambda_{\mathfrak{m}}$. The failing product does not involve U_{37}, so the refutation does not depend on U_ℓ; ideal 1 of the same cell ($\mathrm{eis}(2, 5)$, not Kummer–Frobenius) holds. All 18 ideals are Kummer–Frobenius ($w = 6$, $t = a$), so none is a counterexample to the restated form.

Evidence for the restated form (of v1.15.1-pre.7, i.e. the v1.16.0 form; every item at a non-Kummer–Frobenius $\mathfrak{m}$; no item against it until slot 17):

source	ℓ	range	rows / blocks	max Loewy of A (A_1)
census edition 2 (slot-13 WP-9 lattice test)	$7, 11, 13, 691$	$n \le 300$	654 non-reduced rows	2; C-F9 held 654/654 (at 691 the Kummer class B_{12} has onset $k \ge 691 \cdot 680$, far beyond)
WP-M1 route (first cluster, `34551683`)	$7, 11, 13$	$n \le 200$ / 150	1233 non-reduced blocks	$\mathfrak{m}_A^2 = 0$ on all
staged np data set (slot-14 WP-C15)	$37, \dots, 103$	$n \le 700$ (partial)	2008 non-reduced rows	2 everywhere except the 18 Kummer–Frobenius rows at 37; this includes the 120 non-reduced Kummer-predicted rows (Loewy 2) below their onset at $37, 59, 101, 103$ $(14, 70, 19, 17$; onsets $k \ge 37 \cdot 6 + 38t$, $59 \cdot 16$, $101 \cdot 34$, $103 \cdot 80)$
census edition 3, its complete C1b block (v1.16.1; first-cluster `34568172` and RB1–RB4 `34573807`–`10`, slot-15 WP-C18, WP-15)	$61, 67, \dots, 103$ (ten primes)	every even $n \le 700$, 3500/3500 cells	62 113 rows, 2165 non-reduced	2 at every prime (embedding dimension ≤ 1, every block Gorenstein); no Kummer–Frobenius ideal of Loewy ≥ 3 exists there (the only Kummer–Frobenius ideals in range are the two $t = 0$ ideals at $(668, 67)$, Loewy 2 and 1), so 0 violations of (R)
weight-shift A_1 (WP-L0930, `a1_loewy_scan.py`)	$17, 19, 29, 37, 47, 59, 71, 79, 83, 107, 137$	$n \le 1369$ / 180	1388 blocks with a recorded A_1	2, except *exactly* the six Kummer–Frobenius cells $(258, 37)$, $(294, 37)$, $(330, 37)$ (eis$(1, 6)$), $(296, 37)$, $(332, 37)$ (eis$(2, 7)$), $(334, 37)$ (eis$(3, 8)$), each $[1, 1, 1, 1]$
C-F9 lattice at $(654, 37)$, slot-15 WP-13	37	1 cell	ideal 1 (eis$(2, 5)$, not K–F): held; ideal 8 (K–F): failed	—

Process note. The slot-13 and slot-14 evidence for the v1.14.0 statement (the 654/654 and the 1233/1233) lay entirely below every Kummer–Frobenius onset, so it never tested the refuted case. *The open part*: the case $t = 0$ (the untwisted E_w class) is not covered by the data — there the block is the Eisenstein ideal itself and F9's (Mazur–Wiles) shape applies; the next tests are $(736, 67)$ ($67 \mid B_{58}$, $w = 10$, eis$(1, 10)$) and $(372, 37)$ ($t = 4$). The remaining 17 Loewy-4 cells at $\ell = 37$ ($n = 656, \dots, 700$) are lattice-test candidates for the cluster ([X] v1.20.1, WP-C19). (v1.16.1) The rows of the table are, from $\ell = 17$ on, the rows of census edition 3 (§7.2); there (R) has 0 violations on all 80 293 rows present, and the C-F9 lattice test held on all 155 evaluable cheap new-prime rows (9 rows of residue degree > 1 not evaluated). Whether the restated form holds at every regular prime is open. The C++ readings (`cf9_holds()`, `HeckeCensusLatticeValidation`'s summary, `LOEWY_RING_CF9_FINDING`) keep *reporting*, never gating, as before.

The v1.14.0 statement (refuted as stated, v1.15.1-pre.7; kept as published, with its evidence). (v1.14.0-pre.2; candidate C-F9 of the slot-12 finality proposal `docs/implementation/slot12/WP-12_fina` v2 §7, tested by slot-13 WP-9, owner answer Q8) Let $\ell \ge 5$, $n \ge 1$, $\Lambda := H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}_\ell)$, and $\mathfrak{m}$ a maximal ideal of the Hecke algebra $\mathbb{T}$ acting on Λ with residue field $\mathbb{F}_\ell$; write $a_q \in \mathbb{F}_\ell$ for the

eigenvalue of T_q at $\mathfrak{m}$. Then

$$(T_q - a_q)(T_{q'} - a_{q'})\,\Lambda_{\mathfrak{m}} \subseteq \ell\,\Lambda_{\mathfrak{m}} \qquad \text{for all primes } q, q'$$

(with $T_\ell = U_\ell$). *Equivalently*: since $H^2(\Gamma; \mathrm{Sym}^n \mathbb{Z}_\ell)[\ell] = 0$ ([C, `cl:bockstein`](1)), $\Lambda/\ell = H^1(\Gamma; M_n)$ Hecke-equivariantly, so the statement is that the local algebra A of $\mathbb{T}$ on $V = H^1(\Gamma; M_n)_{\mathfrak{m}}$ has $\mathfrak{m}_A^2 = 0$, i.e. Loewy length ≤ 2 — in contrast with $\mathbb{T}_{\mathfrak{m}}/\ell$ acting on $M_k(\mathbb{F}_\ell)$, whose Loewy length reaches 6 on the same census (slot-12 WP-12, the forms side). (v1.14.0-pre.3, referee finding M4.) The mod-ℓ form needs no hypothesis "$\mathfrak{m}$ not of Dickson type" for the *equivalence* (the identification $\Lambda/\ell = H^1(\Gamma; M_n)$ holds for every $\mathfrak{m}$ at $\ell \geq 5$). The *evidence*, however, is entirely non-Dickson: every tested row is `mixed` or `cuspidal`, has a free lattice and `sturm_proof_range = true`, and every Dickson-type block computed so far is reduced ($\mathfrak{m}_A = 0$, where the statement is trivial: the 340 Dickson-type census rows and the 100 Dickson-type blocks of slot-13 WP-11's scan). On a non-reduced Dickson-type $\mathfrak{m}$ the prediction therefore has *zero* evidence — an untested extrapolation. (v1.15.0, slot-14 WP-6, `HeckeRingLoewyValidation`'s route over $\mathbb{Z}/\ell^K$: the 206 Dickson-type (torsion) blocks read on 348 cells — 101 of them in the window $\ell = 7$, $n \leq 200$; $\ell \in \{11, 13\}$, $n \leq 150$ — are all reduced mod ℓ, so the prediction holds trivially on them; a non-reduced Dickson-type block has not occurred, and the extrapolation stays untested.) *Evidence* (computed; census edition 2, §7.2): on all 654 non-reduced H^1 rows ($\ell \in \{7, 11, 13, 691\}$, $n \leq 300$, $\dim V \leq 26$; 128 / 298 / 226 / 2 by ℓ) the statement holds, and fails on none; it was tested twice, independently — on the lattice mod ℓ^2 by the module route ($H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}/\ell^2)$, the $\mathfrak{m}$-part cut out by Fitting projectors, never by the $\mathbb{F}_\ell$ decomposition) and on V over $\mathbb{F}_\ell$ — and the two agree on every row (slot-13 WP-9, `HeckeCensusLatticeValidation`, its all-rows mode `HECKE_CENSUS_LATTICE_ALL=1`; the TSV `docs/implementation/slot13/WP-9_census_lattice.tsv`). (v1.15.0) *A third, independent route*: slot-14 WP-M1's scan (the $\mathbb{Z}/\ell^2$ module route and joint idempotents, with no $\mathbb{F}_\ell$ decomposition; `docs/implementation/slot14/WP-M1_loewy_note.md`) re-derived A on 1229 $\mathbb{F}_\ell$-rational blocks in 250 cells ($\ell = 7$, even $n \leq 200$; $\ell \in \{11, 13\}$, even $n \leq 150$). The 259 non-reduced blocks among them are the census's rows in that window, and all have $\mathfrak{m}_A^2 = 0$. The Δ-block of $(400, 7)$, outside the census window, also has $\mathfrak{m}_A^2 = 0$. The C++ route over $\mathbb{Z}/\ell^K$ (slot-14 WP-6, `HeckeRingLoewyValidation`; `docs/implementation/slot14/WP-6_report.md` §3) re-derived that window block for block and 98 further cells (348 cells): the statement held on 452/452 non-reduced blocks. (v1.15.1-pre.3; P12's cluster half, owner sign-off 2026-09-30) *Beyond the census window*: the WP-M1 route on the first cluster (job `34551683`: $\ell = 7$, $n = 402 \ldots 604$; $\ell \in \{11, 13\}$, $n = 302 \ldots 556$) found $\mathfrak{m}_A^2 = 0$ on all 1233 non-reduced $\mathbb{F}_\ell$-rational blocks of its completed cells (98 / 384 / 751 by ℓ; the one further non-reduced block, recovered from the killed cell $(604, 7)$, also has it), and on none fails; the data set is `docs/matrix-engine/results/noctua2_section_m/` (published 2026-09-30). The forms side (q-expansions, all T_q up to the Sturm bound) re-derived $\mathbb{T}_{\mathfrak{m}}/\ell$ on all 343 blocks of the WP-M1 scans with a $\mathbb{Z}/\ell^K$ algebra and at $(400, 7)$, $(402, 7)$, $(500, 7)$, $(604, 7)$ (slot-14 WP-7, `docs/implementation/slot14/WP-7_referee.md` §3). The tag stays [conjectural]. *What the prediction is not*: the Hecke algebra over $\mathbb{Z}/\ell^K$, reduced afterwards ($A_K \otimes \mathbb{F}_\ell$ on the $\mathfrak{m}$-part of $H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}/\ell^K)$), has Loewy length 3–6 on census rows and exactly 8 at $(400, 7)$. The prediction therefore concerns the reduction of the *module*, not of the algebra. The pairs tested are $q, q' \in S = \{2, 3, 5, 7\} \cup \{\ell\}$; since every row of edition 2 is `final = true` ($A_S = \mathbb{T}_\infty|_V$, §5), $\mathfrak{m}_{A_S}^2 = 0$ is the statement for all q, q'. The 2250 reduced H^1 rows satisfy it trivially ($\mathfrak{m}_A = 0$). *What does not explain it.* The candidate mechanism of the proposal — the Jordan–Hölder filtration of $\mathrm{Sym}^n(\mathbb{F}_\ell^2)$ by Serre weights $F(a, b) = \mathrm{Sym}^a \otimes \det^b$, which does give V a Hecke filtration with scalar graded pieces — does not bound the Loewy length: at most two *distinct* weights occur on any row, the Loewy length of V is 2 on *every* row while the number of pieces with multiplicity ranges over 1–26, and "Loewy length $\leq$ number of distinct weights" **fails on** 89 **rows** (the 87 cuspidal rows at $\ell = 13$, where the weight of Δ occurs two or three times — at $(192, 13)$ and $(194, 13)$ all three copies lie in V — and the two $\ell = 691$ rows, where Sym^n is irreducible). Why extensions between repeated copies of one weight never nest deeper is open. *Not asserted*: any statement at $\ell \in \{2, 3\}$, at residue degree > 1, or on X_i other than X_4; no code gates this prediction beyond the census test itself. *Evidence (slot 19, v1.19.0-pre.5)*: at $(1220, 37)$ nine congruence-free hidden blocks with $A_1 = \mathbb{F}_{37}$ (consistent; $\mathfrak{m}_A^2 = 0$

trivially) and one Kummer block with $\mathfrak{m}_{A_1}^2 \neq 0$ (outside the statement; Remark 4.13). No change of the statement.

Prediction 4.10 (The Kummer onset and its window: the standard and the swapped family, and $g = 2$). [conjectural] (v1.15.1-pre.7; companion of the restated Prediction 4.9, slot-15 WP-12 (E-1); **restated at v1.17.0-pre.1** with the window bound proposed by the $\ell = 37$ report `docs/latex/l37_loewy4_example.tex` (not binding; its `rmk:kfonset`, `conj:window`, `prop:secondwindow`), for the slot-17 referee and the owner's ruling O3 of 2026-10-01. **Generalised at v1.18.0-pre.1** (hecke-extensions slot 18, WP-2; the slot-17 WP-2 flag and the slot-17 referee F1's B-2 and m-E5; the owner's ruling O3 of 2026-10-03): part (A) is the v1.17.0 statement, unchanged in content and reworded only through Claim 4.14; parts (B), the swapped family, and (C), $g = 2$, are new, both [conjectural]; the title read "The Kummer–Frobenius onset, and its window" through v1.17.0. The label is kept; the v1.15.1-pre.7 statement stands below, marked superseded.) Let (ℓ, w_K) be an irregular pair, $w_O := \ell + 1 - w_K$, $g := g_{w_K}$ (Claim 4.16), and $\mathfrak{m}$ a cuspidal Kummer class of the pair, presented by Claim 4.14(2) as $\bar{\rho}_{\mathfrak{m}} \cong \omega^t(1 \oplus \omega^{w-1})$ with $w \in \{w_O, w_K\}$ and $1 \leq t \leq J(w) - 1$: $w = w_O$ is the standard family (there $\mathfrak{m}$ is Kummer–Frobenius in the sense of Prediction 4.9's v1.16.0 form from its onset on, ℓ dividing the companion $B_{\ell+1-w}$), $w = w_K$ the swapped family ($\ell \mid B_w$ itself; never Kummer–Frobenius). Write A for the census algebra on $V = H^1(\Gamma; M_n)_{\mathfrak{m}}$, $k = n + 2$ running through the class.

(A) *(The standard family, $g = 1$; the v1.17.0 statement.)* If $w = w_O$ and $g = 1$ (exactly one cusp form of weight $w_K = \ell + 1 - w$ congruent to E_{w_K} modulo ℓ; this holds at 154 of the 155 irregular pairs with $\ell < 2000$ and fails at $(547, 486)$, slot-17 R17-1 §4; hypothesis added at v1.17.0-pre.3, the slot-17 referee F1's B-2), then $A \cong \mathbb{F}_\ell[x]/x^4$ for every k in the class of $\mathfrak{m}$ with

$$\ell w + t(\ell + 1) \;\leq\; k \;<\; w + t(\ell + 1) + (\ell^2 - 1),$$

and the Loewy length of A rises from at most 2 to 4 at $k = \ell w + t(\ell + 1)$.

(B) *(The swapped family, $g = 1$; new at v1.18.0-pre.1.)* If $w = w_K$ and $g = 1$, the same: $A \cong \mathbb{F}_\ell[x]/x^4$ for every k in the class with $\ell w_K + t(\ell + 1) \leq k < w_K + t(\ell + 1) + (\ell^2 - 1)$, and the Loewy length of A rises from at most 2 to 4 at $k = \ell w_K + t(\ell + 1)$ (the *swapped meeting*; at the same twist t it is the cheaper of the two meetings of a pair exactly when $w_K < w_O$, e.g. at 103 and 691; v1.18.0-pre.2, the referee F1's m-E7: pre.1 omitted "at the same twist").

(C) *($g = 2$, the first window; new at v1.18.0-pre.1.)* If $g = 2$, in either family, then $A \cong \mathbb{F}_\ell[x]/x^3$ for every k in the class with $w + t(\ell + 1) \leq k < \ell w + t(\ell + 1)$ (the first window: there $(m_O, m_K) = (1, 1)$, $\dim V = 1 + (1 + 2g) = 6$ and $\operatorname{rank} \mathbb{T}_{\mathfrak{m}} = 3$ by Claim 4.16).

In (A)–(C) "rises from at most 2" and the bounds $\dim A \leq \operatorname{rank} \mathbb{T}_{\mathfrak{m}}$ are [proved here] by Claim 4.16 (Remark 4.17); what is conjectural is that A is curvilinear of the largest length the rank allows, which includes that $\mathbb{T}_{\mathfrak{m}}$ is saturated there ([C, `prop:cohforms`](5), [C, `rmk:sattor`]). *Nothing is asserted*: for $k \geq w + t(\ell + 1) + (\ell^2 - 1)$ in (A), (B) (the multiplicities grow a second time there, $\dim V = 12$; the forms-side algebra has Hilbert function $(1, 2, 2, 1)$ at every such cell computed, in both families); in (C) from the meeting $k = \ell w + t(\ell + 1)$ on ($\dim V = 4 + 4g = 12$, $\operatorname{rank} \mathbb{T}_{\mathfrak{m}} = 6$; the meeting is Prediction 4.18's case $(m, r_0) = (2, 3)$, computed $\mathbb{F}_{547}[x]/x^6$ on the forms side at $k = 34462$, the meeting of the class $a = 1$, by the slot-18 referee F1, `docs/implementation/slot18/wp3/`; v1.18.0-pre.2, pre.1 read "no datum"); at $g \geq 3$ (no such pair is known; $g = 1$ at 154 of the 155 pairs below 2000, $g = 2$ only at $(547, 486)$); for the two end twists $t = 0$ (the class of E_w: the Eisenstein ideal itself) and $t = J(w) = \ell - w$ (the residual representation is then $1 \oplus \omega^{\ell-w}$, the class of $E_{\ell+1-w}$, the other Eisenstein end; v1.17.0-pre.2 wording, pre.1 read "the swapped configuration"); the definition of Kummer–Frobenius keeps its cap $0 \leq t \leq J(w)$ for Prediction 4.9. Prediction 4.18 states the curvilinearity in the form of the Hecke algebra $\mathbb{T}_{\mathfrak{m}}/\ell$, for these families and for the irreducible congruence classes. *What changed at v1.18.0-pre.1, and why.* (B): the swapped family meets at $\ell w_K + t(\ell + 1)$, and the slot-17 forms-side data there (Prediction 4.9, "Refuted as stated", item 2) are curvilinear $\mathbb{F}_\ell[x]/x^4$ at every onset cell computed; slot-18 WP-2 tested the upper end (evidence below). (C): at $g = 2$ the coefficient $1 + 2g = 5$ enters already in the first window, so Loewy length 3 appears without any doubling of Serre weights; $(608, 547)$ is the cohomology datum (Prediction 4.9, item 1). Nothing is extrapolated beyond a configuration with a datum: (C) is not asserted at the

meeting, where it would read $\mathbb{F}_\ell[x]/x^6$ (Prediction 4.18 states that case separately; v1.18.0-pre.2: it now has one forms-side datum, $(34460, 547)$, $\mathbb{F}_{547}[x]/x^6$, the referee F1's M-2; (C)'s claim stays the first window). (A)'s text is unchanged; its hypothesis "Kummer–Frobenius with seed w and twist t, $1 \le t \le J(w) - 1$" is now the case $w = w_O$ of Claim 4.14(2). *What changed, and why* (v1.17.0-pre.1). (a) *The upper end.* "From $k = \ell w + t(\ell + 1)$ on" is withdrawn. On the forms side the unbounded form is false at $\ell = 37$: at $k = 1412$ ($t = 1$) and $k = 1450$ ($t = 2$) the algebra generated on the block W of $S_k(\mathbb{F}_{37})$ by the T_p, $p \le 23$, has dimension $6 = \dim W$ and Hilbert function $(1, 2, 2, 1)$, so it is $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/37$, and it is not curvilinear (the report's `prop:secondwindow`, by script). For the cohomological A itself the point is open: at $(1410, 37)$ the report's dimension formula gives $\dim V = 12$ on a torsion-free block, so $\operatorname{rank} \mathbb{T}_{\mathfrak{m}} = 6$ ([C, `lem:heckerankone`]) and A is a quotient of $\mathbb{T}_{\mathfrak{m}}/37$ ([C, `prop:cohforms`](5)); either A has the Hilbert function $(1, 2, 2, 1)$ or $\mathbb{T}_{\mathfrak{m}}$ is not saturated there, and it was not computed (the report's open problem `pb:secondwindow`). The new upper bound is where the Jordan–Hölder multiplicities of $\mathrm{Sym}^n \mathbb{F}_\ell^2$ grow a second time: at $\ell = 37$, $w = 6$ the pair (m_1, m_2) of the report's `thm:dimformula` becomes $(3, 3)$ at $k = 6 + 38t + 1368$ (its `comp:jh`, exact, $t = 1, 2, 3$; one period $\ell^2 - 1 = 1368$ in n). (b) *The end twists.* At $\ell = 37$, for $t = J(6) = 31$ the dimension formula gives $\dim V = 3$, 7, 11 from $k = 32$, 1184, 1400 — never the 8 of the window — and for $t = 0$ the block is the Eisenstein ideal of E_6; neither was evidence for the prediction. (c) *At $\ell \ne 37$* the window's upper end is computed on the multiplicity side and extrapolated only in the shape of the algebra (v1.17.0-pre.3, the referee F1's m-E1; pre.2 read "an extrapolation of the $\ell = 37$ pattern"): slot-17 R17-1's committed tables (`docs/implementation/slot17/r171/classes/`) give the multiplicities $(2, 2)$ at $k = \ell w + t(\ell + 1)$ and the next event $(3, 3)$ at exactly $k = w + t(\ell + 1) + \ell^2 - 1$ on 1492 of 1492 standard cuspidal classes of its 13 pairs (v1.18.0-pre.1 correction: pre.3 through v1.17.0 read "2092 of 2092", the referee F1's count; the standard classes of the 13 pairs number $\sum(w_K - 2) = 1492$, recounted by `docs/implementation/slot18/wp2/window_events.py`, which also gives 1906 of 1906 for the swapped family, Remark 4.17); that A is $\mathbb{F}_\ell[x]/x^4$ throughout the window, and stops being so at its upper end, is the $\ell = 37$ pattern extended; at pre.1 no other irregular prime had a computed Loewy-4 cell, and at $\ell = 61 \ldots 103$ the first window lies beyond $n = 700$ (below). (v1.17.0-pre.2) Slot-17 R17-2 tested the window's lower end at 67, 59 and 101 (below); its upper end was then still untested off 37. (v1.17.0-pre.5) On the forms side the window is now tested off 37 up to its last cell: at $(3496, 59)$ and $(4498, 67)$ ($k = 3498$, 4500, $t = 1$; the next cell of the class, $k + \ell - 1$, is the upper bound itself) the block has rank 4 and $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell$ is curvilinear, $\mathbb{F}_\ell[x]/x^4$ (by script, the $\ell = 37$ report's referee F4, `docs/implementation/slot17/r173/f4_edeg_out.txt`); the cells beyond the upper bound were not computed off 37. (d) *Scope* (v1.17.0-pre.2). The prediction concerns the Kummer–Frobenius (standard) family only, at $g_{\ell+1-w} = 1$. The swapped family and every class at a pair with $g \ge 2$ (v1.17.0-pre.3; pre.2 read "the $g \ge 2$ first windows"), also Kummer classes, give curvilinear algebras too (Prediction 4.9, "Observed on the Kummer locus"); they are outside this prediction, and its generalisation to them is a slot-18 item. (v1.18.0-pre.1: that generalisation is now parts (B) and (C); this paragraph is the v1.17.0 scope of part (A), kept as published.) *Evidence for the restated form* (all at $\ell = 37$, $w = 6$, inside the window, $1 \le t \le 30$): the 24 onset cells below (cohomology, census and weight-shift data); on the forms side $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/37 \cong \mathbb{F}_{37}[x]/x^4$ at $k = 260, \ldots, 728$ ($t = 1$) and at $k = 1376$ ($t = 1$, inside the window, 36 below its upper end), by script. In the window $\dim V = 8$ is proved in the ranges computed (the report's `cor:onset`, from the Serre-weight multiplicities), so there $A \cong \mathbb{F}_{37}[x]/x^4$ is equivalent to $\mathbb{T}_{\mathfrak{m}}/37 \cong \mathbb{F}_{37}[x]/x^4$ with $\mathbb{T}_{\mathfrak{m}}$ saturated. (v1.17.0-pre.2) *Off $\ell = 37$*, at the window's lower end (slot-17 R17-2, `docs/implementation/slot17/R17-2_report.md` §2, §6.2): on cohomology at $(736, 67)$ ($67 \mid B_{58}$, $w = 10$, $t = 1$, $k = 738$): $\dim V = 8$, module Loewy layers $[3, 2, 2, 1]$, $A \cong \mathbb{F}_{67}[x]/x^4$ with U_{67} in the operator set, $\mathbb{T}_{\mathfrak{m}}$ saturated ($I_{\mathfrak{m}} = 0$) and the `stable` verdict certified at the system $(21, 25, 45, 40, 0)$ (`HeckeSaturationDefectValidation`); on the forms side, by script, $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell \cong \mathbb{F}_\ell[x]/x^4$ ($\dim A = \dim W = 4$) at $(736, 67)$, $(804, 67)$ ($t = 2$), $(1002, 59)$ ($59 \mid B_{44}$, $w = 16$, $t = 1$), $(1062, 59)$ ($t = 2$) and $(3534, 101)$ ($101 \mid B_{68}$, $w = 34$, $t = 1$). Each was predicted before it was run (R17-1 §7); none failed. (v1.18.0-pre.1) *The C-F9 lattice at the window's cells, $\ell = 37$* (slot-17 `docs/implementation/slot17/LAT37_local_report.md`, the env-armed `HeckeCensusLatticeValidation` on the October records, U_{37} in the operator set; `refused:0`, `fallback:0`): at $(700, 37)$, $(698, 37)$ and $(696, 37)$ exactly one ideal per cell fails the lattice test,

the Kummer–Frobenius ideal of dimension 8 and Loewy length 4 (`first_fail=2,2`, `cf9_lattice = cf9_field = false`, module layers of the $(656, 37)$ shape), and every ideal of dimension 4 holds, as at $(654, 37)$ and $(656, 37)$; $(694, 37)$, $(692, 37)$, $(690, 37)$ run in slot 18 (WP-L37). (At $(696, 37)$ the validation's own (J1)/(J2) self-checks failed for an environmental reason, a removed build tree; the lattice rows are unaffected, report §1.)

Evidence for part (B) (v1.18.0-pre.1; forms side only, $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell$ on the block $W \subseteq S_k(\mathbb{F}_\ell)$ with the generation certificate $\dim A = \dim W$; nothing here is cohomology). At the swapped meeting: $\mathbb{F}_{37}[x]/x^4$ at $b = 1, \ldots, 4$ ($k = 1222$, 1260, 1298, 1336), $\mathbb{F}_{103}[x]/x^4$ at $b = 1, 2$ ($k = 2576$, 2680), $\mathbb{F}_{691}[x]/x^4$ at $b = 1, 2$ ($k = 8984$, 9676) (slot-17 R17-2 §§2–3). At the window's last cell and one cell past it (slot-18 WP-2, `docs/implementation/slot18/wp2/`, by script, `r172_forms.py` unchanged; the predictions committed before the run, `predictions.txt`; 11/11 held): $\mathbb{F}_\ell[x]/x^4$ at $(1400, 37)$ ($b = 1$), $(1438, 37)$ ($b = 2$) and $(10632, 103)$ ($b = 1$), each the last cell below $w_K + t(\ell+1) + \ell^2 - 1$; and $\dim W = 6$, Hilbert function $(1, 2, 2, 1)$, not curvilinear, at the next cell $(1436, 37)$, $(1474, 37)$, $(10734, 103)$, exactly as in the standard family at $k = 1412$, 1450. (v1.18.0-pre.2) The slot-18 referee F1 (`docs/implementation/slot18/wp3/`, the same script, predictions written before each run, `cells_f1.txt`; 5/5 held here): $\mathbb{F}_\ell[x]/x^4$ at $(1364, 37)$ ($b = 1$, mid-window), $(1476, 37)$ ($b = 3$, last cell), $(1514, 37)$ ($b = 4$, last cell) and $(10736, 103)$ ($b = 2$, last cell); $\dim W = 6$, $(1, 2, 2, 1)$, not curvilinear, at $(1512, 37)$ ($b = 3$, first cell past the window). On the multiplicity side the window of (B) is computed on all 1906 swapped classes of R17-1's 13 pairs (Remark 4.17). The census algebra A was not computed at any swapped cell: the cheapest, $(1220, 37)$ (the $\ell = 37$ report's `pb:swappedcoh`), needs an exclusive memory window (slot-18 R18-3, ≈ 20 GB); $(2574, 103)$ and $(8982, 691)$ are beyond local reach. So (B) is [conjectural] with the saturation of $\mathbb{T}_{\mathfrak{m}}$ at these blocks as its untested half. *Evidence for part (C)* (v1.18.0-pre.1). On cohomology: $(608, 547)$, the class $a = 1$ of the standard family at the first cell of its first window, $A \cong \mathbb{F}_{547}[x]/x^3$, $I_{\mathfrak{m}} = 0$, `stable` certified (Prediction 4.9, item 1). On the forms side, $\mathbb{F}_{547}[x]/x^3$ ($\dim W = 3$, generated): $(1156, 547)$ ($a = 2$, R17-2), and (slot-18 WP-2, predicted before the run) $(1154, 547)$, $(1700, 547)$, $(2246, 547)$ (the second to fourth cells of the class $a = 1$) and, in the *swapped* family, $(1032, 547)$ and $(1578, 547)$ (the first two cells of $b = 1$; its first window runs from $k = 1034$ to the swapped meeting $k = 266390$). (v1.18.0-pre.2, the referee F1's M-2; `docs/implementation/slot18/wp3/`, predicted before the run, 3/3 held here; operators $T_2, \ldots, T_{11}$, no T_ℓ): $\mathbb{F}_{547}[x]/x^3$, generated, at $(16988, 547)$ (the 31st cell of $a = 1$), at $(33914, 547)$ (its 62nd and *last* cell, $k = 33916$) and, swapped, at $(11952, 547)$ (the 21st cell of $b = 1$). The first window of $a = 1$ (62 cells, $610 \le k < 34462$) is now sampled at its cells 1–4, 31 and 62, so (C) is stated over the whole first window on evidence at its start, middle and end (pre.1: "the data cover their first two to four cells"); the swapped window of $b = 1$ (from $k = 1034$ to 266390) is sampled at its cells 1, 2 and 21 only. If a cell inside a first window at 547 read a non-curvilinear algebra, (C) would be restricted to what was computed. At the meeting $(34460, 547)$, $k = 34462$, the same run gives $\dim W = 6 = \operatorname{rank} \mathbb{T}_{\mathfrak{m}}$, Hilbert function $(1, 1, 1, 1, 1, 1)$, generated: $\mathbb{F}_{547}[x]/x^6$ — the datum of Prediction 4.18 at $(m, r_0) = (2, 3)$, outside (C).

The v1.15.1-pre.7 statement (superseded at v1.17.0-pre.1 by the window form above; kept as published, with its evidence). On a Kummer–Frobenius $\mathfrak{m}$ (Prediction 4.9), $A \cong \mathbb{F}_\ell[x]/x^4$ from $k = \ell w + t(\ell+1)$ on: the Loewy length of A rises from 2 to 4 at exactly that weight. *Evidence* (computed, not certified as a law): the weight-shift data predict Loewy length 4 at $\ell = 37$ for $\mathrm{eis}(t, t+5)$ exactly at $n = 220 + 38t + 36m$, $m \ge 0$, $t \ge 1$. In the scanned range $n \le 356$ that gives the six cells $(258, 37)$, $(294, 37)$, $(330, 37)$, $(296, 37)$, $(332, 37)$, $(334, 37)$; all six are observed, and no other cell has Loewy length ≥ 3. The cluster window $n = 648 \ldots 700$ gives $t = 1, \ldots, 12$ and $t = 1, \ldots, 6$, the 18 records of Prediction 4.9. So the onset has 24/24 hits and 0 false positives; it was predicted (WP-12 pass 1, §1.2: $(258, 37)$, $(294, 37)$ as "the cheap test") before the weight-shift scan was read. (v1.16.1) *At* $\ell = 61 \ldots 103$ (census edition 3's complete C1b block, slot-15 WP-C18 §3.2): the first Loewy-4 cell of every Kummer class there lies beyond $n = 700$ ($n = 736$ at $67 \mid B_{58}$, $n = 3534$ at $101 \mid B_{68}$, $n = 8342$ at $103 \mid B_{24}$), so the law predicts no Loewy-4 cell and none is observed; the 229 Kummer-predicted rows lie below their onset and have Loewy ≤ 2. This is consistent with the law but **does not test its positive half**; the first cell that would is $(736, 67)$, beyond the C1 grid. (v1.17.0-pre.2, two corrections.) $(736, 67)$ was tested in slot 17 and held (the evidence of the restated

form, above). And $n = 8342$ is the first *Kummer–Frobenius* Loewy-4 candidate at 103, not the first Loewy-4 cell there: the swapped $(2574, 103)$ comes first (slot-17 R17-1 §5.2) and reads $\mathbb{F}_{103}[x]/x^4$ on the forms side (R17-2 §2), outside this prediction and a counterexample to the v1.16.0 form of Prediction 4.9. *Mechanism* ([conjectural], WP-12 §1.2): at $k = \ell w + t(\ell+1)$ the Frobenius direction of the seed E_w enters the block. In the split case it adds a line (module Loewy $[3,1]$, still $\mathfrak{m}_A^2 = 0$); in the Kummer case it doubles the Ribet-congruent block to a curvilinear $\mathbb{F}_\ell[x]/x^4$. Why the doubling is curvilinear is open: Bellaïche–Khare's $A_{\bar\rho} \cong \mathbb{F}_\ell[[x,y]]$ (unobstructed, Vandiver; [3]) bounds only the embedding dimension, by 2. (v1.17.0-pre.1) At the window's upper end the multiplicities grow a second time and the block's rank rises from 4 to 6; on the forms side the algebra there has embedding dimension 2 (at $\ell = 37$, $t = 1, 2$; the report's `prop:secondwindow`). Why the window closes exactly there, and whether the census algebra follows the forms algebra beyond it, is open.

Remark 4.11 (The split / ramified switch is not obstructedness). [proved here] (from [3]; v1.16.3, owner ruling O4 of 2026-10-01, slot-15 WP-12 proposal (E-3) of `docs/implementation/slot15/WP-12_U_E_proposals.m` the reading is `docs/implementation/slot15/WP-12_report.md` §5 and §8.4(e).) Slot-14 WP-C14's I-8 dichotomy for a twisted Eisenstein ideal $\mathrm{eis}(i,j)$, $0 < i < j$, with Bernoulli seeds B_{w_1}, B_{w_2}, $w_1 = j - i + 1$, $w_2 = \ell + i - j$ (the *split* case: both seeds are ℓ-units, two ℓ-adically rational eigenforms, module Loewy $[3,1]$ from k_2 on, §7.2, data label `twisted_eis_split`; the *ramified* case: ℓ divides a seed, one ramified Galois orbit — the Kummer case, compare Prediction 4.10) is **not** Bellaïche–Khare obstructedness ([3] Thm. 22 (i), (iii)). For $\bar\rho = \omega^i \oplus \omega^j$, $a = j - i$, one has $a + 1 = w_1$ and $\ell - a = w_2$, so Thm. 22 (i) is exactly "both seeds are units"; but Thm. 22 (iii) makes $\bar\rho$ unobstructed whenever Vandiver's conjecture holds at ℓ, the Kummer case included (the Kummer class lies in the odd eigenspace, [29], while the obstruction lives in even ones), and Vandiver is verified for every $\ell < 2^{31}$ (Hart–Harvey–Ong [19]). So both cases are unobstructed, $A_{\bar\rho} \cong \mathbb{F}_\ell[[x,y]]$ in both, and the dichotomy is a statement about $\mathbb{T}_{\bar\rho}$ in characteristic 0 (one ramified orbit against two rational forms), not about $A_{\bar\rho}$.

Prediction 4.12 (The Frobenius layer (F1)). [conjectural] (v1.16.3, owner ruling O4 of 2026-10-01: slot-15 WP-12 proposal (E-2) of `docs/implementation/slot15/WP-12_U_E_proposals.md`, accepted as a [conjectural]; the law L-F1 of `docs/implementation/slot15/WP-12_report.md` §2.2) Let $\ell \geq 5$ and let f be an ordinary level-one eigenform of weight $k_0 \leq \ell$ (v1.20.0-pre.7: $k_0 \leq \ell + 1$ through pre.6; the boundary weight $k_0 = \ell + 1$ is excluded after slot-20 WP-SW, below), $\bar\rho_f$ its residual representation. Let $J(k_0)$ be the least $J \geq 0$ with $k_0 + J \equiv 0 \pmod{\ell}$, so that $\theta^j f$ keeps filtration $k_0 + j(\ell+1)$ for $j \leq J(k_0)$. For $1 \leq j \leq J(k_0)$, the block of $\bar\rho_f \otimes \omega^j$ in $H^1(\Gamma; \mathrm{Sym}^n \mathbb{F}_\ell^2)$ goes from $D = 2$ to $D = 4$ at

$$n_V = \ell k_0 + j(\ell+1) - 2,$$

and it is *hidden* ($A_1 = \mathbb{F}_\ell$, $A_K/\ell \cong \mathbb{F}_\ell[x]/x^2$) from n_V on.

Why ([proved here] given two per-cell facts — $D(n_V) = 4$, and the two blocks at n_0 are reduced; WP-12 §2.2). (i) At $n_0 = \ell k_0 - 2$ the $\bar\rho_f$-part of H^1 is $B_{\mathrm{ord}} \oplus B_V$, separated by U_ℓ ($a_\ell \neq 0$ on B_{ord}, $a_\ell = 0$ on B_V, the $V(f) = f^\ell$ direction); each has $D = 2$ and reduced algebra $\mathbb{F}_\ell$ (the per-cell fact). (ii) θ_L^j maps $H^1(\Gamma; M_{n_0})$ to $H^1(\Gamma; M_{n_V})$ with $T_p\theta_L = p\,\theta_L T_p$ and $U_\ell\theta_L = 0$ ([C, `thm:thetales`](2)), injectively off the Eisenstein line ([C, `thm:thetales`](3)); so $\theta_L^j(B_{\mathrm{ord}} \oplus B_V)$ is a 4-dimensional Hecke submodule of the $\bar\rho_f \otimes \omega^j$ block on which every T_p acts as the scalar $p^j a_p(f)$ and U_ℓ as 0. (iii) $D(n_V) = 4$ (the per-cell fact) makes it the whole block; hence $A_1 = \mathbb{F}_\ell$. (iv) The block is torsion-free (non-Dickson), so [C, `lem:heckerankone`] gives a local $\mathbb{Z}_\ell$-algebra of rank $\lceil 4/2 \rceil = 2$, whose reduction is local of dimension 2, i.e. $\mathbb{F}_\ell[x]/x^2$; by [C, `cor:stablecert`] this is A_K/ℓ. (v) So $A_1 \subsetneq A_K/\ell$, a saturation defect ([C, `prop:saturation`](2)): the congruence is hidden. *What stays conjectural*: the onset — that nothing enters the block earlier, so that the doubling happens exactly at n_V — and that the V-block first appears at weight exactly ℓk_0 with $V(f) = f^\ell$ (compare [C, `rmk:thetaforms`](1)). Both held on every tested cell. *Why* $j \leq J(k_0)$ *only*: for $j > J(k_0)$, $\theta^j f$ has dropped filtration, appears early, and at $k_0 + j(\ell+1)$ the block gains $\theta_L^j(B_{\mathrm{ord}})$ glued non-semisimply (not hidden); identifying the dropped form with a companion-form twist needs Gross's companion-form theorem and is not attempted.

Evidence (computed, WP-12 §2.3, `law_f1.py` on the weight-shift data of WP-L0930): at $\ell = 17$, 33 irreducible chains with the first doubling as predicted and 0 failures, 17 first hidden rungs as

predicted (the three $\theta^j(\Delta E_6)$ at $n = 322, 340, 358$ included) and 0 failures; at $\ell = 19$, 49 and 9 (plus $n = 362, 366$ beyond 360), 0 failures; at $\ell = 29, 37, 47$ the doublings $14, 15, 2$ as predicted and no hidden rung predicted or observed in range; at $\ell = 59 \dots 199$ none predicted, none observed; 0 spurious hidden blocks at every ℓ. The predictions of WP-12 §2.4 ($\ell = 17$, $n = 362 \dots 500$; $\ell = 19$; at $\ell = 37$ no hidden block below $n = 480$) were written before any further data and are untested. *Evidence (slot 16, 2026-10-01)*: consistent, no qualification of the statement. Slot-16 WP-5 (`HeckeSaturationDefectValidation`, `docs/implementation/slot16/WP-5_report.md` §3) read the hidden-chain cells $(220, 17)$ ($\theta\Delta$), $(238, 17)$ ($\theta^2\Delta$), $(246, 19)$ ($\theta\Delta$): $D = 4$, $\operatorname{rank}\mathbb{T} = 2$, $A_1 = \mathbb{F}_\ell$, $\mathbb{T}/\ell = \mathbb{F}_\ell[x]/x^2$, $\mathbb{T}^{\mathrm{sat}}/\mathbb{T} = \mathbb{Z}/\ell$, the defect ideal $I = \mathfrak{m}$ principal, 3/3 held; its default arm (H) gates the law's $j = 1$ onset at $\ell = 13$, the cell $(168, 13)$ (slot-16 WP-3/WP-5 audit). Slot-16 WP-11 (`docs/implementation/slot16/WP-11_report.md`) read $(220, 17)$ and $(246, 19)$ with $a_\ell = 0$, $\dim A_1 = 1$ and the `stable` verdict certified ($\mathbb{T}_\mathfrak{m}/\ell$ of Hilbert function $[1, 1]$), and records that there $\Lambda_\mathfrak{m} \not\cong \mathbb{T}^2_\mathfrak{m}$: a hidden block is free of rank 4 over a rank-2 algebra with a saturation defect — a fact beside this prediction, not against it. *Evidence (slot 19, 2026-10-04)*: consistent, no qualification. At $(1220, 37)$ (slot-19 WP-RC's `HeckeSaturationDefectValidation` run, read by slot-19 sat1220, `docs/implementation/slot19/sat1220_analysis.md`, predictions committed first) all nine $\mathbb{F}_{37}$-rational irreducible rungs $n_V \le 1220$ of the weight class are hidden ($D = 4$, $A_1 = \mathbb{F}_{37}$, $I = \mathfrak{m}$) and there is no spurious hidden block: the first test at $\ell = 37$ beyond WP-12's range, 0 failures ([computed and compared]; two rungs at $n_V = 1184$, two at $n_V = 1220$). The tenth block with $\dim I = 1$ there is the rung of a Kummer class, Remark 4.13.

Evidence (slot 20, forms side; v1.20.0-pre.3, scope qualified at v1.20.0-pre.4): consistent with the transport step (ii) of *Why*; **outside the statement's hypotheses** — the family-(c) seed is a non-reduced block of two congruent eigenforms ($\mathbb{T}/\ell = \mathbb{F}_\ell[\epsilon]/\epsilon^2$), not a reduced $D = 2$ pair, and the onset block is predicted saturated, not hidden ([L37, `pb:irrcoh`]). At the family-(c) onsets $(2210, 67)$, $(2278, 67)$, $(3798, 131)$, $(3930, 131)$ ($j = 1, 2, 1, 2$) the θ^j-image of the seed space $S_{k_0}(\mathbb{F}_\ell)$ is $\ker x^2$ of the curvilinear block, so the first-order Hecke data of the doubled block is the θ^j-twist of the seed's $\mathcal{O}_K/\mathfrak{m}^2$: $T_p \mapsto p^j(a_p \bmod \mathfrak{m}^2)$, $t_1(p) = 2^{-j}p^j s(p)$ ([proved here] given the curvilinearity, [computed and compared] at all 1228 primes $p < 10^4$ at each cell; slot-20 R20-2 §3, `docs/implementation/slot20/R20-2_frobenius_shadows.md`; [L37, `sec:fieldsirr`]). Neither the onset nor the hidden conclusion is tested by this; the first-order data is non-scalar, as a non-reduced source demands. *Evidence and an observation (slot 20, cohomology; v1.20.0-pre.6)*: slot-20 WP-SW's three cells $(1258, 37)$, $(1296, 37)$, $(1334, 37)$, read with slot-19 sat1220's `lf1_rungs.py`: at each, the 9 irreducible rungs with $n_V \le n$ are present and hidden ($D = 4$, $A_1 = \mathbb{F}_{37}$), among them the onset $n_V = 1334$ of $k_0 = 32$, $j = 4$ exactly at $(1334, 37)$, and no predicted rung is missing ([computed and compared]). One hidden $D = 4$ block at $(1334, 37)$ is not a predicted rung: $\bar\rho_f \otimes \omega$, f the weight-38 ($= \ell + 1$) form with $(a_2, a_3, a_5, a_7, a_{37}) \equiv (0, 1, 0, 36, 1)$, $J(38) = 36$, predicted onset $n_V = 1442$. Both reduced $D = 2$ copies of $\bar\rho_f$ ($a_{37} = 1$ and $a_{37} = 0$) are present already at $n = 1296$ (in the role of *Why* (i)'s $n_0 = \ell k_0 - 2 = 1404$), and $1296 + (\ell + 1) = 1334$: *Why* (ii)–(v) with $n = 1296$ in the role of n_0 explains the block ([proved here] given the per-cell facts read in the logs). So at the boundary weight $k_0 = \ell + 1$ the statement as written through v1.20.0-pre.6 ($k_0 \le \ell + 1$) is **refuted at this one class**: the doubling is not at $n_V = 1442$ but at 1334, because the V-copy of $\bar\rho_f$ is present at weight $1298 = \ell k_0 - 3(\ell - 1)$, not first at $\ell k_0 = 1406$; the mechanism (ii)–(v) holds. The hypothesis is therefore restricted to $k_0 \le \ell$ (v1.20.0-pre.7, F1′ M-E2), where every tested cell is consistent; the onset at $k_0 = \ell + 1$ is open (why the $a_\ell = 0$ copy appears early is not examined; [L37, `app:ribetdata`]). Tag unchanged, [conjectural].

Remark 4.13 (The Frobenius layer on a Kummer class (v1.19.0-pre.5)). [computed and compared] at one cell, [conjectural] in general (slot-19 sat1220, `docs/implementation/slot19/sat1220_analysis.md` §§1–3). At the rung $n_V = \ell w_K + j(\ell+1) - 2$ of a Kummer class ($\ell \mid B_{w_K}$, Ribet's form ordinary; $k_0 = w_K$ in Prediction 4.12) the block is the swapped family's $D = 8$ block (the swapped onset $k = \ell w_K + b(\ell+1)$ of Prediction 4.10(B) *is* this rung, $j = b$), and the θ-layer hides not all of $\mathfrak{m}$ but $\mathfrak{m}^3$: $\mathbb{T}_\mathfrak{m}/\ell \cong \mathbb{F}_\ell[x]/x^4$, $A_1 \cong \mathbb{F}_\ell[x]/x^3$, $\mathbb{T}^{\mathrm{sat}}_\mathfrak{m}/\mathbb{T}_\mathfrak{m} \cong \mathbb{Z}/\ell$, $I_\mathfrak{m} = \mathfrak{m}^3$. *Computed* at $(1220, 37) = (37 \cdot 32 + 38 - 2, 37)$, the rung of θ^1 on the weight-32 Kummer class, system $(9, 19, 14, 16, 0)$: torsion-free, `stable` certified at $K = 2$, $\mathbb{T}_\mathfrak{m} = \mathbb{Z}_{37}[T_p : p \in \{2, 3, 5, 7, 37\}]$ the full Hecke algebra of the block (the S-primes

generate $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/37$, Nakayama; so no other generating set changes $I_{\mathfrak{m}}$), $\mathbb{T}_{\mathfrak{m}}/37 \cong \mathbb{F}_{37}[x]/x^4$ on the forms side, $A_1 \cong \mathbb{F}_{37}[x]/x^3$, radical layers $[4,3,1]$, $T_2 - 9$ of Jordan type $[3,2,2,1]$; an independent route on $H^1(\Gamma; \mathrm{Sym}^{1220}\mathbb{F}_{37})$ agrees. At v1.19.0-pre.5 not yet verified: that θV_{src} is the socle of V. *Computed at v1.20.0-pre.1* ([computed and compared] as a run, one route; slot-20 WP-RC20 window TS, `docs/implementation/slot20/TS_launch.md`, the console script `theta_socle_1220_37.oc` of sat1220, V_{src} the block of $(1182, 37)$ at $a_p = 1 + p^{31}$): $\dim V = 8$, $\dim V_{\mathrm{src}} = 7$, θ is injective on V_{src}, $\theta V_{\mathrm{src}} = \ker(U_{37}|V)$ of dimension 7 ($\operatorname{rank} U_{37}|V = 1$), θV_{src} is *not* killed by $\mathfrak{m}$ ($T_p - a_p$ has rank 3 on it for $p = 2, 3, 5, 7$), and $\dim \operatorname{soc}(V) = 4$: θV_{src} **is not the socle of** V, so the identification of the defect's θ-layer with the socle (sat1220 §3's reading) is refuted. The facts $\dim I_{\mathfrak{m}} = 1$, $I_{\mathfrak{m}} = \mathfrak{m}^3$, $A_1 \cong \mathbb{F}_{37}[x]/x^3$ above are not touched; a mechanism is open. This is consistent with Prediction 4.12 "Why (ii)" and with [C, `thm:thetales`](2),(3),(5) (v1.20.0-pre.2, F1 m-E6): θ_L is injective on $H^1(M_{1182})$ ($\operatorname{rank}\delta = 0$) and $a_p = 1 + p^{31} \neq 1 + p^{-1}$; $U_\ell \theta_L = 0$ gives $\theta V_{\mathrm{src}} \subseteq \ker U_{37}$, equality by dimension; $T_p \theta_L = p\,\theta_L T_p$ carries the non-semisimple part of V_{src} ($T_p - a_p$ of rank 3) into V. (ii)'s scalar T_p rests on *reduced* $D = 2$ source blocks, which the 7-dimensional V_{src} of a reducible $\bar\rho$ is not; the prediction's hypotheses do not hold at this cell, and no statement of it moves. *Conjectured*: the same ($\dim I_{\mathfrak{m}} = 1$, $I_{\mathfrak{m}} = \mathfrak{m}^3$, $A_1 \cong \mathbb{F}_\ell[x]/x^3$) at every swapped cell $n = \ell w_K + b(\ell+1) - 2$, $1 \le b \le J(w_K)$ (the cells run or planned: at 37, $b = 1, \ldots, 4$ of $J(32) = 5$, the fifth $n = 1372$ predicted `a20b9531` and running; at 103, $b = 1, 2$ of $J(24) = 79$, $(2574, 103)$, $(2678, 103)$; at 691, $b = 1, 2$ of $J(12) = 679$, $(8982, 691)$, $(9674, 691)$); slot-20 WP-SW runs $b = 2, 3, 4$ at 37 ($n = 1258$, 1296, 1334; predictions committed first), the two 103 cells go to a cluster kit and the 691 cells are recorded as untested (owner ruling O10 of 2026-10-04). *v1.20.0-pre.6*: WP-SW **confirms** it at $b = 2, 3, 4$ ([computed and compared]; $D = 8$, `stable` certified at $K = 2$, $\mathbb{T}/37 = \mathbb{F}_{37}[x]/x^4$, $A_1 = \mathbb{F}_{37}[x]/x^3$, $\dim I_{\mathfrak{m}} = 1$, $I_{\mathfrak{m}} = \mathfrak{m}^3$ at each; [L37, `conj:swappedsat`]), so at 37 at the first four of the five cells of the range; [conjectural] at $b = 5$ and beyond 37. *1.20.0*: **confirmed at** $b = 5$ as well, at the boundary cell $(1372, 37)$ ($b = J(32)$; [computed and compared], `docs/implementation/slot20/SW1372_report.md`, prediction `a20b9531` first): the block of system $(33, 22, 18, 10, 0)$ has $D = 8$, `stable` certified at $K = 2$, $\mathbb{T}/37 = \mathbb{F}_{37}[x]/x^4$, $A_1 = \mathbb{F}_{37}[x]/x^3$, $\dim I_{\mathfrak{m}} = 1$, $I_{\mathfrak{m}} = \mathfrak{m}^3$; so at 37 the conjecture holds at all five cells of its range ($b = 1, \ldots, 5$), and it stays [conjectural] beyond 37. Prediction 4.9 (R) is not touched: the class is not congruence-free.

4.4 The dimension of a reducible block at every ℓ

(New at v1.18.0-pre.1, hecke-extensions slot 18, WP-2. Slot-17 R17-1's Lemmas 3.1, 3.2 and Theorem 4.1, `docs/implementation/slot17/R17-1_report.md` §§3–4, become statements of this document, as the slot-17 referee F1 asked (m-E5, "since [E] now argues from Theorem 4.1"); the $\ell = 37$ report's proposal `conj:depth` (`docs/latex/l37_loewy4_example.tex` 0.2.0-pre.2, not binding) becomes Prediction 4.18. Placed at the end of §4, so that no earlier statement number moves.) Throughout $\ell \ge 5$. A level-1 residual representation $\omega^i \oplus \omega^j$ has the *seeds* $w_1 = d+1$, $w_2 = \ell - d$ of Prediction 4.9 ($d \in [1, \ell-2]$, $d \equiv j - i \pmod{\ell-1}$, odd); it is of Dickson type iff a seed equals 2, and otherwise both seeds lie in $[4, \ell-3]$. For an even $w \ge 4$, $J(w) \in [0, \ell-1]$ is defined by $J(w) \equiv -w \pmod{\ell}$, as in Prediction 4.9.

Claim 4.14 (Two seeds per class; the standard and the swapped family)**.** *[proved here]*

1. *Let $4 \le w, w' \le \ell - 3$ be even. Then $1 \oplus \omega^{w-1}$ and $1 \oplus \omega^{w'-1}$ agree up to a twist by a power of ω iff $w' \in \{w, \ell + 1 - w\}$.*
2. *Let (ℓ, w_K) be an* irregular pair *($\ell \mid B_{w_K}$, w_K even, $4 \le w_K \le \ell - 3$) and $w_O := \ell + 1 - w_K$; then $J(w_O) = w_K - 1$ and $J(w_K) = w_O - 1$. The classes with seeds $\{w_O, w_K\}$ — the* Kummer classes *of the pair — are $\omega^a(1 \oplus \omega^{w_O - 1}) = \omega^b(1 \oplus \omega^{w_K - 1})$, $a \in \mathbb{Z}/(\ell-1)$, $b \equiv a + w_O - 1$, and each is exactly one of: the class $a = 0$ (that of E_{w_O}); the class $b = 0$ (that of E_{w_K}); a class of the* standard family*, $1 \le a \le w_K - 2$; a class of the* swapped family*, $1 \le b \le w_O - 2$. Equivalently, a Kummer class other than the two Eisenstein ends has exactly one presentation $\omega^t(1 \oplus \omega^{w-1})$ with $w \in \{w_O, w_K\}$ and $1 \le t \le J(w) - 1$: with $w = w_O$, $t = a$ in the standard family (where ℓ divides the companion $B_{\ell+1-w}$, the Kummer–Frobenius situation of Prediction 4.9's v1.16.0 form) and with $w = w_K$, $t = b$ in the swapped family (where ℓ divides B_w itself).*

3. *A class is a Kummer class of two different irregular pairs only if $\ell \mid B_w$ and $\ell \mid B_{\ell+1-w}$ for one of its seeds w (a* double divisibility*).*

Proof. (1) Characters of the cyclic group $(\mathbb{Z}/\ell)^\times$ are determined by their exponents modulo $\ell-1$, so $\omega^t \oplus \omega^{t+w'-1} \cong 1 \oplus \omega^{w-1}$ iff $\{t, t+w'-1\} \equiv \{0, w-1\}$: either $t \equiv 0$ and $w' \equiv w$, or $t \equiv w-1$ and $w'-1 \equiv -(w-1)$, i.e. $w' \equiv \ell+1-w$; as $w, w', \ell+1-w$ all lie in $[4, \ell-3]$, the congruences are equalities. (2) $J(w_O) = \ell - w_O = w_K - 1$, $J(w_K) = w_O - 1$. Take $a \in [0, \ell-2]$: for $1 \le a \le w_K - 2$, $b = a + w_O - 1 \in [w_O, \ell-2]$; for $a = w_K - 1$, $b = \ell - 1 \equiv 0$; for $w_K \le a \le \ell-2$, $b \equiv a + w_O - \ell \in [1, w_O - 2]$. So the four sets partition $\mathbb{Z}/(\ell-1)$, and by (1) the presentations of a Kummer class with a seed in $[4, \ell-3]$ are exactly the two given. (3) By (1) the seeds of a class determine the pair up to $w \leftrightarrow \ell+1-w$. □

By script (slot-17 R17-1 §3, three independent routes to $B_w \bmod \ell$): no double divisibility for $\ell < 2{\cdot}10^4$ (900 irregular primes, 1153 pairs); so at every reachable cell each Kummer class belongs to one pair. A heuristic (not a statement) expects the first one far beyond.

Claim 4.15 (The self-mirrored seed is regular)**.** *[proved here] (given the [standard, reference] input (i) of the proof) If $\ell \equiv 3 \pmod 4$ and $\ell > 3$, then $\ell \nmid B_{(\ell+1)/2}$: the seed $w = (\ell+1)/2$, the one with $w = \ell+1-w$, is never irregular.*

Proof. Let $\chi = (\cdot/\ell) = \omega^{(\ell-1)/2}$, odd, and $m := (\ell+1)/2$, even, $4 \le m \le \ell-3$ for $\ell \ge 7$ (at $\ell = 7$, $m = 4$). (i) [standard, reference]: the class number formula $h(-\ell) = -B_{1,\chi} = -\frac{1}{\ell}\sum_{a=1}^{\ell-1} a\,\chi(a)$ for $\ell > 3$ (Washington [37, Ch. 4]; the theorem number is **not verified**, rule A2). (ii) $B_{1,\omega^{m-1}} \equiv B_m/m \pmod \ell$, [38, Cor. 5.15, p. 61] (verified on the page image, slot-17 R17-3; the $\ell = 37$ report's `app:verify`); here $\omega^{m-1} = \chi$, so $h(-\ell) \equiv -B_m/m \pmod \ell$. (iii) $0 < h(-\ell) < \ell$: pairing a with $\ell - a$ ($\chi(\ell-a) = -\chi(a)$), $\sum_{a=1}^{\ell-1} a\chi(a) = \sum_{a=1}^{(\ell-1)/2} \chi(a)(2a-\ell)$, of absolute value at most $\sum_{a \le (\ell-1)/2}(\ell-2a) = (\ell-1)^2/4$, so $h(-\ell) \le (\ell-1)^2/(4\ell) < \ell$. Hence $\ell \nmid B_m/m$, and $\ell \nmid m$, so $\ell \nmid B_m$. □

By script (R17-1 §3): the congruence and the bound on all 154 primes $7 \le \ell < 2000$ with $\ell \equiv 3 \pmod 4$.

Claim 4.16 (The dimension of a reducible block, at every ℓ)**.** *[proved here] (given the [standard, reference] inputs named in the proof; two of their locators are **not verified**, see the status paragraph below) Let $\ell \ge 5$, $n \ge 2$ even, $k = n+2$, and $\mathfrak{m}$ a maximal ideal of $\mathbb{T}$ on $H^1(\Gamma; M_n)$ with residue field $\mathbb{F}_\ell$ whose residual representation $\bar\rho_{\mathfrak{m}} \cong \omega^i \oplus \omega^j$ is reducible and not of Dickson type, with seeds $w_1 = d+1$, $w_2 = \ell - d$. For an even w, $4 \le w \le \ell-3$, let g_w be the dimension of the generalised eigenspace of the system $a_p = 1 + p^{w-1}$ ($p \ne \ell$) in $S_w(\mathbb{F}_\ell)$; $g_w = 0$ unless $\ell \mid B_w$. Then*

$$\dim_{\mathbb{F}_\ell} H^1(\Gamma; M_n)_{\mathfrak{m}} \;=\; m_1(n)\,(1+2g_{w_1}) \;+\; m_2(n)\,(1+2g_{w_2}),$$

$$m_1(n) := [M_n : \mathrm{Sym}^{w_1-2}\mathbb{F}_\ell^2 \otimes \det{}^i], \qquad m_2(n) := [M_n : \mathrm{Sym}^{w_2-2}\mathbb{F}_\ell^2 \otimes \det{}^{i+w_1-1}],$$

the Jordan–Hölder multiplicities of $M_n = \mathrm{Sym}^n\mathbb{F}_\ell^2$ as a $\mathrm{GL}_2(\mathbb{F}_\ell)$-module ($\det^s$ *acting on T_p through p^s, the convention of Claim 5.13). If moreover $\mathfrak{m}$ is* cuspidal *— $\bar\rho_{\mathfrak{m}} \not\cong 1 \oplus \omega^{k-1}$, i.e. E_k's system is not in $\mathfrak{m}$ — then the block $\Lambda_{\mathfrak{m}} = H^1(\Gamma; \mathrm{Sym}^n\mathbb{Z}_\ell)_{\mathfrak{m}}$ is torsion-free,* $\dim H^1(\Gamma; M_n)_{\mathfrak{m}}$ *is even,* $\operatorname{rank}\mathbb{T}_{\mathfrak{m}} = \frac{1}{2}\dim H^1(\Gamma; M_n)_{\mathfrak{m}}$*, and the census algebra A of the block has* $\mathrm{Loewy}(A) \le \dim A \le \frac{1}{2}\dim H^1(\Gamma; M_n)_{\mathfrak{m}}$.

Proof. (The $\ell = 37$ report's `thm:dimformula`, steps (1)–(3), with 37 replaced by ℓ and its computer input (ii) replaced by (c) below.) (a) *Additivity.* For $p \ne \ell$ the matrices of determinant p in the double coset of T_p act on M_n through their reductions in $\mathrm{GL}_2(\mathbb{F}_\ell)$, so every $\mathrm{GL}_2(\mathbb{F}_\ell)$-submodule carries the anemic Hecke action and the long exact sequences of Γ-cohomology of a composition series of M_n are Hecke-equivariant ([standard, reference]: [1, Lemma 1.1.1, p. 194], verified on the page image, slot-17 R17-4). $H^q(\Gamma; N) = 0$ for $q \ge 2$ and every $\mathbb{F}_\ell[\Gamma]$-module N (the amalgam argument of [C, `cl:amalgam`](2),(3), which uses nothing about the coefficient module, $6 \in \mathbb{F}_\ell^\times$), and localisation at $\mathfrak{m}$ is exact; so $\dim H^1(M_n)_{\mathfrak{m}} - \dim H^0(M_n)_{\mathfrak{m}} = \sum_\sigma [M_n : \sigma]\big(\dim H^1(\Gamma;\sigma)_{\mathfrak{m}} - \dim H^0(\Gamma;\sigma)_{\mathfrak{m}}\big)$ over the irreducible $\sigma = \mathrm{Sym}^r\mathbb{F}_\ell^2 \otimes \det^s$, $0 \le r \le \ell-1$, $s \in \mathbb{Z}/(\ell-1)$. (b) *$H^0$ at $\mathfrak{m}$.* $H^0(\Gamma;\sigma) = \sigma^{\mathrm{SL}_2(\mathbb{F}_\ell)}$ is non-zero only for $r = 0$, where T_p acts by $(1+p)p^s$, the Dickson system $\omega^s(1 \oplus \omega)$; so

$H^0(\Gamma;\sigma)_{\mathfrak{m}} = 0$, and, by left exactness along the composition series, $H^0(\Gamma;M_n)_{\mathfrak{m}} = 0$. (c) $H^1(\Gamma;\sigma)_{\mathfrak{m}}$. As groups $H^1(\Gamma;\mathrm{Sym}^r \otimes \det^s) = H^1(\Gamma;\mathrm{Sym}^r \mathbb{F}_\ell^2)$, with T_p multiplied by p^s. For $r = 0$ it is 0 ($\Gamma^{\mathrm{ab}} \cong \mathbb{Z}/12$); for r odd, $-I$ acts by -1 and it is 0. For even $2 \le r \le \ell - 1$, $\mathrm{Sym}^r \mathbb{F}_\ell^2$ is irreducible and non-trivial, so $H^0(\Gamma;\mathrm{Sym}^r \mathbb{F}_\ell^2) = 0$, $H^1(\Gamma;\mathrm{Sym}^r \mathbb{Z}_\ell^2)$ is torsion-free and reduces onto $H^1(\Gamma;\mathrm{Sym}^r \mathbb{F}_\ell^2)$ ([C, `cl:bockstein`]); by Eichler–Shimura with its Hecke equivariance ([C, `lem:heckerankone`]'s proof: [14, Thm. 12.2.2, p. 108; Prop. 12.4.10, p. 117], verified, slot-17 L17-5) the $\mathfrak{m}'$-part of the reduction has dimension the number of characteristic-0 eigensystems of M_{r+2} reducing to $\mathfrak{m}'$, the Eisenstein one counted once and each cuspidal one twice. The Eisenstein system of E_{r+2} twisted by ω^s is $\omega^s \oplus \omega^{s+r+1}$; it equals $\bar\rho_{\mathfrak{m}}$ iff $(r,s) = (w_1 - 2, i)$ or $(w_2 - 2, i + w_1 - 1)$ (the computation of Claim 4.14(1); the solution $r = \ell - 1$ needs $d \in \{1, \ell - 2\}$, both of Dickson type; v1.18.0-pre.2, the referee F1's m-E2, pre.1 read "$d = 1$"). A cuspidal system of weight $w = r + 2 \le \ell + 1$ twisted into $\bar\rho_{\mathfrak{m}}$ has reducible residual representation, and *a level-*1 *eigenform* f *of weight* $2 \le w \le \ell + 1$ *with reducible* $\bar\rho_f$ *has* $\bar\rho_f^{\mathrm{ss}} \cong 1 \oplus \omega^{w-1}$: if f is ordinary, $\bar\rho_f|_{I_\ell} \sim \left(\begin{smallmatrix} \omega^{w-1} & * \\ 0 & 1 \end{smallmatrix}\right)$ (Deligne, [standard, reference]: [39, Thm. 2.1.4, p. 561], verified, slot-16 L16-3), and a level-1 $\bar\rho^{\mathrm{ss}}$ is a sum of powers of ω (a character of $G_{\mathbb{Q}}$ unramified outside ℓ with values in $\bar{\mathbb{F}}_\ell^\times$ is a power of ω, [standard, reference]: class field theory over $\mathbb{Q}$), hence $\bar\rho_f^{\mathrm{ss}} \cong 1 \oplus \omega^{w-1}$; if $a_\ell(f) \equiv 0$, $\bar\rho_f|_{I_\ell} \cong \omega_2^{w-1} \oplus \omega_2^{\ell(w-1)}$ (Fontaine, [standard, reference]: [16, Thm. 2.6]; statement and attribution verified in a secondary source, the theorem number pending, see the status paragraph; for $w \le \ell$ also [7, Thm. 4.5 with Prop. 4.4(1)]), irreducible since $\ell + 1 \nmid w - 1$ for $2 \le w \le \ell + 1$, a contradiction. So the cuspidal systems twisted into $\bar\rho_{\mathfrak{m}}$ are those of the cusp forms of weight $w \in \{w_1, w_2\}$ congruent to E_w, g_w of them with multiplicity (the generalised eigenspace in $S_w(\mathbb{F}_\ell)$ counts the characteristic-0 systems reducing to it, $S_w(\mathbb{Z}_\ell)$ being a Hecke-stable lattice; this count, and the count on H^1 above, use the semisimplicity of the Hecke action in characteristic 0 and multiplicity one at level 1, [standard, reference]: the inputs of [C, `lem:heckerankone`], [14, §§3.5–3.6, pp. 54–55], verified, slot-17 L17-5, with Serre's Cours Ch. VII §5 and Diamond–Shurman Thm. 5.5.4 as cited there; v1.18.0-pre.2, the referee F1's m-E2); at $w = \ell + 1$ the ordinary case is of Dickson type. That $g_w = 0$ unless $\ell \mid B_w$ is the Eisenstein-ideal congruence ([standard, reference], Prediction 4.3). Summing, $\dim H^1(\Gamma;\sigma)_{\mathfrak{m}} = 1 + 2g_w$ for σ the Serre weight at the seed w, and 0 for every other σ. For a cuspidal $\mathfrak{m}$: the torsion of $H^1(\Gamma;\mathrm{Sym}^n \mathbb{Z}_\ell)$ meets only the systems of [C, `prop:heckeH0`] (the remark after [C, `prop:cohforms`]), which are of Dickson type, so $\Lambda_{\mathfrak{m}}$ is torsion-free of rank $f_{\mathfrak{m}} = \dim H^1(\Gamma;M_n)_{\mathfrak{m}}$ ([C, `cl:bockstein`]); $\mathfrak{m}$ contains no Eisenstein system of weight k, so $f_{\mathfrak{m}}$ is even and $\mathrm{rank}\, \mathbb{T}_{\mathfrak{m}} = f_{\mathfrak{m}}/2$ by [C, `lem:heckerankone`]; A is a quotient of $\mathbb{T}_{\mathfrak{m}}/\ell$ ([C, `prop:cohforms`](5)), so $\dim A \le f_{\mathfrak{m}}/2$. (A corollary, v1.18.0-pre.2, the referee F1's m-E2(iii): on a cuspidal Kummer class $f_{\mathfrak{m}} = m_O + m_K(1 + 2g) \equiv m_O + m_K \pmod 2$ is even, so $m_O \equiv m_K \pmod 2$ — why the two seeds' multiplicities change together in Remark 4.17.) □

Status of the inputs (the owner's ruling O3 of 2026-10-03: [proved here] only with the literature inputs [standard, reference] and the unverified ones flagged, as [C, `lem:heckerankone`]). Verified on page images: Ash–Stevens Lemma 1.1.1, Diamond–Im Thm. 12.2.2 / Prop. 12.4.10 and §§3.5–3.6, Wiles Thm. 2.1.4. **Edixhoven Thm. 2.6: attribution verified in a secondary source, locator pending** (v1.18.0-pre.2, slot-18 R18-4, `docs/implementation/slot18/R18-4_literature.md` §2; pre.1 read "not verified (the number read from the text layer, slot-17 R17-3)"): the statement (ordinary: Deligne; supersingular, $k \le \ell + 1$: Fontaine, $\bar\rho_f|_{I_\ell} \cong \psi^{k-1} \oplus \psi'^{k-1}$ with the two fundamental characters of level 2) and its attribution, with the proof in [16], are confirmed in Edixhoven's Boston survey *Serre's conjecture* (printed p. 19, the section on "Theorem 4.5 of [15]"); the number "Thm. 2.6" is not confirmed, the paper itself not being in the library (an optional scan item for the owner). A second locator, for $2 \le w \le \ell$: [7, Thm. 4.5, p. 12, with Prop. 4.4(1), p. 11] (read on the arXiv copy, slot-17 R17-4) gives $\bar\rho_f|_{G_{\mathbb{Q}_\ell}} \cong \mathrm{ind}\, \omega_2^{w-1}$ when $v_\ell(a_\ell) > \lfloor (w-2)/(\ell-1) \rfloor$, which for $w \le \ell$ is $a_\ell \equiv 0$ (at $w = \ell+1$ it needs $v_\ell(a_\ell) \ge 2$, so Fontaine's theorem stays the input there). The class-field-theory sentence is textbook and carries no locator. If the Edixhoven input failed, the right-hand side would be a *lower bound*, with equality for every $\mathfrak{m}$ into which no non-ordinary ($a_\ell \equiv 0$) level-1 eigenform of weight $\le \ell+1$ with reducible $\bar\rho_f$ twists (at its Serre weight $\mathrm{Sym}^{w'-2} \otimes \det^s$, which need not be a seed); the census check below (21 690/21 690) is evidence that none does in the ranges computed (v1.18.0-pre.2, the referee F1's m-E1: pre.1 placed the possible extra contribution at the seeds). The multiplicities m_1, m_2 are computed by Glover's recursion (Claim 5.13; slot-17 R17-1's `glover_jh.py`, checked against exact

Brauer characters on every semisimple class for $n \le 1500$ at $\ell \in \{17, 37, 67\}$); the recursion is read off [18] ((4.1), p. 439; (4.2), p. 440; (5.7), p. 454; (6.4), p. 461; held since 2026-10-05, read in the slot-21 literature pass, `docs/implementation/slot21/missing_literature.md` §7; v1.21.0-pre.1, in place of "whose locator is not in the library"); a closed multiplicity formula is *not* in Glover's paper. *Evidence* (census edition 3 against the formula, slot-17 R17-1 §4, `census_check.py`): every reducible non-Dickson H^1 class of every cell, 21 690 of 21 690 (cell, class) pairs at $\ell \in \{7, \dots, 103, 691\}$, 0 missing; and every cohomological and forms-side dimension the project measured at $\ell = 37$ (the report's `cor:onset`).

Remark 4.17 (The windows on the multiplicity side). [proved here] (from Claim 4.16) and computed. For a cuspidal Kummer class presented as $\omega^t(1 \oplus \omega^{w-1})$ by Claim 4.14(2), $g := g_{w_K}$, and (m_O, m_K) the multiplicities at the seeds w_O, w_K: $\dim V = m_O + m_K(1+2g)$ and $\operatorname{rank} \mathbb{T}_{\mathfrak{m}} = \frac{1}{2} \dim V$. By script on slot-17 R17-1's class tables (`docs/implementation/slot17/r171/classes/`, 13 irregular pairs, $\ell \le 691$; recounted by `docs/implementation/slot18/wp2/window_events.py`), on every one of the 1492 standard and 1906 swapped cuspidal classes: (m_O, m_K) becomes $(1,1)$ at $k = w + t(\ell+1)$, $(2,2)$ at $k = \ell w + t(\ell+1)$ (the *meeting*) and $(3,3)$ at $k = w + t(\ell+1) + \ell^2 - 1$, with no other change in between (v1.18.0-pre.2, the referee F1's m-E3: `window_events.py` tests the $(2,2)$ and $(3,3)$ events only; the exact event sequence — these three events and nothing else up to the upper end, and $\dim V = m_O + m_K(1+2g)$ at every event — is checked by `docs/implementation/slot18/wp3/window_check_f1.py`, 1492/1492 and 1906/1906). So in the *first window* $w + t(\ell+1) \le k < \ell w + t(\ell+1)$, $\dim V = 2 + 2g$ and $\operatorname{rank} \mathbb{T}_{\mathfrak{m}} = 1 + g$; in the *second*, $\ell w + t(\ell+1) \le k < w + t(\ell+1) + \ell^2 - 1$, $\dim V = 4 + 4g$ and $\operatorname{rank} \mathbb{T}_{\mathfrak{m}} = 2 + 2g$; then $6 + 6g$. At a regular class ($g_{w_1} = g_{w_2} = 0$) the same windows give rank 1, 2, 3. These are statements about dimensions in the ranges computed (the class tables run to $k_{\max} = \ell \max(w) + \ell^2 + 2\ell$), not about the shape of A.

Prediction 4.18 (The depth of a curvilinear block). [conjectural] (new at v1.18.0-pre.1; the $\ell = 37$ report's `conj:depth` (0.2.0-pre.2, its `sec:fieldsproposals`), taken into this document by slot-18 WP-2 under the owner's ruling O3 of 2026-10-03) Let $\mathfrak{m}$ be a cuspidal, non-Dickson block of $H^1(\Gamma; M_n)$ with residue field $\mathbb{F}_\ell$, and $\mathbb{T}_{\mathfrak{m}}$ the Hecke algebra on the torsion-free $\Lambda_{\mathfrak{m}}$ (so $\mathbb{T}_{\mathfrak{m}}/\ell \cong \mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell$ on the forms side, [C, `prop:cohforms`]). Suppose one of:

(a) $\mathfrak{m}$ is a Kummer class of an irregular pair (ℓ, w_K) at a weight where $(m_O, m_K) = (m, m)$ (Remark 4.17), with *seed rank* $r_0 := 1 + g_{w_K}$ (the standard and swapped families with $g = 1$, and the $g = 2$ classes at 547);

(b) $\bar{\rho}_{\mathfrak{m}} \cong \bar{\rho}_f \otimes \omega^j$, $j \in \{1, 2\}$, at $k = \ell k_0 + j(\ell+1)$ (the Frobenius layer of Prediction 4.12), where f is an ordinary level-1 eigenform of weight $k_0 \le \ell + 1$ whose Hecke algebra on S_{k_0}, completed at f's system, is the ring of integers of a ramified quadratic extension of $\mathbb{Q}_\ell$ (two Galois-conjugate eigenforms congruent modulo ℓ); set $r_0 := 2$, $m := 2$.

If $m \le 2$, then $\mathbb{T}_{\mathfrak{m}}/\ell \cong \mathbb{F}_\ell[x]/x^{mr_0}$. In (a), $\operatorname{rank} \mathbb{T}_{\mathfrak{m}} = m\, r_0$ is [proved here] (Remark 4.17), so the prediction is that $\mathbb{T}_{\mathfrak{m}}/\ell$ is curvilinear of the largest length its rank allows; in (b), $\operatorname{rank} \mathbb{T}_{\mathfrak{m}} = 4$ is computed at every cell below, not proved here. In (b) the block is torsion-free, so that $\mathbb{T}_{\mathfrak{m}}/\ell \cong \mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell$, by the H^0-argument of Claim 4.16's proof (the torsion meets only Dickson-type systems), which does not use reducibility (v1.18.0-pre.2, the referee F1's m-E5).

Evidence (forms side unless said; generated, $\dim A = \dim W$). The $\ell = 37$ report's Table `tab:families`: 57 curvilinear blocks — 29 standard Kummer, 8 swapped, 2 with $g = 2$, 18 irreducible — 26 of them also on cohomology (the 24 cells at 37, $(736, 67)$, $(608, 547)$, each with $\mathbb{T}_{\mathfrak{m}}$ saturated). Slot-18 WP-2 (Prediction 4.10's evidence, `docs/implementation/slot18/wp2/`): 8 more — $(1400, 37)$, $(1438, 37)$, $(10632, 103)$ at $(m, r_0) = (2, 2)$, swapped; $(1154, 547)$, $(1700, 547)$, $(2246, 547)$, $(1032, 547)$, $(1578, 547)$ at $(m, r_0) = (1, 3)$, standard and swapped. At $(m, r_0) = (1, 2)$ the bound $\dim \mathbb{T}_{\mathfrak{m}}/\ell \le 2$ makes the statement automatic. (v1.18.0-pre.2) The slot-18 referee F1 (`docs/implementation/slot18/wp3/`, predicted before each run, 9/9 held): 8 more — $(1364, 37)$, $(1476, 37)$, $(1514, 37)$, $(10736, 103)$ at $(m, r_0) = (2, 2)$, swapped; $(16988, 547)$, $(33914, 547)$, $(11952, 547)$ at $(m, r_0) = (1, 3)$; and $(34460, 547)$, $k = 34462$, the meeting of the class $a = 1$, at $(m, r_0) = (2, 3)$: $\dim W = 6 = \operatorname{rank} \mathbb{T}_{\mathfrak{m}}$, Hilbert function $(1, 1, 1, 1, 1, 1)$, generated, $\mathbb{F}_{547}[x]/x^6$. In all 73 curvilinear blocks on the forms side, 26 of them on cohomology. *Why* $m \le 2$: at $m = 3$ it fails, Hilbert function $(1, 2, 2, 1)$ at $k = 1412$, 1450 (standard, 37) and 1438, 1476, 1514 (swapped,

37), 10736 (swapped, 103). The $(2,3)$ datum separates the two readings of the depth law: the hypothesis is $m \le 2$, not $mr_0 \le 4$ — $(m, r_0) = (2,3)$ is curvilinear of length 6 while $(3,2)$ is $(1,2,2,1)$ (pre.1 read "*Untested*: $(m, r_0) = (2,3)$ … an extrapolation, kept as such"; the referee F1's M-2). *Untested*: a congruence between two Galois orbits rather than a conjugate pair; any cell of (b) on cohomology (the cheapest is $(2210, 67)$, cluster only). Whether $\mathbb{T}_{\mathfrak{m}}$ is saturated at every block of (a), (b) — so that the census algebra A equals $\mathbb{T}_{\mathfrak{m}}/\ell$ — is the report's `q:satfam`: measured yes at the 26 cohomology cells; Prediction 4.10 asserts it, with the curvilinearity, in its windows. *Not taken into this document*: the report's `conj:law` (the reducible quotient modulo ℓ^2, 28/28 cells, unexplained). Slot-18 R18-1 tested it at scale (`docs/implementation/slot18/R18-1_report.md`, final; predictions committed before the grid, `r181/predictions.md`): 601 of 715 predicted cells run at 8 primes. (C-1) *as stated* is **refuted** at all 50 second-window cells at $\ell = 37$ ($k = 1412, \dots, 2708$, $a = 1, 2$; the block is not curvilinear and every $s \in \mathbb{Z}/37$ is admitted for a generating set of $\mathbb{T}_{\mathfrak{m}}$ — the T_p, $p < 100$, at 19 cells, with T_{37} added at the 31 cells $a = 1$, $k \ge 1628$, where the T_p alone do not generate (the report's referee F4, M1; re-run `docs/implementation/slot18/r185/`) — so $\mathbb{T}_{\mathfrak{m}}/I_{\mathrm{red}}$ is not cyclic; one of them, $(1412, 37)$, was seen in a smoke run before the predictions commit). The restricted form (C-1′) — the law wherever $\mathbb{T}_{\mathfrak{m}}/I_{\mathrm{red}}$ is cyclic — held at 548/548 cells at 7 primes $(37, 41, 43, 59, 67, 101, 103)$, ℓ-adically (modulo ℓ^{e-1}) at 20/20, vacuous at the 3 $g = 2$ cells at 547; (C-2), (C-3) held at 433/433. The report [L37] v0.3.0-pre.2 (refereed by F4 in pre.1, repairs applied) carries (C-1′) as its conjecture. A pointer only, no [conjectural] taken in (R18-1's final figures replace F1's preliminary ones, 503/50/3 of 556, in this pre.2 text; slot-18 R18-5).

4.5 The block $(258, 37)$ exactly: the Hecke algebra, its over-orders, the lattice

(New at v1.19.0-pre.1, hecke-extensions slot 19, WP-2; owner proposal of 2026-10-03, `docs/implementation/slot1`
The statements are slot-19 R19-1's (`docs/implementation/slot19/R19-1_report.md` §8, predictions committed before any computation, 10/10 held); the checkable item 4.22 is the gate of slot-19 WP-LC, which merges only after this text is at pre.1 (rule 22). Placed at the end of §4, so that no earlier number moves.) **Setting.** $(n, \ell) = (258, 37)$, $k = 260$, $\mathfrak{m} = \mathfrak{m}_1$ the class $\mathrm{eis}(1,6) = \theta E_6 \equiv \theta^6 E_{32}$ — the first cell of Prediction 4.10's Kummer–Frobenius onset and of the $\ell = 37$ report's table ([L37, `thm:example`]). $\Lambda := H^1(\Gamma; M_n \otimes \mathbb{Z}_{37})$, $\Lambda_{\mathfrak{m}}$ its $\mathfrak{m}$-part: free of rank 8 over $\mathbb{Z}_{37}$, torsion-free, $\Lambda_{\mathfrak{m}}/37 = V$ ([L37, `thm:example`](2)). $\mathbb{T}_{\mathfrak{m}} \subseteq \mathrm{End}(\Lambda_{\mathfrak{m}})$ is the $\mathbb{Z}_{37}$-algebra generated by *all* T_p (with $T_{37} = U_{37}$), $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}$ the same on $S_{260}(\mathbb{Z}_{37})_{\mathfrak{m}}$; $t := T_2 - 29$; $A := \mathbb{F}_{37}[x]/x^4$. **Tags** (here only): [computed and compared] = computed by slot-19 R19-1 from the exact integral Hecke matrices of S_{260}, by two independent routes (python-flint and in-house), at 37-adic precision 37^{32}, far above every valuation read — proof-grade for the statements as written, but not a proof in this document.

Claim 4.19 (The Hecke algebra of the block)**.** *1. [computed and compared]* $\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}} = \mathbb{Z}_{37}[t] \cong \mathbb{Z}_{37}[X]/(P)$, $P \equiv X^4 \pmod{37}$, *the valuations of the coefficients of P (low to high)* $(2,2,2,1,0)$, $v(\mathrm{disc}\, P) = 6$*: every $T_n|_{S_m}$, $n \le 21$ (a $\mathbb{Z}$-basis of the Hecke algebra of S_{260} by a_1-duality with the Miller basis) and $n = 37$, is a polynomial in t with $\mathbb{Z}_{37}$-coefficients. So $\mathbb{Z}_{37}[t]$ is the* full *Hecke algebra of the block, not an anemic one.*

2. *[proved here] (from [C, `prop:cohforms`](2),(3) and [L37, `thm:example`](2))* $\mathbb{T}_{\mathfrak{m}} \cong \mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}$, $T_p \mapsto T_p$ *for every p ($p = 37$ included). In particular $\mathbb{T}_{\mathfrak{m}} = \mathbb{Z}_{37}[t]$ and $\mathbb{T}_{\mathfrak{m}}$ is Gorenstein (monogenic).*
3. *[computed and compared] $P = P_1 P_2$ with P_1, P_2 Eisenstein of degree 2; U_{37} has slopes 1 on K_1 and 5 on K_2 ($K_i := \mathbb{Q}_{37}[X]/(P_i)$, ramified quadratic); the normalisation is $\tilde{\mathbb{T}} := \mathcal{O}_{K_1} \times \mathcal{O}_{K_2}$, $\mathcal{O}_{K_i} = \mathbb{Z}_{37}[\rho_i]$; $v(\mathrm{Res}(P_1, P_2)) = 2$; all six pairwise distances of the four roots are $1/2$, so every congruence between two of the four eigenforms has depth exactly $1/2$ (it holds modulo π, not modulo π^2; attained at a_2).*
4. *[computed and compared] $\tilde{\mathbb{T}}/\mathbb{T}_{\mathfrak{m}} \cong \mathbb{F}_{37}[\varepsilon]/\varepsilon^2$ as a $\mathbb{T}_{\mathfrak{m}}$-module (cyclic, generated by the class of the idempotent e_1: $37e_1 \in \mathbb{T}_{\mathfrak{m}}$, $te_1 \notin \mathbb{T}_{\mathfrak{m}}$; Smith form $(37, 37)$, not 37^2), and $\tilde{\mathbb{T}}/\mathbb{T}_{\mathfrak{m}} \cong \mathbb{Z}_{37}[X]/(P_1, P_2)$, the inter-orbit congruence module. The forms congruence module $S_{\mathfrak{m}}/(S_{\mathfrak{m}}^{K_1} \oplus S_{\mathfrak{m}}^{K_2}) \cong (\mathbb{Z}/37)^2$; the $\mathcal{O}$-span of the four eigenforms has index 37^3 in $S_{\mathfrak{m}} \otimes \mathcal{O}$.*
5. *[computed and compared] The over-orders of $\mathbb{T}_{\mathfrak{m}}$ in $\tilde{\mathbb{T}}$ are exactly $\mathbb{T}_{\mathfrak{m}} \subsetneq R \subsetneq \tilde{\mathbb{T}}$ with $R := \mathbb{T}_{\mathfrak{m}} + \mathbb{Z}_{37}\, te_1$ (of the 38 lattices of index 37 between $\mathbb{T}_{\mathfrak{m}}$ and $\tilde{\mathbb{T}}$ exactly one is a ring). In fact,*

since $\tilde{\mathbb{T}}/\mathbb{T}_{\mathfrak{m}} \cong A/x^2$ *is cyclic by (4), exactly one of the* 38 *is even a* $\mathbb{T}_{\mathfrak{m}}$*-submodule — the socle line* $R = \mathbb{T}_{\mathfrak{m}} + \mathbb{Z}_{37}te_1$ *— so the over-order list* $\{\mathbb{T}_{\mathfrak{m}}, R, \tilde{\mathbb{T}}\}$ *follows from (4) and the single check that* R *is a ring (referee F1, slot 19,* `f1_block258.py`*). R is local and Gorenstein (*$\dim \operatorname{soc} R/37R = 1$*),* $R = (\mathfrak{m} : \mathfrak{m})$ *and* $\mathfrak{m} = tR \cong R$*. As A-modules (*$x \mapsto t$*):* $\mathbb{T}_{\mathfrak{m}}/37 \cong A$*,* $R/37R \cong A/x^3 \oplus A/x$*,* $\tilde{\mathbb{T}}/37\tilde{\mathbb{T}} \cong A/x^2 \oplus A/x^2$*. Every over-order of* $\mathbb{T}_{\mathfrak{m}}$ *(and of its two quotients* $\mathcal{O}_{K_i}$*) is Gorenstein:* $\mathbb{T}_{\mathfrak{m}}$ *is a* Bass order*.*

Proof of (2). Λ is torsion-free ([L37, `thm:example`](2)), so $\mathbb{T}^{\mathrm{coh}} = \mathbb{T}^{\mathbb{L}}$ on the block, and [C, `prop:cohforms`](3) gives $\mathbb{T}^{\mathbb{L}} \cong \mathbb{T}^{\mathrm{f}}$ with $T_p \mapsto T_p$; localise at $\mathfrak{m}$. $\mathbb{T}_{\mathfrak{m}} \cong \mathbb{Z}_{37}[X]/(P)$ is a complete intersection, hence Gorenstein. (R19-1 recorded "$\mathbb{T}_{\mathfrak{m}} \cong \mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}$: forms side only"; this is the proof.) □

Claim 4.20 (The lattice)**.** *[proved here] given Claim 4.19(5) ([computed and compared]), [L37,* `thm:example`*](1) (*$V \cong A \oplus A/x^3 \oplus A/x$*, [computed and compared] at the cell) and* Gorenstein ideal theory *(over a reduced one-dimensional Gorenstein order* S*, a fractional ideal* I *with* $(I : I) = S$ *is invertible; [standard, reference], [5], §6, locator not verified, rule A2 — the paper is not in the project library (slot-19 referee F4, m1) and nothing here rests on it — and proved in the proof below, v1.19.0-pre.2; an invertible ideal of a semilocal ring is principal). Let* $\Lambda_{\mathfrak{m}} = \Lambda^+_{\mathfrak{m}} \oplus \Lambda^-_{\mathfrak{m}}$ *be the* ι*-eigenspace decomposition ([C,* `prop:iotasplit`*];* $\mathbb{T}_{\mathfrak{m}}$*-stable by [C,* `prop:iotacommutes`*]). Then one of* $\Lambda^{\pm}_{\mathfrak{m}}$ *is* $\cong \mathbb{T}_{\mathfrak{m}}$ *and the other is* $\cong \mathfrak{m} \cong R$*; so*

$$\Lambda_{\mathfrak{m}} \;\cong\; \mathbb{T}_{\mathfrak{m}} \oplus \mathfrak{m} \qquad \text{as } \mathbb{T}_{\mathfrak{m}}\text{-modules.}$$

The argument does not decide which sign is the free one; the measurement does: [computed and compared] (slot-19 WP-LC, `HeckeLatticeModuleValidation`*, Check 4.22(d) at* $M = 1, \ldots, 4$*; v1.19.0-pre.3) the* plus *sign is free,*

$$\Lambda^+_{\mathfrak{m}} \cong \mathbb{T}_{\mathfrak{m}}, \qquad \Lambda^-_{\mathfrak{m}} \cong \mathfrak{m}$$

("+" the Eisenstein side of [C, `def:iota`*]). Here* $\Lambda_{\mathfrak{m}}$ *is the* H^1 *lattice. On the modular-symbol lattice* $H^1_c(\Gamma; M_n \otimes \mathbb{Z}_{37})_{\mathfrak{m}}$ *the free sign is the other one (*$-$*, the geometric-even sign; slot-19 R19-8, re-derived by isign), because* $\mathfrak{m}$ *is residually Eisenstein (*$T_p \equiv p + p^6 \bmod 37$*,* $p = 2, 3, 5$*). The natural map* $H^1_{c,\mathfrak{m}} \to H^1_{\mathfrak{m}}$ *is injective with a cokernel of order* 37 *in each sign ([computed and compared]: one cyclic factor per sign by isign (K); the order by R19-9 §4, P9, the dual index* $[H^1(D)_{\mathfrak{m}} : H^1_!(D)_{\mathfrak{m}}] = 37$*, reproduced by slot-19 WP-BIL), so the two lattices are not isomorphic as* $\mathbb{T}_{\mathfrak{m}}[\iota]$*-modules ([computed and compared], isign). So the statement is [proved here] given [computed and compared] inputs, and the sign is [computed and compared].*

Proof. By Eichler–Shimura ([C, `prop:cohforms`], its per-sign multiplicity-one statement in (5) — "one of the $\pm$-parts … free of rank one" — and [L37, `prop:cuspidal`](2): no Eisenstein system in the block in characteristic 0 (the block is residually Eisenstein)) each $\Lambda^{\pm}_{\mathfrak{m}} \otimes \mathbb{Q}$ is free of rank one over $\mathbb{T}_{\mathfrak{m}} \otimes \mathbb{Q} = K_1 \times K_2$, so $\Lambda^{\pm}_{\mathfrak{m}}$ is isomorphic to a fractional ideal $I^{\pm}$ of $\mathbb{T}_{\mathfrak{m}}$. Its multiplier ring $S^{\pm} := (I^{\pm} : I^{\pm})$ is an order containing $\mathbb{T}_{\mathfrak{m}}$, hence contained in $\tilde{\mathbb{T}}$: $S^{\pm} \in \{\mathbb{T}_{\mathfrak{m}}, R, \tilde{\mathbb{T}}\}$ (Claim 4.19(5)), all Gorenstein, so $I^{\pm}$ is invertible over $S^{\pm}$ and $I^{\pm} \cong S^{\pm}$. (The Gorenstein input, for $S = S^{\pm}$ local, i.e. $\mathbb{T}_{\mathfrak{m}}$ or R: fractional ideals over a one-dimensional Gorenstein local ring are reflexive, $(S : (S : I)) = I$; with $I^* := (S : I)$, $(S : II^*) = ((S : I^*) : I) = (I : I) = S$. If $II^* \subsetneq S$ then $II^* \subseteq \mathfrak{m}_S$ and $(S : II^*) \supseteq (S : \mathfrak{m}_S) \supsetneq S$, as $\operatorname{Ext}^1_S(k, S) \neq 0$ in depth 1 — a contradiction; so $II^* = S$, I is invertible, hence principal. For $S = \tilde{\mathbb{T}}$, a product of DVRs, every fractional ideal is principal anyway.) Reducing modulo 37, $V = V^+ \oplus V^-$ with $V^{\pm} \cong S^{\pm}/37S^{\pm} \in \{A,\ A/x^3 \oplus A/x,\ A/x^2 \oplus A/x^2\}$. As the Jordan type of x on a finite $A = \mathbb{F}_{37}[x]/x^4$-module is unique, the only pair whose sum is $A \oplus A/x^3 \oplus A/x$ is $\{A,\ A/x^3 \oplus A/x\}$, i.e. $\{S^+, S^-\} = \{\mathbb{T}_{\mathfrak{m}}, R\}$; and $R \cong tR = \mathfrak{m}$ (t is a non-zero-divisor). (R19-1's route, Bass's structure theorem for torsion-free modules over a Bass order [5] §7, reaches the same conclusion without the ι-split; the split makes the weaker input above enough. R19-1's input is a literature locator that stays unchecked — the paper is not in the project library — and this route does not need it; slot-19 referee F4, m1.) □

Remark 4.21 (An independent check already measured). U_{37} mod 37 has rank 2 on A and rank 1 on $A/x^3 \oplus A/x$ ([computed and compared], R19-1 §5), so Claim 4.20 predicts $\operatorname{rank} U_{37}|_V = 3$ — the value measured on cohomology ([L37, `tab:cells`]; slot-18 R18-3). The excluded alternatives give 4 ($\mathbb{T}_{\mathfrak{m}}^2$) and 2 ($\mathbb{T}_{\mathfrak{m}} \oplus \tilde{\mathbb{T}}$).

Check 4.22 (The lattice comparison at $(258, 37)$; the gate of slot-19 WP-LC). For $M = 1, 2, 3, 4$ let $H_M := H^1(\Gamma; M_n \otimes \mathbb{Z}/37^M)_{\mathfrak{m}}$, computed on the $\mathbb{Z}/\ell^K$ route ([C, `sec:saturation`]; §7.1: Howell and Smith forms over $\mathbb{Z}/37^M$), with the operators $t = T_2 - 29$ and U_{37}, and ι^*. Write $\langle d_1, \dots, d_r \rangle_M$ for the Smith valuations of an operator on H_M (the invariants $\min(d_i, M)$, so that its cokernel is $\bigoplus_i \mathbb{Z}/37^{\min(d_i, M)}$). Then:

(a) [proved here] $H_M \cong \Lambda_{\mathfrak{m}}/37^M$: free of rank 8 over $\mathbb{Z}/37^M$ (torsion-freeness, [L37, `thm:example`](2); $H^2(\Gamma; -)$ has no ℓ-torsion for $\ell \geq 5$, [C, `lem:h2torsion`]; so $H^1(\Gamma; M_n \otimes \mathbb{Z}/37^M) = \Lambda/37^M$, and localisation at $\mathfrak{m}$ commutes).

(b) $\dim_{\mathbb{F}_{37}} H_M/(37, t)H_M = 3$ (the minimal number of $\mathbb{T}_{\mathfrak{m}}$-generators: 1 for $\mathbb{T}_{\mathfrak{m}}$, 2 for $\mathfrak{m}$).

(c) t: $\langle 0, 0, 0, 0, 0, 1, 1, 2 \rangle_M$; $\quad U_{37}$: $\langle 0, 0, 0, 1, 5, 6, 6, 6 \rangle_M$.

(d) Per sign: $H_M = H_M^+ \oplus H_M^-$, each free of rank 4; on one sign $\dim H/(37, t)H = 1$, t: $\langle 0, 0, 0, 2 \rangle_M$, U_{37}: $\langle 0, 0, 6, 6 \rangle_M$ (the free summand $\mathbb{T}_{\mathfrak{m}}$); on the other $\dim H/(37, t)H = 2$, t: $\langle 0, 0, 1, 1 \rangle_M$, U_{37}: $\langle 0, 1, 5, 6 \rangle_M$ (the summand $\mathfrak{m}$). The sign of the free summand is *reported*, not predicted.

(e) $P(t) = 0$ on H_M, and $T_p = Q_p(t)$ on H_M for every prime $p < 100$ ($p = 37$ included), with P and the Q_p the forms-side data of R19-1 modulo 37^6 (`docs/implementation/slot19/r191/data/universal_eig` $T_n = c_0 + c_1 t + c_2 t^2 + c_3 t^3$).

Status. (a) [proved here]. (b)–(e) are [proved here] given the [computed and compared] inputs of Claims 4.19, 4.20 (the Smith valuations of t and U_{37} on each of $\mathbb{T}_{\mathfrak{m}}$, R, $\tilde{\mathbb{T}}$ are R19-1's, §5): (b)–(d) are the invariants of $(\mathbb{T}_{\mathfrak{m}} \oplus \mathfrak{m})/37^M$, (e) is Claim 4.19(2). So a failure of (a) or (e) is a defect of the code; a failure of (b)–(d) with (a), (e) holding refutes a [computed and compared] input or [L37, `thm:example`](1), and is a finding. **Discrimination.** $\mathbb{T}_{\mathfrak{m}} \oplus \tilde{\mathbb{T}}$ shares (b) and t's valuations but has U_{37}: $\langle 0, 0, 1, 1, 5, 5, 6, 6 \rangle_M$ (so already $\operatorname{rank} U_{37}|_{H_1} = 2$, against 3); $\mathbb{T}_{\mathfrak{m}}^2$ has (b) = 2; so (b)–(d) at $M \geq 2$ separate the three candidates. **Cost** (plan): ≤ 6 GB at $M = 4$, predicted from 1.0 GB at $M = 1$. **Run** (v1.19.0-pre.3). [computed and compared] (slot-19 WP-LC, `HeckeLatticeModuleValidation`; predictions committed before the run, `docs/implementation/slot19/wplc/predictions.md` §1, outcomes §3): (a)–(e) *held* at $M = 1, \dots, 4$ — rank 8, free; $\mu = 3$; t: $\langle 0^5, 1, 1, 2 \rangle_M$, U_{37}: $\langle 0^3, 1, 5, 6, 6, 6 \rangle_M$; the free summand on the plus sign (Claim 4.20); (e) at all 25 primes $p < 100$ (the full arm), on the block and on each sign. The planted negative controls $\mathbb{T}_{\mathfrak{m}} \oplus \tilde{\mathbb{T}}$ (U_{37}) and $\mathbb{T}_{\mathfrak{m}}^2$ (μ) are rejected; independent runs at $K = 2, 3$ agree with the $K = 4$ run reduced. Measured cost: ≈ 22 min and 2.9 GB (full arm), ≈ 2 min and 0.8 GB (default arm).

Remark 4.23 (The $\ell = 7$ controls of WP-LC: per-sign freeness fails). [computed and compared] (slot-19 WP-LC, the controls of `HeckeLatticeModuleValidation`; cells chosen by a forms-side scan of every even $n \leq 110$, `docs/implementation/slot19/wplc/l7_pred.py`, and predicted before the run, `wplc/predictions.md` §2; outcomes §3; the over-order identification at $(48, 7)$ closed by the slot-19 referee F4, m5; v1.19.0-pre.3). Three non-Eisenstein blocks at $\ell = 7$, written by $(n, 7)$ and the residual system (a_2, a_3, a_5, a_7): $(48, 7)(1, 0, 0, 0)$, rank 2, Loewy length 2, $\mathbb{T}_{\mathfrak{m}} = \mathbb{Z}_7[t]$; $(90, 7)(3, 3, 1, 0)$, rank 2, Loewy 2, $\mathbb{T}_{\mathfrak{m}} = \mathbb{Z}_7[t]$; $(90, 7)(1, 0, 0, 0)$, rank 4, Loewy 3, $\mathbb{T}_{\mathfrak{m}}$ not monogenic in t. Every block is free over $\mathbb{Z}/7^M$ — measured (all cyclic-decomposition exponents equal K), and also the argument of Check 4.22(a), whose universal-coefficient step [C, `lem:h2torsion`] needs only $\ell \geq 5$ (F4, m7); rank, sign ranks, $P(t) = 0$ and $T_p = Q_p(t)$ for $p \leq 13$ hold, the items forced by [C, `prop:cohforms`]. The prediction "$\Lambda_{\mathfrak{m}}^{\pm} \cong \mathbb{T}_{\mathfrak{m}}$ on both signs" is **refuted on five of the six signs**:

1. $(48, 7)(1, 0, 0, 0)$: minus $\cong \mathbb{T}_{\mathfrak{m}}$ (free); plus $\cong \mathbb{Z}_7[t/7]$. With $s = t/7$, $s^2 + as + b = 0$, $v(a) = v(b) = 0$, $v(a^2 - 4b) = 2$, so $\mathbb{Z}_7[s]$ is *not* maximal: the over-orders form the chain $\mathbb{T}_{\mathfrak{m}} \subset \mathbb{Z}_7[t/7] \subset \mathcal{O}_{\max}$, of index 7 at each step, $\mathcal{O}_{\max} = \mathbb{Z}_7[(s + a/2)/7]$ étale. The Smith valuations of t and μ do not separate $\mathbb{Z}_7[t/7]$ from $\mathcal{O}_{\max}$ (both $\langle 1, 1 \rangle$, $\mu = 2$), but U_7 does: $\langle 1, 5 \rangle$, $\langle 2, 4 \rangle$, $\langle 3, 3 \rangle$ on $\mathbb{T}_{\mathfrak{m}}$, $\mathbb{Z}_7[t/7]$, $\mathcal{O}_{\max}$ (F4, from WP-LC's P and Q_7). The plus reading $\langle 2, 4 \rangle_4$ is $\mathbb{Z}_7[t/7]$ and nothing else; and a lattice over a quadratic order is isomorphic to its multiplier ring (quadratic orders are Gorenstein; the argument in the proof of Claim 4.20), so this identifies the plus sign. The same shape as $(258, 37)$: one sign free, the other the next over-order.

2. $(90,7)(3,3,1,0)$: both signs $\cong \mathbb{Z}_7[t/7]$, which is here the maximal order ($v(\mathrm{disc})=0$, $\cong \mathbb{Z}_7\times\mathbb{Z}_7$; t: $\langle 1,2\rangle$, $\mu=2$); no identification gap.
3. $(90,7)(1,0,0,0)$: plus consistent with $\mathbb{Z}_7[t/7]$ (t: $\langle 1,1,1,2\rangle$, $\mu=4$, U_7: $\langle 2,4,4,4\rangle_4$); minus none of the $\mathbb{Z}_7[t/7^j]$ (t: $\langle 0,0,2,3\rangle_M$, $\mu=2$, U_7: $\langle 1,4,4,4\rangle_4$). No candidate was enumerated here; the readings are pinned, the order is not named.

Scope (F4, m6). This is per-sign freeness of $H^1(\Gamma; M_n\otimes\mathbb{Z}_7)_{\mathfrak{m}}$ over $\mathbb{T}_{\mathfrak{m}}$ failing at $n=48,90$ ($\ell=7\le n$), outside the range $n<\ell-1$ of Claim 9.5(2) (Harder) and of the Fontaine–Laffaille comparison: a [computed and compared] statement about three blocks (five signs), not a theorem about the lattice for $\ell<k-1$. It is qualitatively expected; no literature pointer is offered yet. The validation pins the observed order per sign and prints `prediction_free=refuted`; that is not a re-pin of an item of Check 4.22. Adding $\mathcal{O}_{\max}$ to the code's candidate list is a later code item (F4, m5).

Remark 4.24 (What (258,37) says about the depth law and the next cells). [computed and compared] (R19-1 §6) the universal eigenform $\sum_n T_n q^n \in (\mathbb{T}_{\mathfrak{m}}/37^6)[[q]]$ to $n\le 1000$, the four eigenforms in $\mathcal{O}_{K_1}\times\mathcal{O}_{K_2}$ modulo 37^6 and the $\mathbb{Q}(\alpha)$-eigenform ($\alpha=a_2$, degree 21, Maeda) to the Sturm bound, with Ribet's form $g\in S_{32}$ ($a_n(g)\equiv\sigma_{31}(n)$ modulo 37, not modulo 37^2: depth $1=v(B_{32}/32)$) and $\theta^6 g\equiv\theta E_6\equiv f_i \pmod{\pi}$. The Galois side of the two seeds is slot-18 R18-2's pair of Kummer generators, $X^{37}-\eta_{32}$ (Ribet's H) and $X^{37}-\eta_6$ (`docs/implementation/slot18/R18-2_report.md` §1; [L37] `comp:kummergen`); the number-field layer that reproduces them in C++ is §9.4. *Not claimed*: the cohomology lattice at the other 23 cells of [L37]'s table (the same argument applies wherever Claim 4.19(5)'s over-order data are computed); the small Loewy-2 / 3 controls at $\ell=7$ that the plan asks for were not chosen by R19-1; WP-LC chose and ran them (Remark 4.23, v1.19.0-pre.3).

*Code map (**built**, slot-19 WP-LC; planned at v1.19.0-pre.1, contract first; v1.19.0-pre.3).* `HeckeExt/HeckeLatticeModule` and `HeckeLatticeModuleValidation` (registered `standard`, moved to `slow` by the owner's ruling of 2026-10-03; 301 checks; read-only env arm `HECKE_LATTICE_MODULE_FULL=1`: all $p<100$ and independent $K=2,3$ runs, 585 checks): H_M on the shipped $\mathbb{Z}/\ell^K$ module route (`EquivariantCohomologyOnK`) over $\mathbb{Z}/37^M$, with ι^*, the Howell / Smith forms of §7.1, $M=1\dots4$; Check 4.22 (a)–(e) at $(258,37)$; the eigenform data of (e) read from R19-1's tracked file; the Loewy-2 / 3 controls at $\ell=7$ chosen by WP-LC and predicted before the run (rule 20; Remark 4.23). Value-free: no golden line, digest, record key or census verdict changes.

5 The algorithms, stage by stage, with certificates

Throughout, $d_q=\dim H^q$, $d=\dim H_{\mathrm{tot}}=O(n)$, $D:=d_{\mathfrak{m}}=\dim_{\mathbb{F}_\ell} H_{\mathfrak{m}}$, $k:=\#S$, and costs are in $\mathbb{F}_\ell$-multiplications. Everything is exact linear algebra over $\mathbb{F}_\ell$ on the existing engine (`engine::elim`, `FpPoly`); no floating point, no field extension (§6).

Stage 0 — preconditions and the three additions

Available: bases of X_1, X_4, projectors P_q, $A^{(q)}(T_p)$ for all p (§1.2). Required additions, in order of value:

(A1) *Boundary Hecke action* on X_2, X_5: the formula of [C, `def:hecke`] with Γ_∞ in place of Γ. The relevant set is $B_q:=\{\delta\in\Gamma\,\mathrm{diag}(1,q)\Gamma:\ \delta \text{ upper triangular}\}$ (the elements fixing the cusp; $B_q/\Gamma_\infty\to\Gamma\,\mathrm{diag}(1,q)\Gamma/\Gamma$ is a bijection by column Hermite normal form), and the decomposition to use is the analogue of the *left* cosets $\Gamma g\Gamma=\bigsqcup_i \delta_i\Gamma$ of [C, `def:hecke`]: $B_q=\delta_\infty\Gamma_\infty\sqcup\bigsqcup_{j=0}^{q-1}\delta_j\Gamma_\infty$ with $\delta_\infty=\mathrm{diag}(1,q)$, $\delta_j=\begin{pmatrix} q & j\\ 0 & 1\end{pmatrix}$, so that $T\delta_\infty=\delta_\infty T^q$, $T\delta_j=\delta_{j+1}$, $T\delta_{q-1}=\delta_0 T$. On the circle model the operator is $c\mapsto\sum_i \rho_n(\delta_i)\,c(\tau_\infty(\cdot))$ with a one-dimensional τ (a Γ_∞-twisted chain map on the line), certified by $\partial\tau=\tau\partial$ exactly as [C, `alg:tau`], and by [C, `cl:chi`] as oracle. *Status (v1.4.0; [C] v1.4.1).* This one-dimensional τ_∞ is the Γ_∞-level twisted datum of [C, `obl:taubdry`](v.c), a validation route; the normative operator is [C, `eq:bdrychain`] from the canonical τ_∂ of [C, `alg:taubdry`] (cited, not re-described here; shipped as `Hecke/HeckeBoundaryDatum`). Clause (v) of [C, `obl:taubdry`] is three named comparisons — (v.a) degree 0 at every ν, against $\sum_{\delta\in B_N/\Gamma_\infty}\rho_n(\delta)$ on M^T; (v.b) degree 1

at $\nu = 1$, against [C, `eq:bdrygroupform`] in the coordinate Φ of [C, `prop:bdrychar`](3), $[\varphi] \mapsto [\varphi(\langle v_0, Tv_0\rangle)]$, representative map and lift on that one orientation (the opposite orientation gives a different form); (v.c) every computed degree at $\nu = 1$, the datum above — so at $\nu \geq 2$ clause (v) is *degree* 0 *alone*, H^1 there being covered by (vii) and the operator's own certificates; the general-ν double-coset decomposition of B_N is recorded in [C] as *not stated*. Through v1.3.3 this item read clause (v) as one comparison in every degree at every ν. *Scope of the circle model* (owner decision on the verdict's point 5, 2026-09-15): the boundary group is $\Gamma_\infty = \langle T\rangle \times \langle -I\rangle$ and the boundary cohomology is $H^\bullet(\partial; M) := H^\bullet(\Gamma_\infty; M)$ as defined in [C, `sec:six`]; the circle model is $C^\bullet_{\mathrm{eq}}(K_\partial, \mathrm{Rel}_\partial; L)$ with Rel_∂ the cusp relation $f(v_3) = \rho_n(T)f(v_0)$ *together with the* $-I$ *self-relations* $f(\sigma) = \rho_n(-I)f(\sigma)$ ([C, `prop:circlemodel`], v1.4.0; through v1.3.3 this document, like [C], called it "the single relation"). It computes the boundary equivariant complex $C^\bullet_\partial$ for *every* n and every coefficient ring ([C, `prop:circlemodel`](4)): the single-relation description holds exactly when $\rho_n(-I) = \mathrm{id}$, i.e. n even or $2 = 0$, and for n odd with 2 invertible the circle model is the zero complex ([C, `rmk:bdryoddn`]) — what the code computes. Its cohomology is $H^\bullet(\langle T\rangle; M)$ for n even, and agrees with $H^\bullet(\Gamma_\infty; M)$ *exactly when* $2 \in R^\times$ — for n even $-I$ acts trivially, so $H^\bullet(\Gamma_\infty; M) = H^\bullet(\langle T\rangle; M)^{-I} = H^\bullet(\langle T\rangle; M)$ when 2 is invertible. The circle model is therefore the specialisation of the boundary datum at $2 \in R^\times$, as the 7-vertex complex is the specialisation of the orbifold model at $6 \in R^\times$; at $\ell = 2$ they part company, and substantively so (closing paragraph of (A2)). Under the standing hypothesis $\ell \geq 5$ of §0 nothing in this document is affected. **Not** the upper-triangular right-coset representatives $\beta_j = \begin{pmatrix} 1 & j \\ 0 & q \end{pmatrix}$, $\beta_\infty = \mathrm{diag}(q, 1)$ of [C, `prop:cosets`](1): they represent $\Gamma_\infty\backslash B_q$, and their sum $M_\beta := \sum_j \rho_n(\beta_j) + \rho_n(\beta_\infty)$ is a well-defined endomorphism of M^T *only* — the monomials $x^{n-b\ell}\mathrm{Nm}^b$ are an eigenbasis with eigenvalue χ_{b+1} in place of χ_b — not asserted to be induced by any cochain map of $C^\bullet_\partial$ (a sum over $\Gamma_\infty\backslash B_q$ is not Γ_∞-equivariant as a formula on cochains), and built by the implementation as the fibre-level *negative control* `beta_control_fiber_operator` alone ([C, `prop:bdrychar`](2), v1.4.0; through v1.3.3 this document called it "the det-twisted (transposed) operator"). Equivariance $r^q A^{(q)} = A^{(q)}_\partial r^q$ of the two actions is [C, `obl:taubdry`](vii): asserted at $\ell \geq 5$ (864 (cell, q) pairs, slot 1), and at $\ell \in \{2, 3\}$, where the model complexes compute the orbifold model and not group cohomology, a COMP *record* only ([C, `rmk:conerestrictsmallell`]: 216/216, consistent with a pair homotopy over $\mathbb{Z}$, claimed nowhere, and a failure elsewhere would contradict nothing); the negative control breaks it on H^1 exactly when $\chi_0(p) \neq \chi_1(p)$ ([C, `cl:chi`](3)), which on H^0 of the circle model it cannot. The Eisenstein eigenvalue $1 + q^{n+1}$ on $[y^n] \in X_5$, read in the coordinate Φ, is the oracle.

(A2) *Equivariant mapping cone* of $r^\bullet$: the algebraic cone of [C, `def:pair`] and [C, `lem:conerelative`] (re-homed into [C] from the now-frozen [v5] §8) for `EquivariantCochainComplexMorphismOnK`, giving X_3, X_6 with bases and the maps of [C, `eq:les`] as matrices. $r^\bullet$ is Hecke-equivariant on cochains only *up to homotopy*, so $(\Theta_g, \Theta^\partial_g)$ is not a cochain map of the cone in general. **Open Question 2 is settled** (verdict `docs/implementation/hecke_extensions_OQ2_verdict.md`, Appendix C); the binding text is amendment C-A3 of the implementation plan, which is the mechanism by which [C] gains it. In outline:

- *The datum is a triple over* $\mathbb{Z}$, one level below the cochains, where τ lives: the bulk τ of [C, `def:tau`]; a g-twisted chain datum τ_∂ on $\partial X := \Gamma \cdot i(K_\partial)$, the Γ-orbit of the cusp line, with the same $\Gamma'' = \Gamma_0(q)$, the same s_i and the same δ_i as the bulk; and a *pair homotopy* $\eta : C_*(\partial X) \to C_{*+1}(X)$ satisfying (T1) and $\partial\eta + \eta\partial = j\tau_\partial - \tau j$, certified exactly like τ ([C, `obl:tau`]). τ_∂ and η exist over $\mathbb{Z}$ for *every* ℓ: every simplex of ∂X has stabiliser $\{\pm I\}$, which acts trivially, so no stabiliser order is ever inverted — unlike [C, `lem:tauexists`], which needs $\ell \geq 5$ for the elliptic stabilisers. The bulk τ is the canonical table of [C, `alg:tau`], integral exactly when no fill used the mod-ℓ fallback. [proved here] The canonical τ_∂ is the table of [C, `alg:taubdry`] (v1.4.0; shipped since slot 1): the nearest-line-vertex rule with ties to the larger real part, over $\mathbb{Z}$ with coefficients in $\{0, \pm 1\}$ for every ℓ; other admissible choices are homotopic ([C, `lem:taubdryexists`](3)). It is *not* ι-symmetric ([C, `rmk:taubdrynoiota`]; (A3)). The Γ-level τ_∂ on ∂X is the *normative*

boundary datum, and the one-dimensional τ_∞ of (A1) is its Shapiro transcription and a validation route only (owner decision, 2026-09-15; Appendix C).

- *The cochain homotopy is derived, never chosen freely*: by the coset transport of [C, `thm:welldef`](b), $(hc)(\sigma) := \sum_i \rho_n(\delta_i)\,\varphi_c\big(\eta(s_i \cdot i(\sigma))\big)$, giving $h^i : C^i_{\mathrm{eq}}(K) \to C^{i-1}_{\mathrm{eq}}(K_\partial)$ with $dh + hd = \Theta_g^\partial r - r\Theta_g$, and the cone map $\Phi_g(a,b) = (\Theta_g a,\ \Theta_g^\partial b + ha)$ on $C(r)^i = C^{i+1}_{\mathrm{eq}}(K) \oplus C^i_{\mathrm{eq}}(K_\partial)$ with $D(a,b) = (-da,\, db + ra)$. Every map of [C, `eq:les`] is then equivariant automatically. [proved here] The sign convention is fixed once and for all (owner decision, 2026-09-15; Appendix C): $\partial\eta + \eta\partial = j\tau_\partial - \tau j$, so $h = \tilde{\eta}$ carries *no* sign. The identity $dh + hd = \Theta_g^\partial r - r\Theta_g$ and the cone map are not moved by that choice — they are fixed by D.
- *The ambiguity, and why it is not one here.* Two cochain homotopies satisfying that identity differ by a degree-(-1) chain map k, and the two cone maps differ on $X_3 = H^1_c$ by the composite $X_3 \to X_4 \xrightarrow{k_*} X_2 \to X_3$, i.e. by an element of $\mathrm{Hom}(\ker r^1, \mathrm{coker}\, r^0)$ — at $(n,\ell) = (10, 691)$ a space of dimension 3. For a *free* cochain-level h *every* element of that Hom occurs *over a field* (the realisation argument splits the cochain groups, so it is stated for $R = \mathbb{F}_\ell$ and not claimed over $\mathbb{Z}/\ell^k$ or $\mathbb{Z}/m$), so the operator on X_3 would be genuinely undetermined and X_3 would be lost as a Hecke module. [proved here] For an η-derived h it vanishes identically: since X is contractible and two-dimensional ([standard, reference], Borel–Serre), so that $Z_1(X) = B_1(X)$ and $Z_2(X) = 0$, two pair homotopies differ by $\partial M - M\partial$ for a *unique* $M : C_0(\partial X) \to C_2(X)$, and $\Psi(a,b) := (0, -\widetilde{M}a)$ is then an integral cone homotopy between the two Φ_g. Hence T_q on $H^\bullet_c$ is independent of η over $\mathbb{Z}$ for every ℓ; independent of (τ, τ_∂), of the s_i and of g when 6 is invertible in the coefficient ring; and equal there to the operator of the Hecke correspondence on the Borel–Serre pair, so the Hecke relations hold on $H^\bullet_c$ on the nose. [proved here]
- *What does not decide it.* The on-the-nose case is exactly $\eta \equiv 0$, i.e. $\tau j = j\tau_\partial$ (a strict map of pairs; one can be constructed on this model, but the shipped τ is not strict — measured on every cell, v1.4.0: `cochain_commutes` and `eta_zero` are both false on 315/315 (cell, p, ν) pairs of `HeckeBoundaryParabolicValidation`'s grid, the cusp-support walk finding interior support on 15 of the 21 boundary slots of the canonical τ at $p = 2$ (10 of 12 vertex slots, 5 of 9 edge slots), so $h = 0$ is admissible nowhere and the η-table of E5 is genuinely needed; the slot counts [C] [C, `rmk:conestrict`](1) quotes are those of the pre-re-baseline table, a contract finding for the next status bump). The entrywise cochain identity $r\Theta_g = \Theta_g^\partial r$ alone does *not* license $h = 0$ — it can hold while $\tau j \neq j\tau_\partial$, and then h_η is a chain map that need not induce zero on X_3 (whether it actually does for some (τ, τ_∂) is undecided, [conjectural]) — so it is recorded as a diagnostic and gates nothing. It is τ, not $r^\bullet$, that one could change to obtain strict commutation. [proved here]

All of (A2) transfers verbatim to a free Γ-model of the pair (the Borel construction $(X, \partial X) \times E\Gamma$, or an isotropy-resolution model), whose equivariant cohomology is the orbifold cohomology at $\ell \in \{2, 3\}$ and of which the 7-vertex complex is the specialisation where 6 is invertible; there existence and homotopy uniqueness of $\tau, \tau_\partial, \eta$ hold over $\mathbb{Z}$ for every ℓ. [standard, reference] That general model enters the binding contract *now, as definitions*, with its construction deferred (owner decision, 2026-09-15; Appendix C) — which is what makes "the 7-vertex complex is a specialisation" a theorem rather than a slogan.

The boundary group, and what it costs at $\ell = 2$. The boundary group is $\Gamma_\infty = \langle T \rangle \times \langle -I \rangle$ and $H^\bullet(\partial; M) := H^\bullet(\Gamma_\infty; M)$ ([C, `sec:six`]; owner decision, 2026-09-15, *amending* the verdict, which made no recommendation on this point). Since $\Gamma_\infty \cong \mathbb{Z} \times \mathbb{Z}/2$ and $\langle T \rangle \cong \mathbb{Z}$ has cohomological dimension 1, Lyndon–Hochschild–Serre reduces to

$$0 \longrightarrow H^{q-1}(\mathbb{Z}/2; M)_T \longrightarrow H^q(\Gamma_\infty; M) \longrightarrow H^q(\mathbb{Z}/2; M)^T \longrightarrow 0.$$

When $2 \in R^\times$ the terms with $q \geq 1$ vanish and this returns $H^0 = M^T$, $H^1 = M_T$, $H^q = 0$ for $q \geq 2$: the circle model of (A1). At $\ell = 2$, however, $-I$ acts trivially and $H^q(\mathbb{Z}/2; M) \cong M$ for *every* $q \geq 0$, so $H^q(\Gamma_\infty; M) \neq 0$ for every $q \geq 0$ — $H^\bullet(\partial; M)$ is *unbounded above in degree*. The bulk is no better (the 12-torsion argument of [C, `cl:amalgam`] needs $\ell \geq 5$), and hence

neither is the cone: at $\ell = 2$ the sequence [C, `eq:les`] is a *window* of a long exact sequence that does not terminate, and X_2, X_3, X_5, X_6 are the degree-≤ 2 members of infinite families. What the implementation is to compute there — a degree bound and a truncation policy, and what "exactness of [C, `eq:les`]" asserts at the top of the truncation — is *decided in the contract* ([C] v1.3.0, [C, `open:ell2`], policy [C, `oc1a`]: $q_{\max}$ a recorded run parameter, default 2, cohomology reported only in degrees $\leq q_{\max}$), and its construction is deferred with the free Γ-model above; nothing is decided here. Under this document's standing hypothesis $\ell \geq 5$ (§0) none of this is in force.

(A3) *The involution ι*: the 7-vertex model is symmetric under $z \mapsto -\bar{z}$ ($0 \leftrightarrow 3$, $1 \leftrightarrow 2$, $4 \leftrightarrow 5$, 6 fixed), $\iota T \iota^{-1} = T^{-1}$, $\iota S \iota^{-1} = S^{-1}$, $\iota(ST)\iota^{-1} = (TS)^{-1}$, and $\rho_n(\iota) = \mathrm{diag}((-1)^{n-j})_j$ (note $\iota \notin \Delta$; this is a separate automorphism, not a Hecke operator). Implementable as an `Equivariant CochainComplexMorphismOnK` with a simplicial *automorphism* and orientation signs; $[\iota, T_q] = 0$ is a runtime assertion. *The boundary datum and ι (v1.4.0).* The bulk τ is ι-symmetric by construction ([C, `alg:tau`] step (1b) breaks ties through the ι_K-fixed vertex v_6; [C, `obl:iota`]); the boundary datum τ_∂ is *not*, and at $p = 2$ cannot be made so by any T-equivariant vertex rule, since no vertex of the ∞-line is ι-fixed ([C, `rmk:taubdrynoiota`], [C] v1.4.1; the conjugate datum is admissible and homotopic, so Θ^∂_g on cohomology is unaffected, and nothing records or asserts the symmetry). If the sign split of X_2, X_5 or of the cone ever needs $\iota\tau_\partial\iota = \tau_\partial$ at chain level, that is a *contract addition* (owner close-out decision (a) of slot 1), not an assumption of this document. $H_{\mathrm{tot}} = H^+ \oplus H^-$ halves every matrix and, as §8 shows, is what makes the Ramanujan block cyclic on each sign. *The sign (owner decision OQ3, resolved 2026-09-16; [C] v1.2.0 [C, `def:iota`](iv), [C, `rmk:iotasign`]).* The involution of [C] is $\iota^* := -\iota^*_{\mathrm{geo}}$, the *negative* of the geometric pullback $(\iota^*_{\mathrm{geo}} f)(\sigma) = \rho_n(\iota) f(\iota_K \sigma)$, so that H^+ is *by definition* the eigenspace containing the Eisenstein classes: the v1.1.1 run of `HeckeIotaSignValidation` had found them in the (-1)-eigenspace of the geometric involution on every cell of a 77-cell grid, and the owner negated the definition rather than the record. Consequences to keep in mind when reading $\pm$ here: constants are odd ($\iota^* = -1$ on $H^0(\Gamma; \mathbb{F}_\ell)$), $H^0(\partial)$ is odd and $H^1(\partial)$ even for $\ell > n$, ι^* is the geometric involution twisted by det and, for n even, not a pullback along any lift of $z \mapsto -\bar{z}$, and at $\ell = 2$ the two coincide. Every $H^\pm$ of this document is in terms of ι^* so defined (`docs/implementation/oq3_resolution.md`).

Recommended once: two independent models already run in parallel in `HeckeCrossCheckValidation` — the amalgam / `BarCocycleH1` route and the 7-vertex complex, the latter standing in for the $\Gamma(2)/S_3$-invariants model of [M §1], which does not exist in the tree (inventory) — and the identities of §5.1 become one validation.

Stage 1 — base operators

Compute T_q for $S_0 = \{2, 3, 5, 7, \ell\}$ (and ι) on all available X_i. Cost per T_q: the τ-table has $(q+1) \cdot N_\bullet$ entries and $\Theta^{(q)}$ is applied to d_q basis cocycles, i.e. $O\big((q+1)\, n^2\, d_q\big)$ multiplications through the tape for the Θ-application alone, *plus the fixed cost of the cohomology run* that produces the bases, which dominates the measured wall (T_2 at $n = 128$ in 0.12 s against 0.09 s for the cohomology alone; at $n = 1000$, 14.4 s against 14.8 s, `CLAUDE.md`; v1.7.0: the $n = 1000$ pair has the whole *below* its part, so those two walls differ by less than their run-to-run noise — it supports "dominates" and must not be read as a split). Run immediately: $[T_q, T_{q'}] = 0$, $[T_q, \iota] = 0$, equivariance of every map in [C, `eq:les`], the boundary characters of [C, `cl:chi`], [C, `cl:dimH1`]. Failures are convention or coset bugs, not mathematics.

Stage 2 — maximal ideals

Algorithm 5.1. Per X_i (and per sign once (A3) exists):

1. Run `HeckeModuleDecomposition::ComputeMultiOp` on $\{A(T_q)\}_{q \in S_0}$ per X_i (per sign): blocks V with per-operator $\pi_q^{e_q}$, partitions, lower bound f. Cost: Berkowitz $O(d^4)$ per operator plus factorisation; negligible against Stage 1.

2. *Generic refinement* (new, Claim 3.1): for each block, for the deterministic witness sequence $w_c = \sum_j c_j t_j$ (c from the fixed-seed search already in S11), compute the primary decomposition of w_c on V and split V accordingly; repeat until every block passes the locality certificate below, or until a fixed attempt budget (the setting `locality = budget=<N>`, default $N = 64$ candidates per block, counting the single-operator witnesses, the all-ones combination and the seeded combinations as *one* sequence with one attempt index) is exhausted, in which case the block is recorded with `"local": false`, its `residue_degree_lower_bound`, and `final = false` with reason `budget_exhausted:locality` and `resumable.complete = false` — the loop never spins. (v1.4.0. The reason is not `not_local`: a block *proved* non-local splits and never survives the loop, so a survivor is a block whose locality is *unknown under the budget*; slot-2 WP-1 §7 item 3.) Under `locality = strict` (owner decision OQ16; never the default, never run by the golden harness, byte-identical to budget mode wherever budget mode certifies) the loop continues after the N linear witnesses with the terminating route of Certificate 5.2 and emits `local = true` on every block, `local = false` only by interruption at a step boundary (reason `interrupted`, resumable). Strict mode is specified here, parsed since slot 2 and **implemented since slot 3** (v1.4.2; slot-3 WP-3, `HeckeModuleDecomposition::StrictLocalityRoute`): it no longer throws `not_implemented_in_this_build`, and re-running the whole golden grid under it reaches the terminating route on no block.
3. Read off the character $a(\mathfrak{m}) = (P_q(\bar{X}))_{q \in S_0} \in (\mathbb{F}_\ell[X]/(\pi_w))^{S_0}$ (Claim 3.3) and compare with the atlas (Stage 6): label *cuspidal* (lowest-weight representative and θ-twist), *Eisenstein-type* (all a with $\chi_a \equiv a(\mathfrak{m})$), or *both*. If the atlas certifies that S_0 separates $\Sigma(\ell) \cup \{\chi_a\}$ the decomposition is complete on the anemic side; otherwise add $S_{\mathrm{sep}}(\ell)$ to S_0 and repeat. *In the default run* (v1.8.0-pre.1; slot-7 owner ruling 3b (B)) $\Sigma(\ell)$ exists only where it is built in-process, $\ell \in \{5, 7, 11, 13\}$: there $S_{\mathrm{sep}}(\ell)$ is added to S_0 (a run from the start on $S_0 \cup S_{\mathrm{sep}}(\ell)$ is that repeat) and the label is read off. At every other ℓ this step does not run, no stored atlas is read, and the record says so — `s_sep` and every `label` are the placeholder {`"status": "atlas_unavailable"`} of §7 — never a silent omission and never a label from a different run.

Certificate 5.2 (Locality). [proved here] A block V with generators t_j is $\mathfrak{m}$-primary for $\mathbb{T}_S$ with $[\mathbb{T}_S/\mathfrak{m} : \mathbb{F}_\ell] = f$ iff there is $w \in A_S$ with $\pi_w :=$ squarefree part of $\min_V(w)$ irreducible of degree f, and, with $\hat{w} := w^{\ell^N}$ ($f \mid N$, $\ell^N \geq D$; Claim 3.3), for every j a polynomial $P_j \in \mathbb{F}_\ell[X]_{<f}$ with $n_j := t_j - P_j(\hat{w})$ nilpotent on V. Verification: `CertifyIrreducible`(π_w); $\hat{w}$ by N successive ℓ-th powers; P_j *deterministically*: take any $0 \neq v \in U := \ker \pi_j(t_j) \cap \ker \pi_w(w)$ (an A_S-submodule containing $\operatorname{soc} V$, hence non-zero when V is local). On U both t_j and $w = \hat{w}$ are semisimple and commute, so if V is local, uniqueness of the Jordan–Chevalley decomposition of $t_j|_U = P_j(\hat{w})|_U + n_j|_U$ forces $n_j|_U = 0$, i.e. $t_j v = P_j(\hat{w})v$: a consistent $D \times f$ linear system with a unique solution ($v, wv, \ldots, w^{f-1}v$ are independent because $\mathbb{F}_\ell[w]|_{\ker \pi_w(w)}$ is a field). Then verify $n_j^D = 0$ on V by squaring, $O(D^3 \log D)$ per j. Inconsistency or non-nilpotency proves that V is not local: refine (Algorithm 5.1(2)). Sufficiency: A_S/rad is generated by $\bar{w}$ hence is $\mathbb{F}_\ell[X]/(\pi_w)$, a field. Necessity: a generic element of a local A_S is such a w; more precisely the *linear* witnesses $\sum_j c_j t_j$, $c \in \mathbb{F}_\ell^k$, contain one whenever f has at most ℓ distinct prime factors (the residues $\bar{t}_j$ generate $\mathbb{F}_{\ell^r}$ as a field, so their $\mathbb{F}_\ell$-span lies in no maximal subfield, and an $\mathbb{F}_\ell$-space is not a union of $\leq \ell$ proper subspaces — automatic for $\ell \geq 5$, $f < 30030$), and they separate two non-Galois-conjugate characters of residue degrees f, f' whenever $\min(f, f') \leq \ell$ (the same covering argument over the Galois orbit). Outside these ranges the fixed-seed search may fail, which is why Algorithm 5.1(2) is bounded and reports `"local": false` rather than assuming.

Strict mode (v1.4.0; [X] §3.1, transcribed; owner decision OQ16). When the N linear witnesses fail, locality is decided and a witness found by linear algebra with no search, in three steps on A_S computed as in Algorithm 5.5(1). (1) *The radical.* On a commutative $\mathbb{F}_\ell$-algebra the Frobenius $x \mapsto x^\ell$ is $\mathbb{F}_\ell$-linear, so $\operatorname{rad} A_S = \ker(\mathrm{Frob}^N)$ with $\ell^N \geq D$, the kernel of an explicit $\dim A_S \times \dim A_S$ matrix (N successive Frobenius matrices, each column $b_i \mapsto b_i^\ell$ by square-and-multiply on $D \times D$ matrices). The trace form is not usable here: it vanishes identically on $\mathbb{F}_\ell[x]/(x^\ell)$, a local algebra of dimension ℓ. (2) *The Berlekamp subalgebra decides and splits.* $\bar{A} := A_S/\operatorname{rad} A_S \cong \prod_{i=1}^m \mathbb{F}_{\ell^{f_i}}$ with m the number of maximal ideals of A_S on V, and the Frobenius-fixed subalgebra $B := \ker(\mathrm{Frob}_{\bar{A}} - \mathrm{id}) = \mathbb{F}_\ell^m$ has $\dim_{\mathbb{F}_\ell} B = m$. So V is local iff $\dim B = 1$. If $\dim B \geq 2$, the first non-scalar RREF basis element $\bar{b}$

of B (a fixed choice) lifts to $w \in A_S$ whose semisimple part has ≥ 2 distinct eigenvalues in $\mathbb{F}_\ell$, and the primary decomposition of w on V splits V into ≥ 2 *non-zero* blocks — V is faithful over A_S, so no primitive idempotent of $\bar{A}$ lifts to an element acting as 0 — inserted at the parent's position in `CanonicalLess` order of the eigenvalues; recurse. *Termination*: every split strictly reduces $\dim V$. No witness is searched here. (3) *A residue witness over the whole residue field.* With $\dim B = 1$, $\bar{A} = \mathbb{F}_{\ell^f}$ is a field, and a field is never the union of its proper subfields ($\sum_{r|f \text{ prime}} \ell^{f/r} \leq \omega(f)\,\ell^{f/2} < \ell^f$ for $f \geq 2$, $\ell \geq 2$), so an element of A_S whose residue generates $\bar{A}$ exists although the $\mathbb{F}_\ell$-span of the $\bar{t}_j$ may lie in a union of proper subfields; it is drawn from $\bar{A}$'s own coordinate vectors (lifted to A_S by the same coordinates) by the fixed-seed sequence, each candidate tested by $\deg \min_{\bar{A}}(\bar{w}) = f$ — an $f \times f$ Krylov computation, failure probability $\leq \omega(f)\,\ell^{-f/2}$, expected trials ≈ 1 — with the lexicographic enumeration of the ℓ^f coordinate vectors as the deterministic, terminating fallback; then π_w, N, $\hat{w}$, the P_j and the nilpotency checks exactly as above, which succeed because V is local. Strict mode changes *how the witness is found*, never what is certified: the certificate is the same. The covering example is $(\ell, f) = (2, 30)$: $V = \mathbb{F}_{2^{30}}$ as a 30-dimensional $\mathbb{F}_2$-space, $t_1 = a\cdot$, $t_2 = b\cdot$ with $a \in \mathbb{F}_{2^{15}}$, $b = a + c \in \mathbb{F}_{2^{10}}$, $c \in \mathbb{F}_{2^6}$, so that $\mathbb{F}_2(a, b) = \mathbb{F}_{2^{30}}$ and A_S is a field ($f = 30 = 2 \cdot 3 \cdot 5$, three prime factors $> \ell$), while the three non-zero linear witnesses $a, b, a + b$ have residue degrees $15, 10, 6$: budget mode emits `local = false` with `residue_degree_lower_bound = 30` (the correct budget-mode verdict) and strict mode `local = true`, $f = 30$ ([X] §3.2; the strict-mode arms of `HeckeLocalityValidation`, slot 3 — *verified* (v1.4.2, slot-3 WP-3): built from a certified degree-30 irreducible, budget mode `local = false` with bound 30, strict mode `local = true`, $f = 30$, `phase_reached = residue_search`, in 0.10 s). Cost $O(\dim A_S \cdot D^3 \cdot N \log \ell) \leq O(D^5 \log D \log \ell)$ per block, entered only after N failed linear witnesses — on no computed cell so far. What strict mode records: `level_c.locality` = {`mode: strict`, `budget`, `phase_reached` ∈ {`linear`, `berlekamp`, `residue_search`}} and `local = true` on every block; the golden `decomp_*` hashes pin the budget mode, and strict mode must reproduce them bit for bit. *Per block and per run* (v1.5.0; slot-3 WP-3 finding 6): `phase_reached` is the phase at which *that* block's verdict was reached, and the run's high-water mark over all blocks — split parents included, whose phase belongs to no surviving block — is the separate key `phase_reached_high_water` of §7.

*What a **reported** witness must satisfy (v1.5.0; slot-3 WP-3 finding 3).* The sufficiency argument above reads "A_S/rad is generated by $\bar{w}$", which presumes $w \in A_S$. That is automatic for a *linear* witness $w = \sum_j c_j t_j$, which a record reports as the coefficient list (c_j) and a verifier re-forms inside A_S; it is **not** checkable from a witness reported as a *matrix*, which strict mode's step (3) produces (a polynomial in the t_j, not a linear combination, so no coefficient list can carry it). A verifier working from the reported data alone therefore checks the weaker hypothesis w **commutes with** $\{t_j\}_{j \in S}$, and that hypothesis still yields locality: with π_w irreducible of degree f, $K := \mathbb{F}_\ell[\hat{w}] \cong \mathbb{F}_{\ell^f}$ is a field, the $n_j = t_j - P_j(\hat{w})$ are nilpotent, and they commute with $\hat{w}$ and with each other (each t_j does, and w does by hypothesis), so $A_S \subseteq K[n_j : j]$, a local ring whose maximal ideal is its nil radical; hence A_S meets that ideal in nilpotents only and $A_S/\operatorname{rad} A_S$ embeds in K as a finite domain, i.e. is a field, and A_S is local. **Conformance**: a record reporting a matrix witness must carry it as such and the commutation is a certificate clause of its own; a record reporting a linear witness keeps the coefficient list. Implemented as `PrimaryBlock::witness_matrix` with `certify()` code 30 (slot-3 WP-3), empty in budget mode, so no published value moves.

The sub-block order after a split is per mode, not intrinsic (v1.5.0; slot-3 WP-3 finding 2; owner confirmation of 2026-09-21, item 1(c)). Both modes insert the sub-blocks at the parent's position in `CanonicalLess` order *of the splitting element's factors*, and both are deterministic; but the two modes split by *different* elements — a linear witness and the Berlekamp lift of strict mode's step (2) above — so the correspondence factor $\leftrightarrow$ maximal ideal, and with it the resulting *order* of the list, may differ (measured: 9 of 9 cells agree as an unordered partition with bases entrywise equal, 6 of 9 also in order). **What is mode-independent is the partition, not the list.** A statement that two modes produce "the same blocks" is therefore a statement about the partition; a `decomp_*_U` hash, which reads the list, is a function of the mode and is golden only at the default. An intrinsic order — by each sub-block's own π_w, say — was considered and **declined**: it would change the split path of *both* modes, and the default mode's off-grid order with it, for no consumer that exists.

Claim 5.3 (The residue degree of a local block, and the sharpness of the budget-mode bound)**.**

[proved here]Let V be a block on which $A_S = \mathbb{F}_\ell[t_p : p \in S]$ is local, and let π_p be the squarefree part of $\min_V(t_p)$. *Then*

$$\dim_{\mathbb{F}_\ell} \bar{A} \;=\; \operatorname{lcm}_{p\in S} \deg \pi_p, \qquad \bar{A} := A_S/\operatorname{rad} A_S.$$

Consequently the residue degree $f = [\mathbb{T}_\mathfrak{m}/\mathfrak{m} : \mathbb{F}_\ell]$ of a local *block equals* $\operatorname{lcm}_p \deg \pi_p$, *which is exactly the quantity budget mode reports as* `residue_degree_lower_bound` *when its witness family is exhausted:* ***on a block that is in fact local, that lower bound is sharp.***

Proof. A_S local means $\bar{A}$ is a field, and it is generated over $\mathbb{F}_\ell$ by the residues $\bar{t}_p$, so $\bar{A} = \mathbb{F}_\ell(\bar{t}_p : p \in S)$ is the compositum inside $\bar{A}$ of the subfields $\mathbb{F}_\ell(\bar{t}_p)$. The minimal polynomial of $\bar{t}_p$ over $\mathbb{F}_\ell$ is π_p: the squarefree part generates the radical of $(\min_V(t_p))$, so $\pi_p(t_p)$ is nilpotent and $\pi_p(\bar{t}_p) = 0$, while π_p is irreducible in $\mathbb{F}_\ell[X]$ — if $\min_V(t_p) = \prod_i f_i^{e_i}$ had two distinct f_i, the Chinese remainder theorem would make $\mathbb{F}_\ell[t_p] \cong \prod_i \mathbb{F}_\ell[X]/(f_i^{e_i})$ and so exhibit a non-trivial idempotent *in* A_S, which a local ring does not have (v1.5.1: through v1.5.0 this step read "a proper factorisation would give a non-trivial idempotent decomposition of the field $\mathbb{F}_\ell[\bar{t}_p]$", which it does not — inside a field a proper factorisation of π_p says only that $\min(\bar{t}_p)$ is a proper divisor of π_p, i.e. the very alternative to be excluded; the idempotent lives one level down, in A_S) — so no proper divisor annihilates $\bar{t}_p$. Hence $\mathbb{F}_\ell(\bar{t}_p) = \mathbb{F}_{\ell^{\deg \pi_p}}$. Inside a finite field the compositum of $\mathbb{F}_{\ell^a}$ and $\mathbb{F}_{\ell^b}$ is $\mathbb{F}_{\ell^{\operatorname{lcm}(a,b)}}$, and the statement follows by induction on $|S|$. For the consequence: $\bar{A} = \mathbb{T}_\mathfrak{m}/\mathfrak{m}$ when the block is $\mathfrak{m}$-primary for $\mathbb{T}_S$ (Claim 3.3), and the bound budget mode emits on an exhausted family is $\operatorname{lcm}_p \deg \pi_p$ by construction — an *equality* exactly when the block is local, and a *lower* bound on every residue degree of a block that survives the refinement loop, because such a block has every π_p **irreducible** (a candidate with two distinct irreducible factors splits the block, Algorithm 5.1(2)), whence $\deg \pi_p$ divides the degree d_i of every factor K_i of $\bar{A} = \prod_i K_i$ and so does $\operatorname{lcm}_p \deg \pi_p$. (v1.5.1, correcting v1.5.0's "a lower bound in general, because a non-local block's $\bar{A}$ is a product and the lcm bounds only its largest factor": without the irreducibility of the π_p that is false in both directions — for $\bar{A} = \mathbb{F}_\ell \times \mathbb{F}_\ell$ with $S = \{p\}$ and $t_p = (0, 1)$ the lcm is $\deg X(X-1) = 2$ while both residue degrees are 1, and for $\bar{A} = \mathbb{F}_{\ell^{12}} \times \mathbb{F}_{\ell^6}$ with t_p of local degrees $4, 2$ and t_q of local degree 3 in both it is $\operatorname{lcm}(6, 3) = 6 < 12$.) (Slot-3 WP-3 finding 4, where the implementation asserts it. It is also why the covering example of Certificate 5.2 has bound 30 and not merely ≥ 30.) □

Certificate 5.4 (Exhaustiveness of MaxSpec). [proved here] $\sum_\mathfrak{m} \dim H_\mathfrak{m} = d$ (all blocks certified local, pairwise distinct characters up to Galois, U invertible and every $U^{-1}A(T_q)U$ block-diagonal): this is `certify()` plus the locality certificate. Completeness against *uncomputed* operators is Stage 4.

Stage 3 — the local algebra and its tangent space

Algorithm 5.5. For each $\mathfrak{m}$ (matrices of size D):

1. A_S as an $\mathbb{F}_\ell$-subspace of $\operatorname{End}(V) \cong \mathbb{F}_\ell^{D^2}$: start from $\{I\}$, multiply the current basis by the t_j, row-reduce, until stable. By commutativity the products needed are monomials $\prod t_j^{a_j}$ of total degree $< \nu \leq D$; cost $O(k \cdot \dim A_S \cdot D^3)$ products and an RREF on $\dim A_S \times D^2$. Bound: $\dim A_S \leq D$ once V is known cyclic (Stage 4a), $\leq D^2/4 + 1$ in general (Schur).
2. The coefficient field $K = \mathbb{F}_\ell[\hat{w}]$, $\hat{w} = w^{\ell^N}$, and $n_j := t_j - P_j(\hat{w})$ for *every* j, the witness included ($n_w = w - \hat{w}$; Claim 3.3); then $A_S = K[n_j]_j$ and $\mathfrak{m}_S = (n_j)_j \subseteq A_S$ as an ideal (span of the generators times the basis of A_S; no separate $\pi_w(w)$ generator), $\mathfrak{m}_S^2$, $\mathfrak{m}_S^i$; ν; $\operatorname{soc} A_S = \operatorname{Ann}_{A_S}(\mathfrak{m}_S)$ by a linear system; Gorenstein iff $\dim_{\mathbb{F}_\ell} \operatorname{soc} A_S = f$.
3. Tangent-space generation test: the images $\bar{n}_j$ in $\mathfrak{m}_S/\mathfrak{m}_S^2$, which is a K-vector space through $\hat{w}$; record which subset of S spans it *over* K (the K-span of $\bar{n}_j$ is the $\mathbb{F}_\ell$-span of $\hat{w}^i\bar{n}_j$, $0 \leq i < f$), and $\dim_{\mathbb{F}_{\ell^r}} \mathfrak{m}_S/\mathfrak{m}_S^2 = \dim_{\mathbb{F}_\ell}(\mathfrak{m}_S/\mathfrak{m}_S^2)/f$. Over $\mathbb{F}_\ell$ one direction would count as f, and with $P_j(w)$ in place of $P_j(\hat{w})$ the witness's own direction is lost (Claim 3.3). Expectation (Claim 2.13): ≤ 2 anemic directions plus possibly the T_ℓ direction at cuspidal $\mathfrak{m}$; Eisenstein-type and the dominant $\mathfrak{m}$ for $n \gg \ell$ are where surprises would show.

As implemented (v1.4.0; slot-2 WP-4, `HeckeExt/HeckeLocalAlgebra`, `ComputeLocalAlgebra`, namespace `hecke_ext`): $\hat{w}$ and the P_j are *consumed* from the locality certificate on its matrix path — the $k = 1$ analytic path of [X] §3.1 never forms $\hat{w}$, so Stage 3 forces the matrix path — with

$\pi_w(\hat{w}) = 0$ re-asserted as an identity (Frobenius is a ring endomorphism, so its failure is a bug, not a finding) and N re-derived; $\mathfrak{m}_S = (n_j)_j$ with no separate $\pi_w(w)$ generator, since $n_w = w - \hat{w} = \sum_j c_j n_j$ is automatic, and $\mathfrak{m}_S^{i+1} = \sum_j \mathfrak{m}_S^i n_j$ (only the non-zero powers stored); the tangent test over K is the $\mathbb{F}_\ell$-span of $\{\hat{w}^i \bar{n}_j : 0 \le i < f\}$ modulo $\mathfrak{m}_S^2$ in one row reduction, `minimal_active` greedy in *ascending* p (hence deterministic), embdim $= \dim_{\mathbb{F}_\ell}(\mathfrak{m}_S/\mathfrak{m}_S^2)/f$; the output carries both the $\dim A_S \times D^2$ coordinate matrix and the basis as $D \times D$ matrices, so membership is one solve and a certificate compares entrywise (the RREF of a span is unique). A non-local input is *reported*, never thrown — E0b's loop is what acts on it. Stage 2's cross-block evaluation test is `SameMaximalIdeal`: blocks B, B' carry the same $\mathfrak{m}$ iff, with B's own c, π_w and $\{P_j\}$ and N' the least multiple of f with $\ell^{N'} \ge \dim B'$ (taken for the *tested* block), $\pi_w(w'|_{B'})$ and every $t_j|_{B'} - P_j(\hat{w}')$ are nilpotent; it is what the exhaustiveness certificate of E2 calls for pairwise distinctness.

Stage 4 — certifying $A_S = \mathbb{T}_\mathfrak{m}$

Certificate 5.6 ((a) Cyclicity). [proved here] If $\dim_{\mathbb{F}_\ell} V/\mathfrak{m}_S V = f$ (Nakayama: V is cyclic over A_S; deterministic — no random vectors needed; *some* $v \notin \mathfrak{m}_S V$ generates, and every such v does) then $\mathrm{End}_{A_S}(V) = A_S|_V$, and since every T_q commutes with A_S and preserves V (it commutes with the polynomials in A_S that cut V out), $\mathbb{T}_\mathfrak{m} = A_S|_V$: *no further operator can change the module structure at* $\mathfrak{m}$. Cost $O(kD^3)$. Cyclicity fails exactly at multiplicity > 1 (e.g. every cuspidal $\mathfrak{m}$ on X_4 without the ι-split, [C, `cl:dimH1`]: holomorphic and antiholomorphic copies), which is why (A3) is on the list.

The generator is canonical (v1.4.1; stated here rather than only recorded, which is what v1.4.0's implementation note asked for). Which generator a certificate *reports* is not free: a record is verified from the data it carries, so the choice must be a function of the data. **The generator of this certificate is the least standard basis vector of V that lies outside $\mathfrak{m}_S V$** — least in the fixed index order of the stored basis of V, which exists because $\mathfrak{m}_S V \subsetneq V$ by Nakayama. It is canonical given $(V, \mathfrak{m}_S)$ and the stored basis, it needs no search beyond one pass over D vectors, and a verifier recomputes it and compares. Any other $v \notin \mathfrak{m}_S V$ generates equally well and the certificate is *true* for it; but a record emitting one is non-conforming, and `verify()` rejects it (clause 13 of its fifteen). This is strictly stronger than "any $v \notin \mathfrak{m}_S V$" and is deliberately so: it is what makes two runs on the same block emit the same certificate. *Implemented* (v1.4.0; slot-2 WP-4, `HeckeExt/HeckeLocalAlgebra`).

Proof. $V = A_S v$; an A_S-endomorphism ϕ is determined by $\phi(v) = av$, $a \in A_S$, and then $\phi = a\cdot$. Commutation and invariance as stated. □

Certificate 5.7 ((b) Commutant bound). [proved here] $C := \mathrm{End}_{A_S}(V) = \{X : Xt_j = t_j X \ \forall j\}$ is the kernel of a linear map $\mathbb{F}_\ell^{D^2} \to \mathbb{F}_\ell^{kD^2}$. Then $A_S \subseteq \mathbb{T}_\mathfrak{m} \subseteq C$ and $\mathbb{T}_\mathfrak{m}$ is commutative. If $\dim C = \dim A_S$, done. Otherwise every further T_q lies in the commutative subalgebra of C generated by A_S and the $T_{q'}$ already known; testing one new T_q is a membership test in the current A_S (Stage 4c). Cost: naive $O(kD^6)$; with w in rational canonical form $V \cong \bigoplus_i \mathbb{F}_\ell[X]/(g_i)$, $\mathrm{End}_{\mathbb{F}_\ell[w]}(V)$ has the explicit basis of dimension $\sum_{i,i'} \deg \gcd(g_i, g_{i'})$ and only the centralisers of the remaining t_j inside it need solving. *The sharp count (v1.4.0).* The element put in rational canonical form is not the witness w (which may be semisimple, hence scalar on the block, and then gives the naive D^2 unknowns) but a *cyclic-most* $a \in A_S$: the first of the RREF basis elements $b_1, \ldots, b_{\dim A_S}$ and their prefix sums whose minimal polynomial on V has maximal degree — a choice that affects the cost, never the verdict. A_S being local, a is π_a-primary with invariant factors $g_i = \pi_a^{s_i}$, $i = 1, \ldots, m$, and the basis has exactly

$$u = \sum_{i,i'} \deg \gcd(g_i, g_{i'}) = \deg \pi_a \sum_{i,i'} \min(s_i, s_{i'}) \ \le \ mD$$

elements, so the k systems $Xt_j = t_j X$ have $u \le mD$ unknowns instead of D^2, cost $O(kuD^3)$ to form and $O(kD^2u^2)$ to reduce; u can reach D^2 only when $m \approx D$ (through v1.3.3 this read "$O(m^2D)$ unknowns" and carried a [to be verified computationally] on the cost, discharged by slot-2 WP-5, which implements the count and pins $\dim C$ entrywise against the naive commutant). The system is budgeted (`kCommutantBudget`, a named constant; formation gated up front, rows charged one by

one) and resumable from the row reached (`CommutantState`, R3 instance 3); an unaffordable shape ends `over_budget` and never falls back to the naive route.

The socle bound (v1.4.0; underspecified through v1.3.3 — this certificate carried no statement of when *(b) can succeed; [X] §2.2 Stage 4(b), slot-2 WP-5).* Let $t := \dim_{\mathbb{F}} V/\mathfrak{m}_S V$ (the minimal number of generators, (D5)) and $s := \dim_{\mathbb{F}} \operatorname{soc} V$, $\operatorname{soc} V = \{v : \mathfrak{m}_S v = 0\} = \bigcap_j \ker n_j$, both in units of f. Then

$$\dim_{\mathbb{F}_\ell} C \;\geq\; \dim_{\mathbb{F}_\ell} A_S + f\,t\,s - \dim_{\mathbb{F}_\ell} \operatorname{soc} A_S,$$

so $\dim C = \dim A_S$ **requires** $f\,t\,s \leq \dim_{\mathbb{F}_\ell} \operatorname{soc} A_S$, and $f\,t\,s > \dim_{\mathbb{F}_\ell} \operatorname{soc} A_S$ is a *proof* that (b) fails (outcome `excluded_by_socle_bound`, the four numbers recorded). *Proof.* Every $\mathbb{F}_\ell$-linear map $V \twoheadrightarrow V/\mathfrak{m}_S V \to \operatorname{soc} V \hookrightarrow V$ whose middle map is K-linear is A_S-linear, both ends being $A_S/\mathfrak{m}_S$-modules, so $C \supseteq \operatorname{Hom}_K(V/\mathfrak{m}_S V, \operatorname{soc} V)$, of $\mathbb{F}_\ell$-dimension fts; and $A_S \cap \operatorname{Hom}_K(V/\mathfrak{m}_S V, \operatorname{soc} V) \subseteq \{a \in A_S : a\,\mathfrak{m}_S V = 0\} = \operatorname{soc} A_S$ because V is faithful. For cyclic V ($t = 1$, $V \cong A_S$) the bound is an equality and (a) already certifies; for $t \geq 2$ it forces $\dim_{\mathbb{F}} \operatorname{soc} A_S \geq 2$, i.e. A_S *not* Gorenstein. Hence (b) is never attempted on a field, on any $\mathbb{F}[\varepsilon]/(\varepsilon^2)$ or on any Gorenstein A_S — in particular not at $(10, 691)$, where $t = s = 2$, $f = 1$, $\dim \operatorname{soc} A_S = 1$ give $\dim C \geq 2 + 4 - 1 = 5$, the exact value, and not on any Kummer block — and it can beat (a) only on co-cyclic-like modules over a non-Gorenstein A_S, e.g. $V = A_S^\vee$ for $A_S = \mathbb{F}[x, y]/(x, y)^2$ ($t = 2$, $s = 1$, $\dim \operatorname{soc} A_S = 2$, $\operatorname{End}_{A_S}(A_S^\vee) = A_S$), which `HeckeExtAlgebraValidation` pins. Where (b) is not attempted or is `over_budget`, the ideal falls through to (c); a computed $\dim C > \dim A_S$ is the same proof that (b) fails. In the default configuration an ideal that reaches closure by no route is emitted `final = false` with reason `closure_unreachable` (the socle bound or a computed $\dim C > \dim A_S$, with (a) failing and (c) off its proof range) or `budget_exhausted:commutant`, never withheld (owner decision OQ8). `verify()` re-derives $\dim C$ from the stored (g_i) and the change of basis R, a quantity that is basis-independent: the rational canonical basis is *not* pinned, any valid R verifies, and a tamper must break the companion relation (an image column of R), not perturb R (slot-2 WP-5 §4).

Certificate 5.8 ((c) Sturm sweep — fallback). [proved here] *given a recorded $B'(k, \ell)$, on the range of Claim 2.10* First compute $B'(k, \ell)$ (Definition 2.8: Hecke matrices on the Miller basis of $M_k(\mathbb{Z})$ reduced mod ℓ, then the algebra closure — independent of the cohomology pipeline and trivial in cost) and record it with its certificate (the cap, the closure dimension reached, $\dim M_k$). If it does not exist below the cap (`none_at_cap`; equivalently $\ell \mid [\mathbb{T}^{\mathrm{sat}} : \mathbb{T}_{\mathbb{Z}}(M_k)]$ once the cap is large enough), (c) is *unavailable*: in the default configuration `sturm_unavailable = nonfinal` the ideal is `final = false` with reason `sturm_unavailable`; under `sturm_unavailable = heuristic` the sweep runs to the cap anyway and the ideal is `final = "heuristic"`, with the primes actually swept in `sturm_primes` and the mode in `sturm.mode` — never in the default configuration (v1.4.0; slot-1 owner confirmation 6, [X] §1.3.1 R3.2). Otherwise, for all primes $q \leq B'(k, \ell)$ and $q \in S_{\mathrm{sep}}(\ell) \cup \{\ell\}$ (Claim 2.5), compute T_q (Stage 1 cost each; embarrassingly parallel over q and over τ-table entries — the natural GPU stage), restrict to V (a projection already at hand), and test $T_q|_V \in A_S$ by a linear solve against the basis of A_S ($O(\dim A_S \cdot D^2)$). This *proves* $A_S = \mathbb{T}_{\mathfrak{m}}$ on $X_{4,\mathfrak{m}}$ for every non-Dickson-type $\mathfrak{m}$ (Claim 2.10); on Dickson-type $\mathfrak{m}$ and on X_2, X_3, X_5, X_6 the sweep proves nothing, whatever its margin, and the record says so (v1.4.0): `"certificate": "sturm(B')"` versus `"cyclic"` / `"commutant"`, and `final = false` with reason `sturm_heuristic` under the default `final_mode = closure`, or `final = "heuristic"` with `sturm_primes` under the user-selected `final_mode = sturm_heuristic`. A U_ℓ omitted by the coset budget (`u_ell.reason = u_ell_over_budget`) never makes a proof-range record non-final: the proof list is the primes $\leq B'$. With $B(k) = \lfloor k/12 \rfloor + 1$ in place of B' the sweep would certify $A_S = \mathbb{T}_{\mathfrak{m}}$ *falsely* at $(n, \ell) = (22, 5)$ and $(26, 7)$, on the pipeline's own H^1 (§8). Number of operators $\approx B'/\ln B'$.

Certificate 5.9 ((d0) Dickson — the X_1 blocks). [proved here] *(v1.13.0; Proposition D of the slot-12 finality proposal, `docs/implementation/slot12/WP-12_finality_proposal.md` v2 §3.1, owner-approved 2026-09-26)* Let $\ell \geq 5$, $n \geq 1$ even, $V \subseteq X_1 = H^0(\Gamma; M_n)$ a certified local $\mathbb{T}_S$-block (Stage 2) **with** $\ell \in S$, and let $\Lambda(n, \ell)$ be the set of classes a mod $(\ell - 1)$ of the Dickson monomials $L^a D^b$ of degree $n = a(\ell + 1) + b(\ell^2 - \ell)$. If every $t_q = T_q|_V$, $q \in S \setminus \{\ell\}$, is a scalar λ_q and *exactly one* $a \in \Lambda(n, \ell)$ satisfies $(1 + q)q^a = \lambda_q$ for all $q \in S \setminus \{\ell\}$, then $\mathbb{T}_\infty|_V = \mathbb{F}_\ell[U_\ell|_V] = A_S$, i.e. $A_S = \mathbb{T}_{\mathfrak{m}}$ on V in the sense of Claim 2.2(b). In particular this holds whenever S contains a prime q that

is a primitive root mod ℓ with $q \not\equiv -1$ (at $\ell = 5, 11, 13$ the prime 2, at $\ell = 7$ the prime 3, all in S_0). The record (`hecke_ext/v4`) writes `certificate_routes.d0` $\in$ {`passed`, `failed:<class>`, `not_applicable:u_ell`, `not_run`}: `failed:<a>` names the second class of $\Lambda(n, \ell)$ the S-eigenvalues do not separate from the first; `not_applicable:u_ell` is written when U_ℓ was not computed ($\ell \notin S$, e.g. `u_ell_over_budget` or `u_ell = off`); `not_run` when (a)–(c) already passed or the block is not an X_1 block of the unsplit record. A passed (d0) names the certificate `"dickson"`; a failed one leaves the verdict to the reasons (a)–(c) produced. `verify()` re-derives (d0) from the record alone: the stored t_p over `operator_primes`, n and ℓ.

Proof. By [C, `prop:heckeH0`](1) (no hypothesis on the datum) the $L^a D^b$ span $X_1 = M_n^\Gamma$ and are simultaneous eigenvectors of every T_p, $p \neq \ell$, with eigenvalue $(1+p)p^a$, which depends on a mod $(\ell-1)$ only; so $X_1 = \bigoplus_{c \in \Lambda(n,\ell)} X_1^{(c)}$ (the class spans), every T_p, $p \neq \ell$, acts on $X_1^{(c)}$ by the scalar $(1+p)p^c$, and the T_S-primary decomposition of X_1 is the grouping of the $X_1^{(c)}$ by their S-eigenvalue systems. A block V whose system (λ_q) is attained by exactly one c is therefore contained in $X_1^{(c)}$, so every $T_p|_V$, $p \neq \ell$, is the scalar $(1+p)p^c$. $\mathbb{T}_\infty$ is generated by the T_p over all primes and $T_\ell = U_\ell$ ([C, `def:operators`]; the diamonds $T(p,p)$ act by the scalar p^n), so $\mathbb{T}_\infty|_V = \mathbb{F}_\ell[U_\ell|_V]$, and $U_\ell|_V \in A_S$ because $\ell \in S$. If $q \in S$ is a primitive root with $1 + q \neq 0$ in $\mathbb{F}_\ell$, $(1+q)q^a = (1+q)q^{a'}$ forces $a \equiv a' \pmod{\ell - 1}$, so the uniqueness hypothesis holds. *No boundary map, no r^0 and no transcription enter.* □

Certificate 5.10 ((d1) Boundary — the X_4 blocks). [proved here] *(v1.13.0; Proposition F′ of the slot-12 finality proposal v2 §3.4; unconditional since [C] v1.12.0 proves Proposition T as [C, `prop:bdrytranscription`] and Lemma B′ as [C, `lem:bdrytorus`]; owner-approved 2026-09-26)* Let $\ell \geq 5$, $n \geq 1$ even, $V = X_{4,\mathfrak{m}}$ a certified local $\mathbb{T}_S$-block of $X_4 = H^1(\Gamma; M_n)$ **with** $\ell \in S$, A_S its local algebra, and $R(\ell)$ the least prime $p \neq \ell$ of every class of $(\mathbb{Z}/\ell)^\times$ (key `boundary_reps`, in class order $1, \ldots, \ell - 1$). If (H1) $r^1|_V$ is injective ($\operatorname{rank}(r^1 V) = \dim V$) and (H2) $t_p = T_p|_V \in A_S$ for every $p \in R(\ell)$, then $\mathbb{T}_\infty|_V = A_S$. The record writes `certificate_routes.d1` $\in$ {`passed`, `not_injective`, `failed:<p>`, `not_run:budget` (v1.13.1), `not_applicable:u_ell`, `not_run`} — `failed:<p>` the first $p \in R(\ell)$ with $t_p \notin A_S$ — and `boundary_reps` $= R(\ell)$ exactly when the route reached its representatives (`passed` or `failed:<p>`) (the representatives are *tested*, never generators: `generators_certified` stays the Stage-4 $S_\mathfrak{m}$). A passed (d1) names the certificate `"boundary"`. *No shortcut*: all $\ell - 1$ classes are tested, also at $\ell = 5$. The driver computes r^1 from the boundary pair (`SL2ZBoundaryPair`, the restriction alone unless `with_cone` builds the cone on it), and the representatives outside S by the anemic-pool operators, streamed beyond $q \leq 31$ (from $\ell = 11$ on). Its certificate object re-derives (H1) from the stored block basis and r^1, and (H2) by the independent algebra-closure route of `HeckeAlgebraInvariants`, and the driver refuses a record whose (d1) does not re-derive. *The stream's cost cap (v1.13.1; slot-12 FIN audit G1; [X] §1.3.1 R3).* After (H1) and before any representative is computed, the route checks $\sum_{p \in R(\ell),\, p > 31} \mathrm{PredictUEllRssMb}(n, p) \leq 4 \cdot B_{u_\ell}$ — the frozen u_ℓ RSS model (`u_ell_rss/v2`) applied to each streamed T_p ($p + 1$ cosets) against four times the `u_ell` budget in force; over the cap it writes `not_run:budget` and computes nothing. No settings key: the cap is fixed and tied to `u_ell`, so the route is resumed by re-running with a larger `u_ell` budget. Every census cell ($\ell \leq 13$, $n \leq 200$) is within it at the default 2048 MB (worst $(200, 13)$: $\approx$ 2.5 GB of 8 GB); $(10, 691)$ ($\approx$ 391 GB predicted) and every $\ell = 10007$ cell are not. *The model is kept (v1.15.1-pre.2; owner ruling Q-D, 2026-09-29).* `u_ell_rss/v2-2026-09-17` is the pre-lever fit; since slot-14 Q8 the U_ℓ levers are on by default and the measured RSS is 2.7–25× below it, so `u_ell_over_budget` verdicts are conservative (slot-14 WP-C10). The owner ruled the frozen model stays: a re-fit would be a value change (verdicts, records, hashes) and must bump the model version. *What `verify()` re-derives from the record* (v1.14.0-pre.1; audit G2 repaired — it supersedes v1.13.1's "structure only", which was a recorded deviation from the slot-12 proposal v2 §3.4 item 3): (d0) in full (code 31), the structure of (d1) (code 30: the closed list, the group, `boundary_reps` $= R(\ell)$, the cost cap from (n, ℓ,`u_ell.budget_mb`)) *and*, on an ideal with `d1 = passed`, (d1) itself (code 32) from the record's own data. There the ideal carries `boundary_data` = {`rank_r1`, `r1`, `t`} (§7 item (j)): `r1` $= r^1|_V = r^1 V$, an $h^1(\partial) \times D$ matrix over $\mathbb{F}_\ell$ ($D = \dim V$; rows in the stored basis of $H^1(\partial)$, columns in the block basis of V), `rank_r1` its rank, and `t` the $\ell - 1$ matrices $t_p = T_p|_V$ ($D \times D$),

$p \in R(\ell)$ in class order, in the *same* block basis as the ideal's stored `t_p` ($p \in S$). Code 32 refuses: `boundary_data` present iff `d1 = passed` violated; a shape that is not $h \times D$ / $D \times D$ or keys that are not `boundary_reps`; an entry outside $[0, \ell)$; rank(`r1`) $\neq$ `rank_r1` or `rank_r1` $\neq D$ (H1); a t_p with $p \in R(\ell) \cap S$ differing from the stored `t_p`; a t_p not commuting with the stored generators; and a t_p outside A_S (H2), decided as in the certificate object: $\dim \mathbb{F}_\ell[t_q\,(q \in S),\, t_p] = \dim \mathbb{F}_\ell[t_q\,(q \in S)]$ by the independent algebra-closure route of `HeckeAlgebraInvariants` over the record's stored generators. The t_p are *tested*, never generators; nothing in the record is recomputed from the pipeline. *Size*: only on passed ideals — the census's 382 passed rows have $D \leq 26$ and $\ell - 1 \leq 12$. What the record still does not carry is V and the ambient $A^{(1)}(T_p)$: that t_p is *the* restriction of T_p to V and $r^1|_V$ *the* restriction of r^1 is the in-process certificate's clause (code 4 / code 2 there) and the driver's refusal, exactly as the stored `t_p` of (d0) are taken as the driver's restrictions.

Proof. V is $\mathbb{T}_\infty$-stable ([C, `obl:algebra`], [C, `thm:relations`]). The restriction intertwines: $r^1 A(T_g) = A^\partial(T_g) r^1$ for every g, by [C, `thm:conewelldef`](a) for a Hecke datum of the pair, which exists by [C, `lem:tauexists`](2) and [C, `lem:etaexists`] ($6 \in \mathbb{F}_\ell^\times$ as $\ell \geq 5$). Let $p \neq \ell$ be prime and $p' \in R(\ell)$ with $p' \equiv p \pmod{\ell}$. By [C, `lem:bdrytorus`](2) in degree 1 — which rests on [C, `prop:bdrytranscription`], the Shapiro transcription of the pipeline's T_p^∂ on $H^1(C_\partial^\bullet)$ to the group form [C, `eq:bdrygroupform`] — the boundary operators satisfy $T_p^\partial = T_{p'}^\partial$. Hence for $v \in V$, $r^1(T_p - T_{p'})v = (T_p^\partial - T_{p'}^\partial) r^1 v = 0$, and (H1) gives $T_p|_V = T_{p'}|_V \in A_S$ by (H2). $T_\ell = U_\ell \in A_S$ as $\ell \in S$, and the diamonds are scalars. So every generator of $\mathbb{T}_\infty$ restricts into A_S, i.e. $\mathbb{T}_\infty|_V = A_S$. *Injectivity, not Eisenstein-ness, is the hypothesis*: on the `mixed` block $(34, 7)$ ideal 2 ($\dim V = 4$, $\operatorname{rank} r^1 = 2$) every representative lies in A_S and the boundary law $t_q = t_{q'}$ fails; (d1) refuses it (`not_injective`), which `HeckeBoundaryFinalityValidation` arm (X) pins. □

Scope and order (v1.13.0). (d0) and (d1) certify *exactly* the predicate of (a)–(c) — Claim 2.2(b) at $\mathfrak{m}$, $\mathbb{T}_\infty|_V = A_S$ — not a weaker "the ideal does not split", and they run *only* on the blocks where (a)–(c) did not pass. They reach precisely the range (c) cannot (Claim 2.10's Dickson-type clause, and X_1, which Certificate 5.8's heuristic list does not name): on census editions 1 and 2 every `sturm_heuristic` row is discharged — editions 1 / 2 hold 270 / 376 such rows (184 / 184 at $\ell = 5$, of them 36 on X_1; 68 / 150 at 7; 13 / 37 at 11; 5 / 5 at 13), all certified by the validation's independent computation (slot-12 WP-14, arm (H)) and by the driver (S12-FIN, arm (R) by default, arm (H) on all 376). A block of X_4 on which r^1 is not injective *and* (c) fails would stay non-final. None occurs in census editions 1–3 (edition 3: 80 293 rows, all final; its 3302 non-injective H^1 rows are all off the Dickson range, $S_V = 0$, and final by (c)). None occurs in the mixed-finality scans either: $\ell = 5$, $n \leq 400$; $\ell = 7$, $n \leq 500$; $\ell \in \{11, 13\}$, $n \leq 340$ on the Dickson cells; $\ell \in \{17, \ldots, 43\}$ on the Dickson cells with $n \leq 200$ (v1.18.0; slot-18 WP-19c's 64-cell probe). There, no block has $\ker r^1|_V \neq 0 \neq S_V$. On such a block, slot-13 WP-11's Proposition M (refereed, [proved here] from [C, `cl:bockstein`], [C, `thm:conewelldef`], [C, `lem:bdrytorus`] and Claim 2.10's proof) would certify $\mathbb{T}_\infty|_V = A_S$. It does so under the hypotheses (H2)–(H5), by gluing the Sturm law modulo the torsion image S_V to the boundary law modulo $K = \ker r^1|_V$. It is *specified only*: certificate (d2) lands with the next record event, and only once a row needs it.

Practical order: (a) on every $\mathfrak{m}$; (b) on the survivors the socle bound admits; (c) only where (b) leaves room and only on those $\mathfrak{m}$; (d0) / (d1) (v1.13.0) only on the X_1 / X_4 blocks (a)–(c) leave non-final. In the regime $n \gg \ell$ the dominant $\mathfrak{m}$ will need (c). Record per $\mathfrak{m}$ the certified generating set $S_\mathfrak{m}$, the certificate type, and $\dim \mathfrak{m}/\mathfrak{m}^2$. *The answer to "which T_q do we need for (n, ℓ)" is $\bigcup_\mathfrak{m} S_\mathfrak{m}$ with its certificates.*

Stage 5 — module structure at $\mathfrak{m}$

Algorithm 5.11. With $\mathbb{T}_\mathfrak{m} = A_S$ certified, for each X_i and $V = X_{i,\mathfrak{m}}$:

1. Loewy series: $\mathfrak{m}^{j+1} V = \sum_a n_a\, \mathfrak{m}^j V$ over a basis $\{n_a\}$ of $\mathfrak{m}_S$ (spans and RREF, $O(\dim \mathfrak{m}_S \cdot D^3)$); socle series: $\operatorname{soc} V = \bigcap_a \ker n_a$ (one stacked kernel), $\operatorname{soc}^{j+1} V$ the preimage of $\operatorname{soc}(V/\operatorname{soc}^j V)$. Layer dimensions divided by f.
2. Jordan type of each n_j on V: the S10 partition machinery, re-used.

3. Structure maps θ of (D4) for each consecutive pair of layers: the $\mathbb{F}_\ell$-linear map $\mathfrak{m}_S/\mathfrak{m}_S^2 \to \mathrm{Hom}_{\mathbb{F}_\ell}(\mathrm{top}, \mathrm{bottom})$ induced by $n \mapsto (\bar{b} \mapsto nb)$; its $\mathbb{F}_{\ell^r}$-rank is its $\mathbb{F}_\ell$-rank divided by f (Claim 5.12). Active directions and the minimal subset of $S_\mathfrak{m}$ spanning them.
4. Canonical submodules (D6): for W each of $H^1_!$, $\mathrm{soc}(X^+_{4,\mathfrak{m}})$ (the Eisenstein line, when defined), X_1, r^0X_1, $L \cdot H^1(M_{n-\ell-1})$, and the kernels and images of the maps of [C, `eq:les`], compute $\dim(W \cap \mathfrak{m}^j V)$ and the image of W in each layer. ($L\cdot$ needs the $(n-\ell-1, \ell)$ run, Hecke-equivariant up to the known twist [M §2].)
5. Cyclic/free (D5): cyclic iff $\dim V/\mathfrak{m}_S V = f$; free of rank one iff moreover $\dim V = \dim A_S$. For X_4 versus X_3, compare with the duality prediction for D_n (Prediction 4.6).

Claim 5.12. *[proved here] If $\phi : U \to W$ is $\mathbb{F}_{\ell^r}$-linear between $\mathbb{F}_{\ell^r}$-spaces, its rank as an $\mathbb{F}_\ell$-linear map is $r \cdot \mathrm{rank}_{\mathbb{F}_{\ell^r}} \phi$; all $\mathbb{F}_{\ell^r}$-dimensions of (D2)–(D5) are therefore computable over $\mathbb{F}_\ell$ as $\mathbb{F}_\ell$-dimensions divided by $r = f$, once V is known $\mathfrak{m}$-primary with residue field $\mathbb{F}_{\ell^r}$ (the $\mathbb{F}_{\ell^r}$-structure on the layers is the one through $A_S/\mathfrak{m}_S$).*

Proof. $\mathrm{im}\,\phi$ is an $\mathbb{F}_{\ell^r}$-subspace; $\dim_{\mathbb{F}_\ell} = r \dim_{\mathbb{F}_{\ell^r}}$. □

Stage 6 — the per-ℓ atlas

Once per ℓ: run Stages 1–5 for all even $n \le \ell - 1$ (where M_n is irreducible, no Dickson classes, $d = O(\ell)$), collect the anemic systems with their a_ℓ and Galois-orbit structure, close under θ-twist $a_q \mapsto q^j a_q$, $j \in \mathbb{Z}/(\ell-1)$; this is $\Sigma(\ell)$ (Claim 2.4; weight $\ell+1$ must be included: $\Delta \bmod 11$ is a cuspidal level-1 system of weight $12 = \ell+1$ that is neither Eisenstein-congruent nor a twist of a lower weight). Greedily choose $S_{\mathrm{sep}}(\ell)$ (typically 2–4 primes). For each cuspidal $\bar{\rho} \in \Sigma(\ell)$, from a weight with multiplicity ≥ 2 ($n = w - 2 + \ell^2 - 1$, or any weight in the $E_{\ell-1}$-tower) compute $\dim \mathfrak{m}/\mathfrak{m}^2$ and a generating pair of anemic operators (Claim 2.13); store it as the default $S_\mathfrak{m}$ (Stage 4 still certifies). Attach to each (n, ℓ) the predictions of §4.3 before computing; every record confirms or refutes each, and flags the unpredicted.

The atlas against the literature (v1.15.1-pre.1, a citation and a check, not a claim of this document). Citro–Ghitza [9] enumerate the level-one eigensystems mod p one by one over $\overline{\mathbb{F}}_p$: $(p-1)^2/4$ Eisenstein ones (their Cor. 13) and, for every $p < 2000$, the number of the others (the table of their appendix). $\Sigma(\ell)$ stores Galois orbits of anemic systems instead, so the comparison is: Eisenstein orbits (all $\mathbb{F}_\ell$-rational) against $(\ell-1)^2/4$, and the sum of the residue degrees r of the cuspidal orbits against the table. They agree wherever both exist: $\Sigma(5), \Sigma(7), \Sigma(11), \Sigma(13)$ hold $4, 9, 25, 36$ Eisenstein systems and $0, 0, 10, 12$ cuspidal ones (slot-6 WP-8; the table's rows $p = 11, 13$ read $10, 12$), and the stored $\Sigma(691)$ (`hecke_atlas/v1`, FNV `0x73c7210047a81b97`) holds $119\,025 = 690^2/4$ Eisenstein orbits and $883\,200$ cuspidal orbits whose degrees sum to $6\,762\,000$, the table's row $p = 691$ (read from the store on 2026-09-28). The same row's Eisenstein–cuspidal weights, 12 and 200, are the two irregular classes of 691.

Stage 7 — scaling in n for fixed ℓ

Run n in residue classes mod $(\ell-1)$; the $E_{\ell-1}$-tower and the L-maps connect n, $n+\ell-1$, $n+\ell+1$, so records are comparable $\mathfrak{m}$ by $\mathfrak{m}$ (regression test; detection of where an extension is born). Memory: dense $\mathbb{F}_\ell$ matrices of size $d \times d$, $d \approx n/3$ for all six groups and both signs together ($\dim H^1 \approx n/6$); for $n \sim 10^4$ that is $\sim 10^7$ entries per operator — but this estimate applies to level (b) only. Stage 1 works on cochains ($\dim C^1 = 11(n+1)$, peak RSS $\approx n^2$: 1.4 GB at $n = 1000$), so $n \sim 10^4$ is out of reach of the present model *on the reference machine* (inventory §1, Stage 7). *Reframed (v1.4.0; owner decision OQ12):* $n \le 1000$ is a hardware-bounded limit there, not a design limit — the target hardware includes HPC nodes with the memory to push it, the model is kept as the blueprint for higher-rank groups, and a Manin-sized H^1 model (the amalgam presentation of [C, `cl:amalgam`], seeded by `BarCocycleH1`) is a downgraded follow-up, not the direction (through v1.3.3 this read "the practical range today is $n \le 1000$, and beyond it a Manin-sized H^1 model is the way forward"). The first extension campaign is bounded at $n \le 200$ (slot-1 owner confirmation 9(b); E6). The τ-evaluation

of Stage 1 dominates (measured exponent ≈ 2.05 in n for T_2, `CLAUDE.md`); the decomposition layer on legacy `Matrix` (Berkowitz $O(d^4)$) is fine to $d \approx 200$.

Claim 5.13 (The Glover factors of Sym^n, and the convention that fixes them)**.** *[standard, reference] (1) / [proved here] (2), (3), (4) (v1.21.0-pre.5: (3) [standard, reference] → [proved here]). Let ℓ be a prime, $G = \mathrm{GL}_2(\mathbb{F}_\ell)$, $B \subset G$ the upper-triangular Borel and χ_1, χ_2 its two diagonal characters.* ***The convention*** *(v1.5.0; slot-4 WP-7 finding 3, and the one thing item 6 of §5.1 left free):* Sym^n means the standard symmetric power $M_n = \mathrm{Sym}^n(\mathbb{F}_\ell^2)$ *of [C,* `def:rhon`*] — the module on which* $\mathrm{diag}(\alpha, \beta)$ *has eigenvalues $\alpha^{n-t}\beta^t$, $0 \le t \le n$ — and* not *its contragredient. This is not a nicety:* $\mathrm{Sym}^n(V)^* \cong \mathrm{Sym}^n(V) \otimes \det^{-n}$ *(§0.3, [C,* `sec:contragredient`*]), so the two readings give the same n_i and the same congruence while shifting every individual j_i by $-n$.* ***The sentence that fixes it****, in the form an implementation can quote verbatim:* Sym^n is the symmetric power of the **standard** representation — the module ρ_n of [C, `def:rhon`] acts on, in which $\mathrm{diag}(a, d)$ has $\mathrm{entry}(t, t) = a^{n-t}d^t$ — and not of its dual; the dual would shift every det-twist by $-n$.

And this convention* gates *phase E7 *(v1.6.0; slot-4 WP-7 finding 3, the consequence it did not draw). E7's whole subject is the contragredient local system D_n and $\rho_n^\vee$, and the first thing E7 does with the factor multiset is compare the residue characters of $H^1(D_n)$ against θ-twisted systems — which is item 6 of §5.1, and which E6's atlas will evaluate. Under the two readings that comparison differs by θ^{-n} on* every *factor. Both readings are internally consistent, so* ***a green validation suite cannot detect the discrepancy****: every structural invariant an implementation can check — the n_i, the dimension count $\sum_i (n_i + 1) = n + 1$, the central-character congruence, even the Brauer character against a matrix ρ_n, which is computed in whichever convention the matrices are in — is convention-independent or self-consistent. A wrong reading would therefore surface as an E7* finding*, indistinguishable from a mathematical result. E7 may not start against an unfixed convention:* ***[E] v1.6.0 is a precondition of E7's start****, and this paragraph is the thing it is a precondition for.*

1. *The simple $\mathbb{F}_\ell[G]$-modules are $L(r, j) = \mathrm{Sym}^r \otimes \det^j$ with $0 \le r \le \ell - 1$ and $j \in \mathbb{Z}/(\ell - 1)$ (for $\ell = 2$ the twist group is trivial), and the central character forces $r + 2j \equiv n \pmod{\ell - 1}$ on every Jordan–Hölder factor of M_n [18] ((3.1), p. 433; (6.2), p. 459; in Glover's notation $V_m \otimes D^n$ is $\mathrm{Sym}^{m-1} \otimes \det^n$).*
2. *With $\theta := x^\ell y - xy^\ell = \prod_{[a:b] \in \mathbb{P}^1(\mathbb{F}_\ell)} (bx - ay)$, which satisfies $g \cdot \theta = \det(g)\,\theta$, multiplication by θ is an injection of G-modules $\mathrm{Sym}^{n-\ell-1} \otimes \det \hookrightarrow \mathrm{Sym}^n$, and its cokernel has dimension $(n + 1) - (n - \ell) = \ell + 1$ and is the principal series $\mathrm{Ind}_B^G(\chi_1^n \chi_2^0)$ — restriction of a degree-n form to each of the $\ell + 1$ lines of $\mathbb{P}^1(\mathbb{F}_\ell)$.*
3. *$\mathrm{Ind}_B^G(\chi_1^s \chi_2^t) := \mathbb{F}_\ell[G] \otimes_{\mathbb{F}_\ell[B]} \chi_1^s \chi_2^t$, where $\chi_1^s \chi_2^t \left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) = a^s d^t$, has the two Jordan–Hölder factors $\mathrm{Sym}^r \otimes \det^t$ and $\mathrm{Sym}^{\ell-1-r} \otimes \det^s$ with $r = (s - t) \bmod (\ell - 1)$, the trivial/Steinberg pair when $r = 0$ (v1.21.0-pre.5: [proved here] below; [18] states no induced-module result — its (4.1), p. 439, (4.2), p. 440, (5.7), p. 454, (6.4), p. 461, describe Sym^n by explicit bases — so the earlier citation for this item is withdrawn).*
4. *Hence the multiset of Jordan–Hölder factors of M_n, written as pairs (n_i, j_i), is computed by the terminating recursion*

$$\mathrm{JH}(n) = \begin{cases} \emptyset & n < 0, \\ \{(n, 0)\} & 0 \le n \le \ell - 1, \\ \{(r, 0),\ (\ell - 1 - r,\ n)\} \ \cup\ \big(\mathrm{JH}(n - \ell - 1) \otimes \det\big) & n \ge \ell, \quad r := n \bmod (\ell - 1), \end{cases}$$

in $\lfloor (n+1)/(\ell+1) \rfloor$ steps, with at most $2\lfloor (n+1)/(\ell+1) \rfloor + 1$ raw factors before equal pairs are merged, and $\sum_i (n_i + 1) = n + 1$ counted with multiplicity. (v1.5.1, corrected from $\lfloor n/(\ell+1) \rfloor + 1$ steps and $2\lfloor n/(\ell+1) \rfloor + 1$ factors, which undercounts: at $n = \ell$ the recursion emits two raw factors against a bound of 1, and at $n = 2\ell + 1$ four against 3.)

Proof. (2) The product formula for θ and $g \cdot \theta = \det(g)\theta$ are classical; θ is a non-zero divisor in $\mathbb{F}_\ell[x, y]$, so multiplication by it is injective, and it raises degree by $\ell + 1$ and twists the G-action by det, which is the stated map. The cokernel is $\mathbb{F}_\ell[x, y]_n / \theta\, \mathbb{F}_\ell[x, y]_{n-\ell-1}$; since θ is the product of the $\ell + 1$ distinct linear forms cutting out the lines of $\mathbb{P}^1(\mathbb{F}_\ell)$, a form of degree n lies in (θ) iff it vanishes on all of them, so

the quotient is the image of the restriction map into the $\ell+1$ one-dimensional spaces, i.e. the induced module with the stated character. (3) Write $\tilde{\chi} := \chi_1^s\chi_2^t$, $q := \ell-1$, $V := \mathrm{Sym}^r \otimes \det^t \oplus \mathrm{Sym}^{q-r} \otimes \det^s$; both summands are simple by (1) ($0 \le r \le q$). Two finite-dimensional $\mathbb{F}_\ell[G]$-modules with equal Brauer characters have the same Jordan–Hölder factors (the Brauer characters of the simple modules are linearly independent; $\mathbb{F}_\ell$ is a splitting field by (1)), so it suffices to compare Brauer characters on the ℓ-regular, i.e. semisimple, elements g, with Teichmüller lifts $\alpha, \beta \in \mu_{\ell^2-1}$ of the eigenvalues of g. Frobenius' formula $\chi_{\mathrm{Ind}}(g) = |B|^{-1} \sum_{x \in G,\ x^{-1}gx \in B} \tilde{\chi}(x^{-1}gx)$, valid for Brauer characters, gives: $(\ell+1)\alpha^{s+t}$ at a central g ($\alpha = \beta$); $\alpha^s\beta^t + \alpha^t\beta^s$ at a split $g \sim \mathrm{diag}(\alpha, \beta)$, $\alpha \ne \beta$ (the conjugates of g in B have diagonal (α, β) or (β, α), each for $|B|$ values of x); and 0 at a non-split g, which has no conjugate in B. On V, with $\chi_{\mathrm{Sym}^m}(g) = \sum_{i=0}^{m} \alpha^{m-i}\beta^i$: at a central g, $(r+1)\alpha^{r+2t} + (q-r+1)\alpha^{q-r+2s} = (\ell+1)\alpha^{s+t}$, since $\alpha^q = 1$ and $r \equiv s - t$; at a split g, $\alpha, \beta \in \mu_q$ and $\alpha^{\ell-r} = \alpha^{1-r}$, so

$$(\alpha-\beta)\chi_V(g) = (\alpha\beta)^t\big(\alpha^{r+1}-\beta^{r+1}\big)+(\alpha\beta)^s\big(\alpha^{1-r}-\beta^{1-r}\big) = (\alpha\beta)^t\big(\alpha^{r+1}-\beta^{r+1}+\alpha\beta^r-\alpha^r\beta\big) = (\alpha-\beta)(\alpha\beta)^t(\alpha^r+\beta^r),$$

using $(\alpha\beta)^s = (\alpha\beta)^{t+r}$; hence $\chi_V(g) = \alpha^{t+r}\beta^t + \alpha^t\beta^{t+r} = \alpha^s\beta^t + \alpha^t\beta^s$; at a non-split g, $\beta = \alpha^\ell$, $\alpha = \beta^\ell$ and $N := \alpha\beta \in \mu_q$, so $\alpha^{\ell-r} - \beta^{\ell-r} = \beta\alpha^{-r} - \alpha\beta^{-r} = -N^{-r}(\alpha^{r+1} - \beta^{r+1})$ and $(\alpha-\beta)\chi_V(g) = (N^t - N^{s-r})(\alpha^{r+1} - \beta^{r+1}) = 0$. The two Brauer characters agree on every semisimple class, which proves (3); for $r = 0$ the factors are $\det^t$ and $\mathrm{Sym}^q \otimes \det^t$ (as $s \equiv t$), the twisted trivial/Steinberg pair. Item (3) is used below only through these factors, so the realisation of the cokernel in (2) enters (4) only through its Brauer character, $\alpha^n + \beta^n$ at a split g ($= (\alpha^{n+1} - \beta^{n+1} - \alpha\beta(\alpha^{n-\ell} - \beta^{n-\ell}))/(\alpha - \beta)$ with $\alpha^q = \beta^q = 1$), which is χ_{Ind} at $(s, t) = (n, 0)$, and 0 at a non-split g likewise. *Checked* (v1.21.0-pre.5; own Python, exact in $\mathbb{Z}[x]/\Phi_{\ell^2-1}$, the Frobenius sum taken over the explicit group): every $(s, t) \in (\mathbb{Z}/q)^2$ on every semisimple class at $\ell \in \{2, 3, 5, 7, 11, 13\}$, and the multiplicities of M_n, solved over all $\ell(\ell-1)$ simples and verified exactly, equal to the recursion of (4) for $n \le 3\ell(\ell+1)$. (4) Induction on n, the base case being $n \le \ell - 1$, where M_n is irreducible (item (1) with $r = n$): for $n \ge \ell$ the exact sequence of (2) together with (3) at $(s, t) = (n, 0)$ gives the two displayed factors and the twisted factors of $M_n[-\ell - 1]$; each step lowers n by $\ell + 1$, so the chain $n, n - (\ell + 1), \ldots$ contains exactly $\lfloor (n + 1)/(\ell + 1) \rfloor$ values $\ge \ell$, each emitting two raw factors, and terminates in a value of $[-1, \ell - 1]$ emitting one or none — whence the two counts of (4) — and the dimension count is the alternating sum of the same exact sequence. The central character of every emitted pair satisfies $r + 2j \equiv n \pmod{\ell - 1}$: for $(r, 0)$ because $r \equiv n$, and for $(\ell - 1 - r, n)$ because $(\ell - 1 - r) + 2n \equiv n - r + n - n \equiv 0$ is $n \equiv r$ again. □

Implemented (`HeckeExt/GloverFactors`, slot-4 WP-7): the recursion of (4), with the convention of the preamble pinned to the pipeline's own ρ_n (`ComputeSymmetricPowerMatrixFM` branch `[C1]`, $\mathrm{entry}(t, t) = a^{n-t}d^t$), checked against the Brauer character of ρ_n on the ℓ-regular classes of $\mathrm{SL}_2(\mathbb{F}_\ell)$, computed independently from those matrices, for every $n \le 3\ell$ at $\ell \in \{5, 7\}$ (`HeckePredictionsValidation`). Identity 6 of §5.1 is *not* thereby asserted — that needs the atlas (E6) — but the factor list it quantifies over is computed, and its det-twists have one meaning.

5.1 Identities the computation must satisfy

All [standard, reference] unless marked; each becomes an assertion in the validation that will gate this feature.

1. [C, `cl:dimH1`]; $\dim X_2 = \dim X_5 = \lfloor n/\ell \rfloor + 1$; exactness of [C, `eq:les`] with the dimension bookkeeping of [C, `cl:les`]; $\dim X_1 = \#\{a(\ell + 1) + b(\ell^2 - \ell) = n\}$.
2. $[T_q, T_{q'}] = 0$, $[T_q, \iota] = 0$, equivariance of every map in [C, `eq:les`]; the recursion of [C, `thm:relations`] at q^2.
3. Boundary eigenvalues χ_a ([C, `cl:chi`], [to be verified computationally] — the H^0 characters and the $n < \ell$ Eisenstein eigenvalue are asserted, and [C, `cl:chi`](2)'s general-n H^1 multiset recorded, 330/330 cells, by `HeckeBoundaryOperatorValidation` since slot 1; the tag stays until the claim is proved); $1 + q^{n+1}$ on the reduced Harder class; $\tau(q) \bmod \ell$ at $n = 10$; the known cusp forms of [C, `sec:example`].

4. $r^1 = 0$ on the Eisenstein class iff $\ell \mid \mathrm{num}(B_k/k)$ ($n < \ell - 1$; Claim 1.1) — asserted in the form "r^1 vanishes on the joint eigenspace E" on the Kummer list and "$\mathrm{rank}(r^1 E) = 1$" off it; verified 37/37 (`HeckeBoundaryParabolicValidation`, 2026-09-16).
5. Poincaré duality ([C, `cl:pd`]) once D_n runs (E7, scheduled — v1.8.2: *accepted* at the slot-7 close, [C, `cl:pd`](1)–(2) tested on all 54 cells of the E2 case list at $\ell \geq 5$, failing on none; the adjointness of [C, `cl:pd`](2) is the acceptance gate for X_3 at $\ell \geq 5$, `HeckeDualityValidation`): equal characteristic polynomials between $X_3(M_n)$ and $H^1(D_n)$, and the Loewy layers of each equal to the socle layers of the other in reverse order (Prediction 4.6; *not* reversed Loewy series).
6. Compatibility along $L\cdot$ and along the Ash–Stevens filtration [M §5]: the semisimplification of $H^1(\Gamma; M_n)$ is the sum over the Glover factors $\mathrm{Sym}^{n_i} \otimes \det^{j_i}$ ($0 \leq n_i \leq \ell - 1$, $n_i + 2j_i \equiv n \bmod (\ell - 1)$) of θ^{j_i}-twists of weight-$(n_i + 2)$ systems, plus the H^0-corrections. Claim 5.13 supplies the factor multiset itself *and* the convention that fixes the individual j_i; without that convention this item does not have a determinate meaning (v1.5.0). **And this is the item on which the convention gates E7** (v1.6.0): it is here that the det-twists meet D_n, so the two readings of Sym^n differ by θ^{-n} in exactly this comparison and nowhere a validation can see them (Claim 5.13's second preamble paragraph). E7 starts against the standard-module reading or not at all.

   Implemented (`HeckeExt/GloverFactors` for the factor multiset, slot-4 WP-7; bound in the console as `glover_factors(n, ell)`, slot-5 WP-10, which reports the same convention). The *identity* is not thereby asserted — that needs the atlas. *Status* (v1.8.0; the slot-6 WP-9 result, recorded in v1.7.2's changelog and now stated here). The identity **is asserted**, as item 6 of `HeckeIdentityCertificate`, on 108 cells, $\ell \in \{5, 7, 11\}$ times $n \in \{0, 2, \ldots, 60\} \cup \{80, 100, 120, 160, 200\}$, against the in-process $\Sigma(\ell)$ (`HeckeExtEndToEndValidation`, arms [B] and [C]). It holds under the standard-module convention, and the contragredient reading fails on all 63 cells where the two differ. It is **not evaluated** at $\ell = 691$. $\Sigma(691)$ is slot-7 WP-5's, and it is not built at this revision (v1.8.2: built by slot-7 WP-5 and published by hash and build command only; the identity is still not evaluated at $\ell = 691$). Even once built, it is a stored atlas, which neither the record nor golden reads (slot-7 owner ruling 3b (B); §7, "The stage-2 keys" (b)). At $\ell = 691$ the record carries `atlas_unavailable`, and the atlas is joined to the records only outside them, by the campaign's analysis tool.

   v1.12.0-pre.1: the $L\cdot$ half extended by its long exact sequence (E1 identity class 5 extended; slot-11 WP-5 and WP-6; the objects, their names and indexing are §5.2, the mathematics is [C]'s section on the Dickson invariant and θ). Through v1.11.1 the $L\cdot$ half was the one identity $A_n^{(1)}(T_p)\, L_* = p\, L_*\, A_{n-\ell-1}^{(1)}(T_p)$ (`HeckeIdentityCertificate` item 5, slot-5 WP-8). From v1.12.0-pre.1 it is the following list, written with $\theta = \theta_n : H^q(\Gamma; M_n \otimes \det) \to H^q(\Gamma; M_{n+\ell+1})$ (source-indexed, §5.2 (a)), Q_n the quotient local system, $\ell \geq 5$, p a prime $\neq \ell$:

   (6a) *fibre*: $\rho_{n+\ell+1}(A)\, L = \det(A)\, L\, \rho_n(A)$ for every $A \in M_2(\mathbb{F}_\ell)$, singular included, and hence $\rho_{n+\ell+1}(A)(L \cdot \mathrm{Sym}^n) \subseteq L \cdot \mathrm{Sym}^n$, so $\rho_{Q_n}(A)$ is defined for every $A \in M_2(\mathbb{F}_\ell)$ (the well-definedness certificate of `FibreRepresentation::QuotientL`);

   (6b) *exactness*: the six-term sequence $0 \to H^0(M_n) \to H^0(M_{n+\ell+1}) \to H^0(Q_n) \xrightarrow{\delta} H^1(M_n) \xrightarrow{\theta} H^1(M_{n+\ell+1}) \to H^1(Q_n) \to 0$ is exact at every spot, as *subspaces* (a basis of the image and of the kernel at each spot, the inclusion both ways), with δ computed by the snake construction on cochains and never inferred from ranks; the three H^2 terms are 0 ([C, `lem:h2torsion`]);

   (6c) *dimension*: the alternating sum of the six dimensions is 0, and $\mathrm{rank}\,\delta$ equals the closed form in (n, ℓ) of [C, `thm:thetales`];

   (6d) *Dickson ring*: $H^0(\Gamma; M_n) = \mathbb{F}_\ell[L, D]_n$, the span of the explicit invariants $L^a D^b$, $a(\ell + 1) + b(\ell^2 - \ell) = n$, equal to the computed H^0 in dimension and span ([C, `prop:dicksonH0`], with D as [C, `def:dicksoninv`] defines it; consistent with [C, `rmk:dickson`]'s semigroup and with the dimension formula of item 1);

   (6e) *Hecke*: for $p \neq \ell$, $A(T_p)_{n+\ell+1}\, \theta = p\, \theta\, A(T_p)_n$ in degrees 0 and 1, entrywise, with the untwisted identity asserted to *fail* wherever $\theta\, A(T_p)_n \neq 0$ and $p \not\equiv 1 \pmod{\ell}$ (the exponent pinned, as in item 5); π intertwines $A(T_p)_{n+\ell+1}$ with $A(T_p)_{Q_n}$ untwisted, and δ intertwines $A(T_p)_{Q_n}$ with $p\, A(T_p)_n$; and over $\mathbb{F}_\ell$ $A(U_\ell)_{n+\ell+1}\, \theta = 0$, already on cochains (Claim 2.14(1),

[C, `prop:thetatwist`]), and $\delta\, A(U_\ell)_{Q_n} = 0$;

(6f) H^0 *eigenvalues* ([C, `prop:heckeH0`]): $A(T_p)$ acts on L^aD^b by $(1+p)\,p^a$ for $p \neq \ell$; $A(U_\ell)$ on $H^0(\Gamma; M_n)$, $n \geq 1$, has rank ≤ 1 and $A(U_\ell)^2 = 0$, kills every L^aD^b with $a \geq 1$, and is *zero for* $n < \ell^3 - \ell$ but *not* in general: at $n = \ell^3 - \ell$ it is non-zero, the first instance being $(120, 5)$, $U_5 D^6 = -L^{20}$ (the only one with $\ell \in \{5, 7, 11, 13\}$, $n \leq 200$); at $n = 0$, $U_\ell = 1$. What "every Dickson class is non-ordinary" needs is $U_\ell^2 = 0$ ([C, `cor:thetanonordinary`]), which holds at every $n \geq 1$. *v1.12.0-pre.1 correction*: the slot-11 plan (and this item's first draft) said "$U_\ell = 0$ on all of H^0"; [C] refutes it ([C, `rmk:heckeH0plan`]), and the assertion is the one above, with $(120, 5)$ an arm. On $H^0(Q_n)$ (dimension $[(\ell-1) \mid n+2]$, spanned by $\mathbf{1}_n$): $T_p\mathbf{1}_n = (p+1)\mathbf{1}_n$ and $U_\ell\mathbf{1}_n = 0$ ([C, `lem:Qnfunctions`]);

(6g) *kernel*: $\ker(\theta|_{H^1}) = \operatorname{im}\delta$ as subspaces, with its decomposition by systems predicted in writing before the run (slot-11 rule 20);

(6h) *negative seams*: replacing L by $x^\ell y$ (not det-equivariant) fails (6a)'s certificate, and a perturbed δ fails (6b).

Every item is a statement of [C] or follows from one by the twist argument of Claim 2.14(1) (the Hecke operators of $M_n \otimes \det$ are $p\,A(T_p)_n$ and $\ell\,A(U_\ell)_n = 0$, and every map of the sequence commutes with the operators of its own terms); its STATUS is [C]'s. This list names only what the code asserts; where the two differ, [C] wins and the package reports, and a clause [C] does not state is run as a recorded observation, not as a claim (slot-11 rule 21). (6a)–(6d) and (6h) are slot-11 WP-5's (`HeckeDicksonThetaValidation`, `standard`, and its `slow` coverage companion), (6e)–(6g) WP-6's (arms of the same name or a second `standard` name). **They are validation arms, not certificate items**: the item codes 1–7 of `HeckeIdentityCertificate` are its tamper vocabulary and stay as they are (item 5 keeps its one identity), no record carries any of (6a)–(6h), and no golden line is added (slot-11 decision 6 (A)). At $\ell \in \{2, 3\}$ none of them is asserted ([C, `lem:h2torsion`]'s hypothesis $6 \in R^\times$ fails, and the sequence need not stop at $H^1(Q_n)$). *v1.12.0, as built*: (6a)–(6d) and (6h) are `HeckeDicksonThetaValidation` (`standard`, 108 cells, 979 checks) and `HeckeDicksonThetaCoverageSlowValidation` (`slow`, 404 cells, 3590 checks); (6e)–(6g) are `HeckeDicksonThetaHeckeValidation` (`standard`, 5482 checks) and `HeckeDicksonThetaHeckeCoverageSlowValidation` (`slow`, 3836 checks), with (6e) also at cochain level ($n \leq 20$, U_ℓ included). Every item held on every cell. In (6e) the "untwisted identity fails" check adds nothing to the twisted one: for $p \not\equiv 1 \pmod{\ell}$, $pX = X$ iff $X = 0$, so the twisted identity alone pins the exponent (slot-11 WP-14, m-5).

7. Sum over $\mathfrak{m}$ of the Loewy layer dimensions equals $\dim X_i$; the Jordan partition of n_j on V refines the Loewy length ($\nu \geq$ largest part).

5.2 The Dickson–θ objects

This subsection is the specification slot-11 WP-5 and WP-6 build against, written before either commits a claim (slot-11 plan §3, rule 21). *Division of labour*: [C]'s section on the Dickson invariant and θ ([C, `sec:dicksontheta`]; slot-11 WP-2, [C] v1.11.0-pre.1) owns the mathematics — every statement, its hypotheses, its proof or its reference, the complement basis of Q_n and the closed form of the dimensions; this subsection owns the names, the indexing, what the code computes, where each statement is asserted, the hand-off to the census (§7.2) and the refusals. Where the two differ, [C] wins and the package that finds the difference reports it (slot-11 rule 21; WP-2's author repairs).

(a) **Indexing and the map.** Throughout, $\ell \geq 5$, $M_n = \operatorname{Sym}^n(\mathbb{F}_\ell^2)$ in the standard-module convention of [C, `def:rhon`], and $L = x^\ell y - xy^\ell$. The map is *source-indexed*, as in [C] and the slot-11 plan:

$$\theta_n := L_* : \; H^q(\Gamma; M_n \otimes \det) \longrightarrow H^q(\Gamma; M_{n+\ell+1}), \qquad q \in \{0, 1\},$$

induced by $f \mapsto Lf$, which is Δ-equivariant by the fibre identity $\rho_{n+\ell+1}(A)\,L = \det(A)\,L\,\rho_n(A)$ for every $A \in M_2(\mathbb{F}_\ell)$ (the identity $(*)$ of `HeckeExt/HeckeDicksonMultiplication.h`). On Γ the character det is trivial, so $H^q(\Gamma; M_n \otimes \det) = H^q(\Gamma; M_n)$ as groups and the pipeline's own objects at n serve; the twist is visible only to the Hecke operators, which on $M_n \otimes \det$ are $p\,A(T_p)_n$ and $\ell\,A(U_\ell)_n = 0$ over $\mathbb{F}_\ell$. `HeckeDicksonMultiplication` and the console verb

`L_map(H_lower, H)` are *target-indexed* with $m = n-\ell-1$; they compute the same map, θ_m. The census (§7.2) is target-indexed too: at a cell (n, ℓ) the θ-image is $\operatorname{im}\theta_{n-\ell-1} \subseteq H^1(\Gamma; M_n)$, which is 0 for $n < \ell+1$ (in particular at every published $\ell = 691$ cell, $n \le 200$). *Not to be confused* with the atlas label's `theta_twist` (§7), which records a system's twist relative to its atlas base system, a labelling convention; the census reports both and asserts no relation between them.

(b) **The quotient local system.** $Q_n := M_{n+\ell+1}/L \cdot M_n$, of dimension $\ell+1$ for every $n \ge 0$. Its fibre action $\rho_{Q_n}(A)$ is defined for *every* $A \in M_2(\mathbb{F}_\ell)$, singular included (by $(*)$, $L \cdot M_n$ is stable under every A), written on the monomial complement basis $\mathcal{B}_n = (\bar{u}_0, \ldots, \bar{u}_{\ell-1}, \bar{u}_{n+\ell+1})$ that [C, `def:Qn`] fixes, with its class map — the code implements that basis and never chooses one (the normal form $(\bar{u}_0, \bar{u}_{n+2}, \ldots, \bar{u}_{n+\ell+1})$ is *not* it). It enters the pipeline as a third `FibreRepresentation` enumerator, `QuotientL` (`LocalSystemUtils`), appended, with its relation list for `LocalSystemSL2Z` and its arm in `HeckeOperatorOnK` (slot-11 WP-5 only, append-only): `SymN` and `ContragredientN` stay byte-identical (`bench_identity.sh` 45/0, the golden gate). Well-definedness is certified entrywise ((6a) of §5.1) on seeded invertible *and singular* A and on every relation matrix, because WP-6 applies U_ℓ, whose representatives are singular mod ℓ, on all six terms.

(c) **The six terms and five maps; code names.** For $\ell \ge 5$ ([C, `lem:h2torsion`]: the H^2 terms vanish at $6 \in R^\times$) the long exact sequence of $0 \to M_n \otimes \det \xrightarrow{L\cdot} M_{n+\ell+1} \to Q_n \to 0$ is

$$0 \to H^0_n \xrightarrow{\theta^0} H^0_\uparrow \xrightarrow{\pi^0} H^0_Q \xrightarrow{\delta} H^1_n \xrightarrow{\theta^1} H^1_\uparrow \xrightarrow{\pi^1} H^1_Q \to 0,$$

$H^q_n = H^q(\Gamma; M_n)$, $H^q_\uparrow = H^q(\Gamma; M_{n+\ell+1})$, $H^q_Q = H^q(\Gamma; Q_n)$. The code names are fixed here: terms `H0_n`, `H0_up`, `H0_Q`, `H1_n`, `H1_up`, `H1_Q`; maps `theta0`, `pi0`, `delta`, `theta1`, `pi1`. Each map is a matrix over $\mathbb{F}_\ell$ *in the stored bases* of the pipeline's `EquivariantCohomologyOnK` objects of its two terms (column j: the image of the source's j-th stored basis vector, in the target's stored coordinates), the convention of `DicksonInducedMaps`. δ is computed by the snake construction on cochains (lift to $M_{n+\ell+1}$-cochains, apply the coboundary, divide by L), never inferred from ranks. Home: `HeckeExt/HeckeDicksonTheta.{h,cpp}` (new; WP-5, then WP-6), `namespace hecke_ext`, under the layer's one-way dependency.

(d) **The Dickson ring.** $D := \sum_{i=0}^{\ell} x^{i(\ell-1)} y^{(\ell-i)(\ell-1)}$ as [C, `def:dicksoninv`] defines it $(= (x^{\ell^2} y - xy^{\ell^2})/L$, [C, `lem:dicksontwist`](2)), of degree $\ell^2-\ell$, fixed by every A invertible mod ℓ but *not* by singular ones ([C, `lem:dicksontwist`](3)) — which is why U_ℓ need not vanish on H^0 ((6f)); $H^0(\Gamma; M_n) = \mathbb{F}_\ell[L, D]_n$ with basis $\{L^a D^b : a(\ell+1) + b(\ell^2-\ell) = n\}$ ([C, `prop:dicksonH0`]), built as explicit invariants and compared with the computed H^0 in dimension and span ((6d)).

(e) **What is certified, what is asserted.** (6a)–(6h) of §5.1, with the packages and names stated there. The θ-cycle, Ash–Stevens and companion-form instances (slot-11 WP-6 scope 3) are *the list [C, `rmk:thetacheckable`] states as checkable on the model at level 1 and $n \le 200$* ([C, `sec:thetaliterature`]); each is predicted in writing, with its source, in the package's report skeleton and committed before it is run (slot-11 rule 20), and each is compared against an independent object — the label of $\Sigma(\ell)$ (§5) and the T_p-eigenvalues for $p \le 13$. Agreement and disagreement are both results; a disagreement is a finding, reported, never re-fitted.

(f) **The hand-off to the census.** For each published cell (n, ℓ) of `ext_first` with $\ell \le 13$ and $n \ge \ell+1$ and each maximal ideal $\mathfrak{m}$ of its record, slot-11 WP-6 computes $d_\theta(\mathfrak{m}) := \dim_{\mathbb{F}_\ell}\big(E_\mathfrak{m} \cap \operatorname{im}\theta_{n-\ell-1}\big)$, $E_\mathfrak{m} \subseteq H^1(\Gamma; M_n)$ the generalised eigenspace of $\mathfrak{m}$. Since $\operatorname{im}\theta$ is stable under every T_p and killed by U_ℓ ((6e)), it is a Hecke submodule, so $\operatorname{im}\theta = \bigoplus_\mathfrak{m} (\operatorname{im}\theta)_\mathfrak{m}$ and $\sum_\mathfrak{m} d_\theta(\mathfrak{m}) = \operatorname{rank}\theta^1$ — asserted per cell. The run identifies $E_\mathfrak{m}$ with the record's ideal by the matching rule of §7.2 (e) (one-to-one; a mismatch is a finding, never a guess). Output: one line per ideal,

```
THETA_IMAGE n=<n> ell=<ell> ideal=<id> witness=<c0,c1,...> dim_E=<d> theta_image_dim=<t>
                                                          rank_theta_cell=<r>
```

(`witness` the record's `witness_poly`, low→high), printed by WP-6's validation only under the read-only interlock `HECKE_THETA_IMAGE_ALL=1` (not an emitter; the `HECKE_INTEGRAL_CAMPAIGN` pattern), and collected by the census tool into `theta_image.tsv` with the format tag `hecke_census_theta/v1` (§7.2 (h)). Cells with $n < \ell+1$, and every $\ell = 691$ cell, are filled by the tool with $d_\theta = 0$ and the source `n_lt_ell_plus_1`; they are not run.

(g) **What these objects are not.** No member of `hecke_ext/v3`, `hecke_report/v2` or `hecke_ring_report/v1` carries them: the record's per-ideal `canonical` slot, which (D6) reserves for $L \cdot H^1(M_{n-\ell-1})$ among others, keeps emitting {`"status": "unavailable"`}. No settings key is added (eleven stay eleven): θ is a derived object with a validation, not a setting. No golden line is added (slot-11 decision 6 (A)); no `HeckeIdentityCertificate` item code is added.

(h) **Refusals** (§7.1 (b), the v1.12.0-pre.1 paragraph). Every entry point of `HeckeDicksonTheta`, the `QuotientL` relation list and `HeckeOperatorOnK`'s `QuotientL` arm are $\mathbb{F}_\ell$-only in slot 11: under a non-default `coefficient_ring`, or handed a composite object, each raises `hecke::NotImplementedInThisBuild` naming itself (console exit 5). $\ell < 5$ is refused as a mathematical precondition, by `ENGINE_CHECK_MSG` naming [C, `lem:h2torsion`] (a `std::logic_error`, console exit 3), not as a build capability. Odd n is not refused: every term vanishes on this model ($-I$), and the sequence is the zero sequence. θ over $\mathbb{Z}/\ell^m$ is slot 12's, after [C]'s Hida section (slot-11 plan §3.15).

v1.12.0, as built (slot-11 WP-5, WP-6). (a)–(e) and (g) as specified; the code names of (c) are `hecke_ext::DicksonThetaCell`'s. (f): the `THETA_IMAGE` lines cover 384 cells and 1934 ideals, *every* block matched its record ideal one-to-one by `witness_poly` and exact restricted `t_p`, and $\sum_{\mathfrak{m}} d_\theta(\mathfrak{m}) = \operatorname{rank} \theta^1$ held on every cell; the table is committed as `docs/implementation/slot11/WP-6_theta_image.tsv` (`hecke_census_theta/v1`) and copied into the census edition as `theta_image.tsv`. The match uses the block's own `witness_poly` in place of the driver's local-algebra witness, which is exact under the shipped `consume_block_locality = true` (slot-11 WP-14). (h) *differs*: $\ell \in \{2, 3\}$ is refused by `hecke::NotImplementedInThisBuild` (console exit 5), not by an `ENGINE_CHECK_MSG` naming [C, `lem:h2torsion`] (exit 3); the guard's message states the reason (the sequence is not available on the model, [C, `rmk:thetasmallell`]). The ring refusals as built are the first item of §7.1's "Published v1.12.0" paragraph. *v1.12.1 (the slot-11 WP-15 referee's m-1, the ruling on F-2): the specification stands, and the build now follows it.* $\ell < 5$ is a *mathematical precondition*, not a build capability: on this model the equivariant cochain functor is not exact at $\ell \in \{2, 3\}$ and the model does not compute the group long exact sequence there ([C, `rmk:thetasmallell`]), which is definitional unavailability — the class of a field accessor on a ring handle, exit 3 — whereas exit 5 promises an implementation that can never come. Since slot-11 WP-17 `DicksonThetaCell` refuses $\ell < 5$ by `ENGINE_CHECK_MSG` naming [C, `lem:h2torsion`] (a `std::logic_error`, console exit 3), and `HeckeDicksonThetaValidation`'s refusal arm asserts that kind and that name. The "differs" sentence above is the v1.12.0 record of the build before that change.

6 Scalar extension to $\mathbb{F}_{\ell^r}$

6.1 Why, and when

The residue field of $\mathbb{T}_{\mathfrak{m}}$ is $\mathbb{F} = \mathbb{F}_{\ell^r}$, $r = f_{\mathfrak{m}}$; the eigenvalues $a_q(\mathfrak{m})$ live there, and "the" simple $\mathbb{T}_{\mathfrak{m}}$-module is $\mathbb{F}$. Over $\mathbb{F}_\ell$ every object is a Galois *orbit* (irreducible factors, blocks of dimension $r\cdot$(length)). Extension is needed *only* when some $r_{\mathfrak{m}} > 1$, and then only to (i) report eigenvalues as elements, (ii) compare characters across different n in the atlas, (iii) present the structure maps (D4) as $\mathbb{F}$-matrices. Nothing in §5 needs it:

Claim 6.1 (Extension commutes with everything)**.** *[proved here] Let $V = H_{\mathfrak{m}}$ over $\mathbb{F}_\ell$ with $\mathbb{T}_{\mathfrak{m}}/\mathfrak{m} = \mathbb{F}$. Then $\mathbb{T}_{\mathfrak{m}} \otimes_{\mathbb{F}_\ell} \mathbb{F} \cong \prod_{\sigma \in \mathrm{Gal}(\mathbb{F}/\mathbb{F}_\ell)} \mathbb{T}_{\mathfrak{m},\sigma}$ with r local factors permuted simply transitively by* Frob*, each with residue field $\mathbb{F}$, and $V \otimes_{\mathbb{F}_\ell} \mathbb{F} = \bigoplus_\sigma V_\sigma$ accordingly. For every σ: $\dim_{\mathbb{F}} V_\sigma = \dim_{\mathbb{F}_\ell} V/r$, the Loewy and socle layer dimensions of V_σ over $\mathbb{F}$ are those of V over $\mathbb{F}_\ell$ divided by r, the Jordan type of $t_j - a_{p_j}$ on V_σ is the partition whose s-th nullity is $\dim_{\mathbb{F}_\ell} \ker \pi_j(t_j)^s/r$ — i.e. the S10 generalized-Jordan partition of t_j w.r.t. π_j (which is in units of $\deg \pi_j$) with every multiplicity divided by $r/\deg \pi_j$; the two coincide only when $\deg \pi_j = r$ —, the embedding dimension and Loewy length of $\mathbb{T}_{\mathfrak{m},\sigma}$ are those of $\mathbb{T}_{\mathfrak{m}}$, and $\mathbb{T}_{\mathfrak{m},\sigma}$ is Gorenstein iff $\mathbb{T}_{\mathfrak{m}}$ is. In particular* no extension class is created or destroyed by extending scalars*, and the record of §4.2 computed over $\mathbb{F}_\ell$ is the record of every V_σ.*

Proof. $\mathbb{F} \otimes_{\mathbb{F}_\ell} \mathbb{F} \cong \mathbb{F}^r$ via $a \otimes b \mapsto (\sigma(a)b)_\sigma$; tensoring the local ring $\mathbb{T}_\mathfrak{m}$ (an $\mathbb{F}$-vector space through any coefficient-field lift, which exists since $\mathbb{F}/\mathbb{F}_\ell$ is separable) gives the product decomposition, and $\mathfrak{m} \otimes \mathbb{F}$ is the product of the maximal ideals. All listed invariants are dimensions of $\mathbb{F}$-subquotients defined by $\mathbb{T}_\mathfrak{m}$-linear operations, which commute with base change; the Galois group permutes the factors, so all factors have equal invariants, and the $\mathbb{F}_\ell$-dimension of V is the sum over σ of $\dim_\mathbb{F} V_\sigma$. Jordan types: on V_σ only the factor $t_j - a^\sigma_{p_j}$ of $\pi_j(t_j) = \prod_\tau (t_j - a^\tau_{p_j})$ is nilpotent, the others are invertible, so $\ker(t_j - a_{p_j})^s|_{V_\sigma} = \ker \pi_j(t_j)^s|_{V_\sigma}$, and the $\mathbb{F}_\ell$-nullity of $\pi_j(t_j)^s$ on V is r times this common value. □

DECISION. Compute over $\mathbb{F}_\ell$ from start to finish; extend scalars only in the report layer. All arithmetic in the certified pipeline stays `uint32`/`uint64` mod ℓ. This is the route recommended by the capability inventory §4: *no new element type in the engine*; $\mathbb{F}$ is represented as the $\mathbb{F}_\ell$-algebra $\mathbb{F}_\ell[X]/(\pi)$ through the existing `FpPolyT` (`powmod`, `frobenius_powmod`), on the per-maximal-ideal blocks only (dim $\lesssim 200$), and never on a result path. Two tiers: (i) the *block's own field* $\mathbb{F}_\ell[X]/(\pi_w)$, in which the character is already available as the polynomials P_j of Claim 3.3 with zero new arithmetic — always emitted, as `coefficient_field`; (ii) the re-embedding into the canonical field of §6.2 for cross-run comparison, the only place needing polynomials over $\mathbb{F}$ — *since v1.4.0 the default report form* (`split_field_report = canonical`, owner decision OQ10; through v1.3.3 tier (i) was the default), *conditional on E6's performance gate*: at the atlas cells $\ell \in \{5, 7, 11, 13\}$ (every even $n \le \ell - 1$) and at $(10, 691)$, $(160, 691)$, $(198, 691)$ the canonical-field time must satisfy `canonical_field_s` $\le 0.10 \times$ `total_s` on every cell, with the numerator, the denominator and the protocol pinned by [X] §6.2 E6 (d), which is the one governing statement of the gate (v1.7.1: that item is written into [X] at v1.10, the slot-6 close; until then its text is `docs/implementation/e6_gate_brief.md` §6.1, which is binding, `docs/implementation/hecke_extensions_slot6_plan.md` §4.3; v1.7.2: [X] v1.10 carries it, and the interim pointer is superseded) (owner decisions `[OD]` E6-1 and E6-2 of 2026-09-22, `docs/implementation/e6_gate_brief.md`); `own_field_s` is measured and reported beside it and is not a bound. (Through v1.6.2 the gate read `canonical_field_s` $\le 0.10 \times$ `ext_s` *and* $\le 2 \times$ `own_field_s`; since tier (i) is by this document's own words "zero new arithmetic", the second clause compared a certified field computation with a rendering and was not meetable, and the first had a denominator of microseconds on 20 of the 23 cells — both measured on all 23 cells in `docs/implementation/slot5/WP-7_report.md` §4. E6-1 changed the denominator to `total_s`, the wall of `HeckeExtDriver::run()`, and deleted the second clause; the constant 0.10 is unchanged.) If it fails, the default flips to `own_field` for the first campaign (*superseded* by rule Z below; v1.8.1), the measurement goes to the owner, and the $\mathbb{F}_{\ell^r}$ element type is the reopening question ([X] §0.3, §2.2 Stage 6). *The cell list and the failure rule* (v1.8.0; slot-7 owner ruling 4 of 2026-09-23, slot-7 plan §10; this document only points to them). The cells are the 23 above plus, as **gating** cells, the first campaign's highest-r cells, those with `canonical_field_r_max` ≥ 10 in WP-4's $r_{\max}$ scan (`docs/implementation/slot7/WP-4_rmax_ell691.tsv`, read by `e6_measure.sh --cells-from`; option C1). (v1.8.1: the threshold 10 is WP-4's reading of the ruling's "highest-r cells", the default of `e6_measure.sh --r-min`; [X] v1.11 pins it. With it the list is $23 + 19 - 1 = 41$ cells, $(198, 691)$ being both one of the 23 and at $r_{\max} = 10$.) $(210, 691)$ and $(256, 134217757)$ are **reported only**, since neither is a campaign cell ($n = 210 > 200$; $\ell = 134217757$ is not a campaign prime; v1.8.1). The failure rule is option Z: a miss in the verdict window goes to the owner before the campaign's publication run and never flips the report form by itself. This replaces the automatic "flips" of the previous sentence. The owner then chooses among three options: flip the campaign; hold `canonical` with (d) recorded not met; or delay the campaign for a stronger lever. A miss on a reported cell flips nothing. The list and the rule are written into [X] §6.2 E6 (d) at [X] v1.11. Until then, the slot-7 plan's §10 and §2.4 row T3 are binding (v1.8.2: superseded — [X] v1.11 carries both, and pins the threshold 10). *Status*: W1 measured the worst gating cell, $(170, 691)$, at 0.0104 over the 41 gating cells, so the

bound holds and the default stands (Algorithm 6.4's cost note); E6 is declared at the slot-7 close (v1.8.2). The defining polynomial is the setting `field_polynomial = least` (Definition 6.2); `conway` is reserved, parsed and rejected (v1.7.2: parsed and validated as a setting, its fingerprint moving; its consumer `FieldPolynomialFromSettings` raises `hecke::NotImplementedInThisBuild`; v1.8.2: every production entry point — the field's constructor, `--bench ext`, the certificate, the atlas and the record — routes through it since slot-7 WP-4). Working in $\mathbb{F}$ from the start would multiply every cost by r^2–r^3, would need a new engine width (the `ModCtx`/kernel/tape layers are hard-wired to integer residues, inventory §4.1), and would certify nothing that Claim 6.1 does not already give.

Where the two tiers sit in the record (v1.8.0-pre.1): tier (i) is the per-ideal `coefficient_field`, always; tier (ii) is the per-ideal `split`, the `hecke_canonical_split/v1` fragment of Algorithm 6.4 verbatim, present only under `split_field_report = canonical`. Under `own_field` the record carries tier (i) alone. The one statement of the layout, with its presence rule, is the paragraph "The stage-2 keys" of §7; it holds whichever report form the gate above selects.
The coefficient ring is a different axis (v1.9.0-pre.1). This section extends the *residue field* of $\mathbb{T}_{\mathfrak{m}}$ over a run whose coefficients are $\mathbb{F}_\ell$; it says nothing about coefficients in $\mathbb{Z}/\ell^k$. The setting `coefficient_ring` (E8), its grammar, its refusal rule and the objects a $\mathbb{Z}/\ell^k$ run computes are specified in §7.1. Under a non-default ring nothing in this section runs in slot 8: the canonical split, the atlas and the extension driver are field notions, and each refuses by the rule of §7.1 (b).

6.2 Canonical, deterministic choice of $\mathbb{F}$

Definition 6.2 (Project-canonical defining polynomial). $C_{\ell,r} \in \mathbb{F}_\ell[Y]$ is the `CanonicalLess`-least monic irreducible polynomial of degree r (order: degree, then lexicographic on the low→high coefficient vector), found by enumerating candidates in that order and applying `CertifyIrreducible`; $\mathbb{F} := \mathbb{F}_\ell[Y]/(C_{\ell,r})$.

Claim 6.3. *[proved here] $C_{\ell,r}$ is a deterministic function of (ℓ, r) alone, reproducible by any implementation of the stated order, and every step of its construction is certified. It is* not *in general the Conway polynomial (which imposes compatibility across subfields and a different ordering, and is tabulated only for small ℓ, r [24]); when the Conway polynomial is available the record may name it as well, and the record always stores $C_{\ell,r}$ itself, so the identification is part of the data.*

Algorithm 6.4 (Splitting a block, report layer only). Input: a certified block with witness polynomial π_w ($\deg \pi_w = r$) and the polynomials P_j of Claim 3.3.

1. Find a root $\rho \in \mathbb{F}$ of π_w: π_w splits completely in $\mathbb{F}$; use equal-degree splitting over $\mathbb{F}$ with a *fixed* candidate sequence $(\gcd(\pi_w(X), (X+s)^{(\ell^r-1)/2} - 1),\ s = 0, 1, Y, \dots)$, and certify $\pi_w(\rho) = 0$ in $\mathbb{F}$ — the repository's find-then-certify pattern. The r roots are ρ^{ℓ^i}; take ρ_{can} := the `CanonicalLess`-least of them (as coefficient vectors).
2. Split eigenvalues: $a_q(\mathfrak{m}) = P_q(\rho_{\mathrm{can}}) \in \mathbb{F}$ for $q \in S$; the orbit is $\{(P_q(\rho^{\ell^i}))_q\}_i$, all reported.
3. Frob acts on $\mathbb{F}$ by $Y \mapsto Y^\ell \bmod C_{\ell,r}$ and on the split data by $\rho \mapsto \rho^\ell$; it permutes the r conjugate blocks of Claim 6.1 transitively. The record reports both forms: the $\mathbb{F}_\ell$-form (factors, orbit length r) and the split form (one canonical tuple plus the Frobenius orbit).
4. Structure maps in $\mathbb{F}$-coordinates: choose the $\mathbb{F}$-basis of each Loewy layer given by the $\mathbb{F}_\ell$-basis reduced modulo the action of ρ_{can} (the layer is an $\mathbb{F}_\ell[w]/(\pi_w)$-module); the $\mathbb{F}$-matrix of $\theta(n_a)$ is then read off. (This step is optional; ranks are already known by Claim 5.12.)

Only step 1 requires arithmetic in $\mathbb{F}$ (univariate polynomials over $\mathbb{F}$); it is new code confined to the reporting layer, $O(r^3 \log \ell)$ $\mathbb{F}$-operations per block $= O(r^5 \log \ell)$ $\mathbb{F}_\ell$-multiplications with schoolbook arithmetic in $\mathbb{F}$; *not* negligible at high r, see [X] §6.2 E6 (d). (v1.8.0; slot-7 WP-4, commits `fa829c7`, `cab4e38`. Through v1.8.0-pre.1 this sentence read "$O(r^3 \log \ell)$ per block — negligible", with no unit. The unit is $\mathbb{F}$-operations, and "negligible" was **refuted** before the slot-7 fix on the two high-r cells measured, at $r = 14$ and $r = 19$ (v1.8.1; through v1.8.0 this read "at $r \geq 11$", but no cell at $r \in \{11, 12, 13\}$ was measured before the fix). The canonical-field time exceeded the

$0.10 \times$ `total_s` bound of §6 on both high-r cells measured: 0.26 at $(210, 691)$, $r = 14$, and 1.12 at $(256, 134217757)$, $r = 19$ ([X] v1.10 §6.2 E6 (d)). *Neither is a campaign cell* ($n = 210 > 200$, and $\ell = 134217757$ is not a campaign prime), so under slot-7 owner ruling 4 both are **reported only** and are not among the gating cells below (§6; v1.8.1). The first campaign's own $\ell = 691$, $n \leq 200$ cells reach $r = 15$ (`docs/implementation/slot7/WP-4_rmax_ell691.tsv`). WP-4's operation counter, `HeckeCanonicalFieldValidation` arm H *as of commit* `fa829c7`, which counted the unmodified route, found the shipped route at its own count, a fitted exponent of 4.8–5.6 in r in $\mathbb{F}_\ell$-multiplications (over $r = 10..20$ at $\ell \in \{5, 691, 134217757\}$), with no defect above it. The fix keeps the candidate sequence and every value: it evaluates $u \mapsto u^\ell$ in $\mathbb{F}[X]/(f)$ as a precomputed $\mathbb{F}_\ell$-linear map and multiplies in a flat quotient ring, so the bound above is unchanged and its constant drops, at $r = 19$ from 144.1M (arm H at `fa829c7`) to 12.1M (arm H at this revision) $\mathbb{F}_\ell$-multiplications at $\ell = 691$. (v1.8.1: at this revision arm H counts the fixed route and fits 4.89, 4.23 and 3.87 at $\ell = 5$, 691, 134217757 over $r = 10..20$; it counts the pre-fix route only for $r \leq 10$, as its byte-identity oracle, and fits nothing to it.) After the fix, the slot-7 verdict window W1 measured the worst gating cell, $(170, 691)$ at $r = 14$, at ratio 0.0104 over the 41 gating cells of slot-7 owner ruling 4: **E6 (d) MET**. The two reported cells measured 0.0062 at $(210, 691)$ and 0.0151 at $(256, 134217757)$ in the same window, under the bound but not gating (v1.8.1).)

7 Output schema

Two records exist (v1.4.0). The *per-degree* record `hecke_report/v2` (`Hecke/HeckeReport`, since E0b, slot 2 — the golden-visible writer, about the bulk complex) carries level (c) with, per block, `local`, `witness_poly`, `coefficient_field` = {`w_hat_exponent`, `P`}, the corrected multiplicity note, `level_c.locality` = {`mode`, `budget`, `source`, `complete`, `splits`} and the four degree fields below at record level; it carries no resumable state — the two scalars `complete` / `splits` say whether to look for one, and the state is R3's sidecar material (slot-2 WP-2). The *extension* record `hecke_ext/v2` (E2; `v1` until v1.6.0 of this document — the bump is the record-level `u_ell` and the per-ideal `groups` list below, made before the first emission) is one JSON per (n, ℓ) cell, extending that schema and `Hecke/README.md` (conventions kept: polynomial coefficient lists low→high, matrices as arrays of rows, entries in $[0, \ell)$, `residue_degree` versus `residue_degree_lower_bound` never conflated; hand-rolled writer):

```
{ "schema": "hecke_ext/v5", "ell": 691, "n_sym": 10, "weight": 12, "sign": "+|-|all",
  "commit": "<hash>", "model": "sl2z_7vertex", "iota_sign": "eisenstein_in_H1_plus",
  "settings": { "locality": {"value": "budget=64", "source": "default"},
                "u_ell":    {"value": "...", "source": "default|spec|console"},
                ...,        (every REGISTERED key, in registry order; see below)
                "presentation": {"value": "off", "source": "default"},
                "predictions":  {"value": "annotate", "source": "default"},
                "fingerprint": "0000000000000000" },
  "coefficient_ring": "F_ell", "resumable_items": 0,
  "predictions_mode": "annotate",
  "predictions": {"kummer": [{"k": 12, "applicable": true, "divides": true,
                              "k_reduced": 12, "b_over_k_mod_ell": 0, "reason": "direct"}],
                  "glover_factors": {"schema": "hecke_glover_factors/v1", "n": 10,
                                     "ell": 691, "dimension": 11, "length": 1,
                                     "factors": [{"sym": 10, "det": 0, "multiplicity": 1}]}},
  "anemic_pool": [2,3,5,7,11,13,17,19,23,29,31],
  "s_sep": {"status": "atlas_unavailable"},
  "u_ell": {"included": true, "reason": "within_budget", "predicted_mb": 104.06,
            "budget_mb": 2048, "model_version": "u_ell_rss/v2-2026-09-17"},
  "convention_check": {"tau_mod_ell_ok": true, "eis_eigenvalue_ok": true},
  "complexes": {
    "bulk":     {"q_max": 2, "truncated": false, "q_max_source": "derived", "degrees": [0,1,2]},
    "boundary": {"q_max": 1, "truncated": false, "q_max_source": "derived", "degrees": [0,1]},
    "pair":     {"q_max": 2, "truncated": false, "q_max_source": "derived", "degrees": [0,1,2]} },
  "groups": { "H0": {"dim":0}, "H0_bd": {"dim":1, "status":"unavailable"|"computed"},
              "H1c": {"dim":3, "status":"unavailable"}, "H1": {"dim":3}, "H1_bd": {"dim":1, ...},
              "H2c": {"dim":0, "status":"unavailable"},
              "dim_M_k": 2, "dim_S_k": 1, "dim_dickson": 0, "dim_H1_shriek": 2 },
  "les": { "exact": true, "ranks": {"r0": 0, "r1": 1}, "hecke_equivariant": null },
```

```
"operators_computed": [2,3,5,7,691],
"ideals": [
  { "id": 0, "sign": "all", "residue_degree": 1,
    "residue_degree_certified": true, "local": true,
    "groups": [{"group": "H1", "own": true, "match": true, "from": 0,
                "witness": {"pi_w": [24,1], "N": 2, "N_used": 2}, "failure": null}],
    "canonical": {"H1": {"H1_shriek": "codim 1 (ker r^1; rank r^1 = 1)",
                         "eisenstein": "socle E, 2-dim on the unsplit H1; a line per sign",
                         "dickson": "n/a", "L_image": "n/a"}},
    "residue_witness": [[2,1]], "witness_poly": [24,1],
    "coefficient_field": {"w_hat_exponent": 1,
                          "P": {"2": [667], "3": [252], "5": [684], "7": [531], "691": [1]}},
    "split": {"schema": "hecke_canonical_split/v1", "ok": true, "ell": 691, "r": 1,
              "field_polynomial": [0,1], "witness_poly": [24,1], "candidates_used": ..,
              "rho_canonical": [667], "conjugates": [[667]], "primes": [2,3,5,7,691],
              "eigenvalues": [[667],[252],[684],[531],[1]],
              "frobenius_orbit": [[[667],[252],[684],[531],[1]]]},
    "label": {"status": "atlas_unavailable"},
    "char_Fl": {"2": {"pi": [24,1]}, "3": {"pi": [439,1]}},
    "dims": {"H1": 3, "H1c": null, ...},
    "algebra": {"dim": 2, "embdim": 1, "nilpotency": 2, "gorenstein": true,
                "hilbert": [1,1], "basis": [[..],[..]]},
    "generators_certified": [2],
    "certificate": "cyclic"|"commutant"|"sturm(B')"|"dickson"|"boundary",
    "certificate_routes": {"a": "cyclic|not_cyclic",
                           "b": "excluded_by_socle_bound|done|over_budget",
                           "c": "found|none_at_cap|not_applicable|off",
                           "d0": "passed|failed:<class>|not_applicable:u_ell|not_run",
                           "d1": "passed|not_injective|failed:<p>|not_run:budget|not_applicable:u_ell|not_run"},
    "boundary_reps": [],     (R(ell) iff d1 is passed or failed:<p>; v4)
    "boundary_data": {"rank_r1": D, "r1": [[..]], "t": {"<p>": [[..]], ..}},
                             (present iff d1 = passed; p in boundary_reps; v5)
    "final": true|false|"heuristic", "final_reason": null|"<reason>",
    "sturm": {"B_prime": 2, "cap": 24, "closure_dim": 2, "dim_M_k": 2,
              "mode": "nonfinal", "outcome": "found", "sturm_primes": [2],
              "proof_range": true, "proof_range_source": "cell_local",
              "resumable": {"kind": "sturm_bound", "complete": true, "budget": {"cap": 24},
                            "reached": {"B_reached": 2}, "budget_history": [{"cap": 24}]}},
    "tangent": {"2": "active", "3": "active", "5": "active", "7": "active", "691": "active"},
    "modules": { "H1": {"loewy": [2,1], "socle": [2,1], "cyclic": false, "free": false,
                        "min_generators": 2,
                        "jordan": {"2": [2,1], "3": [2,1], "691": [2,1]},
                        "structure_maps": [{"piece": [0,1], "rank": 1,
                                            "active": [2,3,5,7,691],
                                            "matrices": {"2": [[23,0]], "3": [[120,0]],
                                                         "691": [[221,0]]}}]},
                 "H1c": null, ... },
    "predicted": ["eisenstein_cuspidal:k=12"], "confirmed": [true],
    "unpredicted": [] } ],
"identities": {"passed": 7, "failed": 0},
"certificates_ran": ["locality","exhaustive","cyclic|commutant|sturm","module"],
"hashes": {"A1_T2": "fnv64", ...} }
```

`final` is tri-state (v1.4.0; [X] §1.3.1 R3.2; owner decisions OQ8 / OQ15, slot-1 confirmations 6 and 8). Per ideal, `final` $\in$ {`true`, `false`, `"heuristic"`}. `true` iff `local` is true, every applicable certificate verifies with 0, and the generation certificate is `cyclic` or `commutant`, or `sturm(B')` with B' existing on a non-Dickson $\mathfrak{m}$ of X_4 (Claim 2.10), **or (v1.13.0,** `hecke_ext/v4`**)** `dickson / boundary`: Certificate 5.9 (d0) passed on an X_1 block, or Certificate 5.10 (d1) passed on an X_4 block — each a proof of the same predicate as (a)–(c), so `true` in every mode, the default `closure` included; a failed (d0) / (d1) leaves the reason (a)–(c) produced. `"heuristic"` only under the user-selected modes `final_mode = sturm_heuristic` (the heuristic range: Dickson-type $\mathfrak{m}$, X_2, X_3, X_5, X_6) or `sturm_unavailable = heuristic` (B' missing or cap hit): the ideal's only generation evidence is a heuristic sweep, recorded with the primes actually swept (`sturm_primes`)

and the mode (`sturm.mode`) — never in the default configuration. `false` otherwise, with a reason from the fixed list {`not_local`, `sturm_unavailable`, `sturm_heuristic`, `closure_unreachable`, `certificate_failed:<code or certificate name>`, `budget_exhausted:<kind>`, `interrupted`} (**v1.5.0**, corrected from `certificate_failed:<code>`: the payload is *either* the non-zero clause index a `verify()` returned *or* one of the closed list of certificate names {`cyclic`, `commutant`, `sturm`, `locality`, `module`, `identity`, `dickson`, `boundary`} (the last two v1.13.0; the shipped driver lets a failed (d0) / (d1) fall through to the existing reason, so it emits neither). The name form is not a slip: a certificate *route* may report "the sweep ran and its answer was no" — `certificate_failed:sturm`, emitted by Certificate 5.8's route since slot 2 and pinned by `HeckeExtSturmSweepValidation` — with no `verify()` having failed and so no clause index to give. [C] [C, `sec:recordschema`] item 6(d) carries the same rule; slot-4 WP-1 §6 item 1): `budget_exhausted:locality` is the budget-mode locality survivor (Algorithm 5.1(2)), `budget_exhausted:commutant` the budgeted structured system of Certificate 5.7, `closure_unreachable` the proof that (b) fails (the socle bound, or a computed $\dim C > \dim A_S$) with (a) failing and (c) off its proof range, `interrupted` a strict loop stopped at a step boundary; `not_local` is reachable only through strict mode's interruption path, since a block proved non-local splits. A `budget_exhausted` reason always comes with `resumable.complete = false`; `closure_unreachable` and `sturm_heuristic` are not budget outcomes and carry no state. In the default mode (`final_mode = closure`, `sturm_unavailable = nonfinal`) an ideal that cannot reach closure is *emitted* with `final = false`, its reason and its resumable object, listed in the campaign's `analysis.txt`, and never withheld. `final` never reads `resumable`; no field is branched on by the pipeline that wrote it ([C] rule A6). Through v1.3.3 this paragraph read "a record is final at $\mathfrak{m}$ iff ...; otherwise it is marked `"final": false` with the reason" — a two-valued field with no fixed reason list.

Settings, resumable state, coefficient ring (v1.4.0). `settings` carries every mathematical setting in force with its `source` ∈ {`default`, `spec`, `console`} — `final_mode`, `sturm_unavailable`, `sturm`, `commutant_budget`, `locality`, `field_polynomial`, `split_field_report`, `coefficient_ring`, `predictions`, `presentation`, `u_ell`, `integral_fallback`, and `q_max` where settable ([X] §1.5; owner decision R1: these are the *second* family of switches, each of which may change a published value, as no `ENGINE_*` variable may) — and the `fingerprint` of the non-default entries (0 at the defaults), which enters every extension hash input and the campaign job id (the `_s<hex>` suffix), so that only the default configuration is golden and a golden mismatch can never be explained by a setting. (v1.8.0-pre.1: the list just given is [X] §1.5's inventory of the family; the block carries the *registered* members of it, in the shape stated in item (f) of "The stage-2 keys" below, which corrects the example through v1.7.2.) Every budgeted item writes at its own site `resumable` = {`kind` ∈ {`sturm_bound`, `locality`, `commutant`, `u_ell`, `fill`} (v1.9.0-pre.1: and `ladic_level`, §7.1 (d), for the day a record carries it), `complete`, `budget`, `reached`, `budget_history`, `state` | `state_ref` = {`path`, `fnv64`}} — inline if ≤ 64 KiB, else a sidecar named by the record layer and hashed by the one FNV-1a-64 of `Fnv.h` — and the record carries `resumable_items`, the count of incomplete items ([X] §1.3.1 R3.1); a resume with a higher budget is a new recorded setting, a resume under a changed non-budget setting is refused. **Two rules on the sidecar** (v1.5.0; slot-4 WP-1 §6 items 2 and 3; [C] [C, `sec:recordschema`] item 6(b)): `state_ref.path` is *relative to the directory holding the record*, so that the pair of files moves together, and *the name must be injective* — a record may carry several oversize states of the *same* kind, one `commutant` per ideal, so only the first of a kind may take the bare `<record-stem>.state.<kind>.json` and every further one is disambiguated by an index. A writer that has no stem, and so cannot name a file, inlines the state and says so rather than emitting a path that names nothing. `phase_reached` (v1.5.0; slot-3 WP-3 finding 6): where a route reports the phase it reached — strict locality's `linear` / `berlekamp` / `residue_search` — the record carries the *per-block* value, under `phase_reached` inside that block, *and* the run's high-water mark over every block including the parents a split has since removed, under the separate record-level key `phase_reached_high_water`. They are different facts and neither substitutes for the other; both are emitted only where a route that has phases ran. `groups` **holds groups and four scalars** (v1.5.0, stated because a consumer cannot tell them apart from the shape alone; slot-4 WP-1 §6 item 5): every key of `groups` other than the closed list {`dim_M_k`,

dim_S_k, dim_dickson, dim_H1_shriek} is a cohomology group and carries dim and, where it is not computed, status; those four are integers **or** null. A later schema revision may move them to a sibling object; until it does, the closed list is the rule. **null, never dropped** (v1.6.0; slot-5 WP-1 finding 2; [C] [C, sec:recordschema] item 6(c)). The list is a constant of the schema, but dim_H1_shriek is $\dim \ker r^1$, which does not exist until the boundary complex does. A record that has not computed it emits it as null; dropping the key would make the closed list a function of the cell, which is exactly the property that made it a rule, and emitting 0 would say something false.

canonical **is keyed by the ambient group, at *ideal* level** (v1.6.0; slot-5 WP-1 finding 1). Through v1.5.2 the example above nested "canonical" *inside* modules.H1, while the shipped writer emitted it beside modules; the example is corrected to the writer's placement, and the object is a map keyed by the ambient group ("H1", "H1_plus", ...), which carries the same information, reads like groups, and keeps modules a verbatim per-group ModuleStructure serialisation with no string splicing. A group whose canonical positions were not computed is absent from the map, not null inside it — canonical is keyed by what was computed, not by a closed list.

les.hecke_equivariant **is the *action*'s claim** (v1.6.0; slot-5 WP-1 finding 4; [C] [C, sec:recordschema] item 6(g)). The cone knows nothing about T_q, so a record carrying the cone but no cone action has not evaluated the flag. It is therefore tri-state: true asserted, false evaluated and failed, null not evaluated. Through v1.5.2 the example showed true without saying who asserts it, and a cone-only record emitted false, which under [C] rule A6 reads "evaluated and it did not hold" — the one thing it did not mean. The same rule governs exact at a spot the cone did not assert.

A *resumed* run writes the settings block a *fresh* run at the same effective values would write (v1.5.0; slot-4 WP-2 finding 4; [C] [C, sec:recordschema] item 6(a)). The rule is not implied by the paragraph above, and it decides a fingerprint: an implementation that installs the budget it was resumed *to* makes resume(2048) on a refused U_ℓ emit "u_ell": "budget_mb=2048" where a fresh run at the default emits "u_ell": "auto", so the two records differ in their settings text, their fingerprint and every hash downstream of it although they computed the same thing. The binding rule is the equality: a resume to a value *equal to the default* installs the default's canonical text and any other value installs its own, whereupon "a resume to the default budget is byte-identical to a fresh default run, fingerprint included" is a statement the implementation asserts (HeckeExtRamanujanValidation arm 4, slot-4 WP-2).

u_ell **is carried at *record* level, with a single** resumable (v1.6.0, hecke_ext/v2; the repair v1.5.0 booked, approved as owner ruling [OD] slot-4 close-out 5 of 2026-09-22 and made by slot-5 WP-1). The admission decision of §5 Stage 1 is a property of (n, ℓ) and of the budget, not of a maximal ideal, and the record now says so: **one** u_ell object, carrying its verdict and — when U_ℓ is refused — **one** resumable, beside resumable_items, which is accordingly a count of *decisions*. No ideals[] entry carries a u_ell object. A cell that formed no verdict emits the placeholder "u_ell": {"status": "unavailable"} ([C] [C, sec:recordschema] item 6(c)); an included verdict carrying a resumable, a resumable with no verdict, and a reason outside the enumerated three are each refused by the writer's own verify(). *What v1 did, and why this is a version bump*: through hecke_ext/v1 the object and its resumable were attached to *each* ideals[] entry, so a four-ideal cell emitted one decision four times and resumable_items counted it four times — a count of *entries*, which [C] [C, sec:recordschema] item 6(b) still permits a schema that replicates a run-level verdict, provided it says so. The repair changes the *value* of an existing key rather than adding one, so it is a schema version bump and not an amendment in place, and it was made **before** the first emission of an extension record as a golden object, which is what made it free. A consumer distinguishes the two shapes by the schema string alone.

An ideals[] **entry is per *group*, and since** hecke_ext/v2 **it reports the matching** (v1.5.0; v1.6.0 for the addition; slot-4 WP-2 finding 1, taken by slot-5 WP-1). Stage 2 of §5 matches the per-group decompositions by the cross-block evaluation test ([XA M5], SameMaximalIdeal), so a maximal ideal is in principle an object spanning $X_1, \ldots, X_6$; the writer still emits **one entry per (group, block)**, each with a single modules entry — that shape is unchanged, which is why the addition needed no other change — and an entry now additionally carries a per-ideal groups list. Each element names

the ambient group, whether it is the block's *own* group, the verdict of `SameMaximalIdeal`, the ideal the test was run *from*, and the **witness** it used — the reduced π_w low→high, the exponent N, and the N' actually used on *this* block ([XA M5]: nilpotency is bounded on the block the test evaluates *on*, not on the one it comes from) — with the failure code on a non-match. The entry for the block's own group is the identity match, which is what makes two such lists comparable. Through `hecke_ext/v1` the match count was a driver diagnostic that no key reported; [C] [C, `sec:recordschema`] item 6(i) now makes "a matching a record performed is a matching it reports, with its witness" a rule, on the ground that a count cannot be re-checked by a reader who has only the record. `coefficient_ring` ∈ {`F_ell`, `Z/ell^k`} names the ring of the run (`F_ell` today; E8, owner decision OQ18; v1.9.0-pre.1: the setting of that name, its grammar, its refusal rule and the E8 objects are specified in §7.1; the record's top-level key keeps this meaning and reads `F_ell` in every record of slot 8, since every record writer refuses a non-default ring). `P` is an object keyed by the prime, one reduced polynomial per generator — on a single-operator presentation it has one entry (538/538 blocks of the golden grid) — so that Certificate 5.2's clause "for every j" is checkable from the record.

The stage-2 keys (v1.8.0-pre.1; slot-7 WP-12a, spec first). What follows specifies every key that slot-7 WP-3 (stage 2 of the staged re-baseline, [X] §8) and slot-7 WP-2 (`presentation`) will emit, *before* either emits it; a key not specified here is not emitted ([X] §8, slot-7 plan WP-3 "content check"). Each key has its type, its presence rule and its default. All of it rides on [C] [C, `sec:recordschema`] item 6: rule (c) (a required key not computed is `null`; an unavailable object is a placeholder, never an omission), rule (d) (reasons are named from a closed list), and item 6(f), which leaves the inventory to this section.

(a) `sturm.proof_range` (boolean or `null`) and `sturm.proof_range_source` (string), inside every `sturm` object, so present exactly when `sturm` is. `proof_range` says whether the ideal lies in the range on which Claim 2.10 lets Sturm certify, i.e. whether the ideal is *not* Dickson-type; it records that test alone — the claim's other hypotheses (the group, the existence of B') are not part of this field. The source is one of the closed list:

- `cell_local` — **the default, at every** ℓ (slot-7 owner ruling 3a of 2026-09-23, $\ell \geq 691$ included). Decided *at the cell* by the method of Claim 2.10: the ideal's anemic system over the pool `anemic_pool` below, computed at the cell, is compared with the systems of the same cell's H^0 ideals over the same pool ($\operatorname{tors} H^1(\Gamma; M_n(\mathbb{Z})) \otimes \mathbb{F}_\ell \cong H^0$); `proof_range = true` iff it equals none of them. No atlas is read.
- `caller_input` — the value was supplied by the caller (`ExtDriverOptions::proof_range`) and is not verified. Never the default; a record carrying it says so, and a validation arm that forces a branch uses it.
- `method_undefined` — the method cannot be applied at this cell; the one case known is a missing $B'(k, \ell)$ (Claim 2.9). Then `proof_range` is `null`, and the ideal's `final` is governed by `sturm_unavailable` as before.

`dickson_count` is the value every `hecke_ext/v2` record emitted through slot 6 — a caller input under another name (`proof_range` was the unverified default `true`, the M1 qualification of [X] v1.10 §0) — and is **retired**: no record after stage 2 item 3 emits it, and a reader treats it as `caller_input`. *How the decision reaches* ***`final`***: only through the Sturm route's outcome, which the consumer obtains by re-running that route with the decided range — `final` never reads `proof_range` ([C] rule A6; slot-6 WP-9 finding 4: flipping the flag alone and re-evaluating `final` reports no change, which is wrong). `anemic_pool` (record level; list of primes, ascending; always present from the same item on): the pool $T = \{q \leq 31\} \setminus \{\ell\}$ over which the anemic systems of this item and of (b) are compared. The operators of the pool are computed at the cell for that purpose; `operators_computed` keeps its meaning (the operators the decomposition ran on).

(b) **The atlas label and** `atlas_unavailable` (slot-7 owner ruling 3b (B)). $\Sigma(\ell)$ is built in-process at $\ell \in \{5, 7, 11, 13\}$ (the range `HeckeAtlasValidation` pins) and at no other ℓ; the record never reads a stored `hecke_atlas/v1`, so golden stays self-contained, and making an atlas available at another ℓ later is a new owner-gated event, not a correction. Two keys:

- `label`, per ideal, always present. On a local ideal at an atlas ℓ: {`"status": "computed", "key": <the CanonicalLess-least Frobenius conjugate of the ideal's anemic system over anemic_pool, one element of` $\mathbb{F}_{\ell^r}$ `per prime, low→high>, "r": <r>, "system": <its id in` $\Sigma(\ell)$`>, "type": "eisenstein"|"cuspidal", "chi": [a,b]|null, "base": <id>, "theta_twist": <j>`}, the fields being those of the atlas's system record ([WP6-8] §1: `type` is `eisenstein` iff the system equals some $\chi_{a,b}$ on T, `chi` the least such (a, b), and the system is $\theta^j \otimes$ system `base`). `key` is the label proper — a function of the system alone, which is what identifies systems across n; `system` and `base` are ids in the in-process atlas at the same settings, informative only. At every other ℓ: {`"status": "atlas_unavailable"`} and nothing else. On a non-local ideal, which has no character: `null`.
- `s_sep`, record level, always present: at an atlas ℓ, {`"status": "computed", "primes": [...], "added": [...]`} — $S_{\mathrm{sep}}(\ell)$ ascending, and $S_{\mathrm{sep}}(\ell) \setminus S_0$ ascending, the operators Algorithm 5.1(3) added (empty when $S_{\mathrm{sep}}(\ell) \subseteq S_0$, the vacuous case); the added operators appear in `operators_computed`. At every other ℓ: {`"status": "atlas_unavailable"`}.

(c) **The split's location, stated once** (tier (ii) of §6; [OD] OQ10, E6-3). The per-ideal key `split` is the `hecke_canonical_split/v1` fragment of Algorithm 6.4 *verbatim*, as its one renderer writes it (`schema`, `ok`, `ell`, `r`, `field_polynomial` $= C_{\ell,r}$, `witness_poly`, `candidates_used`, `rho_canonical`, `conjugates`, `primes`, `eigenvalues`, `frobenius_orbit`; elements of $\mathbb{F}$ as coefficient vectors of length r, low→high). It is present **iff** the ideal is `local` **and** `split_field_report = canonical`; its `primes` are the keys of `coefficient_field.P`, ascending, and it is computed through one `CanonicalField` per (ℓ, r) under the `field_polynomial` in force. `ok = false` is a certificate failure of the split and is refused by the record's `verify()`. Under `split_field_report = own_field` the record carries `coefficient_field` (tier (i)) only and no `split` key — the key list is then a function of a recorded setting, as for `presentation`, not of the cell. `candidates_used` makes the fragment depend on the root finder's candidate sequence as well as on ρ_{can}, so that sequence is part of the record (slot-7 plan §2.3 item 2). *Retired*: the example through v1.7.2 showed `field` = {`C_ell_r`, `conway_available`, `rho_can`} and `char_split`; no writer emitted either, and `split` replaces both (`conway_available` returns only with the Conway tables that Claim 6.3 permits the record to name).

(d) `predictions_mode` **and the prediction objects** (Stage 6; [X] §1.5, OQ14). `predictions_mode` (record level, string, always present) is the value of the `predictions` setting in force, `annotate` (the default) or `assert`; it is emitted beside `settings` because golden sees `settings` only as its fingerprint (item (f)). `predictions` (record level, always present) holds the cell's prediction *inputs*, attached before the pipeline runs: `kummer`, a list, ascending in k, of Kummer verdicts {`k, applicable, divides, k_reduced, b_over_k_mod_ell, reason`} — `kummer_verdict` of `HeckePredictions`, `k_reduced` and `b_over_k_mod_ell` `null` unless `reason` is `direct` or `kummer_congruence` — holding at least the weight $k = n + 2$ (Claim 1.1); and `glover_factors`, the `hecke_glover_factors/v1` object of Sym^n under the standard-module convention of Claim 5.13. Per ideal, always present: `predicted`, a list of prediction ids `"<name>:<argument>"` with `<name>` from the closed list {`eisenstein_cuspidal`} at this version (Prediction 4.3; `<argument>` is `k=<w>`, the weight whose verdict predicts it); a new name is an edit of this list before it is emitted. `confirmed`, index-aligned with `predicted`, each `true`, `false` (evaluated and refuted) or `null` (not evaluated). `unpredicted`, the ambient groups X_i of this entry with $\mathfrak{m} X_{i,\mathfrak{m}} \neq 0$ (a non-split piece, §4.2) that no entry of `predicted` covers. Under `annotate` a refuted prediction is recorded and the run goes on; under `assert` it fails the run, and `assert` is used by validations, never by a campaign ([X] §1.5, OQ14).

(e) `presentation` (per ideal; slot-7 owner confirmation 15, audit finding m-5). Present **only** under `presentation = on` (default `off`, so absent from the default record and from golden); then on every ideal entry, the per-sign entries carrying the per-sign presentation. Its value is {`"generator_basis": "block", "relation_basis": "algebra_rref", "fragment": <hecke_presentation/v1>`}: the fragment is `HeckePresentation`'s text verbatim. *Encoding*: a generator index j is an index into the **block basis** — the stored basis of $V = X_{i,\mathfrak{m}}$ in which

the driver computed A_S and the entry's `modules`, the basis the Nakayama lift is taken in ([X] §2.2 Stage 5) — and *not* an H^q-column; the object names that basis in `generator_basis`. A relation is a t-tuple over the RREF $\mathbb{F}_\ell$-basis $b_0, \ldots$ of A_S (its entry $i \cdot \dim A_S + j$ is the coefficient of b_j in a_i), named by `relation_basis`. On a non-local entry the fragment is the unit's `available = false` rendering. The presentation is a derived object of stored bases and is **not** a hash input. Turning it on therefore leaves the record's `hashes` and the golden `ext_ideals` and `ext_certify` hashes unchanged. It does *not* leave the record *text* unchanged. Beyond the added objects, `presentation = on` is a non-default setting, so it moves the `settings` block and its fingerprint (item (f)), and with them every hash of the record text. Remove the objects and restore the setting, and the text is the default text again ([X] §5; `HeckePresentationValidation` arm (k4) asserts all three statements, and arm (k3) that the fingerprint moves). (v1.8.1: precisely, (k4) asserts that the three hashes are equal, that the `settings` member of the record text differs, and that the stripped text with the default `settings` member put back *is* the default text; (k3) asserts that the `presentation = on` settings have a non-zero fingerprint. That every hash taken over the text moves — the golden `ext_json_canonical` / `ext_json_full` texts among them, although golden runs at the defaults only — follows from the second and is not separately asserted.) (v1.8.0; slot-7 WP-2 finding. Through v1.8.0-pre.1 this read "leaves every hash of the record unchanged", which is false of any hash taken over the text.)

(f) **The `settings` shape** (audit finding m-12; reconciled to the writer, `MathSettings::to_json`). `settings` maps each key to an object {`"value": "<canonical text>", "source": "<source>"`} and ends with `"fingerprint": "<16 hex>"`: one entry per *registered* key — exactly `MathSettings::keys()`, in registry order, which is append-only — and then the fingerprint, rendered as sixteen lower-case hex digits (`"0000000000000000"` at the defaults: the "0" of the paragraph above). A member of [X] §1.5's inventory that is not registered is absent until its consumer registers it (`coefficient_ring` is today a top-level key, not a settings entry; v1.9.0-pre.1: once slot-8 WP-3 registers it, it is also the *last* settings entry, appended at the tail of the registry, and the top-level key stays, §7.1 (a)). Through v1.7.2 the example showed {`key: value, ..., "source": {...}, "fingerprint": "0"`}, which no writer produced. *Golden*: after slot-7 stage 1 (1) the golden harness reduces the *top-level* `settings` object of its `ext_json_*` texts to {`"fingerprint": ...`} by a path-scoped redaction, so golden sees only the fingerprint and a key registered at its default moves no golden line; the record itself still carries the full block.

(g) `source` **is binding and closed**: `source` ∈ {`default`, `spec`, `console`}, the three values of [C] [C, `sec:recordschema`] item 6(a) and [X] §1.5; a fourth value is an edit of this document and of [X], never a package decision. `default`: the key was not set. `spec`: set by a spec — a campaign spec, *and the golden harness*, which stamps `spec` on every key it installs (slot-7 owner confirmation 13). `console`: set in a console session.

(h) **The schema string.** Items (a)–(d) add keys to the default record (`anemic_pool`, `s_sep`, `label`, `split`, `predictions_mode`, `predictions`, `predicted`, `confirmed`, `unpredicted`), so under slot-7 owner confirmation 10 the string becomes `hecke_ext/v3`, as the example shows, in the last commit of stage 2 (stage 3); a record emitted before that commit is `v2` and carries none of them except the two `sturm` fields, which v2 already emitted. Every new key is covered by the record's `verify()`.

(i) **The finality certificates:** `hecke_ext/v4` **(v1.13.0; slot-12 S12-FIN, owner-approved 2026-09-26).** Every ideal carries `certificate_routes.d0` and `.d1` (Certificates 5.9, 5.10; closed value lists as in the example) and the list `boundary_reps` (empty unless (d1) reached its representatives), after `generators_certified`; `certificate` may be `"dickson"` or `"boundary"`. The routes run on the unsplit record only (a per-sign ideal carries `not_run`, as for (b) and (c)), at $\ell \geq 5$, n even, and only where (a)–(c) did not pass. `verify()` refuses (code 30) a value outside the lists, a route on the wrong group or sign, `passed` disagreeing with the verdict's input, `not_applicable:u_ell` disagreeing with $\ell \in$ `operator_primes`, `boundary_reps` $\neq R(\ell)$ where (d1) reached them, a certificate name the routes do not carry; and (code 31) a (d0) that does

not re-derive from the stored t_p. The string becomes `hecke_ext/v4`; a `v3` record carries none of these keys, and its `final` is the (a)–(c) verdict. The published `v3` campaign sets (`ext_first`, `ext_u691`) are not re-emitted.

(j) **(d1)'s data: `hecke_ext/v5` (v1.14.0-pre.1; slot-13 WP-7, owner-approved 2026-09-26, slot-13 Q3; slot-12 FIN audit G2).** An ideal with `certificate_routes.d1 = passed` carries, after `boundary_reps`, the object `boundary_data` = {`"rank_r1"`: D, `"r1"`: $r^1|_V$ $(h^1(\partial) \times D)$, `"t"`: {`"<p>"`: $t_p|_V$ $(D \times D)$}, $p \in$ `boundary_reps` in class order} — the block basis of the stored `t_p`, matrices as arrays of rows, entries in $[0, \ell)$; every other ideal carries no such key. `verify()` code 32 re-derives (H1) and (H2) from it (Certificate 5.10). The string becomes `hecke_ext/v5`; a `v4` record carries no `boundary_data` and its (d1) is re-derivable in structure only. No verdict moves: `boundary_data` records what the route decided on, never an input to `final`.

The degree fields are per complex (v1.4.0; [C] [C, `sec:recordschema`] item 5, slot-2 owner confirmation 1). `q_max` (the top degree of *that* complex), `truncated`, `q_max_source` ∈ {`"derived"`, `"set"`} (set only where `truncated` is true, E9) and `degrees` = $[0, \ldots, q_{\max}]$ are carried by every complex a record is about, **never as a maximum over a record**: the per-degree `hecke_report/v2`, about the bulk, at record level (bulk 2 / `false` / `"derived"` / $[0, 1, 2]$ on the model); the extension record inside each complex's object (boundary 1 / `false` / `"derived"` / $[0, 1]$; the pair as the bulk). The campaign's job-level `degrees` is the *reported selection*, a different field that keeps its meaning and is part of `Job::id()`. The writer derives the fields from the complex's own degree count and refuses a record that cannot name its complex.

The example's values are those of §8 at $(10, 691)$, *corrected* (v1.4.0): through v1.3.3 this paragraph read "at a Kummer prime $r^1 = 0$ on the Eisenstein class, so $H^1_! = H^1$ (`"all"`)", which the computation refutes — r^1 vanishes on the socle E of the block but $\operatorname{rank} r^1 = 1$, and $H^1_! = \ker r^1$ has codimension 1 ($\dim H^1_! = 2$, hence $\dim X_3 = 3$ and $X_6 = 0$ by [C, `eq:dimbook`]; §8); the Eisenstein line is $\operatorname{soc}(H^1_+) = \mathfrak{m}H^1_+$, the bottom Loewy layer, *per sign*, and undefined on the unsplit H^1 ((D6)); and T_{691} *is* active ($c_{691} = 542 \neq 0$). The fields `local`, `sturm`, `coefficient_field` (the P_j: the character in $\mathbb{F}_\ell[X]/(\pi_w)$, each P_j the reduced representative $X \bmod \pi_w$ of degree $< r$ — a constant at $r = 1$, never the unreduced X — is not reproducible from the π_q alone), the structure-map `matrices` (rows: bottom-layer basis, columns: top-layer basis), `settings` with its fingerprint, `coefficient_ring`, the per-complex degree fields, `commit`, `model` and `iota_sign` are what make a record reproducible by a third party. `iota_sign` *names the normalisation* of the involution relative to the geometric one — `"eisenstein_in_H1_plus"` means $\iota^* = -\iota^*_{\mathrm{geo}}$, [C] v1.2.0 [C, `def:iota`](iv), owner decision OQ3 of 2026-09-16 ((A3)) — and is true by definition; a third party reproduces every ± field of the record from that one formula. It is not a free parameter and is never branched on.

7.1 The coefficient ring and the E8 objects

This subsection is the specification slot-8 WP-3 and WP-4 build against: the key WP-3 registers, the rule by which everything that does not honour it refuses, and the objects a $\mathbb{Z}/\ell^k$ run computes, with their golden names. It is **spec first**, written before any of it is coded (slot-8 plan WP-2), under the owner's rulings of 2026-09-24 on the slot-8 decisions 2 (E8-P1 … P5 adopted), 3 (item 4 split; 4a in slot 8), 4 (refusal first, over the closure) and 5 (the seven E8 points), each at recommendation (A); nothing here is provisional. *Division of labour*: this document owns the key — its grammar, default, canonical text and refusal rule; [C] owns the mathematics of integral coefficients ([C, `prop:coneintegral`], [C, `cl:bockstein`], [C, `cl:amalgam`], [C, `lem:h2torsion`]) and points here for the key; `src/ModArithmetic/ENGINE_API.md` §13 owns the arithmetic and the elimination (`ModCtx::prime_power`, the Howell form, `kernel_module`, `image_module`, `solve_multi_rhs_module`, `length`, `invariant_factors`); where [X] v1.11's E8 item-3 text and §13 differ, §13 wins (slot-8 plan §2.3 item 11). [L] is the E8 investigation, `docs/implementation/e8_ladic_investigation.md`, whose proposals E8-P1 … P5 the owner adopted. [IB] is the integral-line design brief, `docs/implementation/integral_line_design_brief.md`, whose open decisions OD-n this document cites (v1.9.1: the abbreviation was used from v1.9.0-pre.1 on without being defined). *No record*

carries any of this in slot 8: the objects of (d) are computed by `HeckeReductionValidation` and by the golden harness's E8 points only, and are console-unreachable (rule R19 refuses `golden()` at a non-zero fingerprint, and the pipeline verbs refuse by (b)). Campaign rows and record fields for $\mathbb{Z}/\ell^k$ come with C5, in slot 9.

Pre-release v1.10.0-pre.1 (slot-9 WP-2, spec first). Slot 9 lifts four refusals of this subsection and adds one schema, each specified here before it is coded (slot-9 plan WP-2): `EquivariantCohomologyOnK` at $\ell \in \{2, 3\}$ (slot-9 WP-3; (b3), (c1′)); the boundary and the cone over $\mathbb{Z}/\ell^k$, E8 item 4b (slot-9 WP-4; (f)); the campaign's job loop with the job-level schema `hecke_ring_report/v1` (slot-9 WP-8, C5c; (g)); and the console's ring-aware verbs (slot-9 WP-12, C5a; (h)). The lift is listed member by member in (b1). Every text below marked "v1.10.0-pre.1" is binding from this pre-release; slot-9 WP-15 publishes v1.10.0, recording what was built at the same sites. Written under the owner's rulings of 2026-09-24 on the slot-9 decisions 2, 3 (with (iii)), 4 and 6, each at recommendation (A); nothing is provisional. *Record carriage in slot 9*: the objects of (d) are carried by `hecke_ring_report/v1` (the campaign, under a non-default ring only) and by the console handles of (h); the objects of (f) by the golden harness's E8 points (slot-9 WP-4's event R1, *a/s*) and by the validations only — no record and no console verb carries them in slot 9.

Status at v1.10.0-pre.2 (slot-9 WP-15, phase 1). What the pre-release specified is **built as specified**, with the as-built facts added at their sites, each marked "v1.10.0-pre.2". $\ell \in \{2, 3\}$ (decision 4): slot-9 WP-3 (`97d0a8d`, merged at `b53836f`), (b1), (b3), (c1′); `HeckeBredonTreeRingValidation` (`standard`, 1394 checks). **E8 item 4b** (decision 3 with (iii)): slot-9 WP-4 (`d0aca96`, merged at `8c9faf4`; review repairs `2933b65`, `7ddb555`), (f); the event R1 **landed** at `762dbc3`: the 126 entries of (f) at the seven points, predicate-gated (126 added, 91 tier A and 35 tier B, 0 changed, 0 removed; tier A 11 575 → 11 666, tier B 5866 → 5901, entries 17 441 → 17 567), both determinism digests unchanged by construction and checked before the dump (`docs/matrix-engine/rebaseline-2026-09-24-e8-4b-addonly.md`); measured by `HeckeReductionValidation` (69 122 checks at the base, the 4b arms among them), `HeckeReductionCoverageSlowValidation` and `GOLDEN_STRICT=1 GoldenRegressionValidation`. **The console** (decision 2): C5a, slot-9 WP-12 (`b3cd0d3`, merged at `4bf2289`), (a) and (h); C5b, slot-9 WP-13 (`80eb184`, merged at `53066cd`), which binds the extension layer and refuses a non-default ring throughout; their follow-up `f28a0fa`. The campaign (C5c, slot-9 WP-8) and the sites slot-9 WP-6, WP-7 and WP-9 touch were still being built; the next paragraph records them.

Status at v1.10.0 (slot-9 WP-15, phase 2). Everything the pre-releases specified is **built as specified**, with four recorded deviations of (g) and one amendment of (g11); the as-built facts of this phase are marked "v1.10.0" at their sites. Base `9b004f2`, every slot-9 code package merged. **The campaign** (decision 6): slot-9 WP-8, C5c (`ad65bd9`, with the (g11) amendment `f4fc3d2`, merged at `8853544`); (g) as built, with (b1)'s and (b4)'s campaign rows; `CampaignRingValidation` (`standard`, 693 checks); zero movement at the defaults, measured against the base binary. The merge of WP-8 with WP-9 produced one defect, repaired at `158c0be` as the lead ruled: at $\ell \in \{2, 3\}$ the ring job had named `SmallEllPolicy::Bredon`, which WP-9's new scope guard forbids outside `EquivariantCohomologyOnK`; the job loop now builds its relation list from the raw `ModCtx::prime_power` context (the same value), and the Bredon lift stays with `EquivariantCohomologyOnK` ((b1), the scoping row). **The carries of slot-9 WP-9** (`fb8a63a`, merged at `e078600`), three of them closing open review findings of this subsection: review finding m-1 by `SmallEllPolicyScopeValidation` ((b1)); m-2 by `HeckeBredonU64FlatRouteValidation` ((c1′)); the named guard of `RunPipelineModEll` ((b1), "refuses structurally"); and `EquivariantCohomologyOnK::get_prime()` refusing a composite object, with its one composite caller re-pointed to `ring_context()` ((b1), the constant-fibre constructor). `IsotropyModelDeltaSquaredValidation` (review finding m-4) belongs to line 2, not to this subsection. **The item-4b checks trimmed** (slot-9 WP-4t, `45acc4f`, merged at `7e379bd`; owner-approved, test code only): the $(60, 5)$ cell of the 4b arms moved to `HeckeReductionCoverageSlowValidation`,

and rows implied by kept rows were dropped ((b1), the 4b row); `HeckeReductionValidation` is 68 970 checks at this base (the 69 122 above was the pre-release's base), its 4b arms 306. **Slot-9 WP-6 and WP-7** (the elimination driver under the caller's pass options and the effective-policy cache key; the arena first-fit in $O(\log L)$ and the narrower τ lock) are engine and scheduling changes that move no value, record, golden line or digest; no site of this document names them, and [P] v3.7 records the two that reach the console. **E8's declaration, the next paragraph, is unchanged.** Measured at this base: `GOLDEN_STRICT=1 GoldenRegressionValidation` tier A 11 666 failing 0, tier B 5901 mismatching 0 (re-run at this base by this revision), and the three determinism digests unchanged (slot-9 WP-7's gates on this tree).

Phase E8 is complete as specified (slot-9 decision 3 (iii) (A); [X] E8): **items 1–4b as specified; 4b over $\mathbb{Z}/\ell^k$ by route 1; [C, `obl:coneexact`](vii)'s composite half open** (the two cross-routes and `EquivariantCohomologyOnK`'s relative sub-complex constructor refuse a composite ring, with arms). Two further limits are recorded, not open items of E8's specification: [C, `obl:coneexact`](vi) over the ring asserts the commutator and does not compute the $\nu = 2$ recursion ((f)); and at $\ell \in \{2, 3\}$ only the lengths and invariant factors of the Bredon object exist ((b3), (c1′)). [C] v1.9.0 states the same status ([C, `sec:notimplemented`]). (v1.11.0-pre.2: **superseded** by the amended declaration of the "Status at v1.11.0-pre.2" paragraph; this paragraph is v1.10.0's text, kept.)

Pre-release v1.11.0-pre.1 (slot-10 WP-2, spec first). Slot 10 lifts three refusals of this subsection, amends the console's list, and declines one lift by name, each specified here before it is coded (slot-10 plan WP-2), under the owner's rulings of 2026-09-25 on the slot-10 decisions 3, 4, 6 and 8, each at recommendation (A); nothing is provisional. Every text below marked "v1.11.0-pre.1" is binding from this pre-release; slot-10 WP-15 publishes v1.11.0, recording what was built at the same sites. The sites, and the package each binds:

- **[C, `obl:coneexact`](vii) over $\mathbb{Z}/\ell^k$** (decision 3 (A); slot-10 WP-3): (b1)'s "the lift (slot 10)", the three members that move; (f)'s (vii) bullet, the route over the ring, the regime rule and the acceptance statement. *The declaration on acceptance*, which slot-10 WP-15 writes into v1.11.0 in place of the paragraph above, and not before: "**Phase E8 is complete**: items 1–4b as specified; [C, `obl:coneexact`](vii) over the ring **discharged at every E8 point** ($\ell^k < 2^{32}$): route 1 in every regime, the two `uint32_t` cross-routes below 2^{32}, which follow the `uint32_t` regime rule above it." If slot-10 WP-3 stops at the strict-pair route (its recorded risk; slot-10 plan WP-3), the declaration names route 2 only and (vii)'s route-3 half stays open, as the owner rules.
- **The Bredon object at $\ell \in \{2, 3\}$** (decision 4 (A)): (b3)'s "the Hecke action on the Bredon object is declined", and one sentence in (c1′). No code package lifts anything there; the refusals of `HeckeOperatorOnK` and `HeckeReduction` at 2 and 3 stand, with their arms, and their message keeps pointing to (b3).
- **Line 2** (slot-10 WP-10, WP-13): (b1)'s row "outside the closure", the one specifying sentence those packages need from this document — that the free model's objects are not members of (b)'s closure and carry no refusal, and how their read-only use of `HeckeOperatorOnK` relates to (b). Their mathematics is [C]'s ([C, `sec:integral`], [C, `prop:strictlift`], [C, `lem:bulkperiodic`]) and their design [IB]'s; neither is restated here.
- **The console** (decision 6 (ii) (A); slot-10 WP-12): (h)'s "v1.11.0-pre.1" block, the five structural members and the ring-mismatch rule. The other two items of decision 6 — the member-getter position and the campaign verbs' exit 5 — are console behaviour, not ring semantics, and are specified in [P] v3.8-pre.1 (§6, §8), which this document points to.
- **The survey** (decision 8; slot-10 WP-4): (g)'s one sentence, no schema change.

Record carriage in slot 10 is slot 9's, unchanged: the cross-routes of (vii) are validated equal to route 1, whose objects are the emitted values ((f), the event R1), so no record, golden line or console verb carries a cross-route object; line 2 emits nothing (`[IB]` OD-9 (a)). **Two pointers into [C]**, recorded and not patched (slot-10 plan §4.3; slot 9's M-4 pattern): [C, `obl:coneexact`](vii)'s "**its composite half is open**" and [C, `sec:notimplemented`]'s "(vii)'s composite half is open" are **stale**

from slot-10 WP-3's merge until [C]'s next edit (slot-10 WP-15, [C] v1.10.0, decision 12 (A)), which records the discharge; in that window this subsection and the tree are right. [C] v1.10.0 also carries decision 4's remark beside [C, `prop:strictlift`](4) (below, (b3)); until then (b3) is its only statement. (v1.11.0-pre.2: **both pointers closed** — [C] v1.10.0 restates [C, `obl:coneexact`](vii) and [C, `sec:notimplemented`] over the ring, the window running from `c9684e9` to that version's commit, and carries the remark as [C, `rmk:bredonhecke`].)

Status at v1.11.0-pre.2 (slot-10 WP-15, phase 1). What the pre-release specified is **built as specified**, with one recorded deviation in (f) (route 3's transport) and one amendment of the declaration (below); the as-built facts are marked "v1.11.0-pre.2" at their sites. Base `f0640d0`. **(vii) over the ring** (decision 3 (A)): slot-10 WP-3 (`a4abc39`, merged at `c9684e9`); (b1), (f); `HeckeConeCrossRouteRingValidation` (`standard`, 386 checks, new); three rows of `HeckeReductionValidation`'s arms (X) and (X4b) changed in place, its count unchanged (68 970), and `HeckeReductionCoverageSlowValidation` unchanged (3731); the $\mathbb{F}_\ell$ cone byte-identical (`cone_output_diff.sh`, 0 hunks). **Line 2** (outside the closure): slot-10 WP-10 (`ee5bdd4`, merged at `cb1c3c4`; `HeckeStrictLiftValidation`, 1824 checks), WP-13 (`99c6284`, merged at `3e371a2`; `HeckePeriodicityClassValidation`, 2482) and WP-11 (`94ea619`, merged at `8f82016`, an *a/s* package; `HeckeIntegralActionValidation`, 4601); their mathematics is [C] v1.10.0's ([C, `rmk:strictliftbuilt`], [C, `rmk:integralevidence`]). **The Bredon object at** $\ell \in \{2,3\}$ (decision 4 (A)): nothing lifted, (b3); [C] v1.10.0 carries the reason as [C, `rmk:bredonhecke`]. **The console** (decision 6 (ii)): slot-10 WP-12 (`3c96d63`, merged at `6726075`); (h). **The survey** (decision 8): *v1.11.0*: **not run in slot 10** — slot-10 WP-4, the ℓ-adic survey under $\mathbb{Z}/\ell^2$, is optional and belongs to the owner's window W2, which had not run it when this version was written; it is deferred, and (g)'s sentence says the same. Nothing of (g) moves by it. *v1.11.1*: **run in W2** after the close window (owner's consent, 2026-09-25) and published as `docs/matrix-engine/results/ring_survey_l2/` (owner-signed): $\mathbb{Z}/\ell^2$, $\ell \in \{2,3,5,7,11,13,691\}$, even $n \le 64$, $p \in \{2,3\}$ — 462 jobs, 0 failed (132 `ok_partial`, every one at $\ell \in \{2,3\}$, the Bredon object with (b3)'s refusals recorded). At $\ell \ge 5$, $n > 0$: 43 cells with Dickson classes, of which exactly three carry ℓ^2-torsion, $(30,5)$, $(60,5)$, $(56,7)$ — the loci $n = \ell m$ of [C, `rmk:bocksteinexp`](2) inside the grid, and the probe's cells of its clause (3); the bookkeeping identity of (d) and the universal-coefficient shape $H^0 = \{2\}^h\{1\}^b$, $H^1 = \{2\}^{2\dim M_{n+2}-1+h}\{1\}^b$, $H^2 = 0$ hold on all 160 cells; the K-mode at $K = 2$ is uncertified exactly at the three ℓ^2-torsion cells; the O4 rows of [IB] hold; every job object at the five $\mathbb{Z}/\ell^2$ E8 points equals its golden entry (164 objects). A second optional publication ran in the same window: slot-10 WP-5's `results/ext_u691/` (the 45 `u_ell_over_budget` cells of `ext_first` with U_{691}), an $\mathbb{F}_\ell$ campaign outside this section. Nothing of (g) moves by either. **Lane B** (*v1.11.0*; base of this version `47e3ff9`, every slot-10 code package merged). *Slot-10 WP-7* (`2f423af`, merged at `4560624`; `docs/implementation/slot10/WP-7_report.md`) profiled the cone and the sign split at $(160,691)$ and shipped two levers, both scheduling only: **the ι lever** — `HeckeInvolution::compute`'s fifteen independent checks of [C, `obl:iota`](iii) and the per-degree splits of the $\pm$ decomposition run as host tasks in the serial order, the first failure in task order, so the aborting check and its message are the serial loop's (the sign split $1.80 \to 0.79$ s at $(160,691)$, indicative) — and **L4**, route 2's block rows of `HeckeOperatorOnK` on the consumer pool, with a Fenwick tree and an `Executor` per row, every rank, pattern hash, `ShapeKey` and recorded program the serial loop's. **The cone-certificate lever** (`ConeCertificate::verify`'s twelve span tests as host tasks) is **recorded for slot 11, not shipped**. Evidence that no value moves: `cone_output_diff.sh` against the WP-7 base, 0 lines changed and 0 new; `ext_compare_mini.sh`, 84/84 `ext/` records byte-identical with `commit` redacted; `bench_identity.sh`, 45 cases, 0 differ; the golden file and the four digests unchanged; 0 ThreadSanitizer reports. Which site of this subsection names them: (b1)'s honouring list (L4 is inside `HeckeOperatorOnK`, an honouring member; the ι lever is inside `HeckeInvolution`, a refusing member whose guard and arm are untouched). *Slot-10 WP-8* (`00eb1a3`, `aeb8853`, merged at `7551dcc`; `docs/implementation/slot10/WP-8_report.md`): **the ninth bit** of `pass_options_fingerprint()` (bit 256, set iff the effective device policy is `gpu`, so `hybrid` and `gpu` no longer share a program-cache slot; `ENGINE_API.md` §10.6; `EngineFingerprintDeviceBitValidation`,

`fast`, 8 checks) — a cache key, never a value; and **validation carries**, none a change of specification: `HeckeFreeModelValidation`'s (T6) refuses on the 7-vertex complex's own number of degrees rather than on the caller's label, and arm (R) no longer re-runs two registered names in process (803 → 801 checks); `HeckePeriodicityClassValidation`'s ball arm is a genuinely oriented disc, so the fill reaches a second round (2482 → 2483 checks, with the review's m-9); `HeckeConeCrossRouteRingValidation` asserts route 3's non-vacuity (review m-12, (f) below); and `slot10_close_window.sh`. No site of this subsection names WP-8 except (f)'s check count. Slot-10 WP-6 (the thread pool's cache-line placement, merged at `2e5f073`) and WP-9 (`ENGINE_API.md` §13.12, arbitrary k designed and not built, merged at `4664b40`) move no value and no site of this subsection names them; WP-9 *proposes* for a later version of this document the grammar amendment $k \le 63 \to k \le 126$ of (a), with the cell refusal $\ell^k \ge 2^{64} \to \ell^k \ge 2^{127}$, which v1.11.0 does not adopt — and it confirms that (a) already places the word-size refusal at the cell, by (b3), not in `MathSettings::set()`, which enforces the grammar cap only (its finding F-3, a precision for the plan, not an error here). Measured at this base by the packages' own gates (each on its own tree, the merges verified by step 0): `GOLDEN_STRICT=1 GoldenRegressionValidation` tier A 11 666 failing 0, tier B 5901 mismatching 0; the four determinism digests unchanged.

The declaration, as amended (v1.11.0-pre.2; it replaces the "Phase E8 is complete as specified" paragraph below, which stays as v1.10.0's text, and the declaration quoted in v1.11.0-pre.1's paragraph above, which the slot's independent review found to claim more than was built — `docs/implementation/slot10/WP-14_review.md` finding M-1): "**Phase E8 is complete**: items 1–4b as specified; [C, `obl:coneexact`](vii) over the ring **discharged in its first two sentences at every E8 point** ($\ell^k < 2^{32}$) — route 1 in every regime, the two `uint32_t` cross-routes below 2^{32}, which follow the `uint32_t` regime rule above it; **its third sentence**, the Poincaré-duality adjointness gate of [C, `prop:coneroutes`](3), **recorded as unavailable over the ring**, by the clause's own rule, since the E7 pairing and the contragredient refuse a composite ring." The third sentence was never an E8 item ((f)'s "Not lifted"), so E8's completeness does not wait on it. Two limits stay recorded, not open items of E8's specification: [C, `obl:coneexact`](vi) over the ring asserts the commutator and does not compute the $\nu = 2$ recursion ((f)); and at $\ell \in \{2, 3\}$ only the lengths and invariant factors of the Bredon object exist ((b3), (c1′)). [C] v1.10.0 states the same status ([C, `obl:coneexact`](vii), [C, `sec:notimplemented`]).

Pre-release v1.12.0-pre.1 (slot-11 WP-4, spec first). Slot 11 lifts no refusal of this subsection: everything it builds on the pipeline is $\mathbb{F}_\ell$-only. What follows is specified here, before it is coded, so that (b1)'s "no row may be added by a code package" holds (slot-11 plan §3 "WP-4"; owner rulings of 2026-09-25, every decision at recommendation, decision 3 at (A′)); nothing is provisional. Slot-11 WP-15 publishes v1.12.0 with what was built at the same sites.

- **New members of (b1)'s closure** — each *refuses, by a named guard with an arm, setting and input*, raising `hecke::NotImplementedInThisBuild` naming itself (console exit 5): `HeckeExt/HeckeDicksonTheta`, every entry point (slot-11 WP-5, WP-6; §5.2 (h)); `HeckeExt/HeckeCensusSplit` (WP-7; §7.2 (j)); `HeckeOperatorOnK` constructed with `FibreRepresentation::QuotientL` (whatever the other enumerators do over the ring, `QuotientL` refuses: θ over $\mathbb{Z}/\ell^m$ is slot 12's); and `LocalSystemSL2Z`'s `QuotientL` relation list with the ρ_{Q_n} fibre-matrix builder of `LocalSystemUtils` (a composite `ModCtx` or a non-default setting). `SymN` and `ContragredientN` keep their rows of (b1) unchanged. At $\ell \in \{2, 3\}$ the ring refusal (the default `SmallEllPolicy`) comes first; under the default ring the θ entry points refuse $\ell < 5$ as a precondition (§5.2 (h)). *The arms* live in the packages' own validations (`HeckeDicksonThetaValidation`, `HeckeCensusSplitValidation`), not in arm (X) of `HeckeReductionValidation`, whose file no slot-11 package edits (slot-11 plan §3.0, shared-file rule); the as-built list is WP-15's to copy into v1.12.0, and a member a package finds at its base that this list does not name keeps refusing and is reported.
- **The console** (slot-11 WP-16, *a/s*): `dickson_les` and `theta` refuse a non-default ring at dispatch and again in the lazy compute, and a ring handle through `bind::RequireFieldHandle` — exit 5, a surface of (b4) like every C5b verb ([P] v3.9-pre.1 §3.6, §8).

- **Outside the closure**, as (b1)'s row for line 2 says: slot-11 WP-11's `Integral/Model/StrictLift` for $\mathrm{diag}(1, p^2)$ and $\mathrm{diag}(p, p)$ (explicit `ModCtx`, no setting read); and slot-11 WP-18's 128-bit element type (next item).
- **L1-X under decision 3 (A′).** Slot 11 builds `ENGINE_API.md` §13.12's *step 1 only* (slot-11 WP-18, after WP-12's F-2): the engine's 128-bit composite element type (`ModCtx` over `u128`, the `U128_Montgomery` regime, `u256` products, the primitives, Hensel inversion), `DenseBuf` and the scalar kernels, the §13.6 entry points, `ref::howell` on naive arithmetic, the block model's composite surface — *and its certificate / oracle rule*: the `u128` route equal to the `u64` route entrywise on `HeckeReductionValidation`'s elimination matrices, and arithmetic arms at $\ell^k \geq 2^{64}$ against `RefLinAlg`'s naive arithmetic (`EngineU128CompositeValidation`). It is **value-neutral and has no consumer**: no member of (b1)'s closure, no setting, no campaign cell, no console verb and no golden point reaches it. Consequently **(a)'s grammar $k \leq 63$ and (b3)'s cell refusal at $\ell^k \geq 2^{64}$ do not move in slot 11**, and every ring surface keeps refusing $\ell^k \geq 2^{64}$ exactly as at v1.11.1. The grammar amendment $k \leq 126$ with the cell refusal at $\ell^k \geq 2^{127}$ (slot-10 WP-9's proposal, recorded in the v1.11.0 status above) is **slot 12's**, with L1-X's steps 2–4 (the pipeline route, the grammar, campaign and console, the validation and review), beside [C]'s Hida chapter, where the per-cell certified precision bound is written; a later pre-release of this document specifies it, not this one. *F-1* (a Montgomery-64 `prep` / `mulp` pair in the composite row updates of the `u64` Howell / module route; canonical in and out) and *F-2* (`ExactKthRoot`'s trial division replaced by the ring carried as (ℓ, k), or a deterministic Miller–Rabin; the blocker §13.12 names in `Hecke/CoefficientRing.cpp`) are value-preserving: no value, member, refusal kind or exit code of this subsection moves, and a changed refusal *message* is reported by the package.
- **Record carriage in slot 11**: none. θ's objects are carried by validations only (§5.2 (g)); the census by its own artefact `hecke_census/v1` (§7.2), which reads $\mathbb{F}_\ell$ records only. No record, golden line or digest moves.

Pre-release v1.12.0-pre.2 (slot-11 WP-18c, spec first; the (C2) re-ruling). The owner's clarification of 2026-09-25 (latest) puts *arbitrary* k into slot 11, by the (C2) ℓ-adic digit induction of `ENGINE_API.md` §13.11 (slot-11 WP-18b, built, value-neutral) and a pipeline route on it (slot-11 WP-18c). Pre.1's sentence that (a)'s grammar $k \leq 63$ and (b3)'s cell refusal at $\ell^k \geq 2^{64}$ "do not move in slot 11" is **superseded** by the items below; the rest of pre.1 stands. Every text marked "v1.12.0-pre.2" is binding from this pre-release; slot-11 WP-15 publishes v1.12.0 with what was built. The [C] side is [C, `obl:backend`](iv)'s digit-plane sentence ([C] v1.11.0-pre.2): an element of $\mathbb{Z}/\ell^K$ held as $L = \lceil K/m \rceil$ fixed-width words, L fixed per cell, is not the arbitrary-precision arithmetic that clause excludes.

- **(a) The grammar** (v1.12.0-pre.2). `Z/ell^k` with $2 \leq k \leq 4096$; everything else of (a) is unchanged (spelling, reserved values, default, canonical text, registration; eleven keys stay eleven). The cap is a *parse* cap, stated so that the refusal is reviewable; raising it is an owner ruling, not a code change. Whether a value is executable at a cell is decided at the cell, by the next item.
- **The per-cell rule** (v1.12.0-pre.2; it refines (b3)'s word-size refusal). At a cell (n, ℓ) under `Z/ell^k`: (i) $\ell^k < 2^{64}$ — the shipped native route, *unchanged*, with every one of its refusals (in particular the console's `uint32_t` limit); (ii) $\ell^k \geq 2^{64}$ and $\ell \geq 5$ — the **digit route** below; (iii) $\ell^k \geq 2^{64}$ and $\ell \in \{2, 3\}$ — **refused by name** (exit 5): the Bredon object is the flat-coordinate route, C^q_{eq} is not free there and the precision-split lemma (L1) of §13.11.3 does not apply; line 2 serves those primes; (iv) at a digit cell, a working precision with $K \cdot \lceil \log_2 \ell \rceil > 8192$ bits, or a predicted working set above the slot's memory cap (40 GB, rule 19 of the slot-11 plan), is refused by name, exit 5, the message carrying the predicted figure.
- **The digit route** (`Hecke/LadicCell`; one new consumer, instantiating no existing class template for a new element type). For a digit cell with $n > 0$ even (the K-mode's applicability range, (d); elsewhere the route refuses by name) and requested output precision k it computes, over $\mathbb{Z}/\ell^K$ in digit planes: C^q_{eq} (the kernel module of the relation system, asserted free), X_q, Z^q, the (c1)–(c5) cyclic decomposition of H^q and the invariant factors of d^q, all by the §13.11 entry

points, which return the §13.5–§13.6 canonical forms; $\rho_n(g)$ from the exact integral matrices by the division-free Pascal route of §13.10 F4 (the diag$(\ell, 1)$ split at non-units); Θ_g for T_p, $r = 1$, $p \neq \ell$, and for U_ℓ ($p = \ell$), from the shipped integral coset and τ tables, *both routes built and asserted equal*, with assertions (i)–(iii) of the ring route; $A^{(q)}(T_p)$ on the cyclic summands; and $\mathrm{charpoly}^{1,\mathrm{free}}_{A(T_p)}$ (the division-free characteristic polynomial of the free block, as in (d)). *Honoured* at a digit cell: the lengths and invariant factors of $H^q(\mathbb{Z}/\ell^k)$, the K-mode (r_K at the working level, `ladic_type_certified`, `ladic_type`^{q}), the free characteristic polynomial and the free block of $A^{(1)}(T_p)$ at precision k, and the ordinary idempotent (below). *Refused by name at a digit cell*, exit 5 (members of (b1)'s closure that the route does not reach in slot 11): the Bockstein and the reduction maps of (d) other than truncation, the boundary and the cone ((f)), the pairing and the contragredient, the extension driver, T_{p^r} with $r \geq 2$, the operator family and the decompositions, and every stored basis as a console `Matrix`.

- **The per-cell certified precision bound** (v1.12.0-pre.2; the owner's request). For a requested precision k: $e_{\max}$ is read from the *certified K-mode at word size* ([C, `prop:certifiede`](3)): one shipped native run at $K_0 = \max(2, \lfloor 1 + \log_\ell(n/(\ell+1)) \rfloor + 1)$ ([L] §2.5's observed bound, used only as the starting level), escalated once to the largest word level $K_w = \max\{k' : \ell^{k'} < 2^{64}\}$ if K_0 does not certify; if K_w does not certify either, the cell is **refused, resumable**: "precision insufficient", with the R3 state {`kind` = `ladic_precision`, `budget` = $\{K_w\}$, `reached` = $\{r_{K_w}(d^0), r_{K_w}(d^1)\}$, `complete` = `false`} — never a guessed value. $e_{\max}$ is the largest torsion exponent of `ladic_type`^{1} (asserted: `ladic_type`^{2} has no torsion at $\ell \geq 5$). The working precision is $K = k + e_{\max}$; the digit run *re-evaluates* the stopping rule at K (a second, independent certificate) and checks its invariant factors of $H^q(\mathbb{Z}/\ell^K)$ against the universal-coefficient reading of the certified type. The lengths and invariant factors of $H^q(\mathbb{Z}/\ell^k)$ are that reading at k. A digit record states `precision_requested` = k, `precision_working` = K, `e_max`, `word_level` (the level that certified), `precision_certified`, `digit_width` and `digits_L`; the free characteristic polynomial and every Hecke datum on $\Lambda^1 \otimes \mathbb{Z}/\ell^k$ are certified to precision k exactly when `precision_certified` is true.
- **The ordinary idempotent** (v1.12.0-pre.2; a certified datum, not a choice of m). At $p = \ell$ the route computes, on the free block U ($r \times r$) of $A^{(1)}(U_\ell)$ over $\mathbb{Z}/\ell^k$, $e = -p_0^{-1} W\, s(W)$ with $W = U^{rk}$ and $\chi_W(x) = x^a p(x)$, $p(x) = p_0 + x\, s(x)$, p_0 a unit (Fitting: W vanishes on the nilpotent summand and is invertible on the ordinary one, so $Wp(W) = 0$); e is a polynomial in U_ℓ, hence the ordinary projector $\lim U_\ell^{N!}$. It is certified by $e^2 = e$, $eA(T_p) = A(T_p)e$ for the other primes of the cell, $(U(1-e))^r \equiv 0 \pmod{\ell}$, and $\mathrm{rank}_{\mathbb{F}_\ell}(e \bmod \ell) = \mathrm{rank}_{\mathbb{F}_\ell}(U \bmod \ell)^r$. The depth m a Hida computation needs is supplied per question by [C]'s later Hida section; the route's bound $K = m + e_{\max}$ is its input, and no cap on m beyond (a) and the per-cell rule is imposed.
- **The width switch** (v1.12.0-pre.2). `ENGINE_LADIC_WIDTH` $\in \{32, 64, 128\}$ (default 64) selects the digit word; it is an *engine-family* switch — it cannot change a value (the width oracle matrix of §13.11.7 and the route's cross-width arm prove it), enters no fingerprint, job id or record hash, and appears in a digit record only as `digit_width` and `digits_L`.
- **(g) Digit jobs** (v1.12.0-pre.2). The campaign's ring job loop accepts digit cells: the spec's refusal of $\ell^k \geq 2^{64}$ moves to the per-cell rule (a grid cell of case (iii) or (iv) is refused at spec validation, naming the cell). A digit job writes `hecke_ring_report/v1` with the members (g1)–(g16) as a native job does, and in addition `"route":"digit"` and the object `"precision"` holding the bound's keys; its `modulus` is the text `"<ell>^<k>"`; a residue is written as its signed decimal (the representative of least absolute value) when that fits a signed 64-bit integer, else as the string `"ell-adic:[`a_0`,`a_1`,...]"` of its base-ℓ digits, low to high; the Bockstein and the reduction are recorded refusals ((g14)); at $p = \ell$ the record carries `"hida"`. The precision keys appear on digit jobs only; no native job's record, id, file name or summary line moves.
- **(h) Digit handles** (v1.12.0-pre.2). Under `Z/ell^`k with $\ell^k \geq 2^{64}$, $\ell \geq 5$: `symmetric_power`, `cohomology` and `hecke(H, p)` ($r = 1$) return digit handles. On a digit `Cohomology` handle the members `computed`, `q_max`, `truncated`, `q_max_source`, `degrees`, `coefficient_ring`, `lengths`, `invariant_factors`, `ladic_type_certified`, `r_K` and `ladic_type` honour, and the new member

`precision` (a record of the bound's keys) is honoured on digit handles only; the rest of (h)'s ring members refuse, exit 5. *v1.12.0* (slot-11 WP-14 pass 2, M-1, built): on a digit `Cohomology` handle the field members (`dims`, `C`, `Z`, `B`, `H`, `P`) exit 3, as on every ring handle; every other member not on the digit list exits 5. On a digit `HeckeOperator` handle `coefficient_ring`, `p`, `r` and `charpoly_free` honour (its coefficients as Strings in (g)'s residue form); `A` and every other member refuse, exit 5. At $\ell \in \{2, 3\}$ beyond 2^{64}: the per-cell refusal, exit 5.

Published v1.12.0 (slot-11 WP-15): the two pre-release paragraphs as built. Sources: the reports of slot-11 WP-5, WP-6, WP-7, WP-8, WP-12, WP-18a, WP-18b and WP-18c and the slot's independent review, passes 1 and 2 (`docs/implementation/slot11/`). Nothing specified above is withdrawn except where this paragraph says so; where the build differs from the text, the difference is stated here and is a carry, not a silent change.

- **The θ and census members of (b1)'s closure** (pre.1's first item) *refuse*, as specified, but not every one by a guard naming itself: `HeckeExt/HeckeDicksonTheta` (`DicksonThetaCell`) refuses a non-default ring by its own named guard; `HeckeExt/HeckeCensusSplit` refuses through `HeckeExtDriver`'s guard (the message names the driver); `HeckeOperatorOnK` constructed with `QuotientL` over a ring refuses through its non-`SymN` guard, whose message says "contragredient"; the `QuotientL` relation list and fibre builder carry no ring guard of their own and are reached only through those two refusing entry points. No validation arm drives a ring into `DicksonThetaCell` or the split driver (the $\ell \in \{2, 3\}$ arm of `HeckeDicksonThetaValidation` exists). *Carry to slot 12*: a guard naming itself at each of the four sites and one arm each, with θ over $\mathbb{Z}/\ell^m$.
- **The console** (pre.1's second item): **not built**. Slot-11 WP-16 gave way under the owner's re-ruling of decision 3 (slot-11 plan §10); `dickson_les` and `theta` do not exist, and [P] v3.9 records them as specified, not built.
- **L1-X** (pre.1's fourth item): superseded by pre.2 and by the owner's later rulings (slot-11 plan §10: arbitrary k by the (C2) digit induction in slot 11). The native `u128` layer of `ENGINE_API.md` §13.12 is built (slot-11 WP-18a) as the *third digit width and an oracle*, not as a pipeline route: `EngineU128CompositeValidation` (26 587 checks). F-1 and F-2 are built and value-preserving (slot-11 WP-12; `ModCtxMontgomeryPairValidation`, `CoefficientRingKthRootValidation`); no value, member, refusal kind or exit code moved. One pre-existing undefined shift in `ExactKthRoot` at $k < 2$ (reached by no pipeline caller) was made defined with the observed result.
- **The (C2) engine** (`ENGINE_API.md` §13.11, slot-11 WP-18b): `engine::ladic`, header-only, with its width oracle matrix, `EngineLadicInductionValidation` (1888 checks). Domain $2 \le \ell < 2^{31}$, $K \ge 2$, no engine cap on K.
- **The digit route** (pre.2, slot-11 WP-18c, `Hecke/LadicCell`), as specified, with: (i) the caps of the per-cell rule — 8192 bits and 40 GB — checked at the *largest reachable* working precision $K = k + K_w - 1$ (since $e_{\max} < K_w$) at spec validation and at the console's dispatch, so no job is refused inside its run (slot-11 WP-14 pass 2, m-3; at $\ell = 691$: `Z/ell^814` admitted, 815 refused); (ii) $\ell \ge 2^{31}$ with $\ell^k \ge 2^{64}$ refused by name, exit 5 (the engine's domain); (iii) $n = 0$ or n odd refused by name, exit 5 (the K-mode's range); (iv) `ladic_type`^{2} asserted to have free rank 0 and no torsion, and every X_q residual-checked. `HeckeLadicCellValidation` (891 checks) carries the oracles of `ENGINE_API.md` §13.11.11, the M-0 charpolys at $(10, 691)$ beyond 2^{127}, the bound at $(20, 5)$, $(30, 5)$, $(60, 5)$ ($e_{\max} = 1, 2, 2$), a cross-K and cross-route comparison of $\chi(T_2)$ at those three cells, and the forced-uncertified refusal. *v1.18.0* (slot-18 WP-19b, `hecke::ladic_cell::DigitCellSession`): the route is cut at the Hecke step — steps 1–2 (the cell over $\mathbb{Z}/\ell^K$, prime-independent) run once per cell, the Hecke step per prime $p \ne \ell$ on the shared cell; `RunCell` *is* the session run on the request's primes (one code path), and the console keeps a bounded session cache keyed by the cell. Value-free: `HeckeLadicSplitValidation` (56 checks) compares the shared-cell records with fresh `RunCell` records bit for bit.
- **The ordinary idempotent** (pre.2): as specified. Certified at $(10, 691)$ with U_{691} (fully ordinary on the free block, $e = I$) and at $(30, 5)$ with U_5 ($r = 5$, $\chi_W = x^4 p(x)$, $\operatorname{rank}_{\mathbb{F}_5} e = 1$: a genuine

Fitting split). A digit job's record carries `"hida":{"precision", "rank", "rank_e_mod_ell", "ordinary_rank_mod_ell", "certified", "e"}`, `"e"` being the idempotent's $r \times r$ entries, row-major, as residue Strings in (g)'s form (`Campaign/CampaignRing.cpp`). *v1.12.1* (the slot-11 WP-15 referee's M-2): v1.12.0 listed five keys and omitted `"e"`; this paragraph is the record's only inventory of the object, and (b) makes that inventory binding for a non-default-ring record, so the sixth key is named here.

- **(g) as built**: as specified; a digit job books the Hecke time into `t_cohomology` (a timing field, no value). **(h) as built**: as specified, with the field members exiting 3 (the v1.12.0 sentence in (h) above); a digit `HeckeOperator`'s `charpoly_free` is the Record {`precision, coeffs`} with the coefficients as Strings in (g)'s residue form; and `hecke(H, p)` on a digit handle recomputes the whole cell once per operator — deterministic, so no value moves; a cost, recorded here (slot-11 WP-14 pass 2, m-2).
- **Owner item, recorded, not decided: the console's `uint32_t` gap.** By rule (i) the console keeps the native route's `uint32_t` limit below 2^{64}, so it refuses `Z/ell^k` at $\ell = 5$ for $14 \le k \le 27$ and at $\ell = 691$ for $4 \le k \le 6$ while it serves $k \ge 28$ and $k \ge 7$ there by the digit route; the campaign serves every k. Closing it (the console's `uint64_t` instantiation, or the digit route below 2^{64} at the console) is the owner's to rule (slot-11 WP-14 pass 2, m-4; [P] v3.9).

The Hecke algebra over $\mathbb{Z}/\ell^K$, read mod ℓ (v1.15.1-pre.2; slot-14 WP-6; certified verdict and digit route v1.15.1-pre.5, slot-15 WP-5). *Definitional; computed objects, no certificate beyond the checks named and the certified verdict of (ii), which is [C]'s statement, not one of this subsection.*

(i) **Objects.** For $\ell \ge 5$, $n \ge 1$ even, $K \ge 2$ (since v1.15.1-pre.5 any K: $\ell^K \ge 2^{64}$ by the digit route of (v)), let $M_K = H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}/\ell^K)$ by the module route of this subsection and, per $\mathbb{F}_\ell$-rational joint system $\mathfrak{m}$ of $(T_p)_{p \in S}$, $S = \{2, 3, 5, 7, \ell\}$: the joint idempotent $e_\mathfrak{m} \in \mathbb{Z}/\ell^K[T_p]$ (per-p idempotent polynomials, Newton lift $e \leftarrow 3e^2 - 2e^3$; $e^2 = e$ and $eT_p = T_p e$ are checked), the block $B_K = e_\mathfrak{m} M_K \cong \bigoplus_s \mathbb{Z}/\ell^{b_s}$ by `cyclic_decomposition`, A_K the subalgebra of $\mathrm{End}(B_K)$ generated by the T_p (Howell forms), its length, the Hilbert function of $A_K \otimes \mathbb{F}_\ell$ at $\mathfrak{m}$, A_1 the same on B_K/ℓ (the census algebra A of §7.2), and $\varepsilon_K := K \cdot \dim_{\mathbb{F}_\ell}(A_K/\ell) - \mathrm{length}(A_K)$ ($= \varepsilon_K^{\mathrm{sat}}$ of [C, `prop:satdefect`] at the certified step, where $\dim = \lceil f/2 \rceil$; v1.15.1-pre.6, Fable audit M2 — the text before read $K \cdot \mathrm{rank}_{\mathbb{Z}_\ell}(B_K)$, which the code never computed). At $K_{\mathrm{cls}} = \max(2, e_{\max}+1)$ ($e_{\max}$ from [C, `prop:torsionprepass`]) the summands of order $< \ell^{K_{\mathrm{cls}}}$ are the torsion of $\Lambda_\mathfrak{m}$ and the rest its free part of rank f; on a torsion block the free part is read on ℓ^t-scaled generators, t the largest torsion exponent.

(ii) **The `stable` verdict is certified** (v1.15.1-pre.5; owner ruling Q4, 2026-09-30). Until v1.15.1-pre.4 it rested on `docs/implementation/slot14/WP-M1_loewy_note.md`, Proposition 1, and `stable`, `torsion_free_stable` and `torsion_stable` were *readings, not certificates.* [C] v1.15.0 proves the stopping rule — [C, `prop:saturation`](3) (P15), [C, `prop:satdefect`] (P17), [C, `lem:heckerankone`], [C, `lem:freepart`] — and the route's verdict is [C, `cor:stablecert`], in the form of the WP-4 referee's repair B1: the block $e_\mathfrak{m} M_K$ is the block of a maximal ideal $\mathfrak{m}'$ of $\mathbb{T}' = \mathbb{Z}_\ell[T_p : p \in S]$ (the joint $\mathbb{F}_\ell$-rational system of S), which may be a *sum* of blocks of the full algebra $\mathbb{T}^{\mathbb{L}}$; $\mathrm{rank}\, e\mathbb{T}^{\mathbb{L}} = \lceil f/2 \rceil$ for every idempotent e ($f = \mathrm{rank}(e\mathbb{L}')$, at level 1 at most one Eisenstein system in any block), so $\dim \mathsf{A}'_K/\ell \le \mathrm{rank}\, \mathbb{T}'_{\mathfrak{m}'} \le \lceil f/2 \rceil$, and equality certifies $\mathsf{A}'_K/\ell \cong \mathbb{T}'_{\mathfrak{m}'}/\ell$ — Hilbert function and Loewy length included — at this and every larger K. The Hilbert function *printed* is the route's reading of A'_K/ℓ (its sum and $h_0 = 1$ are checked); the certificate is of the isomorphism, which carries it (v1.15.1-pre.6, audit M3). *What it certifies:* $\mathbb{T}'_{\mathfrak{m}'}/\ell$, the algebra generated by the T_p, $p \in S$; *not* $\mathbb{T}^{\mathbb{L}}_{\mathfrak{m}}/\ell$, which needs the separate generation certificate of [C, `cor:stablecert`](2) that this route does not run. The full algebra of a torsion block with $f > 0$ stays uncertified ([C, `cor:stablecert`](3)); its free part (`torsion_free_stable`, [C, `lem:freepart`](2)) and a pure-torsion block (`torsion_stable`, [C, `lem:freepart`](3)) are certified, as is `reduced_rank1` (the case $K = 1$ of [C, `cor:stablecert`](1)). *The certificate* (`CertifyStable`) re-derives, in exact $\mathbb{Z}/\ell^K$ arithmetic and from the module runs themselves, (1) the block: $e_\mathfrak{m}$ a polynomial in

the T_p (so $e_{\mathfrak{m}} \in \mathbb{T}'$), $e_{\mathfrak{m}}^2 = e_{\mathfrak{m}} \neq 0$, the T_p commuting on the block, $T_p - a_p$ nilpotent on it mod ℓ — which make $e_{\mathfrak{m}}$ the idempotent of $\mathfrak{m}'$ — and, as a consistency check only (v1.15.1-pre.6, audit M1), $\mathrm{rank}_{\mathbb{F}_\ell}(e_{\mathfrak{m}} \bmod \ell)$ the joint generalised eigenspace's dimension; the primes of S are checked prime (audit M4); (2) f and the torsion exponents from the *Smith* invariant factors of $e_{\mathfrak{m}} M_{K_{\mathrm{cls}}}$ (a second entry point beside the cyclic decomposition), and on the whole module $M_{K_{\mathrm{cls}}}$ the summands of order $\ell^{K_{\mathrm{cls}}}$ number $\mathrm{rank}_{\mathbb{Z}_\ell} H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}_\ell) = 2d_k - 1$ while the others are exactly the pre-pass torsion of [C, `prop:torsionprepass`] (so $K_{\mathrm{cls}} > e_{\max}$ and no torsion summand is read as free); (3) $\lceil f/2 \rceil$; (4) A'_K by a second construction, the $\mathbb{Z}$-span of the monomials $\prod_p T_p^{i_p}$, $0 \le i_p < D$ (Cayley–Hamilton on the D-generated block), with its length and $\dim \mathsf{A}'_K/\ell$ (on ℓ^t-scaled generators for a free part); (5) the criterion $\dim \mathsf{A}'_K/\ell = \lceil f/2 \rceil$ ($f = 0$ and $K \ge t$ for `torsion_stable`); (6) $\varepsilon_K = \mathrm{level} \cdot \lceil f/2 \rceil - \mathrm{length}(\mathsf{A}'_K) \ge 0$ and, along the recorded levels, $\varepsilon^{\mathrm{sat}}$ non-decreasing ([C, `prop:satdefect`](2)). Each is compared with the route's reading; *any* mismatch throws (`ENGINE_CHECK`) — the verdict is never silently downgraded. Every `LOEWY_RING` line gains *one* appended field, `stable_cert=certified|reading|none` (`reading`: the certificate not run, `LoewyOptions::certify = false`; `none`: no stable verdict); the old text, `verdict=stable` included, is byte-identical.

(iii) **Refusals.** $\ell < 5$, $n = 0$ or odd, $K < 2$ by `ENGINE_CHECK`; beyond 2^{64} the digit route's per-cell rule (§7.1, v1.12.0-pre.2: $K\lceil \log_2 \ell \rceil \le 8192$, the predicted working set, $\ell < 2^{31}$) by `hecke::NotImplementedInThisBuild`; the Hida module runner (an oracle) beyond 2^{64} by name; a non-local algebra by name.

(iv) **Surface.** `HeckeRingLoewyValidation` (`standard`, 181 checks since v1.15.1-pre.6 (169 at pre.5), 123 before; the Python drivers of `wpm1/` as oracle at $(58, 7)$; arms (C1) the certified verdict on every stable verdict met, (C2) negative controls — a perturbed claim (a wrong n, $K_{\mathrm{cls}} \le e_{\max}$ included) and a perturbed A_K are refused, (C3) the digit route); `HECKE_RING_LOEWY_CELLS="n:ell[:KMAX],..."` (read-only), one `LOEWY_RING` line per block, `LOEWY_RING_SUMMARY` (since v1.15.1-pre.5 with an appended `stable_cert` tally), C-F9 failures *reported* as `LOEWY_RING_CF9_FINDING`, never gated. No record key, no golden object, no setting; `coefficient_ring` is not read (the route takes K itself). Evidence: the counts quoted at Prediction 4.9.

(v) **The digit route** (v1.15.1-pre.5; the carry of [X] v1.19 (6)). At $\ell^K \ge 2^{64}$ the module run is the (C2) digit route of this subsection (`hecke::ladic_cell::RunCell` at exactly the working precision K, its level driver and canonicalisation, its UCT check against the pre-pass type [C, `prop:torsionprepass`]), and the algebra reading runs on `engine::ladic`'s ring, Howell form, length, invariant factors and cyclic decomposition — the §13.5–§13.6 objects in digit arithmetic, no new elimination. Every reading is basis-free, so the two routes give the same `LOEWY_RING` lines wherever both apply ((C3): $(58, 7)$, $(30, 5)$, $(34, 5)$ with the digit route forced below 2^{64}); beyond 2^{64}, at $\ell = 5$, $K = 28$ ($5^{28} > 2^{64}$, the NP5R family), the digit module's readings at $(30, 5)$ and $(34, 5)$ are those predicted from the `uint64_t` route at $K = 26, 27$ by [C, `prop:saturation`](3)–(5) and [C, `prop:satdefect`](2) (the plateau $\varepsilon^{\mathrm{sat}}_{26} = \varepsilon^{\mathrm{sat}}_{27} = \delta^{\mathrm{sat}}$, then $\mathrm{length}(\mathsf{A}_{28}) = 28\,r - \delta^{\mathrm{sat}}$), and `CertifyStable` certifies them.

Status at v1.9.0 (slot-8 WP-13). Everything this subsection specifies is **built as specified**; the text below it is the pre-release's, with the as-built facts added at their sites, each marked "v1.9.0". **E8 item 2** (the composite `ModCtx`, the Howell and Smith elimination, the module entry points of §13) is slot-8 WP-1 (commit `9e02013`); **E8 item 3** (the pipeline over $\mathbb{Z}/\ell^k$, the ρ_n repair E8-P5, the key and its refusals) is slot-8 WP-3 (`9071ab0`), with the repairs of the independent review (slot-8 WP-12, pass 2: findings M-1, M-2, m-6, m-7) as slot-8 WP-3r (`28fdb73`); **E8 item 4a** (the reductions, the Bockstein, the identity of (d), the controls, the (D6) arm) is slot-8 WP-4 (`d96d89d`), **accepted**, and its golden emission R1 **landed** in the merge `53a1909`: the 124 entries of (e) at the seven points, predicate-gated (124 added, 0 changed, 0 removed; tier A 11 472 → 11 575, tier B 5845 → 5866), both determinism digests unchanged by construction (`docs/matrix-engine/rebaseline-2026-09-24-e8-addonly.md`). **E8 item 4b** (the boundary and cone halves, [C, `obl:coneexact`](iv)'s non-field half) is slot 9's (slot-8 ruling 3). What measures it: `HeckeReductionValidation` (`standard`; WP-1's elimination arms,

WP-3's arms (K), (B), (P), (C), (T), (U), (X), WP-3r's arm (M) and WP-4's (R0)–(R5), (B1)–(B3), (D6)), `HeckeReductionCoverageSlowValidation` (`slow`; (R0)–(R4) and (B1)–(B3) at $\mathbb{Z}/\ell^2$ on the E2 case list at $\ell \geq 5$, 52 cells, two of them behind the read-only `HECKE_REDUCTION_COVERAGE_LARGE=1`) and `GOLDEN_STRICT=1 GoldenRegressionValidation` (tier A 11 575 failing 0, tier B 5866 mismatching 0). **Every predicted value of (e) held**; the one literal that did not — footnote b's "51 ≠ 52" — is in another model's coordinates, and is restated below.

(a) The key `coefficient_ring`.

- **Grammar.** `coefficient_ring` ∈ {`F_ell`} ∪ {`Z/ell^`k : $2 \leq k \leq 63$}, spelled exactly so: the literal `F_ell`, or the literal `Z/ell^` followed by k in decimal without sign, leading zero or surrounding space. The token `ell` is literal: the setting is run-level and the prime is the cell's, so at a cell (n, ℓ) the value `Z/ell^2` means $\mathbb{Z}/\ell^2$ ($\mathbb{Z}/25$ at $\ell = 5$, $\mathbb{Z}/691^2$ at $\ell = 691$). The field has exactly one spelling: `Z/ell^1` and `Z/ell^0` are *invalid*, as `prime_power(ell, 1)` is rejected (§13.1). The cap 63 is the largest k with $2^k < 2^{64}$; whether ℓ^k fits a machine word is decided at the cell, by (b3). *v1.12.0-pre.2*: the cap is 4096 (the pre-release paragraph "v1.12.0-pre.2" above); the cell decides, by that paragraph's per-cell rule.
- **Reserved values.** `Z/<m>` with a literal integer $m \geq 2$ (a general $\mathbb{Z}/m$; [X] §1.5) is reserved because §13 does not cover it (it needs gcd pivoting, not valuation pivoting). A reserved value is *legal* — a build-capability statement, not a user error — and `MathSettings::set()` refuses it itself, by raising `hecke::NotImplementedInThisBuild` naming the key and the value (console exit 5): no consumer anywhere honours it. *v1.23.0-pre.1* (D6, audit M2): `Z` (integral coefficients; line 2, [IB] OD-10 (a)) is *no longer* reserved in this sense. By [C, `def:ringsemantics`](3) and [C, `sec:recordschema`] item 7 it is stored, with a non-zero fingerprint, and honoured by the integral descent only (the verb `descent`, the campaign's descent cells, the record `integral_descent_record/v1`); every other consumer refuses it by name with `NotImplementedInThisBuild` (exit 5). The text before v1.23.0-pre.1 — `Z` refused inside `set()`, where it was never in force — is superseded by those two statements of [C]. Every other text is invalid and raises `std::invalid_argument`, as for every key.
- **Default and canonical text.** The default is `F_ell`. The canonical text of a value is its one spelling above; it is what `text(key)` returns, what the settings block carries under `value`, and what enters the fingerprint, as `coefficient_ring=Z/ell^<k>`.
- **Registration.** Slot-8 WP-3 registers the key by the four-edit rule of `Hecke/HeckeSettings`, **appended at the tail** of the registry (after `predictions`; the registry is append-only), making eleven keys, with `source` ∈ {`default`, `spec`, `console`} as for every key (item (g) of "The stage-2 keys" above). The record's *top-level* `coefficient_ring` key stays: it names the ring the record was computed over, and reads `F_ell` in every record of slot 8. *v1.9.0, as built* (slot-8 WP-3): registered exactly so, the eleventh key, with the three arms of `set()` (valid, reserved, invalid); the fingerprint is 0 at the defaults (`HeckeReductionValidation` arm (K), `ConsoleSettingsValidation`; the standing measure is `GOLDEN_STRICT=1 GoldenRegressionValidation` at 0 / 0). A fresh golden emission is byte-identical to the reference, and the 22 console transcripts re-emit unchanged; those two are one-off gate runs of slot-8 WP-3's report §4 (c), not validations (v1.9.1: through v1.9.0 they were attributed to the two arms). At the console the reserved values surface with exit 5 on the `set` statement (slot-8 WP-10's binding seam; `ConsoleErrorValidation` arm (K), `HeckeReductionValidation` arm (X)); `--set coefficient_ring=Z` on the command line is exit 1, as every `--set` refusal is ([P] v3.6.1 §6). Since v1.9.1 (slot-8 WP-13, referee finding M-1) that code is pinned by one check in arm (K) of `ConsoleErrorValidation`. Whether it should be exit 5 is an owner question recorded for C5. *v1.10.0-pre.1: ruled* (slot-9 decision 2 (ii) (A), 2026-09-24; it supersedes the lead default S8-D4). A `--set` argument whose value is *legal but unexecutable* in this build — the reserved `Z/<m>` of this key (and, before v1.23.0-pre.1, `Z`; see (a)), which `MathSettings::set()` refuses with `hecke::NotImplementedInThisBuild` — is **exit** 5 from slot-9 WP-12's commit, as on the `set` statement; arm (K) of `ConsoleErrorValidation` changes its pinned 1 to 5 in that same commit, and the report names the change. An *invalid*

value (`Z/ell^1`, any other text) and an undeclared key stay exit 1. Until that commit, exit 1 as above. The same commit makes a `set`-statement refusal carry the statement's `file:line:col` (the second authorised `Console/Core/` edit of that decision, [P] v3.7). *v1.10.0-pre.2, as built* (slot-9 WP-12, `b3cd0d3`): exactly so. The CLI maps a `--set` refusal of kind `NotImplementedInThisBuild` to exit 5, and an invalid value or an undeclared key to exit 1 (`Core/ConsoleCli`); `Evaluator::exec_set` back-fills the statement's position into the refusal (`set coefficient_ring = "Z"` now prints `file:1:1: not implemented in this build: ...`). Arm (K) of `ConsoleErrorValidation` changed its pin 1 → 5 in the same commit and pins the three codes 5 / 1 / 0 and the position. These are the only two `Console/Core/` edits of slot 9.

- **Fingerprint and golden cost.** At the default the key contributes nothing to the fingerprint, which stays 0; a non-default value makes it non-zero and so makes a new campaign job id, as for every key (the paragraph "Settings, resumable state, coefficient ring" above). **Registering the key at its default, and extending its grammar later to admit `Z`, cost no golden line**: since slot-7 stage 1 (1) golden sees the record's settings block only as its fingerprint (item (f) of "The stage-2 keys"), and a key registered at its default is proven key-additive ([X] v1.11 §1.5). That proof covers a key at its default and nothing else: **a record field added for a ring costs golden lines** — emitted at the defaults it moves the 62 `ext_json_*` lines and the ext digest, as `s_sep` did in slot 7 — **unless it is emitted only under a non-default ring**, whose fingerprint-$\neq 0$ texts are not golden. (Line 2's record keys `q_max`, `truncated`, `q_max_source` and `object` are therefore to be emitted only under `coefficient_ring = Z`; v1.9.0: the owner confirmed this, slot-8 ruling 6d, at OD-10 (a) of `docs/implementation/integral_line_design_brief.md`.) Registration still moves every campaign per-job `settings` block, never an id, a file name or a header.
- **Where the ring is read.** The ring is chosen *inside* the pipeline (§13.8 (1)): an object that honours it reads `coefficient_ring` **once, at construction**, from the `MathSettings` in force on the constructing thread, builds `ModCtx(ell)` for `F_ell` and `ModCtx::prime_power(ell, k)` otherwise, and keeps that context for its lifetime; it never re-reads the setting, and an object never changes ring. The composite route is selected by `is_field()` alone, and the ring travels in the context, never as a bare modulus (§13.1, §13.7).

(b) The refusal rule (normative; slot-8 decision 4). Every construction site and every field-only consumer that does not honour a non-default `coefficient_ring` raises `hecke::NotImplementedInThisBuild` naming itself (console exit 5); none computes over $\mathbb{F}_\ell$ while the setting names $\mathbb{Z}/\ell^k$, and none reports a dimension where a length is meant. Precisely:

(b1) *The closure.* The guarded set is the closure of the pipeline under "constructs or consumes": every construction site of `EquivariantCohomologyOnK` or `HeckeOperatorOnK` outside `*Validation.*`, and every consumer downstream of one whose algebra is a field notion (an RREF-defined basis, a dimension, a polynomial over $\mathbb{F}_\ell$, a primary decomposition, a pairing's perfectness, a residue field). The set is *stated as this rule*, not as a list: slot-8 WP-3 derives its members by grep at its base and reports them, an independent reviewer re-derives them (slot-8 WP-12), and WP-13 copies the list into v1.9.0.

v1.9.0: the list. Derived by grep at `3ab190f` (slot-8 WP-3, report §3), re-derived independently at `06bebdc` (slot-8 WP-12, pass 2, which found one member missing, finding m-5, because it was merged after WP-3's base), and re-derived at this revision's base `5c70e95` (slot-8 WP-13, which found one more construction site that the earlier greps did not list, the campaign's job loop; the campaign surface already covered it). Each arm named below is a member of arm (X) of `HeckeReductionValidation` (36 members, plus the console's exit-code checks); "setting and input" means the member refuses both a non-default setting and a composite input object, with an arm for each.

- **Honours**: `EquivariantCohomologyOnK` and `HeckeOperatorOnK` (arms (C), (T) of `HeckeReductionValidation` at nine cell-levels, the U64_Wide cell $(10, 4294967291)$

over $\mathbb{Z}/\ell^2$ among them); `Hecke/HeckeReduction` (slot-8 WP-4: `red_H^q`, the Bockstein, their certificates; it consumes honouring objects only and checks their rings); `HeckeReductionValidation`, `HeckeReductionCoverageSlowValidation`; the golden harness inside `RunE8Point` only.

- **Refuses, by a named guard with an arm**: `HeckeModuleDecomposition` (and with it `FpPoly`'s consumers); `HeckeOperatorFamily` (T_{p^r}, $T(p,p)$, U_ℓ; setting and input); `ContragredientCell` and `PoincareDualityPairing` (E7); `HeckeCrossChecks` (setting and input); `BarCocycleH1`; `HeckeBoundaryOperator`; `SL2ZBoundaryPair` (setting and input); `HeckeInvolution` (setting and input); `EquivariantCochainComplexMorphismOnK`; `HeckeExt/HeckeBoundary` (setting and input); `HeckeConeStrictPairRoute` and `HeckeConeRelativeRoute`; and, inside the honouring class, `EquivariantCohomologyOnK`'s relative sub-complex constructor (the cone-base constructor is guarded too, without an arm of its own: its one caller is the mapping cone, a surface of (b4)).
- **Refuses, by a named guard, unreachable** (no arm): `cochain_commutes` (`HeckeBoundaryDiagnostics`); its `HeckeBoundaryOperator` and `HeckeOperatorOnK` arguments cannot be built over $\mathbb{Z}/\ell^k$ or under a non-default setting.
- **Refuses structurally**: every site below builds its local system with the field generator, so a prime modulus reaches `EquivariantCohomologyOnK`, whose own guard refuses it by name under a non-default ring. The golden harness outside `RunE8Point`; `LocalSystemSL2Z`'s demonstration functions and `LocalSystemPresets`; `Integral/IntegralCharpolyCrt` (`RunPipelineModEll`, slot-8 WP-7, reached only from `HeckeIntegralCharpolyValidation`), because line 2 is CRT over $\mathbb{F}_\ell$ runs by definition (WP-12 finding m-5; it carries no guard naming itself — *v1.10.0*: since slot-9 WP-9, `fb8a63a`, it does: `RunPipelineModEll` opens with a call of `hecke::RequireFieldCoefficientRing` naming itself, so a caller under a composite ring gets (b2)'s named refusal; the refusal was already total); and, behind their surfaces of (b4), the campaign's job loop (`Campaign/CampaignDriver`; it honours from slot-9 WP-8, below), the `--bench` runners and the console bindings (in part, from slot-9 WP-12, below).
- **Unreachable** under a non-default ring, because every input is built by a refusing member (`HeckeModuleDecomposition`, `HeckeInvolution`, `HeckeExtDriver`, or `HeckeCrossChecks`' own local system): `HeckeSturmSweep`, `HeckeSignSplit`, `HeckeExtComplexRecords`, `HeckeAtlas`, `HeckeTangentPair`, `HeckeLocalAlgebra`, `InducedLocalSystem`; and `kernel.cu`'s legacy benchmark, which runs at the process defaults: the argument-less run installs no setting, and no command-line switch sets `coefficient_ring`. (v1.9.1: through v1.9.0 this read "no command-line switch sets a mathematical key outside `--console` and `--campaign`", which is false. `--bench ext --u-ell` installs `u_ell`, source `spec`, through `ScopedUEll` (`BenchDriver.cpp:1273--1276`). It sets no ring, and `--bench` refuses a non-default ring at `RunMode`'s entry in any case.)
- **The seven surfaces** of (b4), each with its arm.

v1.10.0-pre.1: the lift (slot 9). The rule above is unchanged; slot 9 moves the members below, and no other. Each row names the member, its status through slot 8, its status from the named slot-9 package's commit, and the arm that tests it; "arm (X)" is `HeckeReductionValidation`'s. A member that moves to "honours" is tested both ways: an honouring arm over $\mathbb{Z}/\ell^k$, *and* the refusal arms of its field-only downstream consumers kept. **No row may be added by a code package**: a member found at a package's base that this list does not name — the grep includes the template spellings `EquivariantCohomologyOnKT<` and `HeckeOperatorOnKT<` (slot-8 WP-13, finding 1) — keeps refusing, and the package reports it for slot-9 WP-15 to record.

- **Honours at** $\ell \in \{2, 3\}$ (new row; slot-9 WP-3, decision 4): `EquivariantCohomologyOnK`'s ordinary constructor and `compute_cohomology()`, through the flat-coordinate route of (c1′), with the ring's relation list (`GenerateSL2ZRelationsRing`) and ρ_n over $\mathbb{Z}/\ell^k$ (E8-P5's division-free Pascal route, validated entrywise against the division-free oracle on every g the tested cells touch). It reports `lengths_H^q` and `invariant_factors_H^q` only.

Test: `HeckeBredonTreeRingValidation` (`standard`, new) against the frozen tree model over the same ring (O7, [C, `prop:bredon`](2)). *The lift is scoped* (as built, slot-9 WP-3, `b53836f`): `CoefficientRingContext` honours $\ell \in \{2,3\}$ *only* under the policy argument `SmallEllPolicy::Bredon`, which `EquivariantCohomologyOnK` alone passes; under the default policy it keeps refusing at 2 and 3 by name, and that refusal is its own row of (X). Every other member that honours at $\ell \geq 5$ therefore keeps refusing $\ell \in \{2,3\}$ through the default policy, with an arm; inside `EquivariantCohomologyOnK` the four stored-basis accessors and the six K-mode accessors refuse there, and so do the console's pipeline verbs (exit 5). There, **refuse by name, each with an arm**: `HeckeOperatorOnK` (setting and input: a composite object at $\ell \in \{2,3\}$ can now be built, so the input arm is reachable) and `HeckeReduction` (its four entry points), by `hecke::NotImplementedInThisBuild` (a build capability; M1 of line 2 is slot 10's); and the K-mode accessors of `EquivariantCohomologyOnK` (`ladic_type`, `ladic_type_certified`, `ladic_e_max`, `ladic_level_state`, `r_K`, and the sixth WP-3 guards), each by its named refusal. The K-mode is *undefined* there, not merely unbuilt ([C, `prop:certifiede`](5): the model complex over $\mathbb{Z}/\ell^e$ at $\ell \in \{2,3\}$ carries no integral lattice to certify). This pre-release first specified a named `std::logic_error` for it; WP-3 built the default-policy `NotImplementedInThisBuild` (exit 5), which also satisfies (b2), and WP-15 records which kind stands. The flipped arms are named in (b3). *v1.10.0-pre.2: `hecke::NotImplementedInThisBuild` stands* (exit 5 at the console). Both readings are conformant; the built one is kept because the refusal is made where every other member refuses $\ell \in \{2,3\}$ — by the default policy of `CoefficientRingContext`, or by the accessor's own guard naming it — and because a caller cannot tell "undefined here" from "not in this build" by anything it could do differently; the text of the refusal names [C, `prop:certifiede`](5). The slot's review (WP-14, pass 1; v1.10.1: `docs/implementation/slot9/WP-14_review.md`) recommended recording it so. As built (`97d0a8d`): the ten refusing accessors are the four stored-basis ones (`module_degree`, `C_eq_rank`, `H_module_basis`, `H_class_coordinates`) and the six K-mode ones (`r_K`, `ladic_applicable`, `ladic_type_certified`, `ladic_type`, `ladic_e_max`, `ladic_level_state`); the honouring ones are `num_degrees`, `H_lengths`, `H_invariant_factors`, `ring_context`, `flat_module_degree` and `is_flat_route`; the field accessors keep their named `std::logic_error`. `HeckeOperatorOnK` refuses in `init_composite()`; `HeckeReduction` refuses through the stored-basis accessors, and since slot-9 WP-4 its new cone and boundary entry points carry an explicit guard. Arm (X) of `HeckeReductionValidation` went from 36 to 66 members (two changed rows, 30 new), and WP-4's (X4b) adds 21 rows. *Scoping*: `SmallEllPolicy::Bredon` is a public enumerator; that no caller but `EquivariantCohomologyOnK` passes it is held by that class's comment and by arm (X)'s rows for the callers that exist, not by a guard script (review finding m-1, open). *v1.10.0: closed* (slot-9 WP-9, `fb8a63a`): `SmallEllPolicyScopeValidation` (`fast`) scans the tree and allows the token `SmallEllPolicy::Bredon` only in `EquivariantCohomologyOnK.cpp`, `Hecke/CoefficientRing.{h,cpp}`, validation files and (v1.10.1) the guard's own translation unit `Registry/Slot9/WP9.cpp`, and asserts that the one caller's site still exists. It caught the one violation the slot produced: WP-8's ring job named the policy at $\ell \in \{2,3\}$, a merge interaction of WP-8 and WP-9, repaired by `158c0be` as the lead ruled — the job loop builds its relation list from the raw `ModCtx::prime_power` context (the same value), and the Bredon lift stays with `EquivariantCohomologyOnK` alone, whose constructor rederives the context under the policy. The allow-list did not change.

- **Moves from "refuses" to "honours" at** $\ell \geq 5$ (slot-9 WP-4, E8 item 4b, decision 3 (A) with (iii) (A); objects in (f)): `SL2ZBoundaryPair` (the boundary complex, an `EquivariantCohomologyOnK` on K_{bd}, and the restriction); `EquivariantCochainComplexMorphismOnK`'s *constant-fibre* constructor (the restriction $r = j^*$ with fibre I_{d_V}, both complexes over the same ring; a ring mismatch between domain and codomain is a wrong input and raises a named `std::logic_error`; *v1.10.0*: since

slot-9 WP-9, `fb8a63a`, its composite branch reads the modulus from `ring_context()`, because `EquivariantCohomologyOnK::get_prime()` now refuses a composite object by `ENGINE_CHECK` — on one it held ℓ^k, not a prime — the value is unchanged), its induced maps given in the (c5) coordinates of (c); `EquivariantMappingCone` (the dense module route of (f), selected by `is_field()`; the tape route stays the $\mathbb{F}_\ell$ path); `HeckeConeAction`, **route 1 only**; `HeckeBoundaryOperator` (T_p^∂ on the cyclic summands of $H^q(\partial)$, by `solve_multi_rhs_module`); `Hecke/HeckeReduction`, which grows the reductions and the Bockstein on the boundary and the cone (its $\ell \geq 5$ scope unchanged); the golden harness inside `RunE8Point`, which grows the objects of (f) under the (a/s) event R1 only. Tests: arms of `HeckeReductionValidation` at the four $\ell \geq 5$ E8 cells at $k \in \{2, 3\}$ and at the U64_Wide cell $(10, 4294967291)$ over $\mathbb{Z}/\ell^2$; the E2 case list at $\ell \geq 5$ in `HeckeReductionCoverageSlowValidation`. *v1.10.0: the tiers as trimmed* (slot-9 WP-4t, `45acc4f`, owner-approved; test code only): the $(60, 5)$ cell of the item-4b arms, unchanged in content (both levels, both primes, (4b-B/C/R/H/K), the commutators, $K = 3 \to 2$, the Bockstein), runs in `HeckeReductionCoverageSlowValidation` (`slow`, 99 checks on its own `E8_4B_POINT` line), so `--all` gates the ring Hecke action on the cone and boundary at $(6, 5)$, $(16, 5)$, $(30, 5)$ and $(10, 691)$ and the two word-regime cells, and $(60, 5)$'s values stay gated in `--all` by R1; and rows implied by a kept row were dropped (the `coneexact_iv_certified` and `routes_bit_identical` flag reads, the image-length vector and $\ell(H_c^1)$, implied by `ConeModuleCertificate::verify()` $= 0$ with $H_c^0 = 0$ and $H^2 = 0$, and the boundary's equal lengths, implied by its equal invariant factors). The item-4b arms went from 458 to 306 checks, `HeckeReductionValidation` from 69 122 to 68 970, and the coverage list's 4b part from 1482 to 1232 (`HeckeReductionCoverageSlowValidation` 3882 $\to$ 3731). Each of these refuses at $\ell \in \{2, 3\}$ and wherever $\ell^k \geq 2^{64}$ by the same named error, with an arm. *v1.10.0-pre.2, as built* (slot-9 WP-4, `d0aca96`; the closure re-derived by grep at `b53836f`, the template spellings included: guard sites in 28 files, construction sites in 13 non-validation TUs, the same set as this list). Every member above honours, and **only** when its inputs are over the ring in force; every other combination keeps slot 8's named refusal, and every field accessor of an honouring object throws a named `std::logic_error`. **The width**: the route itself — `ModuleConeT`, `ModuleConeActionT`, `HeckeBoundaryOperatorT` — is templated on the ring context's element type and runs in **every** word regime (the `U64_Montgomery` cell $(10, 134217757)$ and the `U64_Wide` cell $(10, 4294967291)$ over $\mathbb{Z}/\ell^2$ run it end to end); the four *classes* `EquivariantMappingCone`, `SL2ZBoundaryPair`, `HeckeConeAction` and `EquivariantCochainComplexMorphismOnK` are `uint32_t`-instantiated, as they always were, so they honour $\ell^k < 2^{32}$ and refuse above it by the ring context's named refusal "in the `uint32_t` instantiation" — a state no caller reaches, because a `uint32_t` bulk object cannot exist there either. At $\ell = 5$ the cone's reductions to $\mathbb{F}_\ell$ run (the $\mathbb{F}_\ell$ cone class being `uint32_t`); at the two word-regime cells they do not, and the dense route, the boundary, both Hecke actions and the boundary reductions do. `SL2ZBoundaryPair` builds the circle model over the ring with a new generator, `GenerateSL2ZBoundaryRelationsRing`, under the default `SmallEllPolicy`. Tests: `HeckeReductionValidation` arms (4b-B), (4b-C), (4b-H), (4b-R), (4b-K) and (X4b).

- **Keeps refusing, with an arm** (decision 3 (iii) (A), slot-9 plan M-3): `HeckeConeStrictPairRoute`, `HeckeConeRelativeRoute`, and `EquivariantCohomologyOnK`'s relative sub-complex constructor. So [C, `obl:coneexact`](vii) over $\mathbb{Z}/\ell^k$ is **route 1 only**: the agreement of the three routes is asserted over $\mathbb{F}_\ell$ as before, and **(vii)'s composite half is open**, recorded here and in [C] by slot-9 WP-15 (v1.11.0-pre.2: closed for the first two sentences of (vii) by slot-10 WP-3 — "the lift (slot 10)" below and the declaration of the status paragraph). Also kept: `EquivariantCochainComplexMorphismOnK`'s `FiberBlockTable` constructor (its callers are the involution and the E7 objects, field notions); `HeckeExt/HeckeBoundary` (setting and input: it consumes `SL2ZBoundaryPair`, which now honours, and keeps its own guard);

every other member of the list above. `EquivariantCohomologyOnK`'s protected (prime, rep_dim) constructor keeps its guard: no class derives from `EquivariantCohomologyOnKT` at the slot-9 base (grep), so it is unreachable, and the cone's module route does not construct through it — a package that must, stops and reports. *v1.10.0-pre.2, as built* (slot-9 WP-4): all kept; the module route did not need the protected constructor. The two cross-routes now also refuse a composite *normative action* handed to them (their guard gained the input argument), with an arm each in (X4b), beside the pre-existing (X) rows.

- **Moves from "unreachable" to "refuses, with an arm"**: `cochain_commutes` (`HeckeBoundaryDiagnostics`). Once `HeckeBoundaryOperator` and `HeckeOperatorOnK` can be built over $\mathbb{Z}/\ell^k$, its named guard is reachable, and slot-9 WP-4 gives it an arm in (X). *v1.10.0-pre.2*: done, a row of (X4b).
- **Moves from "refuses structurally, behind its surface" to "honours"**: the campaign's job loop (`Campaign/CampaignDriver`; slot-9 WP-8, C5c, decision 6), under the schema of (g); its ext pass keeps refusing, at spec validation (b4). *v1.10.0, as built* (slot-9 WP-8, `ad65bd9`; merge fix `158c0be`): the job loop honours $\mathbb{Z}/\ell^k$ at every ℓ with $\ell^k < 2^{64}$ (`uint32_t` below 2^{32}, `uint64_t` above). At $\ell \in \{2, 3\}$ it carries the Bredon object's lengths and invariant factors, built by `EquivariantCohomologyOnK` (the job loop hands it a relation list over the raw `ModCtx::prime_power` context and names no policy), and records `hecke` (`HeckeOperatorOnK`), `bockstein` and `reduction` (`HeckeReduction`) as refused; at $r \geq 2$ it records `hecke` refused at its own dispatch. Tested by `CampaignRingValidation` arms (A), (T), (B), (R), (S).
- **Moves in part**: the console bindings (slot-9 WP-12, C5a, decision 2): exactly the verbs and members of (h) honour; every other pipeline verb keeps exit 5, through a lazy handle too (`ConsoleErrorValidation` arm (K), extended). *v1.10.0-pre.2, as built* (slot-9 WP-12, WP-13, `f28a0fa`): exactly (h)'s list honours at $\ell \geq 5$; every other member refuses through one mechanism (`bind::GateTypeOnRing`), so a member added later refuses until (h) lists it; every C5b verb that runs the pipeline refuses a non-default ring at dispatch and again in its lazy compute, and every C5b verb handed a ring handle refuses it at dispatch (`bind::RequireFieldHandle`), exit 5. Arm (K): 61 single-access scripts under $\mathbb{Z}/\ell^2$ at $(10, 691)$ — 25 honour, 8 exit 3, 28 exit 5 — and five rows at the uncertified $(30, 5)$; 210 checks at the base.

v1.11.0-pre.1: the lift (slot 10). The rule above is unchanged, and so is slot 9's rule for a lift: each row names the member, its status through slot 9, its status from the named slot-10 package's *lift commit* (the last commit of that package's branch, made after this pre-release merges; slot-10 plan §4.2), and the arm that tests it. A member that moves is tested both ways; **no row may be added by a code package** — a member found at a package's base that this list does not name keeps its status (the grep includes the template spellings `EquivariantCohomologyOnKT<`, `HeckeOperatorOnKT<`, `ModuleConeT<` and `ModuleConeActionT<`), and the package reports it for slot-10 WP-15 to record.

- **Moves from "keeps refusing" to "honours" at $\ell \geq 5$, $\ell^k < 2^{32}$** (slot-10 WP-3; decision 3 (A)): `HeckeConeStrictPairRoute`, `HeckeConeRelativeRoute`, and `EquivariantCohomologyOnK`'s *relative sub-complex constructor* (the refusal flipped is its `ring_k_ >= 2` guard). Their route over the ring, and the acceptance, are (f)'s (vii) bullet. *Regime rule* (slot-10 plan audit M-3): both cross-routes consume the `uint32_t`-instantiated `HeckeConeAction` and are **not templated**; they honour exactly where that class honours, $\ell^k < 2^{32}$, and above it refuse by the ring context's named refusal "in the `uint32_t` instantiation", as every `uint32_t` class of the slot-9 row above does — with **one arm** at a 4b word-regime cell (`U64_Montgomery` $(10, 134217757)$ or `U64_Wide` $(10, 4294967291)$ over $\mathbb{Z}/\ell^2$), where route 1 (`ModuleConeActionT`) runs and routes 2 and 3 refuse. The relative constructor is a member of the `EquivariantCohomologyOnKT` template and follows its object's element type; it honours in both instantiations at $\ell \geq 5$, $\ell^k < 2^{64}$, but only its `uint32_t` instantiation has a consumer (route 3). *Kept*: each of the three refuses at $\ell \in \{2, 3\}$ by name (the relative constructor included: its $\ell \geq 5$ module

route does not apply there, and no flat-coordinate relative route is specified), with an arm; the **cone-base constructor** of `EquivariantCohomologyOnK` keeps its guard (the module route of (f) does not derive from it), and so do the protected (prime, rep_dim) constructor, `EquivariantCochainComplexMorphismOnK`'s `FiberBlockTable` constructor and `HeckeExt/HeckeBoundary`; [C, `prop:coneroutes`](3)'s adjointness over the ring is **not** lifted (it needs the E7 pairing, which refuses a composite ring, (b1)'s list). The two cross-routes' *input* guard (a composite normative action under a non-default setting at $\ell \in \{2,3\}$ or above 2^{32}) keeps its (X4b) rows; their *honouring* rows change from "refuses" to "honours" and are reported as *changed* rows, with the reason. Tests: `HeckeConeCrossRouteRingValidation` (new, slot-10 WP-3; its arms are (f)'s), and arms (X)/(X4b) of `HeckeReductionValidation`, whose check count moves only by the changed rows. *v1.11.0-pre.2, as built* (slot-10 WP-3). One guard, `CrossRouteRingGuard`, in both cross-routes: they honour iff the normative action is composite ($k \geq 2$), its k equals the k in force, and `CoefficientRingContext<uint32_t>` admits ℓ^k (the regime refusal); every other combination is slot 8's `RequireFieldCoefficientRing` refusal, as before. The relative sub-complex constructor honours at $\ell \geq 5$ and refuses at $\ell \in \{2,3\}$ by name ("... the cone and its route 3 are not honoured at ell in {2, 3}"); the cone-base guard is untouched. Additive API, all within the three members: a static `RequireRingRegime(ell, k)` on each route; a *validation seam* constructor of the strict route taking an external strict table, non-owning and re-certified in `compute()` (so it cannot make an uncertified table look agreed), used only by the negative control; the relative route's `*_module` accessors, its field accessors now throwing a named `std::logic_error` on a composite route. No member outside the list was found at the base, so no row was reported for this version to record. Arms (X4b) and (X) of `HeckeReductionValidation` (in `HeckeReductionConeValidation.cpp` and `HeckeReductionPipelineValidation.cpp`): three rows changed in place from "refuses" to "honours", each with its reason in the code, the count unchanged; the input-guard rows stay. *v1.11.0* (slot-10 WP-7): **none** in the cone classes (`EquivariantMappingCone*`, `HeckeConeAction`) or in `HeckeSignSplit` — the cone-certificate lever was recorded for slot 11, not shipped. Two levers landed elsewhere in the closure, both value-preserving: L4 in `HeckeOperatorOnK`'s route 2 (an honouring member; rows run concurrently, every key and program the serial loop's, and the E8 golden points, computed through it over the ring, unchanged under `GOLDEN_STRICT=1`), and the ι lever in `HeckeInvolution` (a refusing member; only the order of independent checks on its field path, its guard and arm untouched). Evidence: the status paragraph's lane B.

- **Keeps refusing at $\ell \in \{2,3\}$, now for a stated reason** (decision 4 (A)): `HeckeOperatorOnK` (setting and input) and `HeckeReduction` (its entry points), by `hecke::NotImplementedInThisBuild`. The reason is (b3)'s "v1.11.0-pre.1" block; the rows, arms and messages are unchanged.
- **Outside the closure: line 2's free model** (slot-10 WP-10, WP-13; slot-9 WP-10, WP-11). `Integral/Model/IsotropyModel`, `Integral/Model/IntegralGroups` and slot 10's `Integral/Model/StrictLift` (τ_{Iso}, $\widehat{\Theta}_g$) and `Integral/Model/PeriodicityClass` (e, E_q) — and slot-10 WP-11's `Integral/Model/IntegralHecke`, if it lands — are **not members** of (b1)'s closure: none constructs `EquivariantCohomologyOnK` or `HeckeOperatorOnK` outside `*Validation.*`, none consumes one, and each takes its ring as an explicit `engine::ModCtx` argument and reads no mathematical setting (no settings key, no record key, no golden object). `coefficient_ring` therefore neither selects nor refuses their ring, at any ℓ, 2 and 3 included; what they compute over $\mathbb{Z}/\ell^k$ at $\ell \in \{2,3\}$ is $H^\bullet(\Gamma; V_R)$ ([C, `prop:isomodel`](3)), not the Bredon object of (b3), and they do not refuse there. *The read-only use of an honouring member* (slot-10 WP-10 scope 4): the strict cochain map Θ_g^s is `HeckeOperatorOnK` built, inside `HeckeStrictLiftValidation`, on the `BoundaryStrict` τ table (τ^s, integral, solve regime `"Z"`), over $\mathbb{F}_\ell$ at every ℓ and over $\mathbb{Z}/\ell^k$ at $\ell \geq 5$; that is `HeckeOperatorOnK` honouring as listed, on another admissible table. **No row moves for it**: if the composite constructor refuses that ta-

ble, the refusal stands, slot-10 WP-10 does not edit the class, and its $\mathbb{Z}/\ell^k$ arm drops to $\mathbb{F}_\ell$ and is reported. Over $\mathbb{Z}/\ell^k$ at $\ell \in \{2,3\}$ no Θ_g^s exists in this build (decision 4, (b3)); the strict compatibility of [C, `prop:strictlift`](3) there follows from the exact $\mathbb{Z}$-certificate $\varepsilon \circ \tau_{\mathrm{Iso}} = \tau^s \circ \varepsilon$ over $\mathbb{Z}$, the coset transport [C, `eq:cosettransport`] being functorial in the ring, and is not separately checked. *v1.11.0-pre.2, as built* (slot-10 WP-10, WP-13, WP-11). As specified: `Integral/Model/StrictLift`, `PeriodicityClass` and `IntegralHecke` take an explicit `engine::ModCtx` (`uint32_t` or `uint64_t`), read no setting, add no key, record or golden object, and construct pipeline objects only inside `*Validation.*`; `IsotropyModel` and `IntegralGroups` are unedited. WP-10's step 1 found that `HeckeOperatorOnK`'s composite constructor **accepts** the `BoundaryStrict` table at $k \geq 2$ (its three refusals — the Bredon primes, the contragredient, a τ built by the mod-ℓ fallback — do not apply: the table's solve regime is `"Z"`), so the $\mathbb{Z}/\ell^k$ arm ran at $\ell \in \{5,7\}$, $k \in \{2,3\}$, and no row moved. The sentence above is **completed** (the slot's review, `docs/implementation/slot10/WP-14_review.md`, finding m-1): the identity over $\mathbb{Z}/\ell^k$ at 2 and 3 follows from the proof of [C, `prop:strictlift`](3), a formal computation valid over every commutative ring, whose *three* inputs are certified over $\mathbb{Z}$ or hold by construction — the strict identity $\varepsilon \circ \tau_{\mathrm{Iso}} = \tau^s \circ \varepsilon$, the twisted equivariance (T1) of τ_{Iso} (which makes the coset transport well typed, [C, `lem:cosettransport`](1)), and the Γ-equivariance of ε; the certificate alone is one of the three ([C] v1.10.0 [C, `rmk:strictliftbuilt`](3)).

- **Moves in part: the console** (slot-10 WP-12; decision 6 (ii) (A)): the five structural members of (h)'s "v1.11.0-pre.1" block move from "refuses" (through `bind::GateTypeOnRing`) to "honours" on a ring handle at $\ell \geq 5$; every other unlisted member keeps exit 5; arm (K) of `ConsoleErrorValidation`. *v1.11.0-pre.2, as built* (slot-10 WP-12): as listed — (h)'s "v1.11.0-pre.2" block.

(b2) *Each member is exactly one of three.* It **honours** the ring, with a test over $\mathbb{Z}/\ell^k$; or it **refuses**, by one constructor-entry check raising `hecke::NotImplementedInThisBuild` whose text names the class and the canonical value (for example `"<Class>: coefficient_ring=Z/ell^<k> is not honoured in this build"`), with a test arm; or it is **unreachable** under a non-default ring, with the reason. None is silent. A refusing member refuses when the setting in force is non-default *or* when an input object it is handed was constructed over a composite ring: by (a), an object's ring is its own, whatever is in force when it is consumed.

(b3) *What honours it in slot 8.* `EquivariantCohomologyOnK`, `HeckeOperatorOnK`, `HeckeReductionValidation`, the golden harness **inside** its E8 points only (`RunE8Point`, which installs the ring around the E8 computations and nowhere else, (e)), and optionally one `--bench` switch of WP-3's choosing. An honouring object refuses, by the same named error, at a cell where it cannot honour the value: at $\ell \in \{2,3\}$ (there $6 \notin R^\times$, and [C, `prop:coneintegral`] (2)'s isomorphism $C \otimes R \cong C_R$ is not available, so a $\mathbb{Z}/\ell^k$ run is not the reduction of one integral complex; the finer model is line 2's), and wherever $\ell^k \geq 2^{64}$ (§13's scope; the ℓ-adic induction E8-P6 is deferred) — never by an `ENGINE_CHECK` abort. Over $\mathbb{Z}/\ell^k$ no honouring object reports a `dim`: `length` (§13.6) replaces it, and a dimension accessor on a composite object throws a named `std::logic_error`. *v1.9.0, as built.* WP-3 chose **no** `--bench` switch, so `--bench` refuses every mode (arm (X)). The honouring object's refusals are arms of (X): $(10,3)$ under $\mathbb{Z}/\ell^2$, and, through `CoefficientRingContext`, $\ell = 2$, a `u32` overflow and $\ell^k \geq 2^{64}$. The range $\ell^k \in [2^{63}, 2^{64})$ (U64_Wide) is *honoured and exercised* end to end by the cell $(10, 4294967291)$ over $\mathbb{Z}/\ell^2$ (slot-8 WP-3r, on review finding M-2); no narrowing of this item is needed. The refusal at $\ell \in \{2,3\}$ has its mathematics in [C] v1.8.0: [C, `def:ringsemantics`](2) (the object at $\ell \in \{2,3\}$ is the Bredon one, not a reduction of $C(\mathbb{Z})$) and [C, `prop:certifiede`](5); it was WP-2's choice, put to the owner in WP-2's report, and not objected to (v1.10.0-pre.1: **superseded** by slot-9 decision 4 (A), which supersedes the lead default S8-D2; see the next paragraph). The converse caller error — a local system over ℓ^j, $j \geq 2$, handed to `EquivariantCohomologyOnK` while `F_ell` is in force — raises a named `std::logic_error`, not `NotImplementedInThisBuild`: it is a wrong input, not a build capability (review finding m-6; a member of arm (X)).

v1.10.0-pre.1: $\ell \in \{2,3\}$, and what honours in slot 9 (decision 4 (A); slot-9 plan audit B-1, M-4). From slot-9 WP-3's commit, `EquivariantCohomologyOnK` **honours** $\mathbb{Z}/\ell^k$ at $\ell \in \{2,3\}$ (with $\ell^k < 2^{64}$ and the element-type rule unchanged). The object it computes there is the one [C] names: the Bredon cohomology over $\mathbb{Z}/\ell^k$ ([C, `def:ringsemantics`](2)), which is the tree complex [C, `eq:treemodel`] over the ring in force ([C, `prop:bredon`](2)–(3)) — not a reduction of $C(\mathbb{Z})$ and not group cohomology ([C, `prop:ringinforce`](2)–(3)). It reports the **lengths and invariant factors** of H^q for $q = 0, 1, 2$, by the flat-coordinate route of (c1′), and nothing else: **the certified K-mode refuses there** ([C, `prop:certifiede`](5); by its named refusal, (b1)), and `HeckeOperatorOnK`, `HeckeReduction`, the boundary and cone classes of (f) and every field-only consumer **keep refusing** at 2 and 3 with `NotImplementedInThisBuild`, each with an arm. Its oracle is **O7**, the tree model over the same ring ([IB] §6.2 M0 item 3), frozen from the probe's `model_R` in `HeckeBredonTreeRingValidation`; an in-process tree model over $\mathbb{Z}/\ell^k$ reported beside it shares the Howell code with the pipeline and is a consistency check, not an oracle ([IB] K6). **The flipped arms** (M-4). Two as-built refusal rows of (X) name $\ell \in \{2,3\}$, and both *flip under `EquivariantCohomologyOnK`'s policy* (`SmallEllPolicy::Bredon`) in slot-9 WP-3's commit (`b53836f`), each reported there as a *changed* row: "`CoefficientRingContext` at $\ell = 2$" (the one the slot-9 plan names; under the default policy the refusal stays, as its own row) and "`EquivariantCohomologyOnK` at $(10, 3)$" under $\mathbb{Z}/\ell^2$ (the paragraph above's "$(10, 3)$"), which now honours and asserts O7. The $\ell = 3$ refusal they carried moves to new rows for `HeckeOperatorOnK` and `HeckeReduction` at $(10, 3)$, and for the K-mode accessors there. The `u32`-overflow and $\ell^k \geq 2^{64}$ rows keep refusing. **A pointer into [C]**: [C] [C, `def:ringsemantics`](2)'s clause "the implementation of this build refuses it there" is superseded by this item from v1.10.0-pre.1 — in the tree, from slot-9 WP-3's merge — until [C]'s next edit (slot-9 WP-15), which repairs it under slot-9 decision 10 either way; the window is recorded, not closed by a second [C] editor (slot-9 plan §4.3). *v1.10.0-pre.2*: the window is closed — [C] v1.9.0 (slot-9 WP-15) replaces the clause by what this build computes at $\ell \in \{2,3\}$ and what it refuses. *What honours in slot 9*, beyond slot 8's list above: `EquivariantCohomologyOnK` at 2 and 3 (WP-3); the boundary and cone members of (f) at $\ell \geq 5$ (WP-4); the campaign's job loop (WP-8, (g)); the console verbs of (h) (WP-12) — the full list is (b1)'s "the lift". Still **no `--bench`** switch.

v1.11.0-pre.1: the Hecke action on the Bredon object at $\ell \in \{2,3\}$ is declined (slot-10 decision 4 (A); slot-10 plan §2.3 item 2, audit B-1, M-2, M-6). At $\ell \in \{2,3\}$ the object `EquivariantCohomologyOnK` computes over $\mathbb{Z}/\ell^k$ is the Bredon object $C^{\mathrm{eq}}(\mathbb{Z}/\ell^k)$ ([C, `def:ringinforce`](3), [C, `prop:bredon`](2)–(3)), and two different Hecke objects on it could be lifted. **Neither is, in slot 10**, for two different reasons, stated separately:

(a) *The cochain map Θ_g on $C^{\mathrm{eq}}(\mathbb{Z}/\ell^k)$* — the coset transport [C, `eq:cosettransport`] of an admissible τ (the shipped canonical one, or the strict τ^s), which `HeckeOperatorOnK` would assemble. **Not needed**, and so neither built nor specified. Line 2's Hecke action (slot-10 WP-10, WP-11) reads T_p through $\widehat{\Theta}_g$ on the free model $\widehat{C} = C_{\mathrm{Iso}}(R)$, whose cohomology is $H^\bullet(\Gamma; V_R)$ over every ring ([C, `prop:isomodel`](3)), and never through the Bredon object; the strict compatibility $\varepsilon^* \circ \Theta_g^s = \widehat{\Theta}_g \circ \varepsilon^*$ of [C, `prop:strictlift`](3) is checked in slot 10 only on rings where a Θ_g^s is shipped ($\mathbb{F}_\ell$ at every ℓ; $\mathbb{Z}/\ell^k$ at $\ell \geq 5$), and over $\mathbb{Z}/\ell^k$ at $\ell \in \{2,3\}$ it follows from the exact $\mathbb{Z}$-certificate of $\varepsilon \circ \tau_{\mathrm{Iso}} = \tau^s \circ \varepsilon$, [C, `eq:cosettransport`] being functorial in the ring ((b1), "outside the closure").

(b) *The operator on cohomology $A^{(q)}(T_p)$ of the Bredon object.* **Declined**: it is **not shown to be independent of the choice of τ**. Homotopy uniqueness of admissible τ needs $6 \in R^\times$ ([C, `lem:tauexists`](2)), which fails at $\ell \in \{2,3\}$, and [C, `prop:strictlift`](4) compares Θ_g^s with the shipped Θ_g on cohomology only at $6 \in R^\times$ — at $\ell \in \{2,3\}$ "they may differ, and no statement of [that] chapter reads them there". This is the **absence of a proof of independence, not a proof of dependence**: [C] exhibits no two admissible τ with different induced operators on the Bredon object, and this document claims none. Lifting it would ship the canonical τ's operator with no statement of what it is an invariant of, so no consumer may read it and none is built.

Consequences, binding from this pre-release. `HeckeOperatorOnK`'s refusal at $\ell \in \{2,3\}$ (in `init_composite()`, by `NotImplementedInThisBuild`, citing this item) and `HeckeReduction`'s stand, with their arms; so do the refusals of the boundary and cone classes of (f) and of the cross-routes there ((b1)). The campaign's job loop keeps recording `hecke` (with `bockstein` and `reduction`) as refused at $\ell \in \{2,3\}$ ((g)), and the console lifts nothing there ((h)). [X] §8's re-cut row "`HeckeOperatorOnK` over $\mathbb{Z}/\ell^K$ at $\ell \in \{2,3\}$ (M1's need)" is withdrawn (slot-10 WP-17, [X] v1.14). *If a line-2 package exhibits two admissible τ whose Bredon-object operators differ* at 2 or 3, that is a finding for [C], recorded by slot-10 WP-15 in place of the "not shown" reading — not assumed here. [C] v1.10.0 (slot-10 WP-15, decision 12 (A)) records the same reason in a remark beside [C, `prop:strictlift`](4), in this form: "not shown independent of τ; [C] compares nothing there". *v1.11.0-pre.2, as built.* Nothing was lifted: `HeckeOperatorOnK`'s refusal in `init_composite()` and `HeckeReduction`'s stand with their arms and messages, and no code package of slot 10 edited either class. **No line-2 package exhibited two admissible τ with different operators** at 2 or 3 — none computed an operator on the Bredon object over $\mathbb{Z}/\ell^k$, $k \geq 2$, at all — so the "not shown" reading stands. Item (a)'s last sentence is completed by (b1)'s "outside the closure" as built (the functoriality argument's three inputs, review finding m-1). [C] v1.10.0 records the reason as [C, `rmk:bredonhecke`], in the form quoted, with one precision this item left implicit: at $k = 1$ the shipped $\mathbb{F}_\ell$ pipeline computes (a) and (b) for the *canonical* τ at $\ell \in \{2,3\}$ — a definite object, the canonical datum's operator, which no statement identifies with anything τ-independent there; the declined lift is the one over $\mathbb{Z}/\ell^k$, $k \geq 2$. (*v1.11.0*, the phase-1 referee's M-1 and m-5: "has always computed" is replaced by "computes" — the canonical τ is [C, `alg:tau`]'s since [C] v1.1.0 ([C, `rmk:iotatausym`]), and its operator (b) did not move in that re-baseline and equals the legacy τ's at every $\ell \in \{2,3\}$ golden point, [C, `rmk:tauexistsscope`](1), an observation, not a theorem; [C] v1.10.1 qualifies "compares nothing there" to the operators it names, Θ^s_g, the canonical Θ_g and $\Theta_{\mathrm{Iso},g}$, and points to that observation.)

(b4) *The seven user-facing surfaces* keep named, user-facing refusals of their own, each with a test arm (the `conway` precedent, slot-7 WP-4): `HeckeExtDriver` (at construction); **the campaign** (at spec validation, `Campaign/CampaignSpec`, for the job loop and the ext pass alike, before any job runs); `--atlas` (`HeckeAtlasCli`); `--bench`, every mode other than the ones WP-3 supports; **the console's pipeline verbs** (one dispatch check in `Console/Bindings/`); `HeckeReport` (`hecke_report/v2` never moves); and **the cone** (`EquivariantMappingCone`, `HeckeConeAction`). Every message carries the tag `not_implemented_in_this_build:`. *v1.9.0, as built* (arm (X), one member each): `HeckeExtDriver` at construction; the campaign at spec validation (`CampaignSpec`), before any job, for `Z/ell^2`, `Z/ell^3` and `Z` (refused by every consumer but the integral descent, from v1.23.0-pre.1: (a)); `--atlas` with exit 5; `--bench` in every mode, the guard sitting at `bench::RunMode`'s entry before the mode dispatch (the arm drives `cohom`); the console at the dispatch of `symmetric_power`, `cohomology`, `hecke` and `family`, exit 5 — also through a lazy handle, since slot-8 WP-10's `bind::LazyCompute` ([P] v3.6 §8); `HeckeReport` at `StreamHeckeReportJson`'s entry, the emitted `hecke_report/v2` text being byte-identical; and `EquivariantMappingCone` and `HeckeConeAction`.
v1.10.0-pre.1: the surfaces in slot 9. Four keep their refusal unchanged: `HeckeExtDriver`, `--atlas`, `--bench` (every mode; no switch is added) and `HeckeReport` (`hecke_report/v2` never moves). Three change. **The campaign** (slot-9 WP-8): spec validation *admits* a composite ring for the job loop, which runs under (g). It *still refuses at spec validation, by name and before any job file exists*, a spec with `ext = true` under a non-default ring, and a spec whose grid holds a cell with $\ell^k \geq 2^{64}$ (naming the cell). So no refusal ever follows a job that has run. The refusal of `Z` inside `MathSettings::set()` is superseded (v1.23.0-pre.1): see (a), [C, `def:ringsemantics`](3). **The console** (slot-9 WP-12): the verbs and members of (h) honour, and every other pipeline verb keeps exit 5, through a lazy handle too. **The cone** (slot-9 WP-4): `EquivariantMappingCone` and `HeckeConeAction` (route 1) *honour* at $\ell \geq 5$. So the cone is no longer a refusing surface there; it refuses at $\ell \in \{2,3\}$ and wherever $\ell^k \geq 2^{64}$, and its two cross-routes refuse (b1). *v1.10.0-pre.2, as built.* The console (slot-9

WP-12, WP-13) and the cone (slot-9 WP-4) are as stated, with the class width of (b1) (the cone classes refuse from $\ell^k \geq 2^{32}$). *v1.10.0, as built* (slot-9 WP-8): `CampaignSpec` admits $\mathbb{Z}/\ell^k$ for the job loop, and refuses, by name and before any job file or output directory exists (`--campaign` exits 1), a spec with `ext = true` under a non-default ring ("... is not honoured in this build for the ext pass ...") and a spec whose grid holds a cell with $\ell^k \geq 2^{64}$ (naming `n_sym`, ℓ and the section's line; $\ell = 4294967311$ at $k = 2$ and 3^{41} are refused, 2^{63} is admitted). Arms: `CampaignRingValidation` (X); `HeckeReductionValidation` arm (X), whose `surface campaign (spec validation)` row now carries `ext = true`, the surface that still refuses.

Everything §6, §5 and the extension record compute is a field notion, and is refused under a non-default ring in slot 8 (v1.10.0-pre.1: and in slot 9; the `hecke_ext` record stays $\mathbb{F}_\ell$-only).

(c) The cyclic-summand basis rule ([X] v1.11 E8 finding M3), stated once. $H^q = Z^q_{\mathrm{eq}}/B^q_{\mathrm{eq}}$ over $R = \mathbb{Z}/\ell^k$ is `invariant_factors(ctx, Z, B)` (§13.6), and its cyclic summands get a basis by the following rule, a deterministic function of the input matrices $(G, R_l) = (Z, B)$, as the D5b bases of §8 are of theirs. It is the one definition tier B reads.

(c1) *Presentation.* $\Gamma = [\gamma_1, \ldots, \gamma_s]$ is the Howell generator matrix of $N = \mathrm{colspan}[R_l \mid G]$ (§13.6 step 1), its columns in Howell-row order. The relation matrix Rel ($s \times t$; rows = generators, columns = relations) is $[\,K \mid X\,]$: first the rows of `kernel_module`(Γ), in Howell order, as columns; then the coordinates `solve_multi_rhs_module`(Γ, R_l) of the columns of R_l, in their order. So $M \cong R^s/\,\mathrm{colspan}\,\mathrm{Rel}$. *The coordinates of* (G, R_l) (v1.9.0; slot-8 WP-3's choice, which the pre-release left coordinate-free; review finding m-9). On the pipeline route (G, R_l) is read in C^q_{eq} **coordinates**. C^q_{eq} is `kernel_module`(A_{C^q}), the kernel of the equivariance constraints on the flat cochains, with its Howell rows as the basis: that basis is unique (§13.5), and every pivot is asserted to be a unit, so C^q_{eq} is free ([C, `prop:coneintegral`](1)–(2) at $\ell \geq 5$; v1.9.1: through v1.9.0 this cited (2) alone). X_q is the matrix of d^q from the basis of C^q_{eq} to that of C^{q+1}_{eq}, by `solve_multi_rhs_module`. Then G is the Howell basis of $\ker X_q$ (the cocycles), as columns, and $R_l = X_{q-1}$ (the coboundaries), both in C^q_{eq} coordinates. So (G, R_l), and with it the basis of (c4), is a function of the cell and the ring alone. The flat cochain of b_i is the basis of C^q_{eq} applied to b_i's coordinates. Tier B (`red_H^q`) reads this basis. *v1.10.0-pre.1 (slot-9 plan audit B-1): this item — the unit-pivot assertion, the freeness of* C^q_{eq}*, and the* C^q_{eq} *coordinates of* (G, R_l) *— holds at* $\ell \geq 5$ *only*, where [C, `prop:coneintegral`](1)–(2) and [C, `prop:ringinforce`](1) make C^q_{eq} free. At $\ell \in \{2, 3\}$ it is not free (below), the unit-pivot assertion would fail, and coordinates in a non-free module are not unique; there the route is (c1′).

(c1′) *The flat-coordinate route at* $\ell \in \{2, 3\}$ (v1.10.0-pre.1; decision 4). At $\ell \in \{2, 3\}$, $C^q_{\mathrm{eq}}(\mathbb{Z}/\ell^k)$ is in general **not free**: at odd n the central relation gives $2f = 0$ on every cochain value ([C, `prop:ringinforce`](3)'s computation read over $\mathbb{Z}/2^k$), and at even n, $\ell = 2$, already $n = 2$ gives $(V_R)^{\langle S\rangle} = R(x^2+y^2) \oplus \{b\,xy : 2b = 0\}$ ([C, `prop:ringinforce`](2)'s computation read over $\mathbb{Z}/2^k$). So `EquivariantCohomologyOnK` computes there, in **flat cochain coordinates**, with the flat coboundary $d^q = D_q \otimes I$ and the equivariance constraint matrix A_{C^q} of the pipeline's own relation list over the ring (unchanged):

$$Z^q = \texttt{kernel_module}\begin{bmatrix} A_{C^q} \\ d^q \end{bmatrix}, \qquad B^q = d^{q-1}\,\Gamma_{q-1}, \qquad H^q \cong \texttt{invariant_factors}(\mathrm{ctx}, Z^q, B^q),$$

where Γ_{q-1} is the matrix whose columns are the Howell generators of C^{q-1}_{eq} = `kernel_module`$(A_{C^{q-1}})$ (with $B^0 = 0$), and Z^q is taken as the matrix whose columns are its Howell generators. Both are generator matrices of submodules of the flat cochains, $B^q \subseteq Z^q$, and §13.6's `invariant_factors` takes column spans, so no basis of C^q_{eq} is needed. It yields `invariant_factors_H^q` and `lengths_H^q` = `length`$(\mathrm{colspan}[B^q \mid Z^q]) -$ `length`$(\mathrm{colspan}\,B^q) = \sum_i e_i$, asserted equal, for $q = 0, 1, 2$, and **nothing else**. There is no stored basis of C^q_{eq} or of H^q, so (c2)–(c5) are **not applied** there, and no map on H^q (no `red_H^q`, no $A^{(q)}(T_p)$, no Bockstein) is defined (v1.11.0-pre.1: and slot 10 defines none — the Hecke action on the Bredon object is declined, for the two reasons of (b3)'s "v1.11.0-pre.1" item). There is also no X_q, so no r_K, no `ladic_type`, no free rank and no `charpoly`: those are the K-mode,

which [C, `prop:certifiede`](5) refuses there. The relation list over the ring is the pipeline's, unchanged, and its 2- and 3-torsion is what the tree model predicts ([C, `prop:bredon`](2): $H^0 = V_R^\Gamma$, $H^1 = V_R^{-I}/(V_R^S + V_R^U)$, $H^q = 0$ for $q \geq 2$; the pipeline's $H^2 = 0$ is checked by `HeckeBredonTreeRingValidation`, arms (T) and (A) — v1.10.0-pre.2: through v1.10.0-pre.1 this read "is asserted", but the route itself asserts nothing about H^2, the slot's review, WP-14, finding m-3; v1.10.1: `docs/implementation/slot9/WP-14_review.md`). *As built* (slot-9 WP-3, `compute_cohomology_flat_module()`): exactly this route, with three certificates re-derived from the dense A_{C^q}, d^q and the two Howell lengths, independently of the invariant-factor result — $A_{C^q} B^q = 0$ and $d^q B^q = 0$ entrywise ($B^q \subseteq Z^q$); $\sum_i e_i = \texttt{length}(Z^q) - \texttt{length}(B^q)$; and $\texttt{length}(C^q_{\mathrm{eq}}) = \texttt{length}(Z^q) + \texttt{length}(B^{q+1})$. Measured on 96 cells ($\ell \in \{2,3\}$, $k \in \{2,3,4\}$, even $n \leq 20$, odd $n \leq 9$) against O7, with O7 controls at $k = 1$ (the shipped $\mathbb{F}_\ell$ route, 32 cells) and at $\ell = 5$ (the $\ell \geq 5$ module route, 5 cells). The `uint64_t` instantiation of the route is compiled and reached by no honouring cell (review finding m-2, open). *v1.10.0: closed* (slot-9 WP-9, `fb8a63a`): `HeckeBredonU64FlatRouteValidation` (`fast`) runs one honouring `uint64_t` cell, $\ell = 2$, $n = 4$, $k = 33$, against the frozen O7 row at $(\ell, k, n) = (2, 4, 4)$ (lengths $(2, 6, 0)$, $H^0 = [2]$, $H^1 = [4, 2]$, $H^2 = [\,]$) read through the route's K-scaling: an exponent equal to K is a free summand $\mathbb{Z}/\ell^K$ and moves with K, one below K is ℓ-power torsion and does not. It predicts and finds lengths $(2, 35, 0)$, $H^1 = [33, 2]$ at $K = 33$. **The $\ell \geq 5$ route is byte-identical**: the branch is selected by ℓ at the ring context, and no line of the $\ell \geq 5$ module route moves (slot-9 plan M-2; its oracle is `GOLDEN_STRICT=1` over the 124 entries of (e)).

(c2) *Smith elimination* of Rel by §13.6 step 2's pivot rule on the trailing submatrix (smallest valuation anywhere, then leftmost column, then topmost row), moved to the diagonal by one row swap and one column swap. The pivot is made exactly ℓ^σ by scaling its *column* by `inv(unit_part)` of the pivot (a relation operation). Then the pivot's *row* is cleared first (column operations) and its *column* second (row operations), each by the `divide_exact` quotient; there is no other normalisation. The elimination stops when every trailing entry is 0.

(c3) *Mirroring.* Column operations on Rel act on relations and leave the generators alone. Every row operation is mirrored on Γ by the inverse column operation: $\mathrm{row}_i \leftarrow \mathrm{row}_i - q\,\mathrm{row}_j$ becomes $\gamma_j \leftarrow \gamma_j + q\,\gamma_i$, and a row swap swaps the two columns of Γ. (If $U \,\mathrm{Rel}\, V = D$, the transformed generators are the columns of ΓU^{-1}.)

(c4) *Read-off.* Diagonal position i with entry ℓ^σ, $1 \leq \sigma < k$, gives a summand $\mathbb{Z}/\ell^\sigma$ generated by the transformed γ_i; a generator whose diagonal entry is 0 or missing gives $\mathbb{Z}/\ell^k$; a unit diagonal entry gives nothing (its generator is 0 in M). The summands are then ordered by exponent, *descending*, ties by final generator index, ascending (a stable sort), so that the basis is indexed $b_1, b_2, \ldots$ with b_i generating the $\mathbb{Z}/\ell^{e_i}$ summand and $e_1 \geq e_2 \geq \cdots$. The identity $\sum_i e_i = \texttt{length}(N) - \texttt{length}(\mathrm{colspan}\, R_l)$ is asserted (§13.6 step 4).

(c5) *Coordinates and maps.* A class $x \in M$ has coordinates (c_i) with $x = \sum_i c_i b_i$, the i-th *reduced* into $[0, \ell^{e_i})$; they are unique. A map between such modules (the Hecke matrix $A^{(q)}(T_p)$ on the cyclic summands, WP-3; a reduction map, (d)) is the matrix whose column j holds the reduced coordinates of the image of b_j, entry (i, j) in $[0, \ell^{e_i})$ of the target. Such a matrix is basis-dependent, and its characteristic polynomial is not an invariant of a non-free module ([L] §2.6), which is why (d) defines `charpoly` on the free quotient only.

The rule fixes the two choices §13.6 leaves open — the order of the relation columns and the side on which the pivot is scaled — and nothing else. It is not used at $k = 1$: the $\mathbb{F}_\ell$ path keeps the D5b bases. *v1.9.0, how the rule is tested* (slot-8 WP-3r, on review finding M-1, which found that no arm tested the basis rule directly). Arm (M) of `HeckeReductionValidation` runs over $\mathbb{Z}/4$, $\mathbb{Z}/25$, $\mathbb{Z}/125$ and $\mathbb{Z}/691^2$ (`u32`), and over $\mathbb{Z}/134217757^2$ (Montgomery) and $\mathbb{Z}/4294967291^2$ (U64_Wide). Its cases are fixed-seed known-answer and random (G, R_l). It asserts: the exponents against `invariant_factors` and the naive oracle; the coordinates of the basis are the identity; each b_i has order exactly ℓ^{e_i}; the basis generates; and the coordinates of (c5) are unique and linear. It also rejects three corrupted decompositions, the ones a wrong mirror on U, a wrong mirror on Γ or a wrong read-off position would produce. Each of those three mutations of the code fails the validation hundreds of times. On every pipeline cell, the coordinates of the cells' own basis are asserted to be the identity, for $q = 0, 1, 2$

(arm (C)).

(d) The E8 objects: names and meaning. They are for the golden harness and the validations, not for a record (decision 4). Throughout, C is the integral equivariant cochain complex of the model; for $\ell \geq 5$, $C_{\mathbb{Z}/\ell^k} = C \otimes \mathbb{Z}/\ell^k$ ([C, `prop:coneintegral`] (2)). On the *applicability range* $n > 0$ even, $\ell \geq 5$ ([C, `cl:bockstein`] (1), [C, `cl:amalgam`] (3), [C, `lem:h2torsion`] (1)), $H^1(C)_{(\ell)} \cong \mathbb{Z}_{(\ell)}^r \oplus \bigoplus_{i=1}^t \mathbb{Z}/\ell^{e_i}$ and $H^0(C)_{(\ell)} = H^2(C)_{(\ell)} = 0$, with $r = \dim M_{n+2} + \dim S_{n+2}$ and $t = \dim H^0(\mathbb{F}_\ell)$ the number of Dickson classes; by universal coefficients $H^0(\mathbb{Z}/\ell^k) \cong \bigoplus_i \mathbb{Z}/\ell^{\min(e_i,k)}$, $H^1(\mathbb{Z}/\ell^k) \cong (\mathbb{Z}/\ell^k)^r \oplus \bigoplus_i \mathbb{Z}/\ell^{\min(e_i,k)}$ and $H^2(\mathbb{Z}/\ell^k) = 0$ ([L] §2.1, §2.4). The level of a run is $K := k$, the exponent of the ring in force; slot 8 runs no other K. The K-mode objects (`r_K`, `ladic_type*`, the free-quotient `charpoly`) and the assertion of the Bockstein meaning are defined on the applicability range only, and are absent elsewhere; every E8 point lies in it.

- `lengths_H^q` ($q = 0, 1, 2$; tier A): the composition length of H^q over $\mathbb{Z}/\ell^k$, $\sum_i e_i$ over its invariant factors, so that $|H^q| = \ell^{\text{length}}$. Over $\mathbb{Z}/\ell^k$ it replaces `dim_H^q`. *v1.10.0-pre.1*: this object and the next are the *only* objects of this list defined at $\ell \in \{2, 3\}$. There they are the Bredon object's ([C, `def:ringsemantics`](2)), computed by (c1′); no universal-coefficient statement of this paragraph's preamble applies to them, and every other object below is undefined at $\ell \in \{2, 3\}$. No golden point lies at $\ell \in \{2, 3\}$ in slot 9 (decision 4 (A); (A′) was not ruled).
- `invariant_factors_H^q` ($q = 0, 1, 2$; tier A): the list $e_1 \geq e_2 \geq \cdots \geq 1$ with $H^q \cong \bigoplus_i \mathbb{Z}/\ell^{e_i}$, from `invariant_factors(ctx, Z, B)` (§13.6); $e_i = k$ is a summand that is free over $\mathbb{Z}/\ell^k$. The list is empty for the zero module.
- `r_K(d^0)`, `r_K(d^1)` (tier A) and `ladic_type_certified` (tier A, boolean). $r_K(d)$ is the number of non-zero diagonal entries, i.e. entries ℓ^s with $s < K$, in the Smith form of d over $\mathbb{Z}/\ell^K$ (by (c2)'s pivot rule). The K-mode is **certified** iff $r_K(d^0) = \dim C^0_{\text{eq}}$ and $r_K(d^1) = \dim C^2_{\text{eq}}$ (E8-P2; [L] §2.4). Then every elementary divisor of d^0 and d^1 over $\mathbb{Z}_{(\ell)}$ is $< \ell^K$, and the e_i are read off exactly. Both counts are emitted whether or not the mode certifies. An uncertified level is a loud outcome, never an error. Its state is R3-shaped (kind `ladic_level`, budget $\{K\}$, reached $\{r_K(d^0), r_K(d^1)\}$, `complete = false`), and its continuation is a *fresh* run at a larger k, not a `resume`: `coefficient_ring` is a non-budget setting, and a resume under a changed non-budget setting is refused (§7). *v1.9.0, as built* (slot-8 WP-3, in `EquivariantCohomologyOnK`): $r_K(d^q)$ is the number of invariant factors of X_q (the matrix of (c1)), and the `ladic_level` state is carried by the object. Three things are asserted on every cell by arm (C) of `HeckeReductionValidation`: `ladic_type^q` is checked against `invariant_factors_H^q` by universal coefficients; the cells whose torsion exponents are all < 2 certify at $K = 2$; and the two controls are uncertified at $K = 2$, each with a deficit of exactly one, and certified at $K = 3$. The deficit at $K = 2$ is $r_2(d^0) = \dim C^0_{\text{eq}} - 1$, with $r_2(d^1) = \dim C^2_{\text{eq}}$.
- `ladic_type^q` ($q = 0, 1, 2$; tier A; present **only** where `ladic_type_certified = true`): the $\mathbb{Z}_{(\ell)}$-type $(r_q; e_1^q \geq e_2^q \geq \cdots)$ of $H^q(C)_{(\ell)}$, its free rank and torsion exponents. On the applicability range it is $(0;)$ at $q = 0, 2$ and $(r; e_1, \ldots, e_t)$ at $q = 1$. Write $e_{\max} := \max_i e_i$, and $e_{\max} := 0$ when $t = 0$.
- `charpoly_A(T_p)^1_free_p<p>` (tier A; E8-P4, [L] F-C; slot-8 plan §2.3 item 13), **with its precondition**. It is defined *only* at a level K where `ladic_type_certified = true`, as the characteristic polynomial of $A^{(1)}(T_p)$ on the free quotient

$$H^1(\mathbb{Z}/\ell^K) \,/\, H^1(\mathbb{Z}/\ell^K)[\ell^{e_{\max}}] \;\cong\; (H^1(C)/\text{tors}) \otimes \mathbb{Z}/\ell^{K-e_{\max}},$$

  a free $\mathbb{Z}/\ell^{K-e_{\max}}$-module of rank r, stable under T_p because the $\ell^{e_{\max}}$-torsion is characteristic ([L] §2.6). **The precision $K - e_{\max}$ is part of the object**: the object is the exponent $K - e_{\max}$ together with the monic polynomial of degree r over $\mathbb{Z}/\ell^{K-e_{\max}}$, coefficients low→high in $[0, \ell^{K-e_{\max}})$. It is basis-independent, hence tier A. At an uncertified level it has no definition ([L] §2.4: a summand $\mathbb{Z}/\ell^e$ with $e \geq K$ reads as free), so it is **absent**, and its accessor throws a named `std::logic_error` and never returns a value. It exists for $q = 1$ only: on the applicability range H^0 and H^2 have free rank 0. *v1.9.0, as built* (slot-8 WP-3, `free_charpoly_H1`). It is computed by Berkowitz over $\mathbb{Z}/\ell^K$ on the free block, then reduced to $\mathbb{Z}/\ell^{K-e_{\max}}$. At weight 12 it is *asserted*,

not printed (slot-8 WP-3r, review finding m-7): at every certified torsion-free cell with $n = 10$ and every prime of the cell, the value must be exactly $(x-(1+p^{11}))(x-\tau(p))^2$ over $\mathbb{Z}/\ell^{K-e_{\max}}$. That is $(10, 691)$ over $\mathbb{Z}/691^2$ at $p \in \{2, 3, 5\}$, and $(10, 134217757)$ and $(10, 4294967291)$ over $\mathbb{Z}/\ell^2$ at $p = 2$ (arm (T)). Wherever H^1 is free, its reduction mod ℓ equals the $\mathbb{F}_\ell$ characteristic polynomial. Elsewhere it divides that polynomial.

- `bockstein_rank^0` (tier A): the rank over $\mathbb{F}_\ell$ of the mod-ℓ Bockstein $\beta^0 : H^0(\mathbb{F}_\ell) \to H^1(\mathbb{F}_\ell)$, the connecting map of $0 \to \mathbb{Z}/\ell \to \mathbb{Z}/\ell^2 \to \mathbb{Z}/\ell \to 0$ ([C, `cl:bockstein`]), computed from the stored bases (E8 item 4a). `higher_torsion_count` (tier A) is $\dim H^0(\mathbb{F}_\ell) -$ `bockstein_rank^0`. Hence the **bookkeeping identity**

$$\texttt{bockstein_rank\^0} + \texttt{higher_torsion_count} = \dim H^0(\mathbb{F}_\ell),$$

  true by definition. The content is its *meaning* ([L] §2.4): β^0 is the injective integral Bockstein $H^0(\mathbb{F}_\ell) \cong H^1(C)[\ell]$ followed by reduction mod ℓ, and a summand $\mathbb{Z}/\ell^e$ lies in its kernel iff $e \geq 2$, so

$$\texttt{bockstein_rank\^0} = \#\{i : e_i = 1\}, \qquad \texttt{higher_torsion_count} = \#\{i : e_i \geq 2\},$$

  and the second equality is **asserted wherever the K-mode certifies**, against the e_i of `ladic_type^1`. It is *not* the equality `bockstein_rank^0` $= \dim H^0(\mathbb{F}_\ell)$: that one means "no ℓ^2-torsion", and it fails by design at the two positive controls $(30, 5)$ and $(60, 5)$. Both objects are mod-ℓ objects, and so are the same at every k of a cell. *v1.9.0.* The meaning is now a statement of [C]: [C, `rmk:bocksteinexp`](1), proved in [C] since v1.8.0 (slot-8 WP-9; finding F-A of [L], owner ruling 10), is exactly $\operatorname{rank} \beta^0 = \#\{i : e_i = 1\}$. Its clause (2) explains why the K-mode needs its certificate. A class of $H^0(M_m)$ produces torsion of exponent $\geq j + 1$ at $n = \ell^j m$, so no fixed k reads every exponent correctly ([C, `prop:certifiede`]). *As built* (slot-8 WP-4, `Hecke/HeckeReduction`), β^0 is computed from the stored bases. A D5b class is lifted into $C^0_{\text{eq}}(\mathbb{Z}/\ell^2)$ through the free Howell basis; $X_0 c = \ell w$ gives $\beta^0 = [w]$. Arm (B3) asserts the identity, and asserts the two counts wherever the K-mode certifies; arm (B2) asserts the long exact sequence's length bookkeeping; arm (B1) asserts $\beta \circ A(T_p) = A(T_p) \circ \beta$. At every E8 point `higher_torsion_count` equals its pre-stated value in (e).
- `red_H^q` ($q = 0, 1, 2$; **tier B**; [X] v1.11 E8 finding F3): the reduction map $H^q(\mathbb{Z}/\ell^k) \to H^q(\mathbb{F}_\ell)$ induced by $\mathbb{Z}/\ell^k \to \mathbb{F}_\ell$, as the matrix of (c5) from the cyclic-summand basis of (c) to the D5b basis of the $\mathbb{F}_\ell$ run at the same (n, ℓ), with entries in $[0, \ell)$. Reductions compare as maps, never as bases. On cohomology the reduction is injective after $\otimes\mathbb{F}_\ell$, with a Tor cokernel (universal coefficients), so "the $\mathbb{Z}/\ell^k$ object reduces entrywise to the $\mathbb{F}_\ell$ object" is false even up to a change of basis. The certificates are the map equalities $A(T_p) \circ \text{red} = \text{red} \circ A(T_p)$ and $\beta \circ A(T_p) = A(T_p) \circ \beta$ (item 4a). Cochain-level matrices reduce entrywise and are not objects. *v1.9.0, as built* (slot-8 WP-4, `Hecke/HeckeReduction`). The map is red $= P_q(b_j \bmod \ell)$, with P_q the $\mathbb{F}_\ell$ run's projector to the D5b basis. Five arms of `HeckeReductionValidation` assert it at every E8 cell-level and every prime: (R1) it is the class map; (R2) the map equality above, for $q = 0, 1, 2$; (R3) Θ reduces entrywise; (R4) at $K = 2$ it is exact against β; (R5) functoriality from $K = 3$ to $K = 2$. At the E8 cells its Tor cokernel is non-zero exactly where a summand $\mathbb{Z}/\ell^e$ of H^0 has $e < K$, and it is 0 on H^1 and H^2.

The Hecke matrix $A^{(q)}(T_p)$ over $\mathbb{Z}/\ell^k$ on the cyclic summands (c5) is computed and validated, but is **not** an E8 golden object in slot 8. Nor is (D6)'s refinement at $(10, 691)$, which is a validation arm and a report line (E8 item 4a). At $p = \ell$, T_p is U_ℓ (the harness builds its cosets with $p = \ell$). Over $\mathbb{Z}/\ell^k$ that is item 3's newest route (E8-P5's $\operatorname{diag}(\ell, 1)$ split), validated in slot 8 and not frozen. *v1.9.0: what the (D6) arm measured* (slot-8 WP-4, `HeckeReductionValidation`). At $(10, 691)$ over $\mathbb{Z}/691^2$ the free characteristic polynomial of T_2 is $[252219, 379705, 475480, 1] = (x - 2049)(x + 24)^2$, which is asserted. So the pipeline's normalisation is the classical one: $a_E = 1 + 2^{11}$ and $a_\Delta = \tau(2)$, with $a_E \equiv a_\Delta \pmod{691}$ and $a_E \not\equiv a_\Delta \pmod{691^2}$. $\ker(A - a_E)$ contains a free Eisenstein eigenline of rank 1, and $\ker(A - a_\Delta)$ a free cuspidal eigenplane of rank 2; the arm asserts this $1 + 2$ split. The free eigenline and the free eigenplane together span a submodule of index 691 in H^1: the determinant of their three generators is $415291 = 601 \cdot 691$, of 691-adic valuation 1. That index is recorded, not

asserted. It is the Ramanujan congruence seen integrally. The U_5 route is exercised by arms (P) and (T) at every $\ell = 5$ cell: 115 distinct g per cell against the oracle, the split route among them.

(e) The E8 golden points (slot-8 decision 5). Every E8 object is emitted at a **new point** named

`E8 ring=Z/<ell>^<k> n=<n> ell=<ell>`

with ℓ, k and n in decimal and the ring written with the cell's prime, for example `E8 ring=Z/691^2 n=10 ell=691` (the `S6 sl2z_restrict` naming precedent). This holds at $(10, 691)$ and $(16, 5)$ too, which are also grid points under their `n=... ell=...` names. No existing point gains or loses an entry, and no E8 marker is emitted anywhere else. The E8 point list is its own ($(6, 5)$, $(30, 5)$ and $(60, 5)$ are not grid points and do not join the grid); it is looped from the full golden run only, and the ring is installed (`ScopedMathSettings`, source `spec`) around the E8 computations only. The seven points and their primes: $(10, 691)$ over $\mathbb{Z}/691^2$ with $p \in \{2, 3, 5\}$; $(6, 5)$ and $(16, 5)$ over $\mathbb{Z}/25$, and $(30, 5)$ and $(60, 5)$ over $\mathbb{Z}/25$ and over $\mathbb{Z}/125$, each with $p \in \{2, 3\}$ (at $\ell = 5$, $p = 5$ is U_5, not frozen, (d)).

Predicted values ([L] §2.5's exact probe; the invariant factors follow from it by universal coefficients; slot-8 plan §2.3 item 13 and decision 2). A prediction that fails is a finding for the owner, and is never re-predicted.

cell	K	cert.	r	e ($e_{\max}$)	`charpoly` over	H^0	H^1	β-rank / h.t.c.
$(10, 691)$	2	yes	3	— (0)	$\mathbb{Z}/691^2$	()	$(2, 2, 2)$	0 / 0
$(16, 5)$	2	yes	3	— (0)	$\mathbb{Z}/25$	()	$(2, 2, 2)$	0 / 0
$(6, 5)$	2	yes	1	(1) (1)	$\mathbb{Z}/5$	(1)	$(2, 1)$	1 / 0
$(30, 5)$	2	no[a]	5	—	absent	(2)	(2^6)	0 / 1
$(30, 5)$	3	yes	5	(2) (2)	$\mathbb{Z}/5$	(2)	$(3^5, 2)$	0 / 1
$(60, 5)$	2	no[b]	9	—	absent	$(2, 1)$	$(2^{10}, 1)$	1 / 1
$(60, 5)$	3	yes	9	$(2, 1)$ (2)	$\mathbb{Z}/5$	$(2, 1)$	$(3^9, 2, 1)$	1 / 1

[a] $r_2(d^0) = \dim C^0_{\text{eq}} - 1$: the $\mathbb{Z}/25$ summand reads as free. [b] $r_2(d^0) = \dim C^0_{\text{eq}} - 1$, as at a: a deficit of **exactly one**, the $\mathbb{Z}/25$ summand reading as free (v1.9.0, restated model-independently; slot-8 WP-3's finding, confirmed by review finding m-8). Through v1.9.0-pre.1 this footnote read "$r_2(d^0) = 51 \neq 52 = \dim C^0_{\text{eq}}$ ([L] §2.5)". Those figures are [L]'s, in the coordinates of the amalgam model $V^S \oplus V^U$. They are not the shipped seven-vertex model's, where $\dim C^0_{\text{eq}} = 235$ and $r_2(d^0) = 234$ (and at a, 119 and 118). The invariant content, a deficit of one, is model-free. It is what the arm asserts (`HeckeReductionValidation`, arm (C), which asserts rank -1, not a literal), and it is what golden freezes (`r_K(d^0)` at the two uncertified points). In both, $r_2(d^1) = \dim C^2_{\text{eq}}$. The H^0 and H^1 columns are `invariant_factors_H^q` at $\mathbb{Z}/\ell^K$, exponents with multiplicity; $H^2 = 0$ throughout. $r = \dim M_{n+2} + \dim S_{n+2}$; e and $e_{\max}$ are those of `ladic_type^1`, shown only where certified. At an uncertified level the ℓ^2-torsion reads as free, so `invariant_factors` is correct for $\mathbb{Z}/\ell^K$ but does not determine the $\mathbb{Z}_{(\ell)}$-type, exactly as the certificate says.

The object list at each point. Tier A: `lengths_H^q`, `invariant_factors_H^q`, `r_K(d^0)`, `r_K(d^1)`, `ladic_type_certified`, `bockstein_rank^0` and `higher_torsion_count`, and, at a certified point only, `ladic_type^q` and `charpoly_A(T_p)^1_free_p<p>` for each of the point's primes. Tier B: `red_H^q`. That list, with the prime restriction and the `charpoly` precondition, is the generating rule of WP-4's pre-stated set. The set is WP-4's to generate, and the event is the lead's to land after its gate (slot-8 plan §4.4).

v1.9.0: the landed values (R1, merge `53a1909`; §3 of the re-baseline note named in the status paragraph above). Every value in the table above is the frozen one, line for line. The generated set was 124 entries (103 tier A, 21 tier B): 2 uncertified points $\times 14$, plus 5 certified points $\times 17$, plus 11 `charpoly` objects. `golden_ref_diff.py --expect` held it as a set equality, and `golden_explain.py` found no text or decomposition moved. The emission is byte-identical across three runs, one of them under `ENGINE_THREADS=1`. In the shipped model the counts are as follows (point order as in the table).

- $\dim C^0_{\text{eq}} = 41, 65, 27, 119, 119, 235, 235$ and $r_K(d^0) = 41, 65, 27, 118, 119, 234, 235$.
- $r_K(d^1) = \dim C^2_{\text{eq}} = 55, 85, 35, 155, 155, 305, 305$.

- `lengths_H^1` $= 6, 6, 3, 12, 17, 21, 30$ and `lengths_H^0` $= 0, 0, 1, 2, 2, 3, 3$.

The free characteristic polynomials, coefficients low→high, are listed below. At $(10, 691)$ over $\mathbb{Z}/691^2$ they are $[252219, 379705, 475480, 1]$ ($p = 2$), $[323249, 57149, 299829, 1]$ ($p = 3$) and $[82572, 23641, 342757, 1]$ ($p = 5$). At $(16, 5)$ over $\mathbb{Z}/25$ they are $[18, 21, 8, 1]$ and $[16, 4, 4, 1]$. At $(6, 5)$ over $\mathbb{Z}/5$ they are $[1, 1]$ and $[2, 1]$. At $(30, 5)$, $k = 3$, over $\mathbb{Z}/5$ they are $[1, 1, 3, 3, 1, 1]$ and $[2, 1, 4, 2, 2, 1]$. At $(60, 5)$, $k = 3$, over $\mathbb{Z}/5$ they are $[2, 1, 3, 4, 2, 1, 3, 4, 2, 1]$ and $[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]$. Where two are given, the first is for $p = 2$ and the second for $p = 3$. Measured by `GOLDEN_STRICT=1 GoldenRegressionValidation` (tier A 11 575 / 0, tier B 5866 / 0). At $(6, 5)$, and at $(30, 5)$ and $(60, 5)$ with $k = 3$, the precision $K - e_{\max}$ is 1. So those six polynomials carry nothing beyond their reduction mod ℓ, as (d) says.

(f) E8 item 4b: the boundary and the cone over $\mathbb{Z}/\ell^k$ (v1.10.0-pre.1). Slot-9 decision 3 (A), with its sub-item (iii) (A); built by slot-9 WP-4. *Scope*: $\ell \geq 5$ and $\ell^k < 2^{64}$, as for the bulk. At $\ell \in \{2, 3\}$ every member below refuses by name ((b1), (b3)). *Notation*: $C^\bullet_\partial$ is the circle model of the boundary ([C, `def:pair`], [C, `prop:circlemodel`]), the equivariant complex of K_{bd} in degrees $0, 1$, built by `SL2ZBoundaryPair`; $r = j^*$ is the restriction; and $\mathrm{Cone}(r)$ is [C, `eq:conediff`]. **Everything about the cone is indexed by the degree of H^*_c**, as `EquivariantMappingCone` and the $\mathbb{F}_\ell$ golden objects `dim_Hc^q` are. The stored cochain group in degree q is $Cc^q := \mathrm{Cone}(r)^{q-1} = C^q_{\mathrm{eq}}(K) \oplus C^{q-1}_{\mathrm{eq}}(K_{\mathrm{bd}})$, and $H^q_c = H^q(Cc^\bullet) = H^{q-1}(\mathrm{Cone}(r))$ for $q \in \{0, 1, 2\}$.

- *The mathematics used.* (i) The terms of $C^\bullet_\partial$ and of $\mathrm{Cone}(r)$ over $\mathbb{Z}$ are free ([C, `prop:coneintegral`](1)). The reduction $\mathrm{Cone}(r)_\mathbb{Z} \otimes \mathbb{Z}/\ell^k \to \mathrm{Cone}(r)_{\mathbb{Z}/\ell^k}$ is an isomorphism at $\ell \geq 5$ ([C, `prop:coneintegral`](2)); so is the boundary's, whose stabilisers are $\{\pm I\}$ (the same averaging argument, needing only $2 \in R^\times$). So both $\mathbb{Z}/\ell^k$ complexes are reductions of integral complexes of free modules, and universal coefficients apply to them. (ii) The long exact sequence [C, `eq:coneles`] is exact over every commutative ring ([C, `lem:conerelative`](2)). $H^0_c = 0$ and $H^q_c = 0$ for $q \geq 3$ hold over every ring ([C, `lem:conerelative`](4)), and $H^2(C(\mathbb{Z}/\ell^k)) = 0$ holds at $\ell \geq 5$ ([C, `cl:amalgam`]). (iii) The reductions and the Bocksteins are equivariant on the nose ([C, `prop:coneintegral`](3)).
- *The route.* A **dense module route**, selected by `is_field()` alone (`ENGINE_API.md` §13.6–§13.7: there is no block or tape Howell form). The tape route of the cone stays the $\mathbb{F}_\ell$ path, byte-identical (`cone_output_diff.sh`). The route inherits the ring context's element type and Montgomery regime and uses §13.6's exact `solve_multi_rhs_module`; no new regime and no new bound.
- *Coordinates, and the basis rule.* (c1)–(c5) apply **verbatim** to both complexes, with C^q_{eq} read as follows.
  - *Boundary*: $C^q_{\partial,\mathrm{eq}}$ = `kernel_module`$(A_{C^q_\partial})$. It is free at $\ell \geq 5$, and its unit pivots are asserted.
  - *Cone*: $Cc^q_{\mathrm{eq}} = C^q_{\mathrm{eq}}(K) \oplus C^{q-1}_{\mathrm{eq}}(K_{\mathrm{bd}})$ in the class's flat layout, bulk rows first. Its Howell basis is the concatenation of the two Howell bases, bulk first. Two row modules on disjoint column blocks have, as their sum, the concatenation of their reduced Howell forms, and §13.5's form is unique. This is asserted once per degree.
  - X_q is the complex's own differential in those coordinates. For the cone that is $D(c, c_\partial) = (-dc,\ d_\partial c_\partial + rc)$ of [C, `eq:conediff`], the class's `D_block`.

  So each of $H^q(\partial)$ and H^q_c over $\mathbb{Z}/\ell^k$ has the cyclic-summand basis of (c4). The three map families π^q_*, r^q, ι^q_* of [C, `eq:coneles`], T^∂_p and $A^{(q)}_c(T_p)$ are (c5) matrices between those bases.
- *Certificates* (validation and object level; none of these is a golden object).
  - [C, `obl:coneexact`](i)–(iii) over the ring, as over $\mathbb{F}_\ell$.
  - **(iv)'s non-field half**. At every spot of [C, `eq:coneles`], the composite of consecutive maps is zero. Each Howell generator of ker of the outgoing map is exhibited as the image of a vector under the incoming map, modulo the target's relations, by `solve_multi_rhs_module`. The solve's residual is re-derived: the image minus the generator lies in the colspan of the relations. At every bulk spot, $\mathrm{im}\,\pi^q_* = \ker r^q$ is checked with *both* inclusions by solves. A failure aborts. The E5 certificate's item numbering is kept: the new item is appended,

never renumbered. *v1.10.0-pre.2, as built* (slot-9 WP-4, `ConeModuleCertificate` in `EquivariantMappingConeModule.cpp`): **item 14**. Every composite is solved into the target's relations; the Howell generators of ker of the outgoing map are the first coordinates of `kernel_module`([out | D_N]), each exhibited as in(w_1)+$D_M w_2$ by `solve_multi_rhs_module`, and the witnesses are stored and re-derived by `verify()`. *Why the check is complete* (review finding m-W4-2): `verify()` re-derives each stored witness, not that the witnesses generate the kernel; completeness comes from item 11's length identity at every term together with the composite-zero check — $\operatorname{im}\text{in} \subseteq \ker\text{out}$ and $\ell(\operatorname{im}\text{in}) = \ell(M) - \ell(\operatorname{im}\text{out}) = \ell(\ker\text{out})$ force equality in a finite module — so exactness is certified even if a witness list were short, and the witnesses are a second discharge. Two negative controls in arm (4b-C): a tampered kernel generator fails item 14 (since the review repair `2933b65` it tampers the generator x, compared exactly, not a witness, so it cannot false-alarm), and a wrong stored image length fails item 11.

  - (v) over the ring: every certificate item that states a dimension reads `length` (§13.6), and every rank of a map reads the length of its image. So the first identity of [C, `eq:conedims`] reads $\ell(H^1_c) = \ell(H^0(\partial)) - \ell(H^0) + \ell(\ker r^1)$, which holds unconditionally, and the second reads $\ell(H^2_c) = \ell(H^1(\partial)) - \ell(\operatorname{im} r^1)$, which holds under $H^2 = 0$ and is therefore asserted at $\ell \geq 5$.
  - (vi): **the commutator is asserted; the $\nu = 2$ recursion is not computed over the ring** (v1.10.0-pre.2, narrowed to what is built; through v1.10.0-pre.1 this item read "(vi) is asserted where two primes are computed ($6 \in R^\times$)", which nothing implemented — the slot's review, WP-14, finding M-W4-1; v1.10.1: `docs/implementation/slot9/WP-14_review.md`). As built (review repair `2933b65`): at every level of every 4b cell and for every pair of computed primes, $[A_c(T_p), A_c(T_{p'})] = 0$ for $q \leq 2$ and $[A^\partial(T_p), A^\partial(T_{p'})] = 0$ for $q \leq 1$, over $\mathbb{Z}/\ell^k$ in (c5) form (products over the ring, row i compared modulo ℓ^{e_i} of the target); arm (4b-H) of `HeckeReductionValidation`. [C] [C, `obl:coneexact`](vi) is therefore *partial* over the ring, as [C] v1.9.0 records; the recursion over the ring is a later slot's.
  - (viii)'s two arms stay $\mathbb{F}_\ell$-only in slot 9.
  - The map equalities, as (d) states them for the bulk: $A^\partial(T_p) \circ \text{red} = \text{red} \circ A^\partial(T_p)$ and $A_c(T_p) \circ \text{red} = \text{red} \circ A_c(T_p)$; every map of [C, `eq:coneles`] commutes with red; and $\beta \circ A = A \circ \beta$ on both complexes.
  - The cone's mod-ℓ Bockstein $\beta^1_c \colon H^1_c(\mathbb{F}_\ell) \to H^2_c(\mathbb{F}_\ell)$ is computed and validated like the boundary's, but it is **not** an object in slot 9.

  Pointer into [C]: [C, `obl:coneexact`](iv)'s "until it exists the non-field case is **recorded** as unchecked" is stale from slot-9 WP-4's merge until [C]'s next edit (slot-9 WP-15). *v1.10.0-pre.2*: closed — [C] v1.9.0 restates the clause and records the non-field half as checked over $\mathbb{Z}/\ell^k$, $\ell \geq 5$.
- *[C, `obl:coneexact`](vii) over the ring* (decision 3 (iii) (A), slot-9 plan M-3). **Route 1 only** (`HeckeConeAction`) honours. The strict-pair route, the relative route and `EquivariantCohomologyOnK`'s relative sub-complex constructor refuse, each with an arm ((b1)). The entrywise agreement of the three routes stays an $\mathbb{F}_\ell$ statement, and **(vii)'s composite half is open**. E8 is declared "complete as specified: 4b by route 1; (vii) over the ring open". *v1.10.0-pre.2, as built and declared* (status paragraph above): items 1–4b as specified; 4b over $\mathbb{Z}/\ell^k$ by route 1; (vii)'s composite half open, the strict-pair route, the relative route and the relative sub-complex constructor refusing with arms ((b1)). (v1.11.0-pre.2: superseded by the next item and the amended declaration of the status paragraph.)

  v1.11.0-pre.1: (vii) over the ring, specified (slot-10 decision 3 (A); built by slot-10 WP-3; the three members are (b1)'s "the lift (slot 10)"). **Scope**: $\ell \geq 5$ and $\ell^k < 2^{32}$. Route 1 is unchanged and runs every regime; the two cross-routes run where the `uint32_t` classes they consume run, and refuse above by the regime rule of (b1).
  - *Route 2, the strict-pair route, over the ring.* τ^s (the `BoundaryStrict` table) is integral, with solve regime `"Z"`, so its Θ^s_g over $\mathbb{Z}/\ell^k$ is the reduction of the integral tables, as-

sembled by the honouring `HeckeOperatorOnK` over the ring; the boundary operator is the honouring `HeckeBoundaryOperator` of (f); the strict datum's pair homotopy η^s is asserted 0, as over $\mathbb{F}_\ell$. The strict action $A_c^{s,(q)}$ is assembled by route 1's module machinery (`ModuleConeActionT`) from the strict data, on `ModuleConeT`'s stored generators: the **same** cyclic-summand basis (c4) of H_c^q as route 1's, so the comparison is between (c5) matrices in one basis. The record's fields keep their $\mathbb{F}_\ell$ meaning (`six_invertible` and `asserted` true at $\ell \geq 5$; `solve_regime "Z"`; `eta_zero`).

- *Route 3, the relative route, over the ring.* The relative sub-complex $C^\bullet_{\mathrm{eq}}(K, K_{\mathrm{bd}}) = \ker r$ is built by `EquivariantCohomologyOnK`'s relative constructor on the **module route of the bulk**, with (c1)–(c5) verbatim and C^q_{eq} read as `kernel_module` of the equivariance constraints stacked with the vanishing constraints on the simplices of K_{bd}; its Howell pivots are asserted to be units (it is expected free at $\ell \geq 5$: a simplex's stabiliser is the same in K as in K_{bd}, so the restriction of equivariant cochains is onto, and the kernel of an onto map of free modules over the local ring $\mathbb{Z}/\ell^k$ is free). A non-unit pivot raises a named `std::logic_error` and is a finding for slot-10 WP-15, never a silent fallback. The flat-coordinate route of (c1′) is **not** used at $\ell \geq 5$. Then $A'^{(q)}$ is the (c5) matrix of route 3's cochain map on the cyclic-summand basis of $H^q(K, K_{\mathrm{bd}})$, the correspondence $M^q\colon H^q(K, K_{\mathrm{bd}}) \to H^q_c$ is a (c5) matrix, certified an isomorphism over the ring (equal invariant factors, and every basis element of H^q_c exhibited as an image by `solve_multi_rhs_module` modulo the target's relations), and the transported action $M\,A'\,M^{-1}$ is computed **column by column** — for each b_j of H^q_c a preimage under M by the solve, then A', then M, reduced in (c5) form. No inverse matrix is formed: a (c5) matrix between non-free modules has none in general.
- *Acceptance* ([C, `obl:coneexact`](vii) as written, over the ring). At each of the seven E8 points of (e), in every computed degree, for T_p at the point's primes ($p \in \{2, 3\}$, and $p = 5$ at $(10, 691)$) and for U_ℓ at the cells where route 1's arms compute it, the $A_c^{(q)}$ of route 2 and of route 3 each **equal route 1's entrywise** in (c5) form (row i compared as reduced residues in $[0, \ell^{e_i})$). Both agreements **abort** on failure at $\ell \geq 5$ ($6 \in R^\times$). A negative control: one perturbed entry of τ^s is detected (by the table's own certificate or by the disagreement; the arm says which). One arm at a 4b word-regime cell above 2^{32}: route 1 runs, routes 2 and 3 refuse by the regime refusal. One arm each at $\ell \in \{2, 3\}$: the three members refuse. Measured by `HeckeConeCrossRouteRingValidation` (`standard`; a point whose wall would push the name over its tier runs under a separate `slow` name, not a re-tier). The $\mathbb{F}_\ell$ cone is byte-identical (`cone_output_diff.sh` against the slot-9 close, 0 hunks), and `HeckeReductionValidation` and `HeckeReductionCoverageSlowValidation` keep their check counts except for (b1)'s changed rows.
- *Not lifted*: [C, `prop:coneroutes`](3)'s adjointness over the ring (it needs the E7 pairing, which refuses a composite ring); (vi)'s $\nu = 2$ recursion; (viii)'s two arms, which stay $\mathbb{F}_\ell$-only. *No object*: the cross-routes are validated equal to route 1, whose objects are the emitted ones, so no golden line, record member or console verb is added, and golden does not move.
- *The declaration*, on acceptance, is the one quoted in the "Pre-release v1.11.0-pre.1" paragraph above; slot-10 WP-15 writes it here and in the status paragraph, and [C] v1.10.0 records (vii) over the ring as discharged at $\ell^k < 2^{32}$, every E8 point, with the `uint32_t` regime rule above. *v1.11.0-pre.2*: written, **amended** as the status paragraph says — the first two sentences of (vii) discharged, the third (the adjointness gate this bullet lists as "not lifted") recorded unavailable over the ring (review finding M-1); [C] v1.10.0 records the same.
- *v1.11.0-pre.2, as built* (slot-10 WP-3; `docs/implementation/slot10/WP-3_report.md`). **Route 2** as specified: τ^s and η^s integral exactly as over $\mathbb{F}_\ell$, $\eta^s = 0$ asserted; Θ^s_g from `HeckeOperatorOnK`'s module route on the same composite bulk cohomology; Φ^s from route 1's `ModuleConeActionT` on the same cone and boundary operator; so route 1's item 2 asserts $r\,\Theta^s = \Theta^\partial r$ on the nose, re-derived over the ring and recorded as `cochain_commutes`.

Route 3 with **one recorded deviation**: the transport is not computed column by column by solves, but by an *explicit inverse on classes* — with the cone differential $D_c(c, c_\partial) = (-dc,\ d_\partial c_\partial + rc)$ and the equivariant section ext_0 of r ($r\,\mathrm{ext}_0 = I$, checked), $M\colon [z] \mapsto [(z, 0)]$ and $N\colon [(c, c_\partial)] \mapsto [c + d\,\mathrm{ext}_0 c_\partial]$ are homomorphisms, mutually inverse on classes, asserted $MN = I$ and $NM = I$ in (c5) form, and A' is transported as $M\,A'\,N$; the equal invariant factors of $H^q(K, K_{\mathrm{bd}})$ and H^q_c are then a consistency check. Still no inverse *matrix* is formed. The review verified that N is well defined on classes and that (c5) products of homomorphism matrices are well defined (finding m-11 asks for the argument in the header note). $E = \mathrm{ext}_0 h$ and $\Theta' = \Theta + dE + Ed$, with R3.1–R3.6 as over $\mathbb{F}_\ell$; the relative constructor's pivots were units at every point. **Acceptance, as measured**: at the seven points ($(10, 691)$ with T_2, T_3, T_5; $(6, 5)$, $(16, 5)$ at $\mathbb{Z}/\ell^2$; $(30, 5)$, $(60, 5)$ at $\mathbb{Z}/\ell^2$ and $\mathbb{Z}/\ell^3$, with T_2, T_3), routes 1, 2 and 3 are entrywise equal in every degree $0 \ldots 2$, with non-vacuity rows ($\tau^s \neq \tau$, $A_c \neq 0$, $A_c(T_2) \neq A_c(T_3)$ at every level); H_c's invariant factors equal $H^q(\ker r)$'s, e.g. $() \,|\, (2, 2, 2) \,|\, ()$ at $(10, 691)$. No U_ℓ arm: route 1's arms compute none at these cells. **Negative control**: a strict table perturbed after construction is refused by its own certificate (code 30); a table perturbed *past* its certificate and pushed through by hand aborts at route 1's item 4 ($D_c\Phi \neq \Phi D_c$), not at the final comparison; nothing perturbs route 3, which has no table of its own (review finding m-12 records a stronger seam for a later slot). (*v1.11.0*: slot-10 WP-8 asserts route 3's non-vacuity, which was printed only — $E \neq 0$, $\Theta' \neq \Theta$ and $r\Theta \neq \Theta^\partial r$ on the shipped τ — at every point, level and prime, 3×15 checks more, so 431 by the source; the slot close re-derives the count from the run. The negative seam on route 3 itself stays for slot 11.) **Regime arm** at $(10, 134217757)$ over $\mathbb{Z}/\ell^2$: route 1 (`ModuleConeActionT<uint64_t>`) runs, both `RequireRingRegime` refuse by the named `uint32_t` refusal. `HeckeConeCrossRouteRingValidation`, 386 checks, 0 failing, `standard` (no `slow` sibling was needed). The $\mathbb{F}_\ell$ cone is byte-identical by `cone_output_diff.sh` with 0 hunks against the `bd6d1bb` reference binary (the plan's `2ac9ee8` reference did not exist when WP-3 ran; the slot close re-runs it, review finding m-13).

The objects, all at the seven E8 points of (e) and nowhere else. Tier A:

- `bdry_lengths_H^q`, $q = 0, 1$: the composition length of $H^q(C^\bullet_\partial \otimes \mathbb{Z}/\ell^k)$, as `lengths_H^q`.
- `bdry_invariant_factors_H^q`, $q = 0, 1$: its list $e_1 \geq e_2 \geq \cdots \geq 1$, as `invariant_factors_H^q`. *What is known.* Over $\mathbb{Q}$, $\rho_n(T)$ is unipotent with one Jordan block. So $H^0(C_{\partial,\mathbb{Z}}) \cong \mathbb{Z}$ and $H^1(C_{\partial,\mathbb{Z}})_{(\ell)} \cong \mathbb{Z}_{(\ell)} \oplus \bigoplus_t \mathbb{Z}/\ell^{a_t}$ (even n; the ℓ^{a_t} are the non-unit elementary divisors of $\rho_n(T) - 1$ over $\mathbb{Z}_{(\ell)}$). Universal coefficients give **the same list in both degrees**: $(k, \min(a_t, k) \ldots)$ sorted. So `bdry_lengths_H^0` = `bdry_lengths_H^1`, which is asserted.
- `bdry_bockstein_rank^0`: the rank over $\mathbb{F}_\ell$ of the mod-ℓ Bockstein $\beta^0_\partial\colon H^0(\partial; \mathbb{F}_\ell) \to H^1(\partial; \mathbb{F}_\ell)$, the connecting map of $0 \to \mathbb{F}_\ell \to \mathbb{Z}/\ell^2 \to \mathbb{F}_\ell \to 0$ on $C^\bullet_\partial$, computed from the stored bases. *Why degree* 0. $C^\bullet_\partial$ lives in degrees $0, 1$, so β^1_∂ lands in $H^2(\partial) = 0$, and degree 0 is the only degree in which the object is non-trivial. It is **defined at every point**. It is a mod-ℓ object, the same at every k of a cell, and is computed from the cell's $\mathbb{Z}/\ell^2$ boundary object, as `bockstein_rank^0` is. *Meaning.* It equals $\#\{t : a_t = 1\}$, by the argument of [C, `rmk:bocksteinexp`](1) applied to $C_{\partial,\mathbb{Z}}$:
  - the classes of $H^0(C_{\partial,\mathbb{Z}})/\ell$ lift to $\mathbb{Z}/\ell^2$-cocycles, so β^0_∂ kills them;
  - on $H^1(C_{\partial,\mathbb{Z}})[\ell]$, β^0_∂ is $G[\ell] \to G/\ell G$.

  Since $H^2(C_{\partial,\mathbb{Z}}) = 0$, no certified level is needed: at every $k \geq 2$, `bdry_bockstein_rank^0` is the number of entries equal to 1 in `bdry_invariant_factors_H^1`. This is **asserted at every point**. There is no boundary analogue of `higher_torsion_count`: $H^0(C_{\partial,\mathbb{Z}})$ is not zero, so $\dim H^0(\partial; \mathbb{F}_\ell) - \mathrm{rank}\,\beta^0_\partial$ counts a free class as well, and would misname it.
- `cone_lengths_Hc^q`, $q = 0, 1, 2$: the composition length of H^q_c over $\mathbb{Z}/\ell^k$. This is the slot-9 plan's `cone_lengths_H^q`, spelled `Hc` after the $\mathbb{F}_\ell$ object `dim_Hc^q`, whose degree convention it shares. `cone_lengths_Hc^0` is 0, asserted.
- `cone_invariant_factors_Hc^q`, $q = 0, 1, 2$: its list, as above. This is the plan's `cone_invariant_factors_H^q`.
- `coneexact_iv_certified` (boolean): `true` iff the non-field half of [C, `obl:coneexact`](iv) above

ran at every spot all three of whose terms are reported, and passed. On the model complexes every spot is reported. A failure aborts, so on an emitted point the value is `true`. `false` is the pre-slot-9 state, "recorded as unchecked", and no slot-9 build emits it at an E8 point.

- `cone_les_im_lengths`: the plan's "cone LES length identity", as an object with content. It is the list of the lengths of the images of the eight map instances of [C, `eq:coneles`], in [C]'s order $\pi^0_*, r^0, \iota^0_*, \pi^1_*, r^1, \iota^1_*, \pi^2_*, r^2$ ([C, `obl:coneexact`], preamble). Here $\pi^q_*\colon H^q_c \to H^q$, and $\iota^q_*\colon H^q(\partial) \to H^{q+1}_c$. It is emitted only after **the length identity** is asserted at each of the seven terms $H^0, H^0(\partial), H^1_c, H^1, H^1(\partial), H^2_c, H^2$ of the sequence: the length of the term equals the image length of the incoming map plus that of the outgoing map. This is [C, `eq:coneles`]'s exactness in lengths, with $H^0_c = 0$ and $H^2(\partial) = 0$ at its ends.

Tier B (basis-dependent; (c)'s bases):

- `bdry_red_H^q`, $q = 0, 1$: the reduction $H^q(\partial; \mathbb{Z}/\ell^k) \to H^q(\partial; \mathbb{F}_\ell)$. It is the (c5) matrix from the cyclic-summand basis to the D5b basis of the $\mathbb{F}_\ell$ boundary complex, the one `SL2ZBoundaryPair` builds at the defaults. As in (d), red $= P^\partial_q(b_j \bmod \ell)$.
- `cone_red_Hc^q`, $q = 0, 1, 2$: the same map for H^q_c, into the $\mathbb{F}_\ell$ cone's stored basis `Hc^q`, through its projector `Pc^q`.

Reductions compare as maps, never as bases, exactly as in (d). The Hecke matrices $A^\partial(T_p)$ and $A_c(T_p)$ over $\mathbb{Z}/\ell^k$ are computed and validated, but are **not** objects in slot 9, as $A^{(q)}(T_p)$ over the ring is not.

Hash recipes, as (e):

- a scalar stores its value, and the boolean stores 0 or 1;
- a list hashes its length and then its entries. `rows` is the length and `cols` is 1. `cone_les_im_lengths` has 8 rows;
- a reduction matrix hashes `rows`, then `cols`, then its entries row-major. 0×0 is legal, for example `cone_red_Hc^0`.

The generating rule of slot-9 WP-4's pre-stated set (the event R1, (a/s); the lead lands it, after slot-9 WP-14 pass 1). At each of the seven points of (e) the set is:

- tier A: `bdry_lengths_H^{0,1}`, `bdry_invariant_factors_H^{0,1}`, `bdry_bockstein_rank^0`, `cone_lengths_Hc^{0,1,2}`, `cone_invariant_factors_Hc^{0,1,2}`, `coneexact_iv_certified` and `cone_les_im_lengths`, which is 13;
- tier B: `bdry_red_H^{0,1}` and `cone_red_Hc^{0,1,2}`, which is 5.

No object depends on a prime or on the certified level. Every object is defined at every point, so **the "absent where undefined" rule removes nothing at these seven points**. The set is $7 \times 18 =$ **126** entries, 91 tier A and 35 tier B.

- They are emitted at each point after its existing E8 objects, in the order above; no other point gains or loses an entry.
- Predicted counts after R1: tier A $11\,575 + 91 = 11\,666$, tier B $5866 + 35 = 5901$, entries $17\,441 + 126 = 17\,567$.
- Both determinism digests are unchanged by construction, because the E8 points are looped from `ComputeAll()` only.

Predicted values. A prediction that fails is a finding for the owner, and is never re-predicted. The boundary is predicted exactly, from the elementary divisors of $\rho_n(T) - 1$ over $\mathbb{Z}_{(\ell)}$, computed independently of the pipeline by exact rational elimination (script and output in slot-9 WP-2's report). Write $B :=$ `bdry_lengths_H^0`, $L_q :=$ `lengths_H^q` (the landed values of (e)), and $c_2 :=$ `cone_lengths_Hc^2`. Exactness and $H^2 = 0$ then determine the whole of `cone_les_im_lengths` from c_2:

$$(0,\ L_0,\ B - L_0,\ L_1 - B + c_2,\ B - c_2,\ c_2,\ 0,\ 0), \qquad \texttt{cone_lengths_Hc\^1} = L_1 - L_0 + c_2.$$

Here r^0 is injective ([C, `lem:conerelative`](4)), π^2_* lands in $H^2 = 0$, and ι^1_* is therefore onto H^2_c.

cell	K	$H^0(\partial) = H^1(\partial)$	B	β_∂	cone
$(10, 691)$	2	(2)	2	0	$H_c^1 = (2, 2, 2)$, $H_c^2 = ()$; im-lengths $(0, 0, 2, 4, 2, 0, 0, 0)$
$(16, 5)$	2	$(2, 1, 1, 1)$	5	3	$H_c^1 = (2, 2, 2, m)$, $H_c^2 = (m)$, $m \in \{1, 2\}$
$(6, 5)$	2	$(2, 1)$	3	1	$\ell(H_c^1) = 2 + c_2$
$(30, 5)$	2	$(2, 2, 1^5)$	9	5	$\ell(H_c^1) = 10 + c_2$
$(30, 5)$	3	$(3, 2, 1^5)$	10	5	$\ell(H_c^1) = 15 + c_2$
$(60, 5)$	2	$(2^3, 1^{10})$	16	10	$\ell(H_c^1) = 18 + c_2$
$(60, 5)$	3	$(3, 2^2, 1^{10})$	17	10	$\ell(H_c^1) = 27 + c_2$

The third column is `bdry_invariant_factors_H^q`, the same list at $q = 0, 1$. Exponents are listed with multiplicity. β_∂ is `bdry_bockstein_rank^0`. $H_c^0 = ()$ and `coneexact_iv_certified` $= 1$ at every point.

- *The two complete cone rows.* They use the $\mathbb{F}_\ell$ golden values of the ext cone at the same cell: $\dim H_c^2(\mathbb{F}_\ell) = 0$ at $(10, 691)$, and $\dim H_c^1(\mathbb{F}_5) = 4$, $\dim H_c^2(\mathbb{F}_5) = 1$ at $(16, 5)$.
- *How they follow.* H_c^2 is the top cohomology of the cone, so $H_c^2(\mathbb{Z}/\ell^k) \otimes \mathbb{F}_\ell = H_c^2(\mathbb{F}_\ell)$, and it vanishes by Nakayama when the latter does. The $\mathbb{Q}$-ranks $\operatorname{rk} H_c^1 = 3$ and $\operatorname{rk} H_c^2 = 0$ (weight 12 and 18, one Eisenstein class) then fix the rest by universal coefficients.
- *What is left open.* $m = \min(a, 2)$ with $5^a = |H_c^2(\mathbb{Z})_{(5)}|$ is not predicted. At $(16, 5)$ the im-lengths are $(0, 0, 5, 1 + m, 5 - m, m, 0, 0)$. So $\ker r^1$ over $\mathbb{Z}/25$ has length $1 + m \geq 2$, although $r^1 \otimes \mathbb{F}_5$ is injective: $H^1(\partial; \mathbb{Z}/25)$ has 5-torsion summands, and elements of order 5 can map into them.
- c_2 at the other four rows is not predicted.

v1.10.0-pre.2: the landed values (slot-9 WP-4; R1 at `762dbc3`). **Every prediction above held**: the boundary lists exactly, row by row; the complete $(10, 691)$ row; and the pattern $(0, L_0, B - L_0, L_1 - B + c_2, B - c_2, c_2, 0, 0)$ at the other rows. The values left open are $m = 1$ **at** $(16, 5)$ — $H_c^1 = (2, 2, 2, 1)$, $H_c^2 = (1)$, im-lengths $(0, 0, 5, 2, 4, 1, 0, 0)$, so $|H_c^2(\mathbb{Z})_{(5)}| = 5$: $H_c^3 = 0$ ([C, `lem:conerelative`](4)) makes $H_c^2(\mathbb{Z}/25) = H_c^2(\mathbb{Z}) \otimes \mathbb{Z}/25$, one cyclic summand $\mathbb{Z}/5^a$ by $\dim H_c^2(\mathbb{F}_5) = 1$, and $m = \min(a, 2) = 1$ forces $a = 1$ — and $c_2 = 0$ at $(6, 5)$, $c_2 = 1$ at $(30, 5)$ at both K, $c_2 = 2$ at $(60, 5)$ at both K, with $H_c^2 = (1)$ and $(1, 1)$ there at $K = 2$ and at $K = 3$ alike, so those summands are $\mathbb{Z}/5$ exactly. The cone Bockstein β_c^1 has rank 0 at $(6, 5)$ and $(10, 691)$, 1 at $(16, 5)$ and $(30, 5)$ and 2 at $(60, 5)$, consistent with $\ell(H_c^2(\mathbb{Z}/\ell^2)) = 2 \dim H_c^2(\mathbb{F}_\ell) - \operatorname{rk} \beta_c^1$, which the validation asserts at its `uint32_t` cells (v1.10.1: not at the two word-regime cells, which have no $\mathbb{F}_\ell$ cone class to compare with). Arm (4b-C) pins $m = 1$ at $(16, 5)$ since `7ddb555` (review finding m-W4-4). The landed set is exactly the generating rule's: 126 entries, 91 tier A and 35 tier B, emitted after each point's existing E8 objects; three emissions byte-identical, one of them under `ENGINE_THREADS=1`; `golden_ref_diff.py --expect` holds it as a set equality and `golden_explain.py` exits 0 with no flag. *Four of the eighteen objects per point are structural constants* — `cone_lengths_Hc^0` $= 0$, `cone_invariant_factors_Hc^0` $= ()$, `cone_red_Hc^0` 0×0 and `coneexact_iv_certified` $= 1$ — 28 entries that cannot move without a certificate failing first; they are frozen as specified (review finding m-W4-4). Measured by `GOLDEN_STRICT=1 GoldenRegressionValidation`: tier A 11 666 failing 0, tier B 5901 mismatching 0.

(g) The job-level schema `hecke_ring_report/v1` (v1.10.0-pre.1). Slot-9 decision 6 (A); built by slot-9 WP-8 (C5c). Documented for users in `campaign.md` §15, which slot-9 WP-8 writes. *v1.11.0-pre.1* (slot-10 decision 8 (A)): **no schema change in slot 10** — slot-10 WP-4's ℓ-adic survey under $\mathbb{Z}/\ell^2$ writes `hecke_ring_report/v1` exactly as frozen in v1.10.0, (g1)–(g16) with the four recorded deviations and the (g11) amendment, and a finding that would need a member is reported for a later version, not added. *v1.11.0-pre.2*: no code package of slot 10 touched the schema or `Campaign/`. *v1.11.0*: slot-10 WP-4's survey was **not run in slot 10** (the optional W2; deferred), so nothing here moves. *v1.11.1*: the survey **ran in W2** (`docs/matrix-engine/results/ring_survey_l2/`, 462 records): every record is `hecke_ring_report/v1` exactly as frozen, (g1)–(g16) with the recorded deviations, and no finding of its analysis needs a member — so nothing here moves.

v1.10.0, as built (slot-9 WP-8, `ad65bd9`, with the (g11) amendment `f4fc3d2`, merged at `8853544`; `Campaign/CampaignRing.{h,cpp}`; `campaign.md` §15). `RunOneJob` dispatches to the ring job when-

ever `coefficient_ring` is not `F_ell`; nothing of the field path runs. The members (g1)–(g16) are emitted in that order, with `certify` after (g5); the file is assembled in memory and written once. At $\ell \geq 5$ the job computes the cohomology by the module route; at $\ell \in \{2,3\}$ by the flat-coordinate route of (c1′), through `EquivariantCohomologyOnK`'s own lift: since the merge fix `158c0be` the job loop builds its relation list there from the raw `ModCtx::prime_power` context (the value `CoefficientRingContext` would return) and does **not** name `SmallEllPolicy::Bredon`, which stays with `EquivariantCohomologyOnK` alone ((b1), the scoping row); at every other ℓ the context keeps its refusal rule. The element width is `uint32_t` for $\ell^k < 2^{32}$ and `uint64_t` otherwise. **Four deviations from the text below, recorded as built** (slot-9 WP-8 report, findings 1, 2, 4 and 6): (i) at $\ell \geq 5$ *off the K-mode's range* ($n = 0$ or n odd) `kmode` is `"not_applicable"`, as at $\ell \in \{2,3\}$ ((g7) named 2 and 3 only), and `ladic`, $r_K(d^q)$ and `ladic_type*` are absent, as (d) says, while `bockstein` and `red_H^q` are present (they are mod-ℓ objects, defined there); (ii) the uncertified `ladic.state` of (g9) also carries `need: {r_K0, r_K1}`, the deficit (the console's `ladic_level` state carries the same); (iii) `hecke` is computed iff $\ell \geq 5$, $r = 1$ and $1 \in$ `point.degrees` ((g11)); otherwise, at $\ell \geq 5$, it is absent and **not** refused — at $\ell \in \{2,3\}$ it is refused whatever the degree selection ((g14)); (iv) each refusal's `who` names the member the job loop called (`HeckeOperatorOnK`, `HeckeReduction (BocksteinCounts)`, `HeckeReduction (ReduceHToField)`, `campaign job loop`) and its `message` is that guard's own text, obtained by calling the real guard (at $\ell \in \{2,3\}$ the Bockstein and reduction messages name the Bredon object's accessor reached, `module_degree` or `H_module_basis`); a guard that no longer refuses **fails the job**, so a later lift needs this paragraph amended first. Three further readings, not deviations: `timings_s` keeps `campaign_job/v1`'s nine keys (`cosets`, `tau`, `theta` the $r = 1$ Hecke part; `family` and `external` 0; `report` the objects' own time, including the $\mathbb{F}_\ell$ and $\mathbb{Z}/\ell^2$ comparison objects); `certify.mode` is the spec's certify text verbatim; in `summary.tsv` the `status` column carries `ok_partial` and a degree outside `point.degrees` holds `-`, while `index.json`'s per-job `status` stays `ok` ((g) adds exactly one top-level key there). **(g11) amended** (lead decision of 2026-09-24, before this version froze the schema): the free `charpoly`'s coefficients are decimal strings (below). **Acceptance** is measured by `CampaignRingValidation` (`standard`, 693 checks): arm (A), two ring campaigns at `jobs = 2` over the seven E8 points of (e) with each point's primes (17 jobs), every carried object **equal to the frozen golden entry** (tier, rows, columns, hash) and the object set equal to the point's 4a golden set for the job's p, the (e) table's certified flag, H^0 / H^1 invariant factors, Bockstein rank and higher-torsion count literally, through `summary.tsv`; arm (T), the weight-12 free `charpoly` over $\mathbb{Z}/\ell^2$ at $(10, 691)$ and at the `uint64_t` cell $(10, 134217757)$, in string form; arm (B), four cells at $\ell \in \{2,3\}$ against the frozen O7 rows, `not_applicable`, the three refusals and `ok_partial`; arm (R), $r = 2$; arms (S) and (X), the spec surface of (b4). **The fingerprint-0 byte identity** is measured against the base binary: on `ext_compare_mini.spec` (84 ext records raw-identical; the 84 job files, `index.json` and `summary.tsv` identical once the timing and RSS numbers are blanked), on a reduced field sweep (104 files) and a resident `level_a = always`, `certify = full` spec, by `--dry-run` of the seven shipped specs (byte-identical, so `sweep_full`'s job ids are unchanged), and by the existing campaign validations with their id assertions unchanged. The job loop is clean under `CheckThread` at `jobs = 2`.

- *When.* The campaign's job loop writes this schema **only** under a non-default `coefficient_ring`, and then instead of `campaign_job/v1`.
- *A fingerprint-0 job* writes `campaign_job/v1` with its `hecke_report/v2` reports **byte for byte as before**. The same holds for its `summary.tsv` and `index.json`. No default id, file name, header or shape signature moves.
- *Golden cost: none.* The schema exists only at a fingerprint $\neq 0$, and those texts are not golden ((a), "Fingerprint and golden cost").
- *The ext pass* keeps refusing a non-default ring by name, at spec validation, (b4).

The job file. It is `jobs/<id>.json`, where the id is the existing `Job::id()` with its settings suffix `_s<16 hex>`, because the fingerprint is non-zero; there is no new id rule. It is one JSON object whose members come in this order:

(g1) `"schema": "hecke_ring_report/v1"`.

(g2) `"job_id"`: the id above.

(g3) "point", "model_descriptor", "shape_signature": as in campaign_job/v1 (the same writers). point.degrees is the job's *reported selection*, which is Job::id()'s, not a complex's degree fields.

(g4) "coefficient_ring": the canonical text "Z/ell^<k>". "modulus": ℓ^k as a *decimal string*, since it may exceed 2^{53}. "ladic_level": $K = k$, an integer.

(g5) "settings": the MathSettings::to_json block of campaign_job/v1, with every registered key, its value and source, and then "fingerprint", which is never "0000000000000000" here.

(g6) "degree_fields": the bulk complex's per-complex fields q_max, truncated, q_max_source and degrees (2 / false / "derived" / $[0, 1, 2]$ on the model).

(g7) "kmode": "certified", "uncertified" or "not_applicable". The last is used at $\ell \in \{2, 3\}$ ([C, prop:certifiede](5)) and is **not** a refusal. *v1.10.0, as built*: also at $\ell \geq 5$ off the K-mode's range, $n = 0$ or n odd (deviation (i) above).

(g8) "cohomology": one object per q in point.degrees, ascending, each {"q", "length", "invariant_factors"}. These are (d)'s lengths_H^q and invariant_factors_H^q; the list is descending, and [] is the zero module. **No member anywhere in the file is a dimension.**

(g9) "ladic" (**present only at** $\ell \geq 5$): {"r_K": [$r_K(d^0)$, $r_K(d^1)$], "certified": <bool>, ...}. It carries "type" (a list of {"q", "free_rank", "torsion"} for $q = 0, 1, 2$) *only* when certified. It carries "state", the R3 state of kind ladic_level of (d) (budget $\{K\}$, reached $\{r_K(d^0), r_K(d^1)\}$, complete = false), *only* when uncertified. The state's continuation is a fresh job at a larger k, never resume. *v1.10.0, as built*: the state also carries need: {r_K0, r_K1} (deviation (ii) above).

(g10) "bockstein" (**present only at** $\ell \geq 5$): {"bockstein_rank_0", "higher_torsion_count", "dim_H0_F_ell"}, (d)'s objects with the bookkeeping identity asserted. They are computed from the cell's $\mathbb{Z}/\ell^2$ object and the $\mathbb{F}_\ell$ object, and each of those is built with its own ring installed for its construction only (the RunE8Point precedent). dim_H0_F_ell is a dimension of the *field* run, named as one.

(g11) "hecke" (**present only at** $\ell \geq 5$, $r = 1$, $1 \in$ point.degrees): {"p", "r": 1, "charpoly_free"}. "charpoly_free" is {"precision": $K - e_{\max}$, "coeffs": [...]}, low to high, each coefficient a *decimal string* in $[0, \ell^{K-e_{\max}})$ (it may exceed 2^{53}, as (g4)'s modulus may; v1.10.0, lead decision of 2026-09-24 — through v1.10.0-pre.2 this item wrote JSON numbers), and is *present only when certified*. At $p = \ell$ the operator is U_ℓ. *v1.10.0, as built* (slot-9 WP-8, f4fc3d2): no other member of the file can exceed 2^{53} — under $k \geq 2$, $\ell^k < 2^{64}$ forces $\ell < 2^{32}$; every count is bounded by the complex; hashes and modulus are strings — and CampaignRingValidation arm (T) asserts the string form at $(10, 691)$ and at the uint64_t cell $(10, 134217757)$ (for example "18014406292930825").

(g12) "objects": a list of {"name", "tier", "rows", "cols", "fnv1a64"}, one for every (d) object present, under its golden name of (e) and its hash recipe of (e), in (e)'s order. red_H^q (tier B) is included at $\ell \geq 5$, and "fnv1a64" is sixteen lower-case hex digits. Wherever a job's cell, k and p coincide with an E8 point of (e), each entry equals that point's golden entry. This is the comparison slot-9 WP-8's acceptance (a) makes.

(g13) "refusals": a list of {"part", "who", "message"}, one per part of the job that was refused by name. part is one of "hecke", "bockstein", "reduction". message carries the tag not_implemented_in_this_build:. A refused part is **never silently skipped**.

- At $\ell \in \{2, 3\}$, hecke, bockstein and reduction are refused (HeckeOperatorOnK, HeckeReduction).
- At $r \geq 2$, hecke is refused at the job loop's own dispatch: T_{p^r} over the ring is not validated in this build.

The list is empty when nothing was refused.

(g14) "status": "ok" (no refusal), "ok_partial" (the cohomology computed, and at least one part refused and listed) or "failed" (an error, written as campaign_job/v1 writes a failed job).

(g15) "timings_s", "peak_rss_mb", "peak_rss_scope": as in campaign_job/v1.

(g16) "campaign_job_complete": the same completion trailer, true, or false on failure. So resume

works unchanged.

The `certify` knob of the spec is accepted and recorded as `"certify": {"mode": <spec value>, "applicable": false}` after (g5). Every external cross-check is a field route, and `HeckeCrossChecks` refuses a ring. The in-process certificates always run, and a failure among them fails the job.

`summary.tsv` and `index.json` of a ring campaign.

- *`summary.tsv`.* The 21 columns keep their names and order. The ten field-valued ones hold `-`: `dim_H0`, `dim_H1`, `dim_H2`, `charpoly_A0`, `charpoly_A1`, `charpoly_A2`, `trace_A1`, `rank_A1`, `theta_flat_hash_q1` and `external`. Fifteen columns are appended: `coefficient_ring ladic_level kmode length_H0 length_H1 length_H2 invariant_factors_H0 invariant_factors_H1 invariant_factors_H2 r_K0 r_K1 bockstein_rank0 higher_torsion_count charpoly_free_A1 refusals`.
- *How the new columns are written.* A list is semicolon-separated (the `charpoly` convention: coefficients low to high, then the precision after a `@`). An empty list is `()`. An absent or undefined value is `-`. `refusals` holds the refused parts, semicolon-separated, or `-`.
- *`index.json`* gains one top-level key, `"coefficient_ring"`, only in a ring campaign.
- A campaign with a fingerprint-0 spec writes neither change.

(h) The console under a non-default ring (C5a; v1.10.0-pre.1). Slot-9 decision 2 (A); built by slot-9 WP-12. [P] v3.7 (slot-9 WP-15) records the binding surface. This paragraph fixes **which verbs and members honour** $\mathbb{Z}/\ell^k$, and that **every other pipeline verb keeps exit** 5 by (b), through a lazy handle too.

- *Adding to the list.* A member not listed here refuses with exit 5, naming itself. It may honour only after this list is amended, as a pre-release amendment by the lead or at slot-9 WP-15.
- *Exit codes.* A named `std::logic_error` surfaces as exit 3, and `hecke::NotImplementedInThisBuild` as exit 5.
- $\ell \in \{2, 3\}$. **In slot 9 the console lifts nothing at** 2 **and** 3 (slot-9 plan WP-3, acceptance (b)): under a composite ring, every console pipeline verb at $\ell \in \{2, 3\}$ keeps exit 5. The Bredon object at 2 and 3 is reachable from the validations and from the campaign's job loop, (g).

At $\ell \geq 5$:

- `symmetric_power(Model, n, ell)` **honours**. It builds the ring's relation list, `GenerateSL2ZRelationsRing`, over $\mathbb{Z}/\ell^k$.
- `cohomology(L)` **honours** on a ring `LocalSystem`, and returns a `Cohomology` handle over the ring (the E8 type; [P] §3.6: "`coefficient_ring` becomes a member of `Cohomology`"). Its honouring members:
  - `.coefficient_ring`, the canonical text; also `F_ell` on a field handle, and on every `HeckeOperator`;
  - `.computed`, and the per-complex degree members `.q_max`, `.truncated`, `.q_max_source` and `.degrees`;
  - `.lengths`, a list over $q = 0, 1, 2$, and `.invariant_factors(q)`;
  - the K-mode: `.ladic_type_certified`, `.r_K(q)` for $q \in \{0, 1\}$, and `.ladic_type(q)`, a Record {`free_rank, torsion`}. At an uncertified level `.ladic_type(q)` raises its named `std::logic_error`, which is exit 3;
  - `state(H)` and `save_state(H)`, which return the `ladic_level` state. `resume(H, budget)` on it is refused, by §7's rule for a changed non-budget setting, with that rule's exit code: the continuation is a fresh run at a larger k;
  - `.bockstein_rank` and `.higher_torsion_count`, (d)'s objects, computed as in (g10);
  - `.red(q)`, the tier-B matrix `red_H^q`. Its $\mathbb{F}_\ell$ comparison object is built with `F_ell` installed for that construction only.

  The field members `.dims`, `.Z(q)`, `.B(q)`, `.H(q)`, `.C(q)`, `.P(q)`, `.A_C(q)` and `.A_Z(q)` raise the object's named `std::logic_error`, which is exit 3. There is no `dim` over $\mathbb{Z}/\ell^k$ ((b3)).
- `hecke(H, p)` with $r = 1$ **honours** on a ring `Cohomology`, $p = \ell$ included (then it is U_ℓ). Its honouring members:

  - `.coefficient_ring`;
  - `.A(q)`, the (c5) matrix on the cyclic summands;
  - `.charpoly_free`, the Record {`precision, coeffs`} of (d), defined only at a certified level; otherwise it raises the named `std::logic_error`, which is exit 3.

  Every other member exits 5.
- `iota(H)` **honours** on a ring `Cohomology` (v1.20.0-pre.6, an amendment of this list by slot-20 WP-LV3(d); Remark 9.18), at $\ell \geq 5$, when the ring in force equals the handle's (a mismatch refuses at dispatch, exit 5, as `hecke(H, p)`; a digit handle refuses when forced). Its honouring members: `n`, `ell`, `status`, `coefficient_ring`, `certificate`, `json` (schema `hecke_involution_ring/v1`), and the module members `lengths_plus`, `lengths_minus`, `invariant_factors_plus(q)`, `invariant_factors_minus(q)` (also on $\mathbb{F}_\ell$ handles, as $k = 1$). The field members `dims*` raise a named `std::logic_error` (exit 3); the field methods `matrix`, `plus`, `minus`, `sign_split`, `eisenstein_line` exit 5. Nothing changes on an $\mathbb{F}_\ell$ handle. Script `049_lv3_iota_ring.oc`.

Everything else keeps exit 5 under a non-default ring:

- `hecke(H, p, r)` with $r \geq 2$, refused at the verb's dispatch;
- `family`, `decompose`, `report` and `crosschecks`;
- every extension-layer verb of C5b (slot-9 WP-13), including the reserved `boundary_cohomology(H)` and `compact_support(H)`. **The objects of (f) are console-unreachable in slot 9**, as E8's were in slot 8;
- the reserved `reduce(H, k)` and `bockstein(H)`, which stay reserved and unbound.

`golden()` still refuses at a non-zero fingerprint (R19). The harness verbs lift the session's settings to the baseline as before. *Tests*: arm (K) of `ConsoleErrorValidation`, extended: each honoured verb and member succeeds, and every other one exits 5 (or 3, where named above). The first-freeze scripts `023_...` assert the values against `HeckeReductionValidation`'s printed ones at the E8 points. [P]'s "no edit to `Console/Core/`" holds, except the two named exceptions of (a).

v1.10.0-pre.2, as built (slot-9 WP-12, `b3cd0d3`; follow-up `f28a0fa`). The list above honours, exactly, at $\ell \geq 5$; nothing is lifted at $\ell \in \{2, 3\}$. The ring is read once, at the verb's dispatch, and kept in the handle; a lazy handle installs exactly that ring around its construction, so a handle's ring never depends on who forces it. The comparison objects of `.red` and of the Bockstein members (the $\mathbb{F}_\ell$ object and, at $k \neq 2$, the $\mathbb{Z}/\ell^2$ one) are each built with their own ring installed for their construction only, the `RunE8Point` precedent. `.ladic_type(q)` and `.charpoly_free` are Records ({`free_rank, torsion`}, {`precision, coeffs`}); the `ladic_level` state is the handle's R3 state (`budget_setting = coefficient_ring`, `resumable` exactly when uncertified); `save_state(H)` forces a declared ring handle first, `state(H)` reports once it has computed, and `resume` is refused — by the core's "is complete" at a certified level, by the type's non-budget-rule text (exit 3) at an uncertified one. The `hash()` of a List of Ints follows the golden list recipe (length, then entries), so `hash(H.invariant_factors(q))` equals `invariant_factors_H^q` byte for byte (the empty list hashed identically under both recipes, so no earlier value moved). Scripts `023_e7_duality_10_691`, `024_e8_ring_points` (the seven E8 points against the literals `HeckeReductionValidation` prints, and by golden hash where a console value has the golden recipe) and `025_e8_console_surface`, each a first freeze; the 22 earlier transcripts are byte-unchanged. **Four as-built facts this paragraph did not state** (WP-12 report §4):

- *A ring mismatch is exit* 5. A verb handed a handle built over a different ring refuses at dispatch with `hecke::NotImplementedInThisBuild`, naming both rings — the pipeline's own classification (`HeckeOperatorOnK`); likewise a C5b verb handed a ring handle after the setting has returned to `F_ell` (`f28a0fa`).
- *`.A(q)` over the ring is a List of Int rows, not a `Matrix`*: row i lives modulo ℓ^{e_i}, and the `Matrix` type's field algorithms would be meaningless on it.
- *Unlisted structural members exit* 5 *on a ring handle*, as "Adding to the list" says: `LocalSystem.n`, `.p`, `.d_V`, `.relations`; `HeckeOperator.p`, `.r`; `Cohomology.ell`, `.d_V`, `.dim_C`. Listing any of them (with its code) is an owner item for a later amendment of this list; it is not made here.
- *A member getter's error carries no source position.* The gated types convert their members'

exceptions into console errors in the bindings, so the exit codes are right (5 and 3), but the core's `Evaluator::object_member` does not wrap a getter's exception as a method call's is wrapped, and a getter has no position; the type-level `resume` refusal lacks one for the same reason. The fix is one `try/catch` in `Console/Core/`, the analogue of (a)'s `exec_set` back-fill, and is recorded for the next authorised Core batch ([P] v3.7).

The E7 verbs `dual` and `pairing` and every C5b verb (slot-9 WP-13) refuse a non-default ring with exit 5, as listed above.

v1.11.0-pre.1: the structural members, and the ring-mismatch rule (slot-10 decision 6 (ii) (A); built by slot-10 WP-12, from its lift commit). The list above is amended, as "Adding to the list" requires, by **five members that honour on a ring handle** at $\ell \geq 5$:

- `LocalSystem.n`: n, an Int, as on a field handle.
- `LocalSystem.d_V`: $n+1$, an Int — over $\mathbb{Z}/\ell^k$ the rank of the free module $\mathrm{Sym}^n(R^2)$, not a dimension over a field; the number is the same.
- `LocalSystem.coefficient_ring`: **a new member of the type**, present on every handle, as `Cohomology`'s and `HeckeOperator`'s are: the canonical text `"Z/ell^<k>"` on a ring handle, `"F_ell"` on a field handle. It is the one member slot 10 adds to a type; `docs/console/console.md` is regenerated by the lead after slot-10 WP-12 merges.
- `HeckeOperator.p`: the Hecke prime, an Int ($p = \ell$ is U_ℓ).
- `HeckeOperator.r`: an Int, always 1 on a ring handle (`hecke(H, p, r)` with $r \geq 2$ is refused at dispatch, above).

Each is a read of the handle's stored data: it forces no computation, so on a declared lazy handle it returns without computing, as on a field handle. **Every other unlisted member keeps exit** 5 through `bind::GateTypeOnRing`, in particular `LocalSystem.p` (it would read ℓ while the ring is $\mathbb{Z}/\ell^k$; `.coefficient_ring` carries both) and `.relations`, `HeckeOperator`'s other unlisted members, and `Cohomology.ell`, `.d_V`, `.dim_C`. At $\ell \in \{2, 3\}$ no ring handle can be built (the verbs refuse at dispatch, above), so nothing changes there. **The ring-mismatch rule**, recorded as built by slot-9 WP-12 (its finding 4, "four as-built facts" above) and **binding from this pre-release**: a verb handed a handle built over a ring other than the one in force refuses *at dispatch* with `hecke::NotImplementedInThisBuild` naming both rings, exit 5 — the pipeline's own classification (`HeckeOperatorOnK`) — and so does a C5b verb handed a ring handle while `F_ell` is in force; a *member* read on an existing handle is not a dispatch and follows the handle's own ring, whatever is in force ((b2): an object's ring is its own). *Tests*: arm (K) of `ConsoleErrorValidation` — its four rows `L.n`, `L.d_V`, `T.p`, `T.r` change from exit 5 to "honours", each with its value checked, and are reported as changed rows; `L.coefficient_ring` is a new row, on a ring handle and on a field handle; the other unlisted members (`L.p`, `L.relations`, `H.ell`, `H.d_V`, `H.dim_C(1)`, ...) keep their exit-5 rows; new scripts from `026_`, first freezes. *Positions*: a member getter's error also carries its `file:line:col` from slot-10 WP-12's commit (the one named `Console/Core/` exception of slot-10 decision 6 (i)); that is console behaviour, and [P] v3.8-pre.1 §2 C7, §6 and §8 specify it. No exit code of this paragraph changes by it.

v1.11.0-pre.2, as built (slot-10 WP-12, `3c96d63`, merged at `6726075`; `docs/implementation/slot10/WP-12_report.md`). **The five members** honour exactly as listed: `LocalSystem` gains `coefficient_ring` (`"F_ell"` on a field or preset handle), and `bind::GateTypeOnRing`'s rules admit `LocalSystem.n`, `.d_V`, `.coefficient_ring` and `HeckeOperator.p`, `.r`; each is a read of the payload, and a declared lazy handle stays `declared` after the reads (pinned). Every other unlisted member keeps exit 5. Arm (K) of `ConsoleErrorValidation`: `L.n`, `L.d_V`, `T.p`, `T.r` changed from exit 5 to 0, each with its value; `L.coefficient_ring` new; a new arm (W) (24 scripts) pins the member positions, the handle's own ring, and **the mismatch rule in both directions** ($\mathbb{Z}/\ell^3$ in force against a $\mathbb{Z}/\ell^2$ handle, $\mathbb{F}_\ell$ against a ring handle, a ring against an $\mathbb{F}_\ell$ handle) — the rule held at every handle-taking verb at the base, so no code changed for it; 218 → 294 checks, tier `fast`. Script `026_c5_leftovers_ring_members` is a first freeze; no existing transcript moved; `docs/console/console.md` regenerated (one row, `LocalSystem.coefficient_ring`). **Positions and one exit code**: the named `Console/Core/` exception adds a member access's `file:line:col` to a getter's error ([P] v3.8 §2 C7). Every gated getter of this paragraph's types

already converted `NotImplementedInThisBuild` to exit 5, so no code of this paragraph moved; but an *unwrapped* getter raising `hecke::NotImplementedInThisBuild` was exit 3 at the base (it escaped as a plain `std::exception`) and is **exit** 5 now, the kind classifier deciding as for a method — no shipped getter was found to raise it unwrapped, an audit WP-12 records as not exhaustive over the extension layer's types ([P] v3.8 §8).

7.2 The congruence-census catalogue `hecke_census/v1`

This subsection is the specification slot-11 WP-7 (the tool, the computed split, the catalogue), WP-6 (the θ-image column, §5.2 (f)), WP-8 (the second edition) and WP-19 (the examples document [EX], which selects from (g)) build against, written before any of them commits a result. The catalogue classifies every maximal ideal of the published extension records with the owner's column list (slot-11 plan §3 "WP-7" scope 3). It is a **derived artefact with its own format tag**, `hecke_census/v1`: *not* a record, not a campaign output and not golden. **No record schema changes**: `hecke_ext/v3`, `hecke_report/v2` and `hecke_ring_report/v1` are untouched, the published directories `ext_first/`, `ext_u691/` and `ring_survey_l2/` stay byte-unchanged, and no digest or golden line moves. [CEN] below is the day-zero census note, committed verbatim at `docs/implementation/slot11/census/` (`congruence_census.md`, `census.py`), which the tool must reproduce.

(a) **Inputs and the supersession rule.** Under `docs/matrix-engine/results/`, the records `ext/*.json` of `ext_first/` (505 cells, $\ell \in \{5, 7, 11, 13, 691\}$, even $n \leq 200$) and of `ext_u691/` (45 cells), read-only. *Supersession*: a cell (n, ℓ) of `ext_first` is *superseded* iff `ext_u691` holds a record of the same (n, ℓ) and `ext_first`'s record has `u_ell.included = false`; a superseded record contributes no row and is listed, with its hash, in the provenance. The tool asserts that the superseded set is exactly the 45 cells $\ell = 691$, $n = 112, \ldots, 200$ of [CEN]; any other outcome stops the tool. The second edition adds WP-8's new directory by the same rule (a cell in two directories is a finding unless a supersession rule of that edition's `README.md` names it). Auxiliary inputs: `ring_survey_l2/` for (d5); §7.1 (e)'s table for the E8 points' exponents; the θ-image and split lines of (e).

(b) **Rows.** One row per entry of a record's `ideals` list after supersession — [CEN]'s unit, 2569 rows in the first edition. A row's key is (`source_set`, n, ℓ, `ideal_id`). Its *own group* is the group whose `modules` entry the record carries (`H1`, else `H0`), as [CEN]'s `census.py` reads it; its *groups* are the record's per-ideal `groups` list.

(c) **Columns**, in this order; the TSV header names them exactly so. "Record" means the row's ideal entry, copied verbatim (lists as compact JSON); every cell holds a value or one of the explicit tokens `not_computed`, `not_surveyed`, `not_evaluated`, `not_applicable`, `undecided` — never an empty field.

column(s)	source	rule
`edition`, `source_set`, `record_path`, `record_sha256`	the file	path relative to `docs/matrix-engine/results/`; sha256 of the record's bytes
`n`, `ell`, `weight`, `ideal_id`	record	`weight` $= n + 2$
`own_group`, `groups`	record	(b)
`witness_poly`	record	low→high
`boundary_split`, `rank_r1`, `dim_E`	computed, (d1), (e)	`eisenstein`, `cuspidal`, `mixed`, `undecided`, `not_applicable`
`sign`, `dim_plus`, `dim_minus`	record; computed, (d2), (e)	the owner's ± column
`residue_degree`, `residue_degree_certified`	record	a `residue_degree_lower_bound` is written `>=<f>`, never as a degree (§7)
`length_A`, `reduced`, `loewy_A`, `embdim`, `gorenstein`, `hilbert`	record `algebra.*`	(d3)
`module_D`, `module_length`, `module_loewy`, `module_loewy_length`, `min_generators`, `cyclic`, `free`	record `modules.<own group>.*`	(d3)

column(s)	source	rule
`label_status`, `label_type`, `label_system`, `label_theta_twist`	record `label`	verbatim; `label_type` is the atlas's (reducible / irreducible $\bar{\rho}$), *not* (d1)
`predicted`, `confirmed`, `unpredicted`	record	verbatim; only the Kummer predictor exists ([CEN]), so `unpredicted` is not informative at $\ell \leq 13$ — stated in the edition's `README.md`, not per row
`kummer_tag`	record `predicted`, cell `predictions.kummer`	`eisenstein_cuspidal:k=<k>` with its `confirmed` flag, else `none`; the two $\ell = 691$ occurrences cross-referenced (tag `pair_691`, (g))
`theta_image_dim`, `theta_image`	computed, (d4)	`none`, `partial`, `all`, `not_computed`
`zl_depth`, `zl_depth_source`	(d5)	
`glover_multiplicity`, `glover_main`, `glover_correction`, `glover`	computed, (d6), (e)	`equal`, `differs_by:<d>`, `class_level:<ids>`, `not_evaluated`, `not_applicable`
`final`, `final_reason`, `certificate`, `certificate_routes`	record	verbatim; (d7)
`sturm_B_prime`, `sturm_cap`, `sturm_proof_range`, `sturm_outcome`	record `sturm.*`	`not_applicable` when the record has no `sturm` object
`certified`	derived, (d7)	`true`, `heuristic`
`extremal`	derived, (g)	a JSON list of tags, possibly empty

(d) **The rules.**

(d1) *The boundary split* (the one computed classification; slot-11 WP-7, `HeckeExt/HeckeCensusSplit`). For a row whose own group is `H1`: $E_{\mathfrak{m}} \subseteq H^1(\Gamma; M_n)$ its generalised eigenspace over $\mathbb{F}_\ell$ on the unsplit H^1 (every published record has `sign = "all"`), $\dim E_{\mathfrak{m}}$ = `dim_E`, and `rank_r1` := $\operatorname{rank}_{\mathbb{F}_\ell}\left(r^1|_{E_{\mathfrak{m}}}\right)$ with $r^1 : H^1(\Gamma; M_n) \to H^1(\partial; M_n)$ the restriction `HeckeBoundary` builds (its joint-eigenspace use at `HeckeBoundary.h:293` is the precedent). Verdict: `eisenstein` iff `rank_r1` = $\dim E_{\mathfrak{m}}$; `cuspidal` iff `rank_r1` = 0; `mixed` iff $0 <$ `rank_r1` $< \dim E_{\mathfrak{m}}$. A row whose own group is `H0`: `not_applicable` (its H^1 part, if any, is its own row). *Where computed*: every cell with $\ell \leq 13$, and the two 691-pattern cells $(10, 691)$, $(198, 691)$; at every other $\ell = 691$ cell the verdict is `undecided`, and the row keeps its `kummer_tag`. *What the verdict means*: it is a mod-ℓ statement about $\ker r^1 = H^1_!(\mathbb{F}_\ell)$, not about reductions of characteristic-zero forms — §8 records that for $\ell \leq n+1$ "$H^1_!$ is the cuspidal part" fails in both directions, so `cuspidal` means "inside $\ker r^1$" and `eisenstein` "injects into the boundary". *Expected at the benchmark*: at $(10, 691)$ the Eisenstein ideal is `mixed`, $\dim E_{\mathfrak{m}} = 3$, `rank_r1` = 1 (Claim 8.1; §8, the ten Kummer cells: $\operatorname{rank} r^1 = 1$ on the block). The owner's question — an Eisenstein-labelled ideal that is `mixed` — is the conjunction `label_type` = `eisenstein` (at $\ell \leq 13$) or a confirmed `kummer_tag` (at $\ell = 691$) with `boundary_split` = `mixed`. *Invariants* (per cell, asserted by the split driver): $\sum \dim E_{\mathfrak{m}} = \dim H^1$ over the `H1` rows; $\sum$ `rank_r1` $= \operatorname{rank} r^1 = \dim H^1 - \dim H^1_!$.

(d2) $\pm$. `sign` is the record's, verbatim (`all` in every published record, so it carries no information there). The owner's $\pm$ content is `dim_plus` := $\dim(E_{\mathfrak{m}} \cap H^1_+)$ and `dim_minus` := $\dim(E_{\mathfrak{m}} \cap H^1_-)$ for the production involution ι^* (`HeckeSignSplit`; ι^* commutes with the Hecke family, so $E_{\mathfrak{m}}$ is ι^*-stable), computed by the split driver with the verdict of (d1) and where it is; asserted `dim_plus` + `dim_minus` = $\dim E_{\mathfrak{m}}$ and $\sum$ `dim_plus` = $\dim H^1_+$ of the record's `groups`. `not_computed` where (d1) is `undecided`; `not_applicable` on `H0` rows.

(d3) *Algebraic columns* (the keys [CEN]'s `census.py` reads). `length_A` := `algebra.dim/residue_degree`, the length of the local algebra A over itself; `loewy_A` := `algebra.nilpotency`, its Loewy length; `reduced` := (`loewy_A` = 1), equivalently A a field; `embdim`, `gorenstein`, `hilbert`

verbatim. On the module: `module_D` := `D`, `module_length` := $\mathtt{D}/f$ (the semisimple multiplicity of the ideal's system in its own group; it may exceed `length_A`, the module being neither cyclic nor free in general), `module_loewy` := `loewy`, `module_loewy_length` its number of layers. "Length" in the owner's list is `length_A`; `dim_E` is the $\mathbb{F}_\ell$-dimension of the ideal's block, never called a length.

(d4) *θ-image.* `theta_image_dim` := $d_\theta(\mathfrak{m})$ of §5.2 (f) (cell target-indexed: the image of $\theta_{n-\ell-1}$); `theta_image` is `none` iff $d_\theta = 0$, `all` iff $d_\theta = \dim E_\mathfrak{m}$, `partial` otherwise. $d_\theta = 0$ by definition at $n < \ell+1$ (every $\ell = 691$ row), with source `n_lt_ell_plus_1`. `not_applicable` on `H0` rows (the H^0-level image of θ^0 is the Dickson ring's $L\cdot$, (6d), not a census column). `not_computed` where no `THETA_IMAGE` line exists (e.g. a second-edition cell WP-6's producer has not run on).

(d5) $\mathbb{Z}/\ell^k$ *depth* — the exponents of the ℓ-primary torsion of $H^1(\Gamma;\mathrm{Sym}^n\mathbb{Z}^2)$ localised at $\mathfrak{m}$. By [C, `cl:bockstein`](2)–(3) (Hecke-equivariant Bockstein, $\ell \geq 5$, $n > 0$) that torsion is non-zero at $\mathfrak{m}$ iff $H^0(\Gamma; M_n)_\mathfrak{m} \neq 0$, i.e. iff the row's own group is `H0` or its `groups` list carries an `H0` entry with `matched = true`. Values: `0` (source `bockstein`) for every other row — at every $\ell \geq 5$, $n > 0$, surveyed or not; `not_applicable` at $n = 0$ ($H^0(\Gamma;\mathbb{Z}) = \mathbb{Z}$ is not torsion). For a row meeting H^0 at a cell of `ring_survey_l2` ($\mathbb{Z}/\ell^2$, even $n \leq 64$): the H^0 invariant factors give each cyclic summand's exponent as `1` (factor 1) or `>=2` (factor 2; the `higher_torsion_count` summands), refined to the exact exponent at the E8 points over $\mathbb{Z}/5^3$ by §7.1 (e)'s table ($(30,5)$: (2); $(60,5)$: $(2,1)$), which the tool carries as a frozen table citing that site and never recomputes (source `e8_point`; at a ring level k an exponent is exact below k and written `>=k` at k). The multiset is *attributed to the row* iff the cell has exactly one `H0`-own row (the row and every row matched to it carry it) or all the cell's exponents are equal (each H^0-meeting row carries the common value); otherwise every H^0-meeting row reads `cell_level:{<multiset>}` (source `ring_survey_l2`). At an unsurveyed cell a row meeting H^0 reads `>=1` with source `not_surveyed`. Expected in the first edition (from `ring_survey_l2`'s analysis): ℓ^2-torsion exactly at $(30,5)$, $(60,5)$, $(56,7)$.

(d6) *Glover reconciliation* (Claim 5.13; identity 6 of §5.1). For an `H1` row at $\ell \in \{5,7,11,13\}$ (the in-process atlas range): the ideal's residue class c (one Galois orbit of anemic systems over the canonical split field, as `HeckeIdentityCertificate` item 6 defines it); `glover_multiplicity` := $\dim_{\mathbb{F}_\ell}(E_\mathfrak{m})/r$, the semisimple multiplicity of c in H^1; `glover_main` := the coefficient of c in $\sum_i \theta^{j_i}[H^1(M_{n_i})]$ over the Glover factors $\mathrm{Sym}^{n_i}\otimes\det^{j_i}$ of `glover_factors(n, ell)` (the record's `hecke_glover_factors/v1`, re-derived by the certificate's own recursion, standard-module convention); `glover_correction` := the coefficient of c in the H^0 (and H^2) corrections of identity 6. `glover` is `equal` iff `glover_multiplicity` = `glover_main`, else `differs_by:<d>` with d = `glover_multiplicity` − `glover_main`; identity 6 then requires d = `glover_correction`, and a row where it does not is a **finding** (tag `identity_6_mismatch`), never re-fitted. If c is shared by several rows of the cell (ideals that differ only outside the anemic family), each reads `class_level:<ids>` with the class's numbers. `not_evaluated` at $\ell = 691$ (identity 6 is not evaluated there, §5.1 item 6); `not_applicable` on `H0` rows. Computed by the split driver through `HeckeIdentityCertificate`'s item-6 evaluation, never by a second implementation. This column is where [CEN]'s inference "the $\ell = 5$ multiplicity-4 ideals at $n = 34\ldots44$ are Glover multiplicities" is confirmed or refuted.

(d7) *Certification.* Every column read from the record's `algebra` or `modules` is certified iff `final` = `true`; the row's `certified` is `true` then and `heuristic` otherwise (every non-final published ideal has `final = false`, `final_reason = "sturm_heuristic"`), and the `sturm_*` columns carry its proof range against the Sturm bound. A statement of the catalogue about a heuristic row ("reduced", "Loewy length 2") is marked heuristic wherever it is made.

(e) **The computed columns' line formats and the matching rule.** The split driver runs each cell's $\mathbb{F}_\ell$ pipeline and the shipped ext driver's decomposition (seconds per cell at $\ell \leq 13$), and `HeckeCensusSplitValidation` (`standard`, ≤ 60 s) asserts the invariants of (d1), (d2)

and identity 6 on a grid, with $(10, 691)$'s `mixed` among them. Under the read-only interlock `HECKE_CENSUS_SPLIT_ALL=1` (not an emitter) it runs every census cell of (d1) and prints, per `H1` row,

```
CENSUS_SPLIT n=<n> ell=<ell> ideal=<id> witness=<c0,...> dim_E=<d> rank_r1=<r> verdict=<v>
 dim_plus=<a> dim_minus=<b> glover_mult=<m> glover_main=<g> glover_corr=<c> glover=<token>
```

and the θ producer prints the `THETA_IMAGE` lines of §5.2 (f). *Matching*: a run's block is the record's ideal iff its `witness_poly` and its restricted `t_p` matrices equal the record's (the pipeline and driver are deterministic, so equality is exact); the matching must be one-to-one on every cell, and a block or ideal left unmatched is a **finding** that stops the cell, never a nearest guess.

(f) **The category table.** The tool reproduces [CEN]'s category table (ideals, reduced, non-reduced, maximal nilpotency, maximal `length_A`, maximal `embdim`, non-Gorenstein, $f > 1$, maximal reduced module multiplicity, not final, per ℓ) *exactly* from the published records — the acceptance test of the tool — and adds the `boundary_split` and `theta_image` counts per ℓ below it.

(g) **Extremal selection** (the list [EX] selects its examples from; a row carries every tag it meets; a heuristic row's tag carries the suffix `_heuristic`): `len_ge_3` (`length_A` ≥ 3); `embdim_ge_2`; `non_gorenstein`; `loewy_ge_3` (`loewy_A` ≥ 3 or `module_loewy_length` ≥ 3 — none in the first edition by [CEN], kept for the second); `family:<key>` for a non-reduced row whose key $(\ell, \texttt{length_A}, \texttt{embdim}, \texttt{gorenstein}, \texttt{hilbert})$ occurs at ≥ 3 values of n (the recurring families); `reduced_high_mult` (`reduced` and `module_length` ≥ 4); `mixed` (every `mixed` row); `pair_691` (a non-reduced $\ell = 691$ row with a confirmed `kummer_tag`: $(10, 691)$ and $(198, 691)$ in the first edition, the pattern $A \cong \mathbb{F}[\varepsilon]/\varepsilon^2$, $\dim E_{\mathfrak{m}} = 3$, module Loewy layers $[2, 1]$); `theta_partial` ($0 < d_\theta < \dim E_{\mathfrak{m}}$); `zl_depth_ge_2`; `identity_6_mismatch`. The selection is the union; [EX] chooses among it and says why.

(h) **The `results/` layout**, published owner-signed as the directory `congruence_census/` of `docs/matrix-engine/results/`:

`README.md`	the index of editions (the one file a later edition edits)
`edition1/README.md`	the format tag, the edition, the inputs with each directory's `index.json` sha256, (a)'s rule, the tool's version and command line, the binary's commit, the owner's sign-off line
`edition1/census.tsv`	every row, the columns of (c) in order; first line `# schema=hecke_census/v1 edition=1 ...`
`edition1/extremal.tsv`	the rows with a non-empty `extremal`, same columns
`edition1/category_table.txt`	(f)
`edition1/inputs.sha256`	one line per record read (`<sha256> <relpath>`), superseded ones marked
`edition1/split_lines.txt`	the `CENSUS_SPLIT` lines, verbatim
`edition1/theta_image.tsv`	`hecke_census_theta/v1`: the `THETA_IMAGE` lines as a table
`edition1/findings.txt`	the answers of slot-11 WP-7 scope 5, each with its rows
`edition2/`	the same files over edition 1's inputs and WP-8's directory

A later edition never rewrites an earlier one's files. The tool is `congruence_census.py` under `docs/matrix-engine/tools/` (WP-7; WP-8 appends its second-edition mode), read-only on every input; it writes only into the edition directory it is given.

(i) **Non-default settings.** A record computed under a non-default mathematical setting (WP-8's `ell = 5` cells under a raised `sturm`) is *not golden-comparable*: its `settings.fingerprint` is non-zero, and the edition's `README.md` says so and names the fingerprint. The catalogue carries the record's fingerprint in the provenance, not as a column.

(j) **Refusal.** The split driver and the θ producer are $\mathbb{F}_\ell$-only (§7.1 (b), the v1.12.0-pre.1 paragraph); the census tool reads only $\mathbb{F}_\ell$ records and stops on a record whose `coefficient_ring` is not `F_ell`.

v1.12.0, as built (slot-11 WP-7, WP-8). The tool, `HeckeCensusSplit` and the first edition are built to (a)–(k), with these differences, each recorded in the edition's `README.md`:

- **(e)** `CENSUS_SPLIT` carries one field after the ones listed, `t_p=<fnv64>`: the FNV-1a-64 of the canonical text `"<p>:[[a,b],...];..."` of the restricted operators, ascending in p. The tool

recomputes it from the published record, which makes (e)'s matching exact without printing matrices; its negative controls (one changed entry, one changed prime) move it.

- **(f)** The acceptance is *cell-exact* (the 55 cells of [CEN]'s table compared with whitespace stripped, 55/55, and its two summary lines by value), not byte-exact: [CEN]'s padding is hand-made. Two negative controls stop the tool.
- **(c), (d)** `kummer_tag` is written `eisenstein_cuspidal:k=<k>:confirmed` (or `:unconfirmed`); at the $\ell = 691$ cells other than $(10, 691)$ and $(198, 691)$ the computed columns read `not_computed` and the split `undecided`. (d6)'s Glover classes run over the cell's anemic primes $\{2, 3, 5, 7\} \setminus \{\ell\}$ ($S_{\mathrm{sep}}(\ell)$ lies inside it at every $\ell \leq 13$); a class shared by several rows is checked against $\sum \dim E_{\mathfrak{m}} = \mathrm{mult} \cdot f$ by the review on all 298 classes of edition 1, but not asserted in the code (a carry, slot-11 WP-14 pass 2, m-6). `HeckeCensusSplitValidation` is `standard` (117 checks); under `HECKE_CENSUS_SPLIT_ALL=1` it runs the edition-1 grid (406 cells, 5643 checks).
- **(i)** Edition 1 is published: `docs/matrix-engine/results/congruence_census/edition1/`, 2569 rows, owner-signed on 2026-09-25 (commit `bb7c860`; the edition's `README.md` line still reads "pending" and is the slot close's to update). **Edition 2 is not built in slot 11**: the owner moved it to slot 12 (slot-11 plan §10, ruling (i)). In its place slot-11 WP-8 published the W2 campaigns `ext_census_l7111` (112 cells, $\ell \in \{7, 11\}$, $n \leq 300$) and `ext_sturm_l5` (101 cells, $\ell = 5$ under a raised `sturm`, fingerprint `29cc925d211962b3`, *not golden-comparable*, (j)) and a *data set*, not an edition: `congruence_census/w2_dataset/`, format tag `hecke_census_dataset/v1` (the tool's dataset mode: the record-derived columns only; the split, $\pm$, θ-image and Glover columns read `not_computed` / `not_evaluated`), owner-signed on 2026-09-25 (commit `63f2191`). `HECKE_CENSUS_SPLIT_ALL` hard-codes the edition-1 grid; widening it is edition 2's code change.

A prediction read off edition 2 (v1.14.0-pre.2): on every non-reduced H^1 row the local algebra has $\mathfrak{m}_A^2 = 0$ — Prediction 4.9 ([conjectural]), tested on the lattice mod ℓ^2 by slot-13 WP-9 (v1.14.0-pre.3: on non-Dickson rows only; on a non-reduced Dickson-type block it is untested; v1.15.0: a third route and slot-14 WP-6's 348 cells, no non-reduced Dickson-type block among them; v1.15.1-pre.7: **refuted as stated** at $\ell = 37$ beyond the census window, on Kummer–Frobenius ideals, and restated with their exclusion — every edition-2 row lies below every Kummer–Frobenius onset; v1.17.0-pre.2: the v1.16.0 form refuted on cohomology at $(608, 547)$, beyond every edition, and restated again on Kummer-free classes; v1.17.0-pre.4, owner ruling on the referee F1's B-1: that form contradicted on the forms side at irreducible congruence classes, the regular $\ell = 139$ among them, and (R) restated on congruence-free classes; the rows of editions 2 and 3 unaffected).

Edition 2 reproduced on a second machine (v1.15.0, provenance): the first cluster run (job `34550470`, commit `07f413f`; `docs/matrix-engine/results/noctua2_run1/cpu2_arms.txt`, published 2026-09-28) recomputed edition 2's `CENSUS_SPLIT` lines (the union of its two split cases is exactly edition 2's 2568-line `split_lines.txt`), the θ-image table of slot-11 WP-6 (1934 rows) and the forms census of slot-12 WP-12 (655 rows), each byte-identical to the tracked reference.

The depth window beyond $n = 64$ (v1.15.0, pointer): (d5)'s $\mathbb{Z}/\ell^k$ depth is surveyed at even $n \leq 64$ (`ring_survey_l2`). The first cluster run extended the survey over $\mathbb{Z}/\ell^2$ to even $66 \leq n \leq 300$ at $\ell \in \{7, 11, 13\}$ and $66 \leq n \leq 200$ at $\ell \in \{5, 691\}$ (`docs/matrix-engine/results/noctua2_run1/depth_window_l2.ts`
published 2026-09-28; [C, `rmk:bocksteinexp`](3); [EX] `nc:depth`). No edition carries it: a `zl_depth` column for $n > 64$ would be a later edition's (census edition 3, deferred to slot 15). (v1.16.1) Edition 3 (below) carries a `depth_window` column from the $\mathbb{Z}/\ell^2$ window at $\ell \in \{5, 7, 11, 13, 691\}$ only; at its new primes the column reads `not_surveyed`.

Census edition 3 (v1.16.1; slot-15 WP-15, owner request of 2026-10-01; a pointer and its evidence, no statement of this document changes). `docs/matrix-engine/results/congruence_census/edition3/`, format tag `hecke_census_dataset/v2` (edition 1's 58 columns unchanged, then 13 trailing columns), is published *beside* edition 2 (a sibling; editions 1 and 2 are untouched and still reproduced byte for byte, `EDITION2_NATIVE rows=3285 identical=3285`). It has 80 293 rows: edition 2's 3285 and 77 008 at the 21 primes $\ell = 17 \ldots 103$, in 4891 cells, the new ones all `hecke_ext/v5` records under `u_ell = budget_mb=16384` (settings fingerprint `43364ef4020ac327`, so *not golden-comparable*); every row is `final = true`. **Coverage, stated honestly**: the C1 grid is every even $n \leq 700$ at each new prime (7350 cells); edition 3 is **complete at** $\ell = 61 \ldots 103$ (350 cells each, the C1b block, 62 113

rows) and **partial at** $17\ldots59$ ($n \le 120$ only at $17\ldots31$; the heavy end of C1a and the local smoke at $37\ldots59$): 3064 of the 7350 C1 cells are missing (`missing.txt`); they are to be added, add-only, as edition 3.1. A statement "at ℓ" about edition 3 means "on the rows present". Columns not computed on every row: the split columns (604 local-smoke cells only), `hidden_congruence` (3035 new rows joined to a weight-shift block, no `true` among them), `saturated` / `mult_one` (321 new rows; arm (T) was not run at $n > 500$ and its lines were dropped on 230 cells at $61\ldots103$, $n \ge 302$, where its default `u_ell` leaves U_ℓ out — a slot-16 repair, then edition 3.1). *What edition 3 adds as evidence* (computed, on the rows present):

- Prediction 4.9 (R): `loewy_A` ≥ 3 occurs on exactly 18 rows, all at $\ell = 37$ and all Kummer–Frobenius, so (R) has 0 violations; at $\ell = 61\ldots103$, where the coverage is complete, all 2165 non-reduced rows have Loewy length ≤ 2, consistent with (R). The C-F9 lattice test held on all 155 evaluable cheap new-prime rows (`cf9.tsv`).
- Prediction 4.10: the onset law predicts no Loewy-4 cell at $\ell = 61\ldots103$ inside $n \le 700$, and none occurs; the law's positive half is untested there.
- Kummer agreement (`kummer_exact_agree`): true on 55 015 rows of edition 3, false on none; on the complete C1b block 40 713/40 713.
- The k2 law of slot-14 WP-C14 I-8 (a twisted Eisenstein ideal $\mathrm{eis}(i,j)$, $0 < i < j$, outside the degenerate family, is non-reduced $[3,1]$ iff $k \ge \ell(j-i+1)+i(\ell+1)$): 174/174 on the unpredicted twisted-Eisenstein ideals at $61\ldots103$, no miss, none early (slot-15 WP-C18 §3).
- The rationality clause of slot-15 WP-12's Proposition D on the dihedral $\mathbb{Q}(\sqrt{-\ell})$ blocks: 3, 2, 1 rational systems per cell at $\ell = 71$, 79, 83 (55, 45, 39 cells), and at $\ell = 103$ one block of residue degree 2 with a character of order 5 (28 cells), exactly as predicted; none at $\ell \equiv 1 \pmod 4$ or at 67.
- F9: the $\mathrm{eis}(0, k-1)$ ideal is non-reduced in every $v_\ell(B_k/k) \ge 1$ cell present, the depth-2 cells $(282, 37)$ and $(226, 103)$ included.

Report: `docs/implementation/slot15/WP-15_report.md`; the C1b analysis `WP-C18_noctua2_rb_report.md`; the edition's `README.md` and `findings.txt` (E3-1 … E3-8).

Census edition 3.1 (v1.17.0-pre.1; slot-16 WP-4 and WP-5, published at the slot-16 close by the owner's default ruling S16-N4; a pointer and its evidence, no statement of this document changes). `docs/matrix-engine/results/congruence_census/edition3/3.1/` is an **add-only** extension of edition 3: the same 80 293 rows and the same 70 non-edition columns, byte for byte (`addonly.txt`: `ADDONLY rows=80293 ... violations=0`), with `edition` = 3.1; `saturated` and `mult_one` filled on the 2544 residue-degree-1 non-reduced rows that edition 3's arm (T) had dropped (all 2544 read `saturated = true`; with edition 3's, 2865 of 2865 new-prime rows are saturated); and three trailing columns from slot-16 WP-5's `HeckeSaturationDefectValidation` (`sat_defect_dim`, `sat_defect_loewy`, `sat_length`: the $\dim I$, the Loewy length of I and $\mathrm{length}(\mathbb{T}^{\mathrm{sat}}/\mathbb{T})$ of [C, `rmk:sattor`], read on a torsion-free block at a stable K), **partial**: computed on the cells counted by `sat_cells_done` in the edition's `FINGERPRINT.txt` (897 of 1459 at its publication, 2026-10-02; 560 when it was staged, 2026-10-01), `not_computed:no_log` on the other rows and `not_computed:torsion_block` on Dickson-locus blocks. Edition 3's 3064 missing C1 cells are still missing; the cluster returns become edition 3.2. A statement "at ℓ" about edition 3.1 means "on the rows present", as for edition 3. Report: `docs/implementation/slot16/WP-4_report.md`; the edition's `README.md`.

Census edition 3.2 (v1.18.0-pre.1; a pointer and its evidence only, no statement of this document changes; slot-17 WP-9, `docs/implementation/slot17/WP-9_report.md`). `docs/matrix-engine/results/congr` is **staged, not published** (owner-gated; the slot-18 plan's ruling O7: it is completed after the cluster returns and published add-only then). It has edition 3.1's 80 293 rows and 74 columns; the only changes are the three saturation-defect columns filled on 176 further cells (the cells (n, ℓ) with $546 \le n \le 608$ of `satdef_cells.tsv`), so `sat_cells_done=1073 of 1459` in its `FINGERPRINT.txt`; the add-only proof against 3.1 reads `violations=0` (`addonly.txt`; 4982 fills per column, one reason refined from `not_computed:no_log` to `not_computed:torsion_block`). The remaining 386 cells ($608 \le n \le 700$, the edition's `README.md`) read `not_computed:no_log`. (v1.18.0-pre.2, the referee F1's m-E6: pre.1 read "$n \ge 608$" for the remaining cells; the ranges are in n and share $n = 608$, a

cell being a pair (n,ℓ): the 7 cells $(608,\ell)$ filled in 3.2 are among the 176.) Until it is published this document cites edition 3.1.

8 First test cases and expected outcomes

8.1 Ramanujan: $(n,\ell)=(10,691)$

Campaign data: $\dim H^1=3$, one block (3:1) for each of $p=2,3,5$; oracle O6: $\min(T_2)=(X+24)^e$, $e\geq 2$ [C, `sec:example`].

Claim 8.1 (The local algebra at the Ramanujan ideal)**.** *[proved here] (given [C, `cl:bockstein`], Claim 2.7 and Eichler–Shimura over $\mathbb{Z}_{691}$) Let $\mathfrak{m}$ be the unique maximal ideal of $\mathbb{T}_\infty$ on $H^1(\Gamma;M_{10}(\mathbb{F}_{691}))$. Then $\mathbb{T}_\mathfrak{m}\cong\mathbb{F}_{691}[\varepsilon]/(\varepsilon^2)$ with*

$$T_q \;=\; \sigma_{11}(q)+c_q\,\varepsilon, \qquad c_q\equiv\frac{\tau(q)-\sigma_{11}(q)}{691}\pmod{691},$$

for every prime q, $q=\ell=691$ included ($U_{691}=T_{691}$, [C, `cor:ulrecursion`]); concretely $c_2=-3$, $c_3=-256$, $c_5\equiv-174$, $c_7\equiv-137$, and $c_{691}\equiv 542$ from $\tau(691)=-2747313442193908$, $\sigma_{11}(691)\equiv 1$. Hence the matrix identities *on all of H^1*

$$3\,(T_3-252)=256\,(T_2+24), \qquad 3\,(T_5-4830)=174\,(T_2+24),$$
$$3\,(T_7+16744)=137\,(T_2+24), \qquad 3\,(U_{691}-1)=-542\,(T_2+24),$$

every $T_q-\tau(q)$ has Jordan type $(2,1)$, embedding dimension 1, $\mathbb{T}_\mathfrak{m}$ Gorenstein, every T_q with $c_q\neq 0$ is an active direction, and $H^1_\mathfrak{m}\cong\mathbb{T}_\mathfrak{m}\oplus\mathbb{F}_{691}$ (Loewy layers $(2,1)$, socle of dimension 2, not cyclic; cyclic on each ι-sign once (A3) exists: $H^1_+\cong\mathbb{T}_\mathfrak{m}$ free of rank one, $H^1_-\cong\mathbb{F}_{691}$).

Proof. $T_\mathbb{Z}(M_{12})\subseteq\mathbb{Z}\times\mathbb{Z}$, $T_m\mapsto(\sigma_{11}(m),\tau(m))$, is the subring generated by $(1,1)$ and these pairs; $\sigma_{11}(m)\equiv\tau(m)\pmod{691}$ for all m (Ramanujan), and $\gcd\big((\tau(2)-\sigma_{11}(2))/691,(\tau(3)-\sigma_{11}(3))/691\big)=\gcd(3,256)=1$, so $T_\mathbb{Z}(M_{12})=\{(a,b):a\equiv b\ (691)\}=:R$, a free $\mathbb{Z}$-module on $(1,1)$ and $\varepsilon:=(0,691)$ with $\varepsilon^2=691\varepsilon$. $R\otimes\mathbb{F}_{691}=\mathbb{F}_{691}[\varepsilon]/(\varepsilon^2)$ is local. Since $10<690$, $H^1(\Gamma;M_{10}(\mathbb{Z}))$ has no 691-torsion ([C, `cl:bockstein`]), the Hecke algebra on $H^1(\Gamma;M_{10}(\mathbb{Z}_{691}))$ is $R\otimes\mathbb{Z}_{691}$ (both torsion-free, equal after $\otimes\mathbb{Q}$), and $\mathbb{T}_\mathfrak{m}$ is its reduction, a quotient of $\mathbb{F}_{691}[\varepsilon]/(\varepsilon^2)$; O6 (non-scalar T_2) excludes the quotient $\mathbb{F}_{691}$. The coefficients: $(a,b)=a(1,1)+\frac{b-a}{691}\varepsilon$. On the 3-dimensional $H^1_\mathfrak{m}$, $\varepsilon^2=0$ and $\varepsilon\neq 0$ force $\operatorname{rank}\varepsilon=1$, whence the Jordan type, the Loewy layers and $H^1_\mathfrak{m}\cong\mathbb{T}_\mathfrak{m}\oplus\mathbb{F}$; the sign split is Eichler–Shimura ($H^1_+\cong M_{12}$, $H^1_-\cong S_{12}$ [M §5]). □

Verified. On the existing $A^{(1)}(T_q)$ (campaign run at $(10,691)$, $p\in\{2,3,5,7\}$, 2026-09-14) the stored matrices are $T_q=a_qI+c'_qE_{31}$ with $(a_q,c'_q)=(667,23),(252,120),(684,643),(531,129)$, and all three anemic identities hold exactly in $M_3(\mathbb{F}_{691})$; moreover $c'_q/c'_2\equiv c_q/c_2$ for $q=3,5,7$ (ratios $546,58,276$), which is stronger. **That one line is basis-dependent, and the rest of this paragraph is not** (v1.5.0; slot-4 WP-2 finding 6; [XA m19] draws the distinction and the code honours it): a_q, the ratios, the Jordan type, $\operatorname{rank}\varepsilon$ and the U_{691} identity below are invariants of the module, whereas the individual c'_q — and with them the shape $a_qI+c'_qE_{31}$ — are read off *the stored H^1 basis*, a golden *tier-B* object. A re-derivation on another admissible basis reproduces every other number here and need not reproduce the c'_q; a claim that it must would be a claim about the basis choice and not about $\mathbb{T}_\mathfrak{m}$. Rerun for this revision with $p=691$ (`--campaign`, `certify = none`): $A^{(1)}(U_{691})=I+221\,E_{31}$, and $221\equiv 542\cdot 23\cdot(-3)^{-1}\pmod{691}$ exactly as predicted, so $3(U_{691}-1)=-542(T_2+24)$ holds too — the strongest single test in this document, since it exercises U_ℓ and the value $\tau(691)$. The Jordan type $(2,1)$ and $\operatorname{rank}\varepsilon=1$ are read off directly. This is the first acceptance test of the programme, passed with no new code.

8.2 Two further campaign rows at $\ell = 691$

(n, ℓ)	campaign data ($p = 2, 3, 5$; `results_table.tsv`)	expected
$(198, 691)$	$\dim H^1 = 33$; blocks 2:1, 3:1, 4:2, 4:2, 20:10; the 3:1 block is the Eisenstein class merged with one $\mathbb{F}_{691}$-rational cuspidal system ($(691, 200)$ is an irregular pair [37])	[to be verified computationally] As Claim 8.1, $\mathbb{T}_{\mathfrak{m}} \cong \mathbb{F}[\varepsilon]/(\varepsilon^2)$, *provided* the congruence has index one: $691 \parallel \mathrm{num}(B_{200}/200)$ holds (recomputed for this revision: $\lvert \mathrm{num}(B_{200}/200) \rvert \equiv 217665 \pmod{691^2}$; $B_{200} < 0$, and the signed numerator is $\equiv 259816$, v1.12.0). **Uniqueness: closed at v1.12.0.** Exactly one $\overline{\mathbb{Q}}_{691}$-eigenform of weight 200 is congruent to E_{200}, with $\mathbb{F}_{691}$-rational reduction: at $(198, 691)$ $H^0 = 0$, so the block is torsion-free ([C, `cl:bockstein`]), and $\dim E_{\mathfrak{m}} = 3$ with $\pm$ content 2/1 (the census, §7.2) is one Eisenstein class plus two per congruent eigenform (Eichler–Shimura), which leaves room for exactly one; independently, on S_{200} (dimension 16) the characteristic polynomial of T_p modulo 691 factors with degrees $1, 1, 2, 2, 10$ for $p = 2, 3, 5, 7$ — the census's five blocks 2:1, 3:1, 4:2, 4:2, 20:10 — and the joint generalised eigenspace of $1 + p^{199}$ is one-dimensional (slot-11 WP-19's reading of the record, recomputed from q-expansions by its referee, WP-20, `docs/implementation/slot11/WP-20_referee.md`; the examples document [EX] §5.1). Through v1.11.1 this read: the uniqueness "is *not checked* and the possible ambiguity is documented here (v1.4.0; owner decision OQ13: uniqueness does not concern this stage — cohomology over $\mathbb{Z}$ and $\mathbb{Q}$ with functoriality is the long-term route; an LMFDB lookup … is a to-do …; through v1.3.3 'still to be checked')". Loewy $(2, 1)$, $r^1 = 0$ on the Eisenstein class in the sense of Claim 1.1 — on the socle E, with $\mathrm{rank}\, r^1 = 1$; the other blocks semisimple with $\mathbb{T}_{\mathfrak{m}}$ a field; the 20:10 block one Galois orbit of residue degree 10 with multiplicity 2, cyclic per sign.
$(160, 691)$	T_5 alone has a 4:1 block where T_2 and T_3 each have 2:1, 2:1, 2:1, 4:2	**Verified 2026-09-16** (v1.3.3, `HeckeLocalityProbe`, recorded, not asserted): the `ComputeMultiOp` run on $\{T_2, T_3, T_5\}$ gives seven blocks 2:1, 1:1, 2:1, 2:1, 4:2, 6:3, 10:5 on $\dim H^1 = 27$, all seven certified local (`certify() = 0`); T_2's 4:2 is a residue-degree-2 system which certifies local with $f = 2$. This is the value `HeckeExtSeparationValidation` (E2) asserts. **The second alternative, not the first** (v1.7.0; measured by that validation's `SEPARATION_SINGLE` line, slot-5 WP-8): T_2 alone, T_3 alone and $\{T_2, T_3\}$ each give exactly the family's seven blocks, and T_5 alone gives the *same* seven-block partition, differing only in reporting $f = 1$ on the four-dimensional block — T_5's 4:1 *is* the family's 4:2, seen with a coarser invariant. Through v1.6.2 this row said the first alternative holds (T_5's 4:1 two $\mathbb{F}_{691}$-rational systems merged at a_5 and split by the refinement); that cannot be right, since a split would give eight blocks and every family gives seven (the family's decomposition refines each single operator's; the validation compares $(\dim : f)$ lists and block counts, not subspaces — v1.7.1). So the block partition at $(160, 691)$, with its residue degrees, is already T_2's and T_3's, and only T_5 alone under-reports a residue degree (Claim 2.13; v1.7.0's "no single T_q certifies the residue degrees that $\{T_2, T_3\}$ does" is refuted by the same line). Every $\mathbb{T}_{\mathfrak{m}}$ here is a field except where the residue degree exceeds 1 — the row now *is* evidence for the refinement logic; Claim 2.13's isomorphism stays recalled.

8.3 Dickson and boundary phenomena at small ℓ

- $\ell \in \{5, 7, 11\}$, $n = \ell + 1$: first Dickson class L, character χ_1 ([C, `cl:chi`]); $\dim H^0 = 1$, $\dim H^1 = \dim M_k + \dim S_k + 1$ ([C, `cl:dimH1`]; the known loci $(6,5), (12,5), (8,7)$ of `CLAUDE.md`); check its gluing in the cone once (A2) exists. [to be verified computationally] *Recorded 2026-09-16* (`HeckeBoundaryParabolicValidation`, slot-2 WP-7): $\dim H^1_! = \dim H^1 - \operatorname{rank} r^1$ with $\operatorname{rank} r^1 \le \lfloor n/\ell \rfloor + 1$, and the slot-2 plan's expectation "$\dim H^1_! \ge 2 \dim S_k$ at a Dickson locus" — never a claim of this document — fails on 3 of 6 cells: $(6,5), (12,5), (8,7)$ give $0 = 2 \dim S_k$, $(16,7)$ gives $1 < 2$, $(20,5), (24,5)$ give $0 < 2$; for $\ell \le n + 1$ the classical "$H^1_!$ is the cuspidal part, of dimension $2 \dim S_k$" fails in both directions (25 cells), with $H^1_! = 0$ at $(10,5), (14,5), (16,5), (14,7)$ where $\operatorname{rank} r^1 = 3$; $\dim H^1_!$ is even on all 39 cells with $\ell > n$.
- $\ell = 5$, $n \in \{4, 8\}$: $(\ell - 1) \mid n$, $\chi_0 \equiv \chi_1$, the boundary characters collapse (Prediction 4.5). $\ell = 5$, $n = 6$: $E_8 \equiv 1 \pmod 5$, Eisenstein character $1 + q^{-1}$, and simultaneously the Dickson locus. [to be verified computationally]
- $\ell = 5$, $n = 34$ and $\ell = 7$, $n = 58$ ($k = 12 + \ell^2 - 1$): multiplicity 2 for Δ's full system, since $\tau(5) = 4830 \equiv 0$ (5) and $\tau(7) = -16744 \equiv 0$ (7) (Claim 2.6). Measured at $(34, 5)$ (inventory, Stage 9): T_2 and T_5 are each semisimple on H^1, hence so is $\mathbb{F}_5[T_2, T_5]$ (commuting semisimple elements over a perfect field generate a reduced algebra); the block is split unless some other T_q acts non-semisimply — expected *split* [conjectural], to be settled by Stage 4.
- $\ell = 5$, $n = 58$ and $\ell = 7$, $n = 82$ ($k = 12\ell$): the U_ℓ self-extension of Claim 2.6: T_ℓ with a Jordan block of size 2 at Δ's ideal, the anemic operators inactive there. [to be verified computationally]
- The Sturm counterexamples of Claim 2.7(b) on the pipeline's own H^1 (rerun for this revision, $p \le 13$): at $(n, \ell) = (22, 5)$ and $(26, 7)$, $\dim H^1 = 5$, the $\mathbb{F}_\ell$-algebra generated by $A^{(1)}(T_2), A^{(1)}(T_3)$ has dimension 2, the one generated by all $A^{(1)}(T_q)$, $q \le 13$, has dimension $3 = \dim M_k$: Certificate 5.8 with $B(k) = 3$ would have certified falsely, with $B' = 5$, resp. 7, it is sound. At $(32, 131)$ the algebra generated by T_2 alone is already everything ($B' = 2$) although the $\mathbb{Z}$-span of T_1, T_2, T_3 is not: the certificate tests *algebra* membership, so that point breaks the $\mathbb{Z}$-module statement only. [to be verified computationally] (the B' side is computed; the H^1 side measured)
- Small-ℓ Kummer cases with $n < \ell - 1$ (Claim 1.1): (n, ℓ) one of $(14, 3617)$, $(16, 43867)$, $(18, 283)$, $(18, 617)$, $(20, 131)$, $(20, 593)$, $(22, 103)$, $(22, 2294797)$ (all $k \le 24$) and $(24, 657931)$: exactly one non-split Eisenstein–cuspidal $\mathfrak{m}$ in X_4^+; all other $\mathfrak{m}$ cyclic per sign. [to be verified computationally] (needs E3). **Refuted and corrected, 2026-09-16.** Through v1.3.3 this item continued "$r^1 = 0$ on it, X_3 picking up the boundary class", and §7's example drew $H^1_! = H^1$ from it. `HeckeBoundaryParabolicValidation` (slot-2 WP-7), on all ten Kummer cells: r^1 vanishes on the socle E of the block ($\operatorname{rank}(r^1 E) = 0$ — Claim 1.1's Kummer direction, 10/10; the converse $\operatorname{rank}(r^1 E) = 1$ on 27 non-Kummer cells with $n < \ell - 1$) but *not* on the block: $\operatorname{rank} r^1 = 1$ and $\dim H^1_! = \dim H^1 - 1 = 2 \dim S_k$ on every cell ($(10, 691)$: 3/2/1; $(22, 103)$: 5/4/1; the other eight 3/2/1), because r^1 factors through the cosocle of the $\mathfrak{m}_{\mathrm{Eis}}$-block and the class restricting non-trivially is its *top*, not its socle. Consequently, by [C, `eq:dimbook`], $X_6 = H^2_c = 0$ and $\dim X_3 = \dim H^1$ at a Kummer cell — not $X_6 \cong H^1(\partial)$ — while X_3 does pick up the boundary class ($X_2 \hookrightarrow X_3$ regardless). At $(10, 5)$ and $(10, 7)$, outside $n < \ell - 1$, the criterion is not asserted: E is a line with $\operatorname{rank}(r^1 E) = 1$, and $H^1_! = 0$ resp. $\dim H^1_! = 1$ (recorded).
- D_n versus M_n for one $n \ge \ell$ (e.g. $(n, \ell) = (12, 5)$): the Poincaré-duality reversal of Loewy series (Prediction 4.6), once D_n is a local system in the code. [to be verified computationally]

9 The L-value interface, and its Eisenstein instance

(New at v1.19.0-pre.1, hecke-extensions slot 19, WP-2; owner ruling O6 of 2026-10-03 and the owner's interface requirement of the same day, `docs/implementation/slot18/lead_carry.md` C9. Placed after §8 and before the appendices, so that no section number moves. Rule 21: the code, slot-19 WP-LV, merges only after this text is at pre.1; its normalisations are fixed here *before* any claim.) This section fixes (i) an abstract interface for computing L-values, or their ℓ-adic valuations, from the cohomology the project computes, general enough to carry later Birch's GL_2 modular symbols (linear

functionals on H^1 from cycles, twisted values $L(f,\chi,1)$) and Hida's $\mathrm{GL}_2\times\mathrm{GL}_2$ pairing (a bilinear form on H^1, $L(1,\mathrm{Ad}f)$, Hida's congruence criterion), and its own first instance, a Hecke character with its congruence module (v1.19.0-pre.2); (ii) its first implemented instance, the Eisenstein / Bernoulli one (since v1.19.0-pre.6 the linear and bilinear instances are implemented too, Remark 9.2), with its exact index statements; (iii) the normalisation table. **Not used**: Harder's route to rationality through the boundary spectral sequence (owner, 2026-10-03). Throughout $\ell \geq 5$ and $v = v_\ell$; [U] is `docs/latex/hida_hecke_algebra.tex`, cited as [U, `label`].

9.1 The interface

Definition 9.1 (An L-value route). An *L-value route* is a tuple $\mathcal{R} = (D, \kappa, \mathcal{N}, \mathrm{ev}, \mathcal{I})$:

1. **Domain** D: a named model of [C] and a coefficient ring $\mathsf{R} \in \{\mathbb{Z}_\ell, \mathbb{Z}/\ell^m\}$ — $H^1(\Gamma; M_n \otimes \mathsf{R})$ at level one, H^1_c (the cone of [C, `sec:cone`]), or the character model $H^1(\Gamma_0(N); V_{\mathsf{R}}(\chi))$ of [C, `def:nebmodel`] — with its Hecke operators, ι^* ([C, `def:iota`]; "+" is the Eisenstein side) and a named Hecke-stable subquotient H' (a sign, an $\mathfrak{m}$-part, the cuspidal part) on which the route lives.
2. **Kind** $\kappa \in \{\mathsf{linear}, \mathsf{bilinear}, \mathsf{congruence}\}$ (the third added at v1.19.0-pre.2, referee F1 M-2): linear — a R-linear functional $\lambda\colon H' \to \mathsf{R}$ (Birch's modular symbols, (B) of Remark 9.2); bilinear — a R-bilinear pairing $\langle\,,\rangle\colon H' \times H'' \to \mathsf{R}$ with a stated adjunction of the Hecke operators ($\langle T_p x, y\rangle = \langle x, T_p^* y\rangle$, T_p^* named) (Hida's pairing, (H)); congruence — a *Hecke character* $e = (e_p)_p$, i.e. a system of Hecke eigenvalues given by its values $e_p \in \mathcal{O}$ on the T_p (a ring homomorphism from the abstract Hecke algebra $\mathsf{R}[T_p : p]$ to $\mathcal{O}$, $\mathcal{O}$ the coefficient ring R or a named finite extension of it; e need *not* occur in H'), together with the Hecke algebra $\mathbb{T}(H')$ (the R-subalgebra of $\mathrm{End}(H')$ generated by the T_p), the ideal $I_e \subset \mathbb{T}(H')_{\mathcal{O}} := \mathbb{T}(H') \otimes_{\mathsf{R}} \mathcal{O}$ generated by the $T_p - e_p$, and the *congruence module*
$$Q_e \;:=\; \mathbb{T}(H')_{\mathcal{O}} \big/ \big(I_e + \mathrm{Ann}_{\mathbb{T}(H')_{\mathcal{O}}}(I_e)\big).$$
When e does not occur in $H' \otimes \mathbb{Q}_\ell$ and $\mathbb{T}(H')$ is torsion-free, $\mathrm{Ann}(I_e) = 0$ and $Q_e = \mathbb{T}(H')_{\mathcal{O}}/I_e$ (the Eisenstein instance (E)); when $e = \lambda_f$ is the system of an eigenform f occurring in H', $I_e = \ker\lambda_f$ and $Q_e \cong \mathcal{O}/\lambda_f(\mathrm{Ann}(\ker\lambda_f))$ is Hida's congruence module of f.
3. **Normalisation record** $\mathcal{N}$: the L-function; the point s_0; the twist character χ (conductor, values); the sign at infinity $\varepsilon \in \{+,-\}$ (the ι-eigenvalue); the Euler factors removed, each as an exact rational at s_0; the period convention (none; $\Omega_f^{\pm}$; the Petersson norm); every normalising constant (Gauss sums, powers of $2\pi i$, factorials) as an exact rational or an algebraic number with its ℓ-valuation; and the *ambiguity* allowed — an index statement is an equality of ideals of $\mathbb{Z}_\ell$ (of ℓ-valuations), never of ℓ-adic units unless the record says so.
4. **Evaluation** ev: $\lambda(c)$, or $\langle c, c'\rangle$, for named classes, a value in R; for the congruence kind, the exact values e_p for p in a named finite set S (with a certificate that the T_p, $p \in S$, generate $\mathbb{T}(H')$, §5) and the finite module Q_e (its elementary divisors).
5. **Index statement** $\mathcal{I}$: an assertion $\mathrm{length}_{\mathbb{Z}_\ell}(Q) = v(L^{\mathrm{alg}})$, Q a named finite module computed from D (a congruence module, the cokernel of λ on a lattice, the discriminant of $\langle\,,\rangle$, the module Q_e of a congruence route), $L^{\mathrm{alg}} \in \mathbb{Q}$ (or in a number field, with a place over ℓ) given exactly by $\mathcal{N}$; over $\mathsf{R} = \mathbb{Z}/\ell^m$ it reads $\mathrm{length}(Q \otimes \mathbb{Z}/\ell^m) = \min(m, v(L^{\mathrm{alg}}))$ when Q is cyclic. Every index statement carries its own status tag.

An instance fixes all five before anything is computed (rule 20); the code carries $\mathcal{N}$ as data and prints it with every value.

Remark 9.2 (The three shapes). [specification] (E) *Eisenstein / Bernoulli*, congruence (ring form; its module form is a linear reading on H^+), §9.2: implemented by WP-LV. (B) *Birch / Manin–Stevens modular symbols*, linear: $D = H^1_c$ (the modular symbols of weight k are $H^1_c(\Gamma; M_n)$, [1]), $\lambda_{\chi,j}(\varphi) = \sum_{a \bmod m} \bar\chi(a)\,\langle\varphi(\{a/m\} - \{\infty\}), P_{j,a}\rangle$ with $P_{j,a} := \rho_n^\vee(\mathrm{adj}\, B_{a,m})\, u_{n-j}^\vee \in D_n$ (the u_{n-j}-coordinate of $\rho_n(B_{a,m})\Phi(\{a/m,\infty\})$, [C, `prop:pairing`](3); R19-8 §6; F1-delta m-E2) (at pre.1 written as a polynomial $(mX - aY)^j$, of the wrong degree; F1 m-E5 written $(mX - aY)^j$, of the wrong degree; F1 m-E5), giving the critical values $L(f,\chi,j+1)$, $0 \leq j \leq k-2$, divided by $\Omega_f^{\pm}$, in the normalisation $L(f,\chi,j+1)/\Omega_f^{\pm} = \chi(-1)\,(-2\pi i)^{j+1}\tau(\chi)\,\lambda_{\chi,j}/(j!\binom{n}{j}\,m^{j+1})$ [26] (§8; v1.19.0-pre.4, slot-19

R19-8 §1.4 as confirmed by isign; v1.19.0-pre.5: the formulas (8.3), (8.6), p. 10, read on the page by R19-8, the constant derived there from them). Here $\pm$ is the *geometric* sign $\chi(-1)(-1)^j$, which in the convention of [C] is $-\chi(-1)(-1)^j$; the earlier constant was off by $\binom{n}{j}\tau(\bar\chi)/(\chi(-1)\tau(\chi))$ (R19-8 tested only quadratic and trivial χ; for non-real χ the factor is derived, not tested; v1.21.0-pre.3: for every primitive χ of conductor 25, 49, 121 at $k = 12$ and 25, 49 at $k = 16$, non-real ones included, [computed and compared] by slot-21 P4, Remark 9.29(1); for non-real χ of any other conductor (a prime, or not a prime power) still derived, not tested). The functional $\lambda_{\chi,j}$ lives on H^1_c and descends to $H^1_{\mathfrak m}$ only at a residually non-Eisenstein $\mathfrak m$. *Prototype of (B)* (v1.19.0-pre.5; slot-19 R19-8, `docs/implementation/slot19/R19-8_report.md` with its errata, Python, in-house, predictions committed first; not code of this project): on [C]'s model a class (c, c_∂) of $H^1_c = H^0(\mathrm{Cone})$ has the modular symbol $\Phi(\{0,\infty\}) = (\rho_n(S)-1)\big(c(v_0v_4)+c(v_4v_6)-c_\partial(v_0)\big)$, well defined, equivariant, zero on coboundaries ([proved here] there), and $H^1_c(\text{model}) \to \{\text{modular symbols}\}$ is an isomorphism at 8 cells over $\mathbb{F}_p$ ([computed and compared]); the normalisation above matches the numerically computed $L(f,\chi,j+1)$ at 945 critical values (the weights $k \le 26$ with $\dim S_k = 1$, 9 twists, all j) to 58 digits ([computed and compared]; trivial and quadratic χ only); leg (c) in modular-symbol form holds at the 8 Eisenstein cells with $\dim S_k = 1$, each to depth exactly $v(B_k/k) = 1$ ([computed and compared]); at $(258, 37)$ the free sign of H^1_c is the geometric-even one (Claim 4.20) and the critical values of the four eigenforms agree modulo π and not modulo π^2 ([computed and compared]). The instance (`BirchModularSymbolRoute`; gates G1–G5; index statements (L1) [computed and compared] at depth 1, [conjectural] at depth ≥ 2 (v1.20.0-pre.6: [computed and compared] at depth 2 at $(282, 37)$, $(226, 103)$, slot-20 WP-LV3, Claim 9.14), (L2) [proved here] from (L1), (L3) [computed and compared] at $(258, 37)$) was specified in R19-8 §6 for slot 20 and is *implemented* in slot 19 (v1.19.0-pre.6, WP-LIN): on the C++ cone's H^1_c (the field cone over $\mathbb{F}_\ell$, the module cone over $\mathbb{Z}/\ell^m$) $\Psi(c, c_\partial) := \Phi(\{0,\infty\})$ is an isomorphism $H^1_c(\mathsf{R}) \cong W(\mathsf{R}) = \ker(1+\rho_n(S)) \cap \ker(1+\rho_n(U)+\rho_n(U)^2)$ — lengths and invariant factors, including the torsion summand at $(72, 37)$ over $\mathbb{Z}/37^3$ — at 7 cells over $\mathbb{F}_\ell$ and $\mathbb{Z}/\ell^3$ and at $(258, 37)$ over $\mathbb{F}_{37}$ (and $\mathbb{Z}/37^2$, env arm) ([computed and compared]); [C]'s T_p on H^1_c *is* the symbol T_p, $\Psi \circ T_p^c = T_p^{\mathrm{sym}} \circ \Psi$ ($p = 2, 3$, over $\mathbb{Z}/\ell^2$; no adjoint, no scalar), and $\Psi \circ \iota^*_{[C]} = -\rho_n(\iota) \circ \Psi$ read through $H^1_c \to H^1$ — a geometric-even symbol is [C]-minus ([computed and compared]); *torsion* (v1.21.0-pre.2; slot-21 F4 on [U] part 4, S2, S3): $W(\mathbb{Z}/\ell^R)$ need not be free — $H^2_c(\Gamma; M_n \otimes \mathbb{Z}_\ell)[\ell^\infty] \ne 0$ iff $(D_n \otimes \mathbb{F}_\ell)^\Gamma \ne 0$, e.g. at $(72, 37)$ — so slot-21 P8 reads the $\mathfrak m$-blocks, at $\mathfrak m_{\mathrm{Eis}}$ on *both* signs, as the image of $W(\mathbb{Z}/\ell^{R+d})$ in $W(\mathbb{Z}/\ell^R)$, which kills torsion of exponent $\le d$ (a reading convention of the code, [specification]; no statement here changes); the sign rule (the functional vanishes on the sign opposite to $\chi(-1)(-1)^j$, geometric) holds also for three non-real χ ([computed and compared]); R19-8's frozen numbers (the eigenlines $r^\pm$ at $k = 12, \dots, 26$, 108 twisted rows) are reproduced by the C++ symbols modulo ℓ^2 ([computed and compared]); leg (c), Remark 9.3. (H) *Hida's pairing*, bilinear: $D = H^1_! \otimes \mathbb{Z}_\ell$ (or $\mathbb{Z}/\ell^m$) for $n < \ell$ (so that M_n is self-dual, [C, `prop:pairing`]), $\langle x, y\rangle = (x \cup \iota^* y)[\Gamma\backslash\mathbb{H}]$ with $T_p^* = T_p$; for a newform f, $\langle \delta_f^+, \delta_f^-\rangle$ for integral generators of the f-eigenlines of the two signs is, up to a unit, the congruence number of f, and equals $L(1, \mathrm{Ad}f)/(\Omega_f^+\Omega_f^-)$ times explicit factors (Hida's congruence criterion, [21]). The factors (v1.19.0-pre.5; slot-19 R19-9, `docs/implementation/slot19/R19-9_report.md` §2): at level one, with $r_f^\pm = \Omega^\pm\delta^\pm$,

$$L(k, \mathrm{Sym}^2 f)/(\Omega^+\Omega^-) \;=\; -(-1)^{k/2}\, i\, \frac{(2\pi)^{k+1}}{(k-1)!}\, \langle \delta^+, \delta^-\rangle_!,$$

$L(k, \mathrm{Sym}^2 f) = L(1, \mathrm{Ad}f)$ in the unitary normalisation, *no Euler factor removed*: [standard, reference] there ([21, Thm. 5.1, pp. 247–248, and (3.2), p. 237] at $N = 1$, read on the page by R19-9) and the identity $\langle f, f\rangle = -2(2i)^{1-k}\Omega^+\Omega^-\langle \delta^+, \delta^-\rangle_!$ ([computed and compared] at the 10 level-one eigenforms of weights 12–28, to $2.4\cdot10^{-15}$; the Haberland / Kohnen–Zagier form of it is [standard, reference], recalled, not read, so the constant is [standard, reference] (Hida, read; Haberland, recalled) and [computed and compared] (10 forms), not [proved here]; F1-delta M-1). For $\ell \ge k$ the constant is an ℓ-unit. *Prototype of (H)* (v1.19.0-pre.5; R19-9, Python, in-house, predictions committed first; not code of this project): the pairing $\langle \Phi, c\rangle := \langle \Phi(\{0,\infty\}), y_c\rangle$, y_c the potential of $c \in H^1(\Gamma; D_n)$, is the Poincaré pairing $H^1_c(\Gamma; V_n) \times H^1(\Gamma; D_n) \to \mathbb{Z}/\ell^m$ for every even n and $\ell \ge 5$ ([proved here] there); on [C]'s model it equals -1 times the cone-level cup form of `Hecke/HeckePairing.h` ([computed and compared], 9

cells, $n \geq \ell$ among them), with $T_p^* = T_p$ ($p = 2, 3$, and $p = 5$: slot-19 WP-BIL, [computed and compared] at the same cells, predicted from $\Gamma\alpha\Gamma = \Gamma\,\mathrm{adj}(\alpha)\Gamma$ at level one) and ι anti-isometric ([computed and compared], 27 cells and $(258, 37)$). Its value on integral eigenlines is the length of the congruence module *of the lattice it is computed on* (a lemma, [proved here] there): on Hida's $H^1_!$ it sees exactly the cusp–cusp congruences (144169 at $k = 24$; 131, 139 at $k = 28$) and none of the Eisenstein ones, on the mixed lattice $H^1_c(V) \times H^1(D)$ it adds $v(B_k/k) = 1$ at the 8 Eisenstein cells on the sign carrying the boundary class ([computed and compared]). At $(258, 37)$ ($n \geq \ell$; modulo 37^{10}) the mixed value of each of the four eigenforms is $v_\pi = 3$ on the free sign and 2 on the $\mathfrak{m}$-sign, and 3 on both through $H^1_!(D)$ (index 37 in $H^1(D)_\mathfrak{m}$ per sign) ([computed and compared]): $v_{37} = 3/2$ per eigenform, not the pairwise depth 1/2, and $\sum_f v_{37}(\eta_f) = 6 = v(\mathrm{disc}\, P)$ — the pairing reads the discriminant, not $\tilde{\mathbb{T}}/\mathbb{T}_\mathfrak{m}$. The mixed reading at depth 2 (e.g. $(282, 37)$) was [conjectural] through v1.20.0-pre.5 (v1.20.0-pre.1: Claim 9.14(H); v1.19.0-pre.6: still; the C++ $\mathfrak{m}$-block reader was specific to R19-1's quartic at $(258, 37)$, a generic $\mathfrak{m}_{\mathrm{Eis}}$-block of rank > 2 needs the eigensystem split over a general $\mathcal{O}$); **v1.20.0-pre.6**: [computed and compared] at $(282, 37)$, $(226, 103)$ — $v = 2$ on the p_0 sign, 0 on the other — by slot-20 WP-LV3's generic reader (Check 9.19(a), (b)). The instance (`HidaPairingRoute`; gates G1–G5; index statements (H1)–(H3)) was specified in R19-9 §6 for slot 20 and is *implemented* in slot 19 (v1.19.0-pre.6, WP-BIL): $\langle \Phi, c\rangle = \langle \Phi(\{0, \infty\}), y_c\rangle$ over $\mathbb{Z}/\ell^m$ on three named lattices — the mixed perfect form $H^1_c(\Gamma; V_n) \times H^1(\Gamma; D_n)$ (every even n, $\ell \geq 5$), Hida's self-dual form on $H^1_!$ ($n + 1 < \ell$, $G_!(P, Q) = \langle P, \theta\iota_! Q\rangle = -\frac{1}{6}B(P, Q)$, radical $\mathbb{Z}_\ell p_0$), and $H^1_c(V_n) \times H^1_!(D_n)$ at $(258, 37)$. Its gates (`HeckeLValueValidation`, [computed and compared]): on the code's `PoincareDualityPairing` the cone-level Gram matrix G^1 equals $-\langle \Psi(c, c_\partial), y(v_4 v_6)\rangle$ at R19-9's 9 cells ($n \geq \ell$ among them) and at $(258, 37)$ (env arm), and the route reproduces it; $A_D(T_p) = A_c(T_p)^{\mathsf{T}}$ for $p = 2, 3, 5$ and ι anti-isometric at the 27 cells of weights 12–28 and at $(258, 37)$ modulo 37^{10}; R19-9's valuations at all 27 cells and at $(258, 37)$ (mixed 3/2, through $H^1_!(D)$ 3/3, $\sum_f v_{37} = 6$); every value equals the length of the congruence module of its lattice read from $A_c(T_2)$ alone (the lemma's right-hand side), and the Eisenstein values equal $v(B_k/k)$ of the congruence kind's Bernoulli routine. The change-of-lattice constant for $n \geq \ell$ (between $\theta(H^1(V))$ and $H^1(D)$ on the f-line) was not computed in slot 19; the normalisation record printed it as open (v1.20.0-pre.1: defined in Definition 9.15; v1.20.0-pre.6: its value is Proposition 9.16, [proved here], with the values of Remark 9.17, and the record computes it, slot-20 WP-LV3). (B) and (H) were *declared* in the code (throwing `NotImplementedInThisBuild`) through v1.19.0-pre.5; since v1.19.0-pre.6 both are implemented (slot-19 WP-LIN, WP-BIL), on [C]'s seven-vertex model at level one only (other models and levels refuse by name). The implementations reproduce the research prototypes and are gated; no statement of this document outside this remark, Remark 9.3 and §9.3 rows (B), (H) rests on them. The three kinds are distinct and each has its instance: (B) linear, (H) bilinear, (E) congruence; (H)'s index statement "discriminant = congruence number" equates a bilinear evaluation with the module Q_{λ_f} of the congruence kind read on $\mathbb{T}(H')$ (Definition 9.1(2), $e = \lambda_f$), so a route may carry two kinds that its index statement compares (v1.19.0-pre.2). At the block of §4.5 the congruence modules of the block are computed already, by other means: Hida's congruence module of each eigenform f_i, $\mathcal{O}_{K_i}/(P'(\rho_i)) \cong \mathcal{O}_{K_i}/\pi^3$ (order 37^3: three root distances 1/2, so $v_\pi(P'(\rho_i)) = 3$; it includes the Galois conjugate), and the inter-orbit congruence module $\mathbb{T}/\mathbb{T}_\mathfrak{m}$ (order 37^2) of Claim 4.19(4) (v1.19.0-pre.2, F1 m-E4). The first is what (H)'s prototype reads on the free sign ($v_\pi = 3$, R19-9; v1.19.0-pre.5). (M) (v1.20.0-pre.1; owner ruling O5 of 2026-10-04) *The Mazur–Tate–Teitelbaum measure of an ordinary symbol* is not a fourth shape but a family of linear evaluations of (B), one per ball of $\mathbb{Z}_\ell$: §9.5 (Definitions 9.8, 9.9, Proposition 9.10, Claim 9.11). *Coefficients* (v1.21.0-pre.1; slot-21 P3, `docs/implementation/slot21/P3_spec.md`): (M) is defined for an ordinary eigen-system with values in a local monogenic $\mathbb{Z}_\ell$-order $\mathcal{O} = \mathbb{Z}_\ell[x]/(F)$ — unramified, $W(\mathbb{F}_q)$, or totally ramified — by $\mathcal{O}$-linearity of Definition 9.9 in the symbol ($P_\theta = \sum_j x^j P_j$, P_j with $\mathbb{Z}_\ell$-coordinates), and Proposition 9.10 holds verbatim over $\mathcal{O}$ ([proved here]: its proof is an identity of c-tables, valid over any $\mathbb{Z}_\ell$-algebra); *implemented* (slot-21 P3, `HeckeLValueCoefficientRingValidation`, Remark 9.24); refused by name: residual multiplicity ≥ 2 of the eigenvalue on a sign, and non-ordinary.

Remark 9.3 (Leg (c) in modular-symbol form (v1.19.0-pre.6)). (Slot-19 WP-LIN on R19-8 §§3, 6; predictions `docs/implementation/slot19/wplin/predictions.md`, committed before any run.) Let

$p_0 := u_0 - u_n \in W$ (the image of $H^0(\Gamma_\infty; V_n) = \langle u_0 \rangle$, the symbol of the boundary class; $T_p p_0 = (1 + p^{k-1}) p_0$), and $\mathrm{Bern}_k := \sum_{i \text{ even}, 2 \le i \le k-2} \frac{B_i B_{k-i}}{i!\,(k-i)!} X^{i-1}$, (the interior of the Eisenstein period polynomial; shape recalled, not read; R19-8 §2.2), written in the coordinates u_t and made primitive. At the 8 Eisenstein cells with $\dim S_k = 1$, $(k, \ell) = (12, 691)$, $(16, 3617)$, $(18, 43867)$, $(20, 283)$, $(20, 617)$, $(22, 131)$, $(22, 593)$, $(26, 657931)$, on $H^1_c(\Gamma; V_n \otimes \mathbb{Z}/\ell^3)$ of the C++ module cone ([computed and compared] at all 8; the controls $(10, 37)$, $(22, 37)$ with $v(B_k/k) = 0$ give depth 0):

1. the $\mathfrak{m}_{\mathrm{Eis}}$-block of the geometric-even part has rank 2 (p_0 and the cusp line);
2. **(L1)** the cusp eigenline r^+ satisfies $r^+ \equiv u \cdot p_0 \pmod{\ell}$, u a unit, to depth exactly $v(B_k/k) = 1$ (the route's index statement);
3. $r^- \equiv s \cdot \mathrm{Bern}_k \pmod{\ell}$ to the same depth, with $\mathrm{Bern}_k \in W(\mathbb{F}_\ell) \smallsetminus W(\mathbb{Z}/\ell^2)$ and $T_p \mathrm{Bern}_k \equiv (1 + p^{k-1}) \mathrm{Bern}_k \pmod{\ell}$, $p \le 7$;
4. **(L2)** every twisted value ($\chi \neq \mathbf{1}$, the 8 quadratic characters of R19-8) of r^+ is $\equiv 0 \pmod{\ell}$; on r^- the number of right-sign twisted values that are ℓ-units is R19-8's count.

(L2) follows from (L1) and $\lambda_{\chi,j}(p_0) = 0$ for $\chi \neq \mathbf{1}$ ([proved here] in R19-8 §1.4). Item 2 is the H^1_c form of $\Delta \equiv E_{12}$: the integral cusp vector is congruent to the boundary symbol. At depth ≥ 2 ($v(B_k/k) \ge 2$, e.g. $(282, 37)$) (L1) and item 3 are [conjectural] (not computed: none of the six weights with $\dim S_k = 1$ has $v(B_k/k) \ge 2$ at any ℓ, and the gated reading is on a block with one cusp line). *Evidence* (v1.20.0-pre.1; slot-20 R20-1, `docs/implementation/slot20/R20-1_measures.md` §4.2, Python, one route, predictions committed first): at $(282, 37)$ and $(226, 103)$ ($d = 2$; on the $\mathfrak{m}_{\mathrm{Eis}}$-block separated by T_3) $r^+ \equiv u \cdot p_0$ and $r^- \equiv s \cdot P^{\mathrm{Eis}}$ (Claim 9.11(1),(2)) hold to depth exactly 2, and at $(30, 37)$ to depth exactly 1. Both items **stay [conjectural]** at depth ≥ 2 (lead decision, slot-20 WP-2) until the C++ gate of Check 9.19(b) reads them; Claim 9.14. **v1.20.0-pre.6**: the C++ gate read (L1) — $r^+ \equiv u \cdot p_0$ to depth *exactly* 2 at $(282, 37)$ and $(226, 103)$ (slot-20 WP-LV3, `BirchModularSymbolRoute::index_statement()` on the group-cocycle W over $\mathbb{Z}/\ell^3$ and, in the heavy arm, on the C++ cone over $\mathbb{Z}/37^3$, $\mathbb{Z}/103^3$; the $\mathfrak{m}_{\mathrm{Eis}}$-block cut jointly by T_2, T_3, T_5), so **(L1) is [computed and compared] at depth** 2 at these two cells (R20-1's Python and WP-LV3's C++, two routes). Item 3 was not gated by WP-LV3 and stays [conjectural] at depth ≥ 2 as an index statement of this route (R20-1's reading, one route; the congruence itself is [computed and compared] as a run, Claim 9.11(2)). **v1.22.0-pre.2** (slot-22 P5″, the Fable referee `docs/implementation/slot22/P5pp_referee.md` §§1.3–1.4): (L1) at depth *exactly* 1 at $(66, 37)$ and $(102, 37)$ ($d = 1$, $n > \ell$) — $r^+ \equiv p_0 \pmod{37}$, $r^+ \not\equiv p_0 \pmod{37^2}$, r^+ rebuilt over $\mathbb{Z}/37^3$ on the group-cocycle W from the period relations and Merel's T_2, T_3 with the a_2, a_3 of the verb `lfunction` (`docs/implementation/slot22/p5pp/referee/ref_direct.py`, outputs `referee/out/direct_66_37.txt`, `direct_102_37.txt`): computed, **the referee's Python, one route** — not yet [computed and compared] in this document's two-route sense; `BirchModularSymbolRoute::index_statement()` has not read these cells (no arm of the gate runs them; a follow-up).

9.2 The Eisenstein / Bernoulli instance

Definition 9.4 (The instance (E)). Let $k \ge 4$ be even, $n = k - 2$, $a := k \bmod (\ell - 1)$ with $a \not\equiv 0$. D: $H := H^1(\Gamma; M_n \otimes \mathbb{Z}_\ell)$ modulo torsion, and $H_m := H^1(\Gamma; M_n \otimes \mathbb{Z}/\ell^m)$; H': the minus sign H^- (cuspidal: the Eisenstein class lies in H^+) localised at the Eisenstein maximal ideal $\mathfrak{m}_{\mathrm{Eis}} = (\ell, T_p - 1 - p^{k-1})$. κ: congruence (v1.19.0-pre.2; at pre.1 mislabelled linear, referee F1 M-2) — the Eisenstein character $e\colon T_p \mapsto e_p := 1 + p^{k-1}$ (values in $\mathcal{O} = \mathbb{Z}_\ell$; it does not occur in H^-) against the Hecke algebra $\mathbb{T}^-_k = \mathbb{T}(H')$ of $H^-_{\mathfrak{m}_{\mathrm{Eis}}}$, $I_{\mathrm{Eis}} := I_e = (T_p - 1 - p^{k-1})_p$, and $Q_e = \mathbb{T}^-_k / I_{\mathrm{Eis}} \mathbb{T}^-_k$ ($\mathrm{Ann}(I_{\mathrm{Eis}}) = 0$: $\mathbb{T}^-_k$ is torsion-free and $I_{\mathrm{Eis}} \otimes \mathbb{Q}_\ell = \mathbb{T}^-_k \otimes \mathbb{Q}_\ell$). This is the ring form, Claim 9.5(1); the module form, Claim 9.5(2), reads the same valuation through a linear functional on the *other* sign, $r^1|_{H^+}$ with its ω_∂-coordinate, and is [standard, reference]. $\mathcal{N}$: $L = \zeta$ at $s_0 = 1 - k$, $\zeta(1 - k) = -B_k/k$ (the constant term of E_k; not $L(E_k, s) = \zeta(s)\zeta(s - k + 1)$ at $s = k$, which is a pole; F1 m-E1); the Euler factor at ℓ removed, $(1 - \ell^{k-1})$ (a unit, so it changes no valuation); no period; $\chi = \mathbf{1}$; sign $-$; the ambiguity: $\mathbb{Z}_\ell^\times$. $\mathcal{I}$: Claim 9.5.

Claim 9.5 (The Eisenstein index statements)**.** *In the setting of Definition 9.4:*

1. ***(Ring.)*** *[proved here] from [U, `pr:u-U1`] (itself [proved here] from [standard, reference] (Emerton, Thm. 5, verified in [U])) $\mathbb{T}^-_k/I_{\mathrm{Eis}}\mathbb{T}^-_k \cong \mathbb{Z}_\ell/\ell^{v(B_k/k)}$, for every $k \geq 4$ in a class $a \not\equiv 0$.*
2. ***(Module.)*** *[standard, reference] (Harder's denominator of the Eisenstein class, [20], [locator not verified, rule A2]; Claim 1.1 is its mod-ℓ reading) for $n < \ell - 1$: the primitive integral Eisenstein class $E \in H^+$ restricts to $r^1(E) = \ell^{v(B_k/k)} \cdot u \cdot \omega_\partial$, $u \in \mathbb{Z}_\ell^\times$, ω_∂ a generator of the free part of $H^1(\Gamma_\infty; M_n \otimes \mathbb{Z}_\ell)$; equivalently $[H^+ : H^+_! \oplus \mathbb{Z}_\ell E] = \ell^{v(B_k/k)}$. [conjectural] for $n \geq \ell - 1$.*
3. ***(Finite precision.)*** *[proved here] from (1) and [C, `cor:stablecert`]. Let $\mathsf{A}_m := \mathbb{T}^-_{k,\mathfrak{m}_{\mathrm{Eis}}}/(\mathbb{T}^-_{k,\mathfrak{m}_{\mathrm{Eis}}} \cap \ell^m\mathrm{End})$ be the Hecke algebra of the $\mathfrak{m}_{\mathrm{Eis}}$-block of H^-_m (§7.1). If $\mathbb{T}^-_{k,\mathfrak{m}_{\mathrm{Eis}}}$ is* saturated *in $\mathrm{End}(H^-_{\mathfrak{m}_{\mathrm{Eis}}})$ — the `stable` verdict of [C, `cor:stablecert`](1) at $K = 1$ together with the generation certificate (2), equivalently [C, `prop:cohforms`](5)'s condition; automatic on a rank-1 block — then $\mathsf{A}_m = \mathbb{T}^-_{k,\mathfrak{m}_{\mathrm{Eis}}}/\ell^m$, and for a set S of T_p generating A_m (§5)*

$$\mathsf{A}_m/\big(T_p - 1 - p^{k-1} : p \in S\big) \;\cong\; \mathbb{Z}/\ell^{\min(m,\, v(B_k/k))}.$$

Without saturation (a `stable` verdict first at some $K > 1$; [C, `cor:stablecert`] allows $K \leq c^{\mathrm{sat}} + 1$), with $\ell^{c^{\mathrm{sat}}}\mathbb{T}^{\mathrm{sat}} \subset \mathbb{T}^-_{k,\mathfrak{m}_{\mathrm{Eis}}}$, only $\min(m - c^{\mathrm{sat}}, v(B_k/k)) \leq \mathrm{length} \leq \min(m, v(B_k/k))$ (v1.19.0-pre.2, referee F1 M-1: a `stable` verdict at $K > 1$ does not give $\mathsf{A}_m = \mathbb{T}/\ell^m$ — e.g. $\mathbb{T} = \mathbb{Z}_\ell[\ell e] \subset \mathbb{Z}_\ell \times \mathbb{Z}_\ell$, $e = (1,0)$, is `stable` at $K = 2$ but $|\mathsf{A}_2| = \ell^3 \neq |\mathbb{T}/\ell^2\mathbb{T}| = \ell^4$). Independently of any certificate, $\min_{p \neq \ell} v\big(c_{k,p}(1 + p^{k-1})\big) \geq v(B_k/k)$ for the characteristic polynomials $c_{k,p}$ of T_p on $H^-_{m,\mathfrak{m}_{\mathrm{Eis}}}$ (read modulo ℓ^m; [U, `thm:u-sandwich`], over $p \neq \ell$: at $p = \ell$ a non-ordinary form of the residual class contributes $v(a_\ell - 1 - \ell^{k-1}) = 0$ on the full H^-).

Proof. (1) The Eisenstein system is ordinary ($T_\ell \equiv 1 + \ell^{k-1} \equiv 1$ modulo $\mathfrak{m}_{\mathrm{Eis}}$), so $\mathbb{T}^-_{k,\mathfrak{m}_{\mathrm{Eis}}}$ is the $\mathfrak{m}_{\mathrm{Eis}}$-factor of the ordinary cuspidal algebra $\mathbf{h}_k$ of [U] ([U, `lem:u-levelone`](1): the algebras generated by the $a_p(f)$ and by the unit roots α_f agree, $\alpha \in \mathbb{Z}_\ell[T_\ell]$ by Hensel; and $H^- \otimes \bar{\mathbb{Q}}_\ell$ carries each cuspidal eigensystem once, [C, `prop:cohforms`]). On the $\mathfrak{m}_{\mathrm{Eis}}$-factor, I_{Eis} (with $T_\ell - 1 - \ell^{k-1}$) and [U]'s I^0 (with $U - 1$) generate the same ideal: $T_\ell - 1 - \ell^{k-1} = (\alpha - 1)(1 - \ell^{k-1}/\alpha)$, α the unit root, the second factor a unit (v1.19.0-pre.2, F1 m-E2). By (U1-Λ) $\mathbf{h}(a)/I^0(a) \cong \Lambda/(G_a)$; tensoring with Λ/P_k (right exact; [U, `thm:u-control`]) gives $\mathbf{h}_k/I^0\mathbf{h}_k \cong \mathbb{Z}_\ell/G_a(u^k - 1)$, and $v(G_a(u^k - 1)) = v(B_k/k)$ ([U, `pr:u-U1`]). The quotient is supported at $\mathfrak{m}_{\mathrm{Eis}}$ only. (3) Saturation gives $\mathbb{T} \cap \ell\,\mathrm{End} = \ell\mathbb{T}$, hence by induction $\mathbb{T} \cap \ell^j\mathrm{End} = \ell^j\mathbb{T}$ for all j, so $\mathsf{A}_m = \mathbb{T}^-_{k,\mathfrak{m}_{\mathrm{Eis}}}/\ell^m$; the ideal generated by the $T_p - e_p$, $p \in S$, is the image of I_{Eis} (the T_p, $p \in S$, generate, so I_{Eis} is generated by the $T_p - e_p$, $p \in S$, as an ideal); reduce (1) modulo ℓ^m. Without saturation, $\ell^{c^{\mathrm{sat}}}\mathbb{T}^{\mathrm{sat}} \subset \mathbb{T}$ gives $\ell^m\mathbb{T} \subset \mathbb{T} \cap \ell^m\mathrm{End} \subset \ell^{m-c^{\mathrm{sat}}}\mathbb{T}$, so A_m/I is a quotient of $\mathbb{T}/(\ell^m, I_{\mathrm{Eis}})$ and surjects onto $\mathbb{T}/(\ell^{m-c^{\mathrm{sat}}}, I_{\mathrm{Eis}})$: the two-sided bound. The inequality is [U, `thm:u-sandwich`]'s upper bound (over $p \neq \ell$), $L_k \leq v(c_{k,p}(1 + p^{k-1}))$, with $L_k = v(B_k/k)$. □

A reading of Claim 9.5(2) on the ordinary block (v1.21.0-pre.1; slot-21 F4, `docs/implementation/slot21/P8_F4_r` M3 and §4 item 3; no tag moves). For $n < \ell - 1$, (2) is equivalent to: the ordinary $\mathfrak{m}_{\mathrm{Eis}}$-block of H^+ is free of rank one over $\mathbb{T}_{k,\mathfrak{m}_{\mathrm{Eis}}}$ — this is [U, `pr:u-msym`] (U17)(a), which [U] proves from (2) on the ordinary block ($\ell^{j_k} = [H^+ : H^+_! \oplus \mathbb{Z}_\ell E]$ there, $H^1(\partial; M_n \otimes \mathbb{Z}_\ell)$ being torsion-free for $n < \ell$); for $n \geq \ell - 1$ both stay [conjectural].

Claim 9.6 (The twisted instance through $\Gamma_1(\ell)$)**.** *[proved here] from [U, `pr:u-U1`], [U, `lem:u-twistvalues`] and [standard, reference] (Hida's control at the nebentypus points: for $k \geq 2$* not *in the class a, $\mathbf{h}(a)/P_k \cong \mathbf{h}^{\mathrm{ord}}_k(\Gamma_0(\ell), \omega^{a-k})$ [22] (pp. 249–250: Thm. 3.1 and Cor. 3.2 (3.3b), $h_0(N, a; \mathcal{O}_K)/P_k h_0(N, a; \mathcal{O}_K) \cong h^0_k(\Gamma_1(N_0 p), a; \mathcal{O}_K)$ for* every *$k \geq 2$, $P_k = (1 + X) - (1 + p)^k$, with p. 249's $S_k(\Gamma_1(N_0 p), a; A) = S_k(\Phi, \omega^{a-k}; A)$, $\Phi = \Gamma_0(p) \cap \Gamma_1(N_0)$, ω a character of Φ via $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \mapsto \omega(d)$; here $N_0 = 1$, $p = \ell$; read on the page by referee F1, slot 19). Convention check: [C, `def:nebmodel`] twists by $\chi(a_\gamma) = \bar{\chi}(d_\gamma)$, but [C, `prop:nebcorr`](3) ($T(p,p) = \chi(p)p^n$) and [C, `thm:nebrelations`](2) ($T_{p^2} = T_p^2 - \chi(p)p^{n+1}$) pin the model's eigensystems to those of $S_k(\Gamma_0(\ell), \chi)$ in the standard (d-)convention, which is Hida's. Let $\chi = \omega^j$ with j even (so $\chi(-1) = 1 = (-1)^k$), $j \not\equiv 0 \pmod{\ell - 1}$, and $a := k + j \not\equiv 0$. On the cuspidal cohomology $H^1_!$ (both copies of each eigensystem; Eichler–Shimura) of the character model $H^1(\Gamma_0(\ell); V_{\mathbb{Z}_\ell}(\chi))$ ([C, `def:nebmodel`]), localised at the maximal ideal of the Eisenstein system $p \mapsto 1 + \chi(p)p^{k-1}$ (that*

of $E_k(\mathbf{1},\chi)$), with Hecke algebra $\mathbb{T}^-$:

$$\mathbb{T}^-/I_{\mathrm{Eis}}\mathbb{T}^- \;\cong\; \mathbb{Z}_\ell/\tfrac{B_{k,\chi}}{k}\mathbb{Z}_\ell.$$

Proof. As Claim 9.5(1), with the nebentypus specialisation in place of [U, `thm:u-control`]: $\epsilon_p(x_k) = 1+\omega(p)^{a-1}\langle p\rangle^{k-1} = 1+\omega^{a-k}(p)p^{k-1}$ ([U, `def:u-eischar`]), and $G_a(x_k) = -B_{k,\omega^{a-k}}/k$ ([U, `lem:u-twistvalues`]), $\omega^{a-k}=\chi$. The identification of $\mathbf{h}^{\mathrm{ord}}_k(\Gamma_0(\ell),\chi)$ with the Hecke algebra of the minus sign of the character model is Eichler–Shimura for $\Gamma_1(\ell)$ ([standard, reference], [34] Ch. 8). □

Remark 9.7 (Checkable form and gate cells for WP-LV). **Untwisted.** At $(\ell,k)\in\{(37,32),(691,12)\}$ ($v(B_k/k)=1$), at the regular controls $(37,k)$ with $v(B_k/k)=0$, and at $(59,44)$, $(67,58)$ ($v=1$): Claim 9.5(3) at $m=1,2,3$ — $\mathbb{Z}/\ell^{\min(m,1)}$ or 0 — and the boundary reading of (2), all with $n<\ell-1$. The $\mathfrak{m}_{\mathrm{Eis}}$-blocks there have rank 1 (one cusp form congruent to E_k), so the saturation hypothesis of Claim 9.5(3) is met at all four untwisted cells (a rank-1 algebra $\mathbb{Z}_\ell$ is saturated; v1.19.0-pre.2, F1 M-1). A deeper rung, $v(B_k/k)=2$, is $(282,37)$ ($k=284$) and $(226,103)$ ($k=228$), both with $n\geq\ell-1$: there (1) is [proved here], (3) is [proved here] under its saturation hypothesis, and (2) is [conjectural]. At pre.2 the rank of the $\mathfrak{m}_{\mathrm{Eis}}$-block at these two cells was not established; since [U] 0.3.0-pre.3 it is: (U13), [U, `pr:u-lambda`], is [proved here] at 37 and 103 (from the [computed and compared] reading $d_{\mathrm{Eis}}=1$ at one weight, slot-19 R19-2, and Hida's control; slot-19 referee F4, m8), so the cuspidal component through $\mathfrak{m}_{\mathrm{Eis}}$ is $\mathbb{Z}_\ell[[X]]$, the block has rank 1 at every weight of the two classes, and the saturation hypothesis of (3) is met there (v1.19.0-pre.3). Neither cell is a gate cell of WP-LV; both are cells of WP-LV2's env arm (`HECKE_LVALUE_HEAVY=1`, read-only; v1.19.0-pre.5, Status below); `HeckeLValueValidation` prints the rank and the saturation verdict of every $\mathfrak{m}_{\mathrm{Eis}}$-block it gates, falling back to the two-sided bound of (3) where saturation fails. **Twisted.** Claim 9.6 at $\ell\leq 37$ only (cost), on [C, `sec:nebentypus`]'s model (as built, v1.19.0-pre.5: at 37 in the default arm, at 59, 67 in the env arm). **Bernoulli numbers.** Exact rationals from WP-NF's `BigInt` (§9.4), and independently modulo ℓ^m (the Kummer / Voronoi route); the two must agree. The λ-readings and the first depth-2 rungs that [U] derives from two Bernoulli numbers per class ([U, `prop:u-eistrunc`]: 284 at 37, 228 at 103, and at 691 the rungs $k=342252$ (class 12) and $k=342440$ (class 200), where $v(B_k/k)=2$ is now *computed*, with $v=1$ at $k\mp 690$ — referee F1, power-sum route, `docs/implementation/slot19/wp3/f1_lambda.py` and `.out`) are the cross-check of the Bernoulli code. As built (v1.19.0-pre.3): the exact route is WP-LV's own tangent-number recurrence on a minimal unsigned big integer (`HeckeExt/LValue/Bernoulli`; WP-NF's `BigInt` may replace it, value-free), the modulo-ℓ^m routes are Voronoi's congruence and slot-15's Kummer–Mahler route (`HeckeExt/HeckeHidaComponents`); all three agree, and the depth-2 rungs 284, 228, 342252, 342440 read $v=2$. [U] 0.3.0-pre.3 adds the depth-3 rungs 37580, 914250, 49928412, 159113510, [proved here] from five exact Bernoulli numbers per class ([U, `prop:u-eistrunc`]); they are not gated here. **Status** (v1.19.0-pre.3). [computed and compared] (slot-19 WP-LV, `HeckeLValueValidation`, 311 checks; reproduced on the forms side by the slot-19 referee F4, m18): at the four untwisted cells, $(n,\ell)=(30,37)$, $(10,691)$, $(42,59)$, $(56,67)$, the $\mathfrak{m}_{\mathrm{Eis}}$-block is rank 1 and saturated, $\#Q_m=\ell^{\min(m,1)}$ for $m=1,2,3$, the module form's index is $\ell^1=\ell^{v(B_k/k)}$ ($n<\ell-1$ at all four), and $v_\ell(c_{k,p}(1+p^{k-1}))=1$ at $p=2,3,5,7$; at the regular controls $(10,37)$, $(22,37)$ the module Q_m is 0. The saturated rank-1 path is the only one exercised: the two-sided-bound fallback of Claim 9.5(3) has no test cell on cohomology. **Status** (v1.19.0-pre.5; slot-19 WP-LV2, `HeckeLValueValidation` extended, 537 checks by default, 657 with `HECKE_LVALUE_HEAVY=1`; `docs/implementation/slot19/WP-LV2_predictions.md`, predictions committed before the runs). *Twisted* ([computed and compared]): Claim 9.6 on the character model $H^1(\Gamma_0(\ell);V_{\mathbb{Z}/\ell^K}(\omega^j))$ (slot-18 WP-6's model route; ι^* on the induced model new) at $(\ell;k,j)=(37;4,28)$, $(37;6,26)$ (default arm), $(59;4,40)$, $(67;4,54)$ (env arm): $v(B_{k,\chi}/k)=1$ by two routes (the definition with exact B_i and Teichmüller ω, and Kummer's congruence), the $\mathfrak{m}_{\mathrm{Eis}}$-block of H^- of rank 1 and saturated, $\#Q_m=\ell^{\min(m,1)}$ at $m=1,2,3$, and $\operatorname{rank}H^+=\operatorname{rank}H^-+2$; the controls $(37;4,8)$ and $(37;4,34)$ (the latter with one torsion summand, read through the free part) have $v=0$ and $Q=0$; the conjugate twist and the untwisted character give $Q=0$. The twisted depth-2 rung $(37;62,6)$ has $v(B_{62,\omega^6}/62)=2$ by both Bernoulli routes; its module was *not* run (above 7.3 GiB,

stopped by a 6 GiB watchdog; v1.20.0-pre.1: slot-20 WP-LV3 runs it under a cap taken from a measured growth curve, Check 9.19(e); v1.20.0-pre.6: **run** at $n = 60$, 16.43 GB, 7.3 h — the $\mathfrak{m}_{\mathrm{Eis}}$-block of H^- of rank 1, saturated, $\#Q_m = 37^{\min(m,2)}$, $m = 1, 2, 3$, [computed and compared]). *Deep rungs* ([computed and compared]): at $(282, 37)$ and $(226, 103)$, $H^1(\Gamma; M_n \otimes \mathbb{Z}/\ell^4)$ is free (no ℓ-torsion; the prediction of torsion was wrong), of rank 47 resp. 39 (24+23, 20+19), the $\mathfrak{m}_{\mathrm{Eis}}$-block of H^- has rank 1 and is saturated, $\#Q_m = \ell^{\min(m,2)}$ at $m = 1, 2, 3$, and $v_\ell(c_{k,p}(1 + p^{k-1})) = 2$ at $p = 2, 3, 5, 7$; the module form's index reads $c = 2 = v(B_k/k)$, printed and not gated — (2) stays [conjectural] for $n \geq \ell - 1$. Slot-19 R19-7 read the same two cells independently (console over $\mathbb{Z}/\ell^3$ and an outside reading of the matrices, the joint Eisenstein block, $f = 1$): $\#Q_2 = \ell^2$, $v(a_p - 1 - p^{k-1}) = 2$ at $p = 2, 3, 5, 7, \ell$ ([computed and compared]). *Fallback* of (3): the two-sided bound is tested on planted algebras ($\mathbb{Z}_\ell[\ell e]$, $\mathbb{Z}_\ell[\ell + \ell^2 e]$, with c^{sat}), still on no cohomology cell. A read-only cell hook (`HECKE_LVALUE_CELLS`, level one or $\chi = \omega^j$) runs further cells and gates only the index statement.

9.3 The normalisation table

route	L at s_0, twist	Euler factor, period, sign	index statement	status
(E) level 1	$\zeta(1-k) = -B_k/k$; $\chi = \mathbf{1}$	$(1 - \ell^{k-1})$ removed (unit); none; −	$\mathbb{T}^-/I_{\mathrm{Eis}} \cong \mathbb{Z}_\ell/(B_k/k)$	[proved here] from [standard, reference] (Claim 9.5(1)); module form [standard, reference] for $n < \ell - 1$
(E$_\chi$) $\Gamma_1(\ell)$	$L(1-k, \chi) = -B_{k,\chi}/k$; $\chi = \omega^j$	none needed ($\chi(\ell) = 0$); none; −	$\mathbb{T}^-/I_{\mathrm{Eis}} \cong \mathbb{Z}_\ell/(B_{k,\chi}/k)$	[proved here] given Hida's control at nebentypus points (Claim 9.6)
(KL) ℓ-adic	$L_\ell(1-k, \omega^a) = G_a(u^k - 1)$	$(1 - \omega^{a-k}(\ell)\ell^{k-1})$; — ; —	— (an interpolation identity)	[proved here] from [standard, reference] ([U, `lem:u-twistvalues`])

route	L at s_0, twist	Euler factor, period, sign	index statement	status
(a) Kummer / Coates–Wiles	$\varphi_k(\eta_i) := D^k \log F_{\eta_i}(T)\vert_{T=0}$, $D = (1+T)\frac{d}{dT}$, F the Coleman series of the norm-coherent system $\eta_i^{(n)} = \prod_c \big((\zeta_{\ell^n}^c - 1)/(\zeta_{\ell^n} - 1)\big)^{f_c}$: Kummer's logarithmic derivative in Washington's normalisation ([38, §13.7, pp. 299–300], defined there modulo ℓ), $\mathbb{Z}_\ell$-valued as the Coates–Wiles homomorphism δ_k on norm-coherent units (there, Lemma 13.52; Coates–Wiles' own φ_k, [10, p. 230, (8)], is modulo p); $\varphi_k(\eta_i) = S_{i,k}B_k/k$, $S_{i,k} = \sum_c f_c(c^k - 1) \equiv -1 \pmod{\ell}$ for $k \equiv i \not\equiv 0 \pmod{\ell-1}$; twist ω^i (the radical character of η_i)	$(1-\ell^{k-1})$: the restriction of the measure to $\mathbb{Z}_\ell^\times$, $(1-\ell^{k-1})\varphi_k(\eta_i) = -S_{i,k}L_\ell(1-k,\omega^i)$ (φ_k itself has none); auxiliary factor $S_{i,k}$ (the (c^k-1)-smoothing); no period; even	$v(\varphi_k(\eta_i)) = v(B_k/k)$; modulo ℓ^m a function of the level-m unit $\eta_i^{(m)}$ (the layer-1 unit gives it modulo ℓ only, $1 \le k \le \ell-2$: Washington p. 300's argument); in the pole class $k \equiv 0 \pmod{\ell-1}$ the correspondence $k \leftrightarrow i$ fails	identity [proved here] (elementary; Kummer's computation, [38, pp. 64–65], and §13.7, pp. 299–300, both read on the page); values [computed and compared] at 37 (all even i, levels ≤ 4) and 691 ($i = 12, 200$), four routes, slot-19 R19-4 (v1.19.0-pre.3). *Level m, Bernoulli-free* (v1.19.0-pre.5; slot-19 R19-7): on the ω^w-part, $\nu(a+\ell^m\mathbb{Z}_\ell) = \ell^{-m}\sum_{j=1}^m \mathrm{Tr}_j(\zeta_{\ell^j}^{-a}\lambda_j)$, $\lambda_m = \log_\ell(1-\zeta_{\ell^m})$, $\lambda_j = \mathrm{Tr}_{m/j}\lambda_m$, and $\Psi_k := \sum_a a^k\nu_a \equiv (1-\ell^{k-1})B_k/k \pmod{\ell^m}$ ([proved here] there); [computed and compared] at $m=2$ (37, 59, 67, 103, 691; 5769/5769 rows), $m=3$ (37, 103), $m=4$ (37): the depth-m rung is the unique zero of Ψ modulo ℓ^m — 284, 914, 3292, 228, 342252, 342440 ($m=2$), 37580, 914250 ($m=3$), 1072544 ($m=4$, predicted from [U]'s α before the run); the level-$(m-1)$ datum is right modulo ℓ^{m-1} only. That the level-2 zero is the unramified $\mathbb{Z}/\ell^2$ Kummer lift: [proved here] there from [standard, reference] ([38, Thm. 13.54, p. 311; Prop. 13.51, p. 309; Lemma 13.52, p. 310], read on the page); at level $m \ge 3$ [specification] formula [standard, reference] ([38, Thm. 5.18, p. 63; Prop. 8.12, p. 156], read on the page); reduction [proved here]; values [computed and compared] modulo ℓ^5 at 37 (all even j; $L_{37}(1,\omega^{32}) \equiv 47910634$) and 691 ($j = 2, 4, 12, 100, 200$), three routes, slot-19 R19-4 (v1.19.0-pre.3)
(b) Leopoldt	$L_\ell(1,\omega^j) = -\frac{\tau(\omega^j)}{\ell}\sum_a \omega^{-j}(a)\cdot\log_\ell(1-\zeta^a) = -\sum_c \omega^j(c)x_c$, $\log_\ell(1-\zeta) = \sum_c x_c\zeta^c$ (Iwasawa's log); j even, $j \not\equiv 0$	none ($\omega^j(\ell) = 0$); no period; even	$L_\ell(1,\omega^j) = G_j(0)$; $\equiv G_j(x_k) \pmod{\ell^{1+v(k)}}$, $k \equiv j$; $v \ge 1 \iff \ell \mid B_j/j$ (equality with $v(B_j/j)$ is not general; it holds, $v=1$, at $(37,32)$, $(691,12)$, $(691,200)$)	

route	L at s_0, twist	Euler factor, period, sign	index statement	status
(B) Birch	$L(f,\chi,j+1)$, $0 \le j \le k-2$	none; $\Omega_f^{\pm}$; $\chi(-1)(-1)^j$ (geometric; $-\chi(-1)(-1)^j$ in [C])	$\lambda_{\chi,j}$ integral, v of its value	[specification] (Remark 9.2); research prototype [computed and compared] (slot-19 R19-8), instance spec R19-8 §6 (v1.19.0-pre.5); *implemented* (v1.19.0-pre.6, slot-19 WP-LIN, `BirchModularSymbolRoute`): gates G1–G5 [computed and compared], (L1) [computed and compared] at depth 1 (8 cells), [conjectural] at depth ≥ 2 (v1.20.0-pre.6: [computed and compared] at depth 2 at $(282,37)$, $(226,103)$, slot-20 WP-LV3), (L2) [proved here] from (L1) and [computed and compared] (Remark 9.3)
(H) Hida	$L(1,\mathrm{Ad}f) = L(k,\mathrm{Sym}^2 f)$	none removed ([21, Thm. 5.1, p. 247], $N=1$, read; v1.19.0-pre.5); $\Omega_f^+\Omega_f^-$, $\delta^{\pm}$ primitive in a named lattice; $+\times-$; constant $-(-1)^{k/2}i(2\pi)^{k+1}/(k-1)!$	$v\langle\delta_f^+,\delta_f^-\rangle$ = length of the congruence module of that lattice	[specification] (Remark 9.2); the constant [standard, reference] (Hida, read; Haberland, recalled) and [computed and compared] (10 forms); research prototype [computed and compared] (slot-19 R19-9), instance spec R19-9 §6 (v1.19.0-pre.5); *implemented* (v1.19.0-pre.6, slot-19 WP-BIL, `HidaPairingRoute`): index statements (H1) $H^1_!$, (H2) mixed, (H3) $(258,37)$ [computed and compared] against the congruence module of the lattice; (H2) at depth 2 ($(282,37)$) [conjectural] (v1.20.0-pre.6: [computed and compared] at $(282,37)$, $(226,103)$, slot-20 WP-LV3); the change-of-lattice constant for $n \ge \ell$ open (v1.20.0-pre.1: Definition 9.15, value open until slot-20 R20-3; v1.20.0-pre.6: Proposition 9.16 [proved here], values Remark 9.17 [computed and compared], computed by the record); (H2) at depth 2: Claim 9.14

route	L at s_0, twist	Euler factor, period, sign	index statement	status
(M) MTT measure (v1.20.0-pre.1)	$L_\ell(f,\alpha,\cdot)$: $\int_{\mathbb{Z}_\ell^\times}\bar\chi(x)x^j\,d\mu_f$, $0\le j\le k-2$, $\chi=\mathbf{1}$ or of ℓ-power conductor; branch series $\mathcal{L}_{f,a}(T)$; values in $\mathcal{O}$, the Hecke order of the eigen-system (v1.21.0-pre.1, slot-21 P3: $\mathbb{Z}_\ell$, $W(\mathbb{F}_q)$ or a totally ramified $\mathbb{Z}_\ell[x]/(F)$; Remark 9.24)	$(1-\ell^j/\alpha)(1-\beta/\ell^{j+1})$ ($\chi=\mathbf{1}$), α^{-m} (conductor ℓ^m); $\Omega_f^\pm$ of H^1_c (on the boundary sign ℓ^d times the cuspidal one); geometric $\chi(-1)(-1)^j$	congruence of measures with $-s\,G_{a+1}G_{k-1-a}$ modulo ℓ^d off the two pole branches, with $-s\,(G_{a+1}G_{k-1-a}+\mathrm{polar}_a)$ on them (v1.21.0-pre.3); $\mu^+=d$	measure, moments and the algebraic interpolation [proved here] (Proposition 9.10, slot-20 R20-1); the analytic constant (B)'s status ($\tau(\chi)$ for ℓ-power conductor derived, not tested; [computed and compared] since v1.21.0-pre.3, slot-21 P4, Remark 9.29); the congruence [computed and compared] at $(30,37)$, $(282,37)$, $(226,103)$, the pole branches an exception (Claim 9.11(4)), explained by an explicit polar term since v1.21.0-pre.3 (slot-21 P2, Proposition 9.27); literature (Mazur, Greenberg–Vatsal) held since 2026-10-05, not yet read for this claim (v1.21.0-pre.1); planned code slot-20 WP-MEAS (`measure()`, `HeckeLValueMeasureValidation`, Check 9.13); over $\mathcal{O}$: slot-21 P3 (Check 9.25); the two-variable family by interpolation: slot-21 P9, Remark 9.26 ([conjectural] as a Λ-adic object, [U, `pr:u-msym`] (U16))

9.4 The number-field layer

[specification] at v1.19.0-pre.1; **built** at v1.19.0-pre.3 (slot-19 WP-NF, the seventh static target `NumberField`; owner's no-CAS rule of 2026-10-03). The objects the L-value legs (a), (b) and the Galois side of §4.5 need, in in-house exact arithmetic: $\mathbb{Z}[\zeta_{\ell^k}]$ and, modulo ℓ^M, its decomposition by the idempotents e_{ω^i} of the group ring of $(\mathbb{Z}/\ell^k)^\times$ (*equivariant before global*: the ω^{12}-part at 691^2 is one module of rank 691 over $\mathbb{Z}/691^2$, not a field of degree 476 790); the cyclotomic units $u_c=(1-\zeta^c)/(1-\zeta)$ and $\eta_i:=\prod_{c=2}^{\ell-1}u_c^{f_c}$, $f_c\equiv c^{-i}\pmod\ell$ lifted to $(-\ell/2,\ell/2)$ (slot-18 R18-2 §1; η_i spans the ω^i-part of E/E^ℓ); v_π; the local test [proved here] there: for a unit η with $v:=v_\pi(\eta^{\ell-1}-1)$, $K(\eta^{1/\ell})/K$ is unramified at $\lambda=(1-\zeta)$ iff $v\ge\ell$ and λ splits iff $v\ge\ell+1$; its layer-2 analogue over $\mathbb{Q}(\zeta_{\ell^2})$; $\mathbb{Q}(\alpha)$ arithmetic and ℓ-adic factorisation (Hensel, Newton polygon). *Gate* (`NumberFieldValidation`, `fast`, 117 checks; **held**, v1.19.0-pre.3): R18-2's exact η_{32}, η_6 at 37 (FNV-1a-64 of the 36 decimal coefficients `0xcba3d1938a721f7b`, `0xef292f5a4e321b01`; $v_\pi(\eta^{36}-1)=38,6$), on the equivariant layer-2 route of slot-19 R19-3 $\kappa=284$ the only unramified lift at 37 ($u=y_0=13$), $\kappa=914$ at 59 ($y_0=31$), $\kappa=3292$ at 67 ($y_0=59$), and the layer-1 test at 691 ($v=692$ at $i=12,200$). The $\mathbb{Z}/691^2$ layer itself ($\kappa_0=342252$, 342440; [L37], `sec:fieldsribet`) is slot-19 R19-3's Python computation and is not a gate of the library. The library realises no statement of [C]; of this document it realises the objects above and exact Bernoulli numbers (`BigRational`); the Bernoulli numbers gated in Remark 9.7 come from WP-LV's

own exact route. *Built at v1.21.0-pre.1* (slot-21 T1, T2; STL only, the include guard unchanged; no golden line, digest or record key). **T1**: an in-house LLL (Cohen, GTM 138, Alg. 2.6.3 rational and 2.6.7 integral; Definition 2.6.1's reducedness; the substitute for Lenstra–Lenstra–Lovász, not held) and POLRED (Alg. 4.4.11 / 4.4.12), `NumberField/{Lll,Polred}`, gate `NumberFieldLllValidation` (`fast`, 674 checks: known lattices ($\mathbb{Z}^n$, E_8, A_2, integer relations), random lattices against an own Gram–Schmidt test and a brute-force λ_1, POLRED of $\mathbb{Q}(\zeta_7)^+$, $\mathbb{Q}(\zeta_{37})^+$, $\mathbb{Q}(\zeta_\ell)$ for $\ell = 7, 11, 37$, negative controls). **T2**: multiprecision reals `NumberField/MpReal` and the degree-ℓ Ribet field `NumberField/RibetField`: $P = \mathrm{charpoly}(\theta)$, $\theta = \mathrm{Tr}_{H/F}(\eta^{1/\ell})$, by two numerical precisions and exactly by CRT at split primes (all equal), the Frobenius root counts against the Kummer prediction, at $(37, 32)$ the maximal order ($\mathrm{disc}\,\mathcal{O}_F = +37^{35}$ exactly) and T1's POLRED; at $(691, 12)$, $(691, 200)$ the unreduced polynomial (degree 691, 24 995 resp. 14 669 digits, both routes equal) in the read-only env arm `NUMBERFIELD_RIBET_CELLS`; gate `NumberFieldRibetFieldValidation` (`standard`, 29 checks; `docs/implementation/slot21/T2_report.md`). Status of T2's claims after the slot-21 referee F4 (`docs/implementation/slot21/rc/F4_referee_report.md` §4): $\mathrm{Gal}(H/\mathbb{Q}) \cong \mathrm{AGL}(1, \ell)$ and $\mathrm{disc}\,F = (-1)^{(\ell-1)/2}\ell^{\ell-2}$ with signature $(1, (\ell-1)/2)$ [proved here] (the derivation is in the non-binding `docs/latex/ribet_cells.tex`, `rc:sec:methfield`); the complement formula for θ_j, the Frobenius counts, the CRT route, the maximal order at 37 and the POLRED output [computed and compared]. Nothing of [C] or of this document's earlier statements rests on T1 or T2.

*Code map of §9 (**built**, slot-19 WP-LV and WP-NF; planned at v1.19.0-pre.1, contract first; v1.19.0-pre.3).* WP-LV: `HeckeExt/LValue/` — `LValueRoute` (Definition 9.1: domain, kind $\in$ {`linear`, `bilinear`, `congruence`}, `NormalisationRecord`, `evaluate`, `index_statement`), `EisensteinBernoulliRoute` (a `congruence` route; Definitions 9.4, Claims 9.5, 9.6), `BirchModularSymbolRoute` (`linear`) and `HidaPairingRoute` (`bilinear`) (declared, throwing `NotImplementedInThisBuild`, through v1.19.0-pre.5); `HeckeLValueValidation` (`standard`, 311 checks, $\approx$ 5 s; planned at $\leq$ 4 GB) on Remark 9.7's untwisted cells. At v1.19.0-pre.3 not built: the twisted route of Claim 9.6. **Built at v1.19.0-pre.5** (slot-19 WP-LV2, same files): the twisted route on the character model over $\mathbb{Z}/\ell^K$ (ι^* on the induced model, T_p / U_ℓ by [C, `prop:nebcorr`]), twisted Bernoulli numbers by two routes, the free part $(H/\mathrm{tors})/\ell^K$ at finite precision (torsion-aware), c^{sat} and the two-sided bound of Claim 9.5(3); the character model is accepted at level ℓ, the Moerdijk–Pronk and Shapiro models refused; `HeckeLValueValidation` now 537 checks, $\approx$ 14 s, 0.17 GB; env arm `HECKE_LVALUE_HEAVY=1` (657 checks, $\approx$ 419 s, 0.98 GB) and the read-only cell hook `HECKE_LVALUE_CELLS`. **Built at v1.19.0-pre.6** (Remark 9.2, Remark 9.3): slot-19 WP-BIL, the `bilinear` instance (`HeckeExt/LValue/HidaPairing.{h,cpp}`, `HidaPairingRoute.cpp`; arms of `HeckeLValueValidation` (G1–G5, negative controls), env arm `HECKE_LVALUE_BILINEAR_HEAVY=1` (the G1 gate at $(258, 37)$)), and slot-19 WP-LIN, the `linear` instance (`HeckeExt/LValue/BirchModularSymbol.{h,cpp}`, frozen R19-8 data `BirchFrozenR198.inc`; arms (BS-$*$) of `HeckeLValueValidation`; `HECKE_LVALUE_HEAVY=1` adds $(258, 37)$ over $\mathbb{Z}/37^2$ and G3 at $(72, 37)$); both on [C]'s seven-vertex model at level one, other models refused by name; check counts and timings in `docs/validation-catalogue.md`. WP-NF: the seventh target `NumberField/` (STL only, include guard `check_numberfield_includes.sh`), `NumberFieldValidation` (`fast`, 117 checks). Both value-free with respect to golden, the digests and the record schema. **Planned at v1.20.0-pre.1** (slot 20, contract first, rule 21). WP-MEAS (§9.5): `HeckeExt/LValue/ModularSymbolMeasure.{h,cpp}` (the measure, the branch elements, μ / λ, the Eisenstein symbol, the two Kubota–Leopoldt group-ring routes), `LValueRoute::measure` with its default refusal and `BirchModularSymbolRoute::measure`, the frozen table `MeasureFrozenR201.inc`, `HeckeLValueMeasureValidation` (`standard`, Check 9.13); no addition to `NumberField/CwLayer` was needed (`cw_lambda` + `cw_measure` return ν; v1.20.0-pre.2, WP-MEAS's report). WP-LV3 (§9.6): the generic $\mathfrak{m}_{\mathrm{Eis}}$-block reader replacing WP-BIL's quartic-specific one, the cross-kind gate, c_f^{ε} in `NormalisationRecord`, ι over $\mathbb{Z}/\ell^k$ in `Hecke/HeckeInvolution` and the console, the twisted rung behind `HECKE_LVALUE_HEAVY` (Check 9.19); no new validation name. **Built at v1.20.0-pre.6** (slot-20 WP-LV3, merged `c1fda051`; WP-MEAS merged before it, Check 9.13): `HeckeExt/LValue/MBlockReader.{h,cpp}` (the generic $\mathfrak{m}$-block reader: the Newton polygon of the block's characteristic polynomial through `NumberField`, the roots refined digit by digit in tame, totally

ramified orders $\mathbb{Z}_\ell[\pi]/(\pi^e-\ell u)$, $e \mid \ell-1$, $f=1$; an unramified residue extension, $e \nmid \ell-1$, a non-integral refinement slope and a multiple root refused by name), the route change in `HidaPairingRoute.cpp` (WP-BIL's quartic-specific reader kept as the cross-check arm), `HeckeLValueLV3Arms.cpp`, the joint T_2, T_3, T_5 cut in `BirchModularSymbol.cpp`, c_f^{ε} in `NormalisationRecord`, `Hecke/HeckeInvolution`'s `BuildRingInvolution` (schema `hecke_involution_ring/v1`) and the console's `iota` under $\mathbb{Z}/\ell^k$, script `049_lv3_iota_ring.oc`; `HeckeLValueValidation` 1471 checks by default ($\approx$ 31 s, 0.58 GB), 1614 with `HECKE_LVALUE_HEAVY=1` (729 s, 1.55 GB), the twisted rung behind `HECKE_LVALUE_HEAVY=twist62`; no new validation name, no golden line or pinned digest moved. **Built at v1.20.0-pre.5** (slot-20 stretch S1, S2): `HeckeExt/LValue/LFunctionRecord.{h,cpp}` (the record `lfunction/v1`, §9.7), the console verb `lfunction` (`read_only`), arm (M7) and the env arm (M691) of `HeckeLValueMeasureValidation` with `MeasureFrozenS2.inc`; no new validation name, no golden line, digest or existing record key. Both on [C]'s seven-vertex model at level one, the model taken as a parameter; neither moves a golden line, a pinned digest or a record key.

9.5 Measures: the measure-valued linear evaluation

(New at v1.20.0-pre.1, hecke-extensions slot 20, WP-2; owner ruling O5 of 2026-10-04: *no fourth kind*. Source: slot-20 R20-1, `docs/implementation/slot20/R20-1_measures.md`, Python, in-house, predictions committed before the gated runs (`ffb34f70`); the code is slot-20 WP-MEAS, `docs/implementation/slot20/WP-MEAS_spec.md`, which merges only after this text is at pre.1 (rule 21). Placed at the end of §9, so that no statement number moves.) Throughout: the domain of Remark 9.2 (B) — [C]'s seven-vertex model at level one, $H^1_c(\Gamma; M_n \otimes \mathsf{R}) \cong W(\mathsf{R})$ by Ψ ($W = \ker(1+\rho_n(S)) \cap \ker(1+\rho_n(U)+\rho_n(U)^2)$, [computed and compared], slot-19 WP-LIN), $\mathsf{R} = \mathbb{Z}_\ell$ or $\mathbb{Z}/\ell^R$, $\ell \geq 5$, $n = k-2$ even, $u_t = x^{n-t}y^t$, $\rho_n(A)P(x,y) = P(ax+cy, bx+dy)$ on all of $\mathrm{Mat}_2(\mathbb{Z})$, $\Phi(D)$ the modular symbol of a class with $\Phi(\{0,\infty\}) = \Psi(x)$, and $\Phi(\{a/M,\infty\})$ computed along Manin's continued-fraction path. The geometric sign ε is the eigenvalue of $\rho_n(\mathrm{diag}(-1,1))$ ($u_t \mapsto (-1)^t u_t$); geometric-even is [C]-minus (Remark 9.2 (B)). $u := 1+\ell$, ω the Teichmüller character, $x = \omega(x)\langle x\rangle$, $\langle x\rangle = u^{\ell_u(x)}$ for $x \in \mathbb{Z}_\ell^\times$.

Definition 9.8 (The measure-valued evaluation of a linear route). [specification]. A *measure-valued evaluation* of a linear route $\mathcal{R} = (D, \mathsf{linear}, \mathcal{N}, \mathrm{ev}, \mathcal{I})$ (Definition 9.1) is a family of values $\mu_x(a+\ell^M\mathbb{Z}_\ell) \in \mathsf{R}$, for $x \in H'$, $M \geq 0$, $a \in \mathbb{Z}/\ell^M$, such that (i) for each ball $x \mapsto \mu_x(a+\ell^M\mathbb{Z}_\ell)$ is an R-linear functional on H' — a linear evaluation in the sense of Definition 9.1(2), so Definition 9.1 is unchanged; (ii) it is a *distribution*, $\mu_x(a+\ell^M\mathbb{Z}_\ell) = \sum_{b=0}^{\ell-1} \mu_x(a+b\ell^M+\ell^{M+1}\mathbb{Z}_\ell)$; (iii) it is bounded (values in R). Its normalisation record states the interpolation formula of its moments; its index statements are *congruences of measures*: equalities of the branch elements in $\mathsf{R}[\Gamma_u/\Gamma_u^{\ell^{M-1}}]$ ($\Gamma_u = 1+\ell\mathbb{Z}_\ell$) for every level M (equivalently in $\mathsf{R}[\![\mathbb{Z}_\ell^\times]\!]$) modulo a named ideal. The code carries it as the accessor `measure()` of linear routes; a congruence or bilinear route refuses it (`std::logic_error`), a linear route that does not implement it throws `NotImplementedInThisBuild`. Slot-19 R19-7's level-m Coates–Wiles measure (§9.3 row (a)) has the same shape on the layer, not on cohomology.

Definition 9.9 (The Mazur–Tate–Teitelbaum measure of an ordinary eigensymbol (M)). Let $x \in H^1_c(\Gamma; M_n \otimes \mathsf{R})$ be a T_ℓ-eigenclass, $T_\ell x = a_\ell x$, with $a_\ell \in \mathsf{R}^\times$ (*ordinary*), $P := \Psi(x)$, α the unit root of $X^2 - a_\ell X + \ell^{k-1}$ (unique, $\alpha \equiv a_\ell \pmod{\ell^{k-1}}$), $\beta := \ell^{k-1}/\alpha$. Put $\gamma_{a,m} := \begin{pmatrix} \ell^m & -a \\ 0 & 1 \end{pmatrix}$ and

$$c(a,m) := \left\langle \rho_n(\gamma_{a,m})\,\Phi(\{a/\ell^m,\infty\}),\, u_n^\vee \right\rangle \quad (m \geq 0), \qquad c(a,-1) := P_n,$$

($c(a,m) = -\sum_i P(q_i, q_i')$ over the bottom rows (q_i, q_i') of Manin's matrices of a/ℓ^m, with $P(c,d) := \sum_t P_t c^{n-t} d^t$). The measure (M) is

$$\mu_x(a+\ell^m\mathbb{Z}_\ell) \;:=\; \alpha^{-m}\, c(a,m) \;-\; \alpha^{-m-1}\,\ell^{k-2}\, c(a,m-1) \qquad (m \geq 0).$$

The factor ℓ^{k-2} on the stabilisation term is part of the definition: without it the family is not a distribution (a negative control, Check 9.13(M6); the formula of the slot-20 plan, `00-slot20-plan.md` v1.0 §2 WP-MEAS, omits it — corrected by R20-1).

Proposition 9.10 ((M) is a measure, its moments, its interpolation formula)**.** *[proved here] (elementary; slot-20 R20-1 §§2.1–2.7, re-derived for this text). In the setting of Definition 9.9:*

1. ***(The ℓ-Hecke relation at the ℓ-adic cusps.)*** $\sum_{b=0}^{\ell-1} c(a+b\ell^m, m+1) = a_\ell\, c(a,m) - \ell^{k-2} c(a, m-1)$ *for all $m \geq 0$ and a.*
2. ***(Measure.)*** *μ_x depends only on $a \bmod \ell^m$, is a distribution and is bounded: it is the $u_n^\vee$-coordinate of the $M_n(\mathsf{R})$-valued measure $D(a,m) := \alpha^{-m}\rho_n(\gamma_{a,m})\Phi_\alpha(\{a/\ell^m, \infty\})$, where $\Phi_\alpha := \Phi - \alpha^{-1}V\Phi$, $V\Phi(D) := \rho_n(\mathrm{diag}(1,\ell))\Phi(\ell D)$, is the ℓ-stabilisation: $U_\ell\Phi_\alpha = \alpha\Phi_\alpha$ with $U_\ell\Phi(D) := \sum_b \rho_n\left(\begin{smallmatrix}\ell & -b\\ 0 & 1\end{smallmatrix}\right)\Phi((D+b)/\ell)$, from $U+V = T_\ell$ and $UV = \ell^{k-1}$ on level-one symbols.*
3. ***(Moments.)*** *For compact open $U \subset \mathbb{Z}_\ell$ and $0 \leq j \leq n$: $\langle D(U), u_{n-j}^\vee\rangle = \binom{n}{j}\int_U (-x)^j\, d\mu_x$; the level-$M$ Riemann sum is correct modulo ℓ^M.*
4. ***(Interpolation, algebraic form.)*** *For χ primitive of conductor ℓ^m, $m \geq 1$, and for $\chi = \mathbf{1}$:*

$$(-1)^j\binom{n}{j}\int_{\mathbb{Z}_\ell^\times} \bar\chi(x)\, x^j\, d\mu_x \;=\; E_{\chi,j}\cdot\lambda_{\chi,j}(\Phi), \qquad E_{\mathbf{1},j} = \Big(1-\frac{\ell^j}{\alpha}\Big)\Big(1-\frac{\ell^{n-j}}{\alpha}\Big) = \Big(1-\frac{\ell^j}{\alpha}\Big)\Big(1-\frac{\beta}{\ell^{j+1}}\Big), \quad E_{\chi,j} = \alpha$$

   $\lambda_{\chi,j}$ the functional of Remark 9.2 (B) at modulus ℓ^m (at modulus 1 for $\chi = \mathbf{1}$: $\lambda_{\mathbf{1},j}(\Phi) = P_{n-j}$).
5. ***(Parity, branches.)*** *$\mu_x(-U) = \varepsilon\,\mu_x(U)$. The* branch series *$\mathcal{L}_{x,a}(T) := \int \omega^a(x)(1+T)^{\ell_u(x)}\, d\mu_x$ satisfy $\mathcal{L}_{x,a}(u^j - 1) = \int \omega^{a-j}(x)x^j\, d\mu_x$, vanish for $(-1)^a \neq \varepsilon$, and level M determines $\mathcal{L}_{x,a}$ modulo $\omega_{M-1} := (1+T)^{\ell^{M-1}} - 1$, by the branch element $g_i = \sum_{\delta\in\mu_{\ell-1}} \omega^a(\delta)\mu_x(\delta u^i + \ell^M\mathbb{Z}_\ell)$; so $\mu = 0$ and $\lambda < \ell^{M-1}$ are read at level M, the T^t-coefficients for $t < \ell$ modulo ℓ^{M-1}.*
6. ***(The boundary symbol.)*** *Write $\mu_Q^{(\alpha)}$ and $D_Q^{(\alpha)}(a,m) := \alpha^{-m}\mathrm{vec}_Q(a,m) - \alpha^{-m-1}\ell^{k-2}\mathrm{vec}_Q(a,m-1)$ for the formula of Definition 9.9 applied to an arbitrary symbol $Q \in W$ with a fixed unit α (linear in Q; for $Q = \Psi(x)$ and α the unit root of x it is $D(a,m)$). For $p_0 = u_0 - u_n$ (Remark 9.3) and $m \geq 1$, $\mathrm{vec}_{p_0}(a,m) = \ell^{mn}p_0$, so $D_{p_0}^{(\alpha)}(a,m) = \alpha^{-m-1}(\alpha-1)\ell^{mn}p_0$ (and 0 for $\alpha = 1$, the trivial zero of the Eisenstein measure). Hence if $\Psi(x) \equiv u\cdot p_0 \pmod{\ell^e}$ then, by linearity, $\mu_x(U) \equiv 0 \pmod{\ell^{\min(e,n)}}$ for every compact open $U \subset \mathbb{Z}_\ell^\times$ (v1.20.0-pre.2, F1 M-1: the object of the identity pinned; with (2)'s $D(a,m)$ of the eigenclass p_0 itself, unit root 1, it would be 0).*

Analytic form *(status of (B)'s constant). With Remark 9.2 (B)'s normalisation, (4) reads*

$$\int_{\mathbb{Z}_\ell^\times} \bar\chi(x)\, x^j\, d\mu_f \;=\; (-1)^j\, E_{\chi,j}\, \frac{j!\, M_\chi^{j+1}}{\chi(-1)(-2\pi i)^{j+1}\tau(\chi)}\cdot\frac{L(f,\chi,j+1)}{\Omega_f^{\pm}},$$

*M_χ the conductor, $\pm = \chi(-1)(-1)^j$ geometric; the $\binom{n}{j}$ cancels, so no unit hypothesis on it is needed. The constant has (B)'s status: [computed and compared] for trivial and quadratic χ of conductor prime to ℓ; for χ of ℓ-power conductor the factor $\tau(\chi)$ is [computed and compared] since v1.21.0-pre.3 (slot-21 P4, Remark 9.29(1): every primitive χ of conductor 25, 49, 121 at $k = 12$ and 25, 49 at $k = 16$, and 125, 169 at $k = 12$ in an env arm, against complex $L(f,\chi,j+1)$; the quotient is the conductor-1 $\Omega_f^{\pm}$ to $\leq 10^{-59}$; through v1.21.0-pre.2 it read "*derived, not tested *(no complex $L(f,\omega^i,s)$ was computed)"). The shape of $E_{\mathbf{1},j}$ is MTT's e_p ([26] §11, recalled, [locator not verified, rule A2]; R19-8 read (8.3), (8.6) on the page for the non-ℓ part).*

Proof sketch (R20-1 §2). (1) $T_\ell\Phi(D) = \sum_b \rho_n(\left(\begin{smallmatrix}\ell & -b\\ 0 & 1\end{smallmatrix}\right))\Phi((D+b)/\ell) + \rho_n(\mathrm{diag}(1,\ell))\Phi(\ell D)$ at $D = \{a/\ell^m, \infty\}$, with $\gamma_{a,m}\begin{pmatrix}\ell & -b\\ 0 & 1\end{pmatrix} = \gamma_{a+b\ell^m,m+1}$, $\gamma_{a,m}\mathrm{diag}(1,\ell) = \ell\gamma_{a,m-1}$, $\rho_n(\ell I) = \ell^n$; at $m = 0$ the last term has u_n-coordinate $\ell^n P_n$. (2) $D(a,m) = \alpha^{-m}\mathrm{vec}(a,m) - \alpha^{-m-1}\ell^{k-2}\mathrm{vec}(a,m-1)$ by (1)'s matrix identity; $\gamma_{a+\ell^m,m} = \gamma_{a,m}T^{-1}$ and T-equivariance give the dependence on $a \bmod \ell^m$; additivity is $U_\ell\Phi_\alpha = \alpha\Phi_\alpha$; boundedness because α is a unit. (3) $\gamma_{a,m} = T^{-a}\mathrm{diag}(\ell^m, 1)$ gives $D_j(a,m) \equiv \binom{n}{j}(-a)^j D_0(a,m) \pmod{\ell^m}$; sum over a partition and let $M \to \infty$. (4) For χ of conductor ℓ^m the stabilisation term depends on $a \bmod \ell^{m-1}$ and is killed by $\sum\bar\chi$; for $\chi = \mathbf{1}$, $\sum_{a\in(\mathbb{Z}/\ell)^\times} D(a,1) = D(0,0) - D(0,1)$ with $\gamma_{0,1} = \mathrm{diag}(\ell,1)$. (5) $\Phi(\{-r,\infty\}) = \varepsilon\rho_n(\iota)\Phi(\{r,\infty\})$ and $\iota\gamma_{a,m} = \gamma_{-a,m}\iota$; the rest is the binomial expansion of ω_r ($v(\binom{\ell^r}{t}) = r - v(t)$). (6) The first column of $\gamma_{a,m}g$ is $(0,\ell^m)^\top$ for every Manin matrix g, and p_0 is the symbol of $\Phi(\{r,s\}) = F(s) - F(r)$, $F(\gamma\infty) = \rho_n(\gamma)u_0$. □

Claim 9.11 (The Eisenstein congruence of measures)**.** *[computed and compared] at* $(n,\ell)=(30,37)$, $(282,37)$, $(226,103)$, $d := v(B_k/k) = 1,2,2$ *(slot-20 R20-1 §§3–4,* `r201/run_cells.py`, `out/*_full*.out`*; each value compared with an independent object: exact twisted Bernoulli numbers at every critical moment, and the Kubota–Leopoldt group ring built from the regularised Bernoulli measure* $E_{1,c}$*). The reduction to the Eisenstein symbol — $\mu_{r^-} \equiv s\,\mu^{(1)}_{P^{\mathrm{Eis}}} \pmod{\ell^d}$ on every ball — is [proved here] given its three inputs (R20-1 Prop. 3.1: $r^- \equiv sP^{\mathrm{Eis}}$, $\alpha\equiv 1$ and $\ell^{k-2}\equiv 0$ modulo ℓ^d, then linearity); its Kubota–Leopoldt form (3) is [computed and compared] (the twisted values of P^{Eis}, [standard, reference] recalled, then the runs). The expected literature is Mazur and Greenberg–Vatsal (v1.21.0-pre.1: both held since the library inventory of 2026-10-05,* `docs/implementation/slot21/library_inventory.md`*; neither yet read for this claim; through v1.20.0 this sentence read "literature pending: neither is in the library"). Let f be the ordinary eigenform with $a_p\equiv 1+p^{k-1} \pmod{\ell}$ found by a separating operator ($T_2+3T_3+7T_5$ at $(30,37)$, T_3 at the two depth-2 cells), $r^{\pm}$ its integral eigensymbols of the two geometric signs.*

1. ***(The Eisenstein symbol.)*** *$P^{\mathrm{Eis}}_{n-j} := \binom{n}{j}\frac{B_{j+1}}{j+1}\frac{B_{n-j+1}}{n-j+1}$ for odd j, 0 otherwise ($= n!\,\mathrm{Bern}_k$ of Remark 9.3, up to the primitive scaling), is integral and primitive, and $P^{\mathrm{Eis}} \in W(\mathbb{Z}/\ell^d)^\times \smallsetminus W(\mathbb{Z}/\ell^{d+1})^\times$ ([computed and compared]; that it lies in $W(\mathbb{Z}/\ell^d)$ follows from Zagier's Laurent polynomial $p_k^- \in \hat{W}_k^-$ — read on the page since v1.21.0-pre.4 (F4m): Zagier, Invent. Math. 104 (1991), Prop. (i), eq. (11), p. 453, read on the page; through pre.3 "Zagier's Eisenstein period function, [standard, reference] recalled, not read" (the name is the repository's, not Zagier's; Proposition 9.27(1)): its terms outside M_n are $\frac{B_k}{k(k-1)}(X^{-1}+X^{n+1})$, of valuation d when $\ell\nmid k-1$).*
2. ***(Minus side, values.)*** *On the geometric-odd sign, $r^-\equiv s\cdot P^{\mathrm{Eis}} \pmod{\ell^d}$, s a unit, to depth exactly d.*
3. ***(Minus side, measures.)*** *On every odd branch $a\not\equiv -1$, $k-1 \pmod{\ell-1}$,*

$$\mathcal{L}^-_{f,a}(T) \;\equiv\; -s\,G_{a+1}\big(u(1+T)-1\big)\,G_{k-1-a}\big(u^{k-1}(1+T)^{-1}-1\big) \pmod{\ell^d},$$

$\mathcal{L}^-_{f,a}$ the branch series of μ_{r^-} (Proposition 9.10(5)), G_b of [U, `lem:u-twistvalues`*]: as whole elements of $\mathbb{Z}/\ell^d[\Gamma_u/\Gamma_u^{\ell^{M-1}}]$ at level $M=3$ (37) and $M=2$ (103) (16/16, 16/16, 49/49 branches), and at every critical moment $T=u^j-1$, $0\le j\le n$, against exact twisted Bernoulli numbers (0 failures in 496, 4528, 11123). The group-ring form is stronger than the moments: at $n=30$ the 31 critical values fix a series only modulo (ℓ,X^{31}) ([U,* `thm:u-interp`*]). Negative control: the wrong twist G_{a+3} matches 0/78 branches.* A second derivation *(v1.21.0-pre.3, slot-21 P2; no tag moves): by the reduction above, $\mathcal{L}^-_{f,a}\equiv s\,\mathcal{L}^{\mathrm{poly}}_a \pmod{\ell^d}$, $\mathcal{L}^{\mathrm{poly}}$ the branch series of $\mu^{(1)}_{P^{\mathrm{Eis}}}$; P2 computes $\mathcal{L}^{\mathrm{poly}}_a\equiv\mathcal{L}^{\mathrm{Eis}}_a \pmod{\ell^d}$ off the pole branches (to depth exactly d; Proposition 9.27(5), [computed and compared]) and $\mathcal{L}^{\mathrm{Eis}}_a = -G_{a+1}(u(1+T)-1)\,G_{k-1-a}(u^{k-1}(1+T)^{-1}-1)$ exactly in $\Lambda\otimes\mathbb{Q}$ (Proposition 9.27(3), [proved here] given [standard, reference]), which gives (3) at every cell where (2) and that comparison hold.*
4. ***(The pole branches: an explicit exception.)*** *On the two branches $a\equiv -1$ and $a\equiv k-1 \pmod{\ell-1}$ one factor of (3) is G_0, the pole of ζ_ℓ, and (3) is* ***not*** *claimed there. [computed and compared]: (3)'s right-hand side, read at the critical moments, fails modulo ℓ^d at every point of both branches (62/62, 566/566, 454/454); modulo ℓ^{d-1} (at $d=2$) it fails at exactly 14 of 566 points at $(282,37)$ and at exactly 4 of 454 at $(226,103)$, namely $(a,j)=(23,21)$, $(23,124)$, $(101,102)$, $(101,205)$ — the points where the argument of the G_0-factor (x_{k-1-j} on the branch $k-1$, x_{j+1} on the branch -1) has valuation ≥ 2. At $(282,37)$ the 14 points are $(a,j)=(31;\,24,61,98,135,172,209,246)$ and $(35;\,36,73,110,147,184,221,258)$ — the rule's prediction ($j\equiv k-1 \pmod{37}$ on the branch 31, $j\equiv -1 \pmod{37}$ on the branch 35), committed by slot-20 WP-MEAS before its first run (`374c2394`) and located by its C++ gate ([computed and compared], Check 9.13(M4)(d)); so 18/18 exceptions are located at both cells by one rule. On both pole branches $\mu=\lambda=0$ at all three cells ([computed and compared]).* Explained *(v1.21.0-pre.3, slot-21 P2, Proposition 9.27(4)–(6)): the exact Eisenstein measure has, on the two pole branches, a polar term $\mathrm{Res}/(u(1+T)-1)$, resp. $\mathrm{Res}/(u^{k-1}(1+T)^{-1}-1)$, with $\mathrm{Res}=(1-\ell)(\log_\ell u/\ell)(1-\ell^{k-1})B_k/k$, $v(\mathrm{Res})=d$ (the constant [computed and compared]), and on*

every odd branch, pole branches included, $\mathcal{L}^-_{f,a} \equiv s\,(\mathcal{L}^{\mathrm{Eis}}_a - \mathrm{polar}_a) \pmod{\ell^d}$*; at the critical points* $\mathcal{L}^-_{f,a}(u^j-1) \equiv -s\left(F_1F_2 + \mathrm{Res}/(u^r-1)\right) \pmod{\ell^d}$*,* F_1F_2 *the product of (3)'s two G-factors at* $T = u^j - 1$ *(R20-1's naive product of twisted Bernoulli numbers),* $r = j+1$ *on the branch* -1 *and* $r = k-1-j$ *on the branch* $k-1$ *([computed and compared] with* r^- *at the three cells:* 0/62, 0/566, 0/454 *critical points, Check 9.28(E8)). As* $v(u^r-1) = 1 + v(r)$*, the polar value has valuation exactly* $d-1-v(r)$ *([proved here] from* $v(\mathrm{Res}) = d$*), so the naive product agrees modulo* $\ell^{d-1-v(r)}$ *and no further: the exceptions modulo* ℓ^{d-1} *are exactly the points with* $\ell \mid r$ *— the rule above, and the* $14+4$ *located points, now* derived *(gate:* $\mathrm{Res}\cdot(1+\ell)$ *in place of* Res *fails exactly there). Greenberg–Vatsal and Mazur (held, library inventory 2026-10-05, not yet read for it) are literature context, not a dependency; R20-L is not needed. Through v1.21.0-pre.2 this sentence read: "*Not explained here*: the regularised Eisenstein measure on the pole branches (R20-1 §4.4 reads a polar term of valuation* $d-1-v(r)$ *[...]) is* ***deferred*** *[...]; no statement of this document rests on it." The congruence (3) is still not claimed on the pole branches; the polar-term form is Proposition 9.27(6). At* $(10, 691)$ *the polar-term form is checked on* Δ*'s own eigensymbol at level* 2*, at all* 3795 *critical points and as group-ring elements on every odd branch, pole branches included ([computed and compared], Check 9.28(E8L2); Proposition 9.27(6)).*

5. ***(Minus side, invariants.)*** $\mu = 0$ *and* $\lambda = 0$ *on every odd branch at the three cells ([computed and compared]; off the pole branches this follows from (3), both G-factors being units —* 32 *is the only irregular index at* 37, 24 *at* 103*). The general form is the level-one Greenberg–Vatsal formula* $\lambda^-_a = \lambda(G_{a+1}) + \lambda(G_{k-1-a})$ *off the pole branches, the cell's own irregular index always on a pole branch (Remark 9.31(7), v1.21.0-pre.7, slot-21 P5; [computed and compared] at nine cells, among them* $(10, 691)$ *with* $\lambda^- = 1$ *at* 199, 501*).*

The plus side is Remark 9.12.

Remark 9.12 (The plus side: the boundary, $\mu^+ = d$, and an observation). (Slot-20 R20-1 §§2.7, 4.2, 4.5.) (i) On the geometric-even sign the H^1_c-primitive eigensymbol satisfies $r^+ \equiv u \cdot p_0 \pmod{\ell^d}$, u a unit, to depth exactly d, at the three cells of Claim 9.11 — (L1) of Remark 9.3, computed at depth 2 at $(282, 37)$, $(226, 103)$ (R20-1, one route, predictions committed first); its tag stays [conjectural] at depth ≥ 2 (Remark 9.3). (ii) Given (i), $\mu_{r^+}(U) \equiv 0 \pmod{\ell^d}$ on every $U \subset \mathbb{Z}_\ell^\times$ (Proposition 9.10(6), linearity in the symbol, $n \geq d$; [proved here] from (i)), and the minimum valuation on the level-M unit balls is exactly d ([computed and compared] at the three cells): $\mu^+ = d$, i.e. the H^1_c-period on the boundary sign is ℓ^d times the cuspidal one — the measure form of slot-19 WP-BIL's finding that the mixed lattice adds $v(B_k/k)$ on that sign (Remark 9.2 (H)). (iii) *Observation, not a claim.* The reduced plus measure $\ell^{-d}\mu_{r^+}$ (from $w = (r^+ - p_0)/\ell^d$) has $\mu = 0$ and $\lambda = 1$ on the branch pair $\{4, 26\}$ at $(30, 37)$ and $(282, 37)$ and on $\{46, 62, 78\}$ at $(226, 103)$, $\lambda = 0$ elsewhere; the pairs are $a \leftrightarrow n - a \pmod{\ell-1}$ (the functional equation; 62 self-paired). At the regular controls $(10, 37)$ (Δ) and $(16, 37)$, $\mu = 0$ throughout and $\lambda = 1$ on the branch pairs $\{21, 25\}$, $\{21, 31\}$ (minus) and $\{8, 22, 26, 30\}$ at $(16, 37)$ (plus). No interpretation is proposed; the plus-side Eisenstein "measure" involves the trivial zero $L(\chi, j-n) = 0$, so a reading needs the derivative or boundary term (Greenberg–Vatsal's λ-formula for the other sign is the expected place; held, not yet read; v1.21.0-pre.1). *Partly placed* (v1.21.0-pre.7, slot-21 P5, Remark 9.31): 62 at $(226, 103)$ and 8, 26 at $(16, 37)$ are even self-dual branches, where λ is odd by the functional equation ([proved here], Remark 9.31(3)); the pairs $\{4, 26\}$ and $\{46, 78\}$ are $\{-k, n+k\} \pmod{\ell-1}$, a pattern seen at all nine Eisenstein cells (through pre.8 "recorded, not derived"), since v1.21.0-pre.9 identified with the McCallum–Sharifi forced zero ([computed and compared] at 32 cells; $\lambda^+ \geq 1$ there through v1.22.0-pre.1 a theorem *conditional on Sharifi's conjecture*; since v1.22.0-pre.2 *unconditional at the computed cells* (slot-22 P5″; a per-pair computation, [proved here] modulo [standard, reference] inputs, given [computed and compared] L3 at the pair and [computed and compared] (H1) at the cell; the uniform statement open), Remark 9.31(9)); the controls' other pairs stay unexplained (Remark 9.31(8)).

Check 9.13 (The gate of WP-MEAS: `HeckeLValueMeasureValidation`). [specification] (the checks); the expected values are R20-1's ([computed and compared] or [proved here] as tagged above), frozen in an in-source table `MeasureFrozenR201.inc`, never this code's output. *Registration*: `Registry/Slot20/WPMEAS.cpp`, `cmake/slot20/WPMEAS.cmake`; tier `standard`; quiet budget (not a gate: no timing is a verdict) ≤ 15 s and ≤ 1 GB (estimate 5–8 s, ≤ 0.2 GB); first run 3.5 s, 37 MB, 178

checks, 0 failing (WP-MEAS). *Cells*: $(10,37)$ (Δ, $d=0$, level 2), $(16,37)$ ($d=0$, level 3), $(30,37)$ ($d=1$, level 3), $(282,37)$ ($d=2$, level 3), $(226,103)$ ($d=2$, level 2), all modulo ℓ^4; $(10,5)$ the non-ordinary refusal ($\tau(5)\equiv 0 \pmod 5$). Arms:

(M0) **The route** on the C++ module cone at $(30,37)$ over $\mathbb{Z}/37^4$: `measure()` of the cone classes mapping to $r^{\pm}$ equals the symbol-side measure (FNV equal); linearity on two eigenclasses of different sign; the bilinear and congruence routes refuse `measure()`; $(10,5)$ refuses "not ordinary"; a non-eigenclass refuses "not a T_ℓ eigenclass". **This arm stays in the `standard` tier** (lead decision, slot-20 WP-2: ≈ 3 s of the 15 s budget).

(M1) Proposition 9.10(1)–(5) at every cell and both signs: three-term and distribution failures 0 at $m=0,\dots,M-1$; the wrong-parity branches vanish; the moment identity (4) for $\chi=\omega^{-i}$, $i\in\{0,1,2,3,(\ell-1)/2,\ell-2\}$, every j, modulo $\ell^{\min(M,R)}$, against slot-19 WP-LIN's $\lambda_{\chi,j}$ (`EvAllJ`); the Riemann-sum moment equals the group-ring branch value at three j per cell.

(M2) R20-1's frozen values modulo ℓ^4 at level 2 (r^- scaled to first unit coordinate 1, r^+ to $r_0^+=1$; FNV-1a-64 of $\mu(a+\ell^2\mathbb{Z}_\ell)$, units a ascending, 8 LE bytes each):

cell	a_2	a_3	$a_\ell=\alpha$	s	$\mu^-(2+\ell^2)$	FNV μ^-	$\mu^+(2+\ell^2)$	FNV μ^+
$(10,37)$	1874137	252	1454464	—	272431	`0x721edc172f8e24bd`	847858	`0xcc0c00522e7e0429`
$(16,37)$	1873633	1869877	1454991	—	57498	`0xed8c569e28ac74be`	698836	`0x57a02c809eb73309`
$(30,37)$	1622769	150214	270360	922897	1022643	`0xf6fef89b62e18907`	78662	`0x6ddd3cd8d0e6a6ad`
$(282,37)$	1079424	1744174	1770118	575834	1326561	`0xc1d253ded2d790cd`	1815294	`0xa812126c962c7079`
$(226,103)$	15697592	81971822	50764066	83864971	44186485	`0x5fb753f5ed92f150`	95024813	`0xde51edced2aacde1`

and $\mu^-(\mathbb{Z}_\ell^\times)=0$.

(M3) Claim 9.11(1),(2) and Remark 9.12(i),(ii) at the three Eisenstein cells: $v(a_p-1-p^{k-1})=d$ for $p\in\{2,3,5,7,\ell\}$; (L1) to depth exactly d; $\mu^+=d$; P^{Eis} integral, primitive, in $W(\mathbb{Z}/\ell^d)\smallsetminus W(\mathbb{Z}/\ell^{d+1})$; $r^-\equiv sP^{\mathrm{Eis}}$ to depth exactly d.

(M4) Kubota–Leopoldt: (a) the two group-ring routes for G_b — from $E_{1,c}$ and from `NumberField`'s Coates–Wiles layer (§9.4; $G_b=-(\ell-1)\nu_b$ in `CwLayer`'s index convention) — agree for every even $b\not\equiv 0$ at 37 (level 3) and 103 (level 2), and both reproduce $-(1-\ell^{k-1})B_k/k$ at ≥ 4 weights (a new prediction of WP-MEAS, [proved here] in principle, R19-7 §§1.1–1.2; held at its first run: 17/17 and 50/50 classes, 68/68 and 200/200 values); (b) Claim 9.11(3) at the critical moments; (c) (3) as group-ring elements, with G_b from both routes; (d) Claim 9.11(4): the located failure sets modulo ℓ^{d-1} gated as computed (the 14 at $(282,37)$ predicted and located).

(M5) The invariants of Claim 9.11(5) and Remark 9.12(iii) per branch and sign, as tables (recorded values; gating them gates the code against R20-1, not the observation).

(M6) Negative controls, each gated as *failing* its positive check: no ℓ^{k-2} (distribution fails at $m=1$ at every cell); the wrong root $\alpha(1+\ell)$; the wrong twist G_{a+3}; the wrong sign $+s$; at $(282,37)$ and $(226,103)$ the separator $T_2+3T_3+7T_5$ must be *refused* with multiplicity 2.

(M7) (v1.20.0-pre.5, slot-20 stretch S1.) The record `lfunction/v1` of §9.7 at the five cells, both signs, every branch of the right parity: values, truncation against level $M-1$ (Lemma 9.21), (μ,λ) from the coefficients, determinism (42 checks; the default arm becomes 220 checks, ≈ 4.5 s, 37 MB). Output lines `MEAS_LFUN`. Its checks are not extended to the heavy cells (the level-3 cell at 103 has $G=10\,609$); under `HECKE_LVALUE_MEASURE_HEAVY=1` the 42 default checks run unchanged.

(M691) (v1.20.0-pre.5, slot-20 stretch S2; env-armed, read-only.) `HECKE_LVALUE_MEASURE_691=1` prints `EFFECTIVE HeckeLValueMeasureValidation arm=ELL691 cells=1` and runs Claim 9.23: the two Kubota–Leopoldt routes at 691, level 2 (344 classes); the cell $(10,691)$ at $R=2$ (the in-TU twisted Bernoulli table needs $\ell^{R+4}<2^{62}$), level 2, through (M1), (M3)–(M7) exactly as at the other Eisenstein cells ($d=1$); $a_p\equiv\tau(p) \pmod{691^2}$; the oracle's s, $\mu^{\pm}(2+\ell^2)$ and FNVs (`MeasureFrozenS2.inc`); $v=0$ at the 22 pole points. 43 checks (263 with the default arm), 0 failing at the first run; ≈ 34 s of the 38.5 s and 66 MB (beyond the `standard` budget of 15 s, hence env-armed). Output line `MEAS_691`.

Output lines `MEAS_CELL`, `MEAS_ID`, `MEAS_EIS`, `MEAS_KL`, `MEAS_INV`, `MEAS_NEG`, and `MEAS_ARMS checks=<c> failing=<f>`. A read-only env arm `HECKE_LVALUE_MEASURE_HEAVY=1` prints `EFFECTIVE HeckeLValueMeasureValidation arm=HEAVY cells=3` and adds $(226,103)$ at level 3, $(282,37)$ at

$R = 5$ and every character in (M1) ($\leq 60\,\mathrm{s}$, $\leq 2\,\mathrm{GB}$). *Exit criteria of WP-MEAS*: WP-MEAS's predictions (Q1–Q9 of its spec, §5) committed before its first run; `failing=0`; the roster $181 = 66/98/17$ (`--all` runs 164), `EXPECT_NAMES` 181, the [CLI] tables regenerated in the same commit and again after the merge, the catalogue entry; no golden line, digest, record key or schema key moves; a disagreement with R20-1 is reported as a finding against this section, never re-pinned.

9.6 The depth-2 triangle, the change of lattice, and ι over $\mathbb{Z}/\ell^k$

(New at v1.20.0-pre.1, hecke-extensions slot 20, WP-2; the contract text of slot-20 WP-LV3, which merges only after it (rule 21), and of slot-20 R20-3's constant. Placed at the end of §9.)

Claim 9.14 (The three kinds at depth 2)**.** *At the two depth-*2 *rungs* $(n, \ell) = (282, 37)$ *(*$k = 284$*) and* $(226, 103)$ *(*$k = 228$*),* $d = v(B_k/k) = 2$*,* $n \geq \ell - 1$*, on the* $\mathfrak{m}_{\mathrm{Eis}}$*-block of* H^1*, cut by the* joint *ideal* $\mathfrak{m}_{\mathrm{Eis}} = (\ell,\, T_p - 1 - p^{k-1} : p = 2, 3, 5)$ *(v1.20.0-pre.6, slot-20 WP-LV3:* $(\ell, T_2 - 1 - 2^{k-1})$ *alone is too large at* $(282, 37)$ *— a cusp form there has* $a_2 \equiv 1 + 2^{k-1}$*, and the* T_2*-block has rank* 3 *on the* p_0 *sign):*

- *(C)* congruence: $\#Q_m = \ell^{\min(m,2)}$*,* $m = 1, 2, 3$ *— [computed and compared] (slot-19 WP-LV2 and, independently, R19-7; Remark 9.7);*
- *(L)* linear: *(L1) to depth exactly* 2*,* $r^+ \equiv u \cdot p_0 \pmod{\ell^2}$ *and not modulo* ℓ^3 *— [computed and compared] (v1.20.0-pre.6; [conjectural] through pre.5): slot-20 R20-1 (Python, predictions committed first; Remark 9.12(i), Claim 9.11(2)) and slot-20 WP-LV3 (C++, on* W *over* $\mathbb{Z}/\ell^3$ *and on the C++ cone over* $\mathbb{Z}/\ell^3$*); and Remark 9.3's item 3 to depth exactly* 2 *— [conjectural] as an index statement of this route (R20-1 only, one route, not gated by WP-LV3; the congruence itself is [computed and compared] as a run, Claim 9.11(2));*
- *(H)* bilinear: *(H2) on the mixed lattice* $H^1_c(\Gamma; M_n) \times H^1(\Gamma; D_n)$*, the value on integral eigenlines has* $v = 2$ *on the sign carrying the boundary class* p_0 *and* $v = 0$ *on the other — [computed and compared] (v1.20.0-pre.6, slot-20 WP-LV3's generic reader, every value equal to the congruence module of its lattice read from* A_c *alone, also for the boundary system; [conjectural] through pre.5: slot-19 R19-9 §8's prediction, which it confirms).*

The triangle *([computed and compared] since v1.20.0-pre.6; [conjectural] through pre.5): the three depths are equal, to* d *— WP-LV3's cross-kind gate, congruence = bilinear = linear =* $v(B_k/k)$*, holds at* $(282, 37)$*,* $(226, 103)$ *(*2*) and at the depth-*1 *controls* $(30, 37)$*,* $(10, 691)$ *(*1*), and in the heavy arm against [C]'s* H^- *(*$\#Q_3$ *of the* congruence *route) and (L1) on the C++ cone (Check 9.19, outcome). As a statement for every depth-*2 *rung it stays [conjectural]. At depth* 1 *the three kinds agree at the cells where each was computed ([computed and compared]; Remarks 9.7, 9.3, 9.2 (H)). Gated by Check 9.19; a disagreement there is a finding against this claim and against Remark 9.2, never a re-pin (the first suspect is the lattice: mixed versus* $H^1_!$*; the second the change of lattice, Definition 9.15).*

Definition 9.15 (The change-of-lattice constant for $n \geq \ell$). [specification]. Let $\theta := \Phi_n \colon M_n \to D_n$ be the comparison map of [C, `lem:symdual`] (Gram matrix $(-1)^t t!\,(n-t)!$; the map of Remark 9.2 (H), Hida's form $G_!(P, Q) = \langle P, \theta\iota_! Q\rangle$, whose code `HidaPairing::theta` is $\Phi_n/n!$, the same up to the unit $n!$ when $n < \ell$): an isomorphism after $\otimes\mathbb{Q}$, and over $\mathbb{Z}_\ell$ iff $n < \ell$ ([C, `lem:symdual`]: $\Phi_n \otimes \mathbb{Z}_\ell$ is an isomorphism iff $n! \in \mathbb{Z}_\ell^\times$). Let f be a Hecke eigensystem with values in $\mathcal{O}$ (a finite extension of $\mathbb{Z}_\ell$, uniformiser π), ε a sign, and for a $\mathbb{Z}_\ell$-lattice L in a Hecke module write $L[f]^\varepsilon := L^\varepsilon_{\mathcal{O}} \cap$ (the f-eigenline of $L^\varepsilon \otimes \operatorname{Frac}\mathcal{O}$) (a free $\mathcal{O}$-module of rank 1 where f occurs with multiplicity one in that sign). The *change-of-lattice constant* is

$$c_f^\varepsilon \;:=\; \operatorname{length}_{\mathcal{O}}\Big(H^1(\Gamma; D_n)/\mathrm{tors}\,[f]^\varepsilon \Big/ \theta_*\big(H^1(\Gamma; M_n)/\mathrm{tors}\,[f]^\varepsilon\big)\Big),$$

defined where θ_* carries the f-line of $H^1(\Gamma; M_n \otimes \mathbb{Q}_\ell)$ onto that of $H^1(\Gamma; D_n \otimes \mathbb{Q}_\ell)$. *This holds* (v1.20.0-pre.2, F1 m-E5; [proved here] from [C]): by [C, `lem:symdual`], $\Phi_n \circ \rho_n(g) = \rho_n^\vee(g) \circ \Phi_n$ for every $g \in \mathrm{Mat}_2(\mathbb{Z})$, untwisted, so Φ_n commutes on cochains with T_p formed from the same cosets, i.e. with $A(D_n)(T_p)$ of [C, `sec:symconv`]'s code map (`HeckeOperatorOnK`'s `FibreRepresentation`; `HeckeDualityValidation` asserts the intertwining untwisted), and with ι ([C, `sec:involution`]); $\theta_* \otimes \mathbb{Q}_\ell$ is then a Hecke- and ι-equivariant isomorphism and carries each eigenline of each sign onto

the corresponding one. It is no longer an input (Proposition 9.16(1)); R20-3 checked that its prototype's T_p on $H^1(\Gamma; D_n)$ is that normalisation (`EQUIV` 0 failures at four cells; a det-twist would break it). $c_f^\varepsilon = 0$ for $n < \ell$ ([proved here]: θ is then an isomorphism over $\mathbb{Z}_\ell$). It converts a pairing value on $H_c^1(M_n) \times H^1(M_n)$ (through θ) into one on the mixed lattice of (H2), and is what slot-19 WP-BIL printed as "open". *Status*: at v1.20.0-pre.1 the value was **open**; since v1.20.0-pre.6 it is given by Proposition 9.16(2) and Remark 9.17 (slot-20 R20-3, theory and two prototypes, predictions first, `92336edc`), and slot-20 WP-LV3 computes it in `NormalisationRecord` for $n \geq \ell$ with $\theta = \Phi_n$ (the record's row "change of lattice c_f (`def:lvlattice`; $\theta = \Phi_n$, $n \geq \ell$)", printing $e\, v_\ell(n!)$ and d_f separately; no longer "open").

Proposition 9.16 (The change-of-lattice constant)**.** *[proved here] (v1.20.0-pre.6; slot-20 R20-3, `docs/implementation/slot20/R20-3_lattice_constant.md` §2). Take $\theta = \Phi_n$ of [C, `lem:symdual`] and let $V_n' := \bigoplus_t \binom{n}{t}\mathbb{Z}_\ell u_t = n!\,\Phi_n^{-1}(D_n) \subset M_n$ (a Δ-stable lattice, since D_n is Δ-stable under $\rho_n^\vee$ and Φ_n is Δ-equivariant and injective; $\ell^m M_n \subset V_n' \subset M_n$ with $m = \max_t v_\ell\binom{n}{t}$, the maximal number of base-ℓ carries in $t + (n-t)$). It is* not *the $\mathbb{Z}_\ell$-span of the n-th powers $(ax+by)^n$ for $n \geq \ell+1$: modulo ℓ those span at most $\ell+1$ dimensions, since $a^{n-1}b \equiv a^{n-\ell}b^\ell$ on $\mathbb{F}_\ell^2$ (v1.20.0-pre.7, F1′ M-E1: through pre.6 the definition read "$\sum_{a,b}\mathbb{Z}_\ell(ax+by)^n = \bigoplus_t \binom{n}{t}\mathbb{Z}_\ell u_t$", false for $n \geq \ell+1$; every value and the code concern the diagonal lattice, so nothing computed moves).*

1. *θ_* commutes with every T_p and with ι^*, so the input of Definition 9.15 holds for every f of multiplicity one in its sign ([C, `lem:symdual`]: $\Phi_n \circ \rho_n(g) = \rho_n^\vee(g) \circ \Phi_n$ for all $g \in \mathrm{Mat}_2(\mathbb{Z})$, with $\rho_n^\vee(g) = \rho_n(\mathrm{adj}\, g)^{\mathsf{T}}$; double-coset operators are natural in Δ-module maps).*
2. *$\Phi_n^{-1}(D_n) = (n!)^{-1}V_n'$ (as $t!\,(n-t)! = n!/\binom{n}{t}$), hence*

$$c_f^\varepsilon = e\, v_\ell(n!) - d_f^\varepsilon, \qquad d_f^\varepsilon := \mathrm{length}_{\mathcal{O}}\Big(H^1(\Gamma; M_n)/\mathrm{tors}\,[f]^\varepsilon \Big/ \mathrm{im}\, H^1(\Gamma; V_n')/\mathrm{tors}\,[f]^\varepsilon\Big),$$

   $e = v_\pi(\ell)$, and $0 \leq d_f^\varepsilon \leq e\,m$; in particular $c_f^\varepsilon = e\, v_\ell(n!)$ for every f when all base-ℓ digits of n below the leading one equal $\ell - 1$ ($V_n' = M_n$), and $0 \leq d_f^\varepsilon \leq e$ for $\ell \leq n < \ell^2$.
3. *$d_f^\varepsilon = 0$ whenever $H^1(\Gamma; M_n/V_n')^\varepsilon_{\mathfrak{m}_f} = 0$ (the sequence $H^1(V_n') \to H^1(M_n) \to H^1(M_n/V_n') \to H^2(V_n') = 0$ ($\ell \geq 5$: $\mathrm{SL}_2(\mathbb{Z}) = \mathbb{Z}/4 *_{\mathbb{Z}/2} \mathbb{Z}/6$ and the orders 4, 6 are units, Mayer–Vietoris), localised at $\mathfrak{m}_f$ and taken on the ε-part: a surjection of free modules of equal rank).*

Remark 9.17 (Values). [computed and compared] (v1.20.0-pre.6; slot-20 R20-3, two routes: the transport by Φ_n on H_c^1 (`r203/lattice.py`), and the definition on $H^1(\Gamma; M_n)$ with the group-cocycle T_p (`r203/direct.py`); 49/49 eigenlines agree at four cells; and slot-20 WP-LV3's C++ `NormalisationRecord`, which reads d_f on the D_n side through the perfect pairing with $H_c^1(D_n)$, a third route). On the $\mathfrak{m}_{\mathrm{Eis}}$-blocks $d_f = 0$ on both signs, for the cusp form and for E_k: $c_f^\pm = 7$ at $(282, 37)$, 2 at $(226, 103)$; at $(258, 37)$ $V_n' = M_n$ and $c_f = 6e$ for every f (R19-1's quartic, $e = 2$: $v_\pi = 12$, $v_{37} = 6$, all four forms, both signs); 0 for $n < \ell$ ($(30, 37)$, $(10, 691)$). Generic eigenlines can carry $d_f = 1$: at $(40, 37)$ five ($\lambda \equiv 3, 7, 29$ on $+$, $3, 29$ on $-$, with λ the eigenvalue of $T_2 + 5T_3 + 11T_5 + 13T_7$; WP-LV3's negative control `LV3_LATTICE_NEG`). For $\ell \leq n < \ell^2$, $M_n/V_n' \otimes \mathbb{F}_\ell \cong \bigoplus_j \mathrm{Sym}^{a_j} \otimes \det^{b_j}$ ($a_j = \ell - 3 - n_0 + n_1 - 2j$, $b_j = n_0 + 1 + j$, $n = n_1\ell + n_0$; the module structure recalled from Glover's filtration, [standard, reference], the graded form [proved here], the dimension counts [computed and compared]), and $d_f^\varepsilon = 1$ exactly on the ω^{b_j}-twists of the weight-$(a_j + 2)$ systems, the sign changed by $(-1)^{b_j}$ ([conjectural] in general; [computed and compared] line by line at $(40, 37)$, $(282, 37)$, $(226, 103)$, with $\sum_m (D_m^+ + D_m^-) = \dim H^1(M_n/V_n')$ at all five cells $n \geq \ell$). Through $\theta_{R199} = \Phi_n/n!$ (slot-19 R19-9; the code's `HidaPairing::theta`, not integral for $n \geq \ell$) the same index is $-d_f$. Whole-lattice indices (WP-LV3, `LV3_THETA_INDEX`, $v_\ell[H^1(D_n)/\mathrm{tors} : \Phi_{n,*}H^1(M_n)/\mathrm{tors}]$ / $v_\ell[H^1(M_n)/\mathrm{tors} : \Theta H^1(D_n)/\mathrm{tors}]$, $\Theta = (\Phi_n/n!)^{-1}$): $(10, 5)$ 4/2, $(10, 7)$ 2/1, $(12, 5)$ 2/0, $(24, 7)$ 8/1, $(40, 37)$ 2/5, $(72, 37)$ 11/0, $(30, 37)$ 0/0, $(258, 37)$ 258/0.

Remark 9.18 (ι over $\mathbb{Z}/\ell^k$). Nothing new is claimed: [C] already gives ι^* over every coefficient ring ([C, `def:iota`]), its commutation with every T_p ([C, `prop:iotacommutes`]) and the eigenspace decomposition whenever 2 is a unit ([C, `prop:iotasplit`]) — so over $\mathbb{Z}/\ell^k$, ℓ odd, $H^q = H^{q,+} \oplus H^{q,-}$, Hecke-stably and compatibly with the reductions $\mathbb{Z}/\ell^k \to \mathbb{Z}/\ell^{k'}$ ([proved here] from [C]). What was missing is the *code*: in the slot-19 build the console refuses `iota` under

`coefficient_ring = "Z/ell^k"` (exit 5; R19-7 §6.3 read the sign split off the matrices). Slot-20 WP-LV3(d) makes `Hecke/HeckeInvolution` and the console honour it, gated in Check 9.19(d). *Built* (v1.20.0-pre.6): `hecke::BuildRingInvolution` applies the field object's cochain formula ($\iota^* = -\iota^*_{\mathrm{geo}}$, $\rho_n(\iota) = \mathrm{diag}((-1)^{n-j})$) on the module basis of every degree, with the always-on checks $\iota^2 = \mathrm{id}$ and $\operatorname{length} H^+ + \operatorname{length} H^- = \operatorname{length} H$ ($\ell \geq 5$; $\ell \in \{2,3\}$ refused by name); the record is `hecke_involution_ring/v1` (a new schema, not golden); the console list is §7.1 (h).

Check 9.19 (The gates of WP-LV3). [specification] (the checks; expected values from the sources named, committed as WP-LV3's predictions before its first run). All arms are inside `HeckeLValueValidation` (no new name; its default wall stays $\leq$ its slot-19 figure +10 s, heavier cells behind the read-only `HECKE_LVALUE_HEAVY=1`).

(a) **The generic $\mathfrak{m}_{\mathrm{Eis}}$-block reader.** Given a Hecke block over $\mathbb{Z}/\ell^m$ and the Hensel factor P of $\mathrm{charpoly}(T_{p^*})$ at $\mathfrak{m}_{\mathrm{Eis}}$, it returns the eigensystem over $\mathcal{O} = \mathbb{Z}_\ell[x]/(P)$ (a product of local orders in general; $\mathbb{Q}(\alpha)$ and v_π through `NumberField`, §9.4) and the per-eigenform lines. *As built* (v1.20.0-pre.6): the block is that of $t = T_2 - \lambda_0$, at $\mathfrak{m}_{\mathrm{Eis}}$ the joint block of T_2, T_3, T_5 (Claim 9.14); the roots of its characteristic polynomial $P \equiv x^r \pmod{\ell}$ are read from the Newton polygon of P and its residual polynomials, each in a *tame, totally ramified* order $\mathcal{O} = \mathbb{Z}_\ell[\pi]/(\pi^e - \ell u)$, $e \mid \ell - 1$, residue degree $f = 1$, refined digit by digit; an unramified residue extension ($f > 1$), $e \nmid \ell - 1$, a non-integral refinement slope and a multiple root to the precision are *refused by name*, never read. No global $\mathbb{Q}(\alpha)$ is formed at the gate cells. At $(258, 37)$ $\mathfrak{m} = (37, T_2 - 29)$, WP-BIL's and R19-1's convention. First gate: slot-19 WP-BIL's values at $(258, 37)$ (Remark 9.2 (H): mixed 3/2, through $H^1_!(D)$ 3/3, $\sum_f v_{37} = 6$) reproduced bit for bit; the quartic-specific path stays as a cross-check arm until they match. The reader takes the model as a parameter (no hard-wired seven-vertex model, no trivial centre).

(b) **The triangle** (Claim 9.14) at $(282, 37)$ and $(226, 103)$: $\#Q_2 = \ell^2$; (H2) $v = (2, 0)$ on (the p_0 sign, the other); (L1) depth exactly 2; and their agreement — the *cross-kind gate* — at these two cells and at two depth-1 controls, $(10, 691)$ (a gate cell of all three kinds) and $(30, 37)$ (expected depth 1 in all three readings). A disagreement is a finding for Claim 9.14.

(c) **The constant c^{ε}_f** of Definition 9.15 in `NormalisationRecord` for $n \geq \ell$, against R20-3's predicted values.

(d) **ι over $\mathbb{Z}/\ell^k$** (Remark 9.18): `Hecke/HeckeInvolution` and the console accept $\mathsf{R} = \mathbb{Z}/\ell^k$ (exit 5 removed); a new console script (the next number, frozen `.expected`: an addition, not a re-baseline) reads $H^{\pm}$ at $(282, 37)$ over $\mathbb{Z}/37^k$; expected: free of ranks 24 (+, the Eisenstein side) and 23 (−) (slot-19 WP-LV2), the T_p-blocks equal to R19-7 §5's matrix-read split. *Amended* (v1.20.0-pre.6, lead decision on WP-LV3's report): the script `049_lv3_iota_ring.oc` reads $(30, 37)$ and the torsion cell $(38, 37)$ over $\mathbb{Z}/37^3$ (the $\mathbb{Z}/37^k$ cohomology at $n = 282$ costs ≈ 25 s per run and every numbered script runs several times inside `ConsoleDeterminismValidation`'s 45 s threshold); the $(282, 37)$ reading runs in the heavy arm (`LV3_IOTA_RING_282`). `ConsoleDeterminismValidation`'s pinned C4 (a) digest does not move; the unpinned all-scripts digest moves, and the report says so.

(e) **The twisted depth-2 rung** $(\ell; k, j) = (37; 62, 6)$ of Claim 9.6 ($v(B_{62,\omega^6}/62) = 2$ by two routes, Remark 9.7): first the character model at $n = 20$, 40 (RSS and wall measured), then the curve's prediction for $n = 60$, then the run behind `HECKE_LVALUE_HEAVY` under a cap ≤ 24 GB; expected: rank 1, saturated, $\#Q_m = \ell^{\min(m,2)}$. If the curve predicts more than 24 GB the cell is not run and is recorded with its curve (or handed to a cluster kit).

Exit criteria of WP-LV3: (a) held before (b) is trusted; (b)–(e) held or each failure reported as a finding the same turn; zero golden movement, the four digests unchanged (the console addition excepted as stated in (d)).

Outcome (v1.20.0-pre.6; [computed and compared], slot-20 WP-LV3, `docs/implementation/slot20/WP-LV3_report.md`; `HeckeLValueValidation` 1471 checks by default, 1614 heavy, 0 failing; zero golden movement, the digests unchanged, the console C4 (a) digest unchanged and the unpinned all-scripts digest moved, as (d) said). (a) At $(258, 37)$ the generic reader equals WP-BIL's bit for bit (cell fields, block ranks, the characteristic polynomial modulo 37^8, the Gram pivots; R19-1's factors P_1, P_2 *recovered*, two roots each; per factor (mixed, P9, $v_\pi P'$, content) geo-even $(3, 3, 3, 0)$, geo-odd

$(2,3,3,1)$). (b) The cross-kind gate holds at all four cells: congruence = bilinear (p_0 sign) = linear (L1) = $v(B_k/k)$, 2 at $(282,37)$ and $(226,103)$, 1 at $(30,37)$ and $(10,691)$, the bilinear value 0 on the other sign; in the heavy arm $\#Q_3$ on [C]'s H^- and (L1) on the C++ cone over $\mathbb{Z}/\ell^3$ give 2 at both depth-2 cells. No finding. (c) Remark 9.17's values, all equal to R20-3's predictions. (d) ι over $\mathbb{Z}/37^3$ at $(22,37)$, $(30,37)$, $(38,37)$: `uint32` = `uint64`; at $k=1$ the field object's dimensions; at $(38,37)$ the torsion on +; heavy: at $(282,37)$ $H^1_+ = (\mathbb{Z}/37^3)^{24}$, $H^1_- = (\mathbb{Z}/37^3)^{23}$. (e) The twisted rung $(37;62,6)$ at $n=60$ (character model, $K=4$): predicted from the growth curve ($n=20$: 2.10 GB; $n=40$: 7.48 GB) at 17.5 GB, ≈ 7.3 h, under a 22 GB cap; measured 16.43 GB, 7 h 18 min; rank 386 $(194+192)$, $f^-=1$, saturated ($c^{\mathrm{sat}}=0$), $\#Q_m = 37^{\min(m,2)}$ at $m=1,2,3$, $v(B_{62,\omega^6}/62)=2$, $\operatorname{rank} H^+ = \operatorname{rank} H^- + 2$, the negative controls $Q=0$ — the first twisted rung at depth 2, [computed and compared] for Claim 9.6 (all 35 checks of the arm).

9.7 The truncated one-variable ℓ-adic L-function as a record

(New at v1.20.0-pre.5, hecke-extensions slot 20, stretch S1, the owner-Q3 roadmap's P1; placed at the end of §9, so that no statement number moves. *No new mathematics*: this subsection writes the branch series of Proposition 9.10(5) as a record with a certified precision per coefficient. Predictions `docs/implementation/slot20/s1s2/predictions.md`, committed before the first run (`c361d65c`).) Notation of §9.5; $\mu=\mu_x$ the measure of Definition 9.9 of an ordinary eigensymbol, read at level $M\ge 2$ modulo ℓ^R; $G:=\ell^{M-1}$; $\omega_r(T):=(1+T)^{\ell^r}-1$.

Definition 9.20 (The truncated ℓ-adic L-function of a branch). [specification]. For a branch $a\in \mathbb{Z}/(\ell-1)$ the *truncated ℓ-adic L-function* is the image of $\mathcal{L}_{x,a}(T)=\int \omega^a(x)(1+T)^{\ell_u(x)}\,d\mu_x$ in $\Lambda/(\ell^R,\omega_{M-1}(T))$, $\Lambda=\mathbb{Z}_\ell[\![T]\!]$, written in the basis T^t, $0\le t<G$:

$$\mathcal{L}^{(M,R)}_{x,a}(T)=\sum_{t<G} c_t T^t, \qquad c_t=\sum_{i<G} g_i \binom{i}{t} \bmod \ell^R,$$

g the branch element of Proposition 9.10(5) at level M ($(1+T)^i \leftrightarrow [u^i]$; $\sum_t c_t T^t$ is the unique representative of degree $<G$, since ω_{M-1} is monic of degree G). Its *values* are $\mathcal{L}^{(M,R)}_{x,a}(u^j-1)=\int \omega^{a-j}(x)x^j\,d\mu_x$ modulo $\ell^{\min(M,R)}$ ($0\le j\le n$: the critical moments of Proposition 9.10(4)); its μ and λ are those of Proposition 9.10(5) (read on $t<\min(G,\ell)$), and $\lambda_{\bmod \ell}$ is the first $t<G$ with c_t a unit.

Lemma 9.21 (The certified precision of the coefficients)**.** *[proved here] (elementary). Let $F=\sum_t F_t T^t\in\Lambda$ and $F=Q\,\omega_r+\mathrm{Rem}$ with $\deg \mathrm{Rem}<\ell^r$ (Weierstrass division by the distinguished polynomial ω_r; $Q\in\Lambda$). Then $\mathrm{Rem}_0=F_0$ and, for $1\le t<\ell^r$, $\mathrm{Rem}_t\equiv F_t \pmod{\ell^{\,r-\lfloor \log_\ell t\rfloor}}$. Hence the coefficients c_t of Definition 9.20 agree with the T^t-coefficients of the true series $\mathcal{L}_{x,a}$ modulo $\ell^{\mathrm{prec}(t)}$,*

$$\mathrm{prec}(0)=R, \qquad \mathrm{prec}(t)=\min\bigl(R,\ M-1-\lfloor \log_\ell t\rfloor\bigr) \quad (1\le t<G):$$

modulo $\ell^{\min(R,M-1)}$ for $t<\ell$ (Proposition 9.10(5)) and modulo ℓ for every $t<G$.

Proof. $\mathrm{Rem}_t = F_t-\sum_{1\le s\le t} Q_{t-s}\binom{\ell^r}{s}$ for $t<\ell^r$ (the coefficient $\omega_{r,s}=\binom{\ell^r}{s}$, $1\le s<\ell^r$), and $v\bigl(\binom{\ell^r}{s}\bigr)=r-v(s)\ge r-\lfloor\log_\ell s\rfloor\ge r-\lfloor\log_\ell t\rfloor$. Apply it to $F=\mathcal{L}_{x,a}$, $r=M-1$: the level-M branch element is the image of $\mathcal{L}_{x,a}$ modulo ω_{M-1} (Proposition 9.10(5)), exact modulo ℓ^R. □

Remark 9.22 (The record `lfunction/v1` and the verb `lfunction`). [specification] (slot-20 S1; code `HeckeExt/LValue/LFunctionRecord.{h,cpp}`, the console verb in `HeckeExt/RegisterHeckeExtensionVerbs.cp` [P] v3.11.1 §3.6). One record per (form, sign, truncation), one entry per branch; canonical JSON, keys in this order, no whitespace; identified by the FNV-1a-64 (`Fnv.h`) of that text. It is a new schema: no key of `hecke_ext/v5` or of any golden object moves, and the record is not golden.

key	content
`schema`	`"lfunction/v1"`
`n`, `k`, `ell`, `R`, `M`, `G`, `u`	the cell (n,ℓ), $k=n+2$, values modulo ℓ^R, level M, $G=\ell^{M-1}$, $u=1+\ell$
`sign`, `geometric`	`"minus"` (r^-, odd coordinates) or `"plus"` (r^+); the geometric sign `"odd"` / `"even"` (§9.5)
`form`	`separator` (the operator $\sum c_pT_p$), `target_root` (its root modulo ℓ), `eisenstein_congruent`, `a_p` ($p\in\{2,3,5,7,\ell\}$ and the separator's primes, modulo ℓ^R), `a_ell`, `alpha` (the unit root), `symbol` (the eigensymbol's normalisation: r^- first unit coordinate 1, $r_0^+=1$)
`truncation`	the ring $\mathbb{Z}/\ell^R[T]/(\omega_{M-1}(T))$, ω, the basis, the precision rule of Lemma 9.21 (text)
`normalisation`	`ModularSymbolMeasure::normalisation()` (Proposition 9.10(4) and its analytic form; the ambiguity $\mathcal{O}^\times$ of the eigensymbol's scaling)
`branches[]`	per branch: `a`, `vanishes` (the wrong parity), `mu`, `lambda`, `readable`, `lambda_mod_ell`, `coefficients` ($c_0,\dots,c_{G-1}$ modulo ℓ^R), `precision` ($\mathrm{prec}(0),\dots,\mathrm{prec}(G-1)$), `values_modulus` ($\ell^{\min(M,R)}$), `critical_values` ($\mathcal{L}_{x,a}^{(M,R)}(u^j-1)$, $0\le j\le n$)

The console verb `lfunction(n, ell, a, M = 2, R = 4, sign = "minus", sep = nil)` returns one branch of the record as a `Record` (the keys above, flattened, plus `hash` and an `LFUNCTION` line). The eigensymbol is picked as in Check 9.13: the root of charpoly(sep | W^-) congruent to the Eisenstein value if it occurs, else the first root modulo ℓ, simple modulo ℓ (else refused); the measure is the one `BirchModularSymbolRoute::measure()` returns on the class with this symbol (Check 9.13(M0)). Refusals (exit 3): $\ell<5$ or not prime, n odd, $M<2$, $\ell^R\ge 2^{62}$, $\ell^M>4\cdot 10^6$ balls (the read-only budget), a non-simple root, no unit r_0^+, $a_\ell\equiv 0$. The verb is tagged `read_only` (pure in-memory arithmetic; `Console/Bindings/ReadOnlyTags.cpp` and `Console/serve/read_only_verbs.tsv`).

The gate is Check 9.13(M7) (v1.20.0-pre.5): at the five default cells and both signs, on every branch of the record's parity, (1) $\sum_t c_t(u^j-1)^t$ equals the group-ring branch value at every $0\le j\le n$ (exact modulo ℓ^R); (2) at the three level-3 cells, the level-3 element folded modulo Γ_u^ℓ equals the level-2 element built from the level-2 balls, and the level-2 coefficients agree with the level-3 ones modulo $\ell^{\mathrm{prec}_2(t)}$ for $t<\ell$ (Lemma 9.21 tested against a deeper truncation; the bound is attained — the coefficients differ modulo $\ell^{\mathrm{prec}_2(t)+1}$ — at 628–631 of the 648 coefficients $1\le t<\ell$ of each sign on which $\mu=0$, so it is sharp there; on the plus side of the Eisenstein cells, where $\mu^+=d$, at none); (3) (μ,λ) recomputed from the coefficients equal Proposition 9.10(5)'s, and $\lambda_{\bmod \ell}=\lambda$ where $\mu=0$; (4) the canonical text is deterministic. 42 checks, 0 failing at the first run ([computed and compared]).

Claim 9.23 (Ramanujan's congruence as a congruence of measures: Δ at 691)**.** *[computed and compared] at $(n,\ell)=(10,691)$, $k=12$, $d=v(B_{12}/12)=1$ (slot-20 stretch S2; predictions `docs/implementation/slot20/s1s2/predictions.md` committed first (`c361d65c`); R20-1's Python oracle at this cell committed before the C++ run (`s1s2/oracle691.md`, `da0687d7`); the C++ arm of Check 9.13 (`HECKE_LVALUE_MEASURE_691=1`) at $R=2$, level $M=2$). Δ is ordinary at 691: $\tau(691)\equiv 1 \pmod{691}$, $v(\tau(691)-1)=1$ (the q-expansion, exact). On the seven-vertex model, with the separator $T_2+3T_3+7T_5$:*

1. *W^- has rank 1; the Eisenstein root occurs and is simple; the eigensymbol r^- has $a_p\equiv\tau(p) \pmod{691^2}$ for $p\in\{2,3,5,7,691\}$ (Ramanujan on the symbol side), and $v(a_p-1-p^{11})=1$ at all five; $\alpha\equiv\tau(691) \pmod{691^2}$; (L1) $r^+\equiv u\,p_0$ and $r^-\equiv s\,P^{\mathrm{Eis}}$ to depth exactly 1.*
2. ***(The congruence of measures.)** Claim 9.11(3) holds at this cell: on all 343 off-pole odd branches, $\mathcal{L}^-_{\Delta,a}(T)\equiv -s\,G_{a+1}(u(1+T)-1)\,G_{11-a}(u^{11}(1+T)^{-1}-1) \pmod{691}$ as elements of $\mathbb{Z}/691[\Gamma_u/\Gamma_u^{691}]$, with G_b from both Kubota–Leopoldt routes (which agree for all 344 even classes and give $-(1-691^{k-1})B_k/k$ at 1376/1376 weights), and at the 3773 critical moments against exact twisted Bernoulli numbers (0 failures). Negative controls: the wrong twist G_{a+3} matches 0/342, the wrong sign 0/343.*
3. ***(The pole branches** $a=11$, 689**.)** One factor is G_0 and the other is G_{12}, the Δ-class. The naive product fails modulo 691 at all 22 critical points, and the difference is a unit at all 22 — the valuation $d-1-v(r)=0$ of R20-1 §4.4's polar term (Claim 9.11(4)), predicted before the run. $\mu=\lambda=0$ on both.*
4. ***(Invariants.)** Minus side: $\mu=0$ on every odd branch, $\lambda=1$ exactly on $a=199$ and $a=501$, $\lambda=0$ elsewhere — predicted from (2): these are the two off-pole branches carrying the other irregular class, $G_{200}\cdot G_{502}$ and $G_{502}\cdot G_{200}$, with $\lambda(G_{200})=1$ ([U, `thm:u-691classes`]); $199+501\equiv n \pmod{690}$. Plus side: $\mu^+=1$ on every even branch (Remark 9.12(ii)); the reduced plus measure has $\lambda=1$ exactly on $\{22,350,678\}$, a functional-equation orbit $a\leftrightarrow n-a$*

(350 self-paired) — recorded, not predicted (Remark 9.12(iii)).
The expected literature is the same as for Claim 9.11 (Mazur; Greenberg–Vatsal; held, not yet read; v1.21.0-pre.1). Every value above equals the Python oracle's where the oracle computed it.

9.8 Coefficient rings, the two-variable family, the Eisenstein measure, wild characters and Iwasawa invariants

(New at v1.21.0-pre.1, hecke-extensions slot 21, the ℓ-adic track P3 and P9; placed at the end of §9, so that no statement number moves. *No statement of §§9.5–9.7 changes*: P3 is a change of coefficients of Definition 9.9, P9 assembles the measures of several weights. Specs, predictions committed before the code: `docs/implementation/slot21/P3_spec.md` (`bfd63a34`), `P9_spec.md` (`7ff1cb29`); neither is refereed.)

Remark 9.24 (The measure over the Hecke order of the eigen-system). (Slot-21 P3.) Let $\chi = \mathrm{charpoly}(T_2 \mid W^-)$ and F its Hensel factor over $\mathbb{Z}/\ell^R$ with $F \equiv \bar{g}^e \pmod{\ell}$, $\bar{g}$ irreducible, $\gcd(\bar{g}, \bar{\chi}/\bar{g}^e) = 1$; $\mathcal{O} = (\mathbb{Z}/\ell^R)[x]/(F)$ — for $e = 1$ the unramified $W(\mathbb{F}_{\ell^f})/\ell^R$, $f = \deg \bar{g}$; for $f = 1$, $e \geq 2$ and $F(X+\lambda)$ Eisenstein the totally ramified $\mathbb{Z}_\ell[x]/(F)$. The eigen-symbol P_θ of $\theta = x$ on $W^\pm \otimes \mathcal{O}$ is found by Gauss–Jordan with unit pivots (applicable iff $T_2 - \bar{\theta}$ has rank $r-1$ residually: *residual multiplicity one*, else refused by name), normalised at its first unit coordinate; $a_\ell \in \mathcal{O}^\times$ is required (else refused, *not ordinary*); $\alpha \in \mathcal{O}^\times$ the unit root; then $\mu_{P_\theta} = \sum_j x^j \mu_{P_j}^{(\alpha)}$ by Remark 9.2 (M). Proposition 9.10(1)–(6) hold over $\mathcal{O}$ ([proved here]; the proof is coefficient-free). *Computed* ([computed and compared]: a Python oracle sharing no $\mathcal{O}$-arithmetic with the C++, its values frozen before the C++ ran; q-expansion characteristic polynomials of T_2, T_3 on S_k as a third, symbol-free route) at $(22,23)$, $(22,29)$, $(26,31)$, $(26,37)$, $(28,29)$ ($W(\mathbb{F}_{\ell^2})$), $(34,37)$ ($W(\mathbb{F}_{37^3})$), $(26,131)$, $(26,139)$ (totally ramified, $e=2$; S_{28}, whose Hecke field has discriminant divisible by 131 and 139), with the degree-one controls $(10,37)$, $(16,37)$ bit-identical to Check 9.13(M2)'s FNVs: the identities of Proposition 9.10 with 0 failures; Galois equivariance $\mu_{\sigma\theta} = \sigma \circ \mu_\theta$ (the eigen-symbol of $\sigma\theta$ recomputed, not conjugated); at the ramified cells a second construction through WP-LV3's $\mathfrak{m}$-block reader in $\mathbb{Z}_\ell[\pi]/(\pi^e - \ell u)$, equal to the first through $x \mapsto \lambda_0 + \rho$ for both roots ρ; $(30,29)$ refused, not ordinary. *Hida control modulo* $\mathfrak{m}$ ([computed and compared] at one residually non-split component; the oracle found it before the C++): at $\ell = 29$, $\bar{g} = X^2 + 22X + 15$, the weights $k = 24, 52, 80, 108$ of one class modulo $\ell - 1$ (ordinary rank 2, one Galois orbit at each), the level-2 unit-ball measures of $k = 52, 80, 108$ are proportional modulo $\mathfrak{m}$ to that of $k = 24$ with scalars $19, 4, 21$ (geometric odd) and $1, 1, 1$ (geometric even) in $\mathbb{F}_{29}^\times$, 0/812 balls mismatching each; the residual eigen-system is the same at all four weights. This is the first numerical instance in this project of the Λ-adic coherence of measures at a residually non-split component; its natural statement is [U, `pr:u-msym`] (U16), which stays [conjectural]. *Scope* (finding F-P3-1, P3's survey, [computed and compared] as a census, one route): at level one, $\ell \leq 59$ and $n \leq 130$–150, no ordinary eigen-system has a repeated residual T_2-eigenvalue, so the ordinary ramified cells come from residually distinct, characteristic-zero-ramified Hecke fields (131, 139 at $k = 28$; 144169 at $k = 24$, the env arm); WP-LV3's ramified $(258,37)$ is *not* ordinary (T_{37} nilpotent modulo 37 on W^-; $k = 260 \equiv 8 \pmod{36}$), so it carries no measure.

Check 9.25 (The gate of P3: `HeckeLValueCoefficientRingValidation`). [specification] (the checks); expected values P3's Python oracle's, frozen in `CoefficientRingFrozenP3.inc` before the C++. Tier `standard`; arms (R0) setup and the q-expansion charpolys, (R1) eigen-symbols, (R2) Proposition 9.10 over $\mathcal{O}$, (R3) frozen measures, (R4) Galois, (R5) the ramified route B, (R6) Hida modulo $\mathfrak{m}$, (R7) degree-one controls and base change into $(\mathbb{Z}/37^4)[x]/(x^2-2)$, (R8) negative controls and the refusals by name. First run: 603 checks, 0 failing, 1.2 s, 24 MB; env arm `HECKE_LVALUE_COEFFRING_HEAVY=1` (adds $(22, 144169)$ at its ramified prime and $(34,37)$ at level 3): 674 checks, 0 failing, 138 s, 74 MB. No golden line, record key or digest. *P2/P4* (v1.21.0-pre.3; through pre.2 "pending"): the Eisenstein measure exact in Λ (P2) is Proposition 9.27; wild characters on the L-side (P4) are Remark 9.29, built over $\mathbb{Z}_\ell[\zeta_{\ell^{m-1}}]$ only — the coefficient ring $\mathcal{O} \otimes \mathbb{Z}_\ell[\zeta_{\ell^{m-1}}]$ of a non-trivial Hecke order $\mathcal{O}$ is *not* built (P4 adds no tensor-product constructor; `P4_spec.md` §6).

Remark 9.26 (The two-variable family by interpolation (slot-21 P9)). (Slot-21 P9, `HeckeLambdaTwoVarValidation` `standard`; inputs P7's Λ-adic q-expansions and P8's normalised masses, [U, `pr:u-msym`],

[U, prop:u-msymnorm].) For a class (ℓ, a) and an étale cuspidal component, the normalised unit-ball masses $\nu_B(x_k)$ of $N = 5$ consecutive weights are interpolated in X ([U, thm:u-interp]), and the branch elements of Proposition 9.10(5) and the coefficients of Definition 9.20 are formed in $\Lambda_{K,m}(x_c)$: a two-variable $L_a(X,T)$ modulo $(\ell^K, (X-x_c)^m, \omega_{M-1}(T))$. It is the specialisation family of one $\mathbf{h}(a)_{\mathfrak{m}}$-valued measure *only if* [U, pr:u-msym] (U16) holds; nothing here proves (U16), and the gates resting on it are findings. Assembly and the agreement with P1's own pipeline (§9.7) at the computed weights are [proved here] statements checked by the code; at the cells $(11,2)$, $(13,0)$, $(23,2)$, $(37,32)$ the first run held every gate and fired no finding (71 checks, 41 finding gates; [computed and compared] as a run, one C++ route, against Python predictions committed first for 11a):

1. **(Numerical control.)** The interpolants predict every branch coordinate at three further weights modulo ℓ^5, and P1's `lfunction/v1` at the further weights equals the predicted L_a up to P1's normaliser modulo ℓ^5 at $(13,0)$ and $(37,32)$ — evidence for (U16)(b), [conjectural].
2. **(The Λ-adic Eisenstein congruence of measures.)** At $(37,32)$, on the component through $\mathfrak{m}_{\mathrm{Eis}}$, the cross products of $L_a(X,T)$ with $G_{a+1}(u(1+T)-1)\,G_{k-1-a}(u^{k-1}(1+T)^{-1}-1)$ (Claim 9.11(3) at each weight) vanish at $X = \alpha_{32}$ — the zero of G_{32}, $\alpha_{32} \equiv 281570 \pmod{37^4}$ ([U, pr:u-lambda], (U13)), recomputed from exact Bernoulli numbers — modulo 37^3 on all 592 off-pole odd slots (the full Kubota–Leopoldt precision at level $M_E = 3$), against 37^1 at every integer weight with $L_k = 1$ and 37^2 at $k = 284$; at $\alpha_{32} + 37$ only modulo 37. In the env arm `HECKE_LAMBDA_TWOVAR_ALL=1`: 21 904 slots at level 3, the same 37^3, and at $(103,24)$ modulo 103^2 at $X = \alpha_{24} \equiv 2266 \pmod{103^2}$ on 5047 slots. The expected statement, a congruence of Λ-adic measures modulo $(X - \alpha_a)$ (Mazur–Kitagawa, recalled; the interpolation of Claim 9.11 in k), is [conjectural]; the run is evidence for it and for (U16).
3. **(Weight two.)** At $(11,2)$ (the Δ family) the values predicted at x_2 from $k = 12, \ldots, 52$ equal 11a's own normalised MTT masses (Manin symbols on $\Gamma_0(11)$, no level-one routine) on every unit ball to level 3, both signs, modulo 11^5; the trivial zero $\nu_{\mathbb{Z}_{11}^\times}(x_2) \equiv 0$ and Greenberg–Stevens' derivative identity hold modulo 11^5 ([conjectural]: they rest on (U16) at $k = 2$).
4. **(The $\mathcal{L}$-invariant, three ways.)** $\mathcal{L}_{\mathrm{GS}} := -2\,d\alpha/dk(2)$ from the level-one family alone $\equiv \mathcal{L}_{\mathrm{Tate}}(11a) \pmod{11^8}$ ($\mathcal{L}_{\mathrm{Tate}} \equiv 14882352232 \pmod{11^{10}}$, committed before the C++), and $\mathcal{L}_{\mathrm{MTT}}$ from 11a's plus measure $\equiv \mathcal{L}_{\mathrm{Tate}} \pmod{11^5}$. The theorem $\mathcal{L}_p(E) = -2a_p'(2)$ and the MTT conjecture for 11a are Greenberg–Stevens' ([standard, reference], recalled: their pp. 422–423 not read on the page; `docs/implementation/slot21/gs_reading_notes.md`); the run reproduces them to the stated precision ([computed and compared] as a run).

P1's two limits (a remark for §9.7, found by P9's oracle arm; no statement of §9.7 changes): P1's pipeline (i) refuses a cell where $W(\mathbb{Z}/\ell^R)$ is not free (H^2_c with ℓ-torsion; at $(11,2)$ every weight $k \geq 22$), whereas P8 reads through $W(\mathbb{Z}/\ell^{R+2})$; and (ii) at a class through $\mathfrak{m}_{\mathrm{Eis}}$ takes the Eisenstein root for every separator, so it cannot select another component of that class. Both are recorded per side by P9 (`P1_COVERAGE`), never a failure. *P2/P4* (v1.21.0-pre.3; through pre.2 "pending"): P2 gives the Eisenstein side of item 2 exactly at each *single* weight, pole branches included (Proposition 9.27); the comparison of two Λ-adic objects — $L_a(X,T)$ against an Eisenstein measure interpolated in X (P2's `EisDistribution` as P9's Eisenstein side, `P2_spec.md` §6) — is *not* built, so item 2 is unchanged. P4's wild characters (Remark 9.29) are at single weights as well; none is evaluated on $L_a(X,T)$.

The Eisenstein measure exactly, and wild characters (v1.21.0-pre.3). (New at v1.21.0-pre.3, hecke-extensions slot 21, the ℓ-adic track P2 and P4; appended to §9.8, so that no statement number moves. *No statement of §§9.5–9.7 changes*; Claim 9.11(4) gains its explanation as added text. Specs with predictions committed before the gated runs: `docs/implementation/slot21/P2_spec.md` (v0.1 `2c82a469`, v0.2 with the first runs; code `74c7e86c`) and `P4_spec.md` (v0.1 `8f9d4cf4`; code `f2c07b20`; v0.2 `b7b0e2c3`); Python oracles `p2/run_p2.py` (output of record `p2/out_run_p2.txt`) and `p4/cal_*.py` (`p4/out_cal.txt`), in-house, no CAS; neither spec is refereed. Tags as in the specs: their COMP is [computed and compared], their STD-recalled is [standard, reference] recalled.) Notation of §9.5; in addition $(\ell - 1) \nmid k$ (so the two pole branches $a \equiv -1$, $a \equiv k-1$ differ and $d := v(B_k/k) \geq 0$), $E_r(x + \ell^m\mathbb{Z}_\ell) := \ell^{m(r-1)}B_r(x/\ell^m)/r$ the Bernoulli distributions (Washington §12.1, [standard, reference] recalled), $\bar{B}_1(t) := \{t\} - \frac{1}{2}$, $\iota(x) := x^{-1}$, $x_r := u^r - 1$.

Proposition 9.27 (The Eisenstein measure (slot-21 P2))**.** *Tags per item. Let $\ell \geq 5$, $n = k-2$ even, $(\ell-1) \nmid k$.*

1. ***(The symbol side is a Dedekind–Rademacher sum.)*** *Let $\tilde{P} := n!\, p_k^-$, Zagier's Laurent polynomial $p_k^-(X) = \sum_{-1\leq m\leq k-1,\ m\ odd} \beta_m \beta_{k-2-m} X^m$, $\beta_m = B_{m+1}/(m+1)!$ (read on the page: Zagier, Invent. Math. 104 (1991) 449–465, Prop. (i), eq. (11), p. 453, $p_k^- \in \hat{W}_k^-$; read on the page for P2, `P2_spec.md` §1.2, and again by the referee F4m; the phrase "Eisenstein period function" is the repository's, not Zagier's, which is all that `missing_literature.md` LIT-05 leaves open; the same polynomial is Kohnen–Zagier §1.3, p. 211; Claim 9.11(1) carries the tag since v1.21.0-pre.4). Its interior is P^{Eis}, its outer coefficients are $B_k/(k(k-1))$ at u_{-1}, u_{n+1}, and $\tilde{P}$ is annihilated by $1+S$ and $1+U+U^2$ as a Laurent polynomial in the convention of §9.5, the interior alone is not ([computed and compared]). Feed $\tilde{P}$ into the Manin-path formula of Definition 9.9 and drop the one singular monomial ($u_{-1} = x^{n+1}/y$ at the first Manin matrix, bottom row $(1,0)$: "the boundary"). Then,* as rational numbers, *for every $m \geq 1$ and $a \in [0, \ell^m)$,*

$$\tilde{c}(a,m) = -D(a,m) + \frac{B_k}{k}\cdot\frac{a}{\ell^m}, \qquad D(a,m) := \sum_{y \bmod \ell^m} \bar{B}_1\Big(\frac{ay}{\ell^m}\Big) E_{k-1}(y + \ell^m\mathbb{Z}_\ell).$$

   [computed and compared] (exact, at every residue; Check 9.28(E1)). The term $(B_k/k)a/\ell^m$ is the period around ∞ (not T-invariant). That the Eisenstein modular symbol has Dedekind–Rademacher form is classical (Stevens, Arithmetic on modular curves, *Ch. 2; [standard, reference] recalled, not read; LIT-07 not held); no proof of the identity is claimed here.*
2. ***(The measure.)*** *$\mu^{\mathrm{Eis}}(a + \ell^m\mathbb{Z}_\ell) := -D(a,m) + \ell^{k-2} D(a, m-1)$ (the formula (M) with $\alpha = 1$, the unit root of $X^2 - (1+\ell^{k-1})X + \ell^{k-1}$) is a distribution over $\mathbb{Q}$ and odd ([computed and compared], exact); on $\mathbb{Z}_\ell^\times$ it is a convolution of two Bernoulli distributions,*

$$\mu^{\mathrm{Eis}}|_{\mathbb{Z}_\ell^\times} = -E_1|_{\mathbb{Z}_\ell^\times} \star \iota_* E_{k-1}|_{\mathbb{Z}_\ell^\times}, \qquad \text{at level } M:\ \mu^{\mathrm{Eis}}(a) = -\sum_{x\in(\mathbb{Z}/\ell^M)^\times} E_1(x)\, E_{k-1}(xa^{-1})$$

   ([computed and compared]). Hence, for χ of conductor ℓ and every level M, $\sum_a \chi(a)\mu^{\mathrm{Eis}}(a + \ell^M\mathbb{Z}_\ell) = -B_{1,\chi}B_{k-1,\bar{\chi}}/(k-1)$ exactly, no Riemann sum ([proved here] from the convolution form and the distribution property of the generalised Bernoulli numbers, [standard, reference]; [computed and compared] at every odd $\chi = \omega^i$, (E4)).
3. ***(The branch series.)*** *The branch map is a ring homomorphism, so* $\mathrm{branch}_a(\mu^{\mathrm{Eis}}) = -\mathrm{branch}_a(E_1)\cdot\mathrm{branch}_a(\iota_* E_{k-1})$, *and on every odd branch*

$$\mathcal{L}_a^{\mathrm{Eis}}(T) = -G_{a+1}\big(u(1+T)-1\big)\cdot G_{k-1-a}\big(u^{k-1}(1+T)^{-1}-1\big) \qquad \text{in } \Lambda\otimes\mathbb{Q},$$

   exactly, not modulo ℓ^d ([proved here] given Washington Thm. 12.2, p. 239, read on the page by the referee F4m, v1.21.0-pre.4: $\int \chi\omega^{-1}(a)\langle a\rangle^s dE_{1,c} = -(1-\chi(c)\langle c\rangle^{s+1})\, L_\ell(-s,\chi)$; through pre.3 "[standard, reference] recalled, not re-read for P2", with the leg mis-recalled as "$x\,\zeta_\ell = E_1$, $dE_{k,c} = k\,x^{k-1}dE_{1,c}$" (F4m, M1): in this text's normalisation $E_r = \ell^{m(r-1)}B_r/r$ the correct forms are $E_1 = -x\,\zeta_\ell$ on $\mathbb{Z}_\ell^\times$, ζ_ℓ the pseudo-measure with $\int \omega^b(x)\langle x\rangle^s d\zeta_\ell = G_b(u^s-1)$, and $dE_r = x^{r-1}dE_1$ (Washington's $E_k = i^{k-1}B_k(\{a/i\})$, p. 232, carries no $1/k$, whence his $k\,x^{k-1}$); G_b of [U, `lem:u-twistvalues`], read as in Claim 9.11(3)). Hence each factor carries the sign -1: $\mathrm{branch}_a(E_1)(T) = -G_{a+1}(u(1+T)-1)$ *and* $\mathrm{branch}_a(\iota_* E_{k-1})(T) = -G_{k-1-a}(u^{k-1}(1+T)^{-1}-1)$ *([computed and compared] by F4m at every critical value of the regular odd branches at $(14,7)$, $(30,37)$, $(66,37)$, $(10,691)$, $(282,37)$; `P2_spec.md` through v0.2 and [U] 0.5.0-pre.5 stated them without the sign; the product is unchanged). [computed and compared] at every regular odd branch against both Kubota–Leopoldt routes of Check 9.13(M4)(a), which are independent of this construction; the even branches vanish.*
4. ***(The pole branches: an explicit polar term.)*** *On $a \equiv -1$ and $a \equiv k-1$ one factor is G_0:*

$$\mathcal{L}_{-1}^{\mathrm{Eis}}(T) = \frac{\mathrm{Res}}{u(1+T)-1} + (\Lambda), \qquad \mathcal{L}_{k-1}^{\mathrm{Eis}}(T) = \frac{\mathrm{Res}}{u^{k-1}(1+T)^{-1}-1} + (\Lambda),$$
$$\mathrm{Res} = (1-\ell)\,\frac{\log_\ell u}{\ell}\,(1-\ell^{k-1})\,\frac{B_k}{k},$$

the same residue on both, $\mathrm{Res} = -\rho_\infty G_k(x_k)$ *with* $\rho_\infty := (1-\ell)\log_\ell(u)/\ell$ *(ζ_ℓ's residue in the variable* $u^s - 1$*, [standard, reference] recalled) and* $G_k(x_k) = -(1-\ell^{k-1})B_k/k$ *([U, `lem:u-twistvalues`]); so* $v(\mathrm{Res}) = d$*. The shape is [proved here] for the branch* -1 *up to the constant (elementary, from* E_1*'s ball values); the constant was [computed and compared] (calibrated at* $(12,5)$*, level* 4*, and* $(14,7)$*, level* 3*; then predicted at the other cells) and is [proved here] given (3) and* ζ_ℓ*'s residue (F4m, v1.21.0-pre.4):* $TG_0(T) \in \Lambda$ *with* $(TG_0)(0) = \rho_\infty$*, so* $G_0(T) = \rho_\infty/T + (\Lambda)$*; with* $X := u(1+T) - 1$ *and* $Y := u^{k-1}(1+T)^{-1} - 1$ *one has* $Y - x_k = -u^{k-1}(1+T)^{-1}X$*, so* $G_k(Y) \equiv G_k(x_k) \pmod{X\Lambda}$ *and* $-G_0(X)\,G_k(Y) = -\rho_\infty G_k(x_k)/X + (\Lambda)$*; on the branch* $k-1$ *the same with the two factors exchanged (the pole at* $Y = 0$*, where* $X = x_k$*). At level* M *the polar part is* $(1-\ell)(1-\ell^{k-1})(B_k/k)\ell^{-M}N_u$*,* $N_u = \sum_{i<\ell^{M-1}} u^i[\gamma^i]$ *(branch* -1*; on the branch* $k-1$*,* $\sum_i u^{-(k-1)i}[\gamma^i]$ *with an extra* $1/(k-1)$*); [computed and compared]: the branch element has valuation exactly* $d - M$ *when* $d < M$*, and minus the polar part it is integral.*

5. ***(WP-MEAS's polynomial symbol.)*** *Let* $\mu^{\mathrm{poly}} := \mu^{(1)}_{P^{\mathrm{Eis}}}$ *(Proposition 9.10(6)'s notation) with branch series* $\mathcal{L}_a^{\mathrm{poly}}$*. At every cell with* $d \geq 1$ *of Check 9.28:* $v(\mathcal{L}_a^{\mathrm{poly}} - \mathcal{L}_a^{\mathrm{Eis}}) = d$ *exactly off the pole branches, and* $v\big(\mathcal{L}_a^{\mathrm{poly}} - (\mathcal{L}_a^{\mathrm{Eis}} - \mathrm{polar}_a)\big) = d$ *exactly on them ([computed and compared], (E7)).*
6. ***(The cusp form's pole branches.)*** *Given Claim 9.11(2) and its reduction (R20-1 Prop. 3.1, [proved here] given its inputs), (5) gives, on every odd branch, pole branches included,*

$$\mathcal{L}^-_{f,a}(T) \equiv s\left(\mathcal{L}_a^{\mathrm{Eis}}(T) - \mathrm{polar}_a(T)\right) \pmod{\ell^d},$$
$$\mathcal{L}^-_{f,a}(u^j - 1) \equiv -s\left(F_1F_2 + \frac{\mathrm{Res}}{u^r - 1}\right) \pmod{\ell^d},$$

$r = j+1$ (branch -1), $r = k-1-j$ (branch $k-1$), F_1F_2 the product of the two G-factors at $T = u^j - 1$. [computed and compared] with the eigensymbol r^- at $(30,37)$ (level 3), $(282,37)$ (level 3), $(226,103)$ (level 2), $R = 4$: 0/62, 0/566, 0/454 critical points (E8); with μ^{poly} in place of f (the Python oracle) also at $(66,37)$, $(42,59)$, $(56,67)$, $(22,103)$ and $(10,691)$ (134, 86, 114, 46, 22 points, 0 failures; at $(66,37)$, $k = 68 > \ell$, the two points with $r = 37$ have a polar value of valuation -1 and the identity still holds modulo 37: F4m). The polar value has valuation exactly $d - 1 - v(r)$ ([proved here] from $v(\mathrm{Res}) = d$ and $v(x_r) = 1 + v(r)$), so the naive product $-s\,F_1F_2$ agrees modulo $\ell^{d-1-v(r)}$ and no further: Claim 9.11(4)'s exceptions modulo ℓ^{d-1} are the points with $\ell \mid r$, the $14+4$ located points of $(282,37)$, $(226,103)$ ([computed and compared]: the located sets of (E8) equal the claim's; the control $\mathrm{Res}\cdot(1+\ell)$ fails exactly at them), and at a depth-one cell with $r < \ell$ throughout — $(30,37)$, $(10,691)$ — the naive product fails at every pole point by a unit (Claim 9.23(3): predicted there by R20-1, now derived). At $(10,691)$ the congruence is moreover checked on Δ's own eigensymbol at level 2 *($R = 2$): $\mathcal{L}^-_\Delta \equiv s(\mathcal{L}^{\mathrm{Eis}} - \mathrm{polar}) \pmod{691}$ holds at all 3795 critical points of the 345 odd branches (3773 regular, 22 pole; 0 failing, located set $\emptyset$; negative controls: no polar term 22/22 failing, $-\mathrm{Res}$ 22/22, $\mathrm{Res}\cdot(1+\ell)$ 0/22 — invisible at critical points), and as elements of $\mathbb{Z}/691[C_{691}]$ on every odd branch, pole branches included: $g_a^\Delta \equiv s(g_a^{\mathrm{Eis}} - \mathrm{polar}_a)$ coefficientwise, 0/343 regular and 0/2 pole branches ($s = 286647$ as in S2; at level 2 $\min v(g^{\mathrm{Eis}}) = -1$ on both pole branches, 11 and 689, the polar term making the difference integral; controls: no polar term 2/2, $-\mathrm{Res}$ 2/2, $\mathrm{Res}\cdot(1+\ell)$ 2/2 — here the control fails, though it passes at every critical point — and $s+1$ 345/345; the C++ fingerprint of g^Δ on the pole branches, 11: $(443652, 194679)$, 689: $(443652, 11103)$, equals the oracle's) ([computed and compared], `HeckeLValueEisensteinMeasureValidation` arms E8L2 and E9L2; independent Python oracle `docs/implementation/slot21/p2/p2x_oracle.py`, output `out_p2x_oracle.txt`; predictions committed first, `P2_spec.md` v0.4 §10). At $d = 1$ the critical values modulo ℓ are level-independent ($u \equiv 1$); the level-2 content is the group-ring identity.*

Not claimed *(`P2_spec.md` §1.5): a proof of (1); the plus side (p_0, the trivial zero; Remark 9.12(iii) stays an observation); $(\ell-1) \mid k$ (one pole branch, a double pole). Characters of ℓ-power order: Remark 9.29(6), (7).*

Check 9.28 (The gate of P2: `HeckeLValueEisensteinMeasureValidation`). [specification] (the checks); the expected values are P2's predictions and its Python oracle's, committed first, never this code's output. *Registration*: `Registry/Slot21/P2_eismeas.cpp`, `cmake/slot21/P2_eismeas.cmake`; sources `HeckeExt/LValue/EisensteinMeasure.{h,cpp}`, `HeckeLValueEisensteinMeasure Validation.{h,cpp}`; WP-MEAS's public API only, no existing L-value file edited. Tier `standard`, read-only. Exact rational identities modulo $q_1 = 2^{61} - 1$ and $q_2 = 2^{62} - 57$ (one prime at $\ell \geq 100$); ℓ-adic statements in a scaled representation $\ell^E x \bmod \ell^{R+E}$, $E = 2M + 1 + v(k-1)$, $\ell^{R+E} < 2^{62}$. *Cells*: Eisenstein $(12, 5)$ level 4, $(14, 7)$ level 3, $(16, 11)$, $(20, 13)$, $(10, 37)$ ($d = 0$), $(30, 37)$, $(66, 37)$, $(42, 59)$, $(56, 67)$, $(22, 103)$ ($d = 1$), $(282, 37)$, $(226, 103)$ ($d = 2$) at level 2, $(10, 691)$ at level 1 (E1–E7, E9) and, for Δ's own eigensymbol, at level 2 (E8L2, E9L2); cusp forms (WP-MEAS's) $(30, 37)$, $(282, 37)$ at level 3, $(226, 103)$ at level 2, $R = 4$. Arms: (E0) Zagier's relations at 6 random points modulo q_1, the interior alone fails; (E1) item (1) at every residue $a < \ell^m$, $m \leq M$; (E2) distribution and parity, negatives: the cusp term kept, μ^{poly} over $\mathbb{Q}$; (E3) the convolution form; (E4) the twisted masses; (E5) item (3) by both Kubota–Leopoldt routes ($E_{1,c}$ only at $\ell > 103$), even branches 0; (E6) item (4); (E7) item (5); (E8) item (6) and the located sets; (E9) negatives — no polar term, $-$Res (fail at every point), Res $\cdot (1 + \ell)$ (fails exactly where $\ell \mid r$), the wrong twist G_{a+3} (at level ≥ 2 only: at level 1 a comparison is one residue and coincides with probability $1/\ell$; first run 3/342 at $(10, 691)$). Lines E8L2 (7 checks: the critical-point identity at 3795 points, the group-ring identity on the 345 odd branches, the C++ fingerprints on the pole branches) and E9L2 (7 checks: the four negative controls at critical points and in the group ring). Lines `EISMEAS`, `EISMEAS_POLE`, `P2_ARMS checks=<c> failing=<f>`; `EFFECTIVE ... arm=DEFAULT cells=13`. *First run* ([computed and compared]): 225 checks, 0 failing, 5.3 s, 20 MB (E1 0/2812 at 37, 0/10712 at 103, 0/691 at 691; E6 $d - M$ exactly at $d < M$, 0 at $d = M = 2$; E7 d, d at the eight cells with $d \geq 1$; E8 0/62, 0/566, 0/454; no polar term and $-$Res fail at every point, Res $\cdot (1 + \ell)$ at 0, 14, 4). E5 is vacuous at $\ell = 5$ (both odd branches are pole branches). Read-only env arm `HECKE_LVALUE_EISMEAS_HEAVY=1` (`arm=HEAVY cells=15`): adds $(30, 37)$ at level 3 (E1 0/104118, E3 0/49284, E6 $(-2, 0)$) and $(20, 131)$ at level 2; 259 checks, 0 failing, 11.9 s, 20 MB. *With E8L2 / E9L2* (P2x, [computed and compared]): default arm 239 checks, 0 failing, 6.2 s; heavy arm 273 checks, 12.6 s (E8L2 0.8 s on 16 threads, 3.9 s single; the Python oracle 78 s, 0.22 GB). `GOLDEN_STRICT=1`: A 11827/0, B 6231/0. No golden line, record key or digest.

Remark 9.29 (Wild characters on the L-side (slot-21 P4)). Let $\ell \geq 5$, $m \geq 2$, $F = \ell^m$, ζ of order ℓ^{m-1}, $\mathcal{O}_m = \mathbb{Z}_\ell[\zeta]$, $e = \varphi(\ell^{m-1})$, $\pi = 1 - \zeta$; for $a \in (\mathbb{Z}/F)^\times$, $\langle a \rangle = u^{L(a)}$, $\psi_t(a) := \zeta^{tL(a)}$ ($\ell \nmid t$; conductor ℓ^m), $\chi_{i,t} := \omega^i \psi_t$ — every primitive character of conductor ℓ^m, once ζ is fixed. Tags per item; everything is at single weights, over $\mathcal{O}_m$ (not over $\mathcal{O} \otimes \mathcal{O}_m$, Check 9.25).

1. **(The analytic constant.)** With Remark 9.2 (B)'s normalisation the quotient $\Omega_{\chi,j} := L(f, \chi, j+1)\, j! \binom{n}{j} M^{j+1} / (\chi(-1)(-2\pi i)^{j+1} \tau(\chi) \lambda_{\chi,j})$ is, for every primitive χ of conductor $M = 25, 49, 121$ at $k = 12$ (Δ) and $M = 25, 49$ at $k = 16$, and every j with $\lambda_{\chi,j} \neq 0$, one constant per sign, equal to the conductor-1 $\Omega_f^{\pm}$ to relative 10^{-59}–10^{-67} (16, 36, 100, 16, 36 characters; 176, 396, 1100, 240, 540 values; env arm: $M = 125, 169$ at $k = 12$) — [computed and compared] (L by the two-sided incomplete-Gamma series, 256-bit `nf::MpReal`; $P^{\pm}$ and $\lambda_{\chi,j}$ exactly over $\mathbb{Q}$); $\tau(\bar{\chi})$ in place of $\tau(\chi)$ fails (relative deviation ≈ 2). No ordinarity is needed: the run is at primes where f is not ordinary as well. The root number of $f_\chi \in S_k(M^2, \chi^2)$, in $\Lambda(f_\chi, s) = w\, i^k N^{k/2-s} \Lambda(f_{\bar{\chi}}, k - s)$, $N = M^2$, is $w = \tau(\chi)^2 / M$ ([proved here], elementary, F4m, v1.21.0-pre.4: $f_\chi = \tau(\bar{\chi})^{-1} \sum_{v \in (\mathbb{Z}/M)^\times} \bar{\chi}(v)\, f|_k \left(\begin{smallmatrix} 1 & v/M \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & v/M \\ 0 & 1 \end{smallmatrix}\right) W_{M^2} = \gamma_v \left(\begin{smallmatrix} M & u \\ 0 & M \end{smallmatrix}\right)$ with $\gamma_v = \left(\begin{smallmatrix} v & -x \\ M & -u \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, $u \equiv -v^{-1} \pmod{M}$, $W_{M^2} = \left(\begin{smallmatrix} 0 & -1 \\ M^2 & 0 \end{smallmatrix}\right)$; as $f|_k \gamma_v = f$ and $v \mapsto u$ permutes $(\mathbb{Z}/M)^\times$ with $\bar{\chi}(v) = \chi(-1)\chi(u)$, $f_\chi|_k W_{M^2} = \chi(-1)\, \tau(\chi) \tau(\bar{\chi})^{-1} f_{\bar{\chi}} = (\tau(\chi)^2 / M)\, f_{\bar{\chi}}$ by $\tau(\bar{\chi}) = \chi(-1)\overline{\tau(\chi)}$ and $|\tau(\chi)|^2 = M$, and the Mellin transform turns $f_\chi|_k W_{M^2} = w\, f_{\bar{\chi}}$ into the displayed functional equation; [computed and compared] to 10^{-59}; P4's v0.1 predicted $\chi(-1)\tau(\chi)^2 / M$ from a sign-blind calibration, and the C++ gate refuted it — `P4_spec.md` §8, deviation 1; the constant of Proposition 9.10 does not use w).
2. **(Proposition 9.10(4) at wild χ.)** $(-1)^j \binom{n}{j} \int \bar{\chi}(x) x^j d\mu_f = \alpha^{-m} \lambda_{\chi,j}$ in $\mathcal{O}_m / \ell^{\min(M,R)}$ (level-M Riemann sum) — a check of the code path of a [proved here] statement: [computed and compared] at $(10, 11)$, $(10, 13)$, $(10, 37)$ and $(30, 37)$ ($R = 4$, $M = 3$, $m = 2$), both signs, every i,

$t \in \{1,2\}$, every j (0/2232 per sign at $(30,37)$; env arm: $(10,11)$ at $m=3$, $e=110$); $\alpha^{-(m-1)}$ in place of α^{-m} matches nowhere.

3. **(A functional-equation symmetry.)** $\lambda_{\chi,j}(P) = -\varepsilon\,\lambda_{\bar\chi,n-j}(P)$ modulo ℓ^R, ε the geometric parity of P ([computed and compared] at the cells of (2) and, in the calibration, at $(14,7)$, $(16,7)$, $(20,11)$). [proved here] (F4m, v1.21.0-pre.4; through pre.3 "looks provable from ρ_n of $\left(\begin{smallmatrix}0 & -1\\ F & 0\end{smallmatrix}\right)$; no proof is given"): for $u \equiv -a^{-1} \pmod F$ and $\gamma = \left(\begin{smallmatrix}a & -x\\ F & -u\end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ one has $\{a/F,\infty\} = \gamma\{\infty, u/F\}$, so $\Phi(\{a/F,\infty\}) = -\rho_n(\gamma)\Phi(\{u/F,\infty\})$, and $\gamma_{a,m}\gamma = S\,\gamma_{u,m}$ with $S = \left(\begin{smallmatrix}0 & -1\\ 1 & 0\end{smallmatrix}\right)$, while $\langle \rho_n(S)v, u^\vee_{n-j}\rangle = (-1)^j\langle v, u^\vee_j\rangle$ ($u_t \mapsto (-1)^t u_{n-t}$); summing against $\bar\chi(a) = \chi(-1)\chi(u)$ gives $\lambda_{\chi,j}(P) = -\chi(-1)(-1)^j\lambda_{\bar\chi,n-j}(P)$ for *every* $P \in W$, which is $-\varepsilon\lambda_{\bar\chi,n-j}(P)$ wherever either side is non-zero ($\chi(-1)(-1)^j = \varepsilon$ there); checked by the referee on a generic $P \in W$: 0/176 at $(10,5)$, 0/1100 at $(10,11)$, $m=2$. Consequence ([computed and compared]): the control "χ for $\bar\chi$" coincides with the true value exactly at $j = n/2$, for the χ with $\chi(-1)(-1)^{n/2} = \varepsilon$, and nowhere else.
4. **(Wild values read λ.)** $\int \omega^{b-j}\psi_t x^j d\mu = \mathcal{L}_b(\zeta^t u^j - 1) = \sum_i g_i \zeta^{ti} u^{ij}$ (Proposition 9.10(5) at the character $[u] \mapsto \zeta^t u^j$ of Γ_u; [computed and compared] modulo $\ell^{\min(M,R)}$), and $v_\pi(\mathcal{L}_b(\zeta^t - 1)) = e\mu_b + \lambda_b$ whenever $\lambda_b < e$ ([proved here]: Weierstrass preparation and $v_\pi(\zeta^t u^j - 1) = 1$). [computed and compared] at every readable branch: $\lambda = 1$ at branch 0 of r^+ at $(10,11)$ and at the branches 21, 25 of r^- at $(10,37)$; at $(30,37)$ $\mu = 1$ on every even branch of r^+ ($v_\pi = 36$ or 37) — Remark 9.12(ii)'s $\mu^+ = d$ seen at wild points.
5. **(Kubota–Leopoldt at a wild point.)** With G_b as WP-MEAS's group-ring element (index $i \leftrightarrow [u^i]$), $G_b(\zeta^t u^r - 1) = -B_{r,\omega^{b-r}\psi_t}/r$, $\psi_t(u) = \zeta^t$ (not ζ^{-t}), no Euler factor ([proved here] from [U, `lem:u-twistvalues`]'s reading of Washington Thm. 7.10 and Thm. 5.11, F4m, v1.21.0-pre.4: $G_b(X) = f(u(1+X)^{-1} - 1, \omega^b)$, and at $1+X = \zeta^t u^r$ the argument is $\zeta^{-t}u^{1-r} - 1 = \zeta_\psi(1+q_0)^s - 1$ with $s = 1-r$ and $\zeta_\psi = \psi_t(u)^{-1}$, so $G_b(\zeta^t u^r - 1) = L_\ell(1-r, \omega^b\psi_t) = -B_{r,\omega^{b-r}\psi_t}/r$, the Euler factor $1 - \omega^{b-r}\psi_t(\ell)\ell^{r-1}$ being 1 as ℓ divides the conductor; [computed and compared], by both routes, against the definition of $B_{r,\chi}$, and independently by the referee at $(\ell, m) = (5,2)$, $(7,2)$ with ζ^{-t} failing at every value; the wild extension of [U, `lem:u-twistvalues`], recorded there as a remark). On the pole class $b = 0$, $v_\pi(B_{r,\omega^{-r}\psi_t}/r) = -1$ exactly: at a wild point the pole of ζ_ℓ is a simple π-adic denominator ([proved here] from $G_0(T) = \rho_\infty/T + (\Lambda)$, Proposition 9.27(4), $v(\rho_\infty) = 0$ and $v_\pi(\zeta^t u^r - 1) = 1$; [computed and compared]).
6. **(The Eisenstein measure at wild χ.)** Proposition 9.27(2)'s identity $\sum_a \chi(a)\mu^{\mathrm{Eis}}(a + \ell^M\mathbb{Z}_\ell) = -B_{1,\chi}B_{k-1,\bar\chi}/(k-1)$ holds at wild odd χ (the argument does not use the conductor; [computed and compared] at $(12,5)$, $m = 2,3$, $(14,7)$, $(16,11)$, $(30,37)$), the even masses vanish, and the polar element of Proposition 9.27(4) evaluated at ζ^t has $v_\pi = ed - 1$ exactly, the value minus it being integral ([computed and compared]; 35 at $(30,37)$).
7. **(The pole branches at wild points.)** $\mathcal{L}^-_{f,a}(\zeta^t u^j - 1) \equiv -s\left(F_1F_2 + \mathrm{Res}/(\zeta^{\pm t}u^r - 1)\right) \pmod{\ell^d\mathcal{O}_m}$, $F_1 = B_{j+1,\omega^{a-j}\psi_t}/(j+1)$, $F_2 = B_{n-j+1,\omega^{j-a}\psi_t^{-1}}/(n-j+1)$, $+t$ and $r = j+1$ on the branch -1, $-t$ and $r = k-1-j$ on the branch $k-1$ ([computed and compared]: $(30,37)$ 0/62 at $t=1$ and, env arm, $t=2$; $(282,37)$ 0/566). As $v_\pi(\zeta^{\pm t}u^r - 1) = 1$ for every r, the polar value has $v_\pi = ed - 1$, i.e. ℓ-adic valuation $d - 1/e$, *independent of* r: it is visible modulo ℓ^d at every critical point (dropping it, or $-$Res, fails at 62/62, 566/566; as at the tame points) and invisible modulo ℓ^{d-1} at every point, so Claim 9.11(4)'s located set $\{\ell \mid r\}$ has no analogue at wild points (`P4_spec.md` §0.9); and $\mathrm{Res}\cdot(1+\ell)$ changes the value only by $\pi^{e(d+1)-1}$ times a unit, so that control has no power at wild points (0 failures, recorded, not a gate on failure). The pole-branch identity at wild points (W6) is [computed and compared] also at $(226,103)$ ($d=2$, $\ell = 103$, $M=2$, $m=2$; v1.21.0-pre.7, slot-21 P4x): 0/454, the polar term visible at every point ($v_\pi = ed - 1 = 203$; dropping it, or $-$Res, fails at 454/454), $\mathrm{Res}\cdot(1+\ell)$ invisible (0); the scaled comparison needs $\mathcal{O}/103^{10}$ ($> 2^{62}$), computed in 128-bit arithmetic (`OArith128` through `ModCtx<u128>`, $P = 12$) and cross-checked against the 64-bit route at $(30,37)$, $(282,37)$ (every input and every point's polar / product element equal, the verdict counts identical); independent exact-integer Python oracle `docs/implementation/slot21/p4/p4x_pole_oracle.py` ($P = 13$: 0/454, 454, 454, 0, its $s = 83864971$ equal to the C++ line's); predictions committed first (`P4_spec.md` v0.4 §10, `487057e6`; code `ab5a1ee1`). Through v1.21.0-pre.6 this item read "Not

covered: $(226, 103)$ (the scaled representation exceeds 2^{62})". Not covered: wild characters of non-trivial tame level, $\ell = 2, 3$, $(\ell - 1) \mid k$.

Check 9.30 (The gate of P4: `HeckeLValueWildCharacterValidation`). [specification] (the checks); expected values P4's predictions and calibration, committed first. *Registration*: `Registry/Slot21/P4_wild.cpp`, `cmake/slot21/P4_wild.cmake`; sources `HeckeExt/LValue/WildCharacter.{h,cpp}`, `HeckeLValueWildCharacterValidation.{h,cpp}`; the public APIs of WP-MEAS, P2 and `NumberField` (`CycloModRing`, `MpReal`) only. Tier `standard`, read-only. Arms: (W0) the characters at $(\ell, m) = (5, 2), (5, 3), (7, 2), (11, 2), (37, 2)$, P4's $\mathcal{O}$-arithmetic against `nf::CycloModRing`, negative $t = \ell$ not primitive; (W1) Remark 9.29(5), $r \le 10$, $t \in \{1, 2\}$, both routes, at $(\ell, m, M, R) = (5, 2, 2, 6)$, $(5, 3, 3, 6)$, $(7, 2, 3, 5)$, $(11, 2, 2, 5)$, $(37, 2, 2, 4)$, negative ζ^{-t}; (W2) items (2), (3); (W3) item (4); (W4) item (6); (W5) item (1), negative controls $\tau(\bar{\chi})$ and conductor ℓM; (W6) item (7); (W7) the non-ordinary refusal of `UnitRoot` at $(10, 5)$, $(10, 7)$. Lines `WILD_W0`, `WILD_W1`, `WILD_CUSP`, `WILD_EIS`, `WILD_COMPLEX`, `WILD_POLE`, `WILD_W7`, `P4_ARMS checks=<c> failing=<f>`; `EFFECTIVE ... arm=DEFAULT cells=27`. *First run* ([computed and compared]): 201 checks, 0 failing, 4.0 s, 24 MB. Read-only env arm `HECKE_LVALUE_WILD_HEAVY=1` (`arm=HEAVY cells=31`: W5 at 125, 169; W2/W3 $(10, 11)$ at $m = 3$; W6 $(30, 37)$ at $t = 2$): 238 checks, 0 failing, 7.4 s, 25 MB. `GOLDEN_STRICT=1`: A 11827/0, B 6231/0. No golden line, record key or digest. *One prediction was wrong* and is recorded as such: the root number (item (1)); the calibration had compared $\min(|x - \text{pred}|, |x + \text{pred}|)$, which cannot see the sign it was meant to calibrate (`P4_spec.md` §8, the methodological lesson). *With P4x* (v1.21.0-pre.7, [computed and compared]): W6 takes a 128-bit route at a cell exactly when its demand exceeds the 64-bit representation ($(226, 103)$, line `WILD_POLE cell=(226,103) route=u128 P=12 Pneed=10 ... points=454 fail=0`), and a new route-agreement check (`WILD_ROUTE`) forces it at $(30, 37)$, $(282, 37)$ against the 64-bit route (0 mismatches); default arm `arm=DEFAULT cells=28`, 222 checks, 0 failing, 5.3 s, 0.09 GB; heavy arm `arm=HEAVY cells=32`, 259 checks, 0 failing, 8.9 s. No new env hook or registered name.

Remark 9.31 (Iwasawa invariants of the measure branches, and the tame functional equation (slot-21 P5)). (New at v1.21.0-pre.7. Source: `docs/implementation/slot21/P5_spec.md` v0.2 — predictions v0.1 committed before any run, `0dd6c269`; code and first runs `58c24210`; gate Check 9.32. Tags per item as in the spec; every run [computed and compared]; nothing upgraded. Refereed at v1.21.0-pre.8 by F4 (`rc/F4_referee_p5_report.md`): the pole-branch formula of (7) added, five wording repairs marked "F4". Item (9) extended at v1.21.0-pre.9 by P5′ (the McCallum–Sharifi identification; not refereed), and at v1.22.0-pre.2 by P5″ (the modular-symbol proof at the computed cells; Fable-refereed).) Notation of Proposition 9.10: $\ell \ge 11$, n even, $k = n + 2$, f ordinary of level one with unit root α, $r^{\pm}$ its eigensymbols, $\varepsilon = \pm 1$ their geometric parity (the eigenvalue of $u_t \mapsto (-1)^t u_t$: r^- has odd coordinates, so $\varepsilon(r^-) = -1$; through pre.7 "∓ 1", F4); $\mathcal{L}_b(T) \in \mathbb{Z}_\ell[\![T]\!]$ the branch $b \in \mathbb{Z}/(\ell - 1)$ of $\mu_{r^\pm}$ (non-zero only for $(-1)^b = \varepsilon$), (μ_b, λ_b) its Iwasawa invariants; G_b as in [U, `lem:u-twistvalues`]; $\sigma(T) := u^n(1 + T)^{-1} - 1$.

1. **(The tame functional equation.)** For every $P \in W$ (not only eigenvectors), every Dirichlet character χ of conductor $F \in \{1, \ell\}$ and every $0 \le j \le n$,

$$\lambda_{\chi,j}(P) = -\chi(-1)(-1)^j \, \lambda_{\bar{\chi},n-j}(P) \quad \text{exactly.}$$

   [proved here]: Remark 9.29(3)'s proof uses nothing about F but $\gcd(a, F) = 1$, and holds verbatim at $F = \ell$; at $F = 1$ ($a = 0$, $\gamma_{0,0} = 1$) the identity is $P + \rho_n(S)P = 0$ read on the coordinate u_{n-j}, i.e. $P_{n-j} = -(-1)^j P_j$. At an eigensymbol of parity ε the values vanish unless $\chi(-1)(-1)^j = \varepsilon$, so the non-zero ones carry the sign $-\varepsilon$. [computed and compared] at every cell of Check 9.32 on $r^{\pm}$ and on generic (non-eigen) $P \in W$, every Teichmüller power ω^i and conductor 1 (31413 values at $(282, 37)$, 70143 at $(226, 103)$; 0 failures); the flipped sign matches nowhere; "χ for $\bar{\chi}$" matches exactly at $j = n/2$ on r^- (as in Remark 9.29(3)).
2. **(The branch form.)** $\mathcal{L}_b(T) = -(-1)^b \mathcal{L}_{n-b}\big(\sigma(T)\big)$. [proved here] given Proposition 9.10(4) at every conductor ℓ^m and (1) at every $F = \ell^m$ (Remark 9.29(3)): $E_{\chi,j}$ and $\binom{n}{j}$ are symmetric under $j \leftrightarrow n - j$, so the difference of the two sides vanishes at the points $\zeta u^j - 1$, ζ of ℓ-power order, $\zeta^{-1} u^{n-j} - 1 = \sigma(\zeta u^j - 1)$, infinitely many in the open disc; Weierstrass preparation. At

level M it holds in $\mathbb{Z}/\ell^R[\Gamma_u/\Gamma_u^{\ell^{M-1}}]$ modulo $\ell^{\min(M,R)}$ ([computed and compared], every branch of the measure's parity, both signs; the flipped sign fails at every branch with $g_b \not\equiv 0$).

3. **(Consequences for (μ,λ).)** [proved here] from (2), σ being an automorphism of $\mathbb{Z}_\ell[\![T]\!]$ ($\sigma(T) \in (\ell, T)$ and $\sigma\circ\sigma = \mathrm{id}$; F4) which preserves μ (it is an automorphism of $\mathbb{F}_\ell[\![T]\!]$) and λ (modulo ℓ, $\bar\sigma(T) = (1+T)^{-1} - 1 = -T + T^2 - \dots$ is T times a unit, so ord_T is preserved): *(FE-inv)* $(\mu_b, \lambda_b) = (\mu_{n-b}, \lambda_{n-b})$ for every branch; *(FE-parity)* at a self-dual branch ($2b \equiv n \bmod \ell - 1$; never a pole branch of the Eisenstein measure, since $a \equiv -1$ or $k-1$ with $2a \equiv k-2$ forces $(\ell-1) \mid k$; F4) σ is an involution of the open disc (over $\mathbb{C}_\ell$) with the single fixed point $T_0 = u^{n/2}-1$ ($-u^{n/2} - 1$ is a unit), $\sigma'(T_0) = -1$, the zeros of $\ell^{-\mu}\mathcal{L}_b$ (counted by Weierstrass preparation) off T_0 pair up, and the order r at T_0 satisfies $(-1)^r = -(-1)^b$: λ_b **is odd at an even self-dual branch (plus measure) and even at an odd one (minus measure)** — an ℓ-adic *forced central zero* at $j = n/2$ on the even self-dual branches. [computed and compared]: (FE-inv) at every branch of every cell; at every self-dual branch of every cell $\lambda = 1$ at the even ones ($(10,\ell)$ at $\ell = 11, 19, 23, 31, 43, 47, 59, 67, 103, 131$; $(16, 37)$ at 8 and 26; $(42, 59)$, $(56, 67)$, $(22, 103)$, $(20, 131)$, $(226, 103)$, $(10, 691)$ at 350) and $\lambda = 0$ at the odd ones.
4. **(The near-trivial zero.)** $\mathcal{L}_0(0) = E_{\mathbf{1},0}\,\lambda_{\mathbf{1},0}$ with $E_{\mathbf{1},0} = (1 - 1/\alpha)(1 - \ell^n/\alpha)$; so if $\alpha \equiv 1 \pmod\ell$ then on the plus measure $\mu_0 \geq 1$ or $\lambda_0 \geq 1$, and likewise at $b \equiv n$ by (FE-inv) ([proved here]). $\alpha \equiv \tau(\ell) \equiv 1 \pmod\ell$ for Δ at $\ell = 11$, 23 (the values [standard, reference], recalled; the code reads a_ℓ from the symbol), and at every Eisenstein-congruent cell, where $\mu^+ = d \geq 1$ anyway. [computed and compared]: $\lambda = 1$ at $(10, 11)$ branch 0 (also its even self-dual branch) and at $(10, 23)$ branches 0, 10.
5. **(A ball-free reading of (μ,λ): the values oracle.)** For one branch the critical values $V_j = \mathcal{L}_b(u^j - 1)$ come from the eigensymbol by Proposition 9.10(4) alone (no ball table, no group ring); their Newton divided differences δ_t at nodes $j_0 < \dots < j_L$ with $\ell \nmid \binom{n}{j_s}$ satisfy $v(\delta_t - c_t) \geq \mu_b + 1$, c_t the T^t-coefficient ([proved here]: $v(T_j - T_{j'}) = 1 + v(j - j')$ and the complete homogeneous polynomials h_m have $v \geq m$), so $v(\delta_t) \geq \mu_b + 1$ for $t < \lambda_b$ and $v(\delta_{\lambda_b}) = \mu_b$. [computed and compared]: consistent with the group-ring invariants at every readable branch, integral everywhere, and the only reading where the group-ring invariants are not readable ($\mu \geq M - 1$: the depth-2 cells' plus side, $\mu^+ = 2$ at $M = 2$).
6. **(The Kubota–Leopoldt invariants.)** $\lambda(G_b) = 0$ iff $\ell \nmid B_b/b$ ([proved here] *given* $\mu(G_b) = 0$: $c_0 \equiv G_b(u^b - 1) = -(1 - \ell^{b-1})B_b/b \pmod\ell$, so a unit c_0 gives $(\mu,\lambda) = (0,0)$ unconditionally, while $\ell \mid c_0$ gives $\lambda \geq 1$ only once $\mu = 0$ is known; F4); $\mu(G_b) = 0$ for every even b (Ferrero–Washington, [standard, reference], recalled, [locator not verified, rule A2]); $\lambda(G_b) = 1$ at every irregular pair (ℓ, b) with $\ell < 2^{31}$ (read on the page: [19], abstract item (3), p. 1, "the Iwasawa invariants satisfy $\lambda_p = \nu_p = i_p$", text layer read for P5 and again by F4, who also read §7 of the manuscript, where λ_p, ν_p are the Iwasawa invariants of the p-class groups of $\mathbb{Q}(\zeta_{p^n})$ and the test is four incongruences per irregular index; the step $\lambda_p = \sum_{b \text{ even}} \lambda(G_b)$ is [standard, reference], recalled (Iwasawa's theorem for the minus part through the analytic class number formula, the plus part trivial by Kummer–Vandiver, verified in [19] item (2); locator unverified), and with $\lambda(G_b) \geq 1$ at each irregular index — Ferrero–Washington again — it forces $\lambda(G_b) = 1$). [computed and compared] by two routes — the Kubota–Leopoldt group ring at level 2, and exact Bernoulli numbers alone (divided differences of $G_b(u^r - 1) = -(1 - \ell^{r-1})B_r/r$ at $r = b + s(\ell - 1)$, $s = 0, \dots, 3$), equal at every class (344/344 at 691): at the 16 primes $11, \dots, 47, 59, 67, 103, 131, 691$, $\mu(G_b) = 0$ at every even b, and $\lambda(G_b) = 1$ exactly at $(37, 32)$, $(59, 44)$, $(67, 58)$, $(103, 24)$, $(131, 22)$, $(691, 12)$, $(691, 200)$, 0 elsewhere. *No $\lambda(G_b) \geq 2$ occurs.*
7. **(Greenberg–Vatsal at level one.)** At a cell with $d = v(B_k/k) \geq 1$, on every odd *regular* branch a ($a + 1 \not\equiv 0$, $k - 1 - a \not\equiv 0 \bmod \ell - 1$):

$$\mu_a^- = 0, \qquad \lambda_a^- = \lambda(G_{a+1}) + \lambda(G_{k-1-a}),$$

[proved here] given Claim 9.11(3) (or Proposition 9.27(3), (5) with Claim 9.11(2)) and $\mu(G) = 0$: modulo ℓ the two substitutions are $T \mapsto T$ and $T \mapsto (1+T)^{-1} - 1$, both λ-preserving — the level-one analogue of Greenberg–Vatsal's formula (no Euler factor at level one; Greenberg–Vatsal held, not read for it). As $\ell \mid B_k/k$, $k \equiv b_0$ for an irregular index b_0, and $a + 1 \equiv b_0 \iff k - 1 - a \equiv 0$:

the cell's own irregular index always lands on a pole branch ([proved here], elementary); only a second irregular index at the same ℓ contributes. [computed and compared] at all nine cells with $d \geq 1$ ($(30,37)$, $(66,37)$, $(42,59)$, $(56,67)$, $(22,103)$, $(20,131)$; $(282,37)$, $(226,103)$; $(10,691)$): $(\mu^-,\lambda^-) = (0,0)$ on every regular odd branch, except $(10,691)$, $\lambda^- = 1$ at $a = 199$ and $a = 501$ ($\lambda(G_{200}) = 1$), as predicted. *The pole branches* (through pre.7 a record; [proved here] at v1.21.0-pre.8, F4, given Proposition 9.27(4), (6) — the same inputs as the regular branches plus the polar term — and $\mu(G) = 0$): on both pole branches

$$(\mu^-,\lambda^-) = (0,\ \lambda(G_{b_0}) - 1), \qquad b_0 \equiv k \text{ the cell's own irregular index},$$

i.e. the formula above holds on every odd branch with $\lambda(G_0) := -1$ for the pole. Proof on the branch -1 (the branch $k-1$ is its (FE-inv) partner): with $X = u(1+T) - 1$, $Y = u^{k-1}(1+T)^{-1} - 1$, $x_k = u^{k-1} - 1$ and $G_0(X) = \rho_\infty/X + H(X)$, $H \in \Lambda$, Proposition 9.27(4) gives $\mathcal{L}^{\mathrm{Eis}}_{-1} - \mathrm{polar}_{-1} = -H(X)\,G_k(Y) - \rho_\infty\big(G_k(Y) - G_k(x_k)\big)/X$, both terms in Λ ($Y - x_k \in X\Lambda$). Modulo ℓ: $X \equiv T$, $x_k \equiv 0$, $Y \equiv \bar\sigma_1(T) := (1+T)^{-1} - 1 = -T + \ldots$, and $\bar G_k = \bar G_{b_0}$ has $\mathrm{ord}_T = \lambda(G_{b_0}) \geq 1$ with $\bar G_k(0) = 0$; so the second term has $\mathrm{ord}_T = \lambda(G_{b_0}) - 1$ exactly (its leading coefficient $\pm\bar\rho_\infty\bar c_\lambda$, $v(\rho_\infty) = 0$) and the first has $\mathrm{ord}_T \geq \lambda(G_{b_0})$; by Proposition 9.27(6) $\mathcal{L}^-_{f,-1} \equiv s(\mathcal{L}^{\mathrm{Eis}}_{-1} - \mathrm{polar}_{-1}) \pmod{\ell^d}$ with s a unit, hence $\mu^-_{-1} = 0$ and $\lambda^-_{-1} = \lambda(G_{b_0}) - 1$. [computed and compared] at the nine cells: $(0,0)$ on both pole branches, as $\lambda(G_{b_0}) = 1$ there by (6). (So the own irregular index *is* seen by the minus measure, on the pole branches, as $\lambda(G_{b_0}) - 1$; a cell with $\lambda(G_{b_0}) \geq 2$, none known, would show $\lambda^- \geq 1$ there.) $\mu^+ = d$ on every plus branch (Remark 9.12(ii); at $d = 2$ through (5)).

8. **(The tables, a record.)** $\mu = 0$ on every branch of every control ($d = 0$) cell: $(10,\ell)$ for the 15 primes $11 \leq \ell \leq 131$ of (6) and $(16,37)$ — evidence, at finite level, for Greenberg's conjecture ($\mu = 0$ for residually irreducible $\bar\rho_f$; [standard, reference], recalled), predicted, not proved. The branches with $\lambda = 1$ ($\lambda = 0$ elsewhere; $\mu = 0$ except $\mu^+ = d$ on the Eisenstein plus side; * = even self-dual, forced by (3)):

cell	minus	plus
$(10,11)$	—	0^* (also (4))
$(10,13)$, $(10,17)$, $(10,29)$, $(10,41)$	—	—
$(10,19)$, $(10,31)$, $(10,43)$, $(10,59)$, $(10,103)$	—	14^*, 20^*, 26^*, 34^*, 56^* resp.
$(10,23)$	—	0, 10 ((4)), 16^*
$(10,37)$	21, 25	—
$(10,47)$	13, 43	16, 40, 28^*
$(10,67)$	31, 45	38^*
$(10,131)$	63, 77	70^*
$(16,37)$	21, 31	22, 30, 8^*, 26^*
$(30,37)$, $(66,37)$, $(282,37)$	—	4, 26
$(42,59)$	—	14, 28, 50^*
$(56,67)$	—	8, 48, 28^*
$(22,103)$, $(226,103)$	—	46, 78, 62^*
$(20,131)$	—	42, 108, 22, 128, 10^*
$(10,691)$	199, 501 ((7))	22, 678, 350^*

Every unstarred entry comes in an (FE-inv) pair $\{b, n-b\}$. No $\lambda \geq 2$ and no $\mu > 0$ occur (other than $\mu^+ = d$). The tables equal the frozen records of WP-MEAS (`MeasureFrozenR201.inc`: $(10,37)$, $(16,37)$, $(30,37)$) and S2 (`MeasureFrozenS2.inc`: $(282,37)$, $(226,103)$, $(10,691)$), both signs, no branch skipped. *Unexplained* (recorded, not derived): the $\lambda = 1$ pairs of the controls away from the self-dual and near-trivial-zero branches — $(10,37)$ minus $\{21,25\}$, $(10,47)$ minus $\{13,43\}$ and plus $\{16,40\}$, $(10,67)$ minus $\{31,45\}$, $(10,131)$ minus $\{63,77\}$, $(16,37)$ minus $\{21,31\}$ and plus $\{22,30\}$. Each such pair is one congruence $c_0 \equiv 0 \pmod{\ell}$ on a branch (all of its critical values divisible by ℓ); a branch with unit-sized random coefficients has $\lambda \geq 1$ with probability $1/\ell$, and the pairs then come in (FE-inv) orbits, so over the 32 sign-cells of the controls ($(\ell-1)/4$ orbits each) about $\sum(\ell-1)/(4\ell) \approx 7.7$ orbits are expected by chance; 7 are seen (47 is a regular prime, so no Eisenstein mechanism is available there). The pairs are therefore consistent with chance and carry no claim (F4).

9. **(The plus side: the McCallum–Sharifi forced zero; through pre.8 "an observation, recorded, not derived".)** On the plus side of *every* Eisenstein cell (nine cells, $d = 1, 2$) $\lambda^+ = 1$ at the (FE-inv) pair $\{-k,\, n+k\}$ (mod $\ell-1$), $k \equiv b_0$ the cell's own irregular index ($4 \equiv -32$, $26 \equiv 30 + 32$ at $(30, 37)$; $678 \equiv -12$, $22 = 10 + 12$ at $(10, 691)$, while the second index 200 gives nothing on the plus side); at $(20, 131)$ a further pair $\{22, 128\} = \{k, n-k\}$. This places Remark 9.12(iii)'s pairs. Not a candidate (F4, pre.8; through pre.7 "a derivation through the boundary part $r^+ \equiv u \cdot p_0$ is the natural candidate"): p_0 contributes nothing to these branches — $\lambda_{\chi,j}(p_0) = 0$ for $\chi \neq \mathbf{1}$ (Remark 9.3 (L2)), so μ_{p_0} lives on the branches 0 and n alone, and it is $\alpha^{-m-1}(\alpha-1)\ell^{mn}p_0$ on a level-m ball (Proposition 9.10(6)), $\equiv 0$ (mod ℓ^d). The pattern sits in the reduced measure $\ell^{-d}\mu_{r^+} \bmod \ell$, i.e. in the measure of $w = (r^+ - u\,p_0)/\ell^d \bmod \ell$ (Remark 9.12(iii)), a mod-ℓ symbol on which $T_q - (1+q^{k-1})$ acts by $u\,(a_q - 1 - q^{k-1})\ell^{-d}\,p_0$ — the next ℓ-adic digit of a_q (the constants of [RC]'s Frobenius shadow). Through pre.8 this item ended: "A derivation would need the twisted values of w; not attempted." *Identified* (v1.21.0-pre.9, slot-21 P5′, `docs/implementation/slot21/P5prime_note.md`, commit `08d83ab3`; scripts `p5p/`, outputs `p5p/out/`). Put $h(b) := \ell^{-d}\mathcal{L}_b^+(0) \bmod \ell$ on the even branches. Given $\mu_b^+ = d$, $\lambda_b^+ \geq 1 \iff h(b) = 0$ ([proved here]: every critical point $u^j - 1$ lies in $\ell\mathbb{Z}_\ell$, so $\mathcal{L}_b^+(u^j-1) \equiv \mathcal{L}_b^+(0)$ (mod ℓ^{d+1}), and Weierstrass preparation; $\mu_b^+ = d$ per branch is read at $M = 3$ only, so at the $M = 2$ cells a zero of h reads "$\lambda_b^+ \geq 1$ or $\mu_b^+ \geq d+1$"). Let $x = (x_i)_{i \text{ odd}}$ be the null vector over $\mathbb{F}_\ell$ of McCallum–Sharifi's relations for the pairing values $e_{i,r} = \langle \eta_i, \eta_{r-i} \rangle_r$, r the cell's own irregular index (their relation from $\{\rho_a, \rho_{a-1}\} = 0$, $4 \leq a \leq \ell - 1$ even, and skew-symmetry $x_i + x_{r-i} = 0$; [27] §5, read on the page of the authors' web PDF, pp. 12–14, not held in the project library). Then ([computed and compared], `p5p_sharifi.py`, `p5p_report.py`; 32 cells, 28 irregular pairs (ℓ, r) with $37 \leq \ell \leq 691$, every even branch, 4961 branch values, 0 mismatches, the nullity 1 at every irregular (ℓ, r) and 0 at the regular $r \pm 2$ tried — not a general fact: the P5′ audit finds nullity 10 at the regular $(257, 162)$, where 2 has order 16; cf. McCallum–Sharifi's regular $(73, 38)$; audit v1.21.0-pre.10):

$$h(b) = c \cdot x_{b+1} \quad \text{for every even } b, \text{ one } c \in \mathbb{F}_\ell^\times \text{ per cell}$$

— consistent with Sharifi's conjecture, stated in [32], Introduction, (1.1) (read on the page of the arXiv version by the P5′ audit): $\varphi_f((\alpha_t, \alpha_{k-t})) = c_{p,k} \cdot L_p(f, \omega^{t-1}, 1)$ with $c_{p,k} \in (\mathbb{Z}/p)^\times$ independent of the odd t and of f, f of weight 2, level p, character ω^{k-2} (announced in [33], Introduction, p. 4: (η_i, η_{k-i}) against $L_p(f, \omega^{i-1}, 1)$ modulo $\mathfrak{p}^2$), seen at level one in this normalisation. The index $t \leftrightarrow \omega^{t-1}$ is Sharifi's; that the branch b (at $j = 0$) of the level-one weight-k measure carries the twist ω^{t-1}, $t = b+1$, of the weight-2 form is read off the data, not derived. McCallum–Sharifi show that the cup-product pairing itself has $e_{\ell-r,r} = 0$ ([27], the remark after Cor. 5.2, read on the page: $\eta_{\ell-r}$ is the Kummer generator of Ribet's unramified extension, so it pairs trivially; their §5 assumes Vandiver's conjecture at ℓ, verified for $\ell < 2^{31}$ [19]), and by skew-symmetry $e_{2r-1,r} = 0$; the computed vector has $x_{\ell-r} = x_{2r-1} = 0$ as well (they observe it for $p < 10^4$; [computed and compared] here); under $i = b+1$ these are exactly $b \equiv -k$ and $b \equiv n+k$. **Theorem, conditional on (S)** "$h(b) = c \cdot e_{b+1,r}$ with $c \in \mathbb{F}_\ell^\times$ at a $d = 1$ cell" (Sharifi's conjecture in this level-one normalisation; not [proved here] as a uniform statement; at the computed cells see *The modular-symbol proof* below, v1.22.0-pre.2): *at every level-one Eisenstein cell with $d = 1$, $h(b) = 0$ at $b \equiv -k$ and $b \equiv n+k$* (mod $\ell-1$), *hence $\lambda_b^+ \geq 1$ there wherever $\mu_b^+ = d$* (in general "$\lambda_b^+ \geq 1$ or $\mu_b^+ \geq d+1$"; $\mu_b^+ = d$ is read at $M = 3$, $\ell \leq 157$) — from (S), $e_{\ell-r,r} = e_{2r-1,r} = 0$ and the reduction above: the pair $\{-k, n+k\}$ is the McCallum–Sharifi forced zero. *Status of (S)* (P5′ audit, `docs/implementation/slot21/rc/P5prime_audit_report.md`): Fukaya–Kato [17] (Introduction read on the page of the authors' 2012 preprint, not held) prove Sharifi's conjectures in the Λ-adic form — $\varpi \circ \Upsilon = 1$ and $\Upsilon \circ \varpi \equiv 1$ modulo torsion (Thm. 0.15), $\Upsilon(C) \equiv L^-$ modulo torsion (Thm. 0.20), componentwise (Rmk. 0.17(3)) — under (i) "ξ has no multiple zero" or (ii); (i) holds on the relevant component for $\ell < 2^{31}$ by [19] ($\lambda(G_b) = 1$ at each irregular index, with the [standard, reference] steps of (7)). *Not checked*: the descent from their Λ-adic statement to the finite-level (1.1) modulo $\mathfrak{p}^2$, and from the weight-2 level-ℓ form to level one, weight k, in this normalisation (Hida's control theorem, recalled). The theorem therefore

stays conditional *as a uniform statement*; at the computed cells it no longer needs (S) (below). *No novelty is claimed*: given (1.1), the forced zero is an immediate consequence of McCallum–Sharifi's remark; the audit did not find it written out in the four texts read ([27] §5, [33] and [32] Introductions, [17] Introduction) and did not search later work (e.g. Wake–Wang-Erickson). A second irregular index $c \not\equiv r$ forces nothing on the plus side of the r-cell (its generator $\eta_{\ell-c}$ kills the projection to a different eigenspace, the pairing $\langle\ ,\ \rangle_c$): the comparison with the McCallum–Sharifi vector of every other irregular index fails ([computed and compared] at 691 both ways — $(198, 691)$ sees $\{490, 398\} = \{-200, n + 200\}$, not $\{678, 210\}$ —, 617, 467, 379, 353, 157). The further zeros ($\{22, 128\}$ at $(20, 131)$, $\{12, 54\}$ at $(66, 101)$, $\{196, 438\}$ at $(18, 617)$, $\{118, 196\}$ at $(82, 233)$, ...) are zeros of the McCallum–Sharifi vector itself, properties of (ℓ, r) constant along the Hida family ([computed and compared]), not forced by the relations' visible symmetries (a pointer, not checked: by [32], Introduction, under mild assumptions (α_t, α_{k-t}) vanishes only if the Selmer group over $\mathbb{Q}$ of $T_f(t)$ is non-zero). Audit (v1.21.0-pre.10): the identity re-derived independently (own code from the relations as printed, the measure read through the verb) at $(30, 37)$, $(282, 37)$ ($d = 2$) and at three irregular pairs outside the 28: $(257, 164)$, $(293, 156)$, $(353, 300)$ (cells $(162, 257)$, $(154, 293)$, $(298, 353)$): every even branch, one scalar, 0 mismatches, the zeros exactly $\{-k, n+k\}$ ([computed and compared]). *The modular-symbol proof* (v1.22.0-pre.2; slot-22 P5″, `docs/implementation/slot22/P5pp_note.md` v0.2, scripts `p5pp/`; Fable referee `P5pp_referee.md`: ACCEPT WITH REPAIRS, 0 B / 2 M / 7 m, the repairs R1–R8 applied in v0.2). It uses neither Sharifi's conjecture nor Fukaya–Kato. Write $r^+ = u\,p_0 + \ell^d w$ — the hypothesis (H1), i.e. (L1) of Remark 9.3 at depth d. Then $h(b)$ is the ω^b-Fourier coefficient of the Ash–Stevens image $\Psi = A_{\bar{w}}$, $A_P(c : d) = P(c, d)$, on the Manin symbols $[a : 1] = \{\infty \to a/\ell\}$ of Fukaya–Kato–Sharifi's module $M_1' = H_1(X_1(\ell), C', \mathbb{Z})^+$ (T. Fukaya, K. Kato, R. Sharifi, *Modular symbols in Iwasawa theory*, Iwasawa Theory 2012, Springer 2014, §§2.1.2–2.1.8, read on the authors' web copy; only the soundness of its presentation of M_1' and $\langle a\rangle^{-1}[u : v] = [au : av]$ are used). Ψ is even, of character ω^{2-r}, kills the relations of M_1', and is Eisenstein for Merel's sum on these labels with eigenvalue $q^{2-r} + q$, because the defect $(T_q - 1 - q^{k-1})\bar{w} = \bar{c}_q\,\bar{p}_0$ lies on the boundary symbol, whose image vanishes on M_1' ([proved here] there, from Proposition 9.10(6) and the [standard, reference] inputs below). At each computed pair the functionals with these properties form one line, McCallum–Sharifi's span(x) ("L3": P5″'s script at 33 irregular pairs — P5′'s 28, $(257, 164)$, $(293, 156)$, $(353, 300)$ and the second indices $(617, 174)$, $(617, 338)$ — re-run with independent code by the referee at three of them and three regular controls; a computation per pair). Hence $h(b) = y_{b+1} = c \cdot x_{b+1}$. The two convention-dependent steps (the eigenvalue $q^{2-r} + q$ and the character ω^{2-r}, derived from Atkin–Li and Fukaya–Kato–Sharifi's normalisation) were verified directly by the referee on the object: an operator identity on all of W^+ for $q = 2, 3$, the character $2 - r$ alone, the eigenvalue unique among five candidates, $h(b) = y_{b+1}$ with scalar 1, at $(30, 37)$, $(66, 37)$, $(102, 37)$. (H1) holds: at the 28 cells with $d = 1$, $n < \ell$ it is [proved here] from P5′'s own computed input $\ell \mid \mathcal{L}_b^+(0)$ (every even b), through the cusps of $X_1(\ell)$ over 0 and the injectivity of $P \mapsto A_P$ for $n \le \ell$; at $(282, 37)$, $(226, 103)$ it is [computed and compared] at depth 2 (Remark 9.3); at $(66, 37)$, $(102, 37)$ ($n > \ell$) it is computed at depth exactly 1 by the referee's Python, one route (Remark 9.3; the C++ gate has not read it). **Status (v1.22.0-pre.2), per pair:** at the 30 cells with the project's own readings of (H1), and at $(66, 37)$, $(102, 37)$ once (H1) is gated (so far the referee's Python, one route, the C++ gate has not read it), the theorem above — $h(b) = 0$ at $b \equiv -k$, $n + k$, hence $\lambda_b^+ \ge 1$ there wherever $\mu_b^+ = d$, now at $d = 1$ and $d = 2$ alike — and (S), $h = c \cdot e_{b+1,r}$ with $c \in \mathbb{F}_\ell^\times$ (additionally given $h \not\equiv 0$, computed, and $e \neq 0$, [33] Thm. 1.2), are **[proved here] modulo [standard, reference] inputs, given [computed and compared] L3 at the pair and [computed and compared] (H1) at the cell**. The [standard, reference] inputs (recalled, not read): Manin's presentation of $M_1(\Gamma_1(\ell))$; Merel's sum as T_q on W and on $M_1(\Gamma_1(\ell))$ (Merel, LNM 1585, 1994, §2; the path condition is [C, `lem:merelpath`]); the cusps of $X_1(\ell)$ over 0 and the diamond action on them. *Nothing uniform in (ℓ, r) is claimed*: L3 is a computation per pair, and the statement for every irregular pair stays open (and conditional on (S) as above). (S) is P5′'s level-one statement $h = c \cdot e$, of the shape of Sharifi's (1.1) and *consistent with Sharifi's conjecture (1.1)*;

its identification with (1.1) — the index map $t = b + 1$, the descent from weight 2, level ℓ — is untouched, and (1.1) itself is not claimed here. The $d = 2$ match at $(282, 37)$, $(226, 103)$ (through pre.1 "a record") is thereby explained. No novelty is claimed. Conversely the minus side sees the second index through (7)'s Greenberg–Vatsal formula: $\lambda^- = 1$ at $\{107, 109\}$ of $(60, 157)$ and at $\{47, 61\}$ of $(108, 157)$ ([computed and compared], $M = 3$), as at $(10, 691)$. *Still open*: $\lambda^+ = 1$ exactly at the forced branches ([computed and compared] where $M = 3$ reads it, $\ell \leq 157$; at $M = 2$ only "≥ 1"); the non-forced zeros of x; (S) as a statement uniform in (ℓ, r) (at the computed cells [proved here] modulo [standard, reference] inputs since pre.2, above), and the index map $i = b+1$ against Sharifi's t; the $d = 2$ cells other than $(282, 37)$, $(226, 103)$ (Sharifi's statement is modulo $\mathfrak{p}^2$; at these two the digit $\ell^{-2}\mathcal{L}_b^+(0)$ mod ℓ matches x, through pre.1 a record, since pre.2 explained by the modular-symbol proof); a *uniform* modular-symbol proof (the Eisenstein part of $M_1' \otimes \mathbb{F}_\ell$ one-dimensional at every irregular pair, or a symbolic derivation of McCallum–Sharifi's relations from the Manin and T_2-Eisenstein relations; through pre.1 "not attempted" for any pair, since pre.2 done pair by pair at the 33 computed pairs; the literature's map between Manin symbols and cup products is Sharifi's ϖ, [32] §5, with Fukaya–Kato's Υ the other way; not compared).

Not covered: P5 (a) (the $\mathcal{L}$-invariant of 11a) is P9's V12 (Remark 9.26); characters of ℓ-power conductor enter only through (2)'s proof; $\lambda \geq 2$ is nowhere seen.

Check 9.32 (The gate of P5: `HeckeLValueIwasawaInvariantsValidation`). [specification] (the checks); expected values P5's predictions, committed first (`P5_spec.md` v0.1, `0dd6c269`). *Registration*: `Registry/Slot21/P5_iwasawa.cpp`, `cmake/slot21/P5_iwasawa.cmake`; source `HeckeExt/LValue/HeckeLValueIwasawaInvariantsValidation.{h,cpp}`; the public APIs of WP-MEAS, WP-LIN and WP-LV only, no existing file's code edited. Tier `standard`, read-only. Arms: (I0) Remark 9.31(6) by both routes at 15 primes (691 in the heavy arm); (I1) item (1), negatives: the flipped sign, and "χ for $\bar{\chi}$" counted only on equal-valuation pairs of power $\ell^{R-v} \geq 10^5$; (I2) item (2) in the group ring, negative: the flipped branch sign; (I3) the tables of item (8) by the group-ring invariants and the values oracle (5), (FE-inv), (FE-parity), item (4), $\mu = 0$ at the controls; (I4) item (7) with $\lambda(G)$ from (I0)'s Bernoulli-only route, $\mu^+ = d$; (I5) the tables against `MeasureFrozenR201.inc`, `MeasureFrozenS2.inc`. Lines `IWA_KL`, `IWA_FE`, `IWA_GRFE`, `IWA_TABLE`, `IWA_GV`, `IWA_FROZEN`, `P5_ARMS checks=<c> failing=<f>`; `EFFECTIVE ... arm=DEFAULT cells=22`. *First run* ([computed and compared]): 580 checks, 0 failing (I0 60, I1 132, I2 88, I3 270, I4 24, I5 6), 21.6 s, 0.10 GB. Read-only env arm `HECKE_LVALUE_IWASAWA_HEAVY=1` (`arm=HEAVY cells=25`: the Kubota–Leopoldt table at 691, the cells $(10, 691)$, $(282, 37)$, $(226, 103)$): 658 checks, 0 failing, 72 s, 0.10 GB. `GOLDEN_STRICT=1`: A 11827/0, B 6231/0. No golden line, record key or digest; `--list` 192. Two negative controls were repaired before the first green run, no predicted value changed (`P5_spec.md` §8: the "χ for $\bar{\chi}$" control had no power at high valuation — a match at power ℓ^2 is not evidence; the group-ring negative is vacuous where $g_b \equiv 0 \pmod{\ell^{\min(M,R)}}$ and is then not counted).

A Index of claims with STATUS

The seven claims of §1.1 that v1.2.2 indexed here — the amalgam presentation, the dimension of H^1, the torsion origin of the Dickson classes, the boundary groups, the six-term sequence, the boundary characters and Poincaré duality — moved into [C] with the scope cut and are no longer statements of this document. Their status is [C]'s own STATUS index, where four of them are also *corrected* (§1.1); [C, `cl:pd`] in particular is carried there as OPEN, not as [standard, reference].

ref	status	content
1.1	[standard, reference] / verified (10 Kummer + 27 converse cells, 2026-09-16)	Harder denominator / Ribet; $r^1 = 0$ iff $\ell \mid \mathrm{num}(B_k/k)$, read mod ℓ as "$r^1 = 0$ on the joint eigenspace E", which is two-dimensional at a Kummer cell
2.1	[proved here]	Ext^1 vanishes across maximal ideals
2.2	[proved here]	generation ⇔ separation + tangent-space spanning

ref	status	content
2.4	[standard, reference]	finiteness of $\Sigma(\ell)$ (Serre, Jochnowitz, Ash–Stevens)
2.5	[proved here]	existence of $S_{\mathrm{sep}}(\ell)$
2.6	[standard, reference]	T_ℓ indispensable; $\theta^{\ell-1}$ and $E_{\ell-1}$ doublings
2.7	[standard, reference] / [conjectural] (a) / [proved here] (b)	cuspidal Sturm bound $\lfloor k/12 \rfloor$; the $\mathbb{Z}$-module statement for M_k with $B(k)$ is *false* ($\ell = 5, 7, 131, \dots$)
2.9	[proved here]	the computed bound $B'(k,\ell)$ certifies generation over $\mathbb{Z}_\ell$
2.10	[proved here] (given a recorded B')	Sturm certifies generation on non-Dickson $\mathfrak{m}$ of X_4 only
2.11	[proved here] (v1.15.1-pre.4, WP-19; referee pending)	$\mathbb{T}/\ell$ of M_k has dimension $\dim M_k$ on $M_k(\mathbb{F}_\ell)$, $\ell \geq 5$; $B'(k,\ell)$ always exists
2.12	[proved here] (v1.15.1-pre.4, WP-19; referee pending)	F9 = Eichler–Shimura per block, exact up to $\dim H^0(\Gamma; M_n)_\lambda$ (the torsion guard)
2.13	[standard, reference] (recalled) / verified ($(160, 691)$, 2026-09-16)	Bellaïche–Khare: $\mathbb{T}_{\bar\rho} \cong \mathbb{F}_\ell[[x,y]]$; the campaign example is resolved — seven local blocks (v1.3.3); T_5's 4:1 is the family's 4:2 block, not split by the refinement, and the block partition is already T_2's (v1.7.0, `HeckeExtSeparationValidation`)
2.14	(1) proved in [C] (v1.12.0-pre.1; through v1.11.1 [standard, reference]); (2) [standard, reference]	$U_\ell \circ \theta = 0$ over $\mathbb{F}_\ell$, on cochains ([C, `prop:thetatwist`]; checked by (6e) of §5.1, slot-11 WP-6, v1.12.0); T_ℓ's Jordan type
§5.2	specification, no claim (v1.12.0-pre.1; built, v1.12.0; (h)'s refusal as specified in the build, v1.12.1)	the Dickson–θ objects: names, indexing, the six terms and five maps, the census hand-off, refusals; the mathematics is [C]'s
§7.2	specification, no claim (v1.12.0-pre.1; edition 1 built, v1.12.0)	the congruence-census catalogue `hecke_census/v1`: rows, columns, the boundary-split, $\pm$, θ-image, Glover and $\mathbb{Z}/\ell^k$-depth rules, extremal criteria, layout
5.13	[standard, reference] (1) / [proved here] (2)–(4) (v1.5.0; (3) [proved here] since v1.21.0-pre.5)	the Glover factors of Sym^n by the θ-recursion, and the standard-module convention that fixes the det-twists
3.1	[proved here]	per-operator refinement can miss a split; locality criterion; enforced in the loop since E0b (slot 2)
3.2	[standard, reference]	the coefficient field $K \cong \mathbb{F}_{\ell^r}$ exists and is unique (Hensel)
3.3	[proved here]	$K = \mathbb{F}_\ell[\hat w]$, $\hat w = w^{\ell^N}$; the P_j, $n_j = t_j - P_j(\hat w)$
5.3	[proved here] (v1.5.0)	$\dim_{\mathbb{F}_\ell} \bar A = \mathrm{lcm}_p \deg \pi_p$ on a *local* block, so budget mode's `residue_degree_lower_bound` is sharp there
4.1	[proved here]	$\mathrm{Ext}^1_{\mathbb{T}_\mathfrak{m}}(\mathbb{F}, \mathbb{F}) = (\mathfrak{m}/\mathfrak{m}^2)^\vee$
4.2	[proved here]	Yoneda class = structure map, complete invariant of two-step modules
§4.2	[standard, reference]	wildness for embedding dimension ≥ 2, Loewy length ≥ 3
4.3	[standard, reference] (reworded, v1.12.0; "cuspidal" read in characteristic zero, v1.12.1)	Eisenstein–cuspidal congruences (Ribet, Mazur), for E_k's own class ($U_\ell = 1$), $k \not\equiv 0 \pmod{\ell - 1}$; the literal pre-v1.12.0 "iff" over every twist fails at $(10, 7)$
4.4	[conjectural]	cuspidal–cuspidal extensions, T_ℓ usually active
4.5	[conjectural]	boundary/cone gluings
4.6	[standard, reference]; computed in E7 (accepted, slot 7; v1.8.2)	duality: Loewy layers $\leftrightarrow$ socle layers in reverse order
4.7	[conjectural]	Galois-cohomological interpretation of active directions
4.8	[standard, reference]	concentration at few $\mathfrak{m}$ for $n \gg \ell$

ref	status	content
4.9	[conjectural] as restated a fifth time (v1.25.0-pre.1, owner ruling of 2026-11-06: irreducible classes *of one weight*, replacing "not dihedral"); the v1.22.0-pre.1 form **refuted 2026-11-06** on cohomology at the non-ordinary class of f_{16} at $\ell = 59$ (four census-4.0 rows, $n = 884, 942, 944, 1000$, $(T_2 - a_2)^2 \neq 0$; `slot25/R59_verification.md`); restated a fourth time (v1.22.0-pre.1, owner ruling of 2026-10-09 on census finding F33-2: dihedral classes excluded); the v1.17.0-pre.4 form **refuted 2026-10-09** on cohomology at the dihedral classes at $\ell = 23, 31$ (161 census-3.3 rows, $A \cong \mathbb{F}_\ell[x]/x^3$; the slot-22 referee `F33-2_referee.md`); restated a third time at v1.17.0-pre.4 (owner ruling of 2026-10-02 on the referee F1's B-1); the v1.17.0-pre.2 Kummer-free form **contradicted 2026-10-02** on the forms side at irreducible congruence classes, $(2210/2278, 67)$, $(3798/3930, 131)$, $(4030/4170, 139)$ (139 regular; $\mathbb{F}_\ell[x]/x^4$; congruences confirmed over $\mathbb{Z}$; six further pairs the same; slot-17 R17-2 §7); the v1.16.0 form **refuted 2026-10-02** on cohomology at $(608, 547)$ ($g = 2$ first window, $A \cong \mathbb{F}_{547}[x]/x^3$, saturated; slot-17 R17-2) and contradicted on the forms side at the swapped classes at 37, 103, 691; the v1.14.0 statement **refuted 2026-10-01** at $\ell = 37$ (18 certified C1a records, the forms side, the $\mathbb{Z}/37^2$ lattice at $(654, 37)$)	(R): $\mathfrak{m}_A^2 = 0$ on every $\mathbb{F}_\ell$-rational non-Dickson H^1 block that is congruence-free (for every even $w \leq \ell + 1$ and every twist the generalised eigenspace of the system in $M_w(\mathbb{F}_\ell)$ has dimension ≤ 1; on reducible classes this is Kummer-free) and, when $\bar{\rho}_\mathfrak{m}$ is irreducible, of one weight (exactly one pair (w, j) of that test with a non-zero eigenspace; this excludes the dihedral, the non-ordinary and the companion classes); nothing asserted on the other classes; no counterexample in census 4.0 (672 rows of Loewy length ≥ 3, none in scope; 5805 irreducible rows of one weight with $\dim V \geq 6$, all with $\mathfrak{m}_A^2 = 0$). History: (v1.22.0-pre.1) the same with "not dihedral" ($\bar{\rho}_\mathfrak{m}$ irreducible with $\bar{\rho}_\mathfrak{m} \cong \bar{\rho}_\mathfrak{m} \otimes \omega^{(\ell-1)/2}$) for "of one weight"; no counterexample among the 100 025 non-Dickson, non-dihedral residue-degree-1 rows of census 3.3; (v1.17.0-pre.4) the same on congruence-free classes; (v1.17.0-pre.2) the same on Kummer-free classes (residual irreducible, or reducible with both seeds ℓ-regular), "in particular at every regular ℓ"; (v1.15.1-pre.7, released v1.16.0) $\mathfrak{m}_A^2 = 0$ off the Kummer–Frobenius locus; (v1.14.0-pre.2; held on 654/654 census rows, slot-13 WP-9) C-F9: $\mathfrak{m}_A^2 = 0$ on every $\mathbb{F}_\ell$-rational H^1 block at $\ell \geq 5$ (the lattice nearly split); not explained by the Serre-weight count; evidence non-Dickson only, zero evidence on a non-reduced Dickson-type $\mathfrak{m}$ (v1.14.0-pre.3); a third route and 452/452 blocks on 348 cells, the 206 Dickson-type blocks read all reduced (v1.15.0, slot-14 WP-M1, WP-6); 1233/1233 non-reduced blocks beyond the census window on the first cluster, `noctua2_section_m/` (v1.15.1-pre.3, P12)

ref	status	content
4.10	[conjectural] (v1.15.1-pre.7; restated v1.17.0-pre.1; hypothesis $g_{\ell+1-w} = 1$ added v1.17.0-pre.3)	on a Kummer–Frobenius $\mathfrak{m}$ with $1 \le t \le J(w) - 1$ and $g_{\ell+1-w} = 1$, $A \cong \mathbb{F}_\ell[x]/x^4$ in the window $\ell w + t(\ell+1) \le k < w + t(\ell+1) + (\ell^2 - 1)$ ("from k on" withdrawn: forms side $(1, 2, 2, 1)$ at $k = 1412$, 1450); 24/24 onset hits at $\ell = 37$, 0 false positives; (v1.17.0-pre.2) the lower end held off 37: $(736, 67)$ on cohomology, $(804, 67)$, $(1002, 59)$, $(1062, 59)$, $(3534, 101)$ on the forms side; the first Loewy-4 cell at 103 is the swapped $(2574, 103)$, outside the prediction; (v1.17.0-pre.3) nothing asserted at $g \ge 2$: the pre.2 form failed at $(547, 486)$ ($(608, 547)$ has Loewy length 3 below the onset, $\dim V = 12$ in the window); (v1.18.0-pre.1, generalised, [conjectural]) (A) as above; (B) the swapped family at $g = 1$, the same window with $w = w_K$ (forms side: onsets at 37, 103, 691; last cell and first cell past the window at $(1400, 37)$, $(1438, 37)$, $(10632, 103)$ / $(1436, 37)$, $(1474, 37)$, $(10734, 103)$); (C) $g = 2$, $A \cong \mathbb{F}_\ell[x]/x^3$ in the first window, both families ($(608, 547)$ on cohomology, six forms-side cells; v1.18.0-pre.2: three more, the $a = 1$ window sampled at cells 1–4, 31, 62 of 62; (B) five more at 37, 103, the referee F1); nothing asserted at the $g = 2$ meeting (there Prediction 4.18's datum $\mathbb{F}_{547}[x]/x^6$), at $g \ge 3$, or past the windows
4.14	[proved here] (v1.18.0-pre.1; slot-17 R17-1 Lemma 3.1)	two classes $1 \oplus \omega^{w-1}$, $1 \oplus \omega^{w'-1}$ agree up to twist iff $w' \in \{w, \ell + 1 - w\}$; the Kummer classes of a pair are the two Eisenstein ends, the standard and the swapped family, each with one presentation $\omega^t(1 \oplus \omega^{w-1})$, $1 \le t \le J(w) - 1$
4.15	[proved here] (given the class number formula, [standard, reference], locator not verified; v1.18.0-pre.1; R17-1 Lemma 3.2)	$\ell \equiv 3$ (4), $\ell > 3$: $\ell \nmid B_{(\ell+1)/2}$
4.16	[proved here] (given [standard, reference] inputs; Edixhoven Thm. 2.6: attribution verified in a secondary source, locator pending (v1.18.0-pre.2, R18-4); second locator Berger–Li–Zhu Thm. 4.5 for $w \le \ell$; v1.18.0-pre.1; R17-1 Theorem 4.1)	$\dim H^1(\Gamma; M_n)_\mathfrak{m} = m_1(1 + 2g_{w_1}) + m_2(1 + 2g_{w_2})$ on every reducible non-Dickson $\mathfrak{m}$, $\ell \ge 5$; cuspidal: torsion-free, $\operatorname{rank} \mathbb{T}_\mathfrak{m} = \frac{1}{2} \dim$; census check 21 690/21 690
4.17	[proved here] (from Claim 4.16) and computed (v1.18.0-pre.1)	the first and second windows on the multiplicity side, 1492/1492 standard and 1906/1906 swapped classes of 13 pairs; the exact event sequence checked by `wp3/window_check_f1.py` (v1.18.0-pre.2)
4.18	[conjectural] (v1.18.0-pre.1; the $\ell = 37$ report's `conj:depth`)	$\mathbb{T}_\mathfrak{m}/\ell \cong \mathbb{F}_\ell[x]/x^{mr_0}$ for $m \le 2$ on Kummer classes ($r_0 = 1 + g$) and irreducible congruence classes ($r_0 = 2$); 73 curvilinear blocks on the forms side, 26 on cohomology (v1.18.0-pre.2; pre.1: 65); fails at $m = 3$; $(m, r_0) = (2, 3)$ held at $(34460, 547)$, $\mathbb{F}_{547}[x]/x^6$ (v1.18.0-pre.2, the referee F1; pre.1: untested), so the hypothesis is $m \le 2$, not $mr_0 \le 4$
4.19	[computed and compared] (1),(3)–(5), slot-19 R19-1, two routes; [proved here] (2) from [C, `prop:cohforms`] (v1.19.0-pre.1)	$(258, 37)$: $\mathbb{T}_\mathfrak{m} = \mathbb{Z}_{37}[T_2 - 29]$ the full Hecke algebra, $= \mathbb{T}^{\mathrm{f}}_\mathfrak{m}$; $\tilde{\mathbb{T}}/\mathbb{T}_\mathfrak{m} \cong \mathbb{F}_{37}[\varepsilon]/\varepsilon^2$; depth 1/2; the over-orders $\mathbb{T}_\mathfrak{m} \subsetneq R = (\mathfrak{m} : \mathfrak{m}) \subsetneq \tilde{\mathbb{T}}$, all Gorenstein (Bass)
4.20	[proved here] given [computed and compared] inputs and [standard, reference] (Bass, §6, locator pending, not in the library; the Gorenstein input proved inline at pre.2) (v1.19.0-pre.1); the sign [computed and compared] (WP-LC, pre.3)	$\Lambda_\mathfrak{m} \cong \mathbb{T}_\mathfrak{m} \oplus \mathfrak{m}$; $\Lambda^+_\mathfrak{m} \cong \mathbb{T}_\mathfrak{m}$, $\Lambda^-_\mathfrak{m} \cong \mathfrak{m}$

ref	status	content
4.22	(a) [proved here]; (b)–(e) [proved here] given the [computed and compared] inputs (v1.19.0-pre.1); run by slot-19 WP-LC: held at $M = 1, \ldots, 4$, [computed and compared] (pre.3)	$H^1(\Gamma; M_{258} \otimes \mathbb{Z}/37^M)_{\mathfrak{m}}$, $M \leq 4$: rank, $\mu = 3$, Smith valuations of t and U_{37}, per sign, $T_p = Q_p(t)$
4.23	[computed and compared] (slot-19 WP-LC; F4 m5–m7) (v1.19.0-pre.3)	$\ell = 7$: per-sign freeness over $\mathbb{T}_{\mathfrak{m}}$ fails on 5 of 6 signs at $(48, 7)$, $(90, 7)$; $(48, 7)$ plus $\cong \mathbb{Z}_7[t/7]$, told from $\mathcal{O}_{\max}$ by U_7
9.1, 9.2	[specification] (v1.19.0-pre.1; kind congruence added at pre.2, F1 M-2)	the L-value interface: domain, kind (linear, bilinear, congruence), normalisation record, evaluation, index statement; Birch (linear) and Hida (bilinear) shapes declared at pre.1, implemented at pre.6 (below); Eisenstein = congruence; v1.19.0-pre.5: research prototypes of (B), (H) [computed and compared] (slot-19 R19-8, R19-9), (H)'s constant from Hida Thm. 5.1 (read), slot-20 specs; v1.19.0-pre.6: (B), (H) implemented (slot-19 WP-LIN, WP-BIL), gates [computed and compared]; (H2) at depth 2 [conjectural]
9.3	[computed and compared] at 8 cells, (L2) [proved here] from (L1); [conjectural] at depth ≥ 2 (v1.19.0-pre.6, WP-LIN)	leg (c) on H^1_c: $r^+ \equiv u\, p_0$, $r^- \equiv s\, \mathrm{Bern}_k$ modulo ℓ to depth $v(B_k/k) = 1$; twisted values of $r^+ \equiv 0$; v1.20.0-pre.1: computed to depth 2 at $(282, 37)$, $(226, 103)$ (slot-20 R20-1), still [conjectural] there; v1.20.0-pre.6: (L1) [computed and compared] at depth 2 there (WP-LV3, C++), item 3 still [conjectural]; v1.22.0-pre.2: (L1) at depth exactly 1 at $(66, 37)$, $(102, 37)$ computed (the P5″ referee's Python, one route)
9.8	[specification] (v1.20.0-pre.1, owner O5)	a measure is a family of linear evaluations, one per ball; index statements are congruences of measures; the accessor `measure()`
9.9, 9.10	definition; [proved here] (v1.20.0-pre.1, slot-20 R20-1); the analytic constant has (B)'s status; $\tau(\chi)$ at ℓ-power conductor [computed and compared] (v1.21.0-pre.3, slot-21 P4)	the ℓ-stabilised MTT measure $\alpha^{-m} c(a, m) - \alpha^{-m-1} \ell^{k-2} c(a, m-1)$: a measure, its moments, $E_{\mathbf{1},j} = (1 - \ell^j/\alpha)(1 - \ell^{n-j}/\alpha)$, $E_{\chi,j} = \alpha^{-m}$
9.20, 9.21	definition; [proved here] (v1.20.0-pre.5, slot-20 S1; elementary)	the branch series modulo (ℓ^R, ω_{M-1}) as T-coefficients; $\mathrm{prec}(0) = R$, $\mathrm{prec}(t) = \min(R, M - 1 - \lfloor \log_\ell t \rfloor)$; the record `lfunction/v1`
9.23	[computed and compared] at $(10, 691)$ (v1.20.0-pre.5, slot-20 S2; predictions and the Python oracle committed first)	Δ's MTT measure $\equiv -s\, G_{a+1} G_{11-a} \pmod{691}$ on 343/343 off-pole branches; $\lambda = 1$ at $\{199, 501\}$; pole points a unit off
9.24	over $\mathcal{O}$ [proved here] (coefficient change); the cells [computed and compared]; Hida modulo $\mathfrak{m}$ [computed and compared] at one component; F-P3-1 [computed and compared] as a census (v1.21.0-pre.1, slot-21 P3)	the measure over $W(\mathbb{F}_q)$ and totally ramified $\mathbb{Z}_\ell[x]/(F)$; refused: residual multiplicity ≥ 2, non-ordinary
9.25	[specification] (v1.21.0-pre.1); first run 603 checks, 0 failing; "pending P2/P4" replaced by pointers (v1.21.0-pre.3)	P3's gate `HeckeLValueCoefficientRingValidation` (`standard`), arms (R0)–(R8)

ref	status	content
9.26	assembly and P1 agreement [proved here]; items 1–3 [conjectural] (rest on [U, `pr:u-msym`] (U16)), evidence [computed and compared] as a run; item 4 [standard, reference] (recalled) and [computed and compared] as a run (v1.21.0-pre.1, slot-21 P9)	the two-variable family by interpolation; the Λ-adic Eisenstein congruence at $X = \alpha_a$ modulo 37^3, 103^2; weight two and the $\mathcal{L}$-invariant of 11a
9.27	(1), (2), (4)'s constant, (5), (6) [computed and compared]; (2)'s twisted masses [proved here] from the convolution form and [standard, reference]; (3) [proved here] given [standard, reference] (Washington §12.2, recalled); (4)'s shape [proved here] for the branch -1; (6)'s valuation $d - 1 - v(r)$ [proved here] (v1.21.0-pre.3, slot-21 P2)	the Eisenstein measure: Dedekind–Rademacher symbol, $\mu^{\mathrm{Eis}} = -E_1 \star \iota_* E_{k-1}$, branch series $-G_{a+1}G_{k-1-a}$ in $\Lambda \otimes \mathbb{Q}$, the polar term Res, $v(\mathrm{Res}) = d$; the cusp form's pole branches; the located $14 + 4$ derived
9.28	[specification] (v1.21.0-pre.3); first run 225 checks, 0 failing (heavy 259); with E8L2 / E9L2 239 (273)	P2's gate `HeckeLValueEisensteinMeasureValidation` (`standard`), arms (E0)–(E9)
9.29	(1), (2), (3), (5), (6), (7) [computed and compared]; (4) [proved here] (Weierstrass) and [computed and compared]; (3)'s proof not given (v1.21.0-pre.3, slot-21 P4)	wild characters $\omega^i\psi_t$ of conductor ℓ^m: $\tau(\chi)$ against complex L-values, root number $\tau(\chi)^2/M$; Proposition 9.10(4); $\lambda_{\chi,j} = -\varepsilon\lambda_{\bar{\chi},n-j}$; $v_\pi = e\mu + \lambda$; Kubota–Leopoldt at $\zeta^t u^r - 1$; the polar term visible at every wild critical point, at $(226, 103)$ too since v1.21.0-pre.7 (P4x, 128-bit)
9.30	[specification] (v1.21.0-pre.3); first run 201 checks, 0 failing (heavy 238); one prediction wrong (the root number); with P4x 222 (heavy 259)	P4's gate `HeckeLValueWildCharacterValidation` (`standard`), arms (W0)–(W7)

ref	status	content
9.31	(1), (2), (3), (4), (5) and (7)'s formula [proved here] ((2) given Proposition 9.10(4) and Remark 9.29(3); (7) given Claim 9.11(3) and $\mu(G) = 0$); (6) [proved here] / [standard, reference] (Ferrero–Washington recalled; [19] read); (8), (9) records; every run [computed and compared] (v1.21.0-pre.7, slot-21 P5); (9)'s forced-zero theorem and (S) [proved here] modulo [standard, reference] inputs at the 30 computed cells with the project's own readings of (H1), +2 once (H1) is gated at $(66, 37)$, $(102, 37)$, given [computed and compared] L3 at the pair and [computed and compared] (H1) at the cell (v1.22.0-pre.2, corrected at pre.3, slot-22 P5″; per pair; uniformly still conditional on (S))	the tame functional equation at conductor 1, ℓ; $\mathcal{L}_b(T) = -(-1)^b \mathcal{L}_{n-b}(u^n(1+T)^{-1} - 1)$; (FE-inv), (FE-parity): λ odd at even self-dual branches; the near-trivial zero; the values oracle; $\lambda(G_b) = 1$ at the irregular pairs; level-1 Greenberg–Vatsal; the (μ, λ) tables; $\lambda^+ = 1$ at $\{-k, n+k\}$ observed
9.32	[specification] (v1.21.0-pre.7); first run 580 checks, 0 failing (heavy 658); two negative controls repaired, no predicted value changed	P5's gate `HeckeLValueIwasawaInvariantsValidation` (`standard`), arms (I0)–(I5)
9.11	[computed and compared] at 3 cells (v1.20.0-pre.1, slot-20 R20-1); shape [proved here] given its inputs; the pole branches an exception, computed; unexplained through v1.21.0-pre.2 (R20-L), explained by P2's polar term since v1.21.0-pre.3 (Proposition 9.27(6), [computed and compared])	$\mathcal{L}^-_{f,a} \equiv -s\, G_{a+1} G_{k-1-a} \pmod{\ell^d}$ off the pole branches, $r^- \equiv s P^{\mathrm{Eis}}$; $\mu = \lambda = 0$ on the minus side
9.12	(ii) [proved here] from (L1), [computed and compared]; (iii) an observation, no claim (v1.20.0-pre.1); its pairs $\{-k, n+k\}$ unconditional at the computed cells since v1.22.0-pre.2 (Remark 9.31(9))	$\mu^+ = d$; the reduced plus measure's $\lambda = 1$ branch pairs
9.13	[specification] (v1.20.0-pre.1)	WP-MEAS's gate `HeckeLValueMeasureValidation` (`standard`), arms (M0)–(M6); (M7) and the env arm (M691) (v1.20.0-pre.5, S1 / S2)

ref	status	content
9.14	(C) [computed and compared]; (L), (H) and the triangle [conjectural] (v1.20.0-pre.1); (L1), (H) and the triangle [computed and compared] at the two cells (v1.20.0-pre.6, WP-LV3), (L)'s item 3 and the general statement [conjectural]	at $(282, 37)$, $(226, 103)$ the congruence, linear and bilinear depths are all 2; $\mathfrak{m}_{\mathrm{Eis}}$ the joint ideal in T_2, T_3, T_5
9.15	[specification]; $c_f = 0$ for $n < \ell$ [proved here]; the value open (v1.20.0-pre.1); well-definedness (the f-line carried onto the f-line) [proved here] from [C, `lem:symdual`] (v1.20.0-pre.2); value by Proposition 9.16 (v1.20.0-pre.6)	the change-of-lattice constant between $\theta_* H^1(M_n)$ and $H^1(D_n)$ on the f-line
9.16	[proved here] (v1.20.0-pre.6, slot-20 R20-3)	θ_* Hecke- and ι-equivariant; $c_f^{\varepsilon} = e\, v_\ell(n!) - d_f^{\varepsilon}$, $0 \le d_f^{\varepsilon} \le e \max_t v_\ell\binom{n}{t}$; $d_f^{\varepsilon} = 0$ if $H^1(M_n/V_n')^{\varepsilon}_{\mathfrak{m}_f} = 0$
9.17	[computed and compared] (v1.20.0-pre.6; R20-3 two routes, WP-LV3 C++); the $d_f = 1$ twist rule [conjectural] ([computed and compared] at three cells)	$c = 7$ at $(282, 37)$, 2 at $(226, 103)$, 6 ($v_\pi = 12$) at $(258, 37)$, 0 for $n < \ell$; $d_f = 1$ on five lines at $(40, 37)$
9.18	[proved here] from [C, `prop:iotasplit`], [C, `prop:iotacommutes`] (v1.20.0-pre.1)	ι over $\mathbb{Z}/\ell^k$, ℓ odd: nothing new mathematically; the code honours it (WP-LV3, built at v1.20.0-pre.6; `hecke_involution_ring/v1`)
9.19	[specification] (v1.20.0-pre.1); outcome [computed and compared] (v1.20.0-pre.6, every gate held)	WP-LV3's gates: generic reader, cross-kind gate, c_f, ι over $\mathbb{Z}/\ell^k$, the twisted rung $(37; 62, 6)$
9.5	(1), (3) [proved here] from [U, `pr:u-U1`]; (2) [standard, reference] for $n < \ell - 1$ (Harder, [locator not verified, rule A2]), [conjectural] beyond (v1.19.0-pre.1)	the Eisenstein congruence module of the cuspidal sign is $\mathbb{Z}_\ell/(B_k/k)$; modulo ℓ^m on a saturated block (`stable` at $K = 1$; pre.2, F1 M-1), a two-sided bound otherwise
9.6	[proved here] given [standard, reference] (Hida's control at nebentypus points, ENS 1986 Cor. 3.2 (3.3b), verified on the page by F1 at pre.2) (v1.19.0-pre.1)	twisted: $\mathbb{T}^-/I_{\mathrm{Eis}} \cong \mathbb{Z}_\ell/(B_{k,\chi}/k)$ at $\Gamma_1(\ell)$, $\chi = \omega^j$; v1.19.0-pre.5: [computed and compared] at 4 cells ($\ell = 37, 59, 67$, $v = 1$) and 2 controls (slot-19 WP-LV2); v1.20.0-pre.6: [computed and compared] at the depth-2 rung $(37; 62, 6)$, $n = 60$ (slot-20 WP-LV3(e))
§9.4	[specification] (v1.19.0-pre.1); built (v1.19.0-pre.3, WP-NF)	the number-field layer (WP-NF): objects, the local test of R18-2 ([proved here] there), the gate
4.11	[proved here] (from [3]; v1.16.3, owner O4)	I-8's split / ramified switch is not Bellaïche–Khare obstructedness: both cases unobstructed (Thm. 22 (i), (iii) with Vandiver)

ref	status	content
4.12	[conjectural] (v1.16.3, owner O4); proof [proved here] given $D(n_V) = 4$ and reduced n_0 blocks; the onset [conjectural]	the Frobenius layer (F1, law L-F1): the $\bar{\rho}_f \otimes \omega^j$ block doubles to $D = 4$ at $n_V = \ell k_0 + j(\ell+1) - 2$ and is hidden from there; WP-12 test 0 failures at 17, 19; slot 16: 3/3 hidden-chain cells; slot 19: 9/9 irreducible rungs ≤ 1220 hidden at $(1220, 37)$, 0 spurious (v1.19.0-pre.5); slot 20 (v1.20.0-pre.3): family-(c) onsets, $\theta^j S_{k_0} = \ker x^2$ on forms — the transport step under a non-reduced source, outside the hypotheses; slot 20 (v1.20.0-pre.6): WP-SW's 1258, 1296, 1334 at 37, 9/9 rungs hidden at each; one unpredicted hidden block at 1334 from $k_0 = \ell + 1 = 38$, explained by the mechanism from $n = 1296$ (the early V-copy); v1.20.0-pre.7: refuted as stated at $k_0 = \ell + 1$ (one class, $(1334, 37)$), hypothesis restricted to $k_0 \leq \ell$
4.13	[computed and compared] at $(1220, 37)$; [conjectural] beyond (v1.19.0-pre.5; slot-19 sat1220)	the Frobenius-layer rung of a Kummer class is the swapped family's $D = 8$ block, with $\mathbb{T}/\ell = \mathbb{F}_\ell[x]/x^4$, $A_1 = \mathbb{F}_\ell[x]/x^3$, $I = \mathfrak{m}^3$, $\mathbb{T}^{\mathrm{sat}}/\mathbb{T} \cong \mathbb{Z}/\ell$; v1.20.0-pre.1: θV_{src} is not the socle ([computed and compared], one run); v1.20.0-pre.2: consistent with Prediction 4.12 "Why (ii)" (its reduced-source hypothesis fails here); v1.20.0-pre.6: the swapped cells $b = 2, 3, 4$ at 37 as stated ([computed and compared], WP-SW)
7.1 (WP-6 paragraph)	definitional (v1.15.1-pre.2); the `stable` verdict certified by [C, `cor:stablecert`] (B1 form) and the digit route beyond 2^{64} (v1.15.1-pre.5, slot-15 WP-5); `HeckeRingLoewyValidation` (`standard`, 181 checks)	A_K, $e_{\mathfrak{m}}$, ε_K over $\mathbb{Z}/\ell^K$ read mod ℓ; `stable` verdicts certified ($\mathbb{T}'_{\mathfrak{m}'}/\ell$ for $\mathbb{T}' = \mathbb{Z}_\ell[T_p : p \in S]$; generation not certified)
5.2	[proved here]	locality certificate with the deterministic P_j recipe; witness ranges; strict mode (Frobenius radical, Berlekamp subalgebra, witness over $\bar{A}$, terminates) specified v1.4.0, implemented slot-3 WP-3
5.6	[proved here], implemented (slot-2 WP-4)	cyclicity $\Rightarrow \mathbb{T}_{\mathfrak{m}} = A_S$; the reported generator is canonical — the least standard basis vector outside $\mathfrak{m}_S V$, *stated* v1.4.1
5.7	[proved here] (statement and socle bound); the cost [to be verified computationally] discharged v1.4.0 ($u \leq mD$, slot-2 WP-5)	commutant bound; the socle bound $\dim C \geq \dim A_S + fts - \dim \operatorname{soc} A_S$ decides when (b) can succeed
5.8	[proved here] (given $B'(k, \ell)$; range of Claim 2.10); implemented on real blocks, slot-3 WP-2	Sturm sweep with the computed bound
5.12	[proved here]	$\mathbb{F}_{\ell^r}$-ranks from $\mathbb{F}_\ell$-ranks
6.1	[proved here]	scalar extension commutes with all invariants; Jordan types in units of r
6.3	[proved here]	canonical defining polynomial, deterministic
8.1	[proved here], verified numerically	$\mathbb{T}_{\mathfrak{m}} = \mathbb{F}_{691}[\varepsilon]/\varepsilon^2$; the identities for $q = 3, 5, 7$ and $U_{691} = I + 221E_{31}$ hold on the stored matrices
§8 Kummer cells	**refuted 2026-09-16**, corrected (v1.4.0)	"$r^1 = 0$ on the Eisenstein class, so $H^1_! = H^1$" is false on all ten cells: $\operatorname{rank} r^1 = 1$, $\dim H^1_! = \dim H^1 - 1$; what holds is $r^1 = 0$ on the socle E
(D6)	[standard, reference] (socle semantics kept) / sharpened 2026-09-16	the joint eigenspace on the unsplit H^1 is two-dimensional at every Kummer cell; the Eisenstein line is per sign; $\mathbb{Z}/\ell^2$ is a refinement (E8; v1.9.0: computed since slot 8, in the validations and golden only, §7.1 (d))
§8 others	[to be verified computationally]	expected outcomes of the remaining test cases

B Audit response

Items are numbered as in `hecke_extensions_audit.md` (§1 claim table, §3–§4 findings, §7 amendment list). Every numerical claim of the audit that this revision relies on was recomputed independently (Miller-basis Hecke matrices, lattice indices, $B'(k,\ell)$, $\tau(691)$, Bernoulli numerators, and the pipeline reruns at $(10,691)$, $(22,5)$, $(26,7)$, $(32,131)$); discrepancies are noted.

item	severity	status	where it landed / remarks
B1 Sturm	blocking	ADOPTED WITH CHANGE	Claim 2.7(a)/(b), Definition 2.8, Claims 2.9, 2.10, Certificate 5.8, schema. All indices $(3,9,27,400,2240,40,80,36680,45)$, the cuspidal index 1 and the 2-of-3 spans reproduce. *Change*: B' is defined through the $\mathbb{F}_\ell$-*algebra* generated by the T_q, $q \le B'$ prime (what the certificate tests), not the linear span of T_m, $m \le B'$; hence $B'(34,131) = 2$, and $(32,131)$ breaks the $\mathbb{Z}$-module statement only, while $(22,5)$, $(26,7)$ break the certificate on the pipeline's own H^1 (§8). The audit's remark that 131 is an Eisenstein-congruence prime for $k = 34$ is not adopted: $131 \nmid \mathrm{num}(B_{34}/34)$.
B2 tangent space	blocking	ADOPTED	Remark 3.2, Claim 3.3 ($\hat{w} = w^{\ell^N}$ with $f \mid N$ so that $\hat{w} \equiv w$), Algorithm 5.5(2)–(3), (D4), Certificate 5.2 (the deterministic P_j recipe; the witness-range remark is stated with the corrected condition $\min(f, f') \le \ell$ for separation, and "at most ℓ prime factors of f" for primitivity, in place of the audit's "$f \le 100$"), Algorithm 5.1(2) (bounded attempts, `"local": false`).
M1 boundary	major	ADOPTED	[C, `cl:chi`] (χ_b, same multiset on $H^1(\partial)$, χ_0 for $n < \ell$; the H^0 and H^1 computations re-derived by hand) and (A1) (left-coset representatives $\mathrm{diag}(1,q)$, $\left(\begin{smallmatrix} q & j \\ 0 & 1 \end{smallmatrix}\right)$).
M2 duality	major	ADOPTED	[C, `cl:pd`], Prediction 4.6, identity 5 of §5.1.
M3 Jordan units	major	ADOPTED	Claim 6.1 (statement and proof), (D3).
M4 Eisenstein class	major	ADOPTED (option b)	(D6): the Eisenstein line is $\mathrm{soc}(X^+_{4,\mathfrak{m}})$ when one-dimensional, undefined otherwise; Algorithm 5.11(4); schema field `eisenstein`. A mod-ℓ^2 run is Open Question 18.
M5 schema	major	ADOPTED	§7: `"691": "active"`, `"H1_shriek": "all"`, `"eisenstein": "bottom layer"`; added `local`, `coefficient_field` (P_j), structure-map `matrices`, `sturm` (B'), `commit`, `model`, `iota_sign`.
Am. 8 U_{691}	minor	ADOPTED	Claim 8.1 and its "Verified" paragraph: $c_{691} = 542$, $U_{691} = I + 221E_{31}$ reproduced on the pipeline in this revision.
minor 1	minor	ADOPTED	[C, `cl:bockstein`]: $n > 0$.
minor 2	minor	ADOPTED	Claim 1.1 and §8: $(24,103)$, $(24,2294797)$; both divisibilities recomputed.
minor 3	minor	ADOPTED	[C, `cl:amalgam`]: Brown VII.9; 12-torsion argument for all n.
minor 4	minor	ADOPTED	Claim 2.13 ("for $n \gg \ell$"); $(160,691)$ tagged [to be verified computationally] in §8 and Appendix A.
minor 5	minor	ADOPTED	Stage 6: parenthetical replaced by the Δ mod 11 counterexample.
minor 6	minor	ADOPTED	Certificate 5.8: $S_{\mathrm{sep}}(\ell)$ of Claim 2.5 in place of the undefined $S_{\mathrm{Eis}}(\ell)$.
minor 7	minor	ADOPTED	Stage 7: $d \approx n/3$ for all six groups and both signs.
minor 8	minor	ADOPTED	Claim 3.1: both clauses required for the "iff".
minor 9	minor	ADOPTED	Remark 3.2, referenced from Claims 2.2, 4.2, 6.1.
minor 10	minor	ADOPTED	§4.2: Drozd 1972, Ringel 1975.
minor 11	minor	ADOPTED	Stage 1: Θ cost plus the fixed cohomology cost.
§2.2 items	—	ADOPTED	Stage 0: the 7-vertex complex named as the substitute for the $\Gamma(2)$ model; the other items were already declared.
§6 concision	—	ADOPTED IN PART	§0.3 shortened; the decision box after Claim 3.1 reduced to a pointer; §8 rows at $\ell = 691$ merged into one table; the T_ℓ paragraph kept in Claim 2.6 only.
OQ13	—	SETTLED	$691 \parallel \mathrm{num}(B_{200}/200)$. The audit's residue 259816 is the *signed* numerator modulo 691^2 and ours, 217665, its absolute value ($B_{200} < 0$; $217665 + 259816 = 691^2$): the discrepancy is the sign, not a disagreement (v1.12.0; through v1.11.1 this read "does not reproduce … in both normalisations"). The uniqueness at $(198,691)$ is closed (§8, v1.12.0).

No item is contested. The programme's certificates that changed: (c) now needs a recorded $B'(k,\ell)$ and is otherwise unavailable; the locality certificate and Stage 3 need the coefficient field $\hat{w}$; the locality loop has a bounded exit; Open Question 7 is moot.

C OQ2 verdict response and the owner's decisions

Open Question 2 — the Hecke action on the mapping cone — was reviewed independently; the verdict is `docs/implementation/hecke_extensions_OQ2_verdict.md` (CONFIRM WITH AMENDMENTS), and this revision applies it. The binding statement of the adopted formulation is amendment C-A3 of `docs/implementation/hecke_extensions_implementation_plan.md` §0.4, which is the mechanism by which [C] gains it; this document states it in (A2). **The eight points of the verdict's §8 were decided by the owner on 2026-09-15**, and v1.2.2 applies those decisions: the rows below that v1.2.1 labelled *working assumption* are now plain ADOPTED, with the decision and its date. The owner followed the verdict's recommendation on every point, with three amendments — *both* cone cross-routes are built rather than one; the boundary group is $\Gamma_\infty = \langle T\rangle \times \langle -I\rangle$; and the sign convention is confirmed as already applied. The decision record, one row per point with the site of each decision, is §11 of the implementation plan; the two amendments that bear on this document are written into (A1) and (A2). The status word in the second column records what this document *states*; where a decision leaves a construction open, the row says so.

item	status	where it landed / remarks
formulation (b*)	ADOPTED	(A2): the datum is the triple $(\tau, \tau_\partial, \eta)$ over $\mathbb{Z}$, the cochain homotopy $h = \tilde{\eta}$ is derived, the cone map is $\Phi_g(a,b) = (\Theta_g a, \Theta_g^\partial b + ha)$. Formulations "on the nose" and "strict map of pairs" are the special case $\eta \equiv 0$; the higher-coherence formulation is not needed (the deliverable is cohomology-level).
sign	CONFIRMED	$dh + hd = \Theta_g^\partial r - r\Theta_g$ with $D(a,b) = (-da, db + ra)$, re-derived independently: $D\Phi - \Phi D = (0, (dh + hd + r\Theta - \Theta^\partial r)a)$. The plan audit's opposite sign was wrong.
$\mathrm{Hom}(\ker r^1, \mathrm{coker}\, r^0)$	NEW, STATED	(A2) third bullet and Prediction 4.5: real for a free cochain-level h (*every* element occurs over a field, [proved here]), eliminated by the η-construction ([proved here] over $\mathbb{Z}$ for all ℓ). This closes the gap that v1.1 did not mention.
the entrywise test	DEMOTED	(A2) fourth bullet: $r\Theta_g = \Theta_g^\partial r$ is a recorded diagnostic, never a branch; $h = 0$ is admissible iff $\tau j = j\tau_\partial$, which the shipped τ fails.
boundary datum	ADOPTED (owner decision 2026-09-15, as recommended)	The normative object is the Γ-level τ_∂ on ∂X with the bulk's own s_i, δ_i; the one-dimensional τ_∞ of (A1) is its Shapiro transcription and a validation route. (A1)'s formulas are unchanged and remain the oracle.
integrality	ADOPTED (owner decision 2026-09-15, as recommended)	(A2): τ_∂ and η are integral for every ℓ; the bulk τ is integral exactly when no fill used the mod-ℓ fallback. When all three tables are integral, the matrices are their reductions, so the reduction maps and Bockstein connecting maps are equivariant on the nose. *Policy*: the mod-m fill fallback is **forbidden by default** — the integer tier is widened instead; it stays reachable behind a setting and then records a non-integral cell, but that is not the default escape, and a record claiming integrality must have solved all three tables over $\mathbb{Z}$.
sign convention	ADOPTED (owner decision 2026-09-15, confirming what was applied)	(A2) second bullet: $\partial\eta + \eta\partial = j\tau_\partial - \tau j$, hence $h = \tilde{\eta}$ with no sign. The cochain identity and the cone map are fixed by D and are unmoved by the choice.
cross-routes and the PD gate	ADOPTED WITH AMENDMENT (owner decision 2026-09-15)	Poincaré adjointness ([C, `cl:pd`], once D_n is computed) is the *acceptance gate* for X_3 at $\ell \geq 5$; and **both** cross-routes — the strict-pair τ^s and the relative complex — are built as validation arms, where the verdict recommended one of the two. They are cross-checks of the independence theorem, never definitions.

item	status	where it landed / remarks
boundary group at $\ell = 2$	ADOPTED WITH AMENDMENT (owner decision 2026-09-15; the verdict made no recommendation)	$\Gamma_\infty = \langle T \rangle \times \langle -I \rangle$, i.e. [C, `sec:six`]'s own definition $H^\bullet(\partial; M) := H^\bullet(\Gamma_\infty; M)$; the circle model of (A1) is its specialisation at $2 \in R^\times$. Consequence, stated in (A1) and (A2): at $\ell = 2$ the boundary — and with it the bulk and the cone — is *unbounded above in degree*, so [C, `eq:les`] is a window of a non-terminating long exact sequence. The degree bound / truncation policy that needs was settled in the contract: [C] [C, `open:ell2`], decided by the owner on 2026-09-16 as [C, `oc1a`] ($q_{\max}$ a recorded run parameter on the free Γ-model; this row said "open … not decided" until v1.3.3).
golden tiers	ADOPTED (owner decision 2026-09-15, as recommended)	$\dim X_3$, $\dim X_6$ and the char-poly / rank / trace of $A_c^{(q)}$ in golden tier A (canonical for $\ell \geq 5$); $\Theta^{\partial}_{\text{flat}}$, h_{flat}, the cone bases and the three table hashes in tier B. The cone bases sit in tier B because they are stored-basis dependent, *not* because a change of η or τ_∂ moves them — it does not.
LES in all degrees	ADOPTED; restated in [C]	v1.2 stated the long exact sequence in all degrees and named $H^0_c = 0$ and $H^2(\Gamma; M) = 0$ as the hypotheses under which it collapses to the six-term sequence. [C, `cl:les`] keeps the sequence and the collapse but not that pair of hypotheses: $H^0_c = 0$ is unconditional ([C, `lem:conerelative`](4)) and is dropped, while $H^2(\Gamma; M) = 0$ is kept and shown not removable ([C, `lem:h2torsion`](2)). See item 3 of §1.1.
torsion primes	ADOPTED (owner decision 2026-09-15, as recommended)	(A2), last paragraph: the formulation transfers verbatim to a free Γ-model of the pair (Borel construction), which is the orbifold cohomology at $\ell \in \{2, 3\}$; the 7-vertex complex is its specialisation where 6 is invertible. Moerdijk–Pronk's complex is a *computational model* of it: their charts are effective and the independence proofs for their μ are omitted, so the citation for the general case is [conjectural] and the safe normative definition is the Borel construction they themselves name. The general model enters the binding contract *now, as definitions*, with the construction deferred.

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[19] W. Hart, D. Harvey, W. Ong, *Irregular primes to two billion*, Math. Comp. 86 (2017), no. 308, 3031–3049 (v1.16.3; data as verified in `docs/latex/loewy_and_galois_refs.bib`).

[20] G. Harder, *Cohomology of Arithmetic Groups*, Vol. III, manuscript, 2022 (`docs/literature/`).

[21] H. Hida, *Congruences of cusp forms and special values of their zeta functions*, Invent. Math. 63 (1981), 225–261 (v1.19.0-pre.1: cited for the shape of Hida's congruence criterion only; v1.19.0-pre.5: Thm. 5.1, pp. 247–248, (3.2), p. 237, Thm. 2.5 and (2.8), p. 236, Thm. 6.1, p. 250, read on the page by slot-19 R19-9; PDF page = journal page −224).

[22] H. Hida, *Iwasawa modules attached to congruences of cusp forms*, Ann. Sci. ENS (4) 19 (1986), 231–273 (v1.19.0-pre.1; Thm. 3.1 and Cor. 3.2, p. 250, verified for [U] and [C]; their nebentypus form at weights outside the class, used in Claim 9.6 — Cor. 3.2 (3.3b) for every $k \geq 2$ with p. 249's $S_k(\Gamma_1(N_0p), a; A) = S_k(\Phi, \omega^{a-k}; A)$ — verified on the page by the slot-19 referee F1 (v1.19.0-pre.2)).

[23] N. Jochnowitz, *Congruences between systems of eigenvalues of modular forms*, Trans. AMS 270 (1982), 269–285.

[24] F. Lübeck, *Conway polynomials for finite fields*, online tables; definition in D. Holt, B. Eick, E. O'Brien, *Handbook of Computational Group Theory*, §7.5.

[25] B. Mazur, *Modular curves and the Eisenstein ideal*, Publ. Math. IHES 47 (1977), 33–186.

[26] B. Mazur, J. Tate, J. Teitelbaum, *On p-adic analogues of the conjectures of Birch and Swinnerton-Dyer*, Invent. Math. 84 (1986), 1–48 (v1.19.0-pre.1; §8, the modular-symbol formula for $L(f, \chi, j+1)$, shape only; constants not checked, rule A2).

[27] W. McCallum, R. Sharifi, *A cup product in the Galois cohomology of number fields*, Duke Math. J. 120 (2003), no. 2, 269–310 (v1.21.0-pre.9: page range recalled; v1.21.0-pre.10: confirmed on the Project Euclid listing; §5, the relations and Thm. 5.1, and the remark after Cor. 5.2 read on the page of the authors' web PDF `math.ucla.edu/~sharifi/pairing.pdf`, version of 5 Dec 2003, pp. 12–14; not in the project library).

[28] L. Merel, *Universal Fourier expansions of modular forms*, in: On Artin's conjecture for odd 2-dimensional representations, LNM 1585, Springer, 1994, 59–94.

[29] K. Ribet, *A modular construction of unramified p-extensions of* $\mathbb{Q}(\mu_p)$, Invent. Math. 34 (1976), 151–162.

[30] C. M. Ringel, *The representation type of local algebras*, in: Representations of Algebras (Ottawa 1974), LNM 488, Springer, 1975, 282–305.

[31] J.-P. Serre, *Congruences et formes modulaires (d'après H. P. F. Swinnerton-Dyer)*, Sém. Bourbaki 416 (1971/72); and *Formes modulaires et fonctions zêta p-adiques*, LNM 350 (1973).

[32] R. Sharifi, *A reciprocity map and the two-variable p-adic L-function*, Ann. of Math. (2) 173 (2011), 251–300 (v1.21.0-pre.10, P5′ audit: the Introduction, (1.1), and §5 (the map ϖ, Prop. 5.7, Conj. 5.8) read on the page of the arXiv version `arXiv:0709.3591`; the journal copy not read, not held).

[33] R. Sharifi, *Iwasawa theory and the Eisenstein ideal*, Duke Math. J. 137 (2007), no. 1, 63–101 (v1.21.0-pre.9: page range recalled; v1.21.0-pre.10: confirmed on the Project Euclid listing; Introduction, p. 4, read on the page of the author's web PDF `math.ucla.edu/~sharifi/paireis.pdf`; not in the project library).

[34] G. Shimura, *Introduction to the Arithmetic Theory of Automorphic Functions*, Princeton UP, 1971.

[35] W. Stein, *Modular Forms, a Computational Approach*, GSM 79, AMS, 2007.

[36] J. Sturm, *On the congruence of modular forms*, in: Number Theory (New York 1984–85), LNM 1240, Springer, 1987, 275–280.

[37] L. C. Washington, *Introduction to Cyclotomic Fields*, 2nd ed., GTM 83, Springer, 1997 (table of irregular pairs).

[38] L. C. Washington, *Introduction to Cyclotomic Fields*, 1st ed., GTM 83, Springer, 1982 (v1.18.0-pre.1; Cor. 5.15, p. 61, verified on page images, slot-17 R17-3; v1.19.0-pre.3: pp. 63–65 (Thm. 5.18 and its proof), Prop. 8.12, p. 156, read by slot-19 R19-4; §13.7, pp. 299–300, read by the slot-19 referee F4).

[39] A. Wiles, *On ordinary λ-adic representations associated to modular forms*, Invent. Math. 94 (1988), 529–573 (v1.18.0-pre.1; Thm. 2.1.4, p. 561, verified on the page image, slot-16 L16-3).

D Revision history

This document follows [C]'s practice: a published version's text never changes under its own number. This appendix is the record of the versions. Its first subsection is the release record, one paragraph per version, newest first (up to 1.25.0-pre.1 it was the date field of the title page); the second quotes the title page of 1.25.0-pre.1.

D.1 Release record

Version 1.25.2 — 2026-11-09 (hecke-extensions slot 26, **signed by the owner 2026-11-09**: the owner-signed release of 1.25.2-pre.1; no change of content — header, date and this record only. Companion pointers: [C] 1.26.2, [U] 0.7.1.)
Version 1.25.2-pre.1 — 2026-11-09 (hecke-extensions slot 26, a **pre-release, not signed**: paths only, no change of content (repository layout, owner ruling 2026-11-09): the program's sources moved under `src/`; one line names `src/ModArithmetic/ENGINE_API.md`. No mathematical statement, label, tag, schema key, record key, golden line or digest changes. Companion pointers: [C] v1.26.2-pre.1, [U] 0.7.1.)
Version 1.25.1 — 2026-11-09 (hecke-extensions slot 26, **signed by the owner 2026-11-09**: the owner-signed release of 1.25.1-pre.1; no change of content — header, date and this record only; no mathematical statement, label, tag, schema key, record key, golden line or digest changes. Companion pointers: [C] v1.26.1, [U] 0.7.1.)
Version 1.25.1-pre.1 — 2026-11-09 (slot 26, a **pre-release, not signed**: no change of mathematical content: machine names and private paths removed for the public release (owner ruling 2026-11-09).)
Version 1.25.0 — 2026-11-07 (hecke-extensions slot 25, **signed by the owner 2026-11-07**: the owner-signed release of 1.25.0-pre.2; no change of content — header, date and this record only; no mathematical statement, label, tag, schema key, record key, golden line or digest changes. Companion pointers: [C] v1.25.0, [U] 0.7.0.)
Version 1.25.0-pre.2 — 2026-11-07 (**a pre-release, not signed, not yet seen by the owner**: front matter, section titles and layout only. The title page has a one-line date, the uniform author block and a generic abstract; the date field of 1.25.0-pre.1, one entry per version from 1.25.0-pre.1 back to v1.0, is moved here verbatim as the release record below (every entry boundary became a paragraph break; whitespace only: in 1.25.0-pre.1 the first nine entries were separated by line breaks and the rest ran on), and the earlier author line and abstract, whose paragraphs narrated versions 1.1 to 1.4.0, are quoted in the last subsection. The version tags are removed from eleven section titles; their earlier titles were "The dimension of a reducible block at every ℓ (v1.18.0-pre.1)" (§4.4), "The block $(258, 37)$ exactly: the Hecke algebra, its over-orders, the lattice (v1.19.0-pre.1)" (§4.5), "The Dickson–θ objects (v1.12.0-pre.1)" (§5.2), "The coefficient ring and the E8 objects

(v1.9.0-pre.1; v1.9.0; v1.10.0-pre.1; v1.10.0-pre.2; v1.10.0; v1.10.1; v1.11.0-pre.1; v1.11.0-pre.2; v1.11.0; v1.12.0-pre.1; v1.12.0-pre.2; v1.12.0; v1.12.1)" (§7.1), "The congruence-census catalogue `hecke_census/v1` (v1.12.0-pre.1)" (§7.2), "The L-value interface, and its Eisenstein instance (v1.19.0-pre.1)" (§9), "Measures: the measure-valued linear evaluation (v1.20.0-pre.1)" (§9.5), "The depth-2 triangle, the change of lattice, and ι over $\mathbb{Z}/\ell^k$ (v1.20.0-pre.1)" (§9.6), "The truncated one-variable ℓ-adic L-function as a record (v1.20.0-pre.5)" (§9.7), "Coefficient rings, the two-variable family, the Eisenstein measure, wild characters and Iwasawa invariants (v1.21.0-pre.1, pre.3, pre.7)" (§9.8) and "OQ2 verdict response (v1.2) and the owner's decisions (v1.2.2)" (Appendix C). No statement, label, number or status tag changes; the provenance notes at the start of these sections are unchanged. Companion pointers: [C] v1.25.0-pre.7, [U] 0.7.0-pre.4.)

Version 1.25.0-pre.1 — 2026-11-06 (hecke-extensions slot 25; **a pre-release, not signed**; on top of the signed 1.24.0; **owner ruling of 2026-11-06** on the verification `docs/implementation/slot25/R59_verification.md` of the gallery referee's finding R1, `docs/implementation/slot25/PG140_referee.md`). Prediction 4.9 (R) **restated a fifth time** (a statement change of a [conjectural], the tag kept): on irreducible classes the hypothesis "not dihedral" is replaced by "*of one weight*" — exactly one pair (w, j) of the congruence-free test with a non-zero generalised eigenspace in $M_w(\mathbb{F}_\ell)$; reducible classes unchanged (congruence-free alone). Of one weight implies not dihedral (proved in the statement), and excludes also the non-ordinary and the companion classes. New paragraph "Refuted on cohomology: the v1.22.0-pre.1 form, at a non-ordinary class": the class of f_{16} at $\ell = 59$ (irreducible, not dihedral, congruence-free, $a_{59} \equiv 0$, Serre weights 16 and 46), four census-4.0 rows $n = 884, 942, 944, 1000$ with $(T_2 - a_2)^2 \neq 0$ on V ($\dim V = 6, 8, 6, 8$), and the consistency of the new form with the 672 Loewy-≥ 3 rows and the 5805 irreducible rows with $\dim V \geq 6$ of census 4.0; the last sentence of the dihedral paragraph gains a pointer; the row of Appendix A updated. No label added, renamed or removed; no computed value, schema key, record key, golden line, digest, validation or code path moves. **Not yet seen by the owner as text.** Companion pointers: [C] v1.25.0-pre.7, [U] 0.7.0-pre.3.)

Version 1.24.0 — 2026-11-05 (hecke-extensions slot 25, **signed by the owner 2026-11-05**: the owner-signed release of 1.24.0-rc.2; no change of content — header and date only; no mathematical statement, label, tag, schema key, record key, golden line or digest changes. Companion pointers: [C] v1.24.0, [U] 0.6.0.)

Version 1.24.0-rc.2 — 2026-11-05 (hecke-extensions slot 25, RC124 rc.2; **the release candidate of 1.24.0: not signed**; rc.1 refereed ACCEPT_WITH_MINORS, `docs/implementation/slot25/RC124_referee.md`): Prediction 4.9, item 2 of "Refuted as stated" — "every saturation defect measured so far has length ≤ 1" is scoped to the Kummer classes, with a pointer to the longer defects elsewhere (length 2 at $(540, 23)$ in the same item; δ up to 190 in census 4.0); the referee's F5, **not yet seen by the owner**; the sentence supports an expectation, not a claim, so no tag or label moves. Pre.2 of Claim 2.7 (rc.1 below) was re-audited after A7 by the HeckeZ re-audit A7b (`heckez/A7b-Z0-reaudit.md` §3, ACCEPT; on the branch `s24-wip-heckez-z0`). Nothing else changes. Companion pointers: [C] v1.24.0-rc.2, [U] 0.6.0.

Version 1.24.0-rc.1 — 2026-11-05 (hecke-extensions slot 25, RC124; **the release candidate of 1.24.0: not signed, not yet refereed**, referee `docs/implementation/slot25/RC124_referee.md`): the text of Claim 2.7(a), its status line, Definition 2.8's parenthesis and the claims-table row exactly as in 1.24.0-pre.2 below, taken onto the main line from the branch `s24-wip-heckez-z0` (`hecke_extensions.tex` only; the erratum of pre.1 accepted by the owner, HeckeZ question Q9, 2026-11-04; pre.2's two corrections, after the HeckeZ referee A7 — whose record `heckez/A7-Z0-referee.md` is on that branch, not on the main line —, not yet seen by the owner). Nothing else changes: no other statement, label, tag, schema key, record key, golden line, digest, validation or code path. Companion pointers: [C] v1.24.0-rc.1, [U] 0.6.0.

Version 1.24.0-pre.2 — 2026-11-04 (hecke-extensions slot 24, HeckeZ Z0 fix round after the adversarial referee A7, `docs/implementation/slot24/heckez/A7-Z0-referee.md`; **a pre-release, not signed**): Claim 2.7(a), second sentence — for $k \equiv 2 \bmod 12$, where $B(k) = d + 1$, the condition equivalent to "$B = B(k)$ suffices" is $(a_d(f_0), a_{d+1}(f_0)) \neq (0, 0)$; pre.1's "$a_{d+1}(f_0) \neq 0$" is sufficient, not equivalent (A7-m6); the verified range is extended to even $4 \leq k \leq 70$ (A7-N8; both coefficients

are non-zero there). The status ([conjectural]) and every consequence are unchanged; no computed value, record, golden line, digest, validation or code path moves. Companion pointers: [C] v1.24.0-pre.1 (main line; the HeckeZ text is [C] v1.25.0-pre.0b on the branch `s24-wip-heckez-z0`), [U] 0.6.0.

Version 1.24.0-pre.1 — 2026-11-04 (hecke-extensions slot 24, HeckeZ package Z0; **a pre-release, not signed**, on top of the signed 1.23.0): the erratum to Claim 2.7(a), second sentence, accepted by the owner (HeckeZ plan, question Q9, signed 2026-11-04; `docs/implementation/slot24/heckez/ERRATUM_E_clsturm.md`). The argument via Δg assumed $a_0(g) = 0$ and is withdrawn; $\mathbb{T}_{\mathbb{Z}}(M_k) \otimes \mathbb{Q}$ is spanned by T_m, $m \le B$, for some finite B (proved), and $B = B(k)$ is [conjectural], verified for even $4 \le k \le 58$. Consequential edits: the claim's status line, Definition 2.8's parenthesis, the claims table row. A correction of a proof: no computed value, record, golden line, digest, validation or code path depends on the sentence. Companion pointers: [C] v1.24.0-pre.1 (main line; the HeckeZ text is [C] v1.25.0-pre.0 on the branch `s24-wip-heckez-z0`), [U] 0.6.0.

Version 1.23.0 — 2026-11-03 (hecke-extensions slot 24, **signed by the owner 2026-11-03**: the owner-signed release of 1.23.0-pre.1; no change of content — header and date only; no mathematical statement, label, tag, schema key, record key, golden line or digest changes. Companion pointers: [C] v1.23.0, [U] 0.6.0.)

Version 1.23.0-pre.1 — 2026-10-26 (hecke-extensions slot 24, D6 code audit finding M2; **a pre-release, not signed**). §7.1 (a) "Reserved values" and the campaign paragraph of the as-built list now agree with [C, `sec:recordschema`] item 7 and [C, `def:ringsemantics`](3): `Z` is no longer refused inside `MathSettings::set()`; it is stored and honoured by the integral descent only, every other consumer refuses it by name (`NotImplementedInThisBuild`, exit 5); `Z/<m>` stays reserved. Wording only: no mathematical statement, label, tag, schema key, record key, golden line or digest changes. Companion pointers: [C] v1.23.0-pre.37.

Version 1.22.0 — 2026-10-12 (hecke-extensions slot 22, CLOSE-T: the **release candidate** from 1.22.0-pre.3; **owner sign-off pending**, not yet signed). The text of 1.22.0-pre.1, pre.2 and pre.3 as stated in their entries (referee records: `F33-2_referee.md`, `P5pp_referee.md`, `F-T1_release_referee.md`); the `pre.N` suffix dropped from the current version only; **no statement's mathematics, label, tag or status changes**; no schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.22.0, [U] 0.6.0, [L37] 0.5.2, [RC] 0.2.0.

Version 1.22.0-pre.3 — 2026-10-12 (hecke-extensions slot 22; **a pre-release, not signed**; CLOSE-T T1 repairs of the Fable release referee `docs/implementation/slot22/F-T1_release_referee.md`). MINOR-2: the cell count of Remark 9.31(9), the status row of Appendix A and this date line corrected from 32 to 30 cells ("PROVED modulo STD inputs, given CMPD L3 at the pair and CMPD (H1) at the cell"), +2, $(66, 37)$ and $(102, 37)$, once (H1) is gated there (the referee's one-route computation), as the P5″ referee's §8 states. NIT-11 (the parity clause) and NIT-12 (the $(540, 23)$ rows) applied as wording; NIT-13 (a `.bib` entry for Fukaya–Kato–Sharifi) not applied. No statement, label, tag or value moves. Companion pointers: [C] v1.22.0-pre.17, [U] 0.6.0-pre.3, [L37] 0.5.1, [RC] 0.1.1-pre.2.

Version 1.22.0-pre.2 — 2026-10-09 (hecke-extensions slot 22; **a pre-release, not signed**; slot-22 P5″, `docs/implementation/slot22/P5pp_note.md` v0.2, and its Fable referee `docs/implementation/slot22/P5pp_referee.md`: ACCEPT WITH REPAIRS, 0 B / 2 M / 7 m, the repairs R1–R8 applied in the note; only the tag changes of the referee's §8). Remark 9.31(9) gains the paragraph "The modular-symbol proof": at the 30 cells with the project's own readings of (H1) (and, once (H1) is gated there, at $(66, 37)$ and $(102, 37)$, so far the referee's one-route computation) the forced-zero theorem and (S) are [proved here] modulo [standard, reference] inputs, given [computed and compared] L3 at the pair and [computed and compared] (H1) at the cell — through pre.1 conditional on Sharifi's conjecture; per pair, nothing uniform in (ℓ, r); "consistent with Sharifi's conjecture (1.1)" kept, (1.1), the index map and the descent untouched; the open list updated. Remark 9.12(iii): "conditional on Sharifi's conjecture" → "unconditional at the computed cells". Remark 9.3: (L1) at depth exactly 1 at $(66, 37)$, $(102, 37)$, the referee's Python, one route (the C++ gate has not read them). No label added, renamed or removed; no schema key, record

key, golden line, digest or census verdict changes. Companion pointers: [C] v1.22.0-pre.2, [U] 0.6.0-pre.2, [L37] 0.5.1, [RC] 0.1.1-pre.2.
Version 1.22.0-pre.1 — 2026-10-09 (hecke-extensions slot 22; **a pre-release, not signed**; census 3.3 finding F33-2, the owner ruling of 2026-10-09, `docs/implementation/slot22/owner_ruling_F33-2_1009.md`, with the lead note; wording of the slot-22 Fable referee `docs/implementation/slot22/F33-2_referee.md` §3.3). Prediction 4.9 (R) **restated a fourth time** (a statement change of a [conjectural], the tag kept): the hypothesis is now "congruence-free and *not dihedral*", dihedral meaning $\bar{\rho}_{\mathfrak{m}}$ irreducible and $\bar{\rho}_{\mathfrak{m}} \cong \bar{\rho}_{\mathfrak{m}} \otimes \omega^{(\ell-1)/2}$ (reducible self-twist classes stay inside). New paragraph "Refuted on cohomology: the v1.17.0-pre.4 form, at dihedral classes": the 161 Loewy-3 rows of census 3.3 at $\ell = 23$ (twists of Δ) and $\ell = 31$ (the weight-16 form), $A \cong \mathbb{F}_\ell[x]/x^3$, a computed counterexample to the v1.17.0-pre.4 form; the narrowing removes 929 dihedral rows (161 violating, 768 consistent); on the 100 025 non-Dickson, non-dihedral residue-degree-1 rows no row has Loewy length 3. The evidence sentence "No row in the table violates the restated form" gains the census-3.3 sentence; the prediction's header note and Appendix A's row updated. No label added, renamed or removed; no schema key, record key, golden line, digest or census verdict changes (the 161 rows are kept unchanged). Companion pointers: [C] v1.22.0-pre.2, [U] 0.6.0-pre.2, [L37] 0.5.1, [RC] 0.1.1-pre.1.
Version 1.21.0 — 2026-10-06 (hecke-extensions slot 21, the close; **the release** from 1.21.0-pre.10 under the **owner's decision D16** (release of the pre-release documents at the close, accepted as recommended, 2026-10-06; `docs/implementation/slot21/slot21_status_and_close_plan.md` §5)). Slot-21 release: the text of 1.21.0-pre.10 as published, the `pre.N` suffix dropped from the current version only; **no statement's mathematics, label, tag or status changes**. The release upgrades no referee status: the P2x / P4x / P5 / P5′ additions (pre.6, pre.7, pre.9) stay as stated in their entries below (F4 of pre.8 on P5, P2x, P4x; the adversarial audit of P5′ at pre.10; whatever is marked not refereed, [standard, reference] recalled or conditional stays so). No schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.21.0, [U] 0.5.0, [L37] 0.5.1, [RC] 0.1.0, [LG] 0.1.5, [EX] 1.4.1, [CLI] 1.3.0.
Version 1.21.0-pre.10 — 2026-10-06 (hecke-extensions slot 21; **a pre-release, not signed**): the adversarial audit of P5′ (`docs/implementation/slot21/rc/P5prime_audit_report.md`: ACCEPT WITH REPAIRS), its repairs to Remark 9.31(9): the conditional theorem's conclusion is $h(b) = 0$, hence $\lambda_b^+ \geq 1$ *wherever* $\mu_b^+ = d$ (the hypothesis was dropped in the statement); the forced zero is $e_{\ell-r,r} = 0$ of the cup-product pairing (McCallum–Sharifi under Vandiver, [19]), not of the computed vector; Sharifi's conjecture cited at its statement [32] (1.1) (read on the page), the index $t \leftrightarrow \omega^{t-1}$ Sharifi's; Fukaya–Kato's theorems and hypotheses read (Introduction of the preprint) and the remaining gap named (descent to level one in this normalisation); "nullity 0 at regular $r \pm 2$" restricted to the pairs tried ($(257, 162)$: nullity 10); no novelty claimed; independent re-derivation at five cells (three new irregular pairs). Bibliography: [32] added, journal data of [17], [27], [33] confirmed (Project Euclid listings).
Version 1.21.0-pre.9 — 2026-10-06 (hecke-extensions slot 21; **a pre-release, not signed, not refereed**): P5′ (`docs/implementation/slot21/P5prime_note.md`, `08d83ab3`) — Remark 9.31(9): the plus-side pair $\{-k, n+k\}$ (through pre.8 "recorded, not derived") identified with the McCallum–Sharifi forced zero: $h(b) = \ell^{-d}\mathcal{L}_b^+(0) \bmod \ell$ equals one unit times the McCallum–Sharifi pairing vector x_{b+1} ([computed and compared], 32 cells, 28 irregular pairs, $37 \leq \ell \leq 691$, 4961 branch values, 0 mismatches), and $\lambda^+ \geq 1$ at $\{-k, n+k\}$ at every $d = 1$ cell is a theorem **conditional on Sharifi's conjecture** (Fukaya–Kato, [standard, reference], recalled, not read); the second index silent on the plus side, seen by the minus side at 157; what stays open listed there.
Remark 9.12(iii)'s pointer sentence updated. Three bibliography items ([27], [33]: read on the authors' web PDFs, not held; [17]: recalled). No label added, renamed or removed; no schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.21.0-pre.3, [U] 0.5.0-pre.8, [L37] 0.5.1-pre.6, [RC] 0.1.0-pre.12.
Version 1.21.0-pre.8 — 2026-10-06 (hecke-extensions slot 21; **a pre-release, not signed**; the Fable referee F4 of P5, P2x and P4x (`docs/implementation/slot21/rc/F4_referee_p5_report.md`: P5 ACCEPT WITH REPAIRS, P2x ACCEPT, P4x ACCEPT; 0 B / 1 M / 9 m), its repairs applied).

M1 — Remark 9.31(7): the pole branches of the minus measure are *derived*, no longer a record: $(\mu^-, \lambda^-) = (0, \lambda(G_{b_0}) - 1)$ on both pole branches, b_0 the cell's own irregular index, from Proposition 9.27(4), (6) (the polar term) and $\mu(G) = 0$ — the Greenberg–Vatsal formula with $\lambda(G_0) := -1$ for the pole; $= (0, 0)$ at the nine cells by $\lambda(G_{b_0}) = 1$. Minors: $\varepsilon = \pm 1$ (not ∓ 1) for $r^{\pm}$ in the notation line of Remark 9.31; (3) $\sigma \circ \sigma = \mathrm{id}$ written in, and no self-dual branch is a pole branch; (6) "$\lambda(G_b) = 0$ iff $\ell \nmid B_b/b$" is [proved here] *given* $\mu(G_b) = 0$, and the step $\lambda_p = \sum_b \lambda(G_b)$ behind [19] is [standard, reference] (Iwasawa's theorem for the minus part, with Vandiver from [19] item (2)), not part of what is read on the page; (8) the unexplained pairs occur at the rate of chance; (9) the "natural candidate" corrected: p_0 itself carries no branch but 0 and n (Remark 9.3 (L2), Proposition 9.10(6)), the object is $w = (r^+ - u\,p_0)/\ell^d$ of Remark 9.12(iii). Independent re-derivation by the referee (own Python, exact rationals, no project code, `docs/implementation/slot21/rc/f4p5/`): the Kubota–Leopoldt (μ, λ) at the seven irregular pairs and at every even class of the 16 primes (tangent-number Bernoulli numbers, divided differences); Δ's tables at $(10, 11)$, $(10, 23)$, $(10, 37)$ by the values oracle, both signs, with r^+ as the T_2-eigenvector of Merel's Heilbronn matrices (every entry of (8) reproduced; the parity rule fixes $\varepsilon(r^-) = -1$); the wild pole-branch identity of Remark 9.29(7) at $(30, 37)$, $t = 1, 2$ (0/62, the controls 62/62, 62/62, 0). No label added, renamed or removed; no schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.21.0-pre.3, [U] 0.5.0-pre.8, [L37] 0.5.1-pre.6, [RC] 0.1.0-pre.11.

Version 1.21.0-pre.7 — 2026-10-06 (hecke-extensions slot 21; **a pre-release, not signed, not refereed**): P4x and P5 (b), (c). P4x — Remark 9.29(7) at $(226, 103)$ in 128-bit $\mathcal{O}$-arithmetic (0/454; the "not covered: $(226, 103)$" caveat removed; [computed and compared]) and Check 9.30's counts. P5 — new Remark 9.31 (the tame functional equation $\lambda_{\chi,j} = -\chi(-1)(-1)^j \lambda_{\bar{\chi},n-j}$ at conductor 1 and ℓ, [proved here]; its branch form and the corollaries (FE-inv), (FE-parity), the near-trivial zero, [proved here]; the level-1 Greenberg–Vatsal formula off the pole branches, [proved here] given its inputs; the Kubota–Leopoldt invariants; the (μ, λ) tables as a record; every run [computed and compared]; one plus-side observation recorded, not derived) and new Check 9.32; pointer sentences in Claim 9.11(5) and Remark 9.12(iii); Appendix A rows. Two labels added (`rmk:lviwasawa`, `chk:lviwasawa`), none renamed or removed; no tag of an existing statement moved. Companion pointers: [C] v1.21.0-pre.3, [U] 0.5.0-pre.7, [L37] 0.5.1-pre.6, [RC] 0.1.0-pre.10.

Version 1.21.0-pre.6 — 2026-10-06 (hecke-extensions slot 21; **a pre-release, not signed, not refereed**): P2x — at $(10, 691)$ the Eisenstein congruence of measures is checked on Δ's own eigensymbol at level 2 (Proposition 9.27(6), Claim 9.11(4) and Check 9.28: arms E8L2 / E9L2, 14 checks added; the "level 1, modulo 691" caveat removed; [computed and compared]). No label added, renamed or removed; no other statement changed. Companion pointers: [C] v1.21.0-pre.3, [U] 0.5.0-pre.6, [L37] 0.5.1-pre.6, [RC] 0.1.0-pre.9.

Version 1.21.0-pre.5 — 2026-10-06 (hecke-extensions slot 21; **a pre-release, not signed, not refereed**): Claim 5.13(3) [standard, reference] → [proved here] — a proof by Brauer characters is written in (the Brauer character of $\mathrm{Ind}_B^G(\chi_1^s \chi_2^t)$ from Frobenius' formula equals that of $\mathrm{Sym}^r \otimes \det^t \oplus \mathrm{Sym}^{\ell-1-r} \otimes \det^s$ on every semisimple class), since [18] has no induced-module statement (its (4.1), (4.2), (5.7), (6.4) are explicit bases); [18] now cited for item (1) by locator ((3.1), p. 433; (6.2), p. 459). Tag line and the summary row now [standard, reference] (1) / [proved here] (2)–(4). Checked exactly (own Python, $\mathbb{Z}[x]/\Phi_{\ell^2-1}$, no project code) for every (s, t) at $\ell \le 13$, and the multiplicities of Sym^n against the recursion of (4) for $n \le 3\ell(\ell+1)$. No label added, renamed or removed; no other statement changed. Companion pointers: [C] v1.21.0-pre.3, [U] 0.5.0-pre.6, [L37] 0.5.1-pre.6, [RC] 0.1.0-pre.8.

Version 1.21.0-pre.4 — 2026-10-06 (hecke-extensions slot 21; **a pre-release, not signed**; the Fable referee F4m of the measure side (P2, P3, P4; `docs/implementation/slot21/rc/F4_referee_measure_report.md`: ACCEPT WITH REPAIRS, 0 B / 1 M / 12 m), its repairs applied). **M1** — the literature leg of Proposition 9.27(3) corrected: $E_1 = -x\,\zeta_\ell$ (not $x\,\zeta_\ell$), so each of the two G-factors is *minus* the branch series of E_1, resp. $\iota_* E_{k-1}$ (the product unchanged), and $dE_r = x^{r-1} dE_1$ in this text's normalisation (Washington's E_k carries no $1/k$); Washington §12.1 p. 232 and Thm. 12.2 p. 239 read on the page ([standard, reference] recalled → read on the page). **Tag moves, each with its proof or its locator written in**:

Zagier's $p_k^- \in \hat{W}_k^-$ read on the page (Prop. (i), eq. (11), p. 453; Claim 9.11(1), Proposition 9.27(1)); the residue constant of Proposition 9.27(4) [proved here] given (3) and ζ_ℓ's residue; in Remark 9.29: the root number $w = \tau(\chi)^2/M$ (1), the symmetry $\lambda_{\chi,j} = -\chi(-1)(-1)^j \lambda_{\bar{\chi},n-j}$ for every $P \in W$ (3), the wild Kubota–Leopoldt convention and the pole-class valuation (5), all [proved here] (elementary). Also: the title of §9.8; the level-1 remark at $(10, 691)$ and the $(66, 37)$ remark in Proposition 9.27(6). Independent re-derivation by the referee (own Python, exact rationals, no project code): Proposition 9.27(1)–(6) at $(12, 5)$ $m \le 4$, $(14, 7)$ $m \le 3$, $(30, 37)$ $m \le 2$, $(66, 37)$, $(282, 37)$ $m \le 2$ (the located 14 reproduced), $(10, 691)$; Remark 9.29(3), (5), (6) at $(10, 5)$, $(10, 11)$, $(5, 2)$, $(7, 2)$, $(12, 5)$. No label added, renamed or removed; no statement changed; no schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.21.0-pre.2, [U] 0.5.0-pre.6, [L37] 0.5.1-pre.5, [RC] 0.1.0-pre.8.

Version 1.21.0-pre.3 — 2026-10-06 (hecke-extensions slot 21; **a pre-release, not signed, not refereed**; the results of the ℓ-adic packages P2 and P4 written in). **New** at the end of §9.8 (so that no statement number moves): Proposition 9.27 (slot-21 P2, `HeckeLValueEisensteinMeasureValidation`: Zagier's Laurent polynomial through the Manin-path formula is a Dedekind–Rademacher sum exactly, the Eisenstein measure $\mu^{\mathrm{Eis}} = -E_1 \star \iota_* E_{k-1}$, its branch series $-G_{a+1}G_{k-1-a}$ in $\Lambda \otimes \mathbb{Q}$, the polar term on the two pole branches, the comparison with WP-MEAS's polynomial symbol, and the consequence for the cusp form's pole branches), Check 9.28, Remark 9.29 (slot-21 P4, `HeckeLValueWildCharacterValidation`: characters of conductor ℓ^m, $m = 2, 3$, the constant $\tau(\chi)$ against complex L-values, Proposition 9.10(4) at wild χ, the symmetry $\lambda_{\chi,j} = -\varepsilon\lambda_{\bar{\chi},n-j}$, the reading $v_\pi = e\mu + \lambda$, the pole branches at wild points), Check 9.30. **Added text** (no statement changed; tags as in the sources, none upgraded): Claim 9.11(4) — the pole branches *explained* by P2's polar term, the located $14 + 4$ derived (the points with $\ell \mid r$), R20-L no longer needed for it; Claim 9.11(3) — a second derivation through P2; Proposition 9.10, analytic form, and Remark 9.2 (B) — $\tau(\chi)$ for ℓ-power conductor [computed and compared] (P4; "derived, not tested" through pre.2); the "Pending P2/P4" sentences of Check 9.25 and Remark 9.26; §9.3 row (M); Appendix A (four rows added, three amended). Four labels added (`prop:lveismeas`, `chk:lveismeas`, `rmk:lvwild`, `chk:lvwild`), none renamed or removed. No schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.21.0-pre.2, [U] 0.5.0-pre.5, [L37] 0.5.1-pre.4.

Version 1.21.0-pre.2 — 2026-10-05 (hecke-extensions slot 21; **a pre-release, not signed, not refereed**): one sentence in Remark 9.2 (B) — the torsion of $W(\mathbb{Z}/\ell^R)$ and P8's reading of the $\mathfrak{m}$-blocks through $W(\mathbb{Z}/\ell^{R+d})$ on both signs (slot-21 F4 on [U] part 4, S2, S3; [specification], no statement changes). Companion pointers: [C] v1.21.0-pre.2, [U] 0.5.0-pre.4, [L37] 0.5.1-pre.3.

Version 1.21.0-pre.1 — 2026-10-05 (hecke-extensions slot 21; **a pre-release, not signed, not refereed**; the slot-21 results written in). **New** at the end of §9, so that no statement number moves: §9.8 — Remark 9.24 (slot-21 P3: the MTT measure over the Hecke order $\mathcal{O}$ of the eigen-system, $W(\mathbb{F}_q)$ or totally ramified, Proposition 9.10 verbatim over $\mathcal{O}$, [proved here]; the cells [computed and compared]; Hida control modulo $\mathfrak{m}$ at one residually non-split component, [computed and compared]; finding F-P3-1 as the scope sentence), Check 9.25 (its gate `HeckeLValueCoefficientRingValidation`), Remark 9.26 (slot-21 P9, `HeckeLambdaTwoVarValidation`: the two-variable family by interpolation, the Λ-adic Eisenstein congruence at $X = \alpha_a$ modulo 37^3 and 103^2, weight two and the $\mathcal{L}$-invariant of 11a; everything resting on [U, `pr:u-msym`] (U16) stays [conjectural]; P1's two limits recorded for §9.7). **Added text** (no statement changed): Remark 9.2 (M) (the coefficient sentence), §9.3 row (M) (values in $\mathcal{O}$; pointers to P3, P9), a reading of Claim 9.5(2) on the ordinary block after its proof (slot-21 F4 M3: [U] (U17)(a)), §9.4 (slot-21 T1: LLL and POLRED, `NumberFieldLllValidation`; T2: the degree-ℓ Ribet field, `NumberFieldRibetFieldValidation`, with the referee F4's verdicts on its claims), Appendix A (three rows). **Stale status sentences repaired** (the library inventory of 2026-10-05, `docs/implementation/slot21/library_inventory.md`): Mazur and Greenberg–Vatsal are *held*, not yet read for Claim 9.11, Remark 9.12, Claim 9.23 and row (M) (five sites; "not in the library" and "deferred to R20-L on the owner's scans" were false); Glover is held and read (the remark after the proof of Claim 4.16: the recursion, not a closed multiplicity formula, is in the paper). Four

labels added (`sec:lvslot21`, `rmk:lvcoeffring`, `chk:lvcoeffring`, `rmk:lvtwovar`), none renamed or removed; no earlier tag moves. No schema key, record key, golden line, digest or census verdict changes. *Pending P2/P4*: the exact Λ-adic Eisenstein measure and wild characters. Companion pointers: [C] v1.21.0-pre.2, [U] 0.5.0-pre.3, [L37] 0.5.1-pre.2.

Version 1.20.0 — 2026-10-05 (hecke-extensions slot 20, the close, WP-18; **the release** from 1.20.0-pre.7 under the **owner's ruling O6** of 2026-10-04 (releases after the Fable windows), on the slot-20 referees' verdicts: F1 ACCEPT WITH REPAIRS (repairs at pre.2, `docs/implementation/slot20/WP-3_repairs.md`), F4 ACCEPT WITH REPAIRS (the research results; repairs at pre.4, `docs/implementation/slot20/R20-6_repairs.md`) and F1′ ACCEPT WITH REPAIRS (the pre.2 → pre.6 delta; repairs at pre.7, `docs/implementation/slot20/WP-17_repairs.md`). Over pre.7: Remark 4.13 gains the result at $(1372, 37)$, $b = 5$ (slot-20 WP-SW follow-up, `SW1372_report.md`): the swapped-family statement [computed and compared] at all five cells of its range at 37 (an evidence line; no tag moves, the statement stays [conjectural] beyond 37). No other statement, tag or label changes; no schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.20.0 (the milestone-M4 `\LEAN` markers), [U] 0.4.0, [L37] 0.5.0, [EX] 1.4.0 (unchanged).

Version 1.20.0-pre.7 — 2026-10-05 (hecke-extensions slot 20, WP-17: the repairs of the Fable referee F1′, `docs/implementation/slot20/F1prime_review.md`, ACCEPT WITH REPAIRS, 0 B / 2 M / 6 m, all landed; **a pre-release, not signed**; record `docs/implementation/slot20/WP-17_repairs.md`): M-E1 — Proposition 9.16's lattice is the diagonal lattice $V_n' = \bigoplus_t \binom{n}{t}\mathbb{Z}_\ell u_t = n!\,\Phi_n^{-1}(D_n)$, Δ-stable through Φ_n; the identification with the span of the n-th powers (false for $n \geq \ell + 1$) is removed (no value moves; WP-LV3's code reads the diagonal lattice); M-E2 — **a hypothesis change** (no tag move): Prediction 4.12 is restricted to $k_0 \leq \ell$, the boundary weight $k_0 = \ell + 1$ recorded as refuted as stated at one class, $(1334, 37)$; m-E1 — Remark 4.13's cell list ($J(32) = 5$ at 37, $J(24) = 79$, $J(12) = 679$; [computed and compared] at the first four of five cells at 37, $n = 1372$ running); m-E2 — Check 9.13(M7) under the heavy arm; m-E3 — Claim 9.14(L) and Remark 9.3: item 3 [conjectural] as an index statement, [computed and compared] as a run; m-E4 — Appendix A row `chk:lvmeas`; m-E5 — Proposition 9.16(3) names the reason for $H^2 = 0$; m-E6 — n_0 no longer re-used for 1296. No label added, renamed or removed. Companion pointers: [C] v1.20.0-pre.2, [U] 0.4.0-pre.3, [L37] 0.5.0-pre.6.

Version 1.20.0-pre.6 — 2026-10-04 (hecke-extensions slot 20, the close delta for the Fable referee F1′; **a pre-release, not signed**; list of changed labels `docs/implementation/slot20/close_delta.md`): the text brought into line with slot-20 WP-LV3 (merged `c1fda051`) and R20-3. **New**: Proposition 9.16 (the change-of-lattice constant $c_f^{\varepsilon} = e\,v_\ell(n!) - d_f^{\varepsilon}$, its bounds and a vanishing criterion; [proved here], R20-3) and Remark 9.17 (the values, [computed and compared] by R20-3's two routes and WP-LV3's C++ record; the $d_f = 1$ twist rule [conjectural]). **Changed**: Definition 9.15 (the value no longer open); Claim 9.14 — (L)'s (L1), (H) and the cross-kind equality at depth 2 move from [conjectural] to [computed and compared] (WP-LV3), $\mathfrak{m}_{\mathrm{Eis}}$ cut jointly by T_2, T_3, T_5; Check 9.19 (the reader's scope: tame, totally ramified orders, an unramified residue extension refused by name; (d)'s console script reads $(30, 37)$, $(38, 37)$; an outcome paragraph); Remark 9.18 and §7.1's console list (h) (ι honours $\mathbb{Z}/\ell^k$, schema `hecke_involution_ring/v1`); Remarks 9.3, 9.2, 9.7 (the twisted depth-2 rung run, [computed and compared]), §9.3 rows (B), (H), the code map, Appendix A. Prediction 4.12 and Remark 4.13 gain slot-20 WP-SW's evidence — the swapped cells $b = 2, 3, 4$ [computed and compared], and an observation at the boundary weight $k_0 = \ell + 1$ (an unpredicted hidden block at $(1334, 37)$, explained by the mechanism, not by the onset formula). S1 / S2's text of pre.5 re-read against `docs/implementation/slot20/S1_S2_report.md`: no change. Two labels added (`prop:lvlattice`, `rmk:lvlatticevalues`); none renamed or removed. Tags moved: Claim 9.14(L1), (H), the triangle ([conjectural] → [computed and compared]); Remark 9.3(L1) at depth 2 ([conjectural] → [computed and compared]); Definition 9.15's value (open → Proposition 9.16). No schema key of `hecke_ext/v5`, golden line, digest or census verdict changes. Companion pointers: [C] v1.20.0-pre.2 (code-map text only), [U] 0.4.0-pre.3, [L37] 0.5.0-pre.5.

Version 1.20.0-pre.5 — 2026-10-04 (hecke-extensions slot 20, stretch packages S1 and S2, plan v1.1

§0.6; **a pre-release, not signed**): S1 — §9.7, the truncated one-variable ℓ-adic L-function as the record `lfunction/v1` (Definition 9.20, [specification]; Lemma 9.21, the certified precision of each T-coefficient, [proved here], elementary; Remark 9.22, the schema and the console verb `lfunction`); Check 9.13 gains arm (M7). S2 — Claim 9.23, Ramanujan's congruence $\Delta \equiv E_{12} \pmod{691}$ as a congruence of measures against Kubota–Leopoldt ([computed and compared] at $(10, 691)$; $\lambda = 1$ at the branches 199, 501 predicted and found; the 22 pole points with a unit difference), and the env arm (M691) of Check 9.13. Placed at the end of §9, so that no statement number moves. Five labels added (`sec:lvlfunction`, `def:lvlfun`, `lem:lvlfunprec`, `rmk:lvlfunschema`, `cl:lvmeas691`); none renamed or removed; no earlier claim's tag moves; a new record schema (not golden); no key of `hecke_ext/v5`, golden line, digest or census verdict changes. Companion pointer: [P] v3.11.1.
Version 1.20.0-pre.4 — 2026-10-04 (hecke-extensions slot 20, R20-6 repairs after the Fable referee F4, `docs/implementation/slot20/R20-6_F4_review.md`, ACCEPT WITH REPAIRS for this text, 0 B / 1 M / 1 m; repairs listed in `docs/implementation/slot20/R20-6_repairs.md`; **a pre-release, not signed**): M-E1 — the slot-20 evidence paragraph of Prediction 4.12 is qualified: the family-(c) seed is non-reduced and lies outside the statement's hypotheses, so the paragraph supports the transport step (ii) of *Why* only; m-E1 — the status-table row of Prediction 4.12 carries the slot-20 entry. The statement and its tag do not change. No label added, renamed or removed; no schema key, record key, golden line, digest or census verdict changes. Companion pointer: [L37] 0.5.0-pre.4.
Version 1.20.0-pre.3 — 2026-10-04 (hecke-extensions slot 20, R20-5; **a pre-release, not signed**, for the slot-20 Fable referee F4, R20-6): one evidence paragraph in Prediction 4.12 (slot-20 R20-2 §3, the family-(c) onsets on the forms side; [proved here] given curvilinearity, [computed and compared] at four cells); the statement and its tag do not change. No label added, renamed or removed; no schema key, record key, golden line, digest or census verdict changes. Companion pointer: [L37] 0.5.0-pre.3.
Version 1.20.0-pre.2 — 2026-10-04 (hecke-extensions slot 20, WP-3 repairs after the Fable referee F1, `docs/implementation/slot20/WP-3_F1_review.md`, ACCEPT WITH REPAIRS, 0 B / 1 M / 10 m; repairs listed in `docs/implementation/slot20/WP-3_repairs.md`; **a pre-release, not signed**): M-1 — Proposition 9.10(6) pins its object ($D_Q^{(\alpha)}$, the formula of Definition 9.9 on an arbitrary symbol with a fixed unit α), Remark 9.12(ii) cites it with linearity; m-E1 — Claim 9.11: the reduction to the Eisenstein symbol [proved here] given its inputs, its Kubota–Leopoldt form [computed and compared]; m-E2 — synced with WP-MEAS's first run (the 14 pole points at $(282, 37)$ listed and located, the two Kubota–Leopoldt routes agree 17/17 and 50/50, 178 checks; no `CwLayer` addition); m-E3 — the polar-term valuation's r is the weight index; m-E4 — a cross-reference; m-E5 — Definition 9.15: $\theta = \Phi_n$ of [C, `lem:symdual`], and the carrying of the f-line, an "input" at pre.1, is [proved here] from [C, `lem:symdual`]; m-E6 — Remark 4.13: the θ-socle run is consistent with Prediction 4.12 "Why (ii)"; m-E7 — the wall budget of Check 9.13 is not a gate. No label added, renamed or removed; no earlier claim's tag moves (the well-definedness clause of Definition 9.15 is new text with its own tag). No schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.20.0-pre.1 (unchanged), [U] 0.4.0-pre.2, [L37] 0.5.0-pre.2.
Version 1.20.0-pre.1 — 2026-10-04 (hecke-extensions slot 20, WP-2; **a pre-release, not signed**, for the slot-20 Fable referee F1 (WP-3); release at the slot-20 close only on F1's and F1′'s acceptance and the owner's yes; list of new and changed labels:
`docs/implementation/slot20/WP-2_drafts.md`). **New** at the end of §9, so that no statement number moves: §9.5, the measure-valued linear evaluation (owner ruling O5 of 2026-10-04: no fourth kind) — Definitions 9.8 ([specification]), 9.9 (the Mazur–Tate–Teitelbaum measure, with the factor ℓ^{k-2} the slot-20 plan's formula lacked), Proposition 9.10 (measure, moments, interpolation; [proved here], slot-20 R20-1), Claim 9.11 (the Eisenstein congruence of measures against Kubota–Leopoldt, [computed and compared] at $(30, 37)$, $(282, 37)$, $(226, 103)$; the two pole branches an explicit exception with their computed failure points, the explanation deferred to R20-L), Remark 9.12 (the plus side; the $\lambda = 1$ branch pairs an observation, not a claim), Check 9.13 (WP-MEAS's gate, arm (M0) in the `standard` tier); §9.6 — Claim 9.14 (the three kinds at depth 2, [conjectural] with the slot-20 evidence), Definition 9.15 (the change-of-lattice constant, [specification], value open until R20-3), Remark 9.18 (ι over $\mathbb{Z}/\ell^k$: nothing new mathematically, [proved here] from [C]), Check 9.19 (WP-LV3's gates). **Changed**: Remark 9.3 (items 2–3 at depth 2 stay [conjectural], the new

computation cited), Remark 9.2 (pointers to (M), to Claim 9.14 and to Definition 9.15), Remark 9.7 (the twisted depth-2 rung's plan), Remark 4.13 (the θ-socle run at $(1220, 37)$: θV_{src} is not the socle, [computed and compared]), §9.3 row (M), the code map, Appendix A. Twelve labels added (`sec:lvmeasure`, `def:lvmeasure`, `def:lvmtt`, `prop:lvmtt`, `cl:lvmeascong`, `rmk:lvmeasplus`, `chk:lvmeas`, `sec:lvdepth2`, `cl:lvtriangle`, `def:lvlattice`, `rmk:lviotak`, `chk:lvtriangle`); none renamed or removed; no earlier claim's tag moves. No schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.20.0-pre.1, [U] 0.4.0-pre.1, [L37] 0.5.0-pre.1.

Version 1.19.0 — 2026-10-04 (hecke-extensions slot 19, the close, WP-18; **the release** from 1.19.0-pre.7 under the **owner's ruling O3**, on the slot-19 referees' verdicts: F1 ACCEPT WITH REPAIRS (repairs at pre.2), F4 ACCEPT WITH REPAIRS (the research results; repairs at pre.3) and F1-delta ACCEPT WITH REPAIRS (the pre.3 → pre.6 delta; repairs at pre.7, `docs/implementation/slot19/F1delta_repairs.md`). Release lines only: no statement, tag or label changes over pre.7; no schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.19.0 (the milestone-M3 \LEAN markers), [U] 0.3.0, [L37] 0.4.0, [EX] 1.4.0 (unchanged).

Version 1.19.0-pre.7 — 2026-10-04 (F1-delta repairs, textual only: the (H) constant retagged [standard, reference]+[computed and compared], no tag otherwise moves; before that: hecke-extensions slot 19, WP-2 continued, **a pre-release, not signed**): the bilinear and linear instances of Definition 9.1 **implemented** — Remark 9.2 (H) (slot-19 WP-BIL, `HidaPairingRoute`: G1 on the code's `PoincareDualityPairing`, $T_p^* = T_p$ for $p = 2, 3, 5$, R19-9's valuations reproduced, [computed and compared]; (H2) at depth 2 stays [conjectural]; the change-of-lattice constant for $n \geq \ell$ open) and (B) (slot-19 WP-LIN, `BirchModularSymbolRoute` on the C++ cone's H_c^1: $\Psi\colon H_c^1 \cong W$, [C]'s T_p is the symbol T_p, a geometric-even symbol is [C]-minus, the $(258, 37)$ sign facts, [computed and compared]); a **new** Remark 9.3 (leg (c) in modular-symbol form: $r^+ \equiv u\, p_0$, $r^- \equiv s\, \mathrm{Bern}_k$ modulo ℓ to depth $v(B_k/k)$, twisted values of $r^+ \equiv 0$; [computed and compared] at 8 cells, [conjectural] at depth ≥ 2); §9.3 rows (B), (H), the code map and Appendix A. One label added (`rmk:lvlegc`); none renamed or removed; no earlier claim's tag moves. Companion pointers: [C] v1.19.0-pre.5, [U] 0.3.0-pre.4, [L37] 0.4.0-pre.3.

Version 1.19.0-pre.5 — 2026-10-04 (hecke-extensions slot 19, WP-2 continued: the slot-19 results that landed after pre.3/pre.4 folded in with their own tags, **a pre-release, not signed**; change table `docs/implementation/slot19/WP-2_pre5_changes.md`). Remark 9.7: the twisted Claim 9.6 [computed and compared] at four cells and two controls (slot-19 WP-LV2), the deep rungs $(282, 37)$, $(226, 103)$ [computed and compared] (torsion-free, rank 1, saturated, $\#Q_m = \ell^{\min(m,2)}$; WP-LV2 and, by an independent reading, R19-7), the two-sided bound tested on planted algebras; "the twisted gate is not run" deleted. §9.3 row (a): the level-m Coates–Wiles measure from $\mathbb{Q}_\ell(\zeta_{\ell^m})$ (slot-19 R19-7; [proved here] identity, [computed and compared] values, the bridge to the unramified Kummer lift [proved here] from [38] Thm. 13.54, Prop. 13.51, Lemma 13.52); rows (B), (H) and Remark 9.2: the research prototypes of the linear kind (slot-19 R19-8, Birch / Manin–Stevens) and the bilinear kind (slot-19 R19-9, Hida's pairing) with their slot-20 specifications; (H)'s constant from [21] Thm. 5.1, pp. 247–248, read on the page, no Euler factor removed (the [locator not verified, rule A2] of row (H) is settled). Prediction 4.12: slot-19 evidence at $(1220, 37)$ and a **new** Remark 4.13 (the Frobenius layer on a Kummer class: $I = \mathfrak{m}^3$, [computed and compared] at one cell, [conjectural] beyond); Prediction 4.9: one evidence sentence, no change. The code map of §9: the twisted route **built** (WP-LV2). One label added (`rmk:frobkummer`); none renamed or removed; no claim's tag moves. Companion pointers: [C] v1.19.0-pre.4, [U] 0.3.0-pre.4, [L37] 0.4.0-pre.2.)

Version 1.19.0-pre.4 — 2026-10-04 (hecke-extensions slot 19, isign-fix: Claim 4.20 states that on H_c^1 the free sign is the other one ($\mathfrak{m}$ residually Eisenstein; $\Lambda_{\mathfrak{m}}^+ \cong \mathbb{T}_{\mathfrak{m}}$ on H^1 stands), its proof reads "in characteristic 0"; Remark 9.2(B): the normalisation fold-in of R19-8 §1.4 (`docs/implementation/slot19/isign_resolution.md` §§4–5). No label, tag, code-side value, golden line, digest or census verdict changes.)

Version 1.19.0-pre.3 — 2026-10-04 (hecke-extensions slot 19, WP-2 continued after the Fable referee F4 of the slot-19 research results, `docs/implementation/slot19/R19-6_F4_referee_report.md`

(ACCEPT WITH REPAIRS, 0 blocking, 1 major, 18 minor); **a pre-release, not signed**; change table `docs/implementation/slot19/WP-2_pre3_changes.md`). The accepted results enter with F4's tags (its §3): Claim 4.20 gains the measured sign, $\Lambda_{\mathfrak{m}}^{+} \cong \mathbb{T}_{\mathfrak{m}}$, $\Lambda_{\mathfrak{m}}^{-} \cong \mathfrak{m}$ ([computed and compared], WP-LC), and F4 m1's wording on [5]; Check 4.22's run (held at $M = 1, \ldots, 4$, [computed and compared]); a **new** Remark 4.23, the $\ell = 7$ controls of WP-LC ([computed and compared]; F4 m5–m7: the over-order chain at $(48, 7)$ told apart by U_7, the scope of the per-sign freeness failure, freeness by [C, `lem:h2torsion`]); Remark 4.24's pointer; Remark 9.7: WP-LV's gate cells ([computed and compared], F4 m18), the Bernoulli routes as built, and the cells $(282, 37)$, $(226, 103)$ after [U] 0.3.0-pre.3's (U13); §9.3 rows (a), (b): the [specification] placeholders replaced by slot-19 R19-4's text, with F4 m15's naming of φ_k and its locators ([38] §13.7, [10] p. 230); §9.4 and the code maps of §4.5 and §9: **built** (WP-LC, WP-LV, WP-NF; the twisted Claim 9.6 not built). One label added (`rmk:latcmpseven`); none renamed or removed; tags: rows (a), (b) of §9.3 move from [specification] to their own tags, §9.4 is built; no claim's tag moves. Bibliography: [10]. No code-side value, golden line, digest, record key or census verdict changes. Companion pointers: [C] v1.19.0-pre.3, [U] 0.3.0-pre.3, [L37] 0.4.0-pre.1.

Version 1.19.0-pre.2 — 2026-10-03 (hecke-extensions slot 19, WP-3 repairs after the Fable referee F1, `docs/implementation/slot19/WP-3_F1_referee_report.md`; **a pre-release, not signed**). M-2: Definition 9.1(2) gains the kind congruence (a Hecke character e and its congruence module $Q_e = \mathbb{T}(H')_{\mathcal{O}}/(I_e + \operatorname{Ann} I_e)$), which carries the instance (E) (Definition 9.4); (B) stays linear, (H) bilinear; Remark 9.2, the code map and Appendix A updated. M-1: Claim 9.5(3)'s hypothesis is *saturation* of the $\mathfrak{m}_{\mathrm{Eis}}$-block's Hecke algebra (the `stable` verdict at $K = 1$; automatic on a rank-1 block), with a two-sided bound otherwise, and its inequality runs over $p \neq \ell$; Remark 9.7 says the hypothesis holds at the four untwisted cells and is open at $(282, 37)$, $(226, 103)$. Minor (F1 m-E1 … m-E8): Definition 9.4's record reads $L = \zeta$ at $s_0 = 1 - k$; the proof of Claim 9.5(1) matches I_{Eis} with [U]'s I^0; Claim 9.6's Hida locator verified on the page (ENS 1986 pp. 249–250, Cor. 3.2 (3.3b)) with the character-convention check, and [22]'s note updated; Remark 9.2: the congruence modules of the $(258, 37)$ block, and $P_{j,a}$ of degree n; Claim 4.19(5): the unique intermediate $\mathbb{T}_{\mathfrak{m}}$-lattice (F1); Claim 4.20's proof: Jordan type in place of Krull–Schmidt, the Gorenstein input proved inline, [C, `prop:cohforms`](5) cited; Remark 9.7: the 691 rungs 342252, 342440 computed (F1, tracked in `docs/implementation/slot19/wp3/`); a macro comment moved. Tags unchanged; no label added, renamed or removed; Appendix A rows of Claims 4.20, 9.5, 9.6 updated. Companion pointers: [C] v1.19.0-pre.2, [U] 0.3.0-pre.2.

Version 1.19.0-pre.1 — 2026-10-03 (hecke-extensions slot 19, WP-2; **a pre-release, not signed**, for the slot-19 Fable referee F1 (WP-3); release at the slot-19 close only on its acceptance and the owner's yes). **New**: §4.5 at the end of §4 — the block $(258, 37)$ exactly (slot-19 R19-1): Claim 4.19 ([computed and compared], with (2) $\mathbb{T}_{\mathfrak{m}} \cong \mathbb{T}_{\mathfrak{m}}^{\mathrm{f}}$ [proved here] from [C, `prop:cohforms`]), Claim 4.20 ($\Lambda_{\mathfrak{m}} \cong \mathbb{T}_{\mathfrak{m}} \oplus \mathfrak{m}$, [proved here] given the [computed and compared] over-order data, [L37]'s V and Gorenstein ideal theory — a weaker input than R19-1's Bass structure theorem, through the ι-split), Remark 4.21, Check 4.22 (the gate of WP-LC), Remark 4.24; a new §9 before the appendices (owner ruling O6) — the L-value interface (Definition 9.1, [specification]; the Birch and Hida shapes, Remark 9.2), the Eisenstein / Bernoulli instance (Definition 9.4; Claim 9.5: ring form [proved here] from [U]'s (U1), module form [standard, reference] for $n < \ell - 1$; the twisted Claim 9.6, [proved here] given Hida's control at the nebentypus points), the normalisation table (§9.3; legs (a), (b) [specification], R19-4), the number-field layer (§9.4, [specification]), and the planned code maps of WP-LC, WP-LV, WP-NF. Harder's boundary-spectral-sequence route is not used. New tags [computed and compared], [specification]; new environment *Check*; bibliography: [5], [21], [22], [26]. Appendix A: seven rows added. No earlier statement, tag or label changed; no schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.19.0-pre.1, [U] 0.3.0-pre.1, [EX] 1.4.0 (unchanged), [L37] 0.3.0 (unchanged).

Version 1.18.0 — 2026-10-03 (hecke-extensions slot 18, the close, WP-18; **the release** from 1.18.0-pre.2 under the **owner's ruling O3 of 2026-10-03**, on the slot-18 referee F1's ACCEPT WITH REPAIRS, repairs at pre.2). Over pre.2, two pointer texts of the slot's stretch packages, no statement or tag change: Stage 4's (§5) "Scope and order" paragraph now cites census editions 1–3

and the mixed-finality scans, including slot-18 WP-19c's 64-cell probe ($\ell \in \{17, \dots, 43\}$ on the Dickson cells, $n \le 200$: no block with $\ker r^1|_V \neq 0 \neq S_V$), and states that Proposition M stays specified only (certificate (d2) lands with the next record event that needs it; `docs/implementation/slot18/WP-19c_report.md` §5); the digit route's item records slot-18 WP-19b's cut at the Hecke step (`DigitCellSession`, value-free, `HeckeLadicSplitValidation`). Census edition 3.2 stays staged, not published (owner ruling O7: published on the return that completes it). No label added, renamed or removed; no schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.18.0, [EX] 1.4.0, [U] 0.2.1 (unchanged).
Version 1.18.0-pre.2 — 2026-10-03 (hecke-extensions slot 18, WP-3 repairs; **a pre-release, not signed**; release as 1.18.0 at the slot-18 close on the owner's yes, ruling O3). The slot-18 Fable referee F1 (`docs/implementation/slot18/WP-3_F1_referee_report.md`) was **ACCEPT WITH REPAIRS**, 0 blocking; its major M-2 and minors m-E1–m-E7 are applied here (table: `docs/implementation/slot18/WP-3_repairs.md`). **Evidence, not statements** (the referee's nine forms-side cells, predicted before each run, 9/9 held; tracked at `docs/implementation/slot18/wp3/`): Prediction 4.18's "untested" $(m, r_0) = (2, 3)$ now has a datum, $(34460, 547)$, $\mathbb{F}_{547}[x]/x^6$, so the hypothesis $m \le 2$ (not $mr_0 \le 4$) is the one the data support; Prediction 4.10 (C)'s first window sampled at its last and middle cells, (B) at five more cells; (C) still asserts nothing at the meeting. **Wording** (no tag change): Claim 4.16 — the Edixhoven fallback (a lower bound; m-E1), $d \in \{1, \ell - 2\}$ and the semisimplicity / multiplicity-one input named (m-E2), the parity corollary $m_O \equiv m_K$; the Edixhoven status "attribution verified in a secondary source, locator pending" (slot-18 R18-4), Berger–Li–Zhu as a second locator for $w \le \ell$ (new bibliography item [7]); Remark 4.17 cites the exact event-sequence check (m-E3); Prediction 4.18 (b)'s torsion-freeness (m-E5); census 3.2's ranges (m-E6); "at the same twist" (m-E7). The `conj:law` pointer: R18-1's final figures, nothing taken in (M-2(d), m-E4; first written with F1's preliminary figures, replaced by R18-5). Appendix A: four rows updated. No label added, renamed or removed (one bibliography key added); no schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.18.0-pre.2, [EX] 1.4.0-pre.2, [U] 0.2.1 (unchanged).
Version 1.18.0-pre.1 — 2026-10-03 (hecke-extensions slot 18, WP-2; **a pre-release, not signed**, for the slot-18 Fable referee F1 (WP-3); release at the slot-18 close only on its acceptance and the owner's yes, ruling O3 of 2026-10-03). **New statements**: §4.4 at the end of §4 — Claim 4.14 ([proved here]; slot-17 R17-1 Lemma 3.1, with the standard / swapped partition), Claim 4.15 ([proved here] given one [standard, reference]; R17-1 Lemma 3.2), Claim 4.16 ([proved here] given [standard, reference] inputs, Edixhoven's locator not verified; R17-1 Theorem 4.1, the referee F1's m-E5), Remark 4.17 (the windows on the multiplicity side) and Prediction 4.18 ([conjectural]; the $\ell = 37$ report's `conj:depth`). **Prediction 4.10 generalised** (the label kept, the title changed): (A) the v1.17.0 statement; (B) the swapped family at $g = 1$, same window form; (C) $g = 2$, $\mathbb{F}_\ell[x]/x^3$ in the first window; nothing asserted where no datum exists (the $g = 2$ meeting, $g \ge 3$, past the windows); new forms-side evidence by slot-18 WP-2 (11/11 predictions held, `docs/implementation/slot18/wp2/`) and the slot-17 LAT37 lattice cells $(696, 37)$–$(700, 37)$. **Corrections**: Prediction 4.10 (c)'s "2092 of 2092" standard classes is 1492 of 1492. **Pointers**: §7.2 edition 3.2 (staged, not published); Prediction 4.9's "Observed on the Kummer locus" and "Where the excluded locus sits" point to the new statements; the report's `conj:law` is *not* taken in (slot-18 R18-1 has not reported). Bibliography: [14], [16], [38], [39]. Appendix A: one row restated, five added. No label renamed or removed; no schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.18.0-pre.1, [EX] 1.4.0-pre.1, [U] 0.2.1 (unchanged).
Version 1.17.0 — 2026-10-02 (hecke-extensions slot 17, the close, WP-18; **the release** from 1.17.0-pre.5 under the **owner's ruling O3 of 2026-10-01**, which releases 1.17.0 on the slot-17 referee F1's acceptance). F1 (`docs/implementation/slot17/WP-3_referee_F1.md`) was ACCEPT WITH REPAIRS, 2 blocking (B-1, B-2), 6 minor: B-2 and the minors applied at pre.3, B-1 by the owner's ruling of 2026-10-02 at pre.4 ((R) restated on congruence-free classes); Prediction 4.10's window bound stands as a restatement of the [conjectural] (the unbounded "from k on" refuted on the forms side at $\ell = 37$, $k = 1412, 1450$). Over pre.5 nothing changes but this line: no statement, tag or label added, renamed or removed; no schema key, record key, golden line, digest or census

verdict changes. Census edition 3.2 is staged at the slot-17 close and **not** published (owner-gated); this version cites edition 3.1 only. Companion pointers: [C] v1.17.0 (the first \LEAN markers), [EX] 1.3.0, [U] 0.2.1.
Version 1.17.0-pre.5 — 2026-10-02 (hecke-extensions slot 17, WP-2; **a pre-release, not signed**; pointers only, no statement or tag change): Claim 2.4 gains the verified locators [23, Thm. 4.1, p. 276] ("weight $\leq \ell + 1$") and [1, Thm. 2.2, p. 204] (finiteness), slot-17 R17-4; Prediction 4.10 (c) records the window's last cell off 37 on the forms side — $(3496, 59)$, $(4498, 67)$ still curvilinear of rank 4 (the $\ell = 37$ report's referee F4). No label added, renamed or removed. Companion pointers: [C] v1.17.0-pre.5, [EX] 1.3.0-pre.4, [U] 0.2.1-pre.2.
Version 1.17.0-pre.4 — 2026-10-02 (hecke-extensions slot 17, WP-2; **a pre-release, not signed**; **owner ruling of 2026-10-02 on finding B-1** of the referee F1). Prediction 4.9 **restated a third time** (a statement change of a [conjectural], the tag kept): (R) is now asserted on *congruence-free* classes (for every even $w \leq \ell + 1$ and every twist j, the generalised eigenspace of $(p^{-j}a_p)_{p\neq\ell}$ in $M_w(\mathbb{F}_\ell)$ has dimension ≤ 1; on reducible non-Dickson classes this equals Kummer-free), as the referee proposed; the sentence "(R) at every regular ℓ" is deleted. The v1.17.0-pre.2 Kummer-free form is recorded as **contradicted on the forms side** by the six cells $(2210/2278, 67)$, $(3798/3930, 131)$, $(4030/4170, 139)$ ($\mathbb{T}^{\mathrm{f}}_{\mathfrak{m}}/\ell \cong \mathbb{F}_\ell[x]/x^4$ at irreducible classes of two congruent eigenforms of weight $\leq \ell + 1$; reproduced with tracked tooling, slot-17 R17-2 §7; congruences confirmed over $\mathbb{Z}$; 139 regular; six further pairs the same), beside the v1.16.0 form and the v1.14.0 statement, both kept. Followers: the prediction's evidence paragraph (no evidence lost), §7.2's clause, Appendix A's row. No label added, renamed or removed; no schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.17.0-pre.4, [EX] 1.3.0-pre.4, [U] 0.2.1-pre.2.
Version 1.17.0-pre.3 — 2026-10-02 (hecke-extensions slot 17, WP-2 resumed; **a pre-release, not signed**: the repairs of the slot-17 Fable referee F1, `docs/implementation/slot17/WP-3_referee_F1.md`, [E] ACCEPT WITH REPAIRS, 2 blocking, 6 minor). **B-2** applied: Prediction 4.10 gains the hypothesis $g_{\ell+1-w} = 1$ (refuted as stated at $g = 2$ by the document's own $(608, 547)$: Loewy length 3 below the onset, $\dim V = 12$ in the window), "nothing asserted at $g \geq 2$", item (d) widened to every class at a pair with $g \geq 2$, Appendix A's row; a restatement of a [conjectural] under its label. **B-1** *not applied, awaiting the owner's ruling*: (R) of Prediction 4.9 is contradicted on the forms side at irreducible classes ($\ell = 67$, 131 and the regular 139; reproduced by slot-17 R17-2 §7 with tracked tooling, six further pairs the same); its statement, the regular-prime sentence and the "Refuted" list are unchanged at this version. Minors: m-E1 (item (c): the window's upper end computed on the multiplicity side at 13 pairs, 2092/2092 classes; only the algebra's shape extrapolated), m-E2 (the embedding-dimension sentence located), m-E3 (one sentence in (R)'s evidence: $\dim V \leq 4$ forces $\mathfrak{m}_A^2 = 0$), m-E4 (item 2's "$\dim I \geq 2$"), m-E5 (R17-1's Theorem 4.1: its inputs VERIFY there, a slot-18 statement); m-E6 needs no edit. No label added, renamed or removed; no schema key, record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.17.0-pre.3, [EX] 1.3.0-pre.3, [U] 0.2.1-pre.2.
Version 1.17.0-pre.2 — 2026-10-02 (hecke-extensions slot 17, WP-2, second pass; **a pre-release, not signed**, for the same referee F1 (WP-3); **owner ruling O-R17-2a of 2026-10-02**: restate (R) now, before the referee). (1) Prediction 4.9 **restated a second time** (a statement change of a [conjectural]): (R) is now asserted on *Kummer-free* classes only (residual representation irreducible, or reducible with both seeds ℓ-regular), in place of "not Kummer–Frobenius". The v1.16.0 form is **refuted on cohomology** at $(608, 547)$ (the $g = 2$ first window: $A \cong \mathbb{F}_{547}[x]/x^3$, saturated, `stable` certified; slot-17 R17-2) and contradicted on the forms side at the swapped classes at $\ell = 37$, 103, 691 ($\mathbb{F}_\ell[x]/x^4$) and at $(1156, 547)$; it is kept below the new form, marked refuted, with its evidence; new paragraphs on where the excluded locus sits, what held, and what is observed (not asserted) on the Kummer locus. (2) Prediction 4.10: the slot-17 evidence at the window's lower end off 37 ($(736, 67)$ on cohomology; 67, 59, 101 on the forms side), a scope item (d), one parenthetical reworded (the $t = J(w)$ class is the Eisenstein end $b = 0$ of the swapped family), and a correction: the first Loewy-4 cell at 103 is the swapped $(2574, 103)$, not $(8342, 103)$. (3) §7.2: one clause on the second restatement; edition 3.1's `sat_cells_done` quoted at publication (897 of 1459; the brief's E-2, mechanical). Appendix A: two rows. No label added, renamed or removed; no schema key,

record key, golden line, digest or census verdict changes. Companion pointers: [C] v1.17.0-pre.2 (one citation repair in [C, `lem:heckerankone`]), [EX] 1.3.0-pre.2.

Version 1.17.0-pre.1 — 2026-10-01 (hecke-extensions slot 17, WP-2, drafted during the slot-16 close; **a pre-release, not signed**, for the slot-17 Fable referee (WP-3) and the owner's ruling O3 of 2026-10-01, which releases 1.17.0 at the slot-17 close on the referee's acceptance).

(1) Prediction 4.10 **restated** (a statement change of a [conjectural], hence the X-type version): the window $\ell w + t(\ell+1) \le k < w + t(\ell+1) + (\ell^2 - 1)$ for $1 \le t \le J(w) - 1$ in place of "from $k = \ell w + t(\ell+1)$ on", which the forms side refutes at $\ell = 37$, $k = 1412$, 1450 (the $\ell = 37$ report `docs/latex/l37_loewy4_example.tex`, `rmk:kfonset`, `conj:window`); the superseded statement is kept below it, its evidence unchanged; the mechanism paragraph gains one sentence; Appendix A's row restated. (2) §7.2: a paragraph on **census edition 3.1** (add-only over edition 3; evidence and pointer only). No label added, renamed or removed; no schema key, record key, golden line, digest or census verdict changes. Not in this version: slot-17 R17-1's double-divisibility observation (not merged when this was drafted). Companion pointers: [C] v1.17.0-pre.1 (in the tree; v1.16.0 released), [EX] 1.3.0-pre.1, [U] 0.2.1-pre.1, [X] v1.21.

Version 1.16.3 — 2026-10-01 (hecke-extensions slot 16, **owner ruling O4, 2026-10-01**: a *patch* over 1.16.2 applying two proposals of `docs/implementation/slot15/WP-12_U_E_proposals.md`; **no existing statement, tag, label, schema key, record key, golden line, digest or census verdict changes**). (1) (E-3) *accepted*: Remark 4.11 ([proved here] from [3]) — the split / ramified switch of slot-14 I-8 is not Bellaïche–Khare obstructedness; with it the bibliography entry [19]. (2) (E-2) *accepted as a* [conjectural]: Prediction 4.12, the Frobenius layer (F1, WP-12's law L-F1), with its proof given two per-cell facts, its WP-12 evidence and a dated slot-16 evidence remark (WP-5, WP-11: consistent, no qualification). Appendix A: two rows added. Prediction 4.10 is untouched (its window-bound change is deferred to slot 17).

Version 1.16.2 — 2026-10-01 (hecke-extensions slot 16, the close: a *patch* over 1.16.1, one locator only, **no statement, tag, label, schema key, record key, golden line, digest or census verdict changes**). Claim 2.11's proof: Serre's weight congruence is cited as LNM 350, n° 1.3, Théorème 1, p. 198, verified on a public scan by slot-16 WP-8 (`docs/implementation/slot16/WP-8_citations.md` §2), in place of "§1"; the paraphrase now gives the source's own hypotheses ($p \ge 3$, rational coefficients, $v_p(f - f') \ge v_p(f) + m$), which contain both uses. Companion pointers: [C] v1.16.0, [EX] 1.2.2, [X] v1.20.2.

Version 1.16.1 — 2026-10-01 (hecke-extensions slot 15, post-close documentation package after census edition 3; a *patch* release over the owner-signed 1.16.0: evidence and pointers only, **no statement, tag, label, schema key, record key, golden line, digest or census verdict changes**; every addition is marked "v1.16.1"). (1) §7.2: a paragraph on **census edition 3** (slot-15 WP-15, `docs/matrix-engine/results/congruence_census/edition3/`, published beside edition 2, 80 293 rows, complete at $\ell = 61 \ldots 103$, partial at $17 \ldots 59$), with its coverage and what it adds as evidence. (2) Prediction 4.9: one evidence row (the complete C1b block, 2165 non-reduced rows at $\ell = 61 \ldots 103$, Loewy ≤ 2) and the edition-3 pointer. (3) Prediction 4.10: one sentence — at $\ell = 61 \ldots 103$ the onset law predicts no Loewy-4 cell inside $n \le 700$ and none is observed, so the law's positive half is *not* tested there. Companion pointers: [C] v1.15.0, [EX] 1.2.1, [X] v1.20.2.

Version 1.16.0 — 2026-10-01, **owner-signed release** (hecke-extensions slot 15, post-close; owner ruling Q13 of the slot-15 plan — release [E] 1.16.0 and [EX] 1.2.0 at the close — unblocked by the ruling S15-O2 of 2026-10-01, which accepted every open item and recommendation of `docs/implementation/slot15/WP-18_close_report.md`). *Released* from v1.15.1-pre.7 with no further edit of the text: v1.15.1 was never released, and because pre.7 changes the statement of a Prediction the release is an X-type version, 1.16.0. The sites keep their "v1.15.1-pre.N" markers as provenance. What v1.16.0 contains over v1.15.0, gathered in one place: (pre.1, slot 14) Citro–Ghitza's count of the level-one eigensystems mod p checked against $\Sigma(5)$, $\Sigma(7)$, $\Sigma(11)$, $\Sigma(13)$, $\Sigma(691)$ (§5), and its bibliography entry; (pre.2, slot-14 audit, owner rulings Q-C, Q-D) the definitional paragraph "The Hecke algebra over $\mathbb{Z}/\ell^K$, read mod ℓ" at the end of §7.1, and the sentence keeping the frozen u_ℓ RSS model `u_ell_rss/v2`; (pre.3, slot 14, P12's cluster half) the first-cluster evidence sentence of Prediction 4.9 (job `34551683`, 1233 blocks, `noctua2_section_m/`);

(pre.4, slot-15 WP-19, owner Q15) Claims 2.11 and 2.12, and the existence of $B'(k,\ell)$ at every $\ell \geq 5$; (pre.5, slot-15 WP-5, owner Q4) the *certified* `stable` verdict ([C, `cor:stablecert`] in the referee's B1 form) and the digit route beyond 2^{64} in that paragraph; (pre.6, slot-15 WP-5 Fable audit M1–M3) ε_K as the code computes it, the eigenspace comparison as a consistency check, one scope sentence; (pre.7, owner S15-O2) Prediction 4.9 **refuted as stated** at $\ell = 37$ and restated as (R) with the Kummer–Frobenius exclusion, and the new Prediction 4.10. No schema key, record key, golden line, digest or census verdict moves in any of the seven. Companion pointers: [C] v1.15.0, [EX] 1.2.0, [U] 0.1.1-pre.6 in the tree (unchanged; binding v0.1.0), [X] v1.20.1.

v1.15.1-pre.7 — 2026-10-01 (slot-15 post-close, **a pre-release**; owner ruling S15-O2 of 2026-10-01, which accepted proposal (E-1) of `docs/implementation/slot15/WP-12_U_E_proposals.md` and the Fable referee `docs/implementation/slot15/WP-4_referee.md` §8): Prediction 4.9 is **refuted as stated** — at $\ell = 37$ the 18 certified C1a records $\mathrm{eis}(a, a+5)$, $n = 654 \ldots 700$, have `loewy_A` $= 4$, Hilbert function $[1,1,1,1]$, module Loewy $[3,2,2,1]$, confirmed on the forms side and on the lattice mod 37^2 (slot-15 WP-13: $(654, 37)$ ideal 8, `CENSUS_LATTICE status=failed`, `first_fail=(2,2)`) — and restated, under its label, as the conjecture (R) with the Kummer–Frobenius exclusion past the onset weight $\ell w + t(\ell+1)$, with its evidence table; the v1.14.0 statement and its evidence stay in the text, marked refuted. New: Prediction 4.10 ([conjectural], the onset). Consistent with [C] v1.15.0 [C, `prop:cohforms`](5). Appendix A: one row changed, one added. The tag on Prediction 4.9 stays [conjectural] (for the restated form); no schema key, golden line, digest or verdict moves; the C++ readings keep reporting, never gating.

v1.15.1-pre.6 — 2026-09-30 (slot-15 WP-5 Fable audit repairs, **a pre-release, not signed**; `docs/implementation/slot15/WP-5_fable_audit.md` §3, M1–M3): in §7.1's paragraph "The Hecke algebra over $\mathbb{Z}/\ell^K$, read mod ℓ" — (i) ε_K defined as the code computes it, $K \cdot \dim_{\mathbb{F}_\ell}(A_K/\ell) - \mathrm{length}(A_K)$ (M2); (ii) the eigenspace-dimension comparison stated as a consistency check, the identification $e_{\mathfrak{m}} = e_{\mathfrak{m}'}$ resting on $e_{\mathfrak{m}} \in \mathbb{Z}[T_p]$, $e_{\mathfrak{m}}^2 = e_{\mathfrak{m}} \neq 0$ and the nilpotency (M1); one scope sentence, the printed Hilbert function a reading, the certificate one of the isomorphism (M3); (iv) the check count. No statement, label, STATUS tag, schema key, golden line, digest or verdict moves.

v1.15.1-pre.5 — 2026-09-30 (slot-15 WP-5, **a pre-release, not signed**; plan `docs/implementation/slot15/00-slot15-plan.md` §WP-5, owner ruling Q4 of 2026-09-30; the Fable referee `docs/implementation/slot15/WP-4_referee.md` accepted P15 and P17 and [C, `cor:stablecert`] with the statement-scope repair B1): the paragraph "The Hecke algebra over $\mathbb{Z}/\ell^K$, read mod ℓ" at the end of §7.1 — its item (ii) now states the *certified* verdict ([C, `cor:stablecert`] in B1's form: the block is that of a maximal ideal of $\mathbb{T}' = \mathbb{Z}_\ell[T_p : p \in S]$, possibly a sum of blocks of the full algebra), replacing "readings, not certificates" *for this verdict only*; item (i) and a new item (v) take the route beyond 2^{64} (the digit route); items (iii), (iv) and the Appendix-A row follow. Placed after WP-19's text (plan §6 document order). No statement of another claim, no STATUS tag, schema key, record key, golden line, digest or census verdict moves; the `LOEWY_RING` lines gain one appended field `stable_cert`, their old text byte-identical.

v1.15.1-pre.4 — 2026-09-30 (slot-15 WP-19, **a pre-release, not signed**; plan `docs/implementation/slot15/00-slot15-plan.md` §WP-19, owner ruling Q15 of 2026-09-30): two new claims after Claim 2.10 — Claim 2.11 (the forms' Hecke algebra mod ℓ has dimension $\dim M_k$ on $M_k(\mathbb{F}_\ell)$ at every $\ell \geq 5$, including $(\ell-1) \mid k$; the M_k claim is *proved*, not weakened) and Claim 2.12 (the F9 identity with its exact torsion term, which gives the torsion guard of plan WP-20) — and one paragraph after Claim 2.10 recording that $B'(k,\ell)$ exists at every $\ell \geq 5$. Both proofs are refereed in the WP-4 window. Placed before WP-5's text on the certified verdict (plan §6 document order). No statement of another claim, no schema key, golden line, digest or verdict moves. Appendix A gains two rows.

v1.15.1-pre.3 — 2026-09-30 (slot-14, **a pre-release, not signed**; the owner signed off proposal P12 on 2026-09-30, `docs/implementation/slot14/WP-D1_binding_doc_proposals.md`): P12's cluster half — one evidence sentence in Prediction 4.9 (the WP-M1 route on the first cluster, job `34551683`: $\mathfrak{m}_A^2 = 0$ on all 1233 non-reduced $\mathbb{F}_\ell$-rational blocks beyond the census window, citing the data set `docs/matrix-engine/results/noctua2_section_m/`, published 2026-09-30), with the forms-side

sentence of the WP-7 addendum, and the Appendix-A row. No statement, label, STATUS tag, schema key, golden line, digest or verdict moves; the tag on Prediction 4.9 stays [conjectural].
v1.15.1-pre.2 — 2026-09-29 (slot-14 audit fixes, **a pre-release, not signed**; owner rulings Q-C and Q-D of 2026-09-29, `docs/implementation/slot14/AUDIT_contract_currency_2026-09-29.md` §3.2): one definitional paragraph, "The Hecke algebra over $\mathbb{Z}/\ell^K$, read mod ℓ", at the end of §7.1 (E3 of the audit — the objects, refusals and surface of slot-14 WP-6's route, whose `stable` verdicts are *readings, not certificates* until the statements P15 and P17 are refereed in slot 15), one sentence at the ($d1$) cost cap recording the owner's ruling that the frozen u_ℓ RSS model `u_ell_rss/v2` is *kept*, and one Appendix-A row. No statement of §7.2, no label of another section, no STATUS tag, schema key, golden line, digest or verdict moves; the tag on Prediction 4.9 stays [conjectural].
v1.15.1-pre.1 — 2026-09-28 (slot-14 references edit, **a pre-release, not signed**): one citation and one check, at §5 — Citro–Ghitza's count of the level-one eigensystems mod p against $\Sigma(5)$, $\Sigma(7)$, $\Sigma(11)$, $\Sigma(13)$ and the stored $\Sigma(691)$, which agree — and its bibliography entry. No statement, label, STATUS tag, schema key, golden line, digest or verdict moves;
v1.15.0 — 2026-09-28, **owner-signed release** (hecke-extensions slot 14, package WP-13, owner decision 2026-09-28 "Go for publication and release", slot-14 plan §9 Q2 and Q3 with the recommended options). It applies the routing of slot-14 plan §8 (`docs/implementation/slot14/WP-D1_binding_doc_proposals.md`) to this document. Every change is evidence, a pointer or provenance; each is marked "v1.15.0" at its site. **P5**, with the local half of **P12**: Prediction 4.9's evidence gains a third, independent route (slot-14 WP-M1: 1229 blocks in 250 cells, all 259 non-reduced with $\mathfrak{m}_A^2 = 0$, and the Δ-block of $(400, 7)$), slot-14 WP-6's C++ route over $\mathbb{Z}/\ell^K$ (452/452 on 348 cells), and the sentence "the prediction concerns the reduction of the module, not of the algebra" ($A_K \otimes \mathbb{F}_\ell$ reaches Loewy length 3–6 on census rows and exactly 8 at $(400, 7)$). **P13**, which supersedes P6: the 206 Dickson-type blocks read by WP-6 are all reduced, so a non-reduced Dickson-type block has not occurred and the extrapolation stays untested. **P11**: edition 2's split lines, the θ-image table and the forms census were reproduced byte for byte on a second machine. **P4**: a pointer from (d5) to the $\mathbb{Z}/\ell^2$ depth window beyond $n = 64$, in the extract `docs/matrix-engine/results/noctua2_run1/`, published on the owner's sign-off. *Not applied here*: P9, which goes to the `mixed` proposal document; P10 and P16, which go to `ENGINE_API.md`; and P12's cluster half (the section-M data, job `34551683`, unpublished), deferred to slot 15. No statement, label, STATUS tag, schema key, golden line, digest or verdict moves, and the tag on Prediction 4.9 stays [conjectural]. Companion pointers: [C] v1.14.0, [P] v3.9.3. Release note `docs/implementation/slot14/WP-13_release_note.md`);
v1.14.0: 2026-09-27, **owner-signed release** (hecke-extensions slot 13, package WP-15 close, owner item 2 of the slot-13 close: *released* from v1.14.0-pre.3 with no further edit — the three pre-releases' text stands as published, the `pre.N` suffix dropped from the version string at every site it appears. What v1.14.0 therefore contains, gathered in one place: (pre.1, slot-13 WP-7, FIN_audit G2 closed) Certificate 5.10's per-ideal `boundary_data` (`rank_r1`, `r1`, `t`) present iff (d1) passed, `verify()` code 32 re-deriving (H1) and (H2) from the record, and the schema `hecke_ext/v4` → `v5` (golden R2: the 62 `ext_json_*` lines, the schema string only); (pre.2, slot-13 WP-9) Prediction 4.9 ([conjectural], candidate C-F9): $\mathfrak{m}_A^2 = 0$ on every $\mathbb{F}_\ell$-rational H^1 block at $\ell \geq 5$, held on all 654 non-reduced census rows of edition 2, with the Serre-weight count recorded as *not* explaining it; (pre.3, slot-13 WP-11, answering the referee's finding M4) the evidence sentence of Prediction 4.9 corrected: the evidence is entirely non-Dickson (all 654 tested rows `mixed` or `cuspidal`), so the extension to a non-reduced Dickson-type $\mathfrak{m}$ is an untested extrapolation, while the *equivalence* $\mathfrak{m}_A^2 = 0 \Leftrightarrow$ nearly split itself needs no such hypothesis. No statement, label, STATUS tag, schema key, golden line, digest or verdict moves beyond what pre.1–pre.3 already record; the tag on Prediction 4.9 stays [conjectural]. Built clean: `pdflatex` × 3, 0 undefined references, 0 undefined citations, 0 multiply-defined labels. Companion pointers: [C] v1.13.0, [P] v3.9.3, [X] v1.18.1. Report `docs/implementation/slot13/WP-15_report.md` (the slot close) and this release note);
v1.14.0-pre.3: 2026-09-27 (hecke-extensions slot 13, package WP-11, **the referee's repair of Prediction 4.9**: finding M4 of `docs/implementation/slot13/WP-11_referee.md`). One sentence of Prediction 4.9 is replaced (marked "v1.14.0-pre.3" at its site), and the §7.2 pointer and the

Appendix-A row say the same: the *equivalence* with $\mathfrak{m}_A^2 = 0$ needs no hypothesis "$\mathfrak{m}$ not of Dickson type", but the *evidence* is entirely non-Dickson — all 654 tested rows are `mixed` (567) or `cuspidal` (87), with a free lattice and `sturm_proof_range = true` — and every Dickson-type block computed so far is reduced ($\mathfrak{m}_A = 0$: 340 census rows and the 100 Dickson-type blocks of slot-13 WP-11's scan), so the extension to a non-reduced Dickson-type $\mathfrak{m}$ has *zero* evidence and is an untested extrapolation. The tag stays [conjectural]. No claim, certificate, schema key, golden line, digest or verdict moves; no label or STATUS tag is removed or renamed (`docs/implementation/slot13/WP-11_response.md`);
v1.14.0-pre.2: 2026-09-27 (hecke-extensions slot 13, package WP-11, **the C-F9 prediction**, owner answer Q8 of the slot-13 plan: "an [E] [conjectural] remark if it holds on all rows" — it held on all 654; a non-normative addition to the X-type pre-release, marked "v1.14.0-pre.2" at its site). (1) §4.3: Prediction 4.9 ([conjectural]): at $\ell \geq 5$, on every local block $V = H^1(\Gamma; M_n)_{\mathfrak{m}}$ with residue field $\mathbb{F}_\ell$, the local Hecke algebra has $\mathfrak{m}_A^2 = 0$ — equivalently the lattice $H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}_\ell)_{\mathfrak{m}}$ is "nearly split" — with its evidence (census edition 2, 654/654 non-reduced H^1 rows, slot-13 WP-9, `HeckeCensusLatticeValidation`) and the record that the Serre-weight count does *not* explain it (Loewy length 2 on every row; the bound by the number of distinct weights fails on 89 rows). (2) §7.2: a pointer to it. No claim, certificate, schema key, golden line, digest or verdict moves; no label or STATUS tag is removed or renamed. Report `docs/implementation/slot13/WP-9_report.md` §5, WP-11 in `docs/implementation/slot13/`);
v1.14.0-pre.1: 2026-09-26 (hecke-extensions slot 13, package WP-7, **spec first: (d1) re-derivable from the record** (slot-12 FIN audit G2's repair, owner-approved 2026-09-26, slot-13 Q3), an X-type change marked "v1.14.0-pre.1" at its sites. (1) Certificate 5.10: on an ideal with `d1 = passed` the record now carries the data (H1) and (H2) are decided on — $\mathrm{rank}(r^1|_V)$, the restriction $r^1|_V$ and the $\ell - 1$ restrictions $t_p|_V$, $p \in R(\ell)$ — and `verify()` re-derives (d1) from the record as it re-derives (d0) (new code 32); the v1.13.1 sentence "(H1) and (H2) in-process only" is superseded. (2) §7: the record becomes `hecke_ext/v5` (item (j), the one new key `boundary_data`, present iff `d1 = passed`). (3) The golden event (slot-13 R2): the 62 `ext_json_*` lines (the schema string only — under the harness's `u_ell = off` no (d1) passes, so no `boundary_data` is emitted there); `docs/matrix-engine/rebaseline-2026-09-26-hecke-ext-v5-d1-data.md`. No claim, label or STATUS tag is removed or renamed; no verdict moves. Report `docs/implementation/slot13/WP-7_report.md`);
v1.13.1: 2026-09-26 (hecke-extensions slot 12, S12-FIN, **the Fable audit's repairs** (`docs/implementation/slot12/FIN_audit.md`, 1 M / 3 m), a Y-type version marked "v1.13.1" at its sites: Certificate 5.10 gains the representative stream's fixed cost cap (G1; the new closed-list value `not_run:budget`, tied to the `u_ell` budget, no settings key) and states what `verify()` re-derives for (d1) (G2: structure only; (H1), (H2) in-process); `boundary_reps`' condition corrected to the implemented one. No golden line, digest or census verdict moves);
v1.13.0: 2026-09-26, **owner-signed 2026-09-26** (hecke-extensions slot 12, package S12-FIN: **the finality certificates land**, an X-type version, every site marked "v1.13.0"; the owner-approved golden changed-line event and record schema event of 2026-09-26). (1) §5: Certificates 5.9 (d0) "dickson" on X_1 (Proposition D of the slot-12 finality proposal v2, from [C, `prop:heckeH0`](1)) and 5.10 (d1) "boundary" on X_4 (Proposition F′, from [C, `prop:bdrytranscription`] and [C, `lem:bdrytorus`], which [C] v1.12.0 proves — so (d1) is *unconditional* and may make a record `final = true` in the default `closure` mode), with statements and proofs, run only where (a)–(c) do not pass; the practical order extended. (2) §7: the record becomes `hecke_ext/v4` — item (i): `certificate_routes.d0` / `.d1`, the per-ideal `boundary_reps`, the certificate names `"dickson"` / `"boundary"` (also admitted after `certificate_failed:`); the definition of `final = true` extended to the two routes; `verify()` codes 30–31. (3) The golden event: the 62 `ext_json_*` lines (schema and the new keys; under the harness's `u_ell = off` no (d0) / (d1) can pass, so no `final` and no raw object moved; `docs/matrix-engine/rebaseline-2026-09-26-hecke-ext-v4-finality.md`). No claim, label or STATUS tag is removed or renamed. Companion pointers: [C] v1.12.0, [P] v3.9.3, [X] v1.17. Report `docs/implementation/slot12/FIN_report.md`);
v1.12.1: 2026-09-25 (hecke-extensions slot 11, package WP-17, the slot close: **the WP-15 Fable**

referee's repairs (`docs/implementation/slot11/WP-15_referee.md`, 0 B / 2 M / 6 m), a Y-type version, every site marked "v1.12.1". (1) §7.1, "Published v1.12.0": the `hida` record object's sixth key `"e"` named (M-2). (2) §5.2 (h): **the specification stands** — $\ell < 5$ is a mathematical precondition, exit 3 — and the build now follows it (`DicksonThetaCell`'s `ENGINE_CHECK_MSG` naming [C, `lem:h2torsion`], slot-11 WP-17; the ruling on F-2, m-1). (3) Prediction 4.3: which "cuspidal" the reworded statement means (m-6). No claim, label, STATUS tag, record member (the key `"e"` was already emitted), golden entry or digest moves. Companion pointers: [C] v1.11.1, [P] v3.9.1, [X] v1.15. Report `docs/implementation/slot11/WP-17_report.md`);
v1.12.0: 2026-09-25 (hecke-extensions slot 11, package WP-15: **v1.12.0 published**, an X-type version, from v1.12.0-pre.2 with what slot 11 built, every site marked "v1.12.0". (1) **§7.1**: the paragraph "Published v1.12.0" — the pre-release paragraphs as built, member by member: the θ and census members of (b1)'s closure with the guards they actually carry, the console's θ verbs *not built* (slot-11 WP-16 gave way), the native `u128` layer as the third width and oracle, the digit route with its caps checked at $K = k + K_w - 1$, the refusal at $\ell \geq 2^{31}$, the `hida` record object, and (h)'s field members exiting 3 on a digit handle (slot-11 WP-14 pass 2, M-1); the console's `uint32_t` gap below 2^{64} recorded as an owner item. (2) **§5.2** and item 6 of §5.1: as built, with the refusal at $\ell \in \{2, 3\}$ being exit 5, not 3. (3) **§7.2**: the catalogue as built — the `t_p=<fnv64>` field of `CENSUS_SPLIT`, the cell-exact acceptance, the published edition 1 and the W2 data set `hecke_census_dataset/v1` in place of edition 2 (slot 12). (4) **Prediction 4.3 reworded**: read literally, its "iff" is false at $(10, 7)$ (Ramanujan's $\tau(n) \equiv n\sigma_3(n) \bmod 7$); it is now stated for E_k's own class, with the exceptional residues named ($w \equiv 0, 2$). (5) §8: the uniqueness item at $(198, 691)$ is **closed** (one congruent eigenform; degrees $1, 1, 2, 2, 10$), and OQ13's residue discrepancy is the sign ($|\,\mathrm{num}\,| \equiv 217665$, signed $\equiv 259816$). No claim, label or STATUS tag is removed or renamed; 4.3 keeps [standard, reference] for the restricted statement. No record member, golden entry or digest moves. Companion pointers: [C] v1.11.0, [P] v3.9, [X] v1.14.1. Report `docs/implementation/slot11/WP-15_report.md`);
v1.12.0-pre.2: 2026-09-25 (hecke-extensions slot 11, package WP-18c, **spec first**, drafted by WP-18c's author for WP-4's editor row: the second pre-release of v1.12.0, an X-type change marked "v1.12.0-pre.2" at its sites, under the owner's clarification of 2026-09-25 (latest; slot-11 plan §10 last row): *arbitrary k by the (C2) digit induction, in slot 11.* It supersedes pre.1's sentence "(a)'s grammar $k \leq 63$ does not move in slot 11". **§7.1**: (a)'s grammar cap $63 \to 4096$; the per-cell rule (native below 2^{64}, the digit route at and above it for $\ell \geq 5$, $\ell \in \{2, 3\}$ refused there by name, a 8192-bit and a memory cap); the digit route's member list, its per-cell certified precision bound $K = k + e_{\max}$ with the record keys that state it, the ordinary idempotent as a certified datum, and the engine-family switch `ENGINE_LADIC_WIDTH`; (g)'s digit jobs; (h)'s digit handles. No existing value, record member, golden entry or digest moves: every $\ell^k < 2^{64}$ keeps the shipped route. Report `docs/implementation/slot11/WP-18c_report.md`);
v1.12.0-pre.1: 2026-09-25 (hecke-extensions slot 11, package WP-4, **spec first**: the first pre-release of v1.12.0, with its own identifier so that no version number is published twice under different text ([C] V1 / V2), written before any slot-11 code package commits a claim (slot-11 plan `docs/implementation/hecke_extensions_slot11_plan.md` §3 "WP-4", rule 21) and under the owner's rulings of 2026-09-25 on every slot-11 decision — all at recommendation, decision 3 at option (A′) (that plan's §10); nothing below is provisional. An X-type change; every addition is marked "v1.12.0-pre.1" at its site. (1) **§5.2** (new): the Dickson–θ objects slot-11 WP-5 and WP-6 build — the indexing, the quotient local system Q_n, the six terms and five maps of the long exact sequence with their code names, what is certified and what is asserted, the hand-off file to the census, the refusals, and what they are *not* (no record member, no settings key, no golden line, no certificate item code). (2) **§5.1 item 6**: its $L\cdot$ half (E1 identity class 5) extended by the identities (6a)–(6h) of that long exact sequence. (3) **Claim 2.14**: split into its two sentences; the first, $U_\ell \circ \theta = 0$ over $\mathbb{F}_\ell$ (already on cochains), upgraded from [standard, reference] to a pointer at [C]'s proof, [C, `prop:thetatwist`] in [C, `sec:dicksontheta`] (slot-11 WP-2), the second (the Jordan type) kept [standard, reference]; and, with [C]'s correction [C, `rmk:heckeH0plan`], (6f) no longer says "$U_\ell = 0$ on all of H^0": U_ℓ has rank ≤ 1 and square zero there, and is zero only for $n < \ell^3 - \ell$ ([C, `prop:heckeH0`]; $U_5 D^6 = -L^{20}$ at $(120, 5)$)); "Dickson and θ-images are non-ordinary" stands

([C, `cor:thetanonordinary`]). (4) **§7.2** (new): the congruence-census catalogue, a derived artefact with its own format tag `hecke_census/v1` — its inputs and supersession rule, its rows, its columns each with its source, the boundary split rule (Eisenstein / cuspidal / mixed by rank r^1 on the ideal's generalised eigenspace), the $\pm$, θ-image, Glover-reconciliation and $\mathbb{Z}/\ell^k$-depth rules, the extremal-example criteria [EX] selects from, the line formats of the two computed columns, and the `results/` layout. (5) **§7.1**: the pre-release paragraph "v1.12.0-pre.1" — the slot-11 members of (b1)'s closure, all refusing a non-default ring by name (θ and the census are $\mathbb{F}_\ell$-only in slot 11), and L1-X under decision 3 (A′): step 1 (the 128-bit engine element type with its oracle rule, and F-1, F-2) is built in slot 11 and is value-neutral; **(a)'s grammar $k \le 63$ does not move in slot 11**; the amendment $k \le 126$ with the cell refusal at $\ell^k \ge 2^{127}$ is slot 12's. No claim, hypothesis, formula or label is removed or renamed and no STATUS tag moves except Claim 2.14's; no member of `hecke_ext/v3`, `hecke_report/v2` or `hecke_ring_report/v1` moves; no settings key is added (eleven stay eleven); no gate, cell list, golden entry or digest moves. Companion pointers: [C] v1.11.0-pre.1 (slot-11 WP-2), [X] v1.14.1, [P] v3.9-pre.1. Report `docs/implementation/slot11/WP-4_report.md`);

v1.11.1: 2026-09-25 (hecke-extensions slot 10, packages WP-4 and WP-5, window **W2**: a Y-type status change. The owner lent the box for W2 after the close window W1 (consent of 2026-09-25), and the two optional publications of decision 8 (A) ran there: slot-10 WP-4's ℓ-adic survey under $\mathbb{Z}/\ell^2$ (`docs/matrix-engine/results/ring_survey_l2/`, 462 jobs, 0 failed) and slot-10 WP-5's second extension campaign (`docs/matrix-engine/results/ext_u691/`, the 45 cells $\ell = 691$, $n = 112, \ldots, 200$ that `ext_first` recorded `u_ell_over_budget`, 0 failed). (1) **The survey** (§7.1, the status paragraph, and (g)): v1.11.0's "not run in slot 10 (deferred)" becomes "run in W2", with what it found; (g) still moves by nothing — the survey wrote `hecke_ring_report/v1` exactly as frozen, and no finding needs a member. Every change is marked "v1.11.1" at its site. No claim, hypothesis, formula, STATUS tag, label, gate, cell list, schema member or constant moves; no golden entry and no digest moved. Companion pointers: [C] v1.10.1, [X] v1.14.1, [P] v3.8. Reports `docs/implementation/slot10/WP-4_report.md`, `WP-5_report.md`);

v1.11.0: 2026-09-25 (hecke-extensions slot 10, package WP-15, **phase 2**: publishes v1.11.0, the first text under this number. It is v1.11.0-pre.2 with every site phase 1 marked filled from the reports of slot-10 WP-7 and WP-8, and with the phase-1 Fable referee's repairs that reach this document (`docs/implementation/slot10/WP-15_referee_phase1.md`: M-1 and m-5 in (b3)). An X-type change, as the pre-releases announced; every addition is marked "v1.11.0" at its site. No claim, hypothesis, formula, STATUS tag or label is added, renamed or removed; no gate, cell list, schema member or constant of an earlier version moves. Base `47e3ff9`: every slot-10 code package merged. (1) **Lane B** (§7.1, the status paragraph): slot-10 WP-7's two levers — the ι sign split's independent checks and splits as host tasks, and L4, route 2's rows of `HeckeOperatorOnK` on the consumer pool — recorded as value-preserving with their evidence, and its cone-certificate lever recorded for slot 11; slot-10 WP-8's ninth fingerprint bit (a cache key) and its validation carries. (2) **(b1)**: no lever in the cone classes or in `HeckeSignSplit`; L4 and the ι lever located in the closure. (3) **(f)**: route 3's non-vacuity asserted (slot-10 WP-8, review m-12), `HeckeConeCrossRouteRingValidation`'s count $386 \to 431$ by the source, re-derived at the close. (4) **The survey** (status paragraph and (g)): slot-10 WP-4 not run in slot 10 (the optional W2; deferred). (5) **(b3)**: "has always computed" becomes "computes", with the canonical τ dated to [C] v1.1.0 and the legacy–canonical observation of [C, `rmk:tauexistsscope`](1) cited (referee M-1, m-5). Companion pointers: [C] **v1.10.1**, [X] v1.13, [P] v3.8. Report `docs/implementation/slot10/WP-15_report.md`);

v1.11.0-pre.2: 2026-09-25 (hecke-extensions slot 10, package WP-15, **phase 1**: the second pre-release of v1.11.0, which records at the sites of v1.11.0-pre.1 what slot-10 WP-3, WP-10, WP-11, WP-12 and WP-13 built (slot-10 plan §3 "WP-15"). It carries its own identifier, so that no version number is published twice under different text ([C] V1 / V2; slot-10 plan §4.3): the packages of the slot's lane B (WP-7, the cone and ι profile with its optional levers, and WP-8, the carries) were still running when it was written, and the sites they may complete were marked in its text, in italics, as bracketed phase-2 sites (the markers are gone since v1.11.0, which filled them); phase 2 fills them and publishes v1.11.0. An X-type change, as the pre-release announced; every addition is marked

"v1.11.0-pre.2" at its site and names the validation that measures it. No claim, hypothesis, formula, STATUS tag or label is added, renamed or removed; no gate, cell list, schema member or constant of an earlier version moves. Base `f0640d0`: WP-2, WP-3, WP-6, WP-9, WP-10, WP-11, WP-12 and WP-13 merged, and the slot's independent review (`docs/implementation/slot10/WP-14_review.md`, passes 1 and 2). (1) **Status** (§7.1, new paragraph "Status at v1.11.0-pre.2"): the three members of [C, `obl:coneexact`](vii) over $\mathbb{Z}/\ell^k$ are built as specified (slot-10 WP-3), and **E8's declaration is written, scoped by the review's finding M-1**: (vii)'s first two sentences discharged at every E8 point ($\ell^k < 2^{32}$), its third — the Poincaré-duality adjointness gate — recorded unavailable over the ring, since the E7 pairing and the contragredient refuse a composite ring; the quoted declaration of v1.11.0-pre.1 is amended accordingly. (2) **(b1) as built**: the cross-routes' guard and regime refusal, the changed rows of arms (X)/(X4b); line 2's free model (WP-10, WP-13, WP-11) outside the closure as specified, the read-only use of `HeckeOperatorOnK` on the strict table at $k \geq 2$ confirmed, and the functoriality argument stated with its three inputs (review finding m-1); the console members. (3) **(b3)**: the decision-4 block as built — nothing lifted, no line-2 package exhibited two τ with different operators, and [C] v1.10.0's [C, `rmk:bredonhecke`] carries the reason. (4) **(f) as built**: route 3's transport by an explicit inverse on classes instead of column-by-column solves (a recorded deviation, verified by the review), the acceptance as measured, the negative control's scope, the regime arm; the pointer into [C] closed. (5) **(h) as built** (slot-10 WP-12): the five members, the mismatch rule pinned both ways, the member-getter position, and one exit code that moves by the Core exception's own rule — an unwrapped getter raising `NotImplementedInThisBuild` is now exit 5, not 3. (6) The two pointers into [C] of v1.11.0-pre.1 are closed by [C] **v1.10.0**. Companion pointers: [C] **v1.10.0**, [X] v1.13, [P] **v3.8**. Report `docs/implementation/slot10/WP-15_report.md`);

v1.11.0-pre.1: 2026-09-25 (hecke-extensions slot 10, package WP-2, **spec first**: the pre-release of v1.11.0 that slot-10 WP-15 completes. It carries its own identifier `1.11.0-pre.1`, so that no version number is published twice under different text ([C] V1 / V2; slot-10 plan §4.3). An X-type change, written *before* the code packages that honour or refuse what it specifies (slot-10 WP-3, WP-10, WP-12, WP-13), under the owner's rulings of 2026-09-25 on the slot-10 decisions, every one at recommendation (A) (slot-10 plan §10); decisions 3, 4, 6 and 8 are the ones used here, and **nothing below is provisional**. No claim, hypothesis, formula, STATUS tag or label is added, renamed or removed; no gate, cell list, schema member or constant of an earlier version moves. Every addition is marked "v1.11.0-pre.1" at its site, in §7.1; the paragraph "Pre-release v1.11.0-pre.1" there is its index. (1) **(b1) and (f): [C, `obl:coneexact`](vii) over** $\mathbb{Z}/\ell^k$ (decision 3 (A)). Three members move from "keeps refusing" to "honours" at $\ell \geq 5$, $\ell^k < 2^{32}$, from slot-10 WP-3's lift commit: `HeckeConeStrictPairRoute`, `HeckeConeRelativeRoute` and `EquivariantCohomologyOnK`'s relative sub-complex constructor, on the dense module route of (f) with (c1)–(c5) verbatim; the acceptance is entrywise equality of each cross-route's $A_c^{(q)}$ with route 1's in the cyclic-summand basis of (f), at the seven E8 points of (e); above 2^{32} both cross-routes keep the ring context's named regime refusal (the `uint32_t` regime rule of (b1)), and route 1 alone runs every regime; $\ell \in \{2, 3\}$ keeps refusing. The declaration E8 reaches on acceptance is stated, for slot-10 WP-15 to write. (2) **(b3) and (c1′): the Hecke action on the Bredon object at** $\ell \in \{2, 3\}$ **is declined** (decision 4 (A)), for two separately named objects: the cochain map Θ on $C^{\mathrm{eq}}(\mathbb{Z}/\ell^k)$ is neither built nor needed (line 2 reads $\widehat{\Theta}_g$ on $\widehat{C}$, [C, `prop:isomodel`](3)); the operator on cohomology is **not shown** independent of τ ([C, `lem:tauexists`](2), [C, `prop:strictlift`](4)), which is the absence of a proof, not a proof of dependence. `HeckeOperatorOnK`'s refusal there stands. (3) **(b1): line 2's free model is outside the closure**: the objects of slot-10 WP-10 and WP-13 (`Integral/Model/StrictLift`, `PeriodicityClass`) take their ring as an explicit `ModCtx` and read no setting, as slot 9's `IsotropyModel` and `IntegralGroups` do; the read-only use of `HeckeOperatorOnK` on the strict table inside a validation is an honouring use of an honouring member, and a refusal it meets stands. (4) **(h): the structural members** (decision 6 (ii) (A)): `LocalSystem.n`, `.d_V` and a new `.coefficient_ring`, and `HeckeOperator.p` and `.r`, honour on a ring handle from slot-10 WP-12's lift commit; every other unlisted member still exits 5; slot-9 WP-12's finding "a ring mismatch is exit 5" is recorded as the rule. (5) **(g): no schema change** (decision 8): slot-10 WP-4's survey

writes `hecke_ring_report/v1` as frozen in v1.10.0. (6) **Pointers**: the staleness of [C] [C, `obl:coneexact`](vii)'s "its composite half is open" and of [C, `sec:notimplemented`]'s matching sentence from slot-10 WP-3's merge until [C]'s next edit (slot-10 WP-15, [C] v1.10.0), recorded and not patched (slot-10 plan §4.3). Companion pointers: [C] v1.9.1, [X] v1.13, [P] **v3.8-pre.1**. Report `docs/implementation/slot10/WP-2_report.md`);
v1.10.1: 2026-09-25 (hecke-extensions slot 9, package WP-17, the slot close: **the Fable referee's repairs to v1.10.0**, a Y-type change — no claim, hypothesis, formula, STATUS tag, label, schema member, constant or rule moves, and every addition is marked "v1.10.1" at its site. (1) **M-1**: the slot's independent review (WP-14, pass 1), which v1.10.0 cites in (b1), (c1′) and (f), is now in the tree as `docs/implementation/slot9/WP-14_review.md` (reconstructed from the reviewer's returned summaries after the original file was lost on 2026-09-25; it keeps its provenance note), and each of those three sites names it. (2) **m-7**: (b1)'s scoping row says that the scope guard also exempts its own translation unit, `Registry/Slot9/WP9.cpp`. (3) **m-8**: (f)'s length identity for β_c^1 is asserted at the `uint32_t` cells, not at the two word-regime cells. (4) **m-10**: v1.10.0's entry below said "nothing else of v1.10.0-pre.2 changed"; it also carried the (g7), (g9) and (g11) pointers and the slot-9 WP-9 sites its items (2)–(5) list, which v1.10.0-pre.2 had announced only as "any site WP-6/7/9 touch" — the sentence is read with that clause (the entry itself is left as written). Companion pointers: [C] **v1.9.1**, [X] **v1.13**, [P] **v3.7.1**. Report `docs/implementation/slot9/WP-17_report.md`);
v1.10.0: 2026-09-25 (hecke-extensions slot 9, package WP-15, **phase 2**: publishes v1.10.0, the first text under this number. It is v1.10.0-pre.2 with every site phase 1 left open filled from the reports of slot-9 WP-6, WP-7, WP-8, WP-9 and WP-4t, and nothing else of v1.10.0-pre.2 changed. An X-type change, as the pre-releases announced (one schema member is amended before the schema is first published, item (2)); every addition is marked "v1.10.0" at its site and names the validation that measures it. No claim, hypothesis, formula, STATUS tag or label is renamed or removed, and no label is added. Base `9b004f2`: every slot-9 code package merged. The Fable referee reads this version; its repairs go to v1.10.1. (1) **Status** (§7.1, new paragraph "Status at v1.10.0"): the campaign (C5c, slot-9 WP-8) built as specified; the merge fix `158c0be`; the carries of slot-9 WP-9; the trimming of slot-9 WP-4t; E8's declaration unchanged. (2) **(g) as built** (slot-9 WP-8): the as-built paragraph — the dispatch, the element width, the routes at $\ell \geq 5$ and at $\ell \in \{2, 3\}$, `CampaignRingValidation`'s arms and the fingerprint-0 byte identity — with **four recorded deviations**: `kmode` `not_applicable` also at $\ell \geq 5$ off the K-mode's range (noted at (g7)); the uncertified state's `need` (noted at (g9)); `hecke` absent and not refused at $\ell \geq 5$ outside (g11)'s condition; each refusal's `who` and `message` from the real guard, a guard that stops refusing failing the job. **(g11) amended** by the lead's decision of 2026-09-24: the free `charpoly`'s coefficients are *decimal strings*, since at a `uint64_t` cell they exceed 2^{53}; no other member can. (3) **(b1) and (b4): the campaign rows as built** — the job loop honours $\mathbb{Z}/\ell^k$ at every ℓ with $\ell^k < 2^{64}$; the ext pass and a cell with $\ell^k \geq 2^{64}$ refused at spec validation, before any output exists. (4) **(b1): review finding m-1 and the `RunPipelineModEll` guard closed** (slot-9 WP-9, `SmallEllPolicyScopeValidation`; the merge fix `158c0be`); the constant-fibre constructor's composite caller of `get_prime()` re-pointed; the item-4b tiers as trimmed by slot-9 WP-4t ($(60, 5)$ in the `slow` sibling; $458 \to 306$ checks). (5) **(c1′): review finding m-2 closed** (slot-9 WP-9, `HeckeBredonU64FlatRouteValidation`, one honouring `uint64_t` cell at $\mathbb{Z}/2^{33}$). Companion pointers: [C] v1.9.0 (unchanged; its pointer already names this version), [X] v1.12, [P] **v3.7**. Report `docs/implementation/slot9/WP-15_report.md`);
v1.10.0-pre.2: 2026-09-24 (hecke-extensions slot 9, package WP-15, **phase 1**: the second pre-release of v1.10.0, which records at the sites of v1.10.0-pre.1 what slot-9 WP-3, WP-4, WP-12 and WP-13 built (slot-9 plan §3 "WP-15", its two-phase start of §4.1 Rule C). It carries its own identifier, so that no version number is published twice under different text ([C] V1 / V2; slot-9 plan §4.3): the packages of the slot's lane B (WP-6 … WP-9, in particular WP-8's campaign support, C5c) were still running when it was written, and the sites they complete were marked in its text for phase 2, which filled them and published v1.10.0 (above); the Fable referee reads v1.10.0. An X-type change, as the pre-release announced; every addition is marked "v1.10.0-pre.2" at its site and names the validation that measures it. No claim, hypothesis, formula, STATUS tag or label is renamed or

removed, and no label is added. Base `7ddb555`: WP-2, WP-3, WP-4 (with its review repairs and the golden event R1), WP-10, WP-11, WP-12 and WP-13 merged. (1) **Status** (§7.1, new paragraph "Status at v1.10.0-pre.2"): E8 item 4b is built by route 1, and **E8 is complete as specified** — items 1–4b as specified, 4b over $\mathbb{Z}/\ell^k$ by route 1, [C, `obl:coneexact`](vii)'s composite half open (the two cross-routes and the relative constructor refuse, with arms); R1 has landed (126 entries, 0 changed, 0 removed; tier A 11 666, tier B 5901). (2) **(a)**: the `--set` exit 5 and the `set`-statement position, as built by WP-12 (`ConsoleErrorValidation` arm (K), $1 \to 5$). (3) **(b1), (b3), (b4) as built**: the scoped lift at $\ell \in \{2, 3\}$; **the K-mode's refusal kind there is** `NotImplementedInThisBuild` (exit 5), recorded as standing; the 4b closure as WP-4 re-derived it, with the `uint32_t` width of the four classes; the console surface (WP-12, WP-13). (4) **(c1′)**: its "the pipeline's $H^2 = 0$ is asserted" is corrected — the route does not assert it; `HeckeBredonTreeRingValidation` checks it (review finding m-3). (5) **(f) as built**: [C, `obl:coneexact`](vi) **narrowed** to what is built — the commutator asserted, the $\nu = 2$ recursion not computed over the ring (review finding M-W4-1); item 14's completeness argument (m-W4-2); the measured values ($m = 1$ at $(16, 5)$, the c_2); the landed set. (6) **(h) as built**: the honouring surface, and WP-12's four findings — member getters carry no source position (a next Core batch), unlisted structural members exit 5 on a ring handle, a ring mismatch is exit 5, and `.A(q)` over the ring is a List of rows. (7) The pointers into [C] of (b3) and (f) now name [C] v1.9.0, which repairs both clauses. Companion pointers: [C] **v1.9.0**, [X] v1.12, [P] **v3.7-pre.1**. Report `docs/implementation/slot9/WP-15_report.md`);

v1.10.0-pre.1: 2026-09-24 (hecke-extensions slot 9, package WP-2, **spec first**: the pre-release of v1.10.0 that slot-9 WP-15 completes. It carries its own identifier `1.10.0-pre.1`, so that no version number is published twice under different text ([C] V1 / V2; slot-9 plan §4.3). An X-type change, written *before* the code packages that honour or refuse what it specifies (slot-9 WP-3, WP-4, WP-8, WP-12), under the owner's rulings of 2026-09-24 on the slot-9 decisions, every one at recommendation (A) (slot-9 plan §10); decisions 2, 3 (with its sub-item (iii)), 4 and 6 are the ones used here, and **nothing below is provisional**. No claim, hypothesis, formula, STATUS tag or label is renamed or removed, and no label is added; no gate, cell list or constant of an earlier version moves. Every addition is marked "v1.10.0-pre.1" at its site, in §7.1 unless noted. (1) **(b3) and (c1) at** $\ell \in \{2, 3\}$ (decision 4; slot-9 plan audit B-1, M-4). `EquivariantCohomologyOnK` honours $\mathbb{Z}/\ell^k$ at 2 and 3, through a *flat-coordinate* module route (new item (c1′)), reporting lengths and invariant factors of the Bredon object only; the certified K-mode, `HeckeOperatorOnK`, `HeckeReduction` and every field-only consumer keep refusing there. (c1)'s "every pivot is asserted to be a unit, so C^q_{eq} is free" is restricted to $\ell \geq 5$. The as-built refusal arm "through `CoefficientRingContext`, $\ell = 2$" of (b3) is recorded as the arm that *flips* to honouring, and its sentence "WP-2's choice … not objected to" is marked superseded. A pointer records that [C] [C, `def:ringsemantics`](2)'s clause "the implementation of this build refuses it there" is superseded by (b3) until [C]'s next edit (slot-9 WP-15). (2) **The 4b objects** (decision 3, with (iii)): new paragraph (f), the boundary complex and Cone(r) over $\mathbb{Z}/\ell^k$ — their coordinates, their tier-A and tier-B objects with golden names at the seven existing E8 points, the generating rule of slot-9 WP-4's pre-stated set (126 entries), predicted values where they follow from the mathematics, and [C, `obl:coneexact`](vii) over the ring as route 1 only, its composite half recorded as open. (3) **The job-level schema** `hecke_ring_report/v1` (decision 6): new paragraph (g), emitted by the campaign's job loop only under a non-default ring; a fingerprint-0 job emits `campaign_job/v1` with `hecke_report/v2` byte for byte as before, and the ext pass refuses a non-default ring by name at spec validation. (4) **The console under a non-default ring** (C5a, decision 2): new paragraph (h), the verbs and members that honour $\mathbb{Z}/\ell^k$, and (a)'s "Registration" sentence rewritten as ruled: a legal but unexecutable `--set` value is exit 5 from slot-9 WP-12's commit. (5) **(b1): the lift list**, one row per member that moves in slot 9, with its package and its arm, and the new "honours at 2 and 3" row; (b4)'s surfaces restated for slot 9. Companion pointers: [C] **v1.8.1**, [X] **v1.12**, [P] v3.6.1, `ENGINE_API.md` §13 (slot-8 close, [WP8-14] §8 item 7). Report `docs/implementation/slot9/WP-2_report.md`);

v1.9.1: 2026-09-24 (hecke-extensions slot 8, package WP-13, the repairs of its Fable referee pass, `docs/implementation/slot8/WP-13_report.md` §7. A Y-type change: **pointers and wording**

only. No claim, hypothesis, formula, STATUS tag or label is added, renamed or removed, and no gate, cell list, constant or rule moves. Every repair is marked "v1.9.1" at its site. (1) **§7.1 (b1), the "Unreachable" bullet** (finding M-2). Its reason for `kernel.cu`'s legacy benchmark said that no command-line switch sets a mathematical key outside `--console` and `--campaign`. That is false: `--bench ext --u-ell` installs `u_ell`. The reason now given is that no switch sets `coefficient_ring`, and that `--bench` refuses a non-default ring at `RunMode`'s entry. (2) **(a) and item (2) of v1.9.0's entry: the `--set` exit code** (findings M-1, m-2). Exit 1 for `--set coefficient_ring=Z` was attributed to arms that did not test it. It is now pinned by one check added to arm (K) of `ConsoleErrorValidation`, and was re-measured with a correct command line. (3) **(a): the emission and transcript facts** (finding m-1) are attributed to slot-8 WP-3's gate runs, not to the arms. (4) **(c1): [C, `prop:coneintegral`](1)–(2)**, not (2) alone (finding m-3). (5) **[IB] is defined** in §7.1's preamble (finding m-4). (6) **(b3): "the two refusals" listed four**; it now reads "the honouring object's refusals" (finding m-7). Companion pointers: [C] v1.8.0, [X] v1.11, [P] **v3.6.1**);

v1.9.0: 2026-09-24 (hecke-extensions slot 8, package WP-13. This revision completes v1.9.0-pre.1 and publishes v1.9.0. It is an X-type change, as the pre-release announced. No claim, hypothesis, formula, STATUS tag or label is added, renamed or removed. No gate, cell list, constant or rule moves. One predicted literal is restated in model-free form, item (7). §7.1 now records what was built against it, at the sites it specifies. Each addition is marked "v1.9.0" and names the validation that measures it. Base `5c70e95`: every slot-8 code package merged. Report `docs/implementation/slot8/WP-13_report.md`. (1) **E8 status** (§7.1, new paragraph "Status at v1.9.0"). Items 2 and 3 are implemented (slot-8 WP-1, WP-3), with the repairs from the independent review (WP-12 pass 2; WP-3r). Item 4a is accepted (WP-4). R1 has landed: 124 entries at the seven points, 0 changed, 0 removed, tier A 11 575, tier B 5866, both digests unchanged. Item 4b is slot 9's. Measured by `HeckeReductionValidation`, `HeckeReductionCoverageSlowValidation` and `GOLDEN_STRICT=1 GoldenRegressionValidation`. (2) **(a) as built.** The key is registered as specified, the eleventh key. At the console the reserved values give exit 5 on the `set` statement (arm (K) of `ConsoleErrorValidation`; arm (X)) and exit 1 as a `--set` argument (v1.9.1: pinned by arm (K); this entry first attributed it to both arms, which did not test it). Line 2's record keys are emitted only under `Z`; the owner confirmed this (slot-8 ruling 6d). (3) **(b1): the list.** The closure is copied from slot-8 WP-3's derivation and re-derived twice: by the review (WP-12), which added `Integral/IntegralCharpolyCrt` (finding m-5; it refuses structurally), and at this revision's base, which added the campaign's job loop, covered by its surface. (4) **(b3) and (b4) as built.** No `--bench` switch was added. The U64_Wide range is honoured and exercised (WP-3r, finding M-2). The refusal at $\ell \in \{2, 3\}$ now points at [C] [C, `def:ringsemantics`](2) and [C, `prop:certifiede`](5). The converse caller error is a named `logic_error` (finding m-6). Each surface's arm is named. (5) **(c): the coordinates of (G, R_l) and the direct test.** (c1) now states that the input is read in C^q_{eq} coordinates (finding m-9). WP-3r's arm (M) tests the rule directly, with mutations (finding M-1). (6) **(d) as built.** The K-mode; the Bockstein, whose meaning now points at [C] [C, `rmk:bocksteinexp`](1)–(2) (proved since [C] v1.8.0; the F-A pointer the slot-8 plan asked for); the free `charpoly`, asserted at weight 12 (finding m-7); `red_H^q` with its arms (R1)–(R5); and what the (D6) arm measured at $(10, 691)$. (7) **(e): footnote b restated, and the landed values.** "$51 \neq 52$" is [L]'s amalgam-model coordinates. In the shipped model it is $234 \neq 235$. The footnote now states the model-free content, a deficit of exactly one (WP-3's finding, confirmed by finding m-8; for the owner to ratify, WP-13 report). The frozen values are listed after the object list. (8) **(D6)**, the item and its row in §A: an E8 run exists, in the validations and golden only, so the published semantics is unchanged. Companion pointers: [C] **v1.8.0**, [X] v1.11, [P] **v3.6**, `ENGINE_API.md` §13, whose pointer repairs (WP-12 pass 1, m-1–m-3) are the slot close's);

v1.9.0-pre.1: 2026-09-24 (hecke-extensions slot 8, package WP-2, **spec first**: the pre-release of v1.9.0 that slot-8 WP-13 completes. It carries its own identifier `1.9.0-pre.1`, so that no version number is published twice under different text ([C] V1 / V2; slot-8 plan §4.3). An X-type change: a mathematical setting gains its grammar, default and refusal rule, and the E8 objects gain their

names and meaning, all *before* slot-8 WP-3 registers the key and WP-4 emits the objects, so that their content gates are checks and not stops. No claim, hypothesis, formula, STATUS tag or existing label is renamed or removed; nothing here is emitted by a record in slot 8. Written under the owner's rulings of 2026-09-24 on the slot-8 decisions 2, 3, 4 and 5, each at recommendation (A) (slot-8 plan §10); nothing below is provisional. New §7.1 at the end of §7: (1) `coefficient_ring`: its grammar `F_ell | Z/ell^<k>`, the reservation of `Z` (line 2, [IB] OD-10 (a)) and of `Z/<m>`, its default `F_ell`, its canonical text, its fingerprint and golden cost (the grammar and the default cost no golden line; a record field added for a ring costs golden lines unless it is emitted only under a non-default ring), and where the ring is read; (2) **the refusal rule** (decision 4): every construction site and every field-only consumer that does not honour a non-default ring raises `hecke::NotImplementedInThisBuild` naming itself (console exit 5); the seven user-facing surfaces by name; the closure as a rule; (3) **the E8 objects** (decisions 2 and 3): `lengths_H^q`, `invariant_factors_H^q`, `ladic_type^q` with `ladic_type_certified` and the two counts $r_K(d^0)$, $r_K(d^1)$, the free-quotient `charpoly` with its precondition and precision $K - e_{\max}$, `bockstein_rank^0` and `higher_torsion_count` with the bookkeeping identity and its meaning, and the reduction maps (tier B); (4) **the cyclic-summand basis rule** ([X] v1.11 E8 finding M3), stated once; (5) **the E8 golden point name** `E8 ring=Z/<ell>^<k> n=<n> ell=<ell>` and the seven points with their primes (decision 5), with the predicted values. Pointers added in §6 and at the two older `coefficient_ring` sentences of §7, each marked "v1.9.0-pre.1". Companion pointers unchanged: [C] v1.7.4, [X] v1.11, [P] v3.5, `ENGINE_API.md` §13);

v1.8.2: 2026-09-23 (hecke-extensions slot 7, package WP-14, the slot close: **status only**. No claim, hypothesis, formula, STATUS tag or label moves; no gate, cell list, constant or rule moves. Close report `docs/implementation/slot7/WP-14_report.md`. (1) **E7 is accepted** (slot-7 WP-1): [C, `cl:pd`](1)–(2) are tested on all 54 cells of the E2 case list at $\ell \geq 5$ and fail on none, and the E7 objects are golden (add-only). The two "E7, scheduled" status phrases (§5.1 item 5, the Prediction table's row for Prediction 4.6) now say so — finding m-7 of the slot-7 referee (`docs/implementation/slot7/WP-13_referee.md`), outside v1.8.0's diff and left to the close. (2) **E6 is declared** at the slot close. §6's cell list and failure rule are now binding in [X] **v1.11** §6.2 E6 (d), which pins the threshold $r_{\max} \geq 10$; the box's "until then, the slot-7 plan's §10 and §2.4 row T3 are binding" is marked superseded. The `conway` fall-back is **routed**: every production entry point raises `hecke::NotImplementedInThisBuild` (slot-7 WP-4). (3) **§5.1 item 6: status at the close.** $\Sigma(691)$ is built (slot-7 WP-5: 1 002 225 systems, byte-identical across two builds) and published by hash and build command only; the identity is still not evaluated at $\ell = 691$, and the record still carries `atlas_unavailable` there (ruling 3b (B)). (4) **The first ext campaign is published** (`docs/matrix-engine/results/ext_first/`, 505 cells, `hecke_ext/v3`, with `analysis.txt`); the record is `hecke_ext/v3` since slot-7 stage 3, exactly as §7 (h) specified. Companion pointers: [C] **v1.7.4**, [X] **v1.11**, [P] v3.5);

v1.8.1: 2026-09-23 (hecke-extensions slot 7, package WP-13, the independent referee pass on v1.8.0; report `docs/implementation/slot7/WP-13_referee.md`). A Y-type change ([C] V2; [OD] slot-4 close-out 4): **pointers and wording only**. No claim, hypothesis, formula, STATUS tag or label is added, renamed or removed, and no gate, cell list, constant or rule moves. Every repair is marked "v1.8.1" at its site. (1) **Algorithm 6.4's cost note: which run of arm H measures what** (finding M-1). v1.8.0 attributed the pre-fix fitted exponent "4.8–5.6" to `HeckeCanonicalFieldValidation` arm H. At this revision arm H counts the *fixed* route and prints its fit (4.89, 4.23, 3.87 at $\ell = 5$, 691, 134217757, over $r = 10..20$), and it counts the pre-fix route only for $r \leq 10$ (no fit, no $r = 19$ count). The pre-fix figures, 4.8–5.6 and 144.1M at $r = 19$, are arm H's output at commit `fa829c7`. The note now names the commit for each figure. (2) **The two high-r cells, framed once** (finding M-2). The note said that the bound failed on "both high-r cells measured" and then that E6 (d) is met on the 41 gating cells, without saying that the two are not among the 41. Both notes now say that neither is a campaign cell ($n = 210 > 200$; $\ell = 134217757$ is not a campaign prime), that both are reported only, and what W1 measured for them after the fix (0.0062 and 0.0151). (3) **"Refuted at $r \geq 11$"** (finding m-1) becomes "refuted on the two high-r cells measured, at $r = 14$ and $r = 19$": no cell at $r \in \{11, 12, 13\}$ was measured before the fix.

(4) **The $r_{\max} \geq 10$ threshold** (finding m-2) is marked as WP-4's reading of ruling 4's "highest-r cells" (`e6_measure.sh`'s `--r-min` default), to be pinned in [X] v1.11. (5) **§6: the superseded automatic flip** (finding m-3) is marked superseded where it stands. (6) **§7, "The stage-2 keys" (e)** (finding m-4): which arm asserts which of the three statements, and that the golden `ext_json_*` texts are text hashes. Companion pointers unchanged: [C] v1.7.3, [X] v1.10, [P] v3.5);
v1.8.0: 2026-09-23 (hecke-extensions slot 7, package WP-12. This revision completes v1.8.0-pre.1 and publishes v1.8.0. It is an X-type change, as the pre-release announced; no claim, hypothesis, formula, STATUS tag or label is added, renamed or removed. Each rewritten claim names the validation or window that measures it. (1) **Algorithm 6.4's cost sentence** (slot-7 WP-4, commits `fa829c7`, `cab4e38`, verdict (ii)). The sentence now gives a unit: $O(r^3 \log \ell)$ $\mathbb{F}$-operations $= O(r^5 \log \ell)$ $\mathbb{F}_\ell$-multiplications with schoolbook arithmetic. "Negligible" is struck: before the slot-7 fix it was refuted at $r \geq 11$ on the high-r cells. The fix and its counts are recorded, measured by `HeckeCanonicalFieldValidation` arm H. After the fix, window W1 measured the worst gating cell, $(170, 691)$, at ratio 0.0104 over the 41 gating cells: E6 (d) MET. The pre-release's wording is quoted in the note, marked as refuted. (2) **§6's decision box: E6 (d)'s cell list and failure rule**, as a pointer (slot-7 owner ruling 4, cells C1 and rule Z). The binding text is [X] §6.2 E6 (d) from [X] v1.11, and the slot-7 plan §10 until then. The W1 status is added. Nothing in this box moves the gate's constant, numerator or denominator. (3) **§7, "The stage-2 keys" (e): `presentation` and the hashes** (slot-7 WP-2 finding). The pre-release's "leaves every hash of the record unchanged" is true of the record's `hashes` and of the golden `ext_ideals` / `ext_certify` hashes. It is false of the record text, which moves through the `settings` block and its fingerprint. The item now states both, as `HeckePresentationValidation` arms (k3) and (k4) assert them. (4) **§5.1 item 6: status**. The identity is asserted on 108 cells at $\ell \in \{5, 7, 11\}$ (slot-6 WP-9, `HeckeExtEndToEndValidation`). It is not evaluated at $\ell = 691$: $\Sigma(691)$ is slot-7 WP-5's and is not built at this revision, and under ruling 3b (B) the record carries `atlas_unavailable` there either way. The body's "not thereby asserted — that needs the atlas" was stale since v1.7.2's changelog; it is kept, and followed by the status. (5) **Addenda requested by slot-7 WP-3's content check**: none. WP-3 has not run at this revision's base (`4b6e88a`). A request that arrives later is a v1.8.1 item for WP-13 or a status item for WP-14. (6) **E8 findings checked, none contradicted here** (slot-7 WP-9's F1–F4 and WP-10's M1–M3, `docs/implementation/slot7/WP-10_review.md`). This document states nothing about `ModCtx::inv`, invariant factors, basis reduction or ρ_n over $\mathbb{Z}/\ell^k$. Its E8 sentences say three things: (D6) says that the $\mathbb{Z}/\ell^2$ Eisenstein datum is a refinement; the cone paragraph says that a claim is *not* made over $\mathbb{Z}/\ell^k$; and §7 says that `coefficient_ring` names the ring. F3's universal-coefficient point (a $\mathbb{Z}/\ell^2$ cohomology reduces injectively into the $\mathbb{F}_\ell$ one, with a Tor cokernel) agrees with (D6)'s "refinement, never a replacement". The findings are [X]'s and E8's, for WP-14. The companion pointer stays at [C] **v1.7.3**);
v1.8.0-pre.1: 2026-09-23 (hecke-extensions slot 7, package WP-12a, **spec first**: the pre-release of v1.8.0 that WP-12 completes. It carries its own identifier `1.8.0-pre.1`, so that no version number is published twice under different text ([C] V1 / V2; slot-7 plan §4.3): v1.8.0 will be WP-12's text, a different identifier. An X-type change, since the record schema gains specified keys; no claim, hypothesis, formula, STATUS tag or label is added, renamed or removed, and nothing here is emitted yet — every key below is written *before* slot-7 WP-3 emits it, so that WP-3's content gate is a check and not a stop (audit finding M-3 of `docs/implementation/hecke_extensions_slot7_plan_audit.md`). In §7, new paragraph "The stage-2 keys" and the corrected example: (1) **`sturm.proof_range` and `sturm.proof_range_source`**: type, presence rule, the closed value list {`cell_local, caller_input, method_undefined`}, the retired v2 value `dickson_count`, and the default `cell_local` at every ℓ (owner ruling 3a of 2026-09-23, slot-7 plan §10), with the record-level `anemic_pool` the decision is made over; (2) **the atlas label and `atlas_unavailable`**: the per-ideal `label` and the record-level `s_sep`, each either computed from the in-process $\Sigma(\ell)$ at $\ell \in \{5, 7, 11, 13\}$ or the placeholder {`"status": "atlas_unavailable"`} at every other ℓ (owner ruling 3b (B)); Algorithm 5.1(3) says what the default run does at an ℓ without an atlas; (3) **the split's location, stated once**: the per-ideal `split` is the `hecke_canonical_split/v1` fragment verbatim, present iff `local` and

`split_field_report = canonical`; under `own_field` the record carries `coefficient_field` only. The example's `field` and `char_split`, which no writer ever emitted, are retired;
(4) `predictions_mode` **and the prediction objects**: the record-level `predictions_mode` and `predictions` (the Kummer verdicts and the Glover factors), and the per-ideal `predicted` / `confirmed` / `unpredicted` with a closed prediction-name list; (5) **the** `presentation` **object's encoding** (audit finding m-5; slot-7 owner confirmation 15): generator indices are indices into the block basis, and the object names that basis; present only under `presentation = on`; (6) **the** `settings` **shape reconciled to the writer** (audit finding m-12): each key is an object {`value, source`}, the block holds exactly the registered keys in registry order, and the fingerprint is sixteen hex digits. After slot-7 stage 1 (1) the golden texts see only {`fingerprint`}; (7) `source` $\in$ {`default, spec, console`} **is binding and closed**, and the golden harness stamps `spec` (slot-7 owner confirmation 13). In §6, one pointer: tier (ii)'s record placement is §7's. The companion pointer stays at [C] **v1.7.3**; [C] [C, `sec:recordschema`] item 6(f) leaves the field inventory to this document, so no [C] edit is forced);
v1.7.2: 2026-09-22 (hecke-extensions slot 6, package WP-14, the slot close: **status only**. No claim, hypothesis, formula, STATUS tag or label moves. (1) **§6's pointer reduced** (slot-6 WP-12 finding M-2, item 5 of its list for the close): [X] §6.2 E6 (d) now exists — [X] v1.10 wrote the text of `e6_gate_brief.md` §6.1 into it — so the interim pointer to `docs/implementation/e6_gate_brief.md` §6.1 is marked superseded. The gate's verdict is not this document's: it is [X] v1.10's and the slot-6 close report's (`docs/implementation/slot6/WP-14_report.md`), which records the two high-r cells above 0.10 and the owner's ruling on them ([X] v1.10, `[OD]` E6-3: a slot-7 fix of the root finder, the default and this box's gate text unchanged, the verdict open, a high-r cell added to the gate's list) and does not declare E6. Finding B-1 of v1.7.1 (the reworded $(160, 691)$ sentence) is **accepted by the owner**, its departure from the slot-6 plan's prescribed wording owner-approved. (2) `conway` **'s status wording** (finding m-4): the value is parsed, validated and moves the fingerprint; its consumer `FieldPolynomialFromSettings` raises `hecke::NotImplementedInThisBuild` (console exit 5 since slot-6 WP-13). "Rejected" now says which. Slot-6 WP-13 found that the two production entry points `CanonicalField(ell, r)` and `SplitInCanonicalField` bypass that consumer, so under `conway` they silently use `least` — latent, since no report consumer exists; an open item of [X] v1.10, recorded here and not ruled. (3) **Status of the E7 and E6 items this document states**: D_n is computable since slot-6 WP-1 (so the slot-3 addendum's last entry is no longer true); the pairings, perfectness and PD adjointness of [C] [C, `cl:pd`] are *tested* on 20 cells (WP-3), socle $\leftrightarrow$ Loewy (Prediction 4.6, identity 7 of the certificate) is asserted (WP-3); identity 6 (§5 item 6, the residue atlas) is asserted on 108 cells at $\ell \in \{5, 7, 11\}$ under the standard-module convention of Claim 5.13, and the contragredient reading fails on all 63 cells where it is distinguishable (WP-9) — the "a green suite cannot detect the discrepancy" of that claim is now backed by a test for identity 6; the atlas $\Sigma(\ell)$ exists for $\ell \leq 13$ (WP-8). The companion pointer moves to [C] **v1.7.3**);
v1.7.1: 2026-09-22 (hecke-extensions slot 6, package WP-12, the independent referee pass on v1.7.0, report `docs/implementation/slot6/WP-12_referee.md`: a Y bump. No claim, hypothesis, formula, STATUS tag or label moves. Three repairs, at two sites of Claim 2.13 / §8 and one of §6. (1) **The** $(160, 691)$ **sentence** (finding B-1; *not* pointers-only). v1.7.0's "no single T_q certifies the residue degrees that $\{T_2, T_3\}$ does" contradicts the sentence before it and the `SEPARATION_SINGLE` line of `HeckeExtSeparationValidation` that it cites: T_2 alone and T_3 alone each give the family's $(\dim : f)$ list exactly, residue degrees included, and only T_5 alone under-reports one ($f = 1$ on the four-dimensional block). Both sites now read "the block partition at $(160, 691)$, with its residue degrees, is already T_2's (and T_3's); T_5 alone gives the same partition and under-reports one residue degree"; the v1.7.0 wording is kept in place, marked as refuted. v1.7.0's item (2) below, which quotes it, is superseded by this item; the Appendix A row made no residue-degree statement and is unchanged. (2) **The same two sites: where the partition identity comes from** (finding M-1; one clause, *not* pointers-only). The validation compares $(\dim : f)$ lists in record order, block counts and dimension multisets, not subspaces; that T_5's partition *is* the family's, and the "eight blocks" argument, rest on the family's decomposition refining each single operator's together with every run

having seven blocks. The clause says so. (3) **§6's pointer** (finding M-2; pointers-only). "[X] §6.2 E6 (d)" names an item that [X] v1.9 does not yet contain; the pointer now says that it is written at [X] v1.10 and that until then `docs/implementation/e6_gate_brief.md` §6.1 is binding (slot-6 plan §4.3). v1.7.0's item (1) is to be read with this pointer. The companion pointer stays at [C] **v1.7.2**);
v1.7.0: 2026-09-22 (hecke-extensions slot 6, package WP-11, on slot-6 WP-7's code side and slot-6 WP-10's ratio-criterion audit: an X bump, since one claim's verification paragraph changes its conclusion and one gate changes its denominator. No claim is added, renamed or removed, no STATUS tag moves, and no label is added or removed. (1) **§6's decision box: the E6 gate restated and reduced to a pointer** (owner decisions `[OD]` E6-1 and E6-2 of 2026-09-22, `docs/implementation/e6_gate_brief.md` §6.3). The gate is now `canonical_field_s` $\leq 0.10 \times$ `total_s` on the same 23 cells, with numerator, denominator and protocol pinned by [X] §6.2 E6 (d), the one governing statement; the $\leq 2 \times$ `own_field_s` half is deleted and `own_field_s` is a reported column. The history of both halves is kept in the box. This matches the code comments slot-6 WP-7 rewrote (`Bench/BenchDriver.cpp`, `Hecke/HeckeSettings.h`, `HeckeExt/HeckeCanonicalField.h`) and the `[E6 gate]` line `--bench ext` prints. Nothing here declares E6 or states a verdict: E6 acceptance is slot 7's ([X] §6.2 E6 (c)), and the high-r cells measured outside the 23 are an owner item, not text of this document.
(2) **Claim 2.13's verification paragraph and §8's** $(160, 691)$ **row repaired normatively** (v1.6.2 item (1); slot-5 WP-8 finding 1). The measurement of `HeckeExtSeparationValidation`'s `SEPARATION_SINGLE` line shows that T_2 alone already gives the family's seven blocks with their residue degrees, and that T_5's 4:1 *is* the family's 4:2 block — the second of Claim 2.13's two alternatives, not the first, which would give eight blocks. The closing sentence now reads "no single T_q certifies the residue degrees that $\{T_2, T_3\}$ does; the block partition at $(160, 691)$ is already T_2's"; the refuted wording is kept in place, marked as such; the Appendix A row follows. Claim 2.13's general "for $n \gg \ell$ no single T_q suffices" is a counting statement and is unchanged; the $(160, 691)$ row is no longer cited as an instance of it. (3) `sec:reportfield`**: recorded as never having existed, and not added.** `git log -S` finds no revision of this document that defined it; the text its citing sites meant is §6's decision box with its tiers (i)/(ii). An alias would give that one text a second name, which is what `[OD]` E6-2's one-authority rule was made to remove, and the code sites already cite `sec:field` (slot-6 WP-7). The remaining documentation sites are outside this package ([X] §0 changelog, `docs/architecture-detail.md`, `HeckeExt/README.md`); an `[E] <label>` citation is not scanned by the label census, so the dangling name was invisible to it. (4) **From slot-6 WP-10's audit** (`docs/implementation/slot6/WP-10_report.md`, rows 2 and 27): the §6 gate was the only cost bound in this document, and it is the one (1) restates — every other cost statement is asymptotic or a record; and Stage 1's quoted $n = 1000$ pair (14.4 s against 14.8 s) has the whole below its part, so it now says it supports "dominates" and is not a split. (5) **A status correction of v1.6.2 item (3)**, which said "with E2, E4b and E5 all accepted in slots 4–5, E7's gates are all met": E4b and E5 are accepted, E2 has not been declared accepted (two of its acceptance items were open at the slot-5 close; `docs/implementation/hecke_extensions_slot6_plan.md` §2.3 item 1). E7 starts on E2's substantive state — the record and the Loewy / socle data, which exist — by the owner's ruling of 2026-09-22 (that plan, §10 ruling 1); the `det`-twist gate of Claim 5.13 is met, as v1.6.2 said. The companion pointer stays at [C] **v1.7.2**. No separate report: the commit message `hecke-ext(S6-WP-11)` carries the package record);
v1.6.2: 2026-09-22 (hecke-extensions slot 5, package WP-13, the slot close: **status only — no claim, hypothesis, formula, tag or label moves.** Three things are recorded. (1) §2.2's closing separation sentence is **refuted by measurement**: slot-5 WP-8 measured the claim on the shipped `ComputeMultiOp` at $(160, 691)$ and $(198, 691)$, and `HeckeExtSeparationValidation` now records rather than asserts it. The repair is a normative question for the next X and is booked there, not taken here. (2) **§6's performance gate is defective as written** and is an owner question: `canonical_field_s` $\leq 2\times$ `own_field_s` cannot be met by any implementation, since `own_field_s` is, by this document's own words, "zero new arithmetic" — 1–6 μs of string formatting — and the other half is a ratio whose denominator is itself microseconds on 20 of the 23 atlas cells. Against

`total_s` at the same 0.10 threshold every one of the 23 cells passes, the worst at 2.9 %. Slot-5 WP-7 measured the table and did *not* flip the default; E6 acceptance is slot 6's. (3) The `det`-twist convention of Claim 5.13, this document's gate on E7, is **met**, and with E2, E4b and E5 all accepted in slots 4–5, E7's gates are all met. The companion pointer moves to [C] **v1.7.2**. Close report: `docs/implementation/slot5/WP-13_report.md`);
v1.6.1: 2026-09-22 (hecke-extensions slot 5, package WP-12, the independent referee pass on v1.6.0: **a Y bump — pointers only. No claim, hypothesis, formula, tag or label moves.** Two repairs, both of v1.6.0's own text, and one of a reading v1.6.0 left standing. (1) The version block said the `det`-twist convention is restated "in the one-sentence form an implementation quotes verbatim (`HeckeExt/GloverFactors.h` and the console's `glover_factors` both do)". *Neither quotes it*: both state the convention correctly, in their own words — `GloverFactors.h` as "this is `Sym^n` of the STANDARD module, not of its dual: the two differ by `det^{-n}`, which would shift every `j_i`" and script `017_predictions.oc` likewise. The sentence is drafted so that they *can* quote it unchanged, and now says so; the mathematics, in all three places, agrees. (2) The last bullet of the slot-3 addendum, "Still not computable: the production ι … the record `hecke_ext/v1`", is correct *of its own date* but is the last addendum in that section, so it reads as the present state; it now points at what slots 4 and 5 built, leaving D_n (E7) as the only entry still true. The companion pointer moves to [C] **v1.7.1**. Report: `docs/implementation/referee_report_2026-09-22_contract_v1_7_0.md`);
v1.6.0: 2026-09-22 (hecke-extensions slot 5, package WP-11: **the convention that gates E7, and §7 at `hecke_ext/v2`** — an X bump, since one claim's preamble acquires a consequence and one schema key changes its value. No claim is added, renamed or removed and no tag moves.
(1) **Claim 5.13's `det`-twist convention is recorded as E7's gate.** The convention itself — Sym^n is the *standard* symmetric power of [C, `def:rhon`] and not its contragredient, the two readings shifting every j_i by $-n$ — has been normative since v1.5.0; what this version adds is the consequence [WP4-7] did not draw: E7's subject *is* D_n and $\rho_n^\vee$, its first use of the factor multiset is item 6 of §5.1, and under the two readings that comparison differs by θ^{-n} on every factor. Both readings being internally consistent, **no green validation suite can detect the discrepancy** — every checkable invariant is convention-independent or self-consistent — so a wrong reading surfaces as an E7 finding indistinguishable from a result. **This document at v1.6.0 is therefore a precondition of E7's start**, and the convention is restated in the one-sentence form an implementation can quote verbatim — `HeckeExt/GloverFactors.h` and script `017_predictions.oc` both state the convention, in their own words, and can now quote this sentence unchanged (v1.6.1: v1.6.0 said "an implementation quotes verbatim … both do", which no file in the tree did). (2) **§7 moves to `hecke_ext/v2`**, making the *two* repairs v1.5.0 booked for "the same slot-5 revision" (owner ruling `[OD]` slot-4 close-out 5 of 2026-09-22; implemented by slot-5 WP-1): `u_ell` is carried at **record level with a single `resumable`**, so `resumable_items` counts *decisions* and no `ideals[]` entry carries the object — a change to an existing key's *value*, hence a version bump, made before the first emission — and an `ideals[]` entry gains the per-ideal `groups` list carrying the `SameMaximalIdeal` witness, `modules` keeping its "one entry per ambient group" shape exactly as v1.5.0 predicted. Three further schema corrections from the same package: `canonical` is keyed by the ambient group at *ideal* level (the example had it nested inside `modules`); a closed-list scalar a record has not computed is `null` and never dropped; and `les.hecke_equivariant` is tri-state, a cone-only record not having evaluated it. The example record is updated to all of these. (3) Claim 1.1's hypothesis-versus-domain paragraph and Claim 5.13's recursion are *unchanged*: both landed at v1.5.0/v1.5.1, so the slot-5 plan's reading that they were outstanding was stale — recorded here so the next reader does not look for them. The companion pointer moves to [C] **v1.7.0** ([C, `obl:coneexact`](iii) at every ℓ with its proof, [C, `sec:recordschema`] item 6's nine rules; v1.6.1 moved this pointer to v1.7.1). Report: `docs/implementation/slot5/WP-11_report.md`);
v1.5.2: 2026-09-22 (hecke-extensions slot 4, package WP-11, the slot close: **status only — no claim, hypothesis, formula, tag or label moves**. What the slot built of this document's programme: **E2** — the extension record `hecke_ext/v1` of §7 is written, for the bulk complex, by a *separate* writer (`HeckeExt/HeckeExtRecord` + `HeckeExtReport`, never an extension of

`Hecke/HeckeReport`), with the one tri-state `Final`, the settings block and its fingerprint, and the resumable sidecar (slot-4 WP-1); and `HeckeExt/HeckeExtDriver` turns a cell into that record through Stages 1–5, with a streamed operator for a Sturm prime outside S_0 and the U_ℓ resume (WP-2). **Claim 8.1 (8.1) is verified end to end through that driver** at $(10, 691)$ — the four exact identities in $M_3(\mathbb{F}_{691})$, *including* $3(U_{691} - 1) = -542(T_2 + 24)$, which exercises U_ℓ and $\tau(691)$ — with the basis-independent content separated from the stored-basis pins, as the claim's "Verified" paragraph now says. **E3** — a *production* ι^* exists (`Hecke/HeckeInvolution`, `HeckeExt/HeckeSignSplit`), so the sign split of every stored H^q and of every computed Hecke matrix, and the Eisenstein line of D6, are pipeline objects and no longer validation-local (WP-3). **E6** — the first two pre-record units ship: `HeckeExt/HeckePredictions` (Claim 1.1's predicate) and `HeckeExt/GloverFactors` (Claim 5.13's recursion), both gated by `HeckePredictionsValidation` against independent formulas — Glover against the *Brauer character* of the real ρ_n on every ℓ-regular class, which is a proof cell by cell. Still open, and unchanged: the atlas, $S_{\mathrm{sep}}(\ell)$, the canonical field, and the boundary / cone / per-sign record objects (slot 5). The [C] pointer moves to v1.6.2. Report `docs/implementation/slot4/WP-11_report.md`);
v1.5.1: 2026-09-22 (hecke-extensions slot 4, package WP-9, the independent referee pass on v1.5.0: **four mathematical repairs, two of them to false statements**, no claim added, renamed or removed and no tag moved. (1) Claim 1.1's new predicate paragraph is rewritten: v1.5.0 named only the two instances $k = \ell - 1$ and $k = \ell$ of the branch $(\ell - 1) \mid k$ and so left every even k with $(\ell - 1) \mid k$ and $k > \ell - 1$ — at $\ell \in \{2, 3\}$ *every* even k, the Kummer branch being vacuous there — with no branch at all, and it answered $k = \ell$ with a fact about *odd* ℓ although the only in-range instance is $(n, \ell) = (0, 2)$, where the verdict is `false` and not the "degenerate `true`" the sentence gave. The general von Staudt–Clausen branch is now stated, and the $B_k = 0$ remark moved to odd k, where it is true. The shipped `kummer_verdict` already decided all of these correctly; the document did not. (2) Claim 5.13(4)'s counts are corrected to $\lfloor (n+1)/(\ell+1) \rfloor$ steps and $2\lfloor (n+1)/(\ell+1) \rfloor + 1$ raw factors: v1.5.0's bound undercounts at $n = \ell$ (2 factors against 1) and at $n = 2\ell + 1$ (4 against 3). (3) Claim 5.3's irreducibility step is moved to A_S, where the idempotent a reducible $\min_V(t_p)$ produces actually lives; inside the residue field the argument assumed what it proves. (4) Claim 5.3's "lower bound in general" is replaced by the surviving-block invariant — every π_p irreducible, hence $\deg \pi_p$ divides every residue degree — v1.5.0's reason being false in both directions on a general non-local block. The companion pointer stays at [C] **v1.6.1**. Report: `docs/implementation/referee_report_2026-09-22_contract_v1_6_0.md`);
v1.5.0: 2026-09-21 (hecke-extensions slot 4, package WP-8: **five open items of this document resolved, and the findings of the slot-4 units it owns** — an X bump, since one claim is added and one certificate gains a hypothesis. Certificate 5.2 states what a *reported* witness must satisfy (w commuting with the family, the hypothesis a matrix witness admits and a coefficient list makes automatic) and that the sub-block order after a split is deterministic *per mode* and not mode-independent — the partition is, the list is not (owner confirmation of 2026-09-21, item 1(c): per mode, explicitly not an intrinsic order); Claim 5.3 is new, $\dim_{\mathbb{F}_\ell} \bar{A} = \mathrm{lcm}_p \deg \pi_p$ on a local block, which is why budget mode's `residue_degree_lower_bound` is sharp there; §7 carries `phase_reached` per block *and* the run's high-water mark under separate keys, the settings under their *registered* names (`sturm`, `sturm_unavailable`, and the new `commutant_budget`), the two sidecar rules (a record-relative path, an injective name), the closed name list behind `certificate_failed:<code or certificate name>` and the four scalars that sit inside `groups`. From the slot-4 units: Claim 5.13 is new — the Glover factors of Sym^n by the θ-recursion, **with the standard-module convention that fixes the** det**-twists**, which item 6 of §5.1 had left free — and Claim 1.1 separates the criterion's hypothesis $n < \ell - 1$ from the divisibility predicate's domain. From the E2b driver (slot-4 WP-2), three further §7 rules and one wording repair: a *resumed* run writes the settings block a *fresh* run at the same effective values would write, which is what makes "resume to the default budget equals a fresh default run" true byte for byte and is otherwise a fingerprint two implementations could disagree on; `u_ell` is carried *per ideal* although the verdict is per *cell*, so one decision is emitted once per ideal and `resumable_items` counts entries; an `ideals[]` entry is *per group*, the cross-group matching of Stage 2 being a driver diagnostic that no key reports;

and Claim 8.1's c'_q line is marked **basis-dependent**, the rest of that paragraph not being. The companion pointer moves to [C] **v1.6.0** (slot-4 WP-8, the eight items v1.5.2 deferred; no label this document cites moved). Report: `docs/implementation/slot4/WP-8_report.md`);
v1.4.2: 2026-09-17 (hecke-extensions slot 3, package WP-11, the slot close: **status only**, a Y bump — no claim, certificate, tag or label moves. Strict locality mode (Certificate 5.2) is *implemented* (slot-3 WP-3), E1b's Sturm sweep with both `unavailable` modes (Certificate 5.8) is implemented on real blocks (WP-2), and the equivariant cone with its Hecke action exists (WP-5, WP-6), so X_3, X_6 are computable on the model complexes; the addendum of §1.2 records it. The companion pointer moves to [C] **v1.5.2** (slot-3 WP-11, status only; no label this document cites moved). Report: `docs/implementation/slot3/WP-11_report.md`);
v1.4.1: 2026-09-17 (hecke-extensions slot 3, package WP-8: **one certificate made precise**, and the companion pointer moved. Certificate 5.6 now *states* that the reported generator is the least standard basis vector outside $\mathfrak{m}_S V$, in a paragraph of its own, in place of v1.4.0's implementation note that the code fixes that choice while the certificate says "any $v \notin \mathfrak{m}_S V$" — the gap slot-2 WP-4 §7.2 reported and slot-2 WP-11 did not take up. **Why a Y bump and not an X**: the mathematical assertion of the certificate is unchanged — every $v \notin \mathfrak{m}_S V$ still generates, and the certificate is true for each — what is added is a *conformance* requirement on what a record may report, which no claim of this document depends on; nothing else in this revision. The companion pointer moves to [C] **v1.5.0** (slot-3 WP-8, the seven items v1.4.2 deferred; no label this document cites moved, so the citations are unaffected). Report: `docs/implementation/slot3/WP-8_report.md`);
v1.4.0: 2026-09-16 (hecke-extensions slot 2, package WP-11: **the consequential revision**, the first since v1.1 in which claims of this document change; it cited the contract at [C] **v1.4.1**, and with it this document agrees with the implementation plan [X] v1.5.1 and is the freeze candidate of [X] §8 (OQ17's $(160, 691)$ verdict being in since v1.3.3). *Two predictions refuted by computation*, each kept with its old text, the evidence and the corrected statement rather than rewritten into its negation: §8 / §7's "at a Kummer prime $r^1 = 0$ on the Eisenstein class, so $H^1_! = H^1$" is *false* on all ten Kummer cells — $\operatorname{rank} r^1 = 1$, $\dim H^1_! = \dim H^1 - 1$ (`HeckeBoundaryParabolicValidation`, slot-2 WP-7, 2026-09-16) — and (D6)'s "both reduce to the same line" is *sharpened*: on the unsplit H^1 the joint $1 + q^{n+1}$ eigenspace is two-dimensional at every Kummer cell, the socle of the block; the socle semantics is kept. *Made precise* from [X] v1.5.1 and the slot-1/slot-2 package reports: §7 — the tri-state `final` with R3.2's reason list, the keys `settings`, `resumable`, `resumable_items`, `sturm.mode`, `sturm_primes`, `coefficient_ring`, the *per-complex* `q_max` / `truncated` / `q_max_source` / `degrees`, `P` keyed by prime, `hecke_report/v2`; Definition 2.8's cap as the setting `sturm` and the record-level `not_applicable` at $n = 0$; Certificate 5.8's `unavailable` branch (`nonfinal` default / `heuristic`); Algorithm 5.1(2) / Certificate 5.2 with the **strict mode** (Frobenius radical, Berlekamp subalgebra, residue witness over $\bar{A}$, termination, the $(2, 30)$ example) and the budget-mode reason `budget_exhausted:locality`; Certificate 5.6's canonical generator; Algorithm 5.5 as implemented; Certificate 5.7's **socle bound** and the sharp unknown count $u \le mD$ (its [to be verified computationally] on the cost discharged); the canonical field as the default report form under the E6 gate (§6); §8's $(198, 691)$ row ("possible ambiguity documented, LMFDB to-do"); Stage 7's reach reframed, the first campaign at $n \le 200$; D_n and the pairing *computed* (E7, scheduled) at §0.3, Prediction 4.6 and identity 5; (D6)'s $\mathbb{Z}/\ell^2$ refinement as a refinement. *The boundary chapter of [C] v1.4.1 folded in* at (A1), (A2), (A3): [C, `obl:taubdry`](v) is (v.a)–(v.c) and degree 0 alone at $\nu \ge 2$; the circle model carries the $-I$ self-relations ([C, `prop:circlemodel`](4)); the coordinate Φ of [C, `prop:bdrychar`](3); M_β on M^T only; [C, `alg:taubdry`] is the canonical τ_∂ and [C, `rmk:taubdrynoiota`] says it is not ι-symmetric; (vii) at $\ell \in \{2, 3\}$ a COMP record. §1.2 gains the slot-2 addendum; [C, `cl:chi`]'s [to be verified computationally] stays; Appendix A follows. Not changed: every [standard, reference] claim is recalled, not re-proved; no claim of §2–§3 moves);
v1.3.3: 2026-09-16 (hecke-extensions slot 1 close; a status pass, no claim of this document changed: the $(160, 691)$ [to be verified computationally] of Claim 2.13 and §8 is *resolved* by the slot-1 probe — all seven blocks local, the shipped per-operator refinement already separates T_5's 4:1 block, the first of the two alternatives holds — and the row says so; Claim 2.9's computed values gain

$B'(258, 691) = 2$ and the values the implementation `HeckeExt/HeckeSturmBound` now pins; the character P_j is written as the reduced representative X mod π_w (constant at $r = 1$) in Claim 3.3 and §7; §1.2 gains a dated addendum on what slot 1 built (the boundary Hecke operator on X_2, X_5, the read-only locality certificate with its all-local grid verdict, the Sturm bound, the identity certificate); the Appendix C row "boundary group at $\ell = 2$" no longer calls the truncation policy open — it is [C] [C, `open:ell2`] / [C, `oc1a`]);
v1.3.2: 2026-09-16 (coherence pass: the (A2) closing sentence of Stage 0 no longer calls the $\ell = 2$ truncation policy "open ... not decided here" — it is decided in [C] v1.3.0, [C, `open:ell2`] / [C, `oc1a`], $q_{\max}$ a recorded run parameter; (A3)'s "not a pullback along any lift" is qualified "for n even"; and the involution is no longer called "not implemented" — two validations build it. No claim of this document changes);
v1.3.1: 2026-09-16 (owner decision OQ3 applied at its two sites, (A3) and the `iota_sign` paragraph of §7: [C] v1.2.0 defines $\iota^* := -\iota^*_{\mathrm{geo}}$, so that H^+ is by definition the Eisenstein side and `iota_sign` names that normalisation; no claim of this document changes);
v1.3: 2026-09-16 (**the scope cut**: the nine statements of the former §1.1–§1.2 and §1.4 — the six groups, the amalgam, the dimension of H^1, the Dickson–Bockstein claim, the boundary groups, the six-term sequence and its display, the boundary characters, Poincaré duality — are removed from this document and cited from `gl2_contract.tex` instead, which is now [C]; the four of them that [C] corrects are named as superseded in §1.1; §1.1 replaces the removed text, Appendix A loses their STATUS rows, and the cone pointers into the frozen `monodromy_v5.tex` become pointers into [C]);
v1.2.2: 2026-09-15 (the owner's decisions on the OQ2 verdict's eight open points applied);
v1.2.1: 2026-09-15 (independent audit of the v1.2 revision applied);
v1.2: 2026-09-15 (OQ2 verdict applied, Appendix C);
v1.1: 2026-09-14 (audit response applied, Appendix B);
v1.0: 2026-09-14, pre-audit

D.2 The title page of 1.25.0-pre.1: author line and abstract, verbatim

Orbifold Cohomology of GL(2) — companion to `gl2_contract.tex`

This document specifies, in the notation and normalisations of the binding mathematical contract `gl2_contract.tex` (cited as [C]), what it means to compute and classify *all* extension classes of Hecke modules occurring in the cohomology groups the project can produce, and how to do it with deterministic certificates at every step. It is the concise LaTeX version of the owner's two notes `hecke_extensions_mathematics.md` ([M]) and `hecke_extensions_computational_strategy.md` ([S]), made rigorous and computation-oriented. The answer to "which T_q are needed" is reduced to a certified local statement, one maximal ideal at a time: the operators computed must generate the local Hecke algebra $\mathbb{T}_{\mathfrak{m}}$, and the extension data are the isomorphism invariants of the $\mathbb{T}_{\mathfrak{m}}$-modules $H^\bullet_{\mathfrak{m}}$ (Loewy and socle series, active tangent directions, Jordan types, positions of the canonical submodules). Every claim carries a STATUS tag; the tags are indexed in Appendix A. The document is accurate about what the code base implements today (the single- and multi-operator primary decomposition with certified residue degrees, Stages S10–S11) and what is new. Version 1.1 applies the independent audit of 2026-09-14 item by item (Appendix B); the two blocking findings changed the Sturm certificate (a per-(k, ℓ) computed bound B', Claim 2.9) and the tangent-space test (the coefficient field, Remark 3.2). Version 1.2 applies the independent verdict on Open Question 2 (the Hecke action on the mapping cone, Appendix C): the cone action is defined from a certified triple $(\tau, \tau_\partial, \eta)$ over $\mathbb{Z}$ with the cochain homotopy *derived*, which removes the $\mathrm{Hom}(\ker r^1, \mathrm{coker}\, r^0)$ ambiguity that v1.1 did not state ((A2), Prediction 4.5). Version 1.2.1 applies the independent audit of that revision, all of it inside (A2): the existence statement is corrected (τ_∂ and η over $\mathbb{Z}$ for every ℓ; the bulk τ is the canonical table of [C, `alg:tau`], integral when no fill used the mod-ℓ fallback), the realisation of the Hom ambiguity is marked as proved *over a field*, the non-vanishing of

h_η on X_3 is marked undecided rather than proved, the contractibility and two-dimensionality of X are stated where they are used, and the two Appendix C rows that record undecided owner points are relabelled as working assumptions. Version 1.2.2 applies the owner's decisions of 2026-09-15 on all eight open points of that verdict: the working assumptions become settled text and Appendix C's rows become plain ADOPTED. Three of the eight were decided as *amendments* to the verdict's recommendation; the one with mathematical content here is the boundary group, fixed as $\Gamma_\infty = \langle T \rangle \times \langle -I \rangle$ (the Borel reading, which is [C, `sec:six`]'s own definition), so that the circle model of (A1) is its specialisation at $2 \in R^\times$ and $H^\bullet(\partial; M)$ is *unbounded above in degree* at $\ell = 2$ ((A1), (A2)). Version 1.3 carries out the scope cut: [C] is now the unified contract `gl2_contract.tex`, which supersedes `hecke_action.tex` and `monodromy_v5.tex`, and the nine statements about the six groups that this document used to state are stated there instead and cited here (§1.1). This document is what the cut left it as: the conjectural and predictive classification specification — the Hecke algebra acting, the separation and generation bounds, the local algebras, the classification data (D1)–(D7), the predictions, the stage-by-stage algorithms with their certificates, the scalar extension and the output schema. Version 1.4.0 (slot 2 of the extension stage) is the first revision since v1.1 in which claims change: two predictions were refuted by the boundary computation of slot 2 and are corrected with the evidence in place (§8, (D6)); the record schema, the Sturm, locality and generation certificates and the local-algebra stage are made precise from what slots 1–2 built; and the boundary chapter of [C] v1.4.1 is folded in.